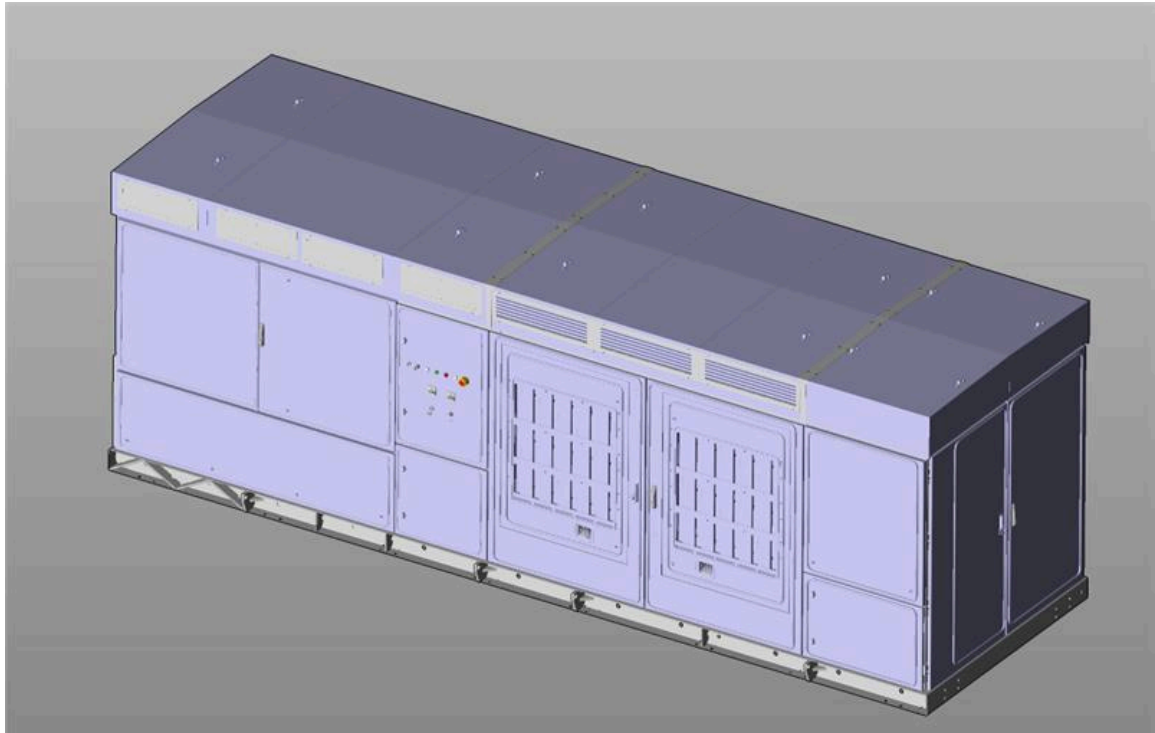


POWER ELECTRONICS

POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

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Project	ESQ_FS4010M10U000012_000		
Description	HEM		
Project Code	FS4010M10U10001100410010020101014		
Designed	I+D 1		
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Structure identifier overview

POWER_IE_cambio_estrc

Plant identifier	Plant description	Mounting location identifier	Mounting location description
MOI	INVERTER MODULE	ETR	REMOVABLE DC/AC STAGE
ZMI	INVERTER ASSEMBLY AREA	MPP	MAIN PROTECTION PANEL
CDT	MV TRANSFORMER CUBICLE	PTA	AUXILIARY TRANSFORMER ASSEMBLY
PDCL		PMV3	FAN PRE-ASSEMBLY 3
BPA		PMV4	FAN PRE-ASSEMBLY 4
		PMV5	FAN PRE-ASSEMBLY 5
		PMV6	FAN PRE-ASSEMBLY 6
		AFC	C FILTER CABINET
		AFC.PFC	PANEL FILTER C
		AFC.PCW	CROWBAR ASSEMBLY
		AFC.PMV7	FAN PRE-ASSEMBLY 7
		EAC	ENCLOSURE CONTROL CABINET
		EAC.PSA	POWER SUPPLY PANEL
		EAC.PCL	CONTROL PANEL
		EAC.PPU	PUSHBUTTONS PANEL
		EAC.PPCU	MAIN CUSTOMER PANEL
		EAC.POCU	CUSTOMER OPTIONS PANEL
		EAC.PTCU	CONTROL USER PANEL
		EAC.PCDU	DU CONTROL PANEL
		ADU	DU CABINET
		ADU.PDC1	DC DISCONNECTOR PRE-ASSEMBLY 1
		ADU.PDC2	DC DISCONNECTOR PRE-ASSEMBLY 2
		ADU.PDC3	DC DISCONNECTOR PRE-ASSEMBLY 3
		ADU.PDC4	DC DISCONNECTOR PRE-ASSEMBLY 4
		ADU.PIND1	INDUCTANCES PRE-ASSEMBLY 1
		ADU.PIND2	INDUCTANCES PRE-ASSEMBLY 2
		ADU.PIND3	INDUCTANCES PRE-ASSEMBLY 3
		ADU.PIND4	INDUCTANCES PRE-ASSEMBLY 4
		ADU.PFU1	FUSES PRE-ASSEMBLY 1
		ADU.PFU2	FUSES PRE-ASSEMBLY 2
		ADU.PADU	CUSTOMER PV INPUT
		ADU.PMV1	FAN PRE-ASSEMBLY 1
		ADU.PMV2	FAN PRE-ASSEMBLY 2

Structure identifier overview

POWER_IE_cambio_estrc

Plant identifier	Plant description	Mounting location identifier	Mounting location description
		ADU.PPD	DC PROTECTIONS PANEL
		ADU.PBP1	BUS PLUS ASSEMBLY 1
		ADU.PBP2	BUS PLUS ASSEMBLY 2
		ADU.PAD1	PANEL DC METER 1
		ADU.PAD2	PANEL DC METER 2
		AMT	MEDIUM VOLTAGE ENCLOSURE
		PMC	PANEL SWITCHGEAR MANEUVER
		AMT.PMC	PANEL SWITCHGEAR MANEUVER
		CMT	MEDIUM VOLTAGE CELL
		CRB	CELL REMOUNT BUSBAR
		PMV8	FAN PRE-ASSEMBLY 8
		PMV9	FAN PRE-ASSEMBLY 9
		PMV10	FAN PRE-ASSEMBLY 10
		PMV11	FAN PRE-ASSEMBLY 11
		CEM	CABINET ECON MODE
		BPA	BUS PLUS ADVANCED
		BPA.PFU3	ADVANCED FUSES PRE-ASSEMBLY
		BPA.PBESS	CUSTOMER BESS INPUT
		PFU3	ADVANCED FUSES PRE-ASSEMBLY
		PBESS	CUSTOMER BESS INPUT
		SPDP	SURGE PROTECTION DEVICE PANEL
		EAC.PKC	HEATING KIT PANEL
		ADU.PTBP	PV- (PT+BP) FUSES PRE-ASSEMBLY
		EAC.PRC	HEATING KIT PANEL
		EAC.PTH	THERMISTOR PANEL
		ADU.PPCU	

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Superior level	Structure	Page	Page description
		1	Title page / cover sheet
		2	Information
		3	Structure identifier overview
		4	Structure identifier overview
		5	Table of contents
		6	Table of contents
		7	Table of contents
		8	Table of contents
		9	Table of contents
		10	Table of contents
		11	Table of contents
		12	Table of contents
		13	Table of contents
		14	Table of contents
		15	Table of contents
		16	Table of contents
		17	Table of contents
		18	Table of contents
		19	Table of contents
		20	Table of contents
		21	Table of contents
		1	Track changes
		10	General layout
		11	General layout -Main AC Busbars
		12	ZMI --> General layout
		13	ZMI --> General layout
		14	Comms RS-485
		15	Earth interconnection

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Superior level	Structure	Page	Page description
		20	Legend connectors
		23	Legend connectors
		24	Legend connectors
		30	Legend terminal blocks
		40	Pushbuttons details
		50	SLD --> HEM Negative Pole Grounded (32 PV) - 1 Protected Pole (S4)
		55	PCB ´s General Layout
		20	Designation terminals and connectors
		100	Interconnection
		101	Interconnection
		102	Interconnection
		103	Interconnection
		2000	Cable overview
		3000	Plug overview : =HEM++MOI+-XC001.1-M1 - =HEM++MOI+-XC046.2
	ETR	1	Title page / cover sheet
	ETR	10	General layout
	ETR	20	Designation terminals and connectors
	ETR	200	Capacitor bus dc-link
	ETR	201	Igbt assembly
	ETR	202	Igbt assembly
	ETR	203	Igbt assembly
	ETR	204	Snubber circuit
	ETR	400	V/I conversor
	ETR	401	PCB IGBT distribution
	ETR	402	PCB control and measures
	ETR	1000	Bill of materials
	ETR	2000	Cable overview
	ETR	3000	Plug overview : =HEM++MOI+ETR-XC010.1 - =HEM++MOI+ETR-XC023.1

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Superior level	Structure	Page	Page description
MOI	ETR	3001	Plug overview : =HEM++MOI+ETR-XC024.1 - =HEM++MOI+ETR-XC032.1
		10	Interconnection Panel ++CDT --> ++ZMI
		11	Interconnection Panel Control --> C Filter and Modules
		12	Interconnection Panel Control --> C Filter and Modules
		13	Interconnection Panel Control --> C Filter and Modules
		14	Interconnection Panel C Filter --> Modules
		15	Interconnection Panel DU --> Modules
		16	Interconnection Panel DU --> Modules
		17	Interconnection Panel DU --> SSAA
		20	Designation terminals and connectors
		100	Interconnections ZMI/CDT - P. Supply
		101	Interconnections ZMI/CDT - Comms/Measures
		102	Interconnections ZMI/CDT - Comms/Measures
		103	Interconnections ZMI/CDT - Comms/Measures
		104	Interconnections PSA//CDT
		105	Interconnections - Fan PCL//CDT
		110	Interconnections - Fan PCL//Modules-AFC
		111	Interconnections F.O. PCL//Modules AC/DC
		112	Interconnections PCL/Modules
		113	Interconnections PCL/MPP Measure AC
		114	Interconnections PPD//PPCU
		120	Interconnections PSA/AFC
		123	Interconnection SPD/Induct - C Filter
		130	Interconnection PSA/ADU
		131	Interconnection SSAA - PSA
		140	Interconnections - Fan-DC Meter PCL//ADU
	141	Interconnections PCL//ADU.PPD/PFU1/2	
	150	Interconnections - Comms RS485 XLOK	

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Superior level	Structure	Page	Page description
ZMI		151	Interconnections - RJ45
		152	Interconnections - RJ45
		200	AC Output
		205	AC Inductance (Power busbar)
		206	AC Inductance (Terminals block)
		220	Earth connection
		221	Earth connection
		1000	Bill of materials
		2000	Cable overview
		3000	Plug overview : =HEM++ZMI+-XC045.1 - =HEM++ZMI+-XC056.1-M4
		3001	Plug overview : =HEM++ZMI+-XC201.1 - =HEM++ZMI+-XC300.2
		3002	Plug overview : =HEM++ZMI+-XC301.2 - =HEM++ZMI+-XC364.1
	MPP	1	Title page / cover sheet
	MPP	10	Layout
	MPP	20	Designation terminals and connectors
	MPP	100	Interconnection PFC/MPP
	MPP	200	AC Main Protection
	MPP	1000	Bill of materials
	MPP	2000	Cable overview
	MPP	3000	Plug overview : =HEM++ZMI+MPP-XC120.1 - =HEM++ZMI+MPP-XC120.1
	PTA	1	Title page / cover sheet
	PTA	200	SSAA transformer
	PTA	1000	Bill of materials
	PMV3	1	Title page / cover sheet
	PMV3	10	Layout PMV3
	PMV3	20	Designation terminals and connectors
	PMV3	200	Power module 4 - Fan
	PMV3	1000	Bill of materials

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Superior level	Structure	Page	Page description
ZMI	PMV3	2000	Cable overview
	PMV3	3000	Plug overview : =HEM++ZMI+PMV3-XC200 - =HEM++ZMI+PMV3-XC201.2
	PMV4	1	Title page / cover sheet
	PMV4	10	Layout PMV4
	PMV4	20	Designation terminals and connectors
	PMV4	200	Power module 3 - Fan
	PMV4	1000	Bill of materials
	PMV4	2000	Cable overview
	PMV4	3000	Plug overview : =HEM++ZMI+PMV4-XC203 - =HEM++ZMI+PMV4-XC204.2
	PMV5	1	Title page / cover sheet
	PMV5	10	Layout PMV5
	PMV5	20	Designation terminals and connectors
	PMV5	200	Power module 2 - Fan
	PMV5	1000	Bill of materials
	PMV5	2000	Cable overview
	PMV5	3000	Plug overview : =HEM++ZMI+PMV5-XC206 - =HEM++ZMI+PMV5-XC207.2
	PMV6	1	Title page / cover sheet
	PMV6	10	Layout PMV6
	PMV6	20	Designation terminals and connectors
	PMV6	200	Power module 1 - Fan
	PMV6	1000	Bill of materials
	PMV6	2000	Cable overview
	PMV6	3000	Plug overview : =HEM++ZMI+PMV6-XC209 - =HEM++ZMI+PMV6-XC210.2
	AFC	1	Title page / cover sheet
	AFC	20	Designation terminals and connectors
	AFC	100	Interconnection AFC - AFC.PFC
	AFC	101	Interconnections - Comms RS485
	AFC	199	Crowbar Output Busbar Connection

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Superior level	Structure	Page	Page description
ZMI	AFC	200	Capacitor filter - AC RFI
	AFC	201	Capacitor filter current measures
	AFC	202	Capacitor filter
	AFC	203	Capacitor filter - PCB ´s discharge
	AFC	300	Capacitor filter - contactors maneuver
	AFC	400	PCB ´s - V/I conversor
	AFC	1000	Bill of materials
	AFC	3000	Plug overview : =HEM++ZMI+AFC-XC215.2 - =HEM++ZMI+AFC-XC215.2
	AFC.PFC	1	Title page / cover sheet
	AFC.PFC	10	Layout
	AFC.PFC	20	Designation terminals and connectors
	AFC.PFC	100	Interconnections PSA/AFC
	AFC.PFC	102	Interconnection SPD/Induct - C Filter
	AFC.PFC	103	Interconnection AFC - AFC.PFC
	AFC.PFC	300	Terminal Block
	AFC.PFC	400	PCB ´s - Inputs/Outputs
	AFC.PFC	1000	Bill of materials
	AFC.PFC	3000	Plug overview : =HEM++ZMI+AFC.PFC-XC120.2 - =HEM++ZMI+AFC.PFC-XC281.2
	AFC.PCW	1	Title page / cover sheet
	AFC.PCW	400	PCB - Crowbar trigger
	AFC.PCW	2000	Cable overview
	AFC.PMV7	1	Title page / cover sheet
	AFC.PMV7	10	Layout PMV7
	AFC.PMV7	20	Designation terminals and connectors
	AFC.PMV7	200	Capacitor filter FAN
	AFC.PMV7	1000	Bill of materials
	AFC.PMV7	2000	Cable overview
	AFC.PMV7	3000	Plug overview : =HEM++ZMI+AFC.PMV7-XC250 - =HEM++ZMI+AFC.PMV7-XC251.2

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Superior level	Structure	Page	Page description
ZMI	EAC	1	Title page / cover sheet
	EAC	100	Interconnections - RJ45
	EAC	101	Optional: Interconnections - RJ45
	EAC	102	Optional: Measure AC meter
	EAC	200	Earth connection
	EAC	300	CAISO
	EAC	1000	Bill of materials
	EAC	2000	Cable overview
	EAC.PSA	1	Title page / cover sheet
	EAC.PSA	10	Layout
	EAC.PSA	20	Designation terminals and connectors
	EAC.PSA	50	Single Line Diagram
	EAC.PSA	100	Interconnections PSA/ADU
	EAC.PSA	101	Interconnection SSAA - PSA
	EAC.PSA	102	Interconnections PSA/AFC
	EAC.PSA	103	Interconnections PSA//PCL
	EAC.PSA	104	Interconnections PSA//CDT
	EAC.PSA	105	Interconnections - PSA//Door
	EAC.PSA	106	Interconnections - PSA//PPCU
	EAC.PSA	107	Interconnections - PSA//Door
	EAC.PSA	108	Interconnections PSA/ POCU
	EAC.PSA	109	Interconnections - PSA//PPCU
	EAC.PSA	110	Interconnections - PSA//PTCU
	EAC.PSA	200	DC/DC Board
	EAC.PSA	201	Three Phase Bridge (Input)
	EAC.PSA	202	Three Phase Bridge (Output)
	EAC.PSA	300	120Vdc Main Bus Distribution
	EAC.PSA	301	Temporitation Circuit PCB DC/DC

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Superior level	Structure	Page	Page description
ZMI	EAC.PSA	400	PCB - DC/DC (PSA)
	EAC.PSA	1000	Bill of materials
	EAC.PSA	3000	Plug overview : =HEM++ZMI+EAC.PSA-XC253.1 - =HEM++ZMI+EAC.PSA-XC414.1
	EAC.PCL	1	Title page / cover sheet
	EAC.PCL	10	Layout
	EAC.PCL	20	Designation terminals and connectors
	EAC.PCL	100	Interconnections - Fan PCL//ADU
	EAC.PCL	101	Interconnections - Fan PCL//Modules-AFC
	EAC.PCL	102	Interconnections - Fan PCL//CDT
	EAC.PCL	103	Interconnections PSA//PCL
	EAC.PCL	104	Interconnections PCL/Modules
	EAC.PCL	105	Interconnections PCL//ADU.PPD/PFU1/2
	EAC.PCL	107	Optional Interconnections MPP//PCL
	EAC.PCL	108	Optional Interconnections PCL//PTCU
	EAC.PCL	200	Secondary Transfomer - SPD
	EAC.PCL	201	Fan distribution
	EAC.PCL	202	24Vdc distribution FPGAs
	EAC.PCL	203	24V DC distribution
	EAC.PCL	204	24Vdc distribution
	EAC.PCL	300	Terminal-strip
	EAC.PCL	301	Switch
	EAC.PCL	400	Panel control board
	EAC.PCL	1000	Bill of materials
	EAC.PCL	2000	Cable overview
	EAC.PCL	3000	Plug overview : =HEM++ZMI+EAC.PCL-XC253.2 - =HEM++ZMI+EAC.PCL-XC401.2
	EAC.PPU	1	Title page / cover sheet
	EAC.PPU	10	Layout
	EAC.PPU	11	Layout

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Superior level	Structure	Page	Page description
ZMI	EAC.PPU	20	Designation terminals and connectors
	EAC.PPU	100	Interconnections - Control Cabinet//Door
	EAC.PPU	101	Interconnections - Control Cabinet//Door
	EAC.PPU	300	Door Control Maneuver
	EAC.PPU	301	Door Control Maneuver
	EAC.PPU	302	General Fault Maneuver
	EAC.PPU	400	PCB ´s - Inputs/Outputs
	EAC.PPU	1000	Bill of materials
	EAC.PPU	2000	Cable overview
	EAC.PPU	3000	Plug overview : =HEM++ZMI+EAC.PPU-XC258.2 - =HEM++ZMI+EAC.PPU-XC260.2
	EAC.PPCU	1	Title page / cover sheet
	EAC.PPCU	10	Layout
	EAC.PPCU	20	Designation terminals and connectors
	EAC.PPCU	50	SLD
	EAC.PPCU	100	Interconnections PPD//PPCU
	EAC.PPCU	101	Interconnections PPCU/ PSA
	EAC.PPCU	200	Power supply advanced UL
	EAC.PPCU	202	Distribution 208/120V
	EAC.PPCU	203	Customer connections
	EAC.PPCU	204	Customer connections
	EAC.PPCU	300	Customer signals
	EAC.PPCU	301	Customer signals
	EAC.PPCU	400	PCB E/S
	EAC.PPCU	401	Internal fault relay
	EAC.PPCU	1000	Bill of materials
	EAC.PPCU	3000	Plug overview : =HEM++ZMI+EAC.PPCU-XC282.2 - =HEM++ZMI+EAC.PPCU-XC364.2
	EAC.PTCU	1	Title page / cover sheet
	EAC.PTCU	10	Layout

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Superior level	Structure	Page	Page description
ZMI	EAC.PTCU	20	Designation terminals and connectors
	EAC.PTCU	100	Interconnections PPCU/ PSA
	EAC.PTCU	101	Interconnections optional BCI PTCU/ PSA
	EAC.PTCU	200	Potencial distribution
	EAC.PTCU	201	Terminal blocks for meter AC -optional
	EAC.PTCU	1000	Bill of materials
	EAC.PTCU	3000	Plug overview : =HEM++ZMI+EAC.PTCU-XC400.1 - =HEM++ZMI+EAC.PTCU-XC414.2
	ADU	1	Title page / cover sheet
	ADU	10	General layout
	ADU	20	Designation terminals and connectors
	ADU	100	Interconnections PSA/ADU
	ADU	101	Interconnections - Fan PCL//ADU
	ADU	102	Interconnections - Voltages measures in PDC1 - PV strings Side
	ADU	103	Interconnections - Voltages measures in PDC1
	ADU	104	Interconnections - Comms RS485
	ADU	105	Interconnections - Comms RS485 (XLOK)
	ADU	106	Interconnections F.O. PCL//ADU
	ADU	107	Interconnections - AC Voltage measure - IMD
	ADU	116	Interconnections PPD//PCDU
	ADU	200	Earth connection
	ADU	1000	Bill of materials
	ADU	2000	Cable overview
	ADU	3000	Plug overview : =HEM++ZMI+ADU-XC416.1 - =HEM++ZMI+ADU-XC554.1
	ADU.PDC1	1	Title page / cover sheet
	ADU.PDC1	10	Layout
	ADU.PDC1	20	Designation terminals and connectors
	ADU.PDC1	100	Interconnection
	ADU.PDC1	200	Power DC Disconnecter (DC+/-)

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Superior level	Structure	Page	Page description
ZMI	ADU.PDC1	201	Busbars PV (+/-)
	ADU.PDC1	300	Disconnecter switch DC 1
	ADU.PDC1	301	Distribution 120 VDC
	ADU.PDC1	400	PCB ´s - Disconnecter - Module
	ADU.PDC1	1000	Bill of materials
	ADU.PDC1	3000	Plug overview : =HEM++ZMI+ADU.PDC1-XC420.2 - =HEM++ZMI+ADU.PDC1-XC471.1
	ADU.PDC2	1	Title page / cover sheet
	ADU.PDC2	10	Layout
	ADU.PDC2	20	Designation terminals and connectors
	ADU.PDC2	100	Interconnection
	ADU.PDC2	200	Power DC Disconnecter (DC+/-)
	ADU.PDC2	300	Disconnecter switch DC 1
	ADU.PDC2	301	Distribution 120 VDC
	ADU.PDC2	400	PCB ´s - Disconnecter - Module
	ADU.PDC2	1000	Bill of materials
	ADU.PDC2	3000	Plug overview : =HEM++ZMI+ADU.PDC2-XC421.2 - =HEM++ZMI+ADU.PDC2-XC476.1
	ADU.PDC3	1	Title page / cover sheet
	ADU.PDC3	10	Layout
	ADU.PDC3	20	Designation terminals and connectors
	ADU.PDC3	100	Interconnection
	ADU.PDC3	200	Power DC Disconnecter (DC+/-)
	ADU.PDC3	300	Disconnecter switch DC 1
	ADU.PDC3	301	Distribution 120 VDC
	ADU.PDC3	400	PCB ´s - Disconnecter - Module
	ADU.PDC3	1000	Bill of materials
	ADU.PDC3	3000	Plug overview : =HEM++ZMI+ADU.PDC3-XC422.2 - =HEM++ZMI+ADU.PDC3-XC481.1
	ADU.PDC4	1	Title page / cover sheet
	ADU.PDC4	10	Layout

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Superior level	Structure	Page	Page description
ZMI	ADU.PDC4	20	Designation terminals and connectors
	ADU.PDC4	100	Interconnection
	ADU.PDC4	200	Power DC Disconnecter (DC+/-)
	ADU.PDC4	300	Disconnecter switch DC 1
	ADU.PDC4	301	Distribution 120 VDC
	ADU.PDC4	400	PCB ´s - Disconnecter - Module
	ADU.PDC4	1000	Bill of materials
	ADU.PDC4	3000	Plug overview : =HEM++ZMI+ADU.PDC4-XC423.2 - =HEM++ZMI+ADU.PDC4-XC486.1
	ADU.PFU1	1	Title page / cover sheet
	ADU.PFU1	10	General layout
	ADU.PFU1	20	Designation terminals and connectors
	ADU.PFU1	100	Interconnections F.O. PFU1//PFU2
	ADU.PFU1	200	DC Input 1 (+)
	ADU.PFU1	201	DC Input 2 (+)
	ADU.PFU1	400	PCB ´s - DU Current measures
	ADU.PFU1	1000	Bill of materials
	ADU.PFU1	2000	Cable overview
	ADU.PFU1	3000	Plug overview : =HEM++ZMI+ADU.PFU1-XC520.1 - =HEM++ZMI+ADU.PFU1-XC520.1
	ADU.PFU2	1	Title page / cover sheet
	ADU.PFU2	10	General layout
	ADU.PFU2	20	Designation terminals and connectors
	ADU.PFU2	100	Interconnections F.O. ADU//PFU1
	ADU.PFU2	101	Interconnections F.O. PFU1//PFU2
	ADU.PFU2	200	DC Input 3 (+)
	ADU.PFU2	201	DC Input 4 (+)
	ADU.PFU2	400	PCB ´s - DU Current measures
	ADU.PFU2	401	RFI DC (+/-)
	ADU.PFU2	1000	Bill of materials

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Superior level	Structure	Page	Page description
ZMI	ADU.PFU2	2000	Cable overview
	ADU.PFU2	3000	Plug overview : =HEM++ZMI+ADU.PFU2-XC490.1 - =HEM++ZMI+ADU.PFU2-XC491.1
	ADU.PADU	1	Title page / cover sheet
	ADU.PADU	10	DC+ - General layout
	ADU.PADU	200	DC Input 1 (+) to PFU1
	ADU.PADU	201	DC Input 2 (+) to PFU1
	ADU.PADU	202	DC Input 3 (+) to PFU2
	ADU.PADU	203	DC Input 4 (+) to PFU2
	ADU.PADU	206	DC Input (-)
	ADU.PMV1	1	Title page / cover sheet
	ADU.PMV1	10	Layout PMV1
	ADU.PMV1	20	Designation terminals and connectors
	ADU.PMV1	200	DU - Fan 1
	ADU.PMV1	1000	Bill of materials
	ADU.PMV1	3000	Plug overview : =HEM++ZMI+ADU.PMV1-XC550 - =HEM++ZMI+ADU.PMV1-XC552.2
	ADU.PMV2	1	Title page / cover sheet
	ADU.PMV2	10	Layout PMV2
	ADU.PMV2	20	Designation terminals and connectors
	ADU.PMV2	200	DU - Fan 2
	ADU.PMV2	1000	Bill of materials
	ADU.PMV2	3000	Plug overview : =HEM++ZMI+ADU.PMV2-XC553 - =HEM++ZMI+ADU.PMV2-XC555.2
	ADU.PPD	1	Title page / cover sheet
	ADU.PPD	10	Layout
	ADU.PPD	20	Designation terminals and connectors
	ADU.PPD	21	Designation terminals and connectors
	ADU.PPD	100	Interconnections PCL//ADU.PPD
	ADU.PPD	102	Interconnections PPD//PCDU
	ADU.PPD	200	Surge protector

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Superior level	Structure	Page	Page description
ZMI	ADU.PPD	201	GFDI DC
	ADU.PPD	300	Distribution 24V DC
	ADU.PPD	400	VOLTAGE PRESENCE PCB
	ADU.PPD	401	Insulation monitoring
	ADU.PPD	402	PCB Resistors IMD
	ADU.PPD	403	PCB Resistors IMD
	ADU.PPD	1000	Bill of materials
	ADU.PPD	2000	Cable overview
	ADU.PPD	3000	Plug overview : =HEM++ZMI+ADU.PPD-XC047.2 - =HEM++ZMI+ADU.PPD-XC557.1
		20	Designation terminals and connectors
		100	Interconnections ZMI/CDT - P. Supply
		101	Interconnections ZMI/CDT - Comms/Measures
		103	Interconnections ZMI/CDT - Comms/Measures
		104	Interconnections ZMI/CDT - Comms/Measures
		106	Interconnections fans (I)
		107	Interconnections fans (II)
		108	Interconnections communication fans (XLOK)
		109	Interconnections protection switchgear
		110	Interconnections protection switchgear
		111	Interconnections AC measurent
		112	Interconnections 120Vdc
		113	Interconnections - 430VAC power supply
		200	Medium voltage transformer
		1000	Bill of materials
		2000	Cable overview
		3000	Plug overview : =HEM++CDT+-XC750.2 - =HEM++CDT+-XC770.1
	AMT	1	Title page / cover sheet
	AMT	200	Earth connection

Table of contents

Superior level	Structure	Page	Page description
CDT	AMT	1000	Bill of materials
	AMT.PMC	1	Title page / cover sheet
	AMT.PMC	10	Layout
	AMT.PMC	11	Interconnection PMC and CDT
	AMT.PMC	20	Designation terminals and connectors
	AMT.PMC	100	Interconnections protection switchgear
	AMT.PMC	101	Interconnections protection switchgear
	AMT.PMC	102	Interconnections AC measurent
	AMT.PMC	103	Interconnections 120Vdc
	AMT.PMC	104	Interconnections fans
	AMT.PMC	105	Interconnections - 430VAC power supply + Econ mode
	AMT.PMC	106	Interconnections CEM/PMC
	AMT.PMC	108	Interconnections Spare Line cell
	AMT.PMC	200	Distribution 120VDC/24VDC
	AMT.PMC	201	General protections
	AMT.PMC	202	Protections Econ mode transformer
	AMT.PMC	300	Auxiliary trip 24Vdc MV switchgear
	AMT.PMC	400	MV measure voltage board
	AMT.PMC	401	Inputs/outputs distributor boards
	AMT.PMC	402	Inputs/outputs distributor board (Econ mode)
	AMT.PMC	1000	Bill of materials
	AMT.PMC	3000	Plug overview : =HEM++CDT+AMT.PMC-XC750.1 - =HEM++CDT+AMT.PMC-XC792.2
	CMT	1	Title page / cover sheet
	CMT	20	Designation terminals and connectors
	CMT	50	MV Single line diagram
	CMT	200	Measures MV protection switchgear
	CMT	201	MV protection switchgear
	CMT	300	MV protection switchgear

Table of contents

Superior level	Structure	Page	Page description
CDT	CMT	301	MV protection switchgear
	CMT	1000	Bill of materials
	CMT	3000	Plug overview : =HEM++CDT+CMT-XC772.1 - =HEM++CDT+CMT-XU
	CRB	1	Title page / cover sheet
	CRB	10	General layout
	CRB	200	Earth connection
	CRB	1000	Bill of materials
	PMV8	1	Title page / cover sheet
	PMV8	10	Layout PMV8
	PMV8	20	Designation terminals and connectors
	PMV8	200	Power module 8 - Fan
	PMV8	1000	Bill of materials
	PMV8	2000	Cable overview
	PMV8	3000	Plug overview : =HEM++CDT+PMV8-XC760 - =HEM++CDT+PMV8-XC761.2
	PMV9	1	Title page / cover sheet
	PMV9	10	Layout PMV9
	PMV9	20	Designation terminals and connectors
	PMV9	200	Power module 9 - Fan
	PMV9	1000	Bill of materials
	PMV9	2000	Cable overview
	PMV9	3000	Plug overview : =HEM++CDT+PMV9-XC763 - =HEM++CDT+PMV9-XC764.2
	PMV10	1	Title page / cover sheet
	PMV10	10	Layout PMV10
	PMV10	20	Designation terminals and connectors
	PMV10	200	Power module 10 - Fan
	PMV10	1000	Bill of materials
	PMV10	2000	Cable overview
	PMV10	3000	Plug overview : =HEM++CDT+PMV10-XC766 - =HEM++CDT+PMV10-XC767.2

Table of contents

Superior level	Structure	Page	Page description
CDT	PMV11	1	Title page / cover sheet
	PMV11	10	Layout PMV11
	PMV11	20	Designation terminals and connectors
	PMV11	200	Power module 11 - Fan
	PMV11	1000	Bill of materials
	PMV11	2000	Cable overview
	PMV11	3000	Plug overview : =HEM++CDT+PMV11-XC769 - =HEM++CDT+PMV11-XC771.1

Revision overview

Revision	Date	Review Overview	Reviewed by
000	12/09/2024	Electric diagram finished from 1.3.11	FFATIGA

General layout

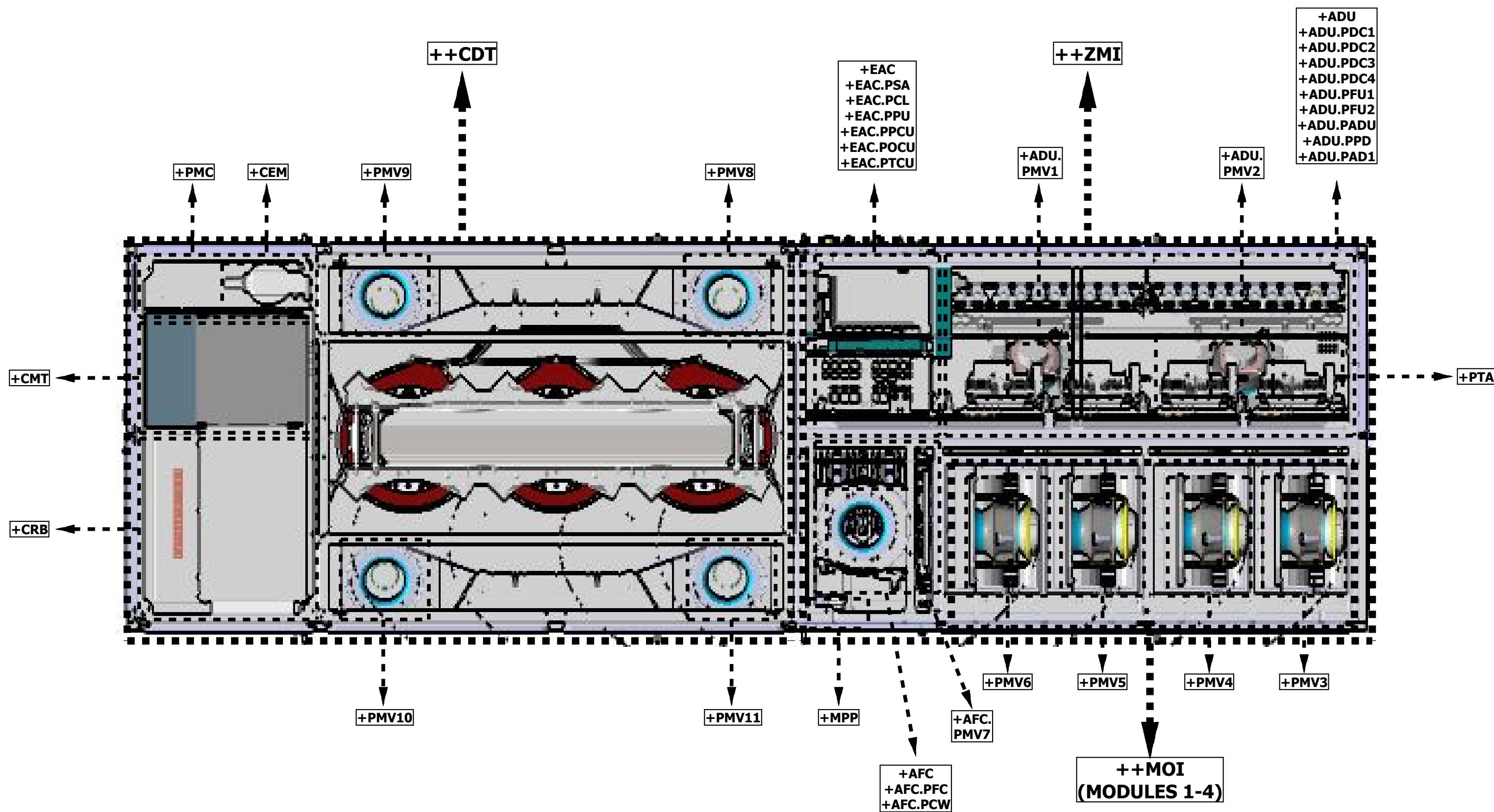
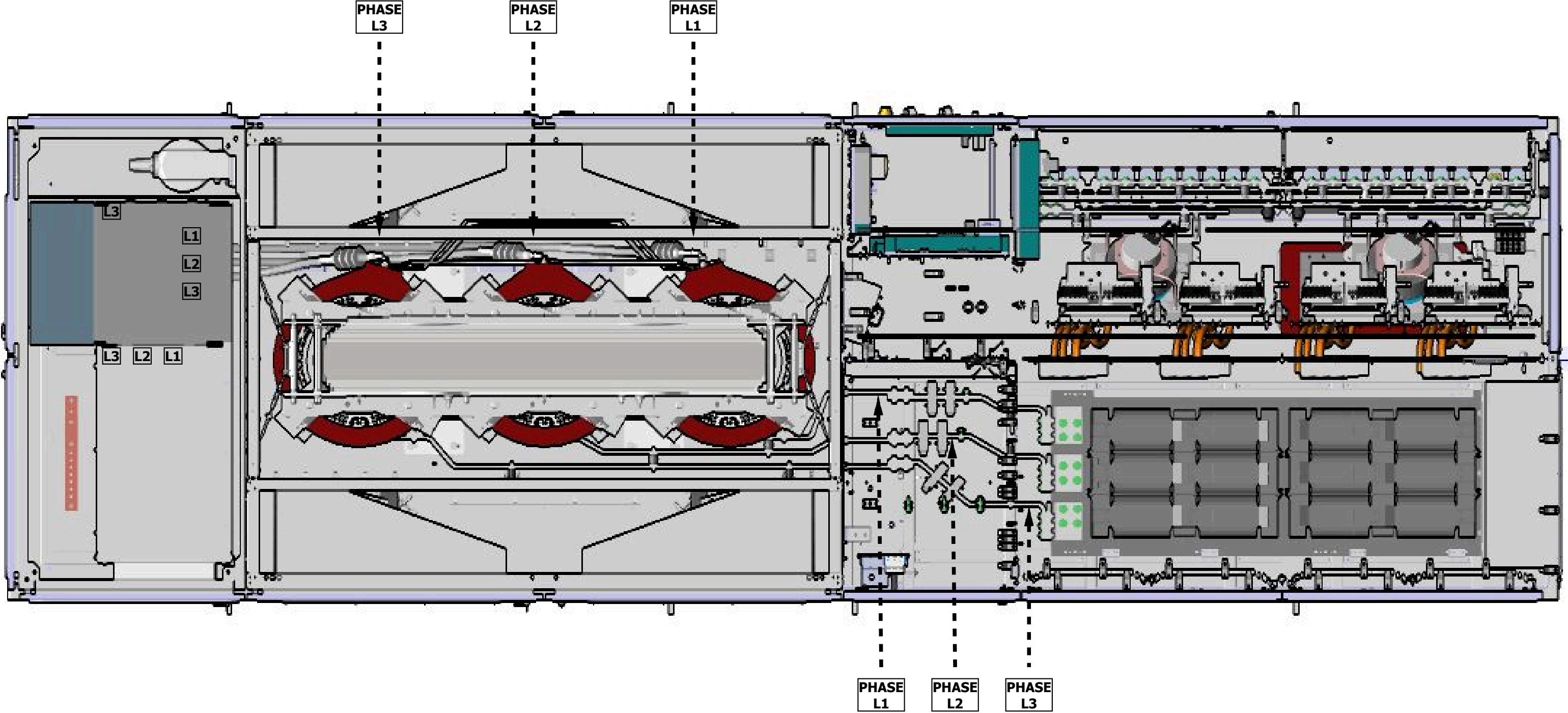


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

General layout -
Main AC Busbars



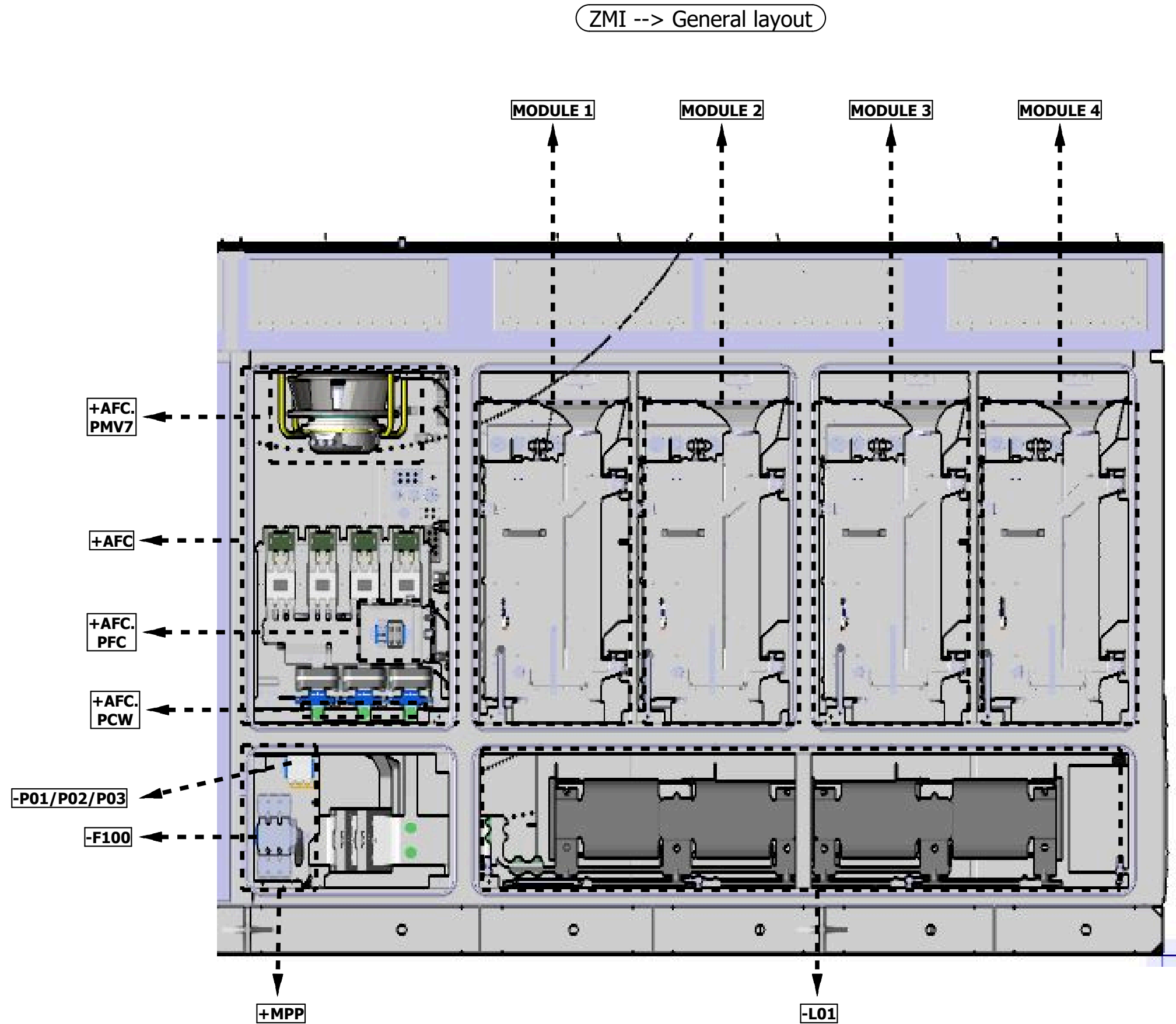


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

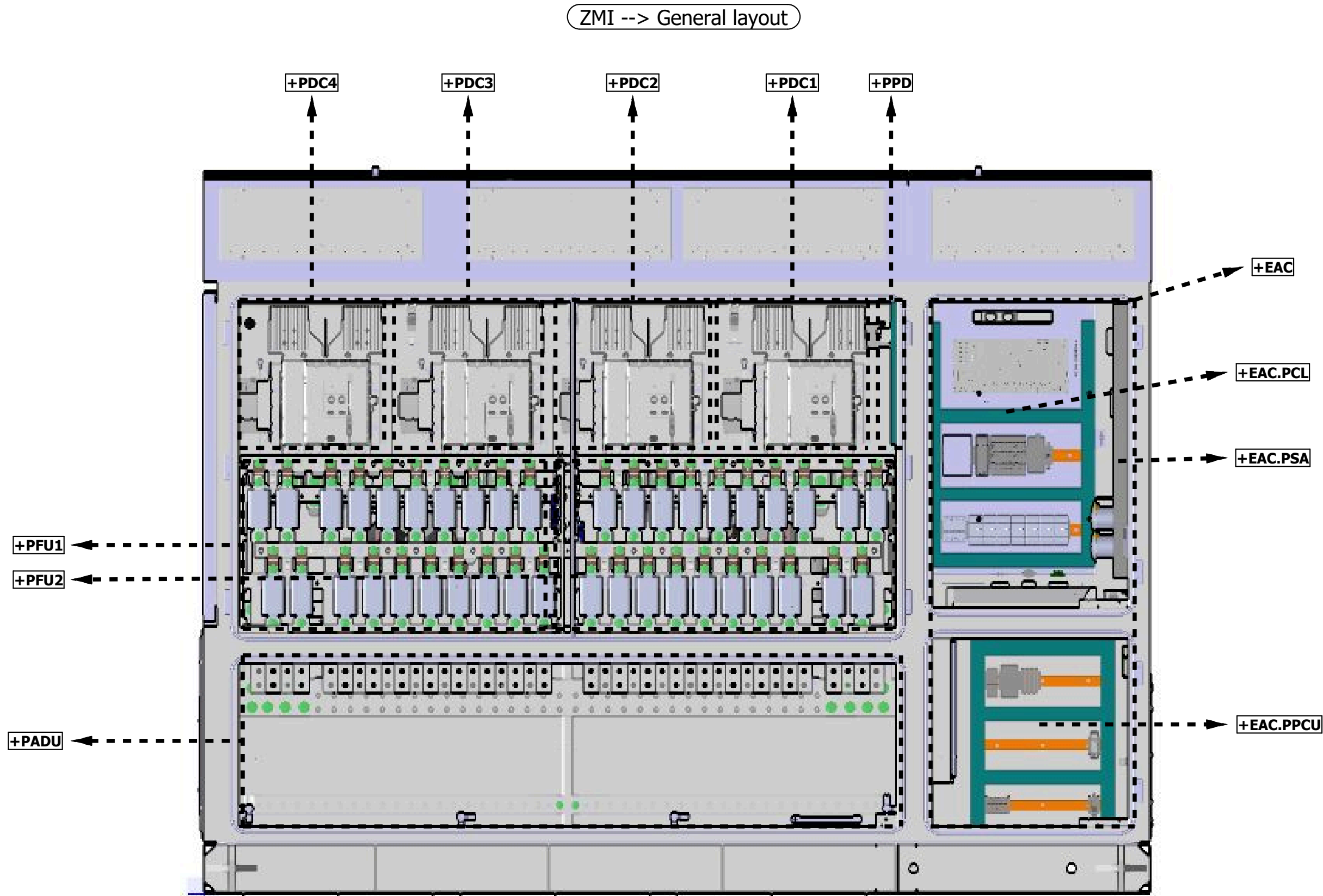
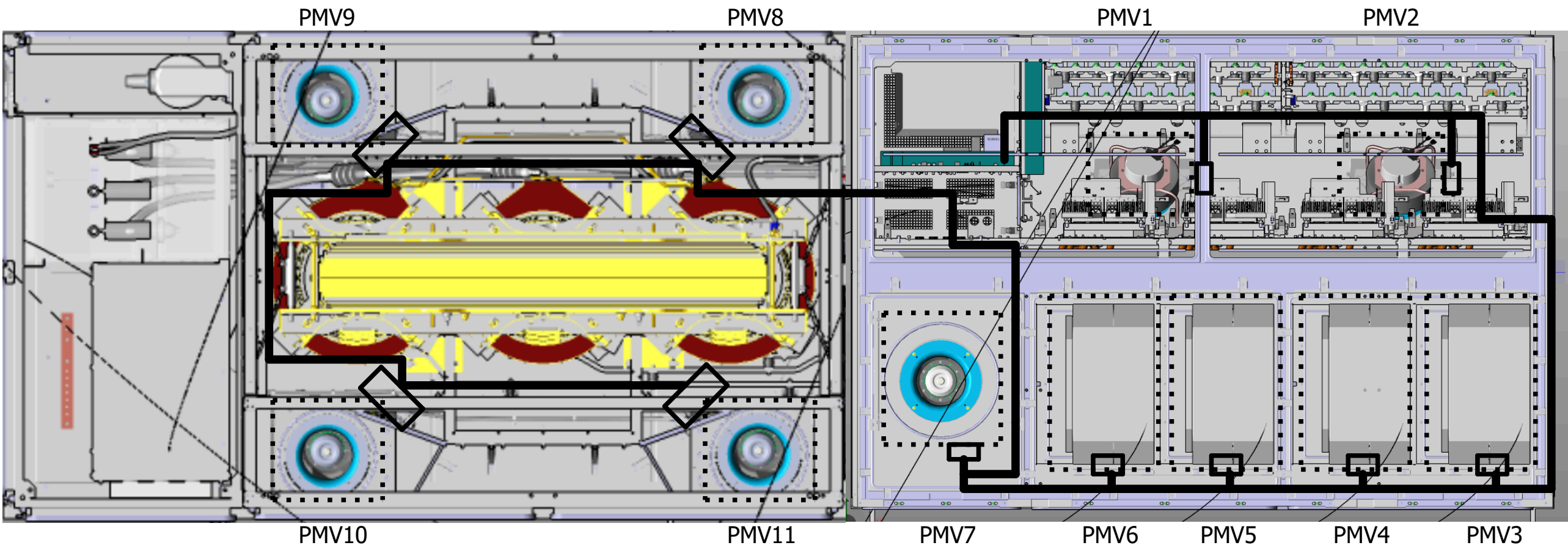


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

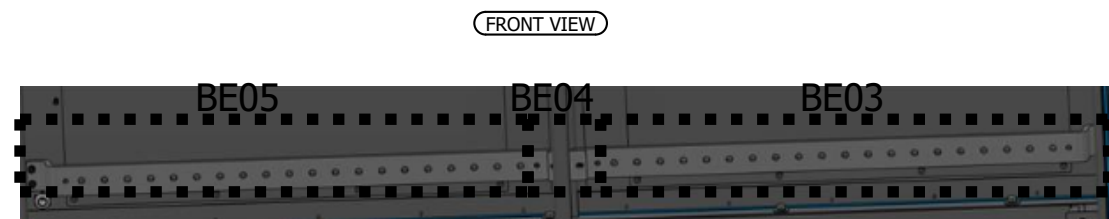
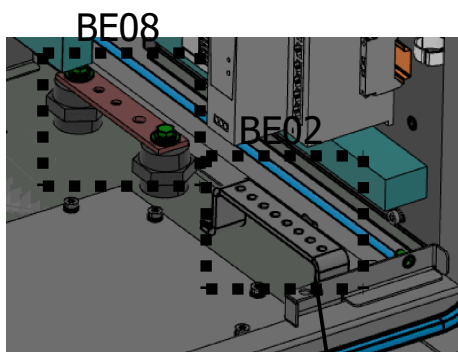
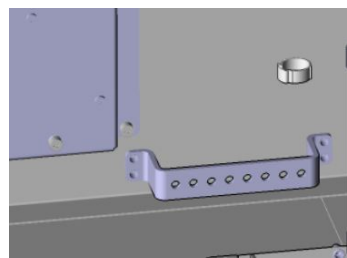
Communications RS-485

CDT

ZMI

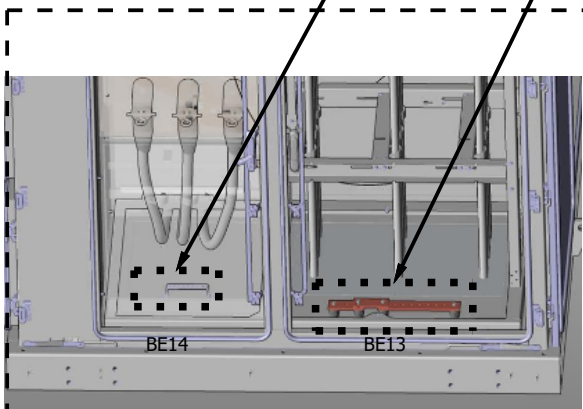
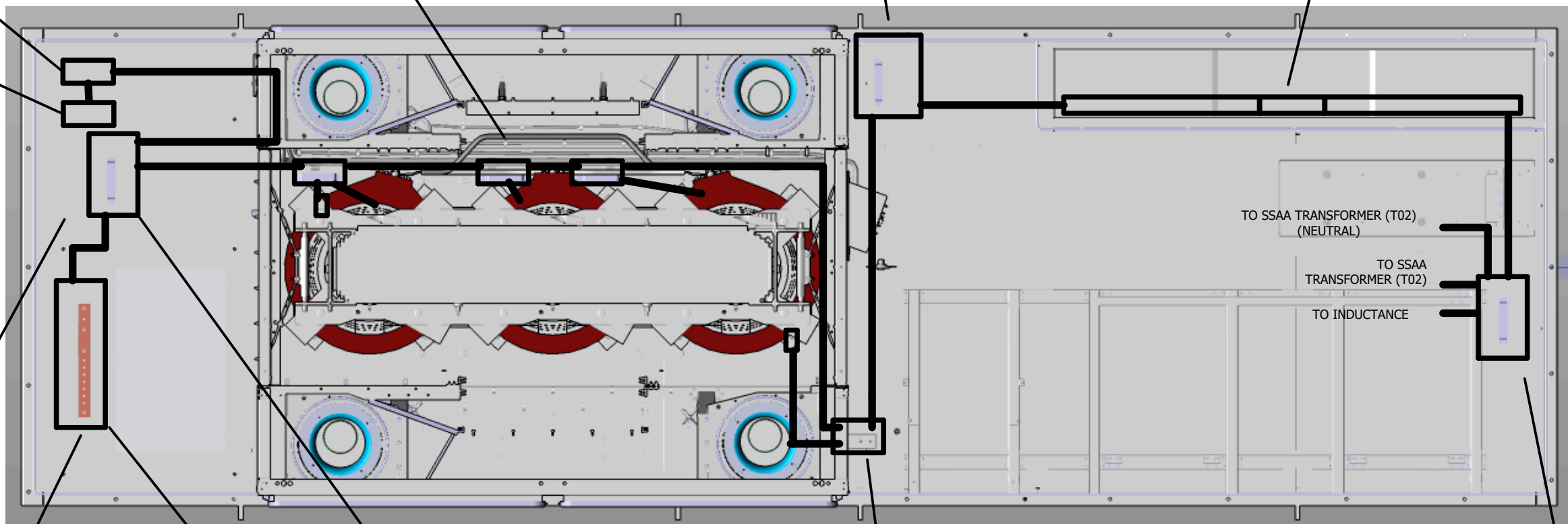


Earth interconnection

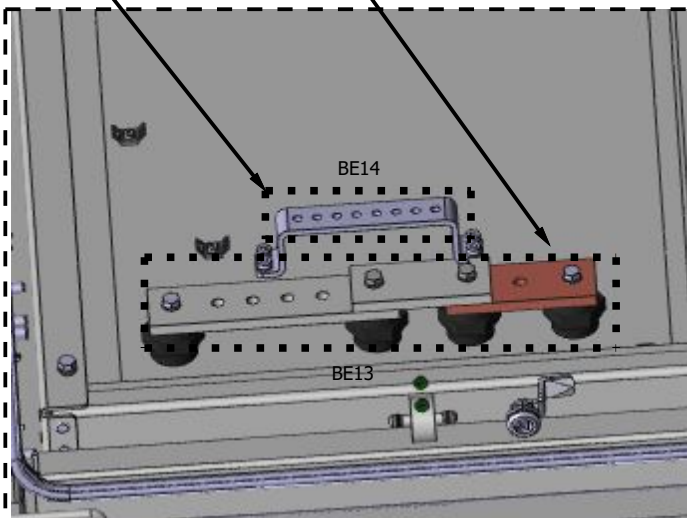


BE16

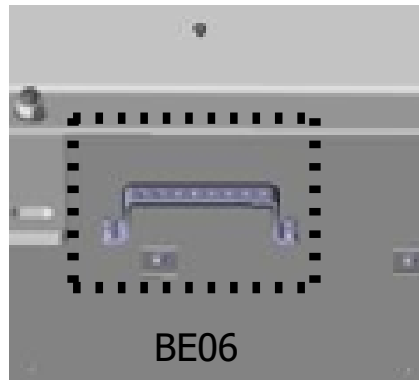
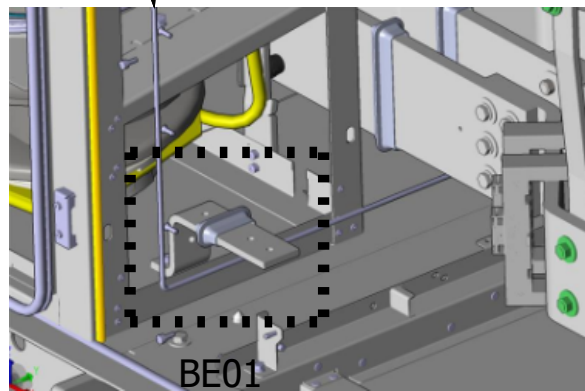
BE15
(Optional Econ mode)



UL OPTION



IEC OPTION



MATE-N-LOK CONNECTORS

Terminal location.
Front side view

E1098

Terminal location.
Front side view

E1099

Terminal location.
Front side view

E1096

Terminal location.
Front side view

E1097

Terminal location.
Front side view

L2003

Terminal location.
Front side view

L2004

Terminal location.
Front side view

L4707

Terminal location.
Front side view

L5777

Terminal location.
Front side view

L5778

Terminal location.
Front side view

L6757

Terminal location.
Front side view

L6758

Terminal location.
Front side view

L7534

Terminal location.
Front side view

L8009

Terminal location.
Front side view

E1374

Terminal location.
Front side view

E1375

Terminal location.
Front side view

L6135

Terminal location.
Front side view

L2476

Terminal location.
Front side view

L2477

POWER CONNECTORS

Example picture

position 1

Terminal location.
Front side view

L7556 - 4 pin connector

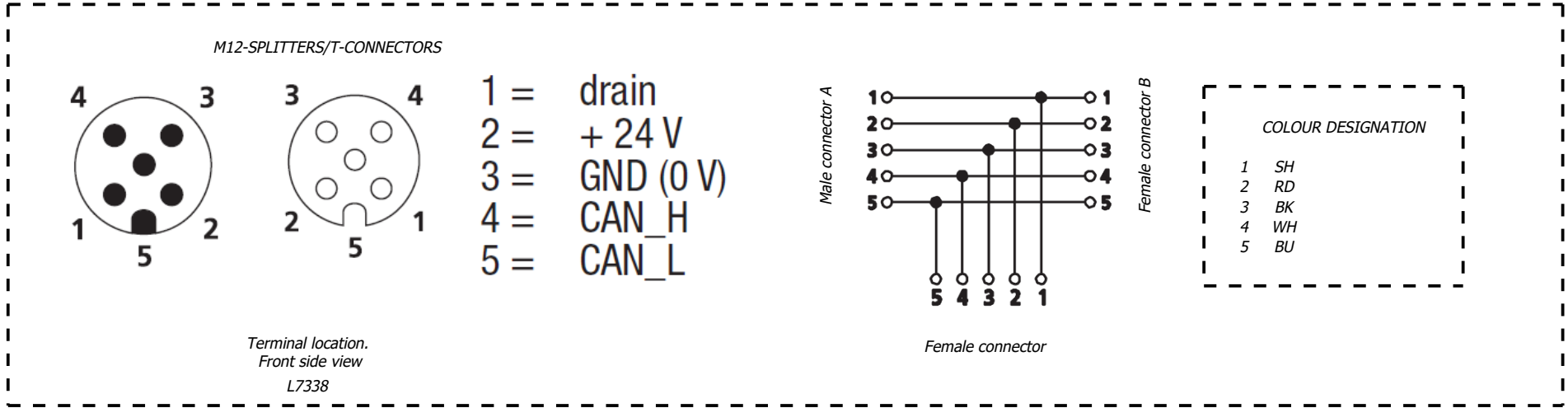
Example picture

position 1

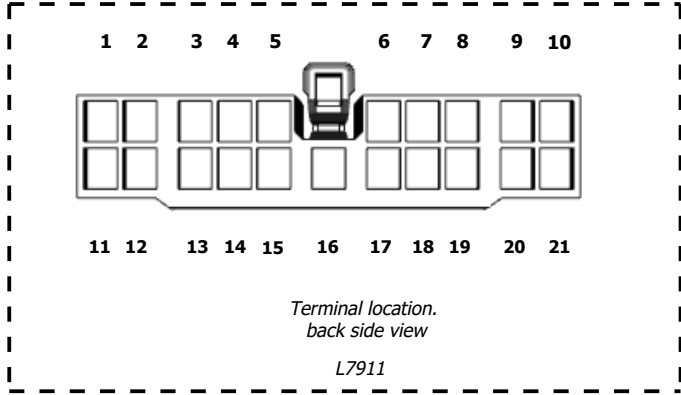
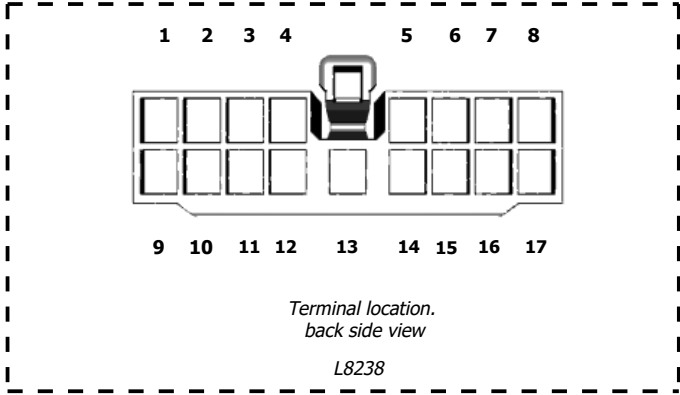
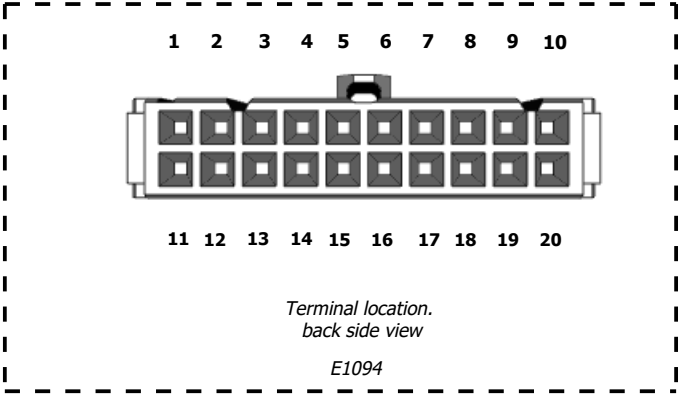
Terminal location.
Front side view

L7555 - 4 pin connector

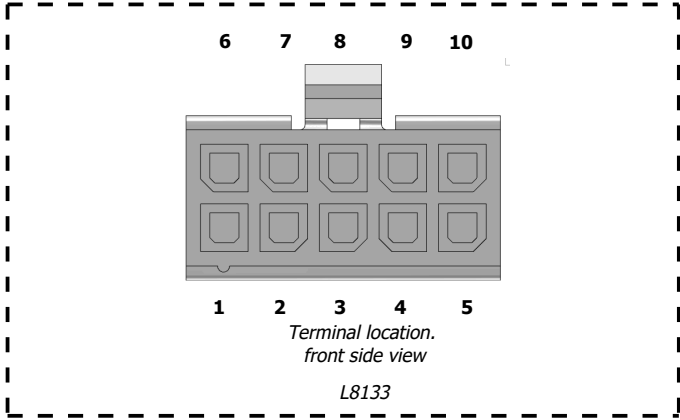
RS485 CONNECTORS

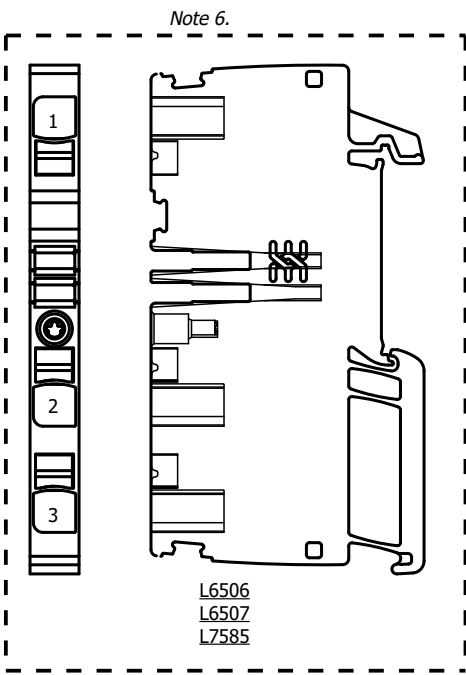
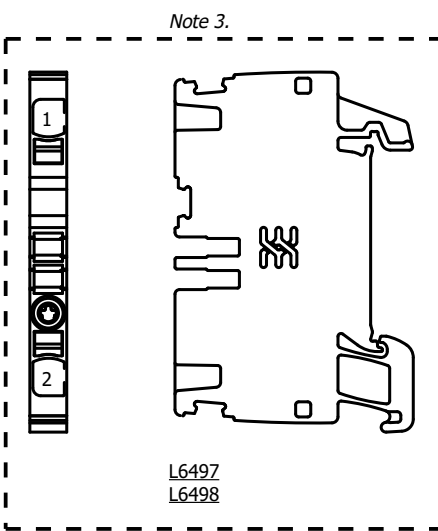
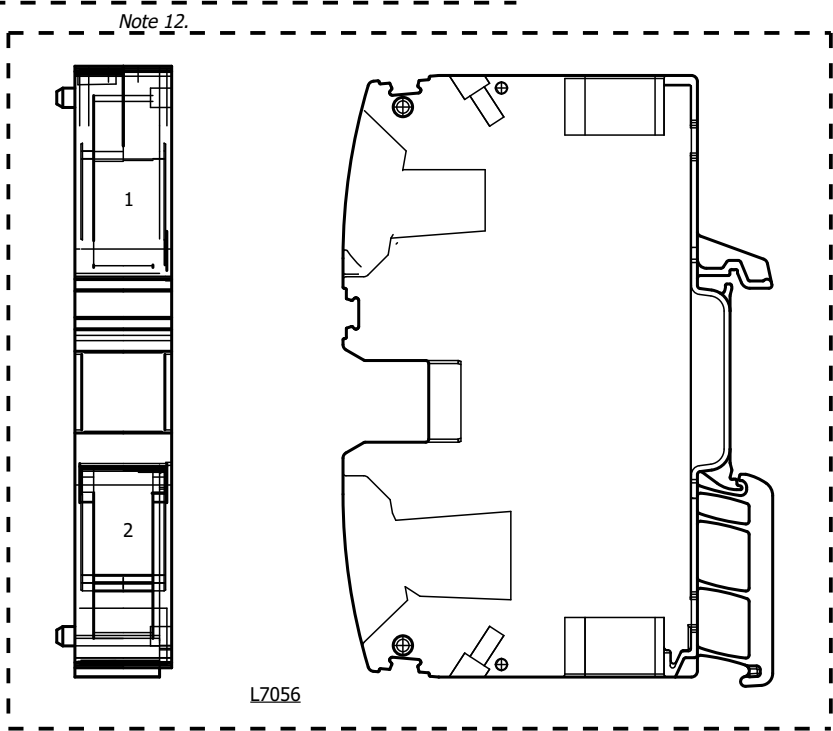
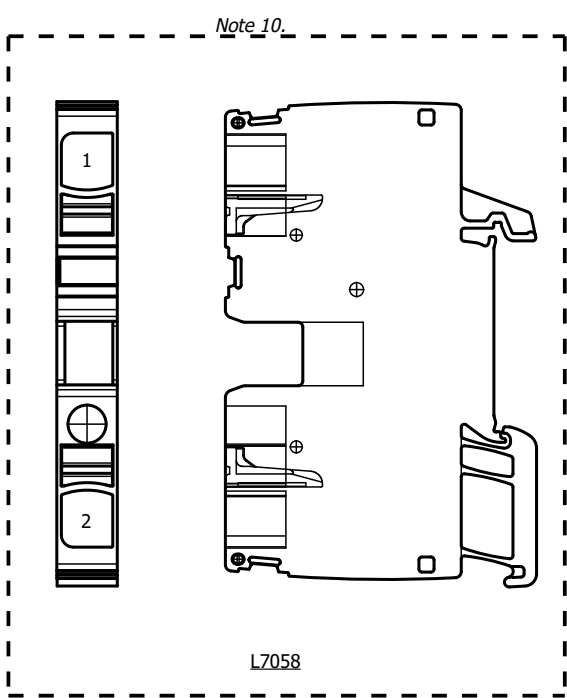
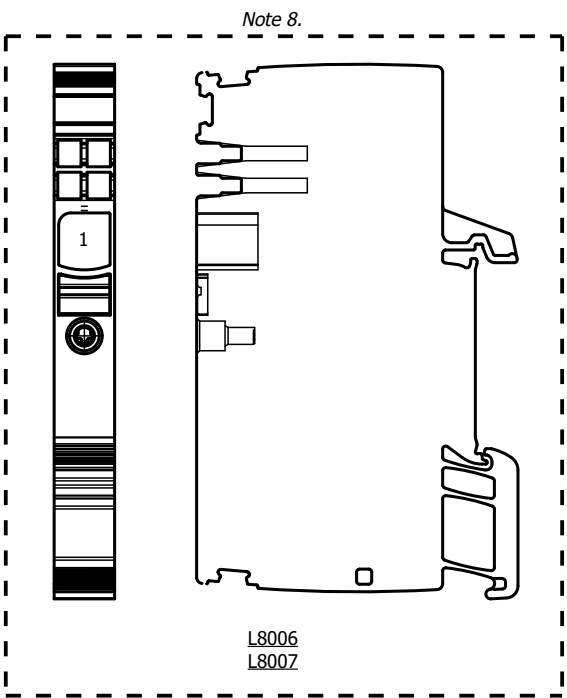
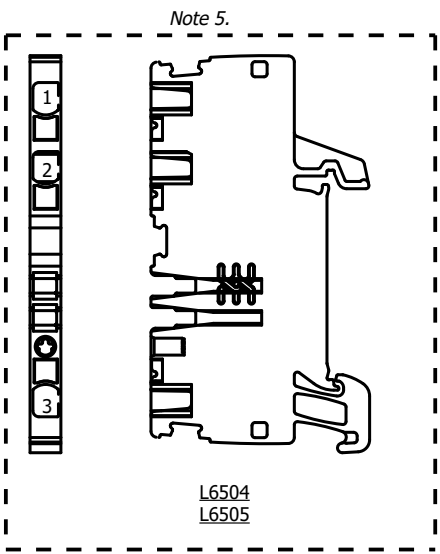
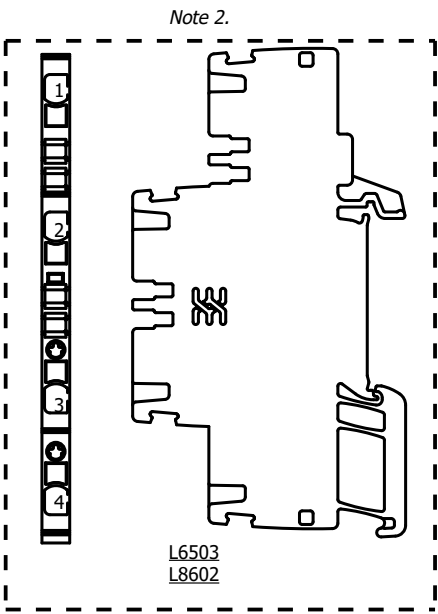
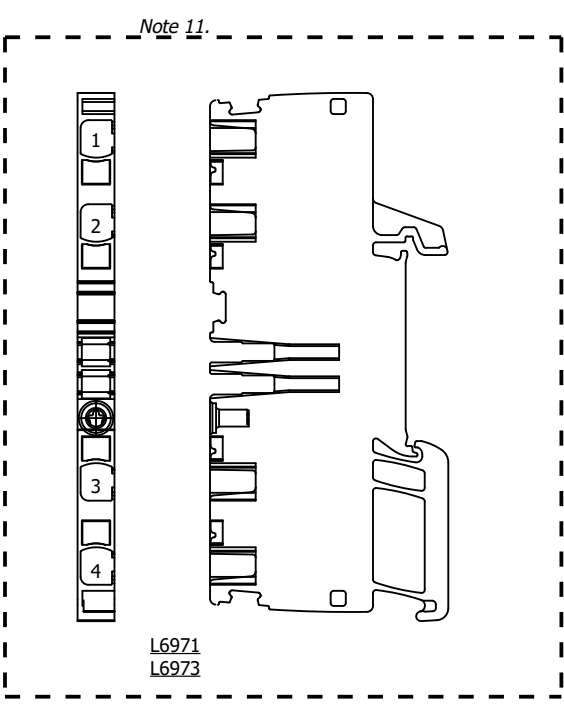
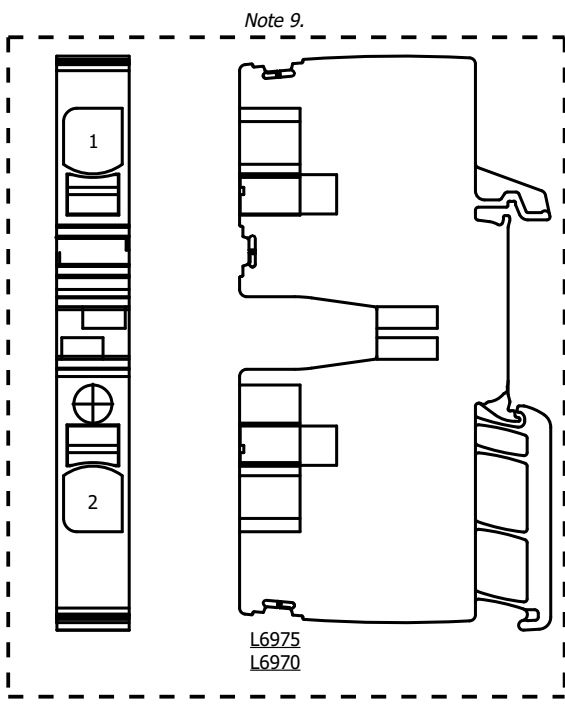
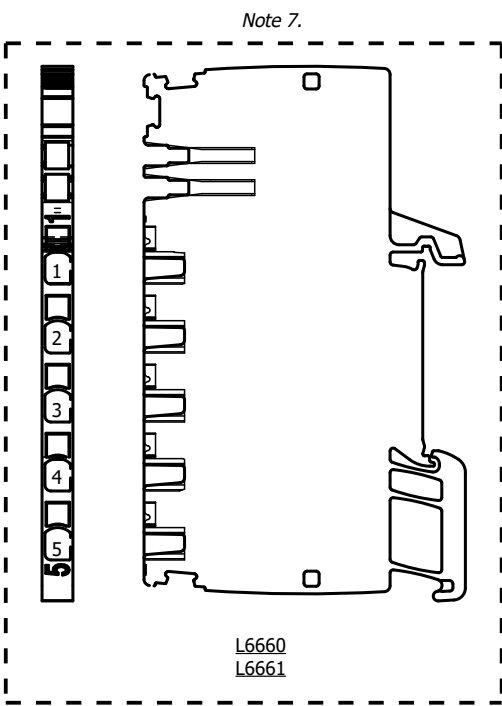
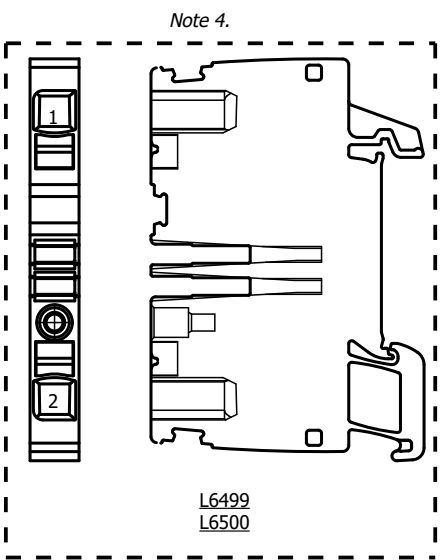
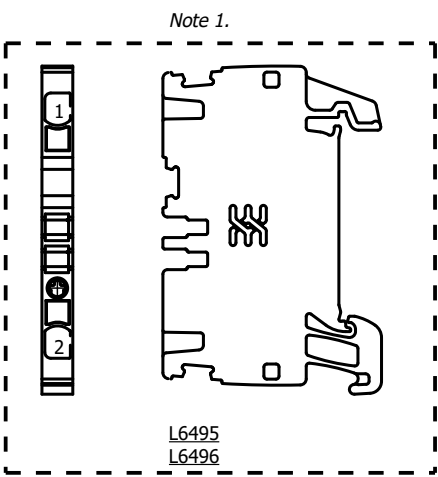


FLAT CONNECTORS

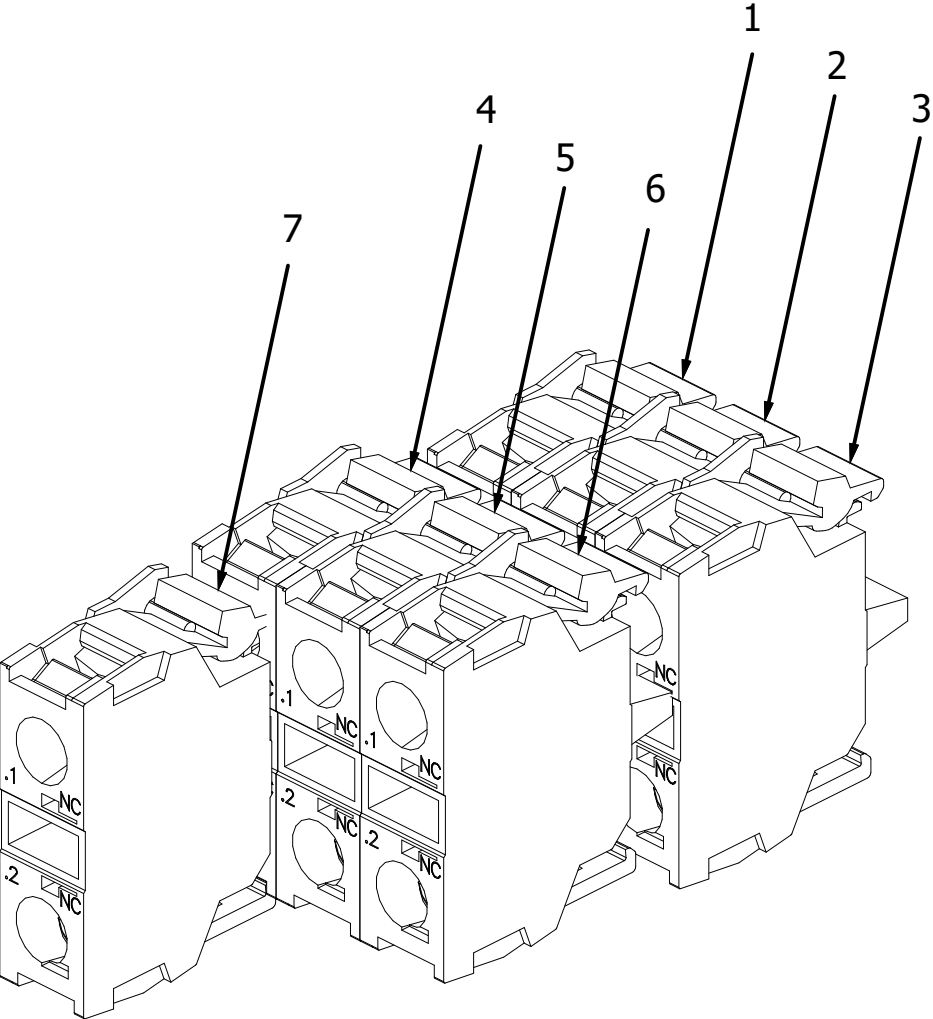


GFDI CONNECTOR (DEPENDS ON DU TYPE)



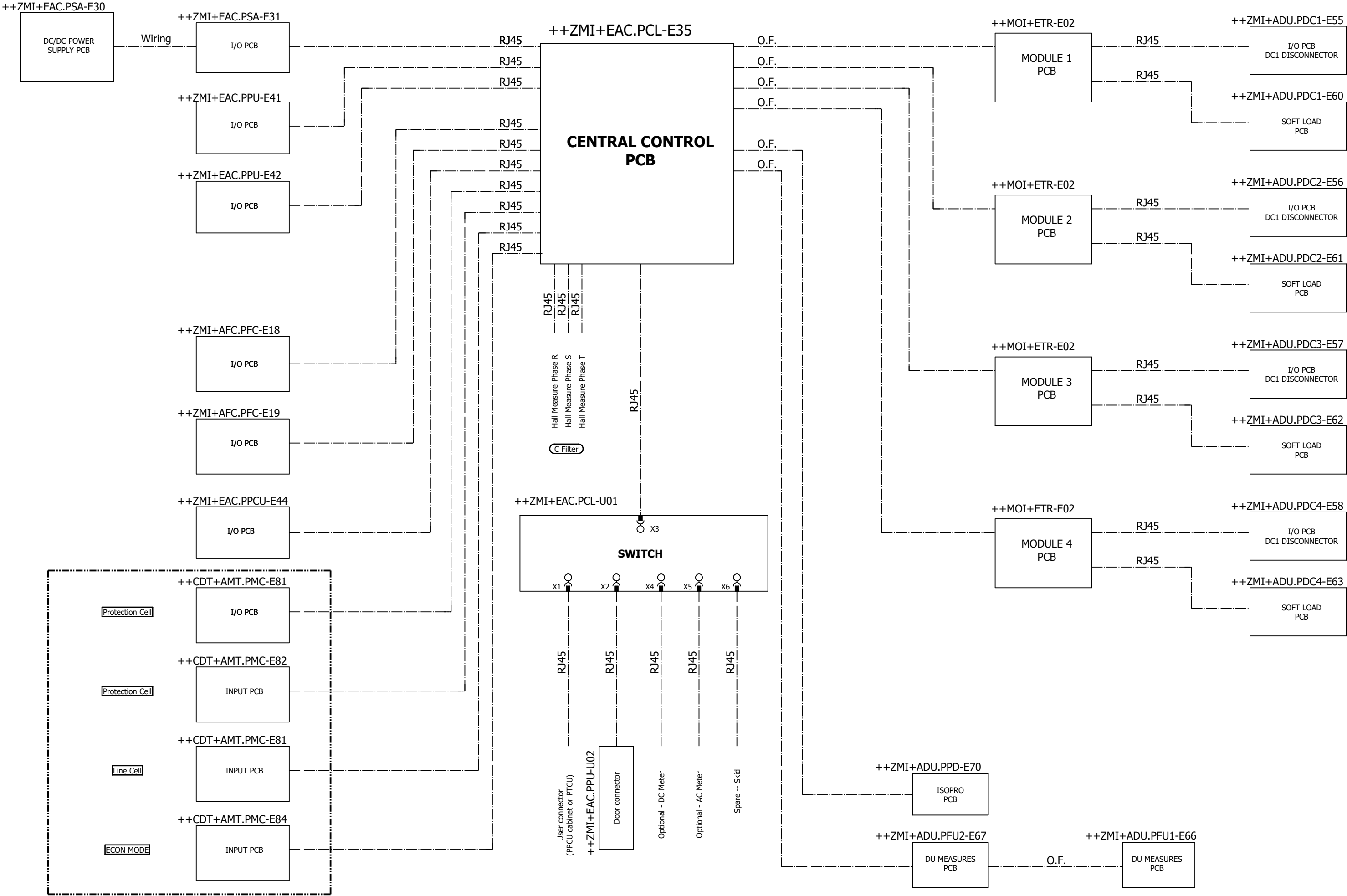


PUSHBUTTONS CONTACTS POSITION



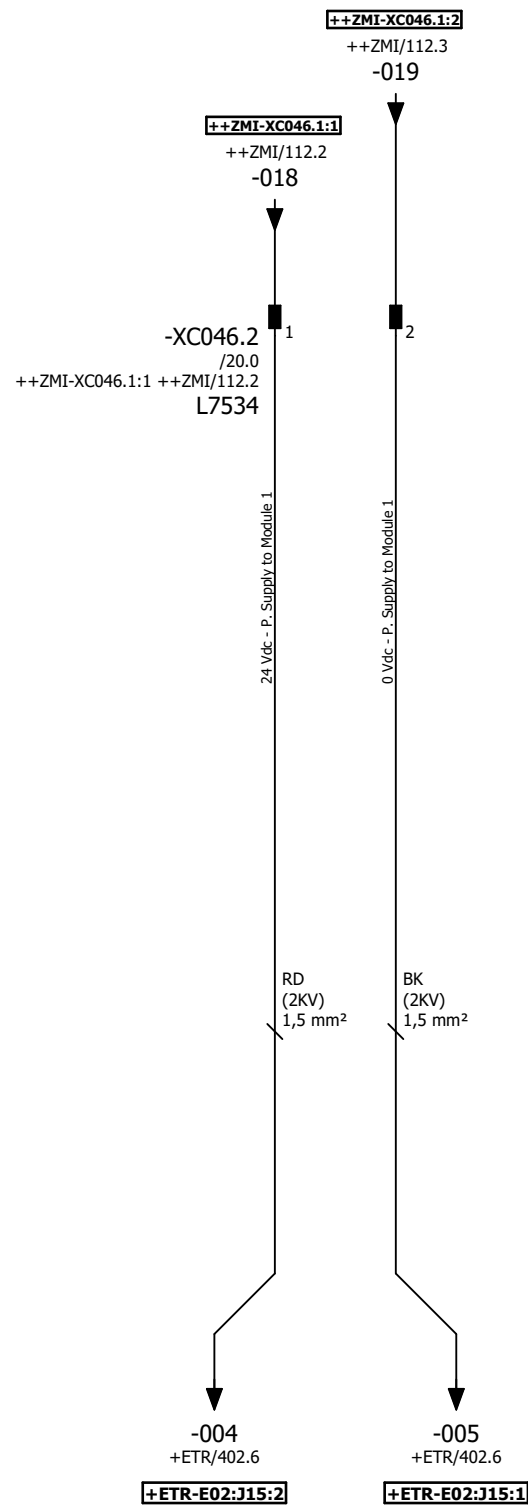
*Normally open contacts are .3 and .4
*Normally close contacts are .1 and .2

PCB ´s General Layout



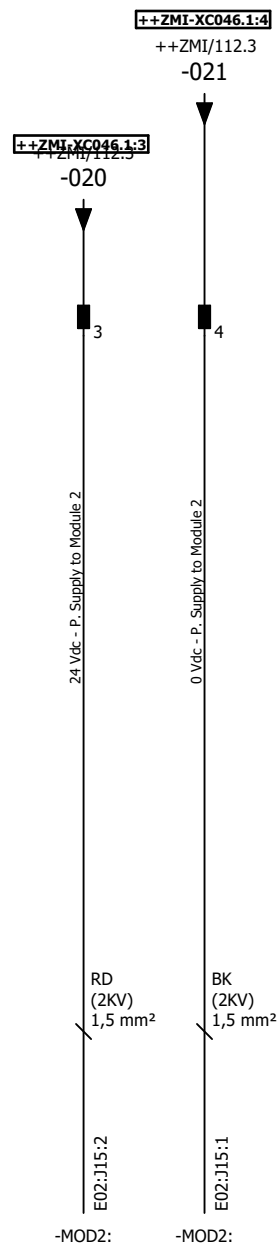
0	1	2	3	4	5	6	7	8	9
-XC001.1-M1 = CONNECTOR OPTICAL FIBER DUPLEX MOD.1 -XC002.1-M2 = CONNECTOR OPTICAL FIBER DUPLEX MOD.2 -XC003.1-M3 = CONNECTOR OPTICAL FIBER DUPLEX MOD.3 -XC004.1-M4 = CONNECTOR OPTICAL FIBER DUPLEX MOD.4 -XC031.1-M1 = CONNECTOR AC MEASURE MOD.1 -XC031.1-M2 = CONNECTOR AC MEASURE MOD.2 -XC031.1-M3 = CONNECTOR AC MEASURE MOD.3 -XC031.1-M4 = CONNECTOR AC MEASURE MOD.4 -XC046.2 = MATE-N-LOK 9P AERIAL CONNECTOR AC/DC MODULES									

MODULE 1



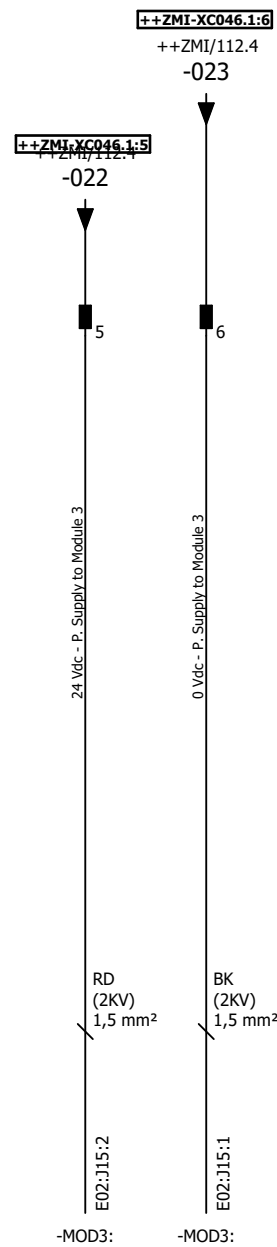
TOWARDS PCB E02 (J15)
MODULE 1

MODULE 2



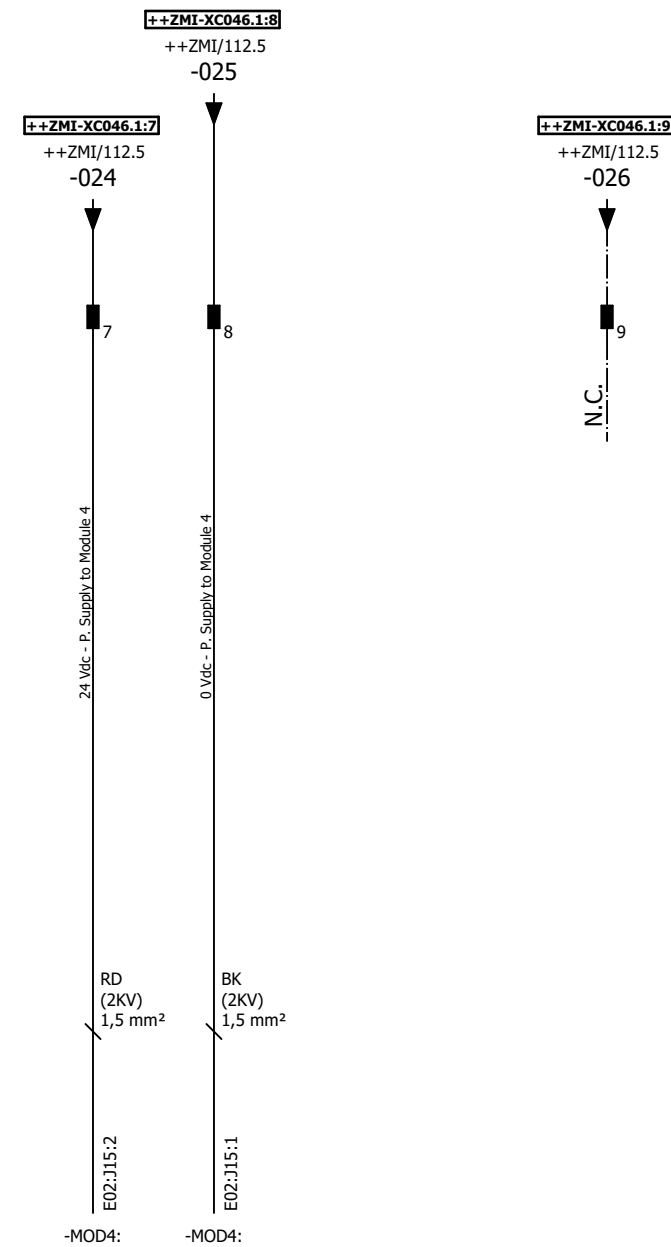
TOWARDS PCB E02 (J15)
MODULE 2

MODULE 3

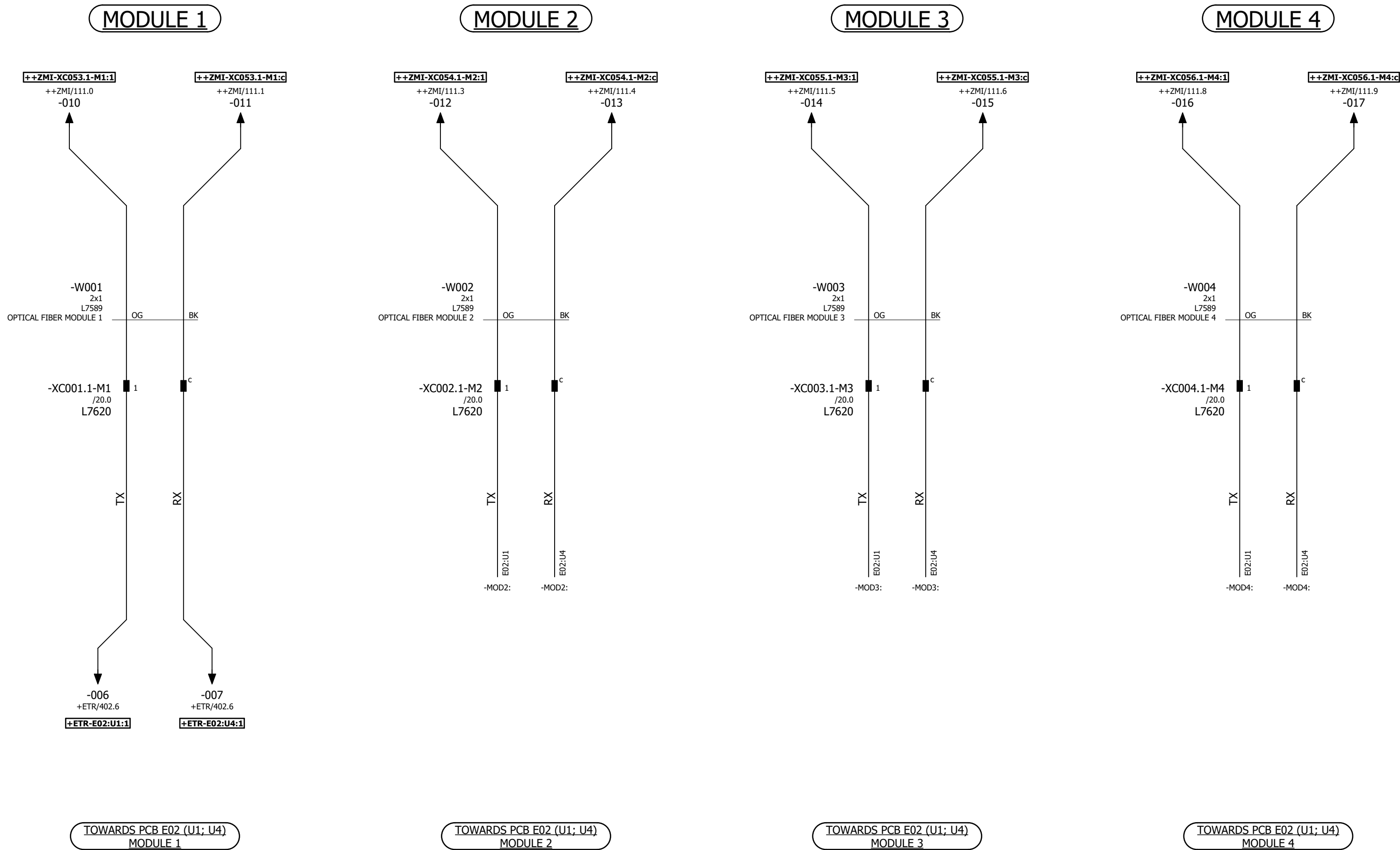


TOWARDS PCB E02 (J15)
MODULE 3

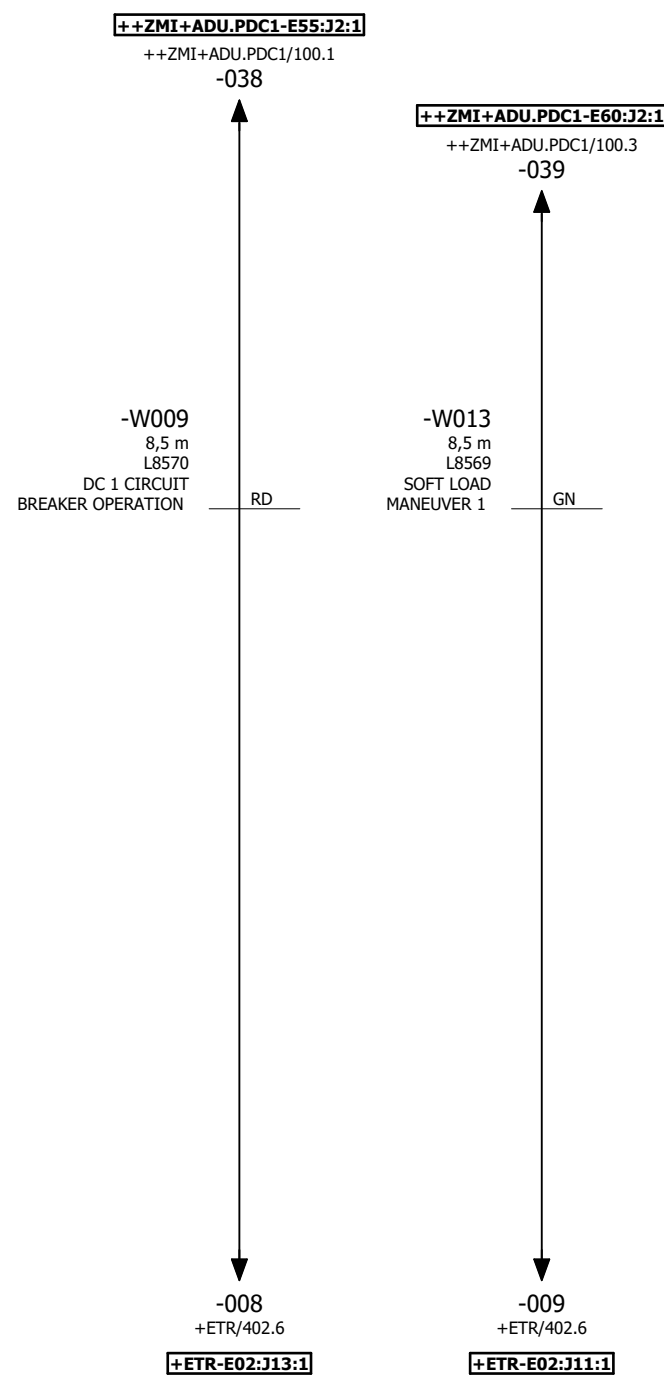
MODULE 4



TOWARDS PCB E02 (J15)
MODULE 4

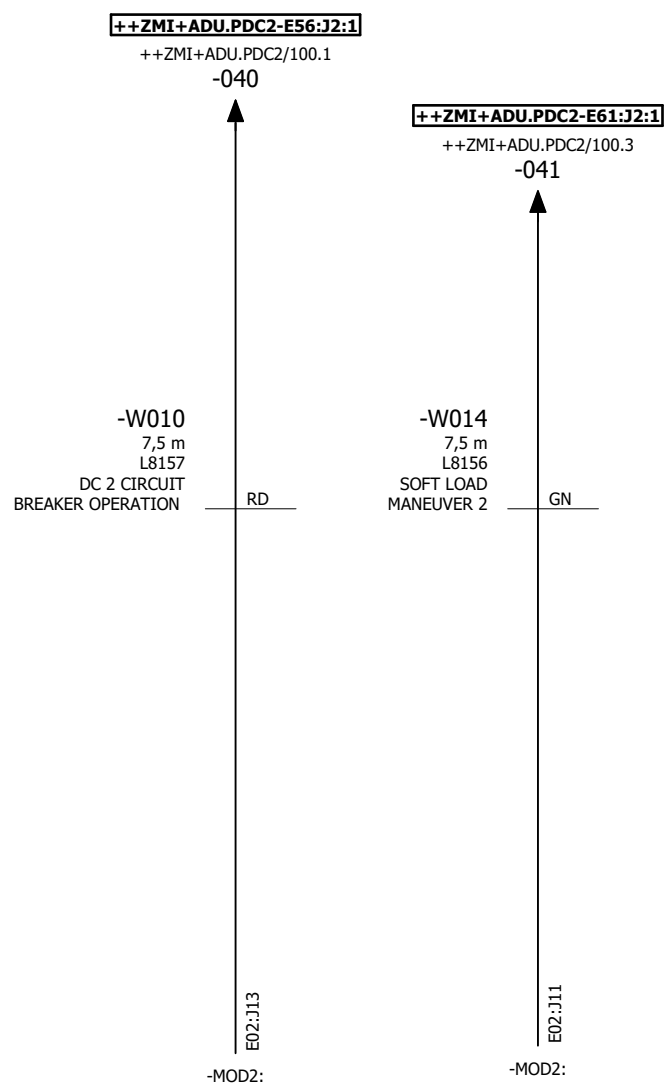


MODULE 1



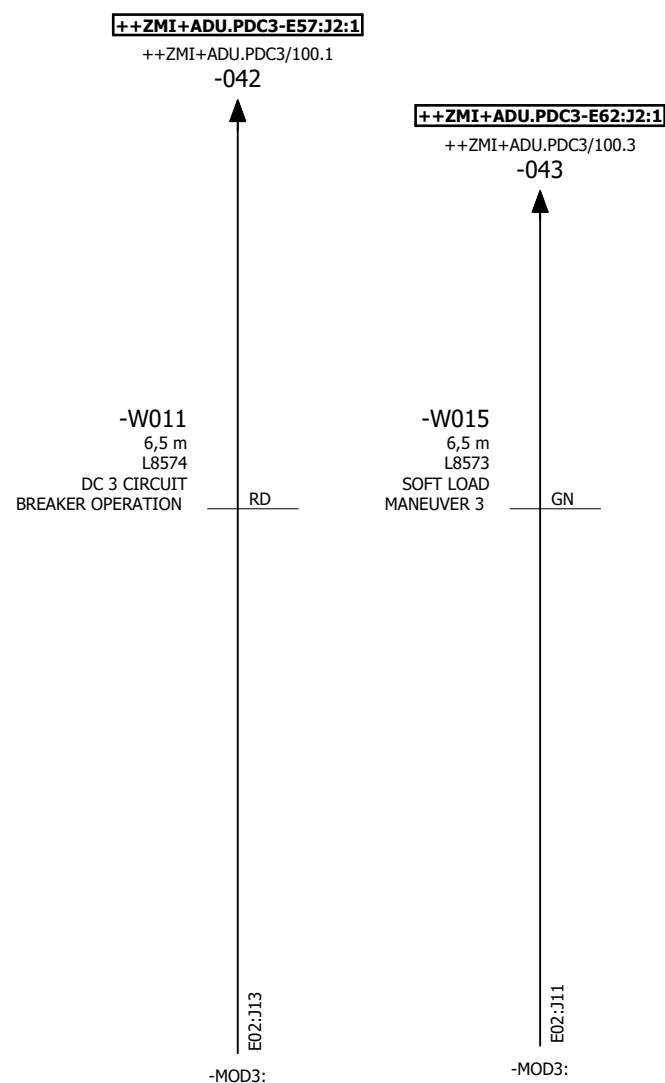
TOWARDS PCB E02 (J11; J13)
MODULE 1

MODULE 2



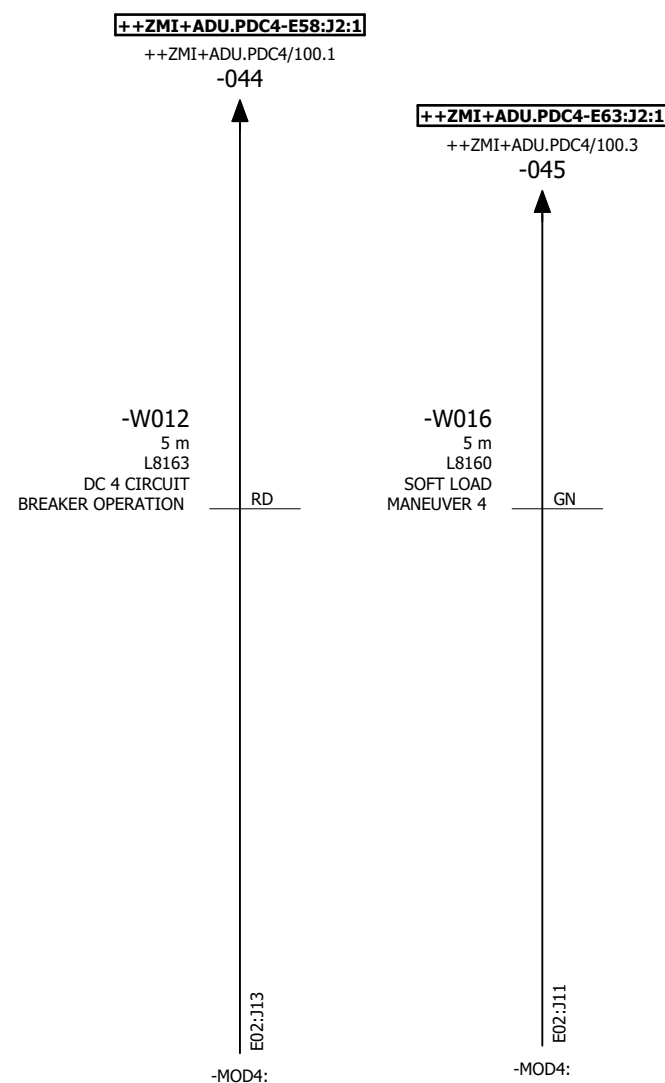
TOWARDS PCB E02 (J11; J13)
MODULE 2

MODULE 3



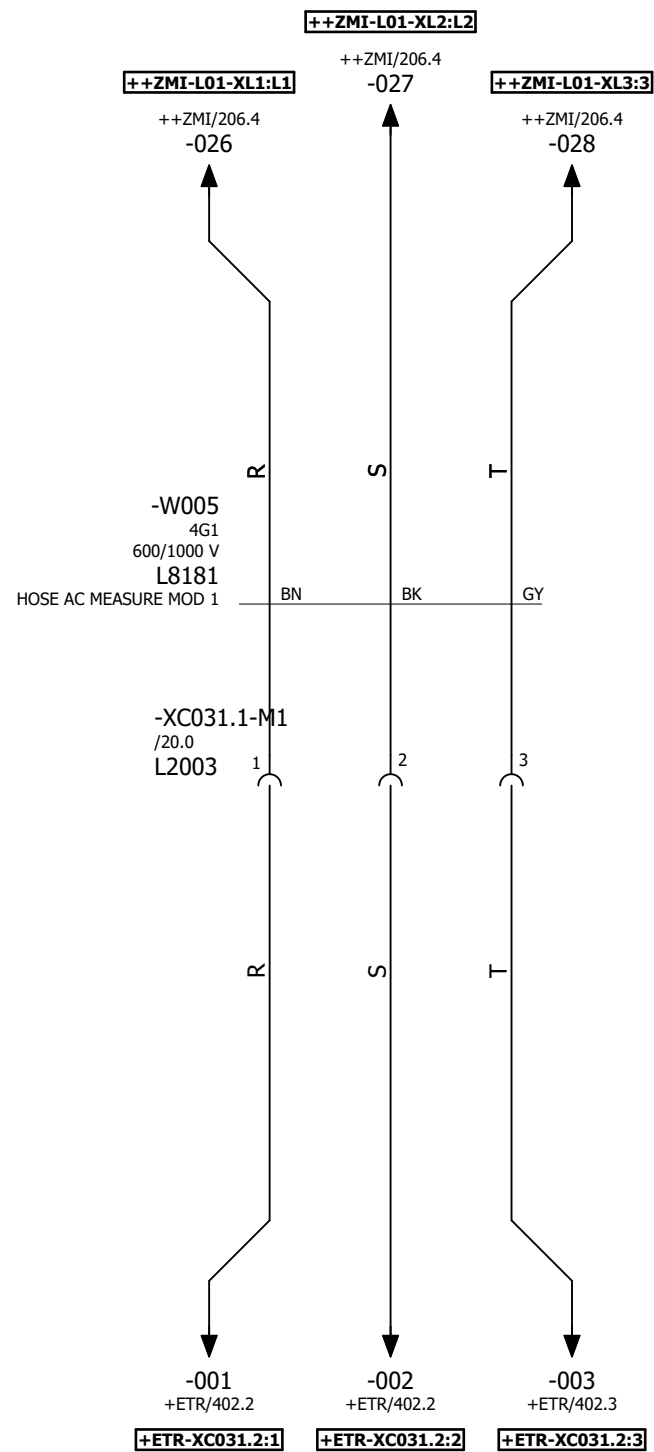
TOWARDS PCB E02 (J11; J13)
MODULE 3

MODULE 4



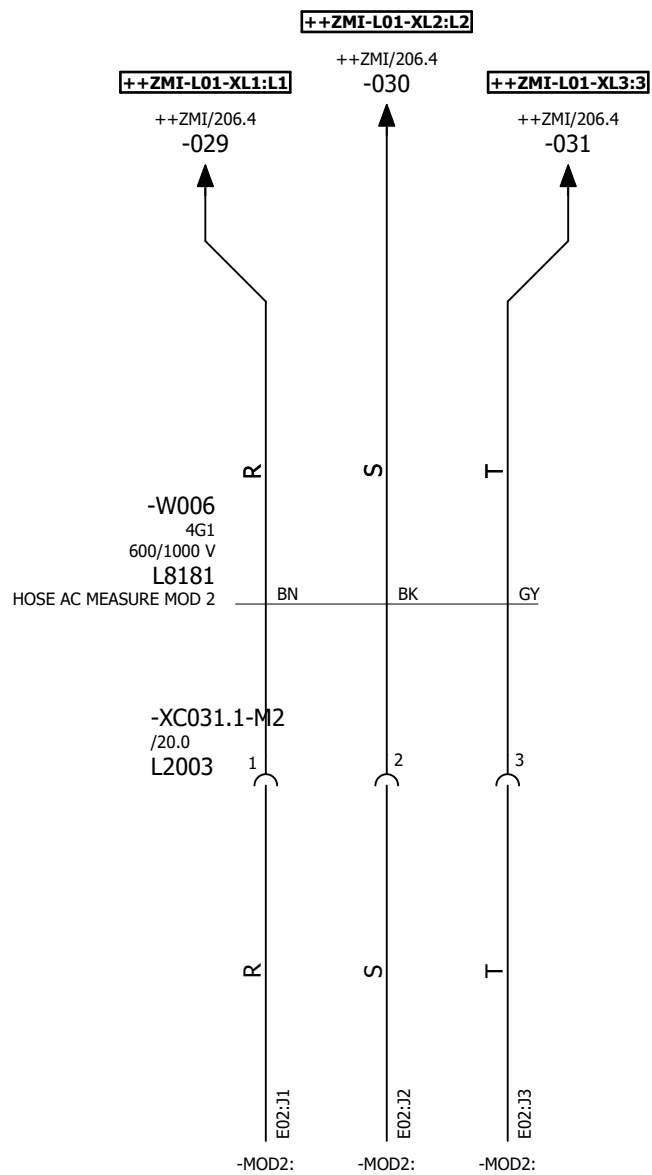
TOWARDS PCB E02 (J11; J13)
MÓDULO 4

MODULE 1



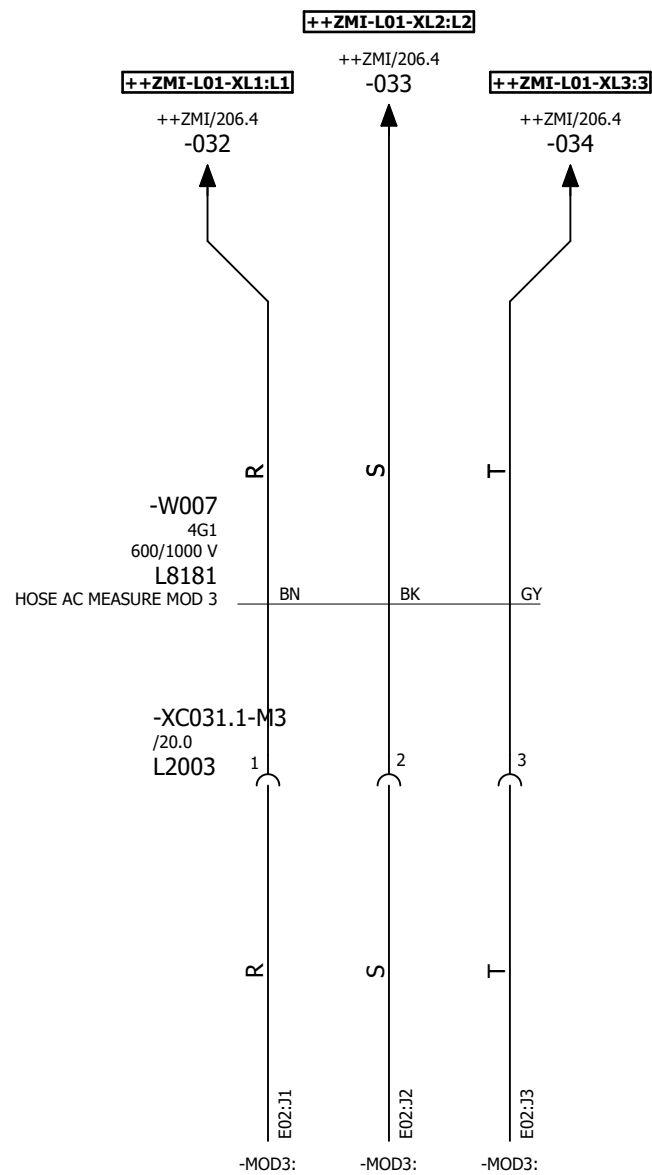
TOWARDS PCB E02 (J1; J2; J3)
MODULE 1

MODULE 2



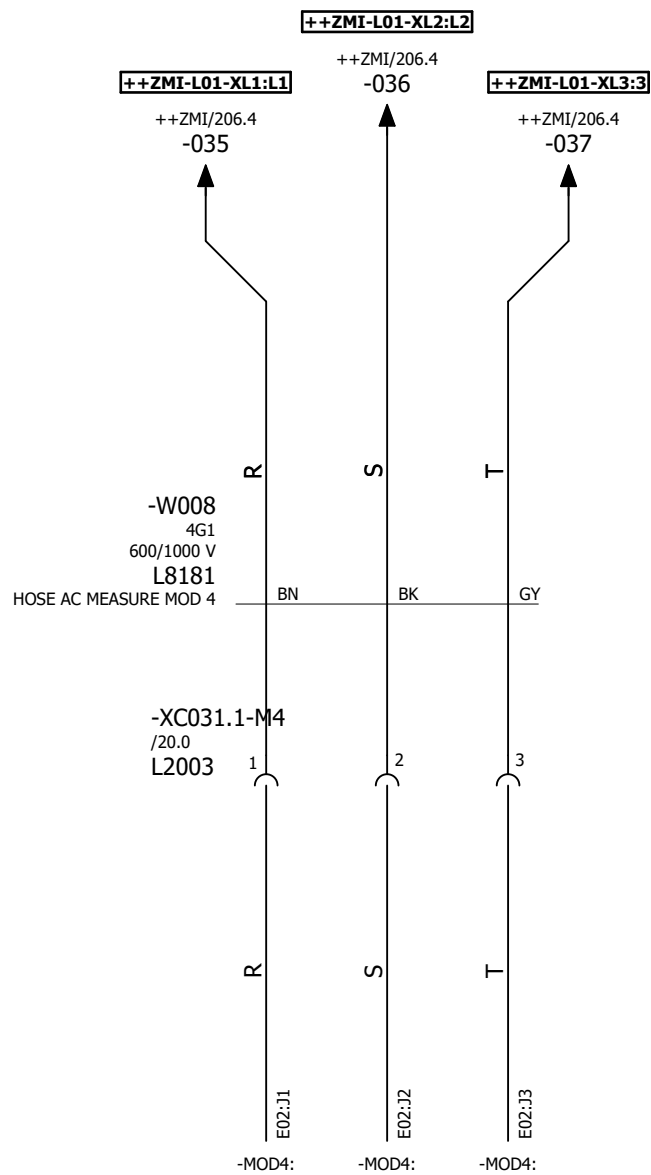
TOWARDS PCB E02 (J1; J2; J3)
MODULE 2

MODULE 3



TOWARDS PCB E02 (J1; J2; J3)
MODULE 3

MODULE 4



TOWARDS PCB E02 (J1; J2; J3)
MODULE 4

Cable overview

Cable designation	Function text	ERP
=HEM++MOI+-W001	OPTICAL FIBER MODULE 1	L7589
=HEM++MOI+-W002	OPTICAL FIBER MODULE 2	L7589
=HEM++MOI+-W003	OPTICAL FIBER MODULE 3	L7589
=HEM++MOI+-W004	OPTICAL FIBER MODULE 4	L7589
=HEM++MOI+-W005	HOSE AC MEASURE MOD 1	L8181
=HEM++MOI+-W006	HOSE AC MEASURE MOD 2	L8181
=HEM++MOI+-W007	HOSE AC MEASURE MOD 3	L8181
=HEM++MOI+-W008	HOSE AC MEASURE MOD 4	L8181
=HEM++MOI+-W009	DC 1 CIRCUIT BREAKER OPERATION	L8570
=HEM++MOI+-W010	DC 2 CIRCUIT BREAKER OPERATION	L8157
=HEM++MOI+-W011	DC 3 CIRCUIT BREAKER OPERATION	L8574
=HEM++MOI+-W012	DC 4 CIRCUIT BREAKER OPERATION	L8163
=HEM++MOI+-W013	SOFT LOAD MANEUVER 1	L8569
=HEM++MOI+-W014	SOFT LOAD MANEUVER 2	L8156
=HEM++MOI+-W015	SOFT LOAD MANEUVER 3	L8573
=HEM++MOI+-W016	SOFT LOAD MANEUVER 4	L8160

Plug overview

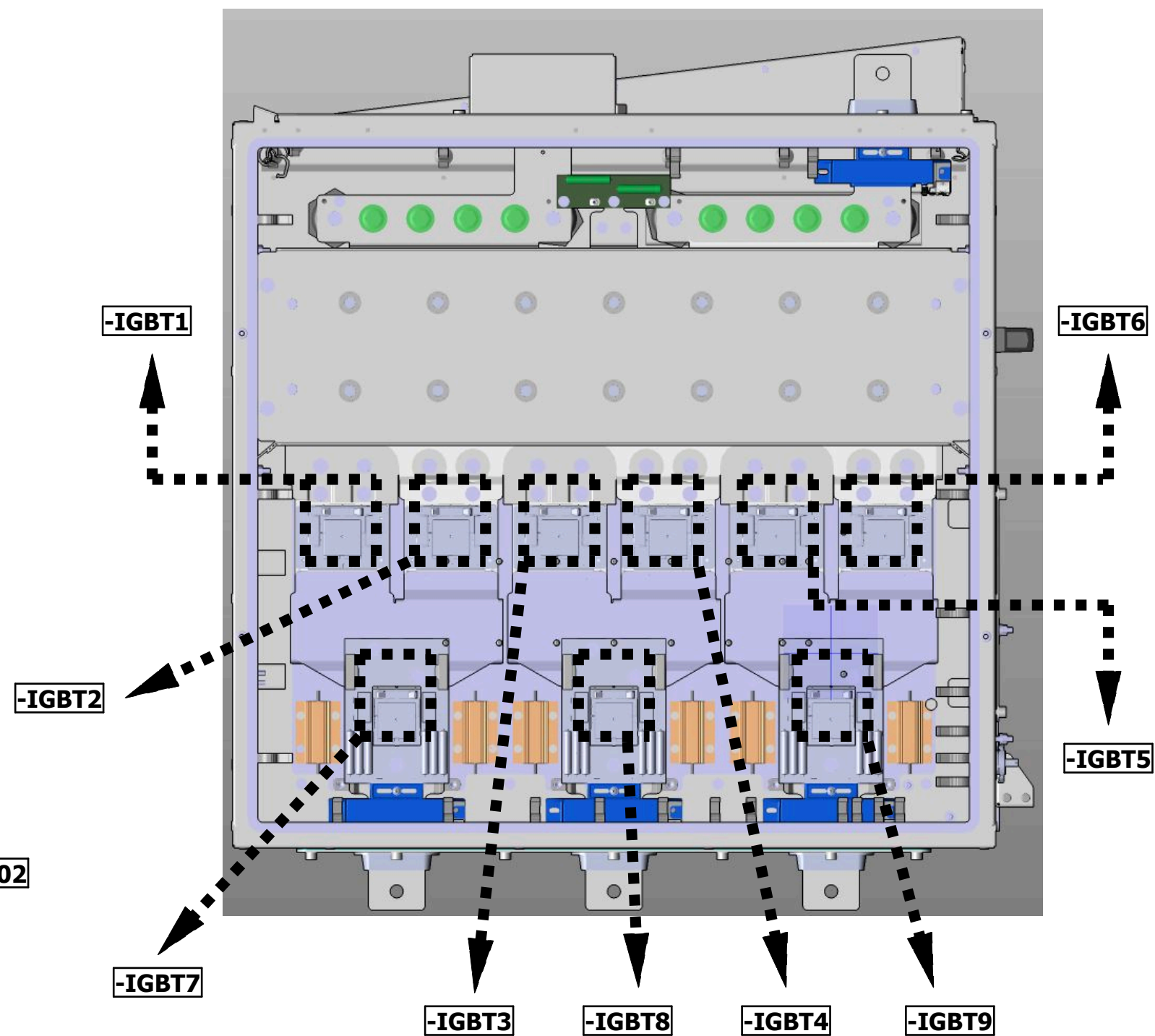
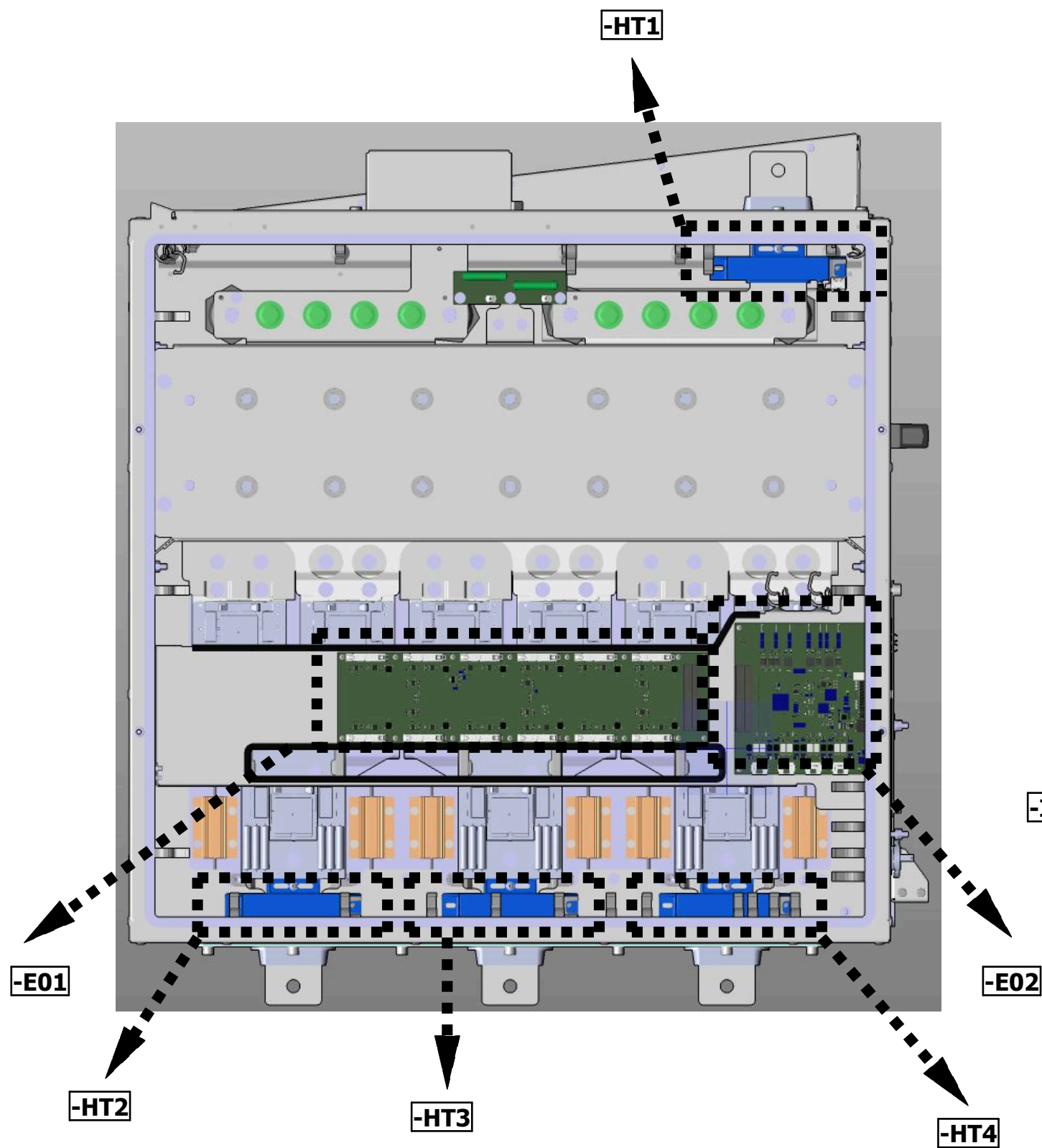
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++MOI+-XC001.1-M1	CONNECTOR OPTICAL FIBER DUPLEX MOD.1	L7620	1
=HEM++MOI+-XC002.1-M2	CONNECTOR OPTICAL FIBER DUPLEX MOD.2	L7620	1
=HEM++MOI+-XC003.1-M3	CONNECTOR OPTICAL FIBER DUPLEX MOD.3	L7620	1
=HEM++MOI+-XC004.1-M4	CONNECTOR OPTICAL FIBER DUPLEX MOD.4	L7620	1
=HEM++MOI+-XC031.1-M1	CONNECTOR AC MEASURE MOD.1	L2003	1
		E1203	3
=HEM++MOI+-XC031.1-M2	CONNECTOR AC MEASURE MOD.2	L2003	1
		E1203	3
=HEM++MOI+-XC031.1-M3	CONNECTOR AC MEASURE MOD.3	L2003	1
		E1203	3
=HEM++MOI+-XC031.1-M4	CONNECTOR AC MEASURE MOD.4	L2003	1
		E1203	3
=HEM++MOI+-XC046.2	MATE-N-LOK 9P AERIAL CONNECTOR AC/DC MODULES	L7534	1
		E1204	8



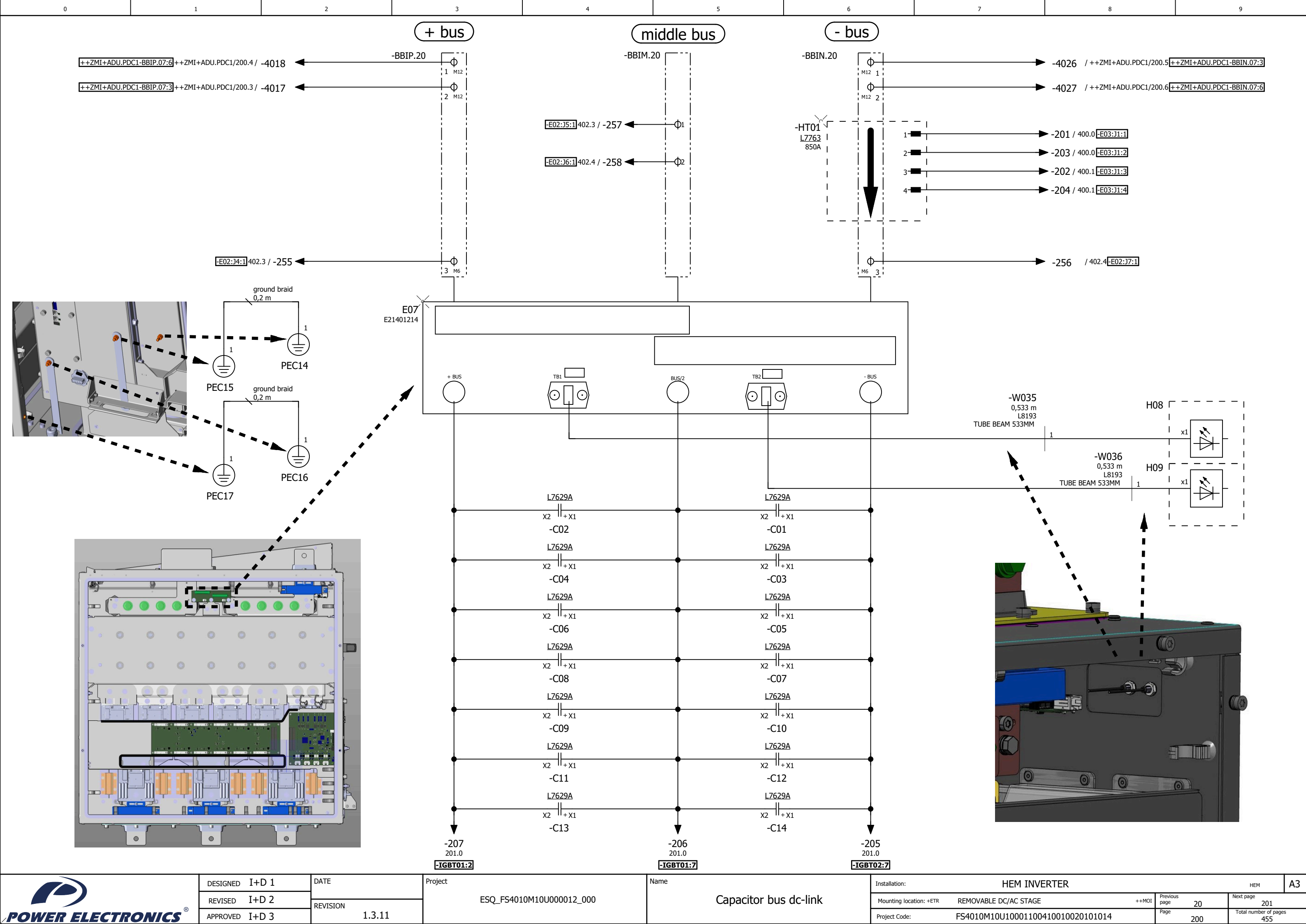
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

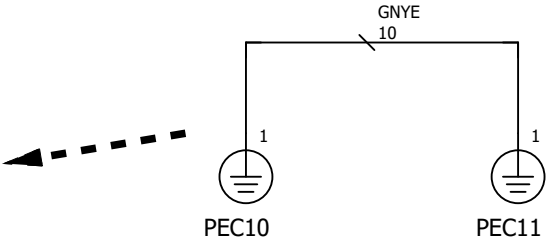
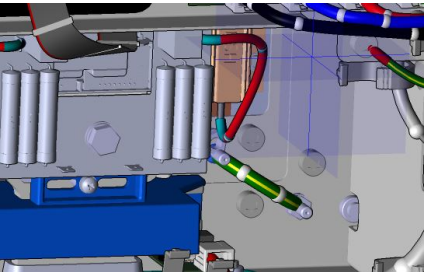
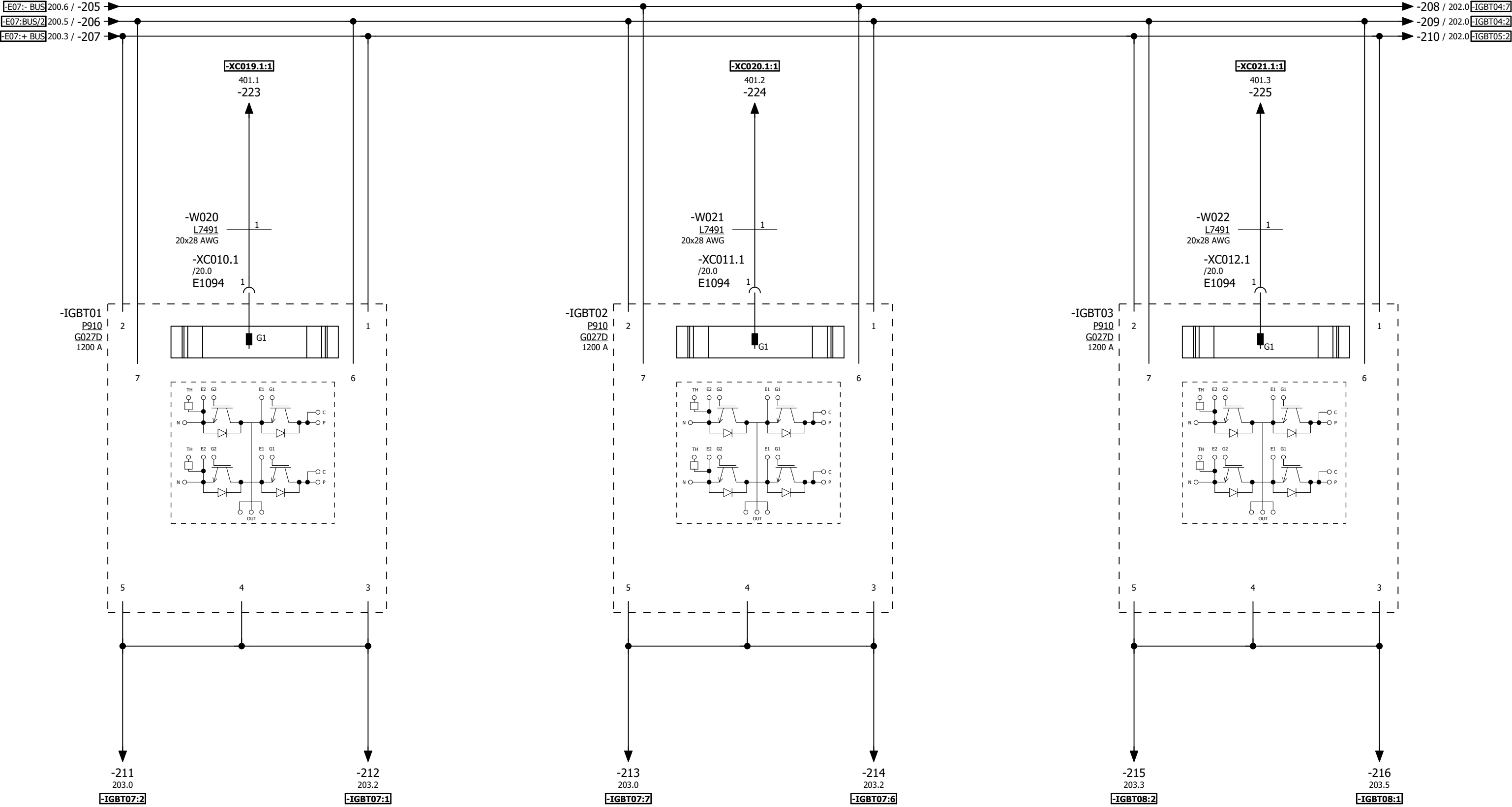
HEM	HEM INVERTER		
MOI	INVERTER MODULE		
ETR	REMOVABLE DC/AC STAGE		
ETR REMOVABLE DC/AC STAGE			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

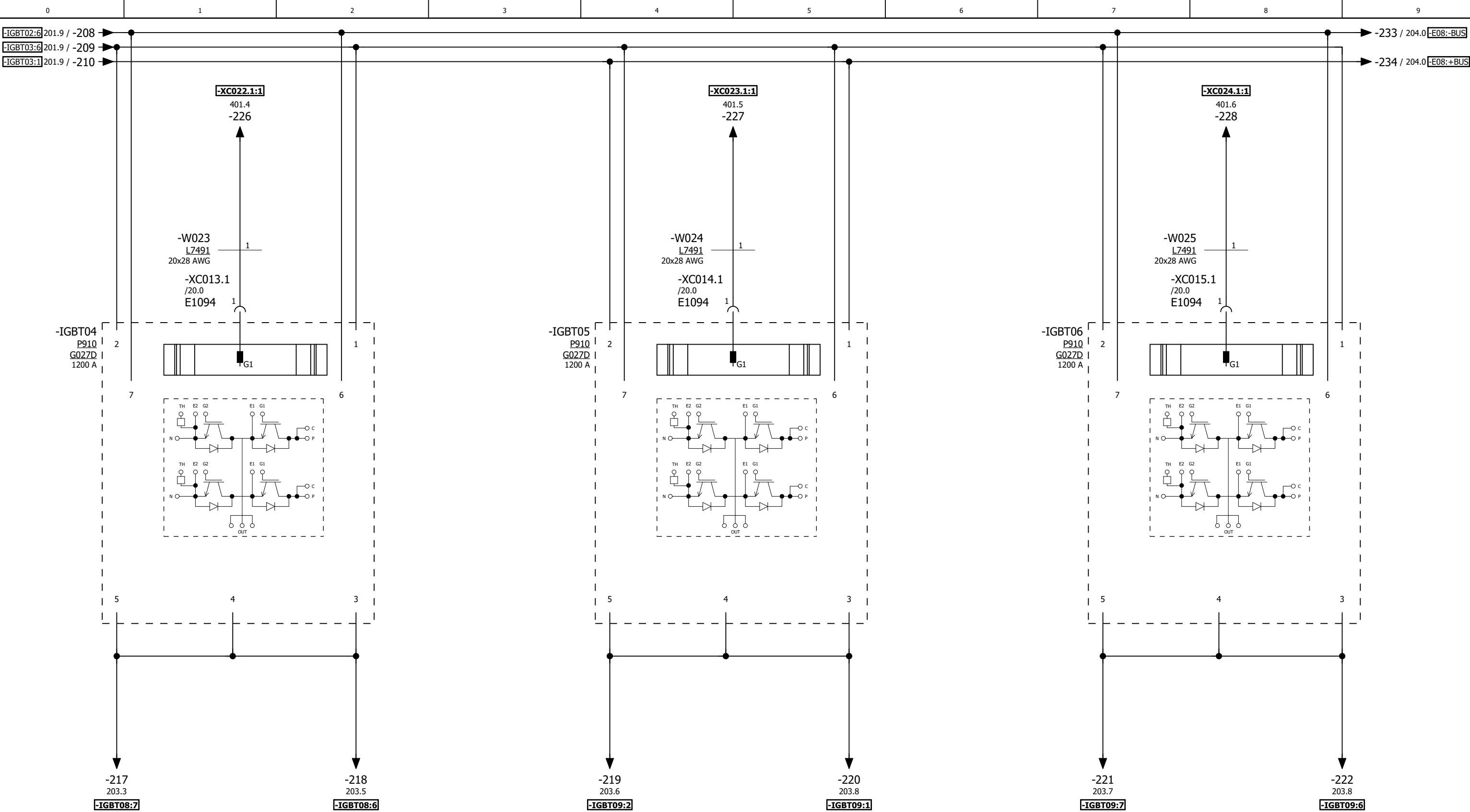
PULL OUT ASSEMBLY (+ETR)

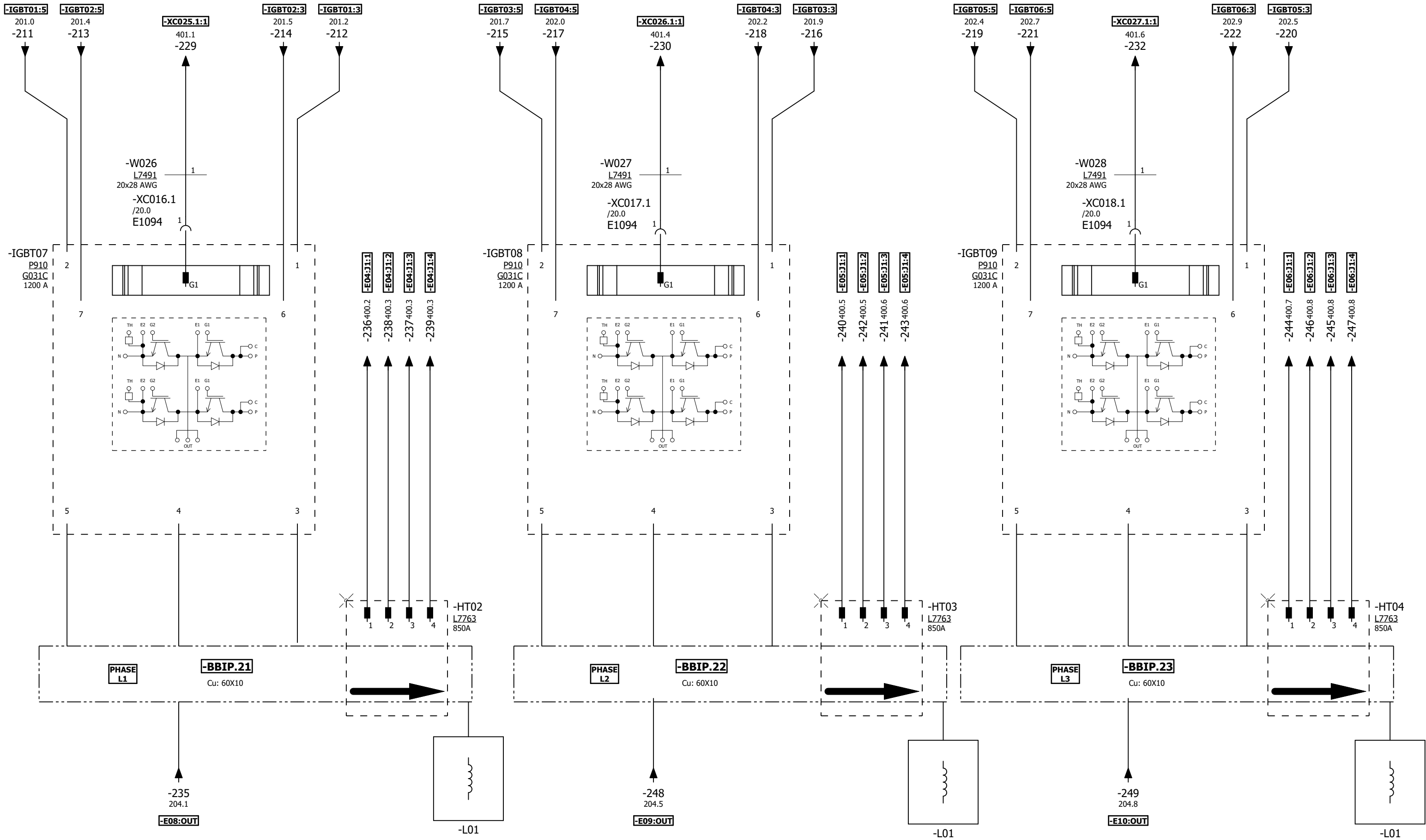


0	1	2	3	4	5	6	7	8	9
-XC010.1 = CONNECTOR FLAT IGBT 1 -XC011.1 = CONNECTOR FLAT IGBT 2 -XC012.1 = CONNECTOR FLAT IGBT 3 -XC013.1 = CONNECTOR FLAT IGBT 4 -XC014.1 = CONNECTOR FLAT IGBT 5 -XC015.1 = CONNECTOR FLAT IGBT 6 -XC016.1 = CONNECTOR FLAT IGBT 7 -XC017.1 = CONNECTOR FLAT IGBT 8 -XC018.1 = CONNECTOR FLAT IGBT 9 -XC019.1 = CONNECTOR FLAT IGBT 1 -XC020.1 = CONNECTOR FLAT IGBT 2 -XC021.1 = CONNECTOR FLAT IGBT 3 -XC022.1 = CONNECTOR FLAT IGBT 4 -XC023.1 = CONNECTOR FLAT IGBT 5 -XC024.1 = CONNECTOR FLAT IGBT 6 -XC025.1 = CONNECTOR FLAT IGBT 7 -XC026.1 = CONNECTOR FLAT IGBT 8 -XC027.1 = CONNECTOR FLAT IGBT 9 -XC031.2 = CONNECTOR AC MEASURE -XC029.1 = CONNECTOR FLAT 50 VIAS -XC030.1 = CONNECTOR FLAT 50 VIAS -XC031.1 = CONNECTOR FLAT 50 VIAS -XC032.1 = CONNECTOR FLAT 50 VIAS									

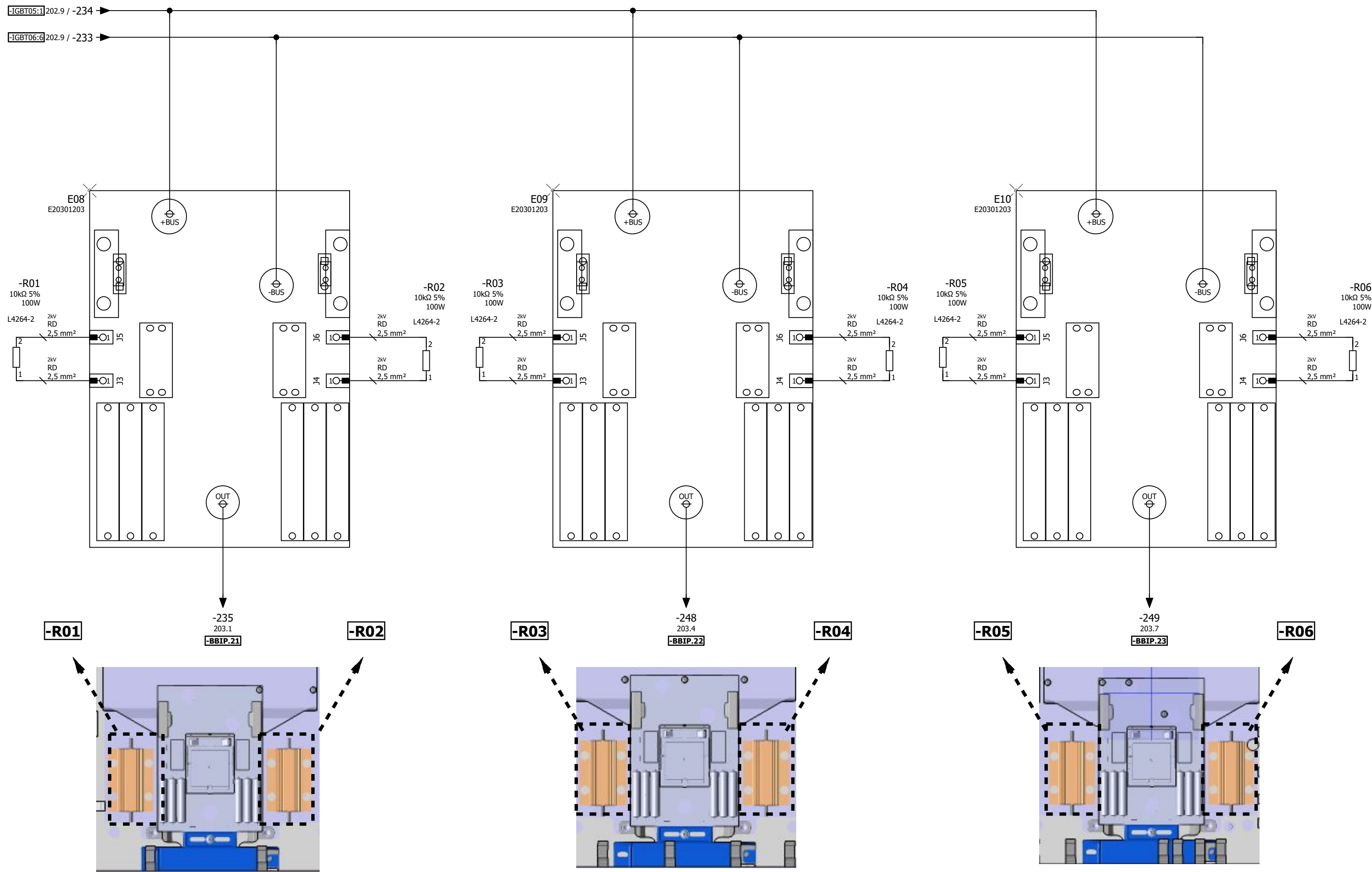


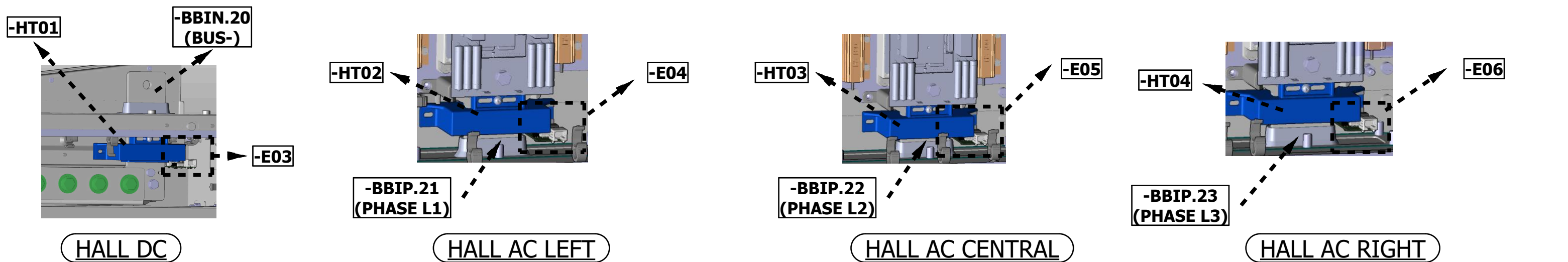


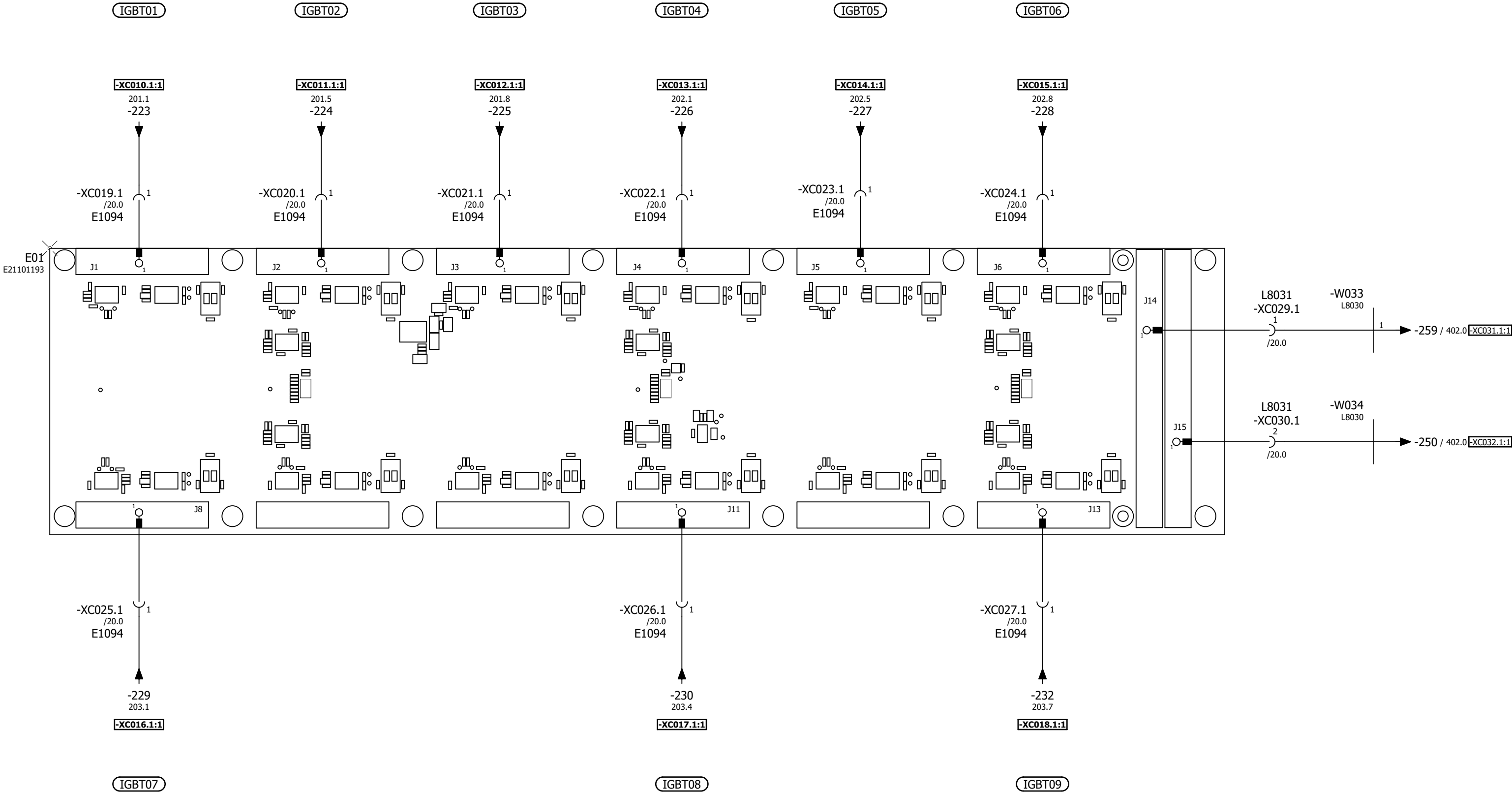


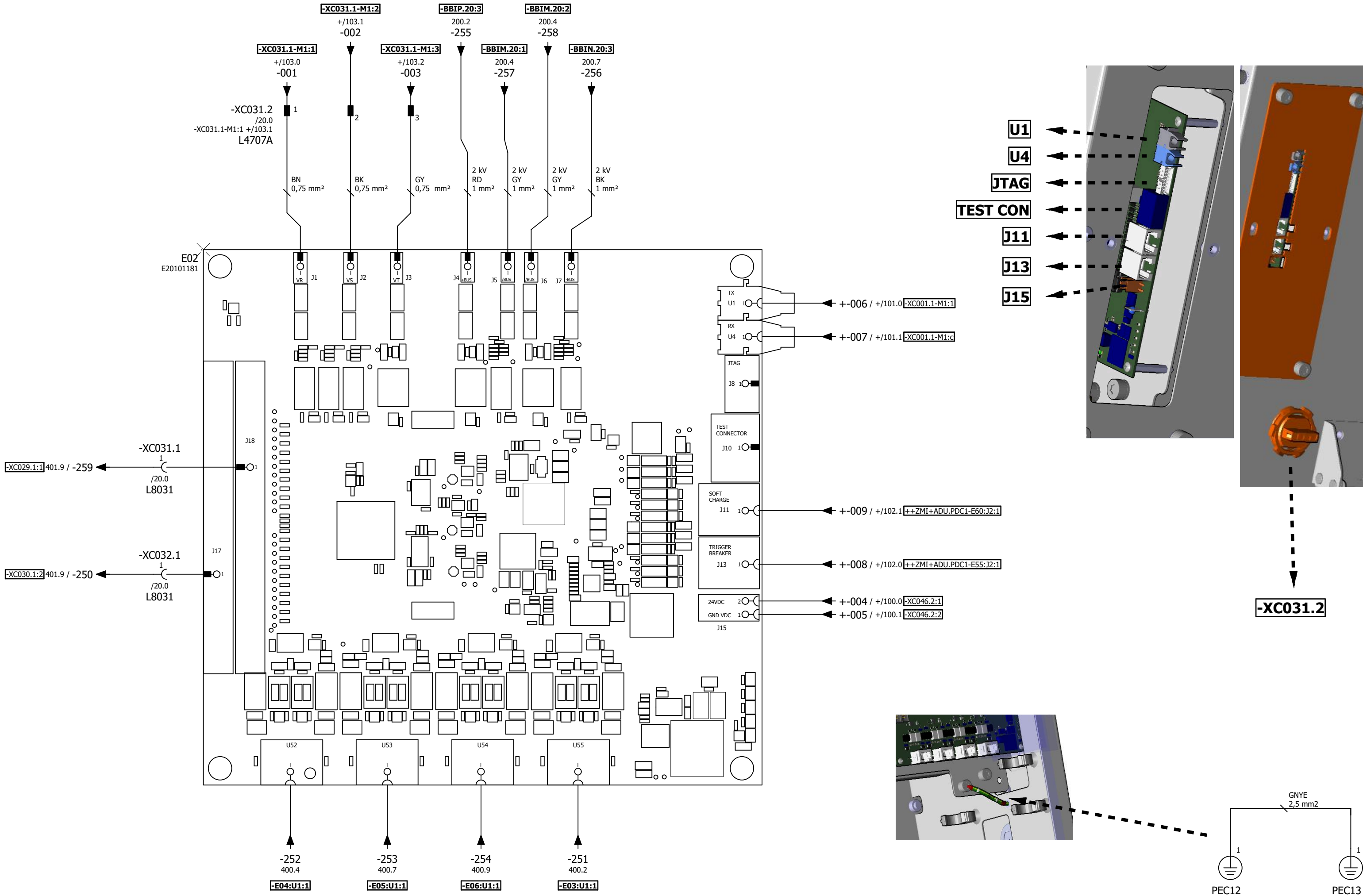


AC OUTPUT TOWARS INDUCTANCE









BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C01	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C02	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C03	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C04	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C05	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C06	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C07	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C08	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C09	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C10	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C11	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C12	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C13	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-C14	L7629A	C44U	CAPACITOR BUS HEM PLUS 14x880uF KEME	1
-E01	PCB E21101193	PCB E21101193	IGBT DISTRIBUTION HEM PLUS PCBA	1
-E02	PCB E20101181	PCB E20101181	CONTROL MODULE HEM PLUS PCBA	1
-E03	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E04	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E05	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E06	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E07	PCB E21401214	PCB E21401214	HEM GEN3 BUS BALANCING PCB BOARD	1
-E08	PCB E20301203	PCB E20301203	TARJETA SNUBBER HEM-MAX	1
-E09	PCB E20301203	PCB E20301203	TARJETA SNUBBER HEM-MAX	1
-E10	PCB E20301203	PCB E20301203	TARJETA SNUBBER HEM-MAX	1
-HT01	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT02	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT03	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT04	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-IGBT01	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT01	G027D	2LG03ACDC11N-01	G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT02	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT02	G027D	2LG03ACDC11N-01	G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT03	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT03	G027D	2LG03ACDC11N-01	G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT04	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT04	G027D	2LG03ACDC11N-01	G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT05	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT05	G027D	2LG03ACDC11N-01	G-Drive LV100 Tamura 2LG03ACDC11N-01	1

-IGBT06	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT06	G027D	2LG03ACDC11N-01	G-Drive LV100 Tamura 2LG03ACDC11N-01	1
-IGBT07	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT07	G031C	2LG04AADC11N-01	GATE DRIVE LV100 TAMURA 2LG04AADC11N-01	1
-IGBT08	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT08	G031C	2LG04AADC11N-01	GATE DRIVE LV100 TAMURA 2LG04AADC11N-01	1
-IGBT09	P910	CM1200DW-24T CGS-7261	IGBT 1200A 1200V LV100 CM1200DW-24T	1
-IGBT09	G031C	2LG04AADC11N-01	GATE DRIVE LV100 TAMURA 2LG04AADC11N-01	1
-R01	L4264-2	WD100 10K 5%	RESISTOR 100W 10K	1
-R02	L4264-2	WD100 10K 5%	RESISTOR 100W 10K	1
-R03	L4264-2	WD100 10K 5%	RESISTOR 100W 10K	1
-R04	L4264-2	WD100 10K 5%	RESISTOR 100W 10K	1
-R05	L4264-2	WD100 10K 5%	RESISTOR 100W 10K	1
-R06	L4264-2	WD100 10K 5%	RESISTOR 100W 10K	1

Cable overview

Cable designation	Function text	ERP
=HEM++MOI+ETR-W020	TWISTED WIRE TRIP IGBT 1	L7491
=HEM++MOI+ETR-W021	TWISTED WIRE TRIP IGBT 2	L7491
=HEM++MOI+ETR-W022	TWISTED WIRE TRIP IGBT 3	L7491
=HEM++MOI+ETR-W023	TWISTED WIRE TRIP IGBT 4	L7491
=HEM++MOI+ETR-W024	TWISTED WIRE TRIP IGBT 5	L7491
=HEM++MOI+ETR-W025	TWISTED WIRE TRIP IGBT 6	L7491
=HEM++MOI+ETR-W026	TWISTED WIRE TRIP IGBT 7	L7491
=HEM++MOI+ETR-W027	TWISTED WIRE TRIP IGBT 8	L7491
=HEM++MOI+ETR-W028	TWISTED WIRE TRIP IGBT 9	L7491
=HEM++MOI+ETR-W029	CABLE ETHERNET HALL HT1	L8567
=HEM++MOI+ETR-W030	CABLE ETHERNET HALL HT2	L8153
=HEM++MOI+ETR-W031	CABLE ETHERNET HALL HT3	L8568
=HEM++MOI+ETR-W032	CABLE ETHERNET HALL HT4	L8155
=HEM++MOI+ETR-W033	CABLE 50 VIAS	L8030
=HEM++MOI+ETR-W034	CABLE 50 VIAS	L8030
=HEM++MOI+ETR-W035	TUBE BEAM 533MM	L8193
=HEM++MOI+ETR-W036	TUBE BEAM 533MM	L8193

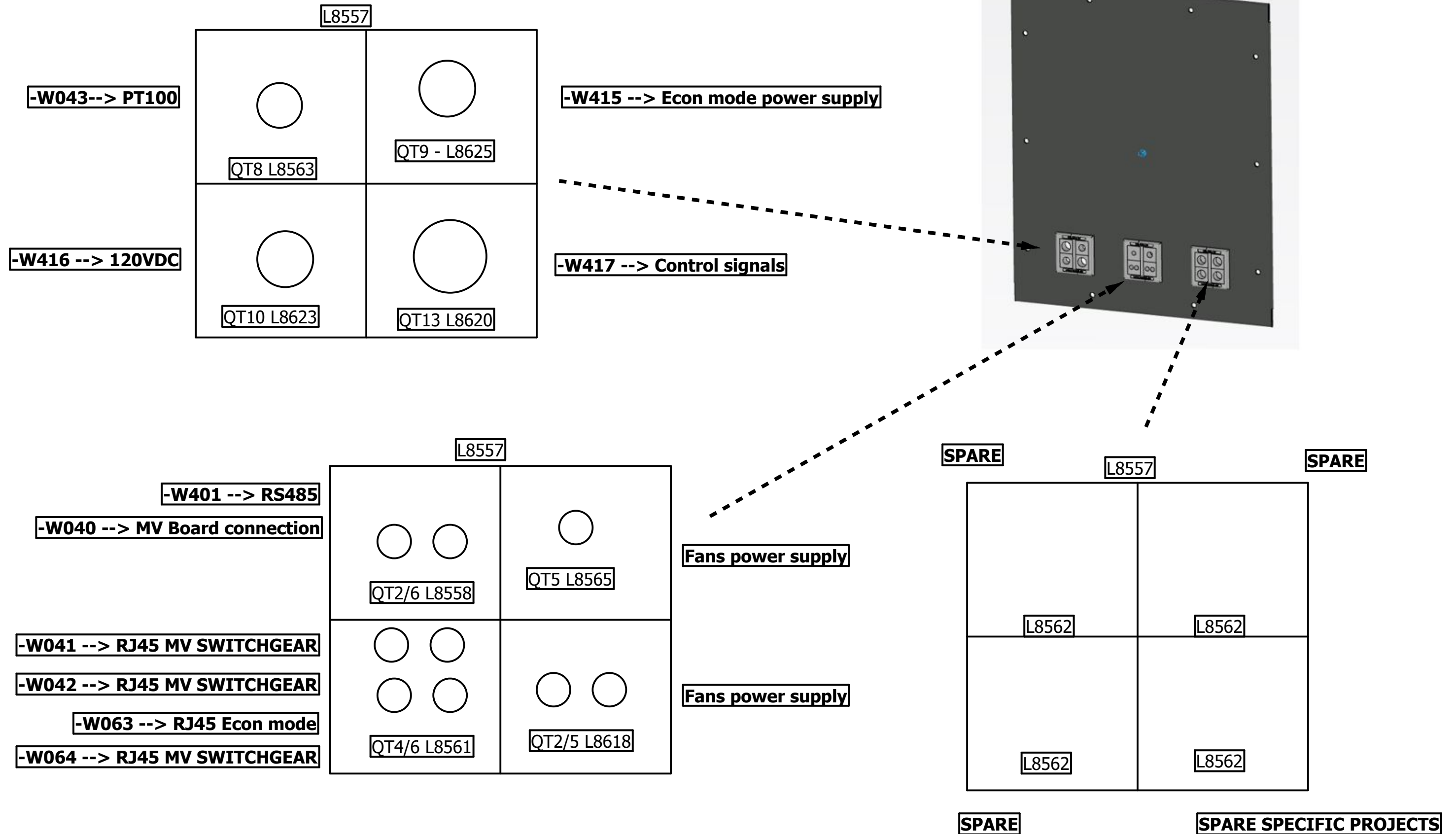
Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++MOI+ETR-XC010.1	CONNECTOR FLAT IGBT 1	E1094	1
		E1095	1
=HEM++MOI+ETR-XC011.1	CONNECTOR FLAT IGBT 2	E1094	1
		E1095	1
=HEM++MOI+ETR-XC012.1	CONNECTOR FLAT IGBT 3	E1094	1
		E1095	1
=HEM++MOI+ETR-XC013.1	CONNECTOR FLAT IGBT 4	E1094	1
		E1095	1
=HEM++MOI+ETR-XC014.1	CONNECTOR FLAT IGBT 5	E1094	1
		E1095	1
=HEM++MOI+ETR-XC015.1	CONNECTOR FLAT IGBT 6	E1094	1
		E1095	1
=HEM++MOI+ETR-XC016.1	CONNECTOR FLAT IGBT 7	E1094	1
		E1095	1
=HEM++MOI+ETR-XC017.1	CONNECTOR FLAT IGBT 8	E1094	1
		E1095	1
=HEM++MOI+ETR-XC018.1	CONNECTOR FLAT IGBT 9	E1094	1
		E1095	1
=HEM++MOI+ETR-XC019.1	CONNECTOR FLAT IGBT 1	E1094	1
		E1095	1
=HEM++MOI+ETR-XC020.1	CONNECTOR FLAT IGBT 2	E1094	1
		E1095	1
=HEM++MOI+ETR-XC021.1	CONNECTOR FLAT IGBT 3	E1094	1
		E1095	1
=HEM++MOI+ETR-XC022.1	CONNECTOR FLAT IGBT 4	E1094	1
		E1095	1
=HEM++MOI+ETR-XC023.1	CONNECTOR FLAT IGBT 5	E1094	1
		E1095	1

Plug overview

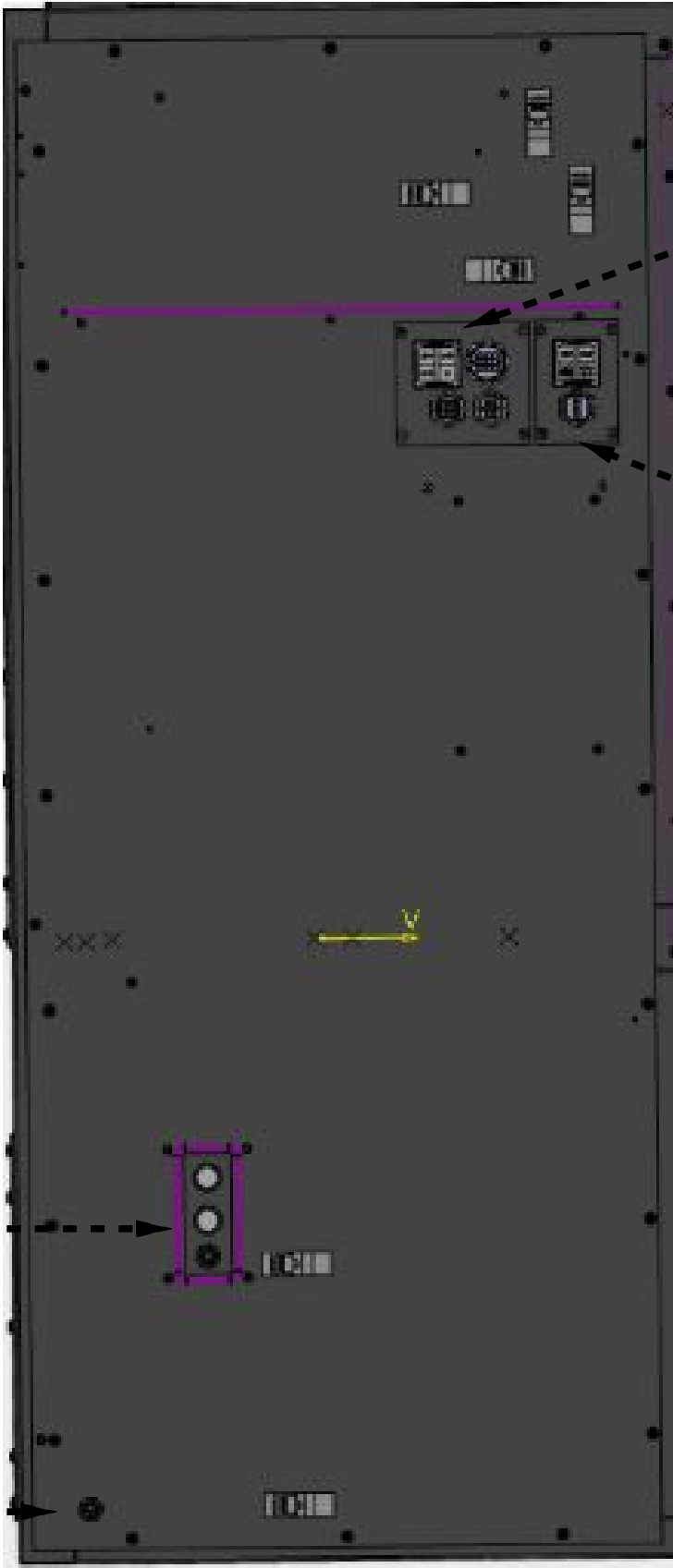
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++MOI+ETR-XC024.1	CONNECTOR FLAT IGBT 6	E1094	1
		E1095	1
=HEM++MOI+ETR-XC025.1	CONNECTOR FLAT IGBT 7	E1094	1
		E1095	1
=HEM++MOI+ETR-XC026.1	CONNECTOR FLAT IGBT 8	E1094	1
		E1095	1
=HEM++MOI+ETR-XC027.1	CONNECTOR FLAT IGBT 9	E1094	1
		E1095	1
=HEM++MOI+ETR-XC029.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=HEM++MOI+ETR-XC030.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=HEM++MOI+ETR-XC031.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1
=HEM++MOI+ETR-XC031.2	CONNECTOR AC MEASURE	L4707A	1
		E1204	3
		L4897	1
=HEM++MOI+ETR-XC032.1	CONNECTOR FLAT 50 VIAS	L8031	1
		L8029	1

Interconnection Panel
++CDT --> ++ZMI



Interconnection Panel

Control --> C Filter and Modules



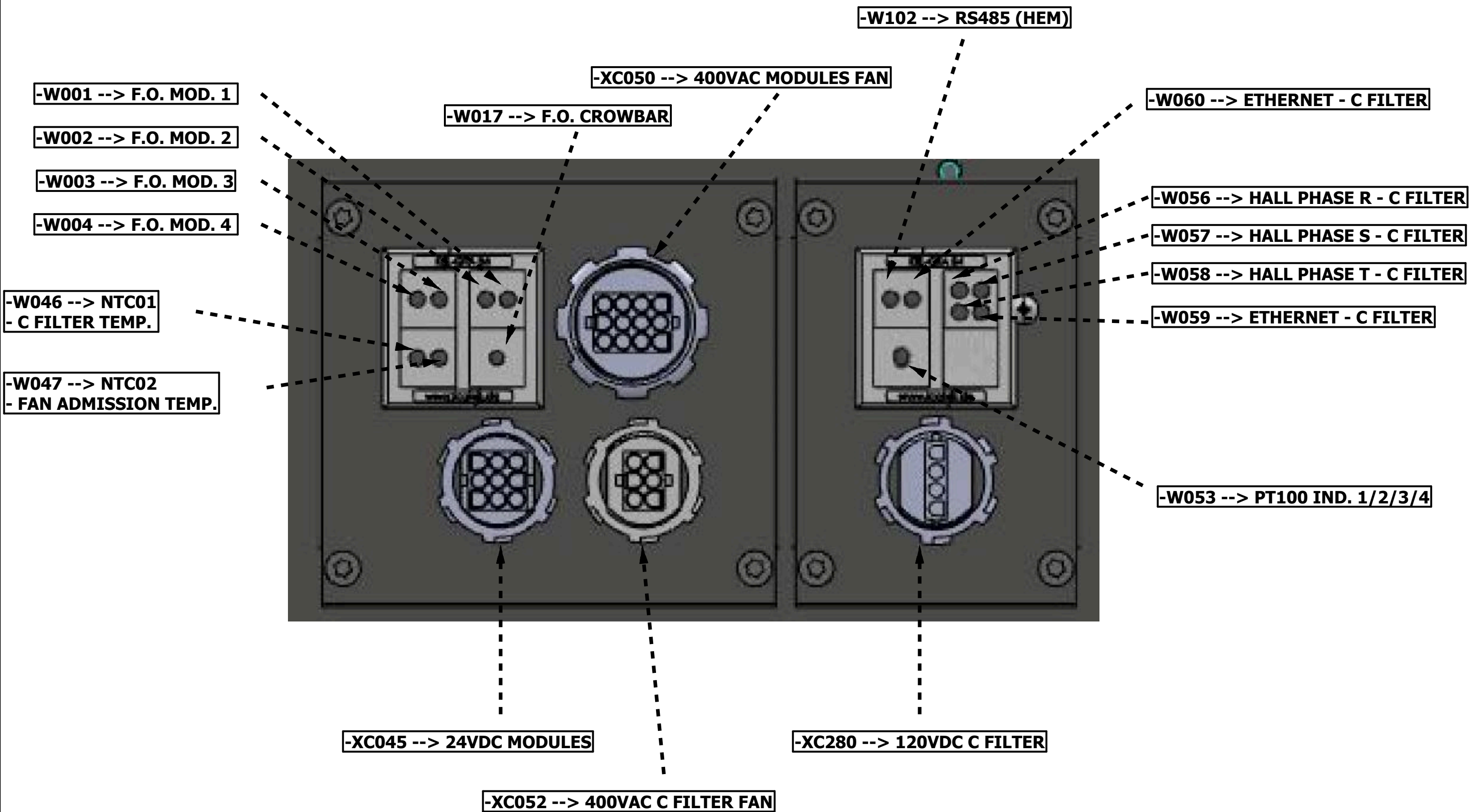
P. SUPPLY & COMMS
FANS AND MODULES

P. SUPPLY & COMMS
C FILTER AND INDUCTANCE

VOLTAGE & CURRENT
MEASURES

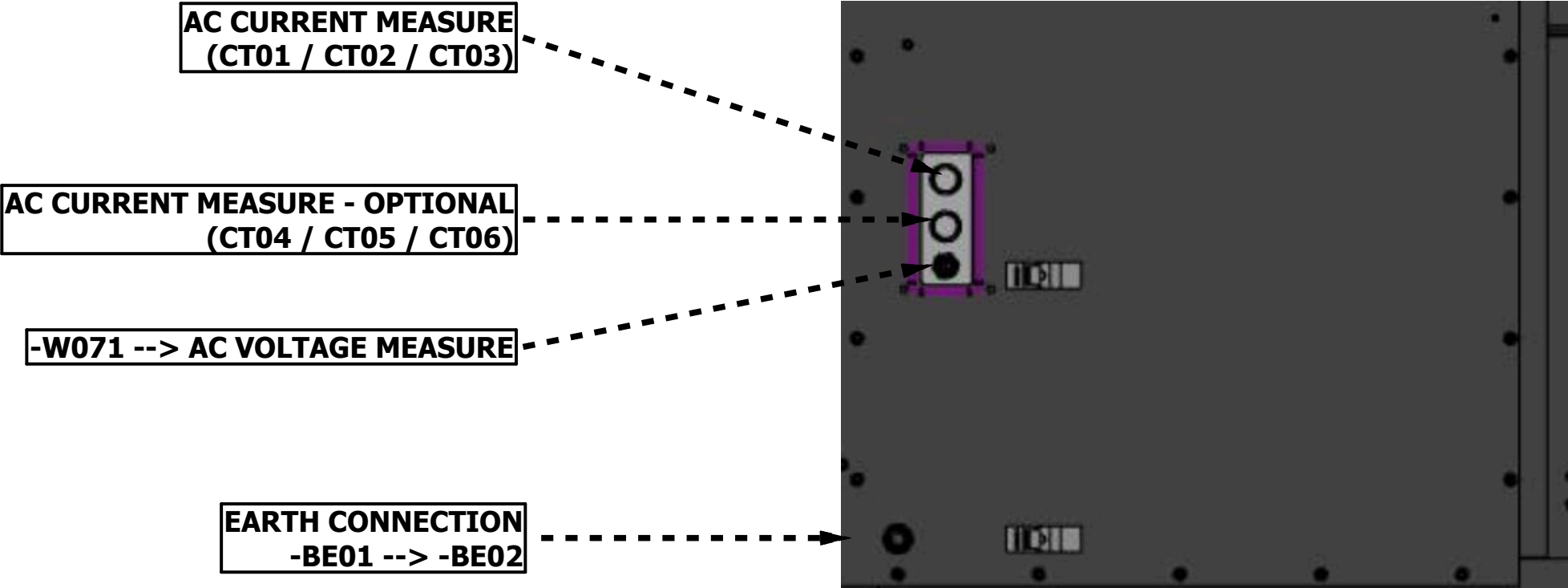
EARTH CONNECTION

Interconnection Panel
Control --> C Filter and Modules



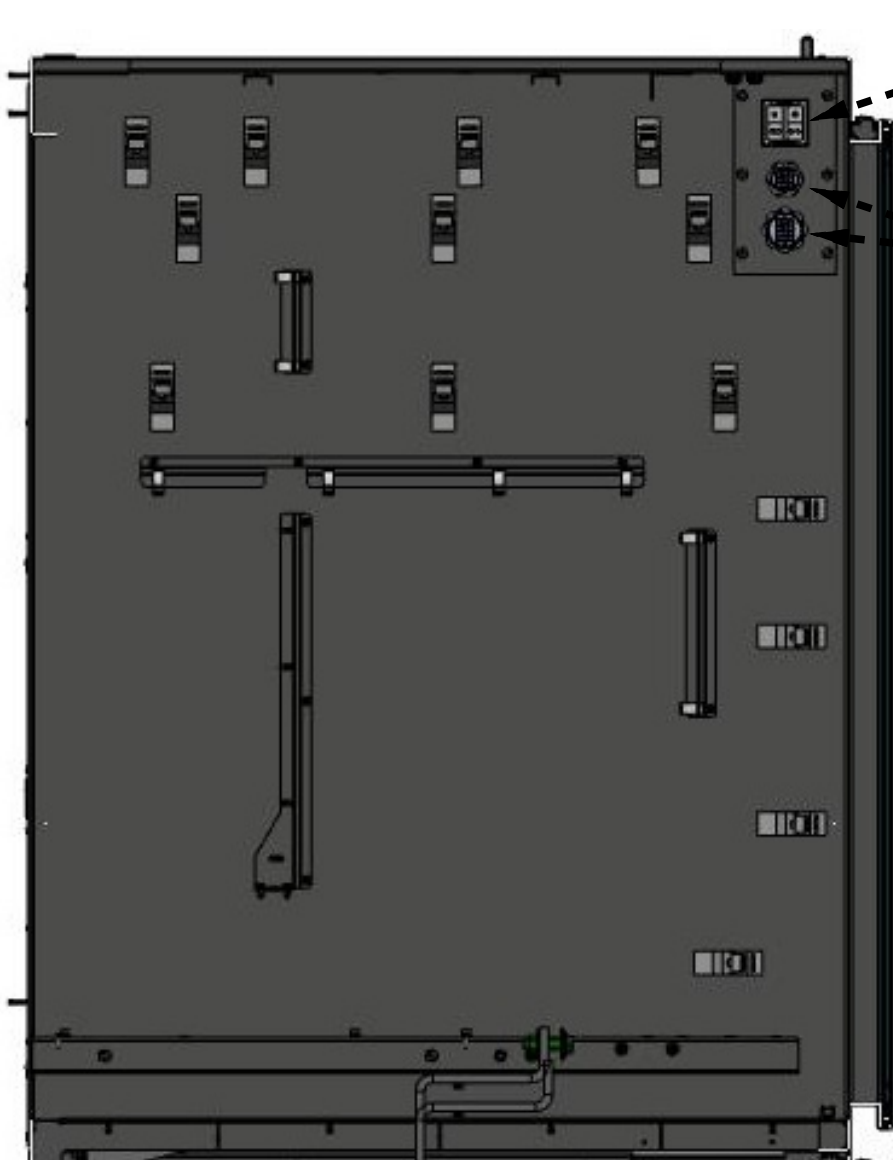
Interconnection Panel

Control --> C Filter and Modules



Interconnection Panel

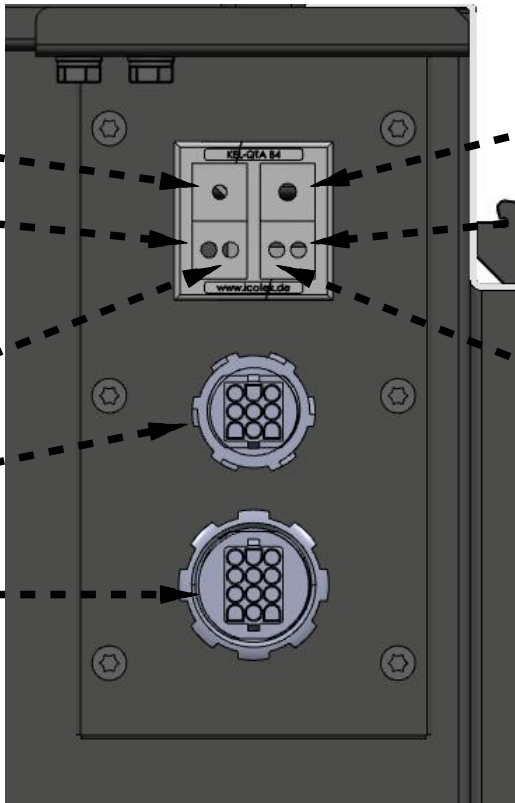
C Filter --> Modules



COMMS
FANS AND MODULES

P. SUPPLY
FANS AND MODULES

DETAIL



-W047 --> NTC02 - FAN ADMISSION TEMP.

-W004 --> F.O. MOD. 4

-W003 --> F.O. MOD. 3

-XC046 --> 24VDC MODULES

-XC051 --> 400VAC MODULES FAN

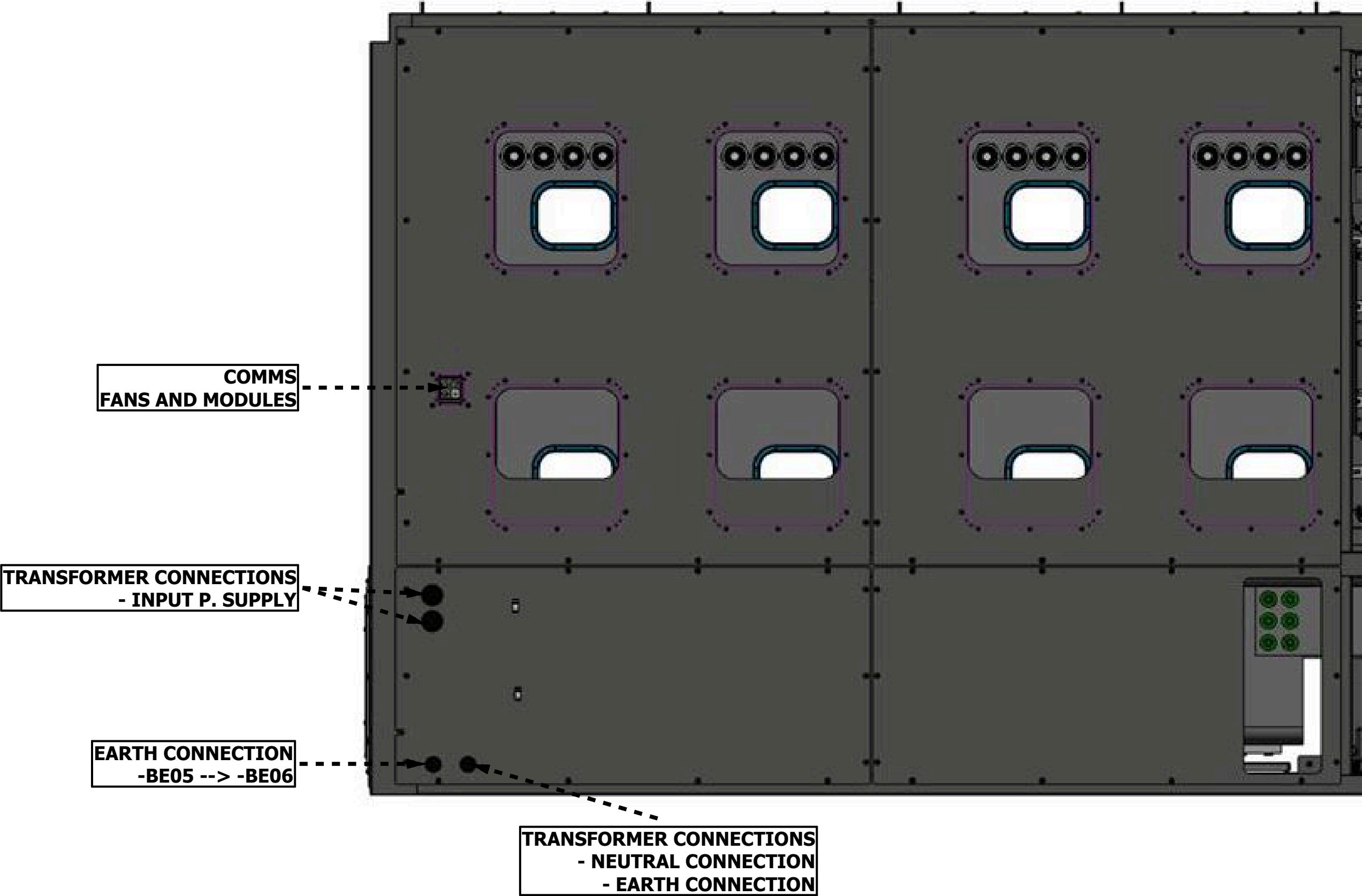
-W051 --> RS485
PMV6 --> PMV7

-W001 --> F.O. MOD. 1

-W002 --> F.O. MOD. 2

Interconnection Panel

DU --> Modules



Interconnection Panel

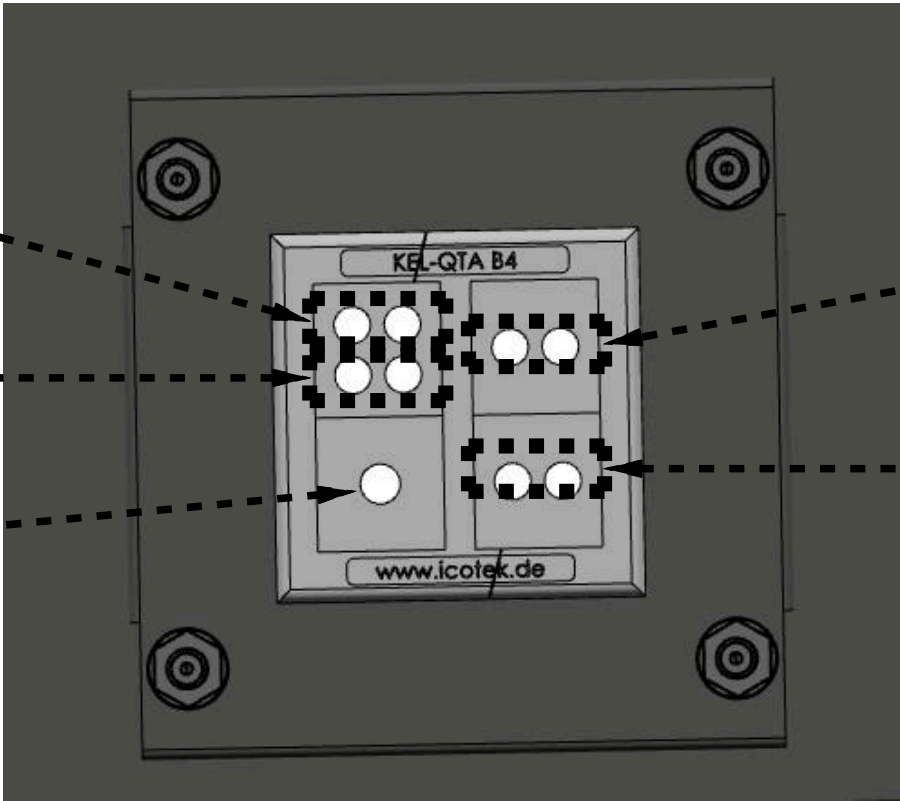
DU --> Modules

DETAIL

-W009 --> RJ45 (TRIGGER BREAKER)
-W013 --> RJ45 (SOFT LOAD)
PDC1 --> MOD. 1

-W010 --> RJ45 (TRIGGER BREAKER)
-W014 --> RJ45 (SOFT LOAD)
PDC2 --> MOD. 2

-W208 --> RS485
PMV2 --> PMV3



-W011 --> RJ45 (TRIGGER BREAKER)
-W015 --> RJ45 (SOFT LOAD)
PDC3 --> MOD. 3

-W012 --> RJ45 (TRIGGER BREAKER)
-W016 --> RJ45 (SOFT LOAD)
PDC4 --> MOD. 4

Interconnection Panel
DU --> SSAA

DETAIL

SECONDARY B - OPTIONAL
400/480 VAC (3PH+N)

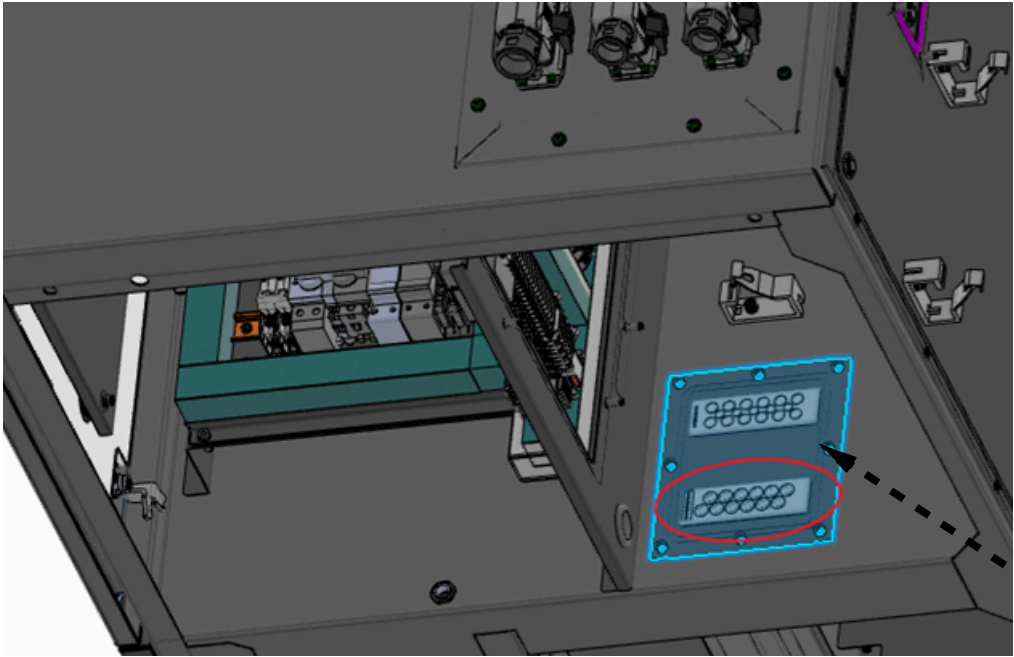
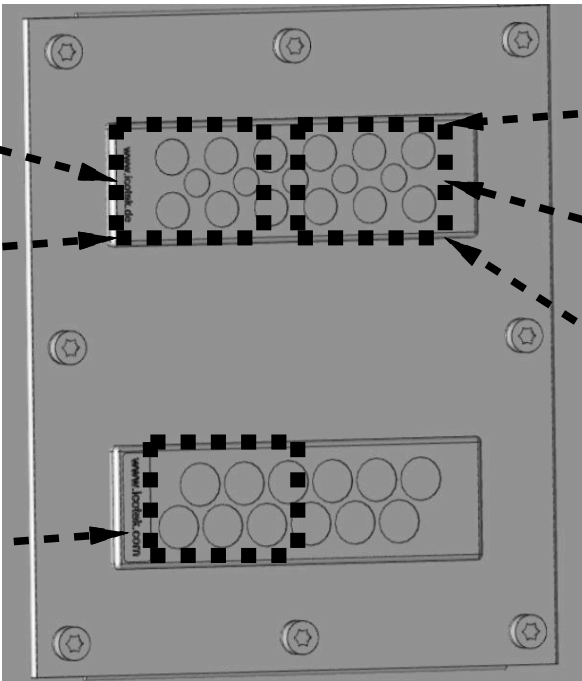
SECONDARY D - OPTIONAL
400/208 VAC (3PH+N)

SECONDARY A - INTERNAL
430 VAC (3PH+N)

SECONDARY C - INTERNAL
90 VAC (1PH+N)

-W070 --> THERMAL
PROTECTION

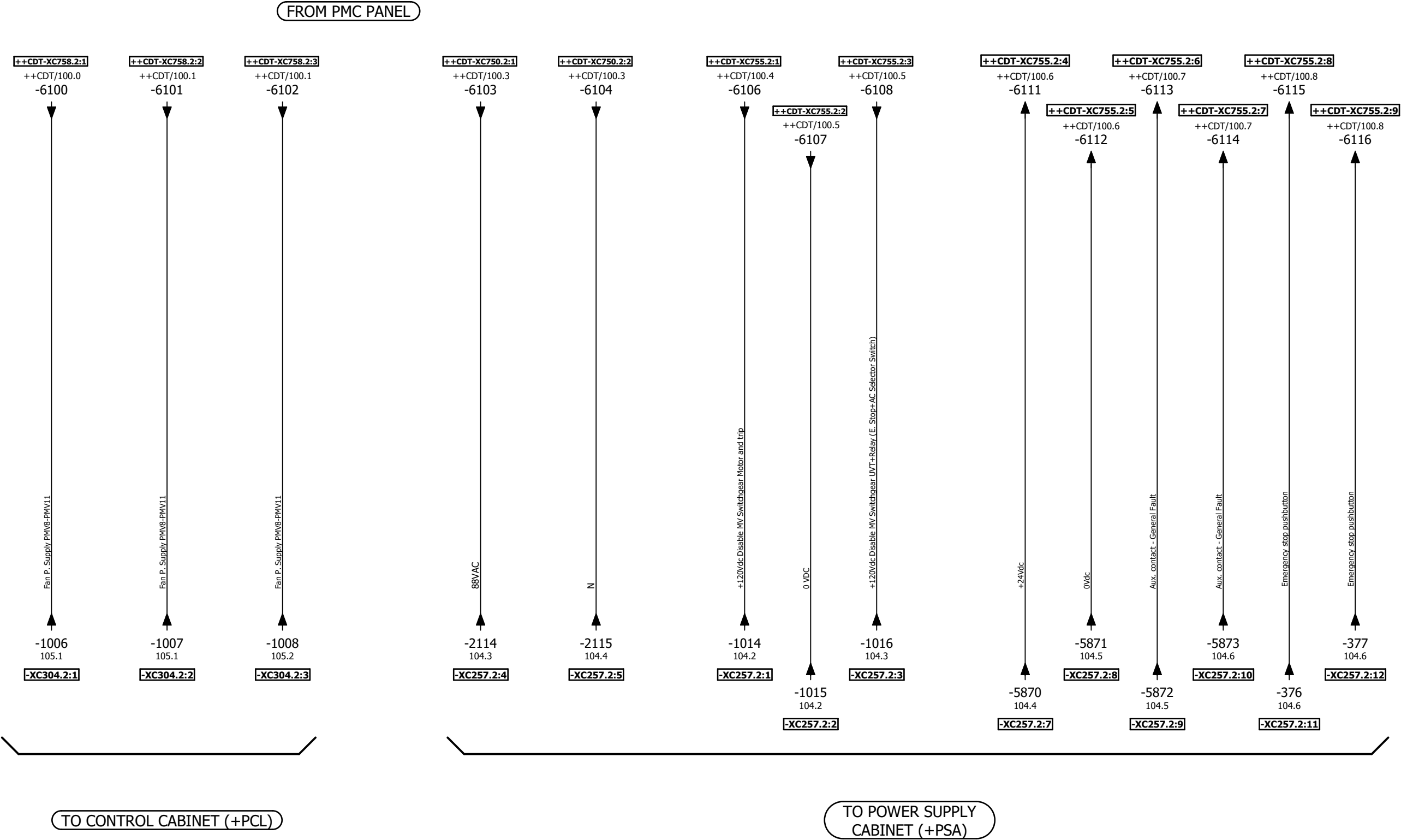
ERCOT/JAPAN OPTIONAL



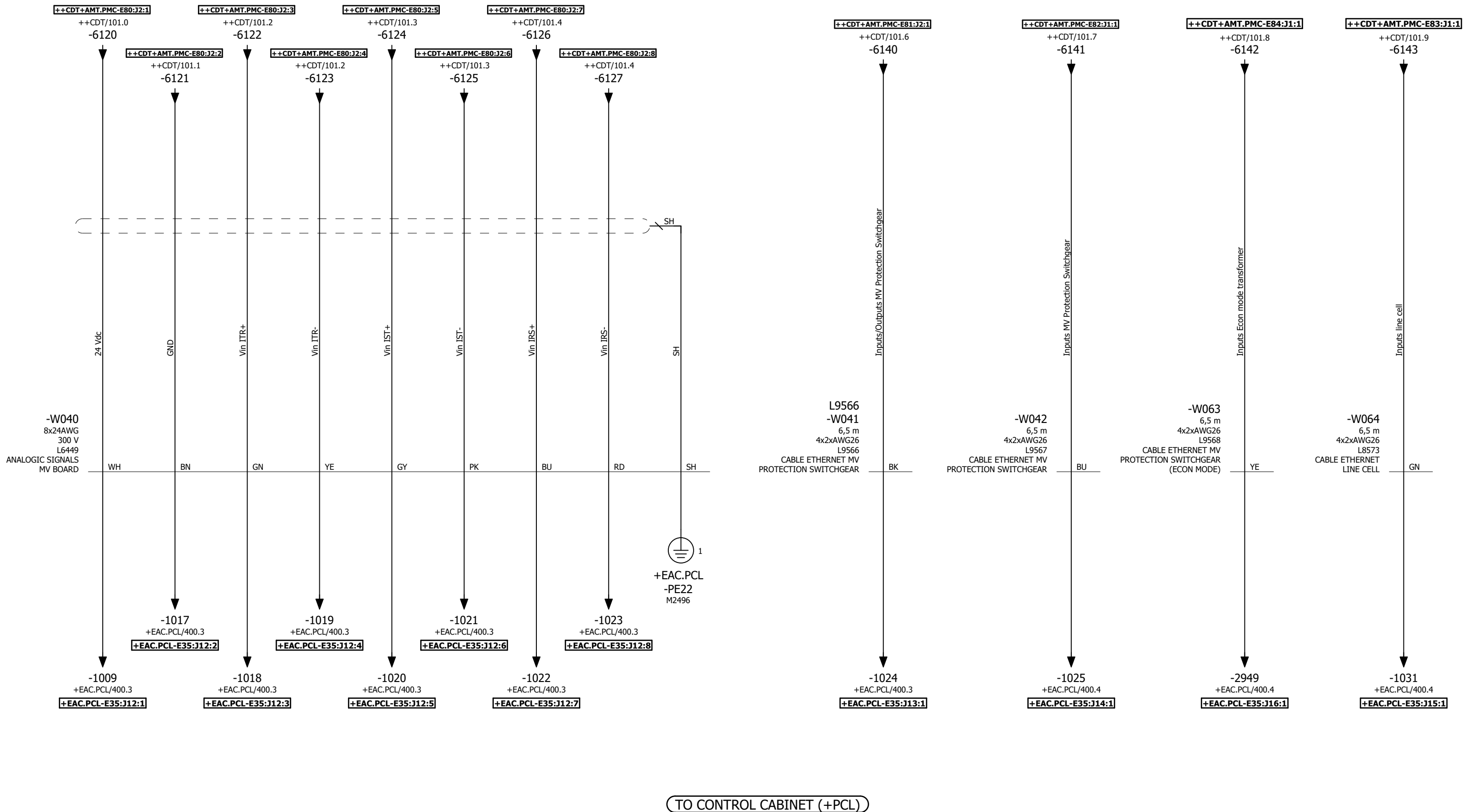
TRANSFORMER CONNECTIONS
- OUTPUT P. SUPPLY

0	1	2	3	4	5	6	7	8	9
-XC254.2 = MATE-N-LOK 6P AERIAL DISABLE DC -XC255.2 = MATE-N-LOK 2P AERIAL FEEDBACK THERMAL TRANS. -XC256.2 = MATE-N-LOK 4P AERIAL DISABLE AC -XC257.2 = MATE-N-LOK 12P AERIAL -XC280.1 = MATE-N-LOK 4P PANEL POWER SUPPLY AFC -XC280.2 = MATE-N-LOK 4P AERIAL POWER SUPPLY AFC -XC281.1 = MATE-N-LOK 2P AERIAL POWER SUPPLY AFC -XC302.2 = MATE-N-LOK 8P PANEL CONNECTOR POWER SUPPLY MOD -XC300.2 = MATE-N-LOK 6P PANEL CONNECTOR FAN SUPPLY -XC301.2 = MATE-N-LOK 12P PANEL CONNECTOR POWER SUPPLY PPD -XC303.2 = MATE-N-LOK 15P PANEL CONNECTOR FAN MODULES -XC304.2 = MATE-N-LOK AERIAL 6mm2 POWER SUPPLY FANS SKID -XC045.2 = MATE-N-LOK 9P AERIAL CONNECTOR MOD INTER -XC045.1 = MATE-N-LOK SEALED 9P PANEL CONNECTOR MOD INTER -XC046.1 = MATE-N-LOK SEALED 9P PANEL CONNECTOR AC/DC MOD -XC047.1 = MATE-N-LOK 9P PANEL CONNECTOR PCL-PPD -XC213.2 = MATE-N-LOK 4P AERIAL THERMAL IND -XC251.1 = MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 7 -XC201.1 = MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 3 -XC204.1 = MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 4 -XC207.1 = MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 5 -XC210.1 = MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 6 -XC050.1 = MATE-N-LOCK PANEL 12P SEALED -XC050.2 = MATE-N-LOCK AERIAL 12P -XC051.1 = MATE-N-LOCK PANEL 12P SEALED -XC051.2 = MATE-N-LOCK AERIAL 12P -XC052.1 = MATE-N-LOCK PANEL CONNECTOR 6P SEALED -XC052.2 = MATE-N-LOCK AERIAL 6P -XC053.1-M1 = CONNECTOR OPTICAL FIBER DUPLEX MOD.1 -XC054.1-M2 = CONNECTOR OPTICAL FIBER DUPLEX MOD.2 -XC055.1-M3 = CONNECTOR OPTICAL FIBER DUPLEX MOD.3 -XC056.1-M4 = CONNECTOR OPTICAL FIBER DUPLEX MOD.4 -XC309.2 = CONNECTOR MATE-N-LOK 3P AC MEASURE -XC364.1 = TRIPLE-LOK 2P ISOPRO FAULT SIGNAL									

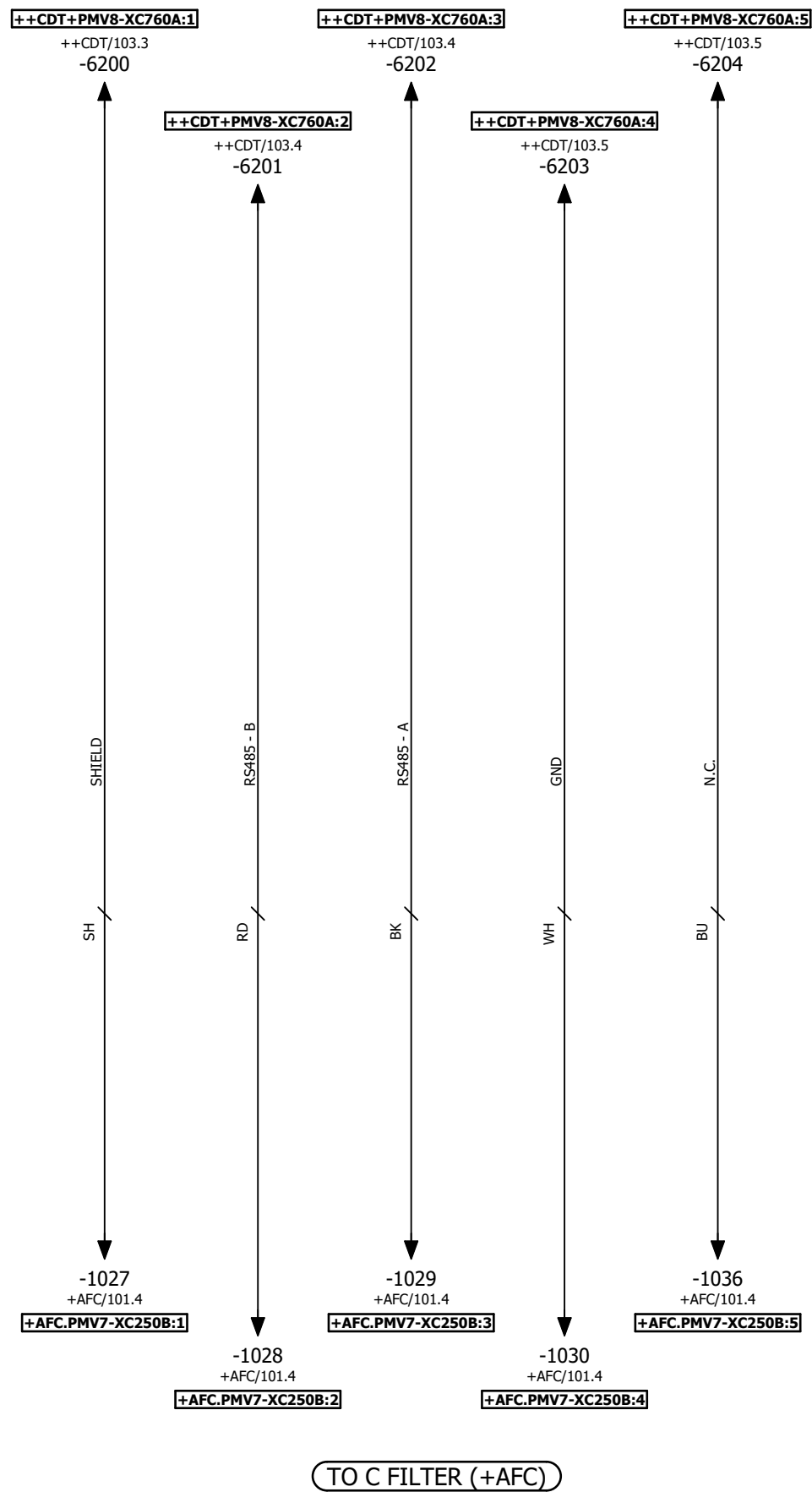
Interconnections ZMI/CDT - P. Supply



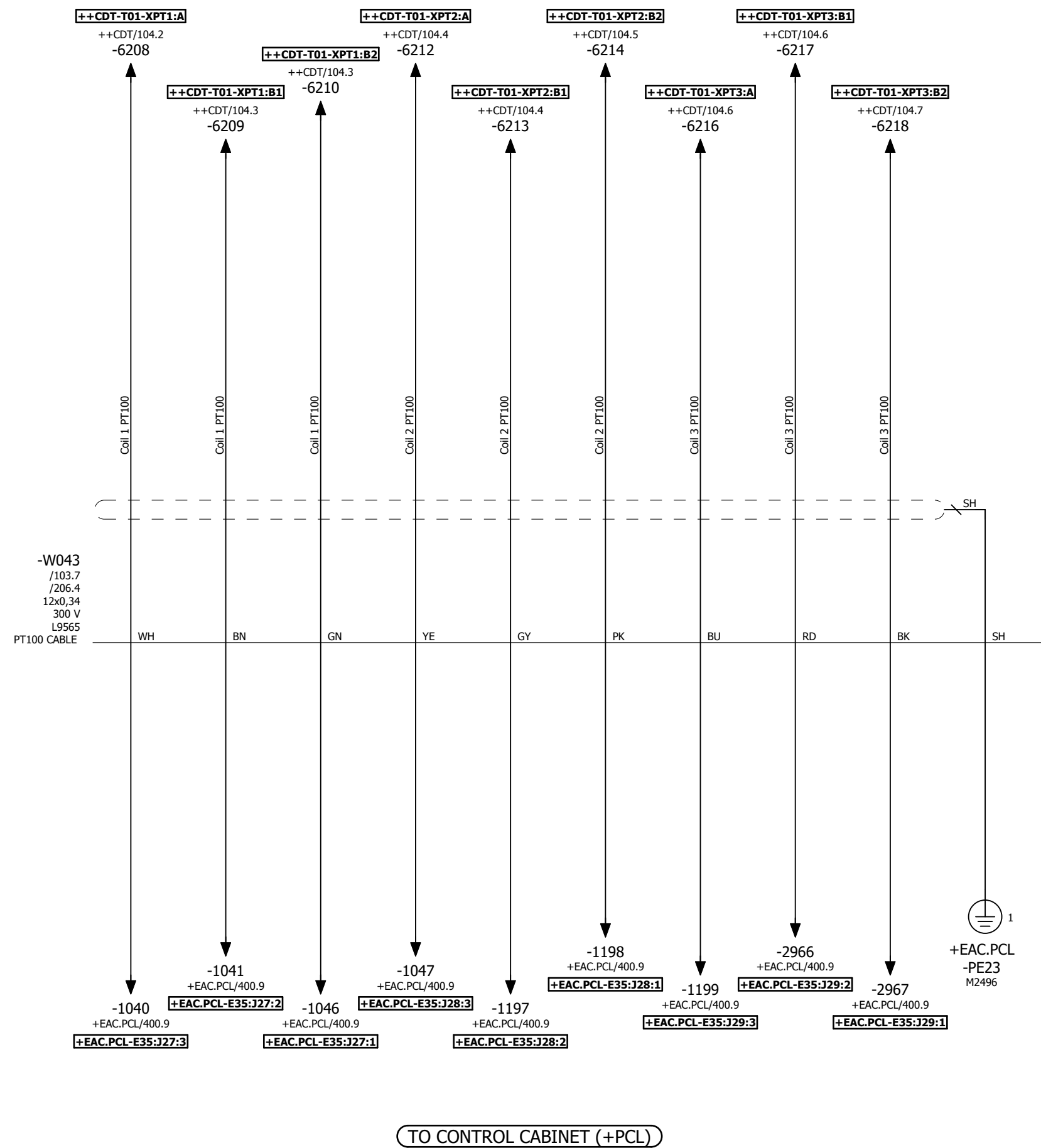
Interconnections ZMI/CDT - Comms/Measures



Interconnections ZMI/CDT - Comms/Measures

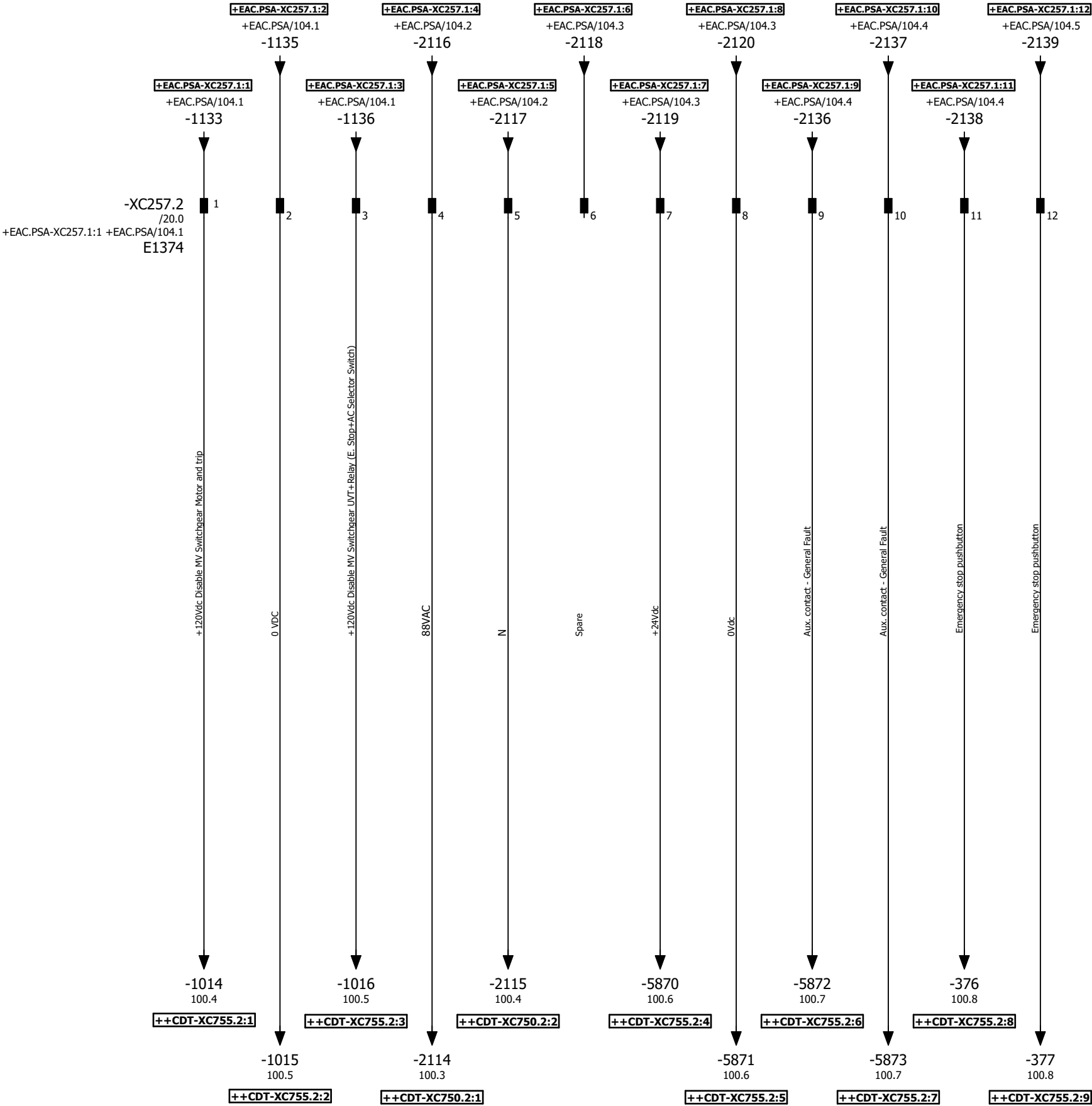


Interconnections ZMI/CDT - Comms/Measures

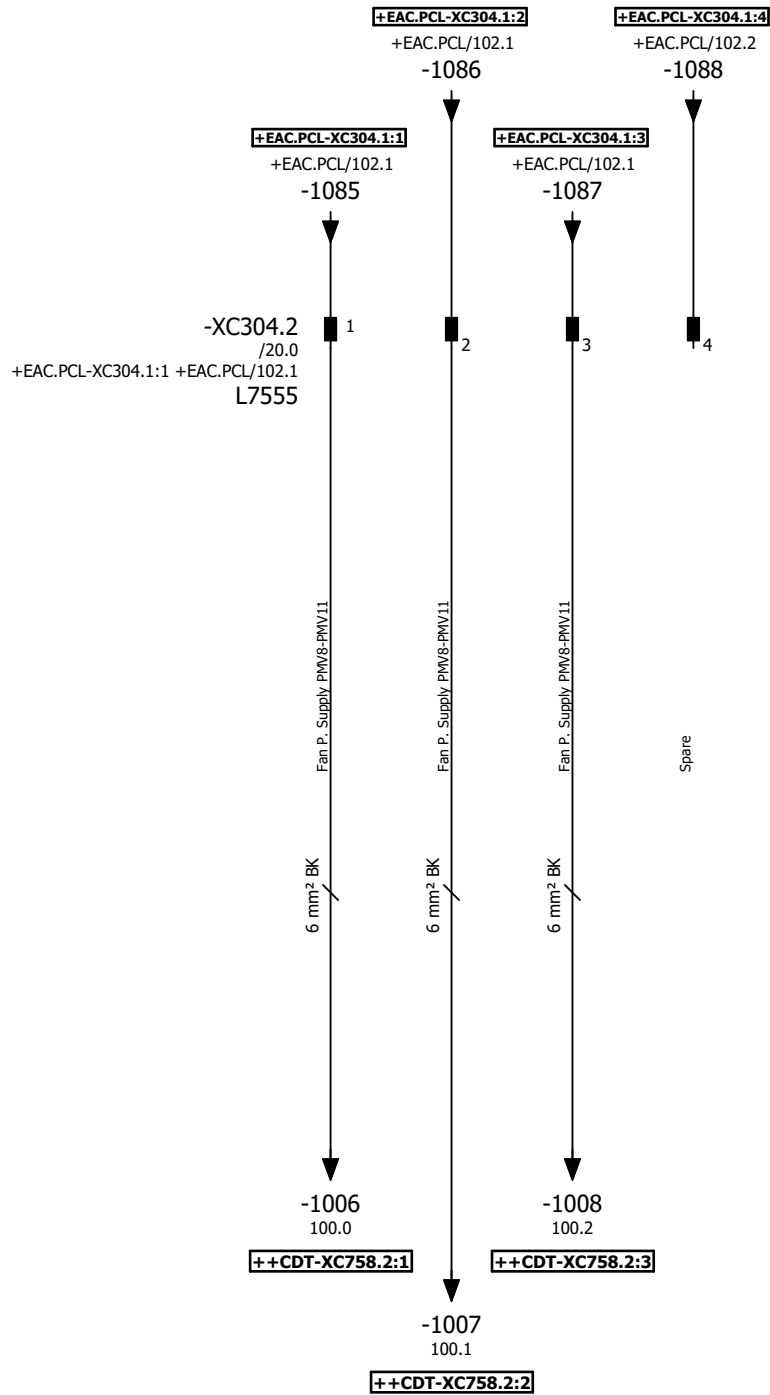


Interconnections PSA//CDT

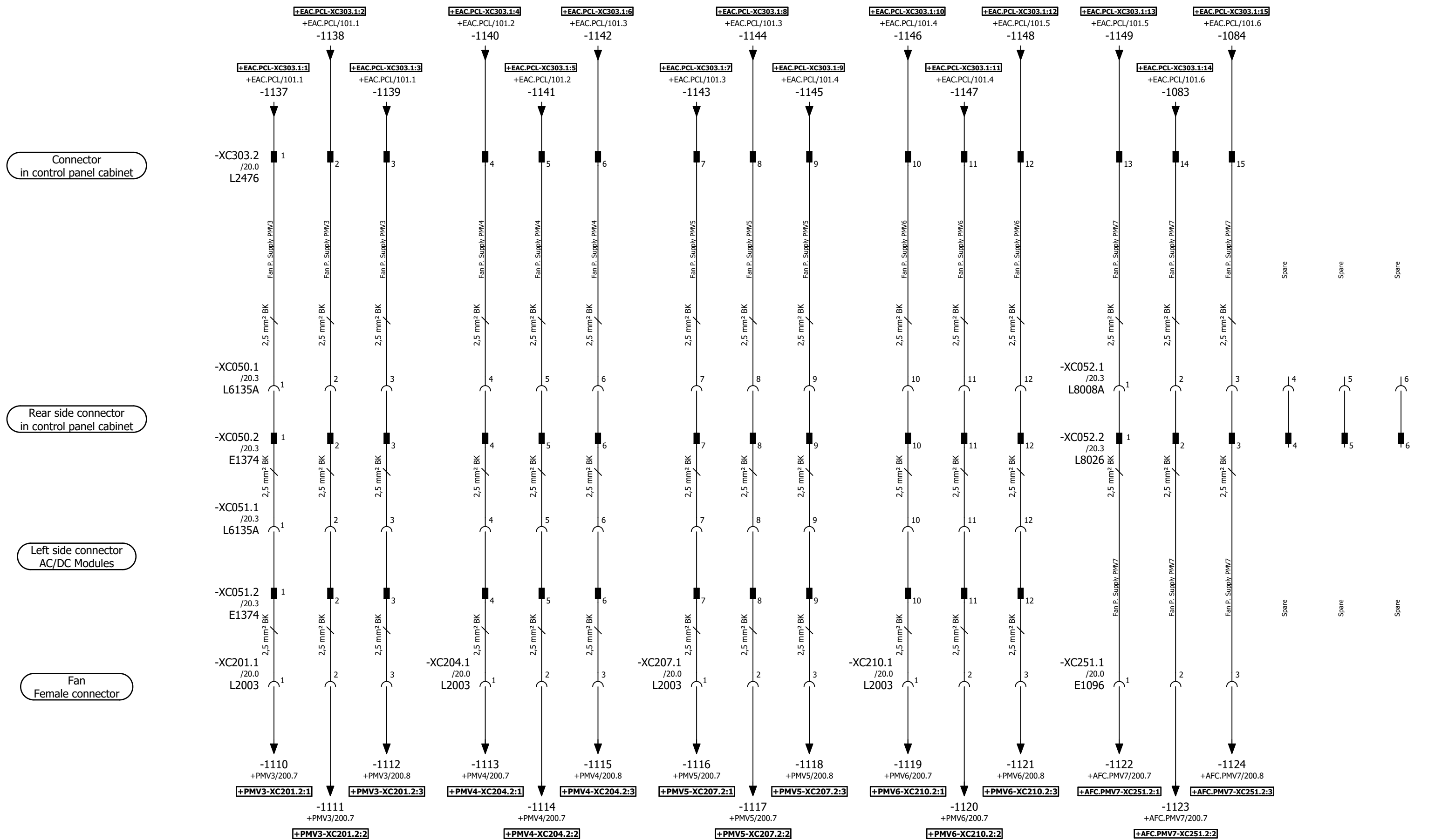
Connector
in PSA panel cabinet



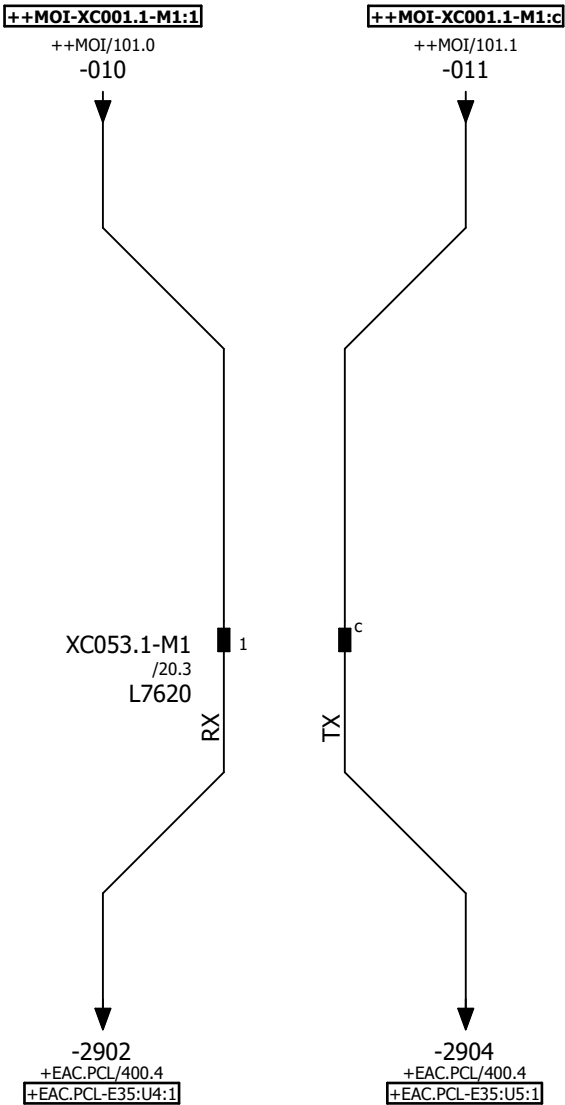
Interconnections - Fan PCL//CDT



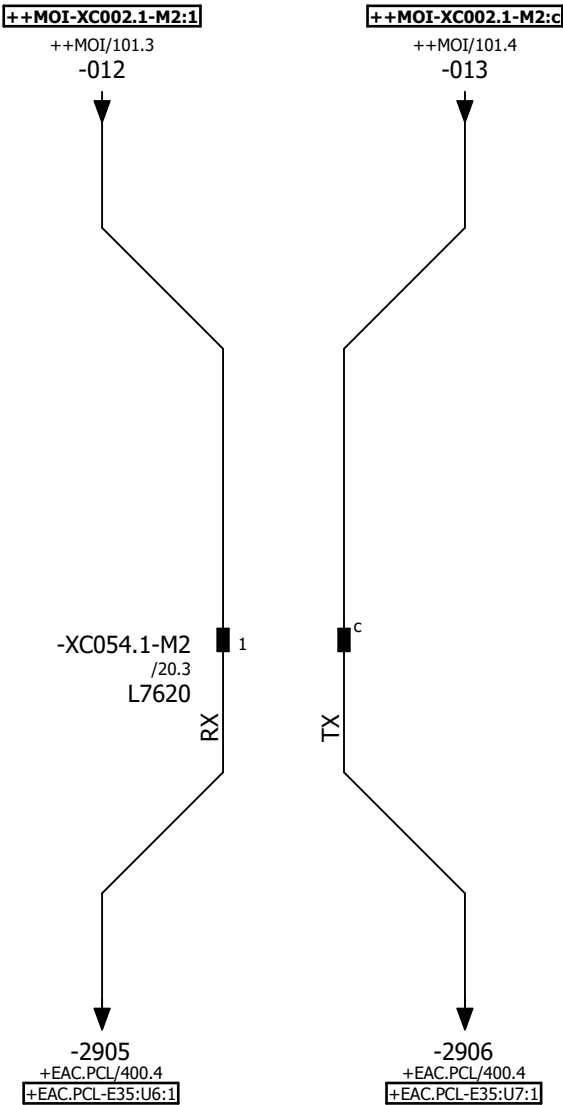
Interconnections - Fan PCL//Modules-AFC



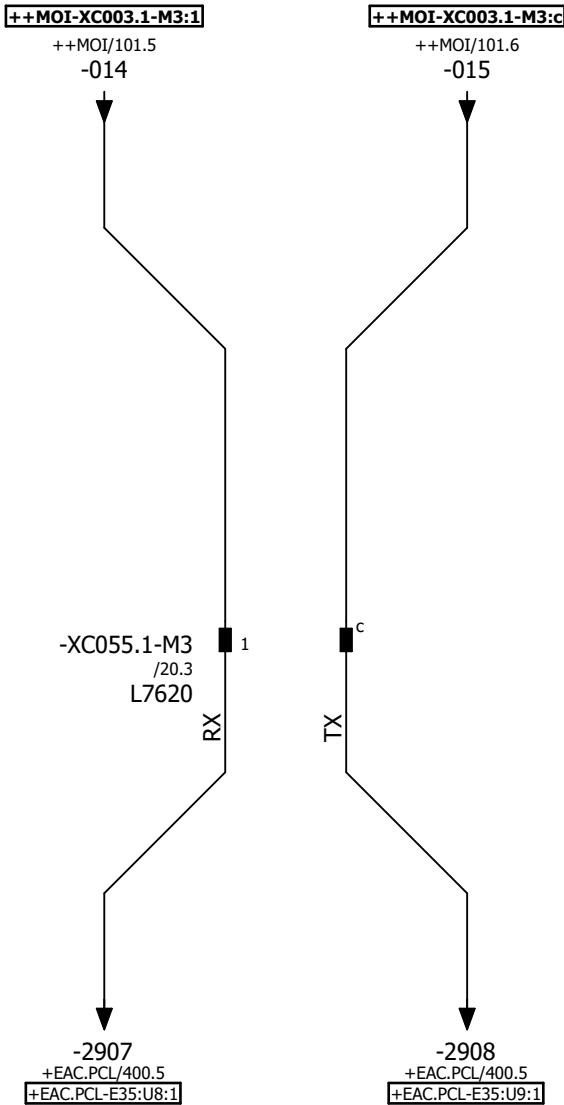
MODULE 1



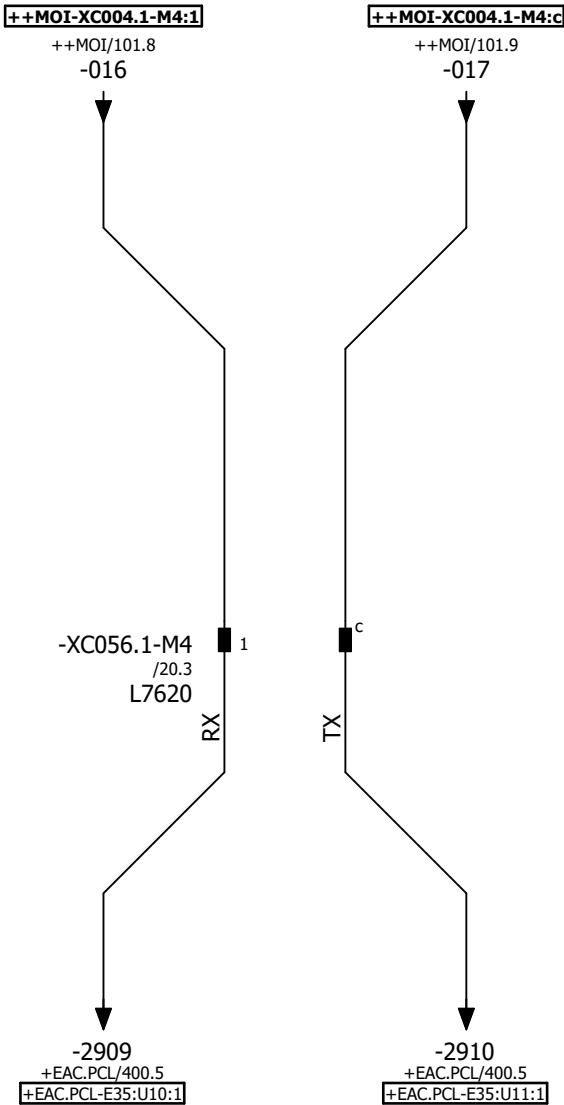
MODULE 2



MODULE 3



MODULE 4



TO PCB E30 (U1; U2)
MODULE 1

TO PCB E30 (U1; U2)
MODULE 2

TO PCB E30 (U1; U2)
MODULE 3

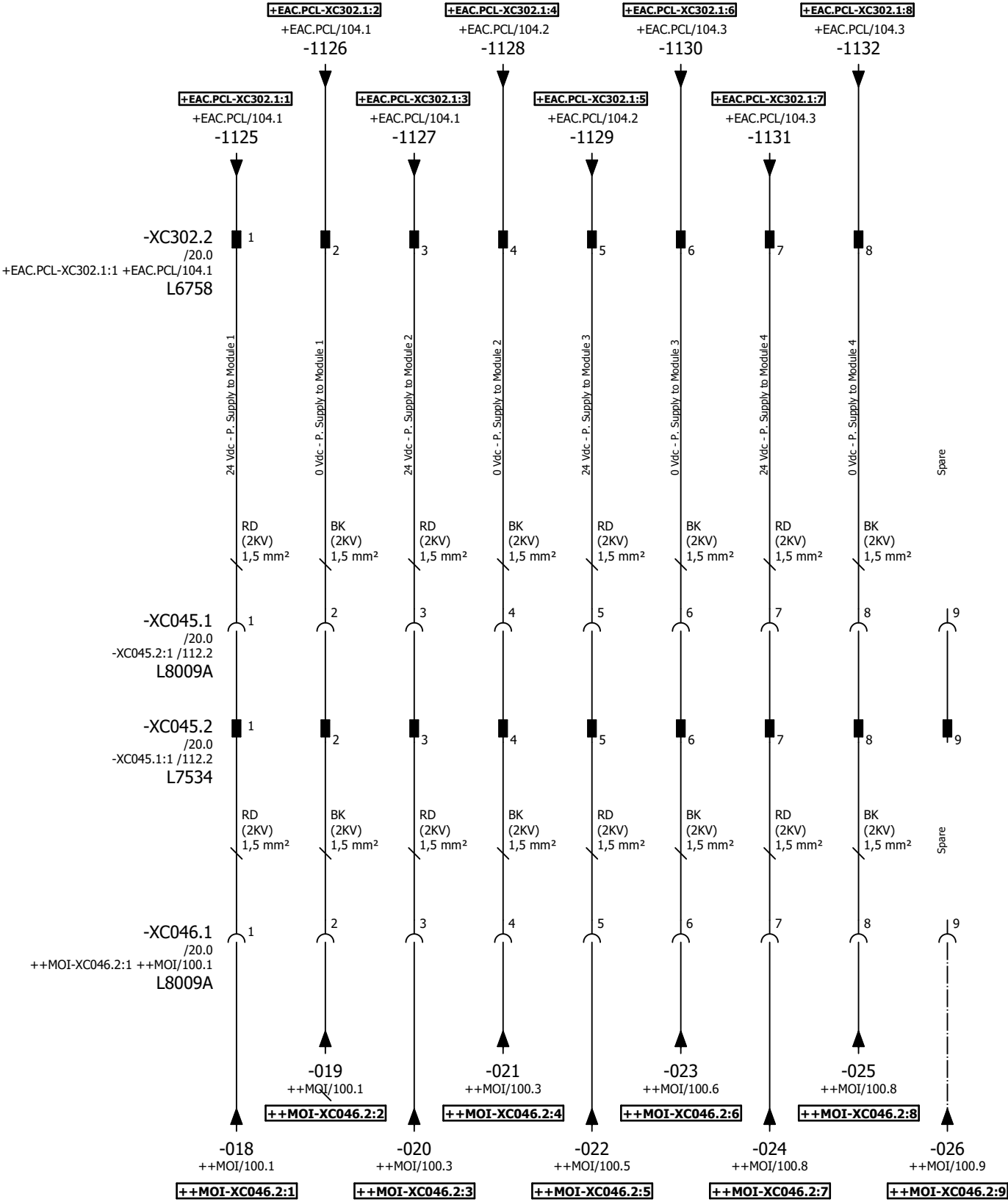
TO PCB E30 (U1; U2)
MODULE 4

Interconnections PCL/Modules

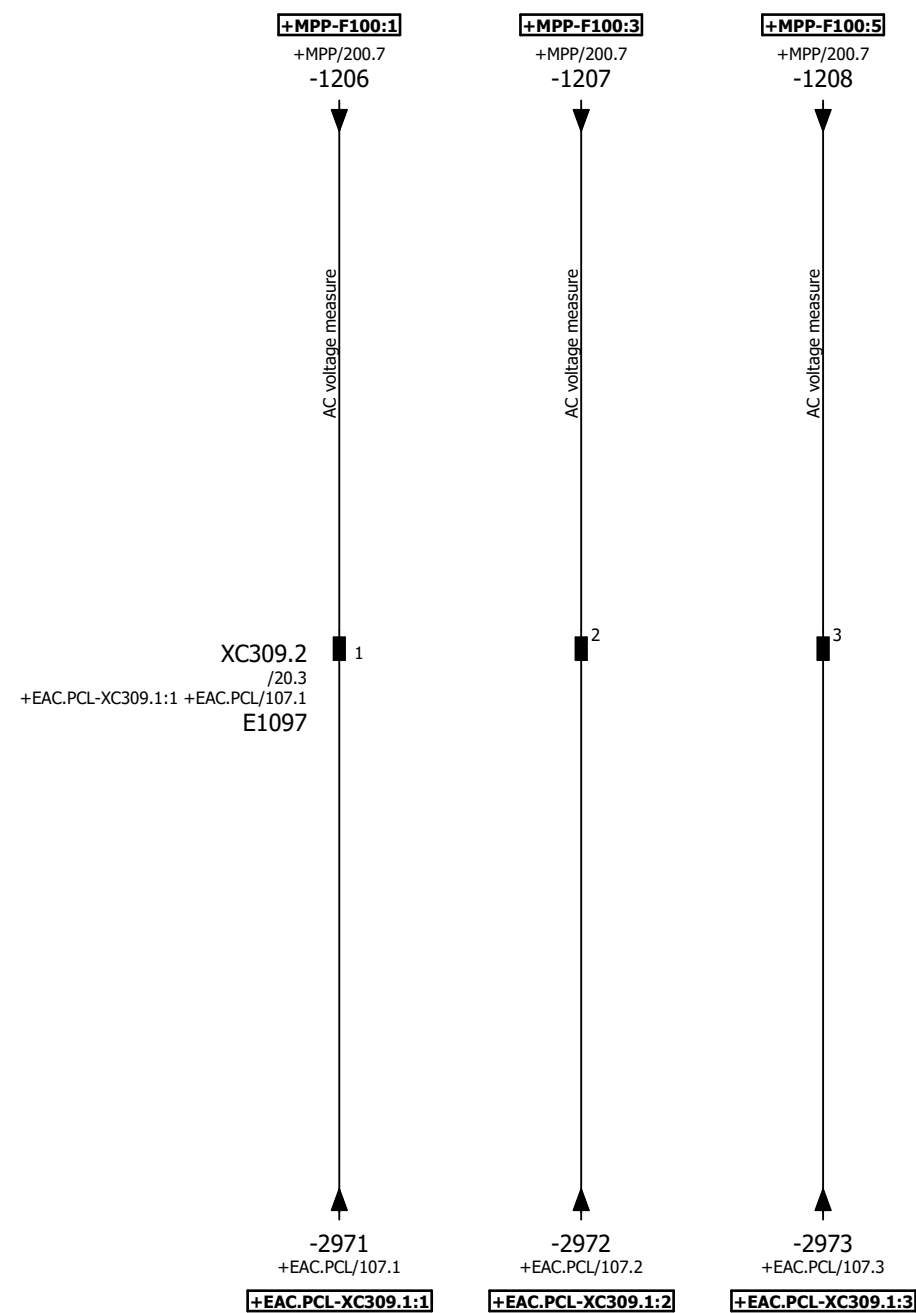
Connector
in PCL panel cabinet

Rear side connector
in control panel cabinet

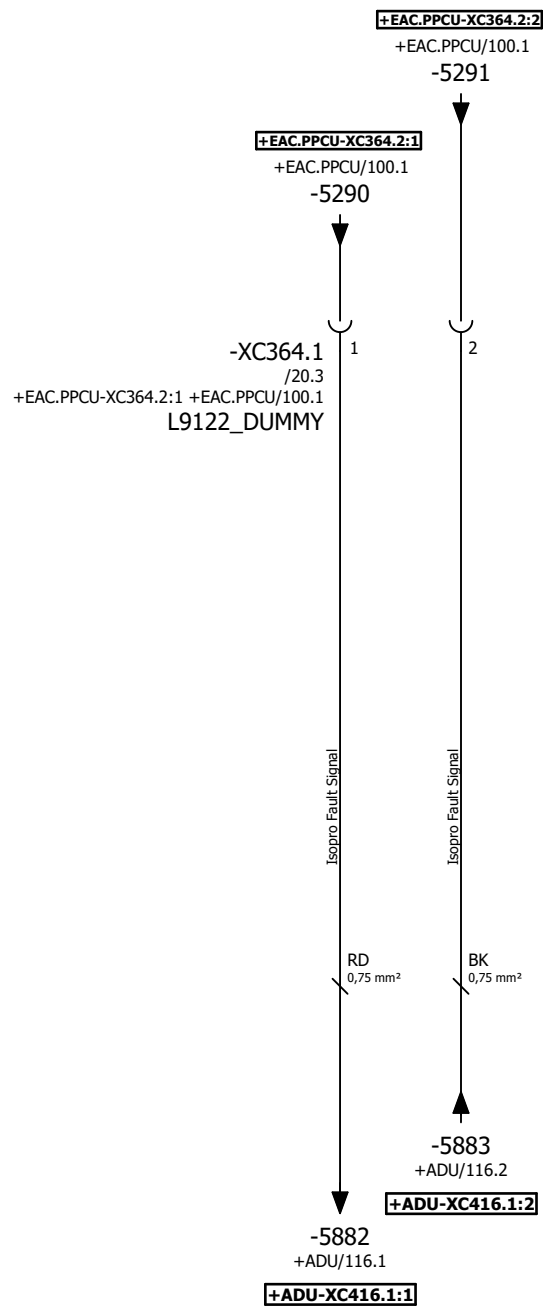
Left side connector
AC/DC Modules



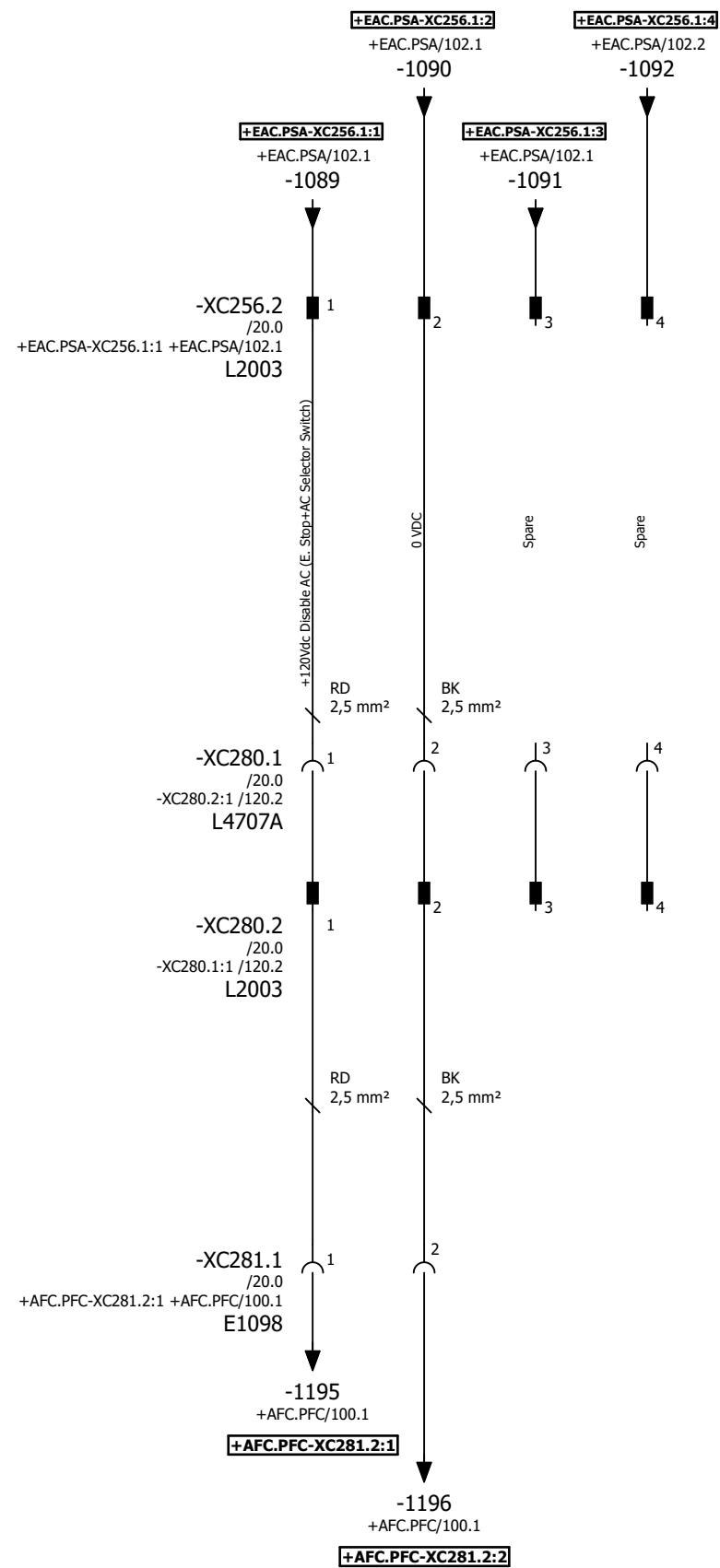
24vdc Power Supply to Modules



Interconnections PPD//PCDU



Interconnections - PSA//AFC



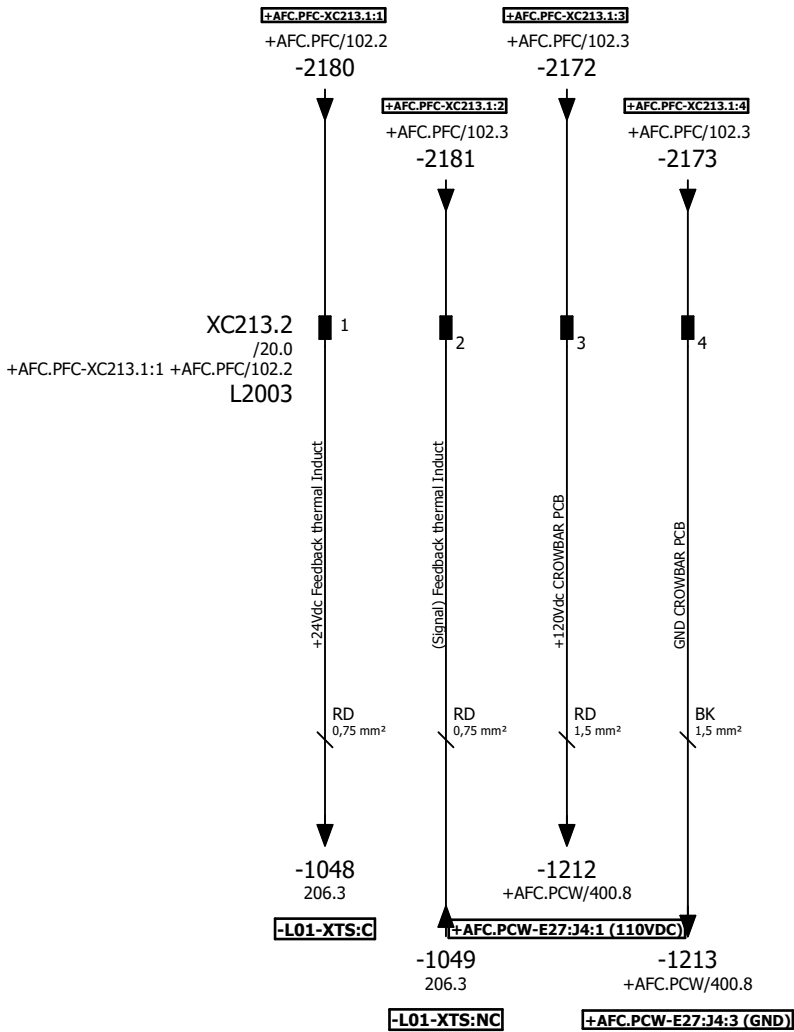
Connector
in PSA panel cabinet

Rear side connector
in control panel cabinet

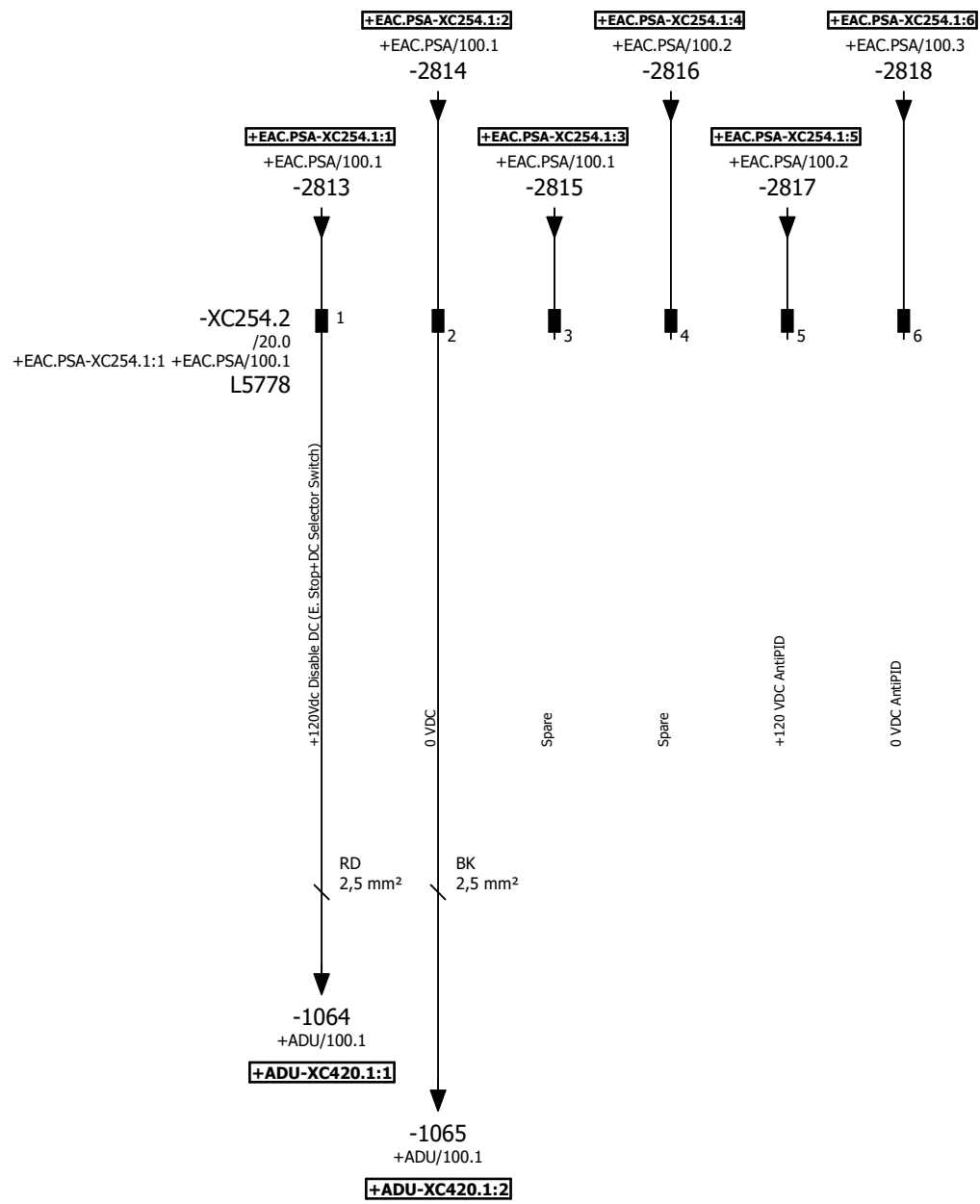
AFC
Female connector

Interconnection

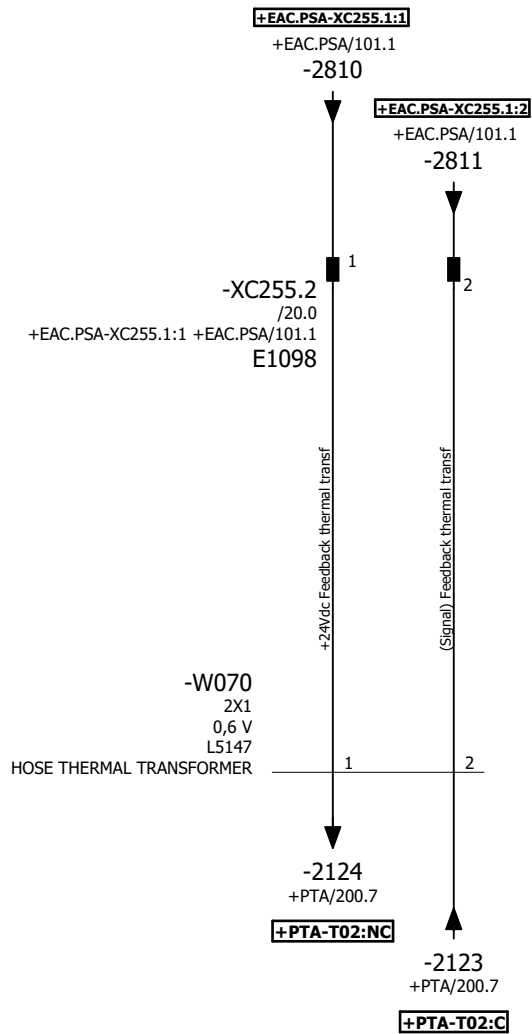
Induct/ Crowbar - C Filter



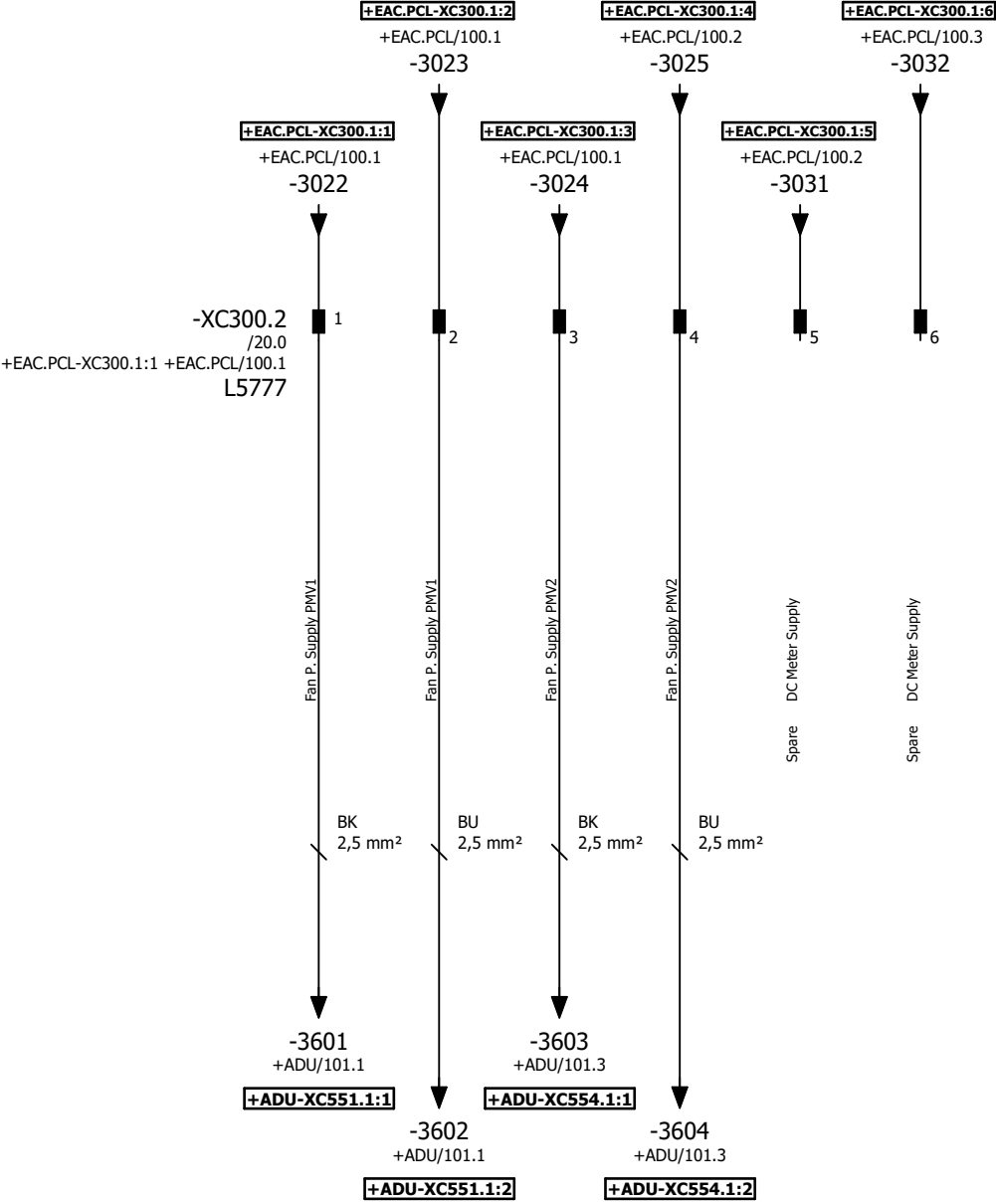
Interconnection PSA/ADU



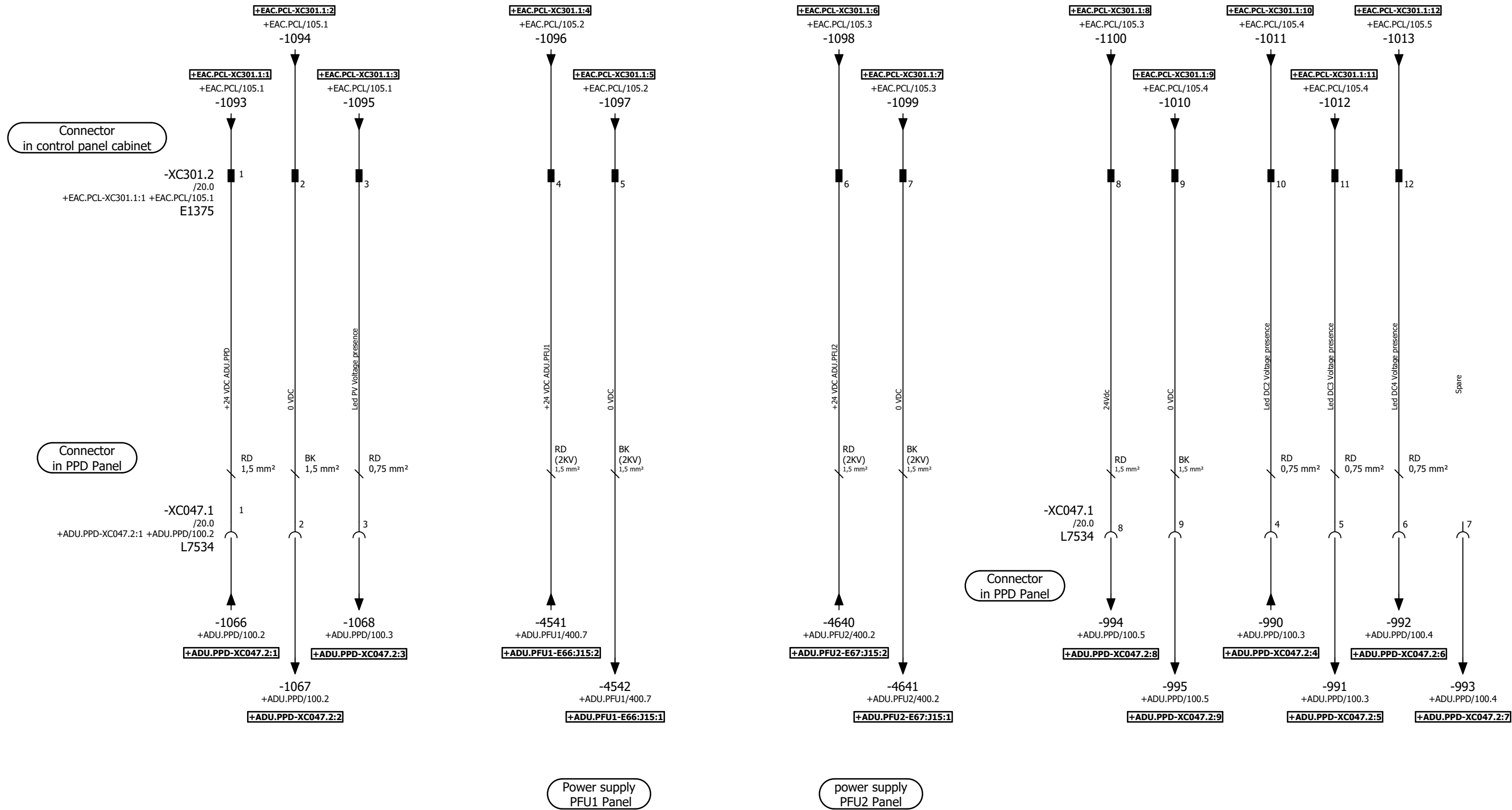
Interconnection SSAA - PSA

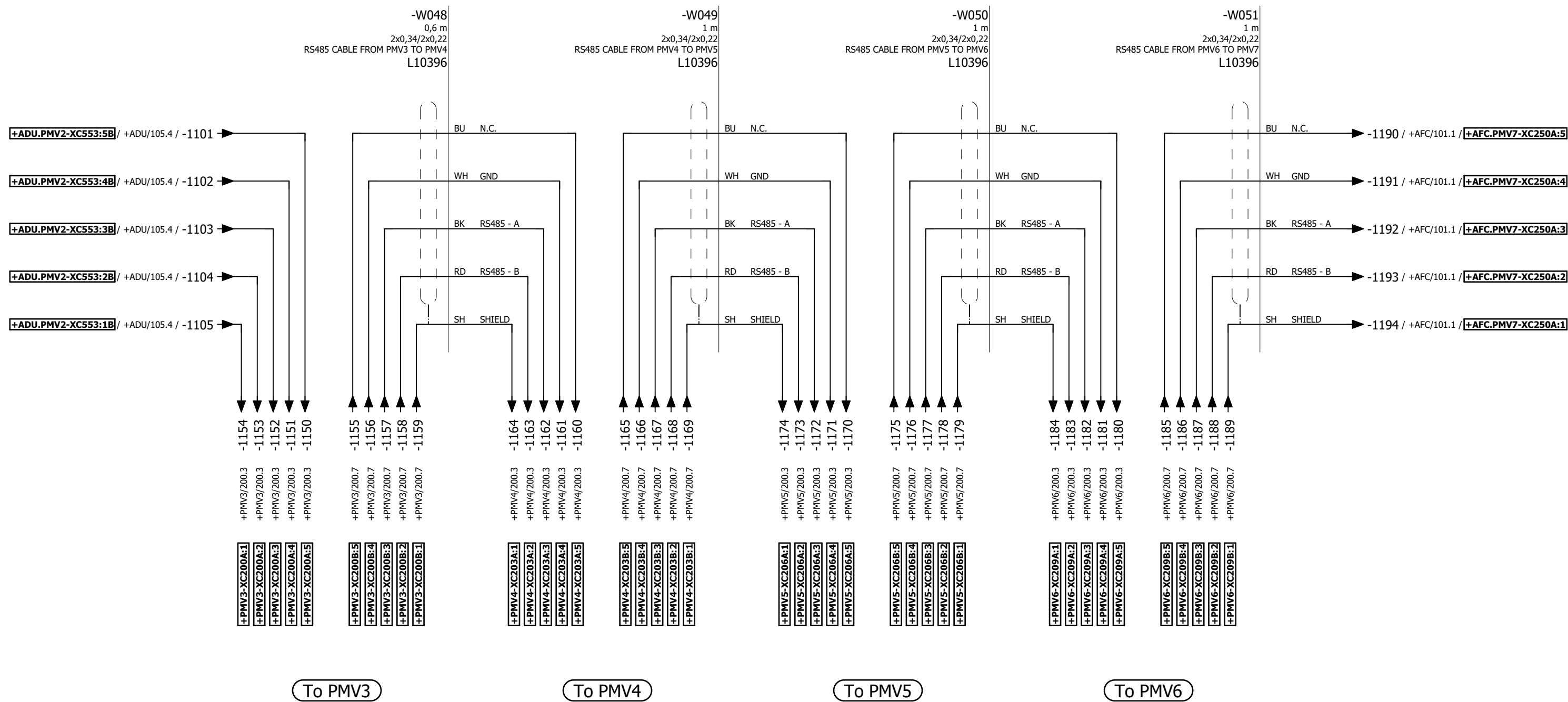


Interconnections - Fan-DC Meter PCL//ADU



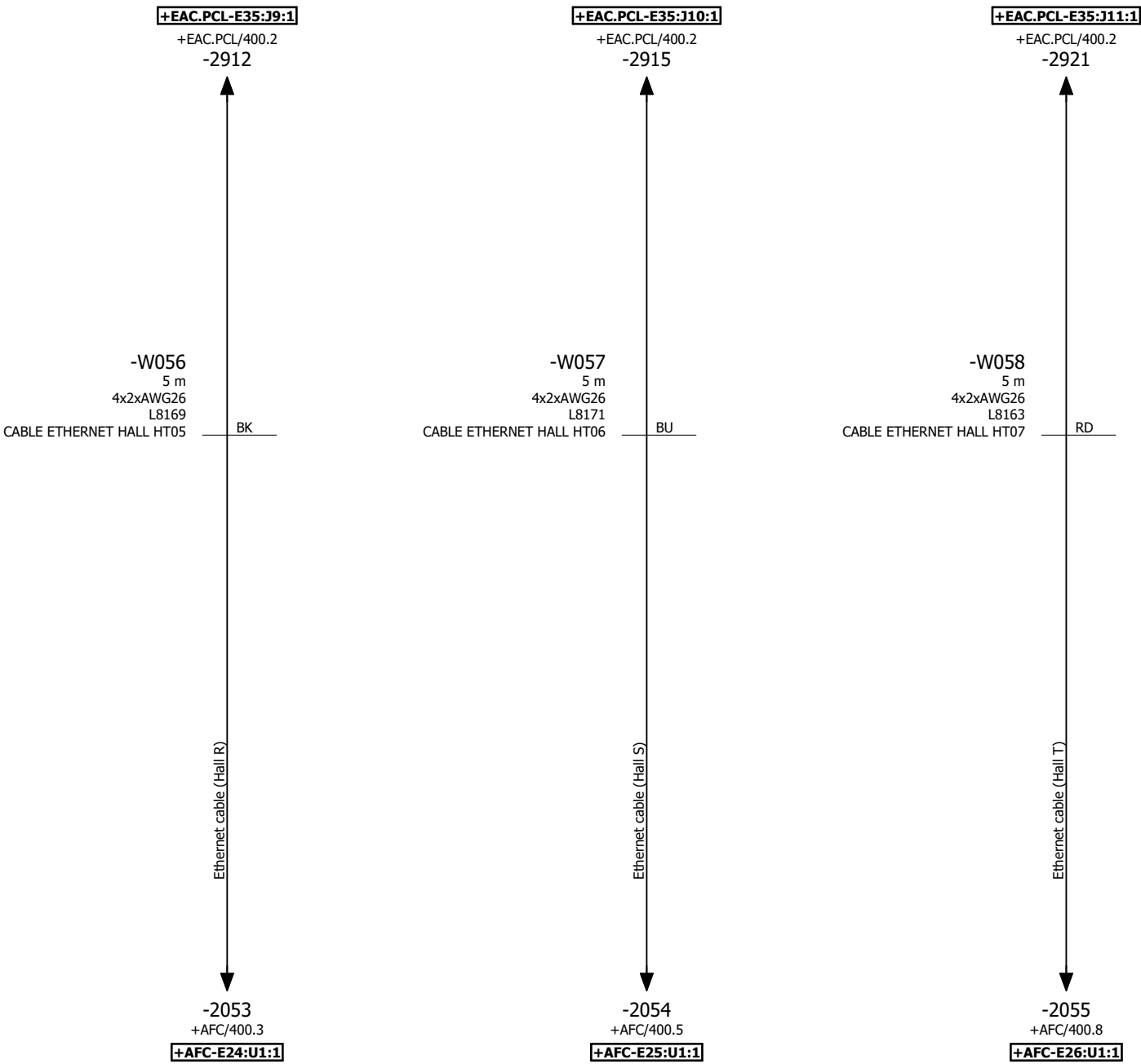
Interconnections PCL//ADU.PPD/PFU1/2





Interconnections - RJ45

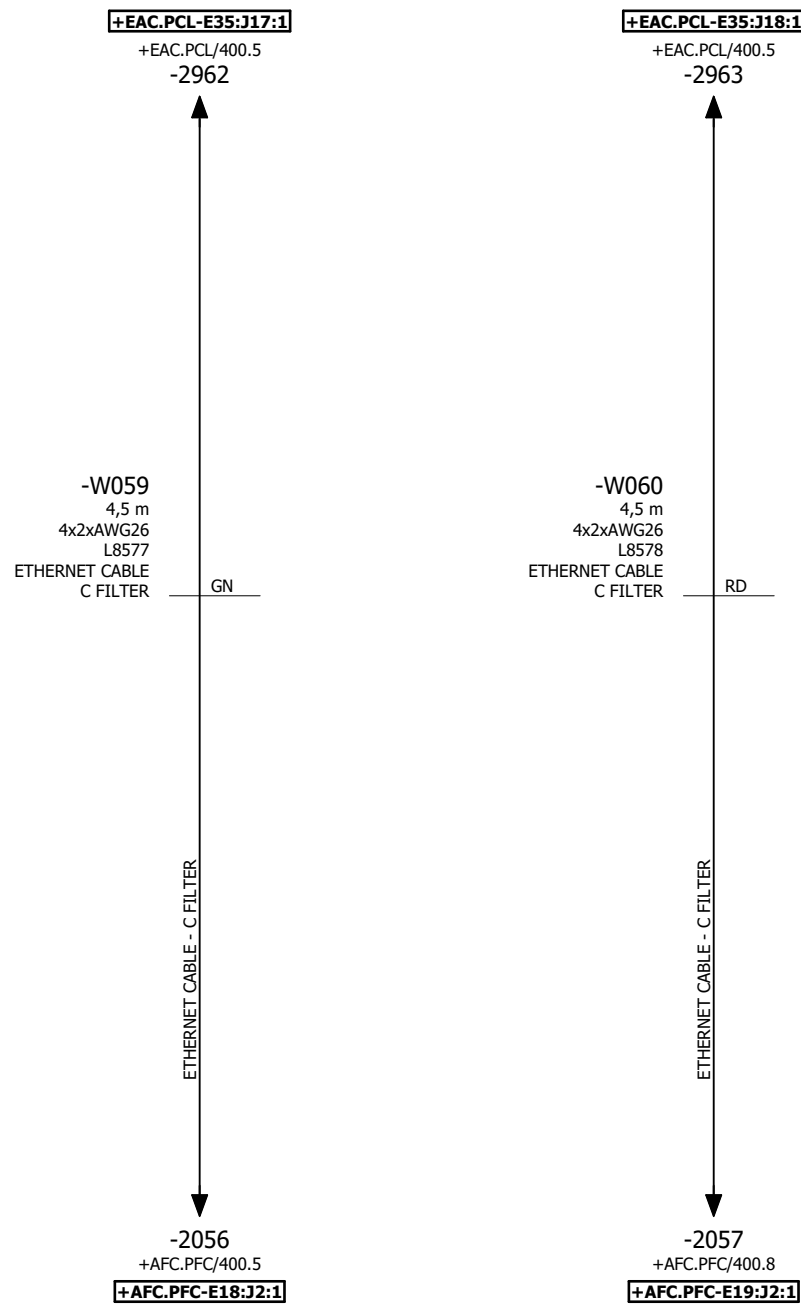
From central control PCB



AC current measurement -
C Filter

Interconnections - RJ45

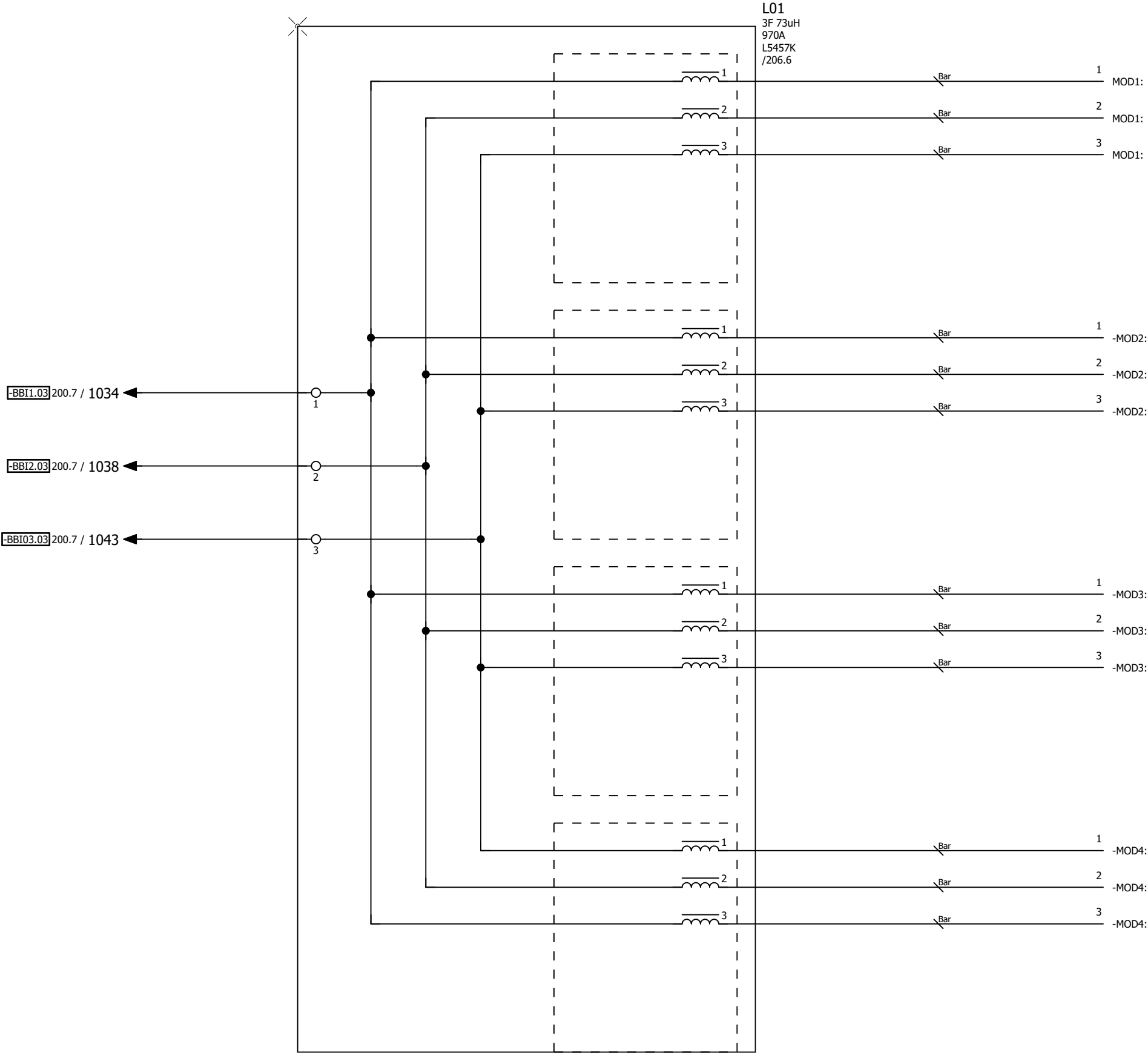
From central control PCB



To PCB - I/O C Filter

To PCB - I/O C Filter

To MV Transformer



MODULE 1

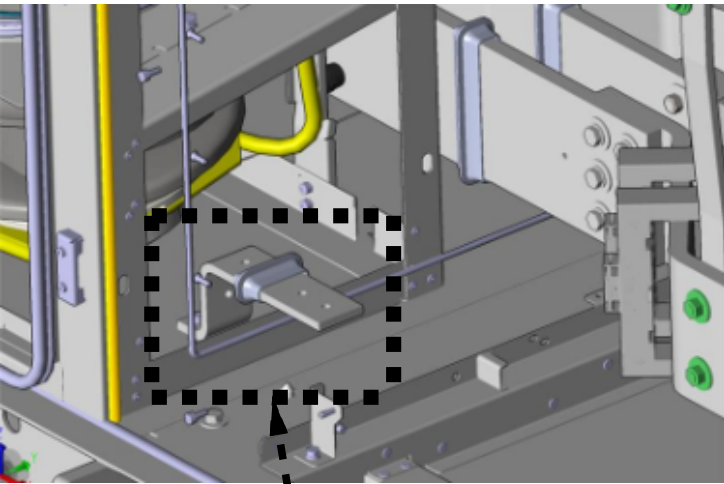
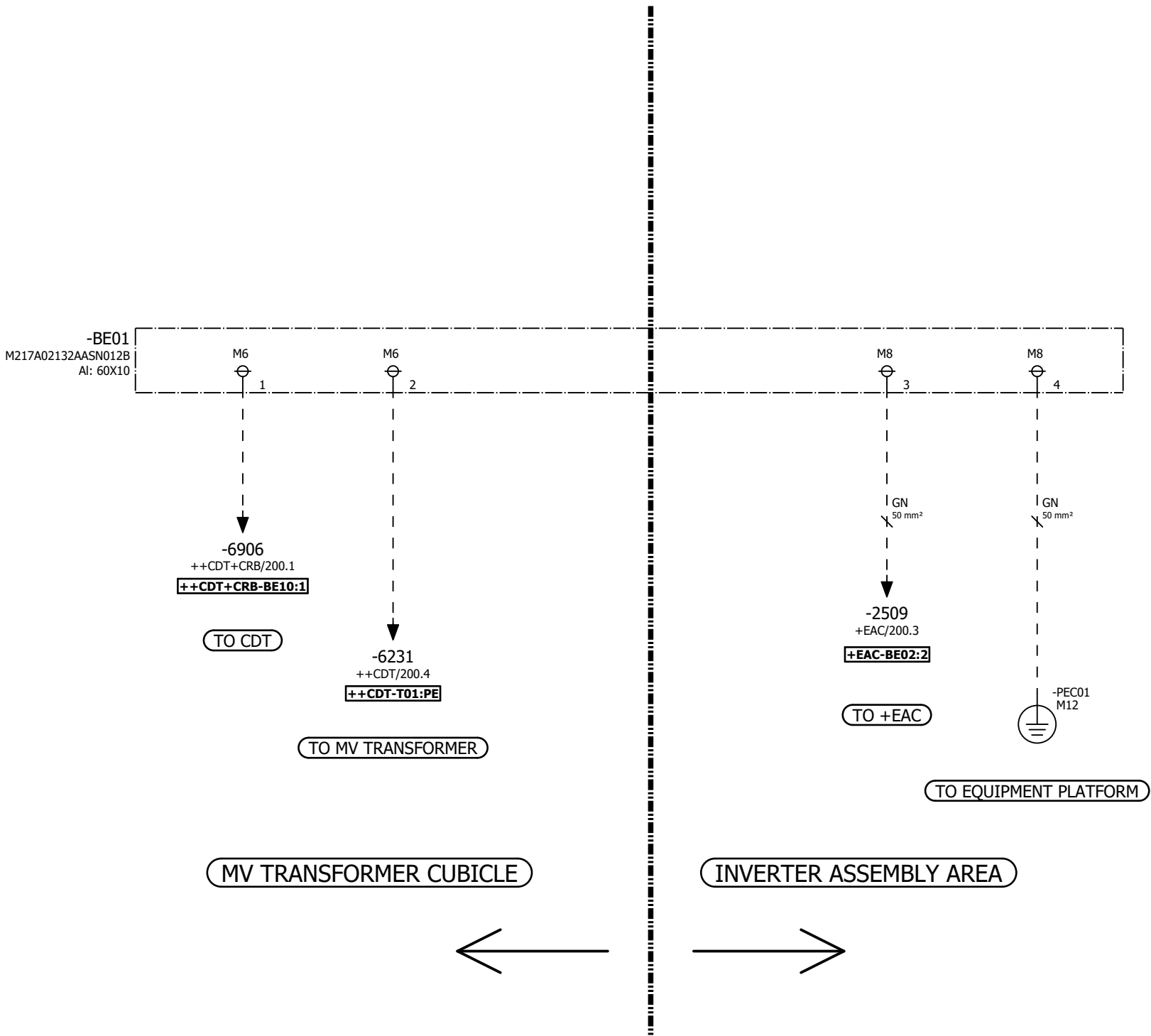
MODULE 2

MODULE 3

MODULE 4

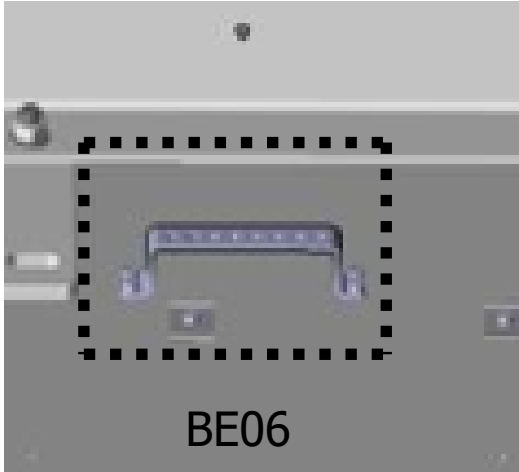
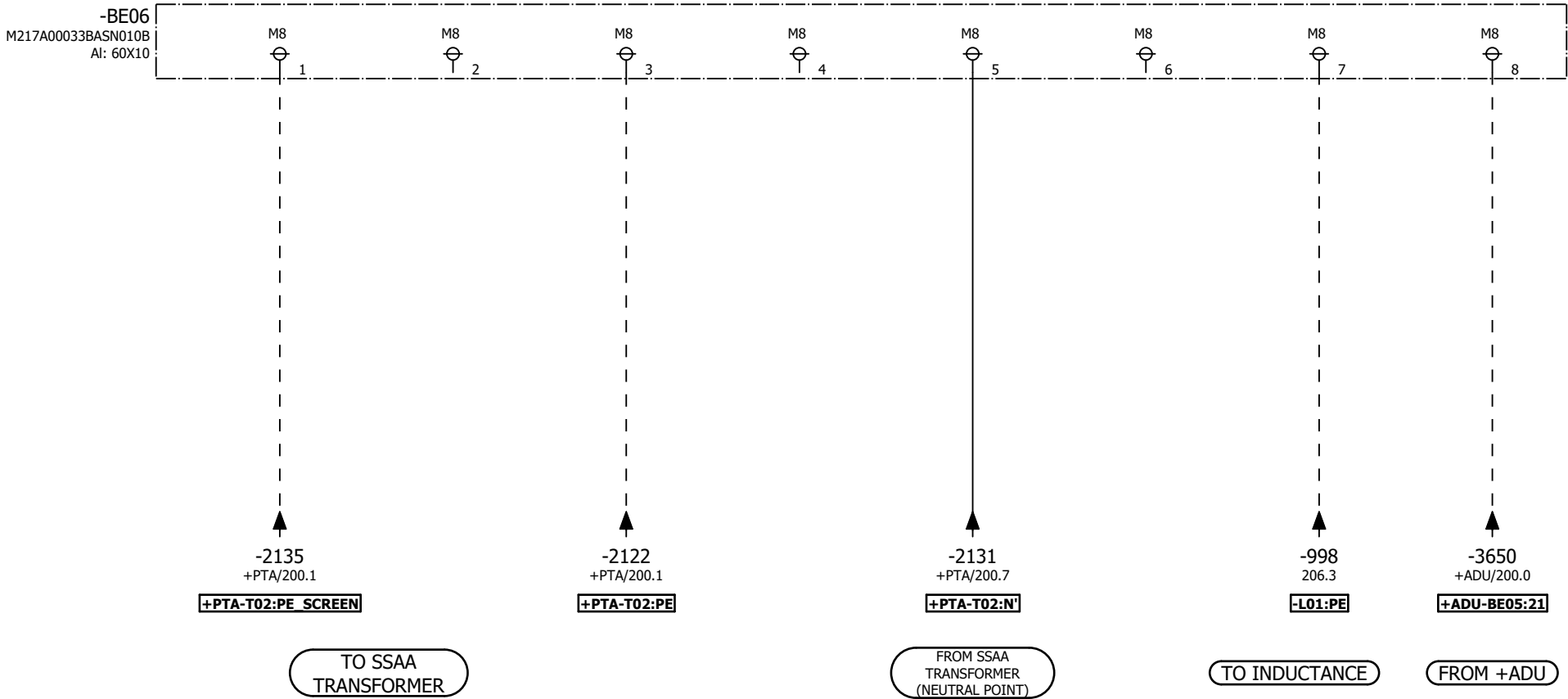


Earth connection



-BE01

Earth connection



BE06

-BE06

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BBI1.02-CT01	L7810A	AST1656 4000/5A CL 1 60VA IAG	CURRENT TRAFO AST1656 4000/5A CL 1 60VA IAG	1
-BBI2.02-CT02	L7810A	AST1656 4000/5A CL 1 60VA IAG	CURRENT TRAFO AST1656 4000/5A CL 1 60VA IAG	1
-BBI3.02-CT03	L7810A	AST1656 4000/5A CL 1 60VA IAG	CURRENT TRAFO AST1656 4000/5A CL 1 60VA IAG	1
-BE01	M217A02132AAS N012B	M217A02132AASN012B	EARTH BUSBAR 4 LEADS M8 DRILL	1
-BE06	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1
-CT04	L7810A	AST1656 4000/5A CL 1 60VA IAG	CURRENT TRAFO AST1656 4000/5A CL 1 60VA IAG	1
-CT05	L7810A	AST1656 4000/5A CL 1 60VA IAG	CURRENT TRAFO AST1656 4000/5A CL 1 60VA IAG	1
-CT06	L7810A	AST1656 4000/5A CL 1 60VA IAG	CURRENT TRAFO AST1656 4000/5A CL 1 60VA IAG	1
-L01	L5457K	L5457K	INDUCTANCE 3F 73uH 970A CUAD	1
-NTC01	L7844	GA10K3435STM010	NTC 10kOhm 3435K PVC	1
-NTC02	L7844	GA10K3435STM010	NTC 10kOhm 3435K PVC	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+-W040	ANALOGIC SIGNALS MV BOARD	L6449
=HEM++ZMI+-W041	CABLE ETHERNET MV PROTECTION SWITCHGEAR	L9566
=HEM++ZMI+-W042	CABLE ETHERNET MV PROTECTION SWITCHGEAR	L9567
=HEM++ZMI+-W043	PT100 CABLE	L9565
=HEM++ZMI+-W046	NTC - C FILTER TEMP.	L4592
=HEM++ZMI+-W047	NTC - FAN ADMISSION TEMP.	L4592
=HEM++ZMI+-W048	RS485 CABLE FROM PMV3 TO PMV4	L10396
=HEM++ZMI+-W049	RS485 CABLE FROM PMV4 TO PMV5	L10396
=HEM++ZMI+-W050	RS485 CABLE FROM PMV5 TO PMV6	L10396
=HEM++ZMI+-W051	RS485 CABLE FROM PMV6 TO PMV7	L10396
=HEM++ZMI+-W053		L9565
=HEM++ZMI+-W056	CABLE ETHERNET HALL HT05	L8169
=HEM++ZMI+-W057	CABLE ETHERNET HALL HT06	L8171
=HEM++ZMI+-W058	CABLE ETHERNET HALL HT07	L8163
=HEM++ZMI+-W059	ETHERNET CABLE C FILTER	L8577
=HEM++ZMI+-W060	ETHERNET CABLE C FILTER	L8578
=HEM++ZMI+-W063	CABLE ETHERNET MV PROTECTION SWITCHGEAR (ECON MODE)	L9568
=HEM++ZMI+-W064	CABLE ETHERNET LINE CELL	L8573
=HEM++ZMI+-W070	HOSE THERMAL TRANSFORMER	L5147

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+-XC045.1	MATE-N-LOK SEALED 9P PANEL CONNECTOR MOD INTER	L8009A	1
		E1203	8
=HEM++ZMI+-XC045.2	MATE-N-LOK 9P AERIAL CONNECTOR MOD INTER	L7534	1
		E1204	8
=HEM++ZMI+-XC046.1	MATE-N-LOK SEALED 9P PANEL CONNECTOR AC/DC MOD	L8009A	1
		E1203	8
=HEM++ZMI+-XC047.1	MATE-N-LOK 9P PANEL CONNECTOR PCL-PPD	L7534	1
		E1203	8
=HEM++ZMI+-XC050.1	MATE-N-LOCK PANEL 12P SEALED	L6135A	1
		L5709	12
=HEM++ZMI+-XC050.2	MATE-N-LOCK AERIAL 12P	E1374	1
		L5710	12
=HEM++ZMI+-XC051.1	MATE-N-LOCK PANEL 12P SEALED	L6135A	1
		L5709	12
=HEM++ZMI+-XC051.2	MATE-N-LOCK AERIAL 12P	E1374	1
		L5710	12
=HEM++ZMI+-XC052.1	MATE-N-LOCK PANEL CONNECTOR 6P SEALED	L8008A	1
		L5709	3
		L4897	3
=HEM++ZMI+-XC052.2	MATE-N-LOCK AERIAL 6P	L8026	1
		L5710	3
=HEM++ZMI+-XC053.1-M1	CONNECTOR OPTICAL FIBER DUPLEX MOD.1	L7620	1
=HEM++ZMI+-XC054.1-M2	CONNECTOR OPTICAL FIBER DUPLEX MOD.2	L7620	1
=HEM++ZMI+-XC055.1-M3	CONNECTOR OPTICAL FIBER DUPLEX MOD.3	L7620	1
=HEM++ZMI+-XC056.1-M4	CONNECTOR OPTICAL FIBER DUPLEX MOD.4	L7620	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+-XC201.1	MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 3	L2003	1
		L5709	3
		L8257	1
=HEM++ZMI+-XC204.1	MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 4	L2003	1
		L5709	3
		L8257	1
=HEM++ZMI+-XC207.1	MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 5	L2003	1
		L5709	3
		L8257	1
=HEM++ZMI+-XC210.1	MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 6	L2003	1
		L5709	3
		L8257	1
=HEM++ZMI+-XC213.2	MATE-N-LOK 4P AERIAL THERMAL IND	L2003	1
		E1204	2
=HEM++ZMI+-XC251.1	MATE-N-LOCK AERIAL POWER SUPPLY FAN ENGINE 7	E1096	1
		L5709	3
=HEM++ZMI+-XC254.2	MATE-N-LOK 6P AERIAL DISABLE DC	L5778	1
		E1204	2
=HEM++ZMI+-XC255.2	MATE-N-LOK 2P AERIAL FEEDBACK THERMAL TRANS.	E1098	1
		E1204	2
=HEM++ZMI+-XC256.2	MATE-N-LOK 4P AERIAL DISABLE AC	L2003	1
		L5710	2
=HEM++ZMI+-XC257.2	MATE-N-LOK 12P AERIAL	E1374	1
		L5710	10
=HEM++ZMI+-XC280.1	MATE-N-LOK 4P PANEL POWER SUPPLY AFC	L4707A	1
		L5709	3
		L4897	1
=HEM++ZMI+-XC280.2	MATE-N-LOK 4P AERIAL POWER SUPPLY AFC	L2003	1
		L5710	4
=HEM++ZMI+-XC281.1	MATE-N-LOK 2P AERIAL POWER SUPPLY AFC	E1098	1
		L5709	2
=HEM++ZMI+-XC300.2	MATE-N-LOK 6P PANEL CONNECTOR FAN SUPPLY	L5777	1
		L5710	4

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+-XC301.2	MATE-N-LOK 12P PANEL CONNECTOR POWER SUPPLY PPD	E1375	1
		E1204	7
=HEM++ZMI+-XC302.2	MATE-N-LOK 8P PANEL CONNECTOR POWER SUPPLY MOD	L6758	1
		E1204	8
=HEM++ZMI+-XC303.2	MATE-N-LOK 15P PANEL CONNECTOR FAN MODULES	L2476	1
		L5710	15
=HEM++ZMI+-XC304.2	MATE-N-LOK AERIAL 6mm2 POWER SUPPLY FANS SKID	L7555	1
=HEM++ZMI+-XC309.2	CONNECTOR MATE-N-LOK 3P AC MEASURE	E1097	1
		E1204	3
=HEM++ZMI+-XC364.1	TRIPLE-LOK 2P ISOPRO FAULT SIGNAL	L9122_DUMMY	1
		L9145	2



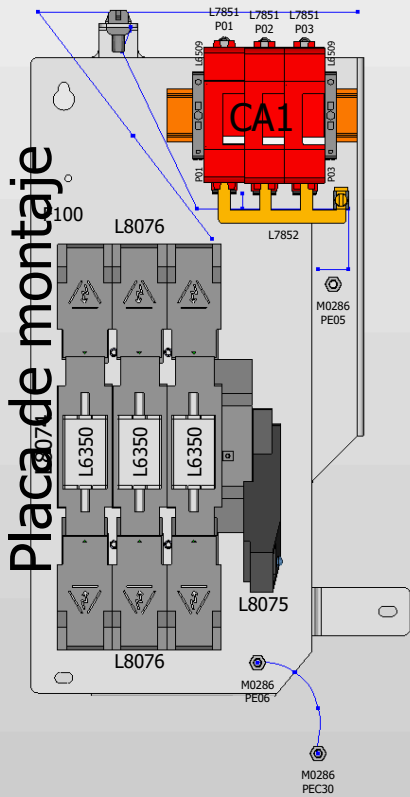
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
MPP	MAIN PROTECTION PANEL		
MPP MAIN PROTECTION PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

LAYOUT PANEL

Enclosure legend

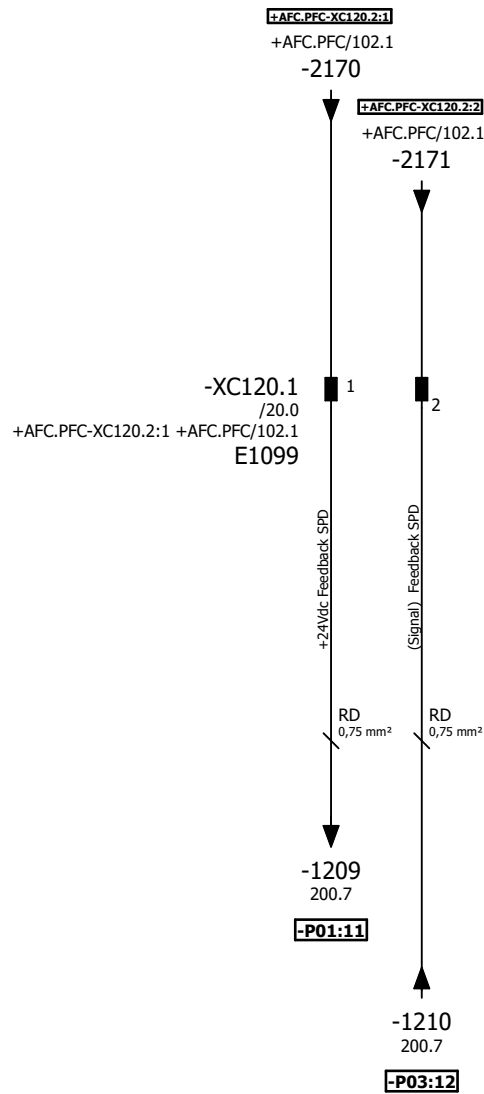
IME	ERP	Description
CA1	L1061	DIN RAIL 35x7,5mm
P01.1	L7852	ACC.P.SOBRETENSION FOR L7851
P01-P03	L7851	Surge arrester Type 2 DEHNguard SE 1-pole Uc 1000V AC, varistor + GDT
F100	L8074	3P NH00 690V FUSE SWITCH FUSERBL
F100	L8075	DIRECT HANDLE FOR L8704
F100	L8076	3P TERMINAL SHROUDS FOR L8074
F100	L6350	FUSE 100A GG 690VAC NH00

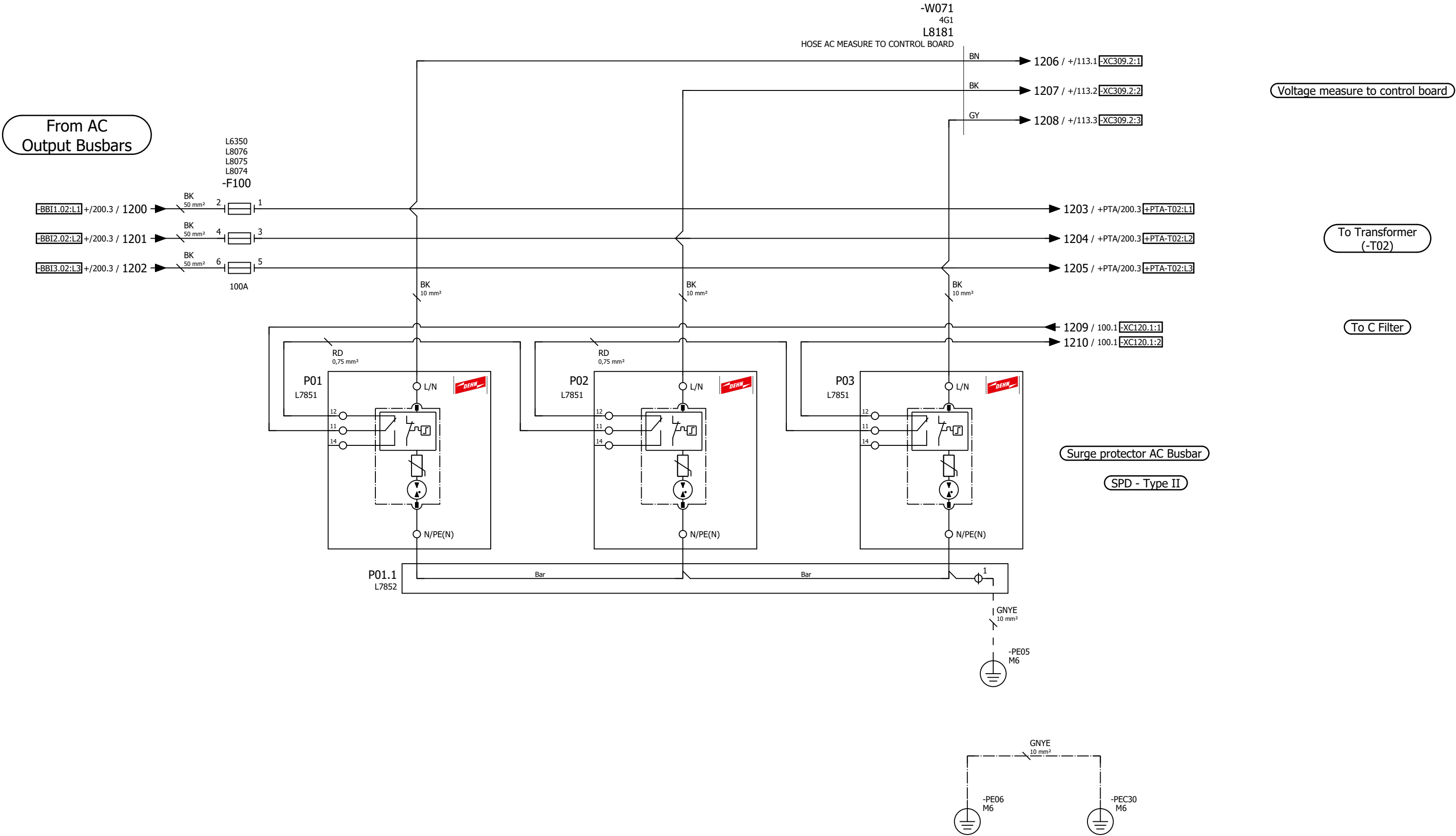


-XC120.1 = MATE-N-LOK 2P SPD



Interconnection PFC/MPP





BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-F100	L8074	36153012	3P NH00 690V FUSE SWITCH FUSERBL	1
-F100	L8075	36297901	DIRECT HANDLE FOR L8704	1
-F100	L8076	39983016	3P TERMINAL SHROUDS FOR L8074	2
-F100	L6350	382065	FUSE 100A GG 690VAC NH00	3
-P01	L7851	952940	Surge arrester Type 2 DEHNguard SE 1-pole Uc 1000V AC, varistor + GDT	1
-P01	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-P01.1	L7852	900 418	ACC.P.SOBRETENSION FOR L7851	1
-P02	L7851	952940	Surge arrester Type 2 DEHNguard SE 1-pole Uc 1000V AC, varistor + GDT	1
-P03	L7851	952940	Surge arrester Type 2 DEHNguard SE 1-pole Uc 1000V AC, varistor + GDT	1
-P03	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+MPP-W071	HOSE AC MEASURE TO CONTROL BOARD	L8181

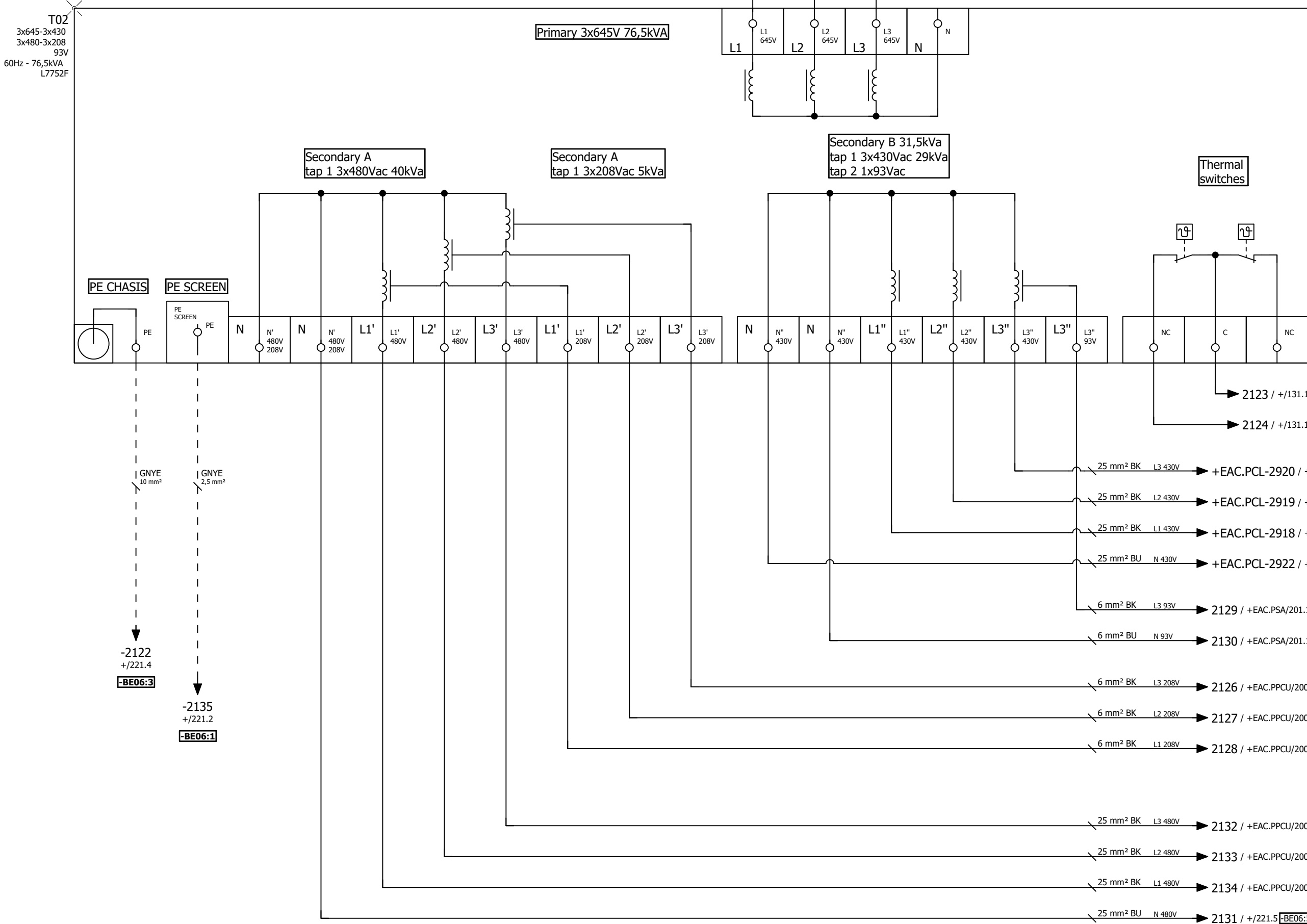
Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+MPP-XC120.1	MATE-N-LOK 2P SPD	E1099	1
		E1204	2



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
PTA	AUXILIARY TRANSFORMER ASSEMBLY		
PTA AUXILIARY TRANSFORMER ASSEMBLY			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		



BILL OF MATERIALS

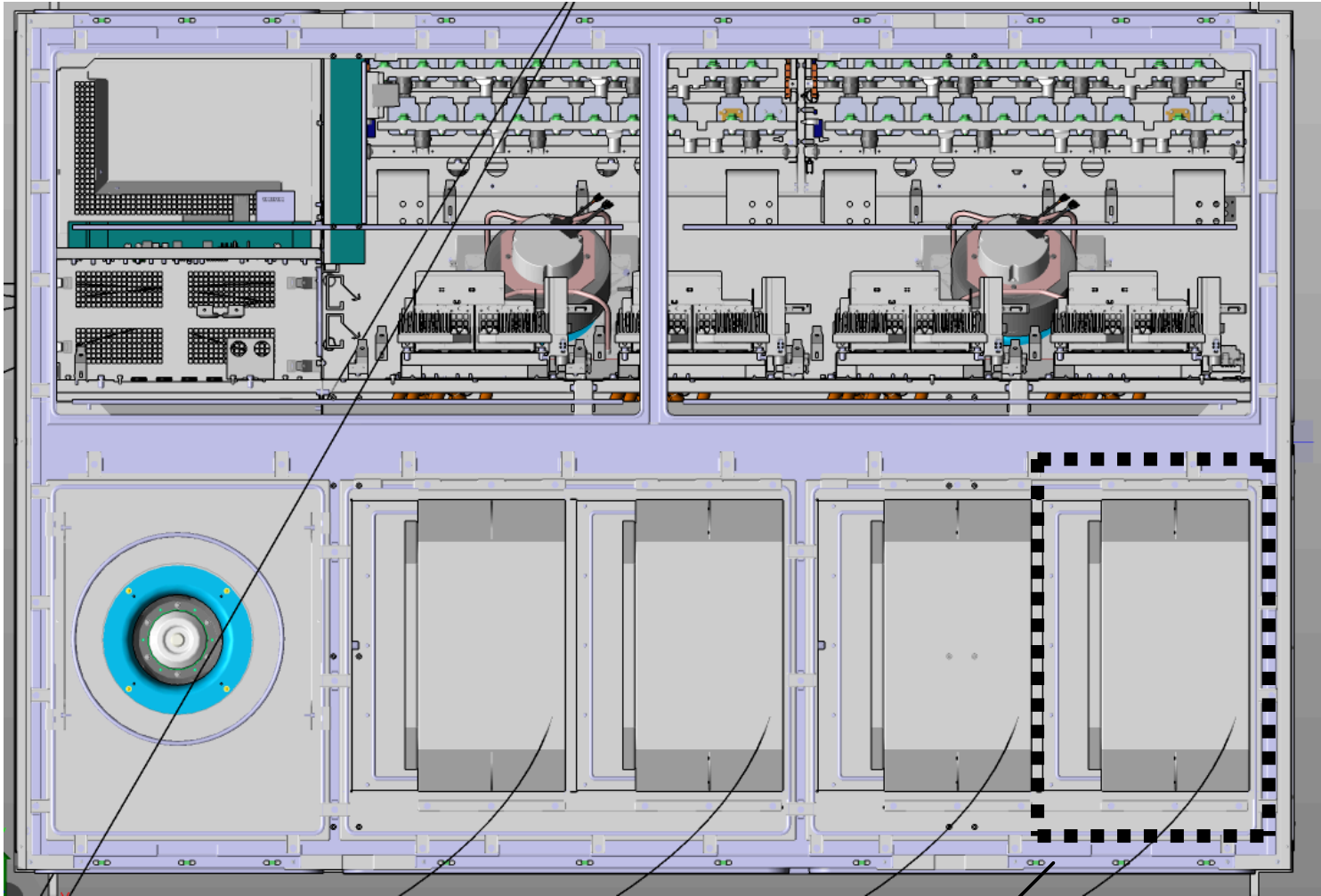
Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-T02	L7752F	L7752F	TRAFO 76,5kVA 645/430-93/480-208 60Hz	1



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
PMV3	FAN PRE-ASSEMBLY 3		
PMV3 FAN PRE-ASSEMBLY 3			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV3



PMV3



-XC201.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 3

-XC200 = T CONNECTOR M12 RS485



DESIGNED I+D 1

REVISED I+D 2

APPROVED I+D 3

DATE

REVISION

1.3.11

Project

ESQ_FS4010M10U000012_000

Name

Designation terminals and connectors

Installation:

HEM INVERTER

HEM

A3

Mounting location: +PMV3

FAN PRE-ASSEMBLY 3

++ZMI

10

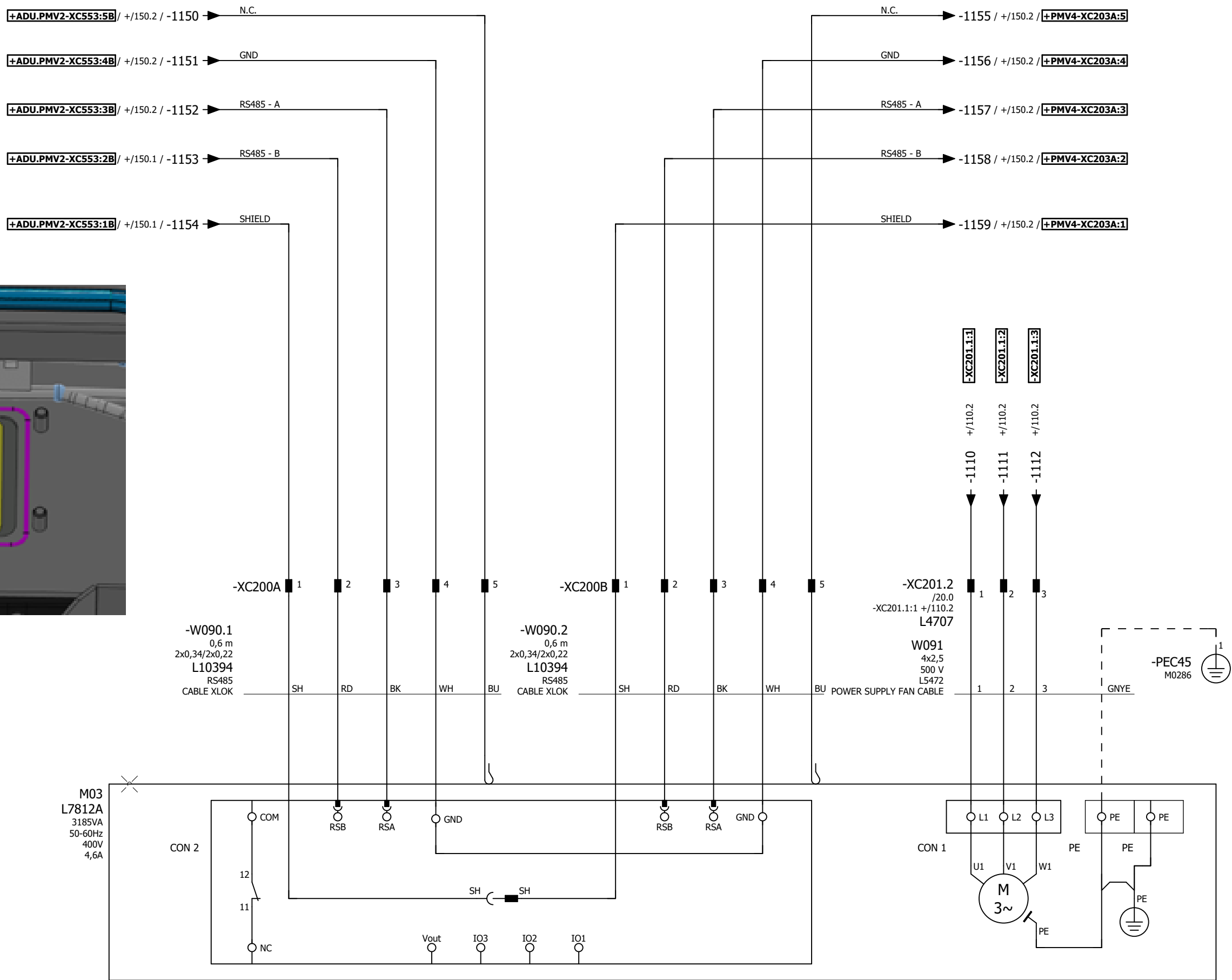
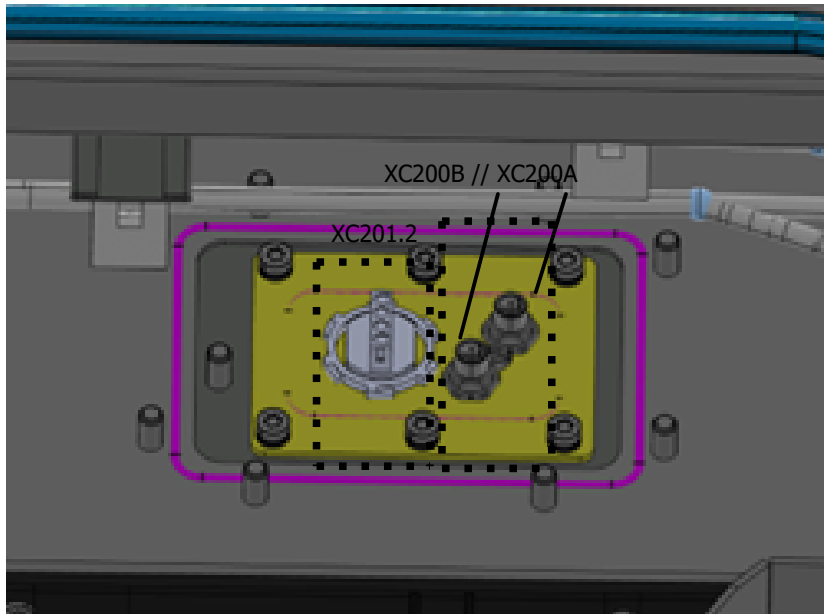
Next page 200

Project Code:

FS4010M10U10001100410010020101014

Page

Total number of pages
455



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M03	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+PMV3-W090.1	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV3-W090.2	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV3-W091	POWER SUPPLY FAN CABLE	L5472

Plug overview

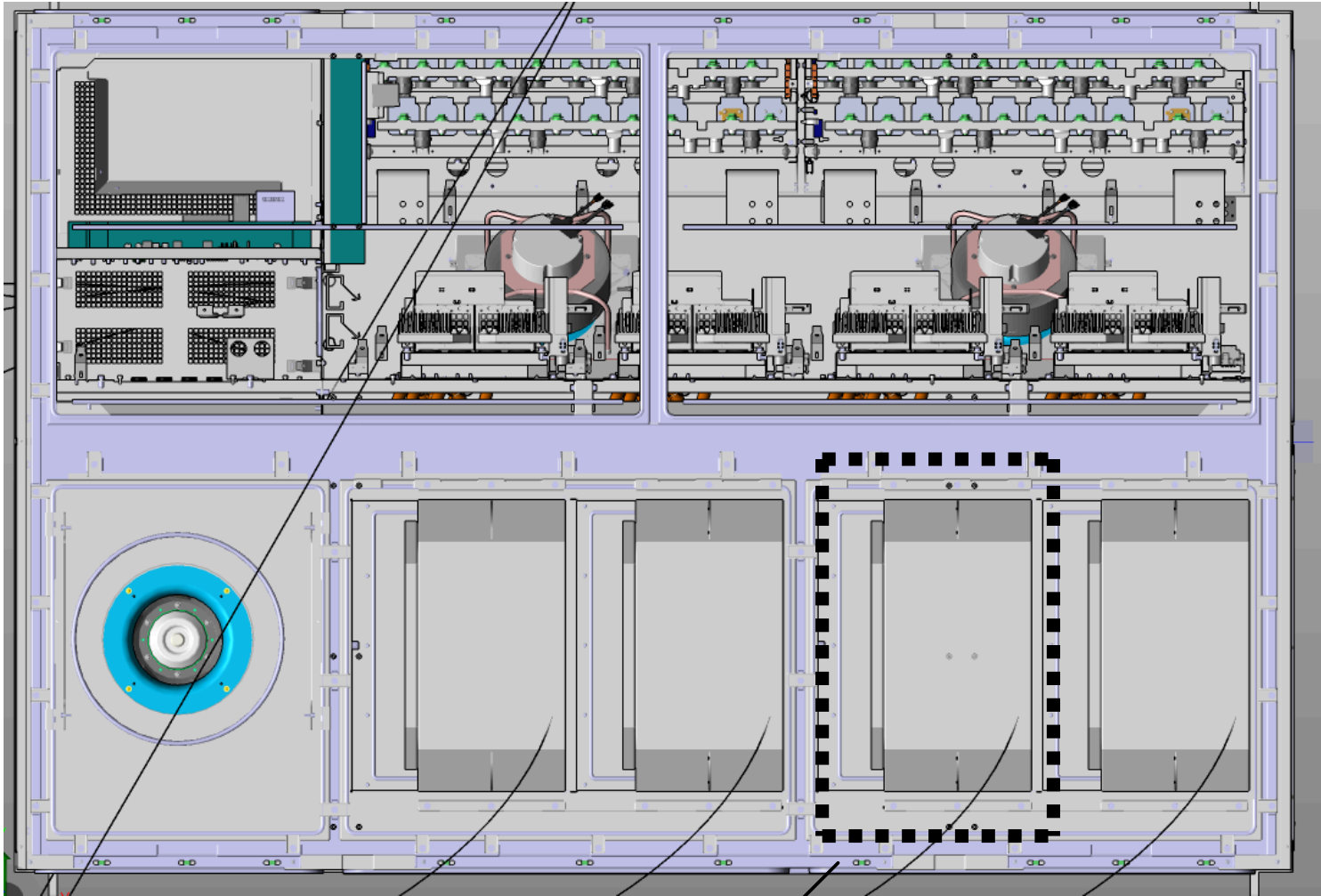
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+PMV3-XC200	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+PMV3-XC200A	SHIELD		
=HEM++ZMI+PMV3-XC200B	SHIELD		
=HEM++ZMI+PMV3-XC201.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 3	L4707	1
		L5710	3
		L8258	1
		L4897	1



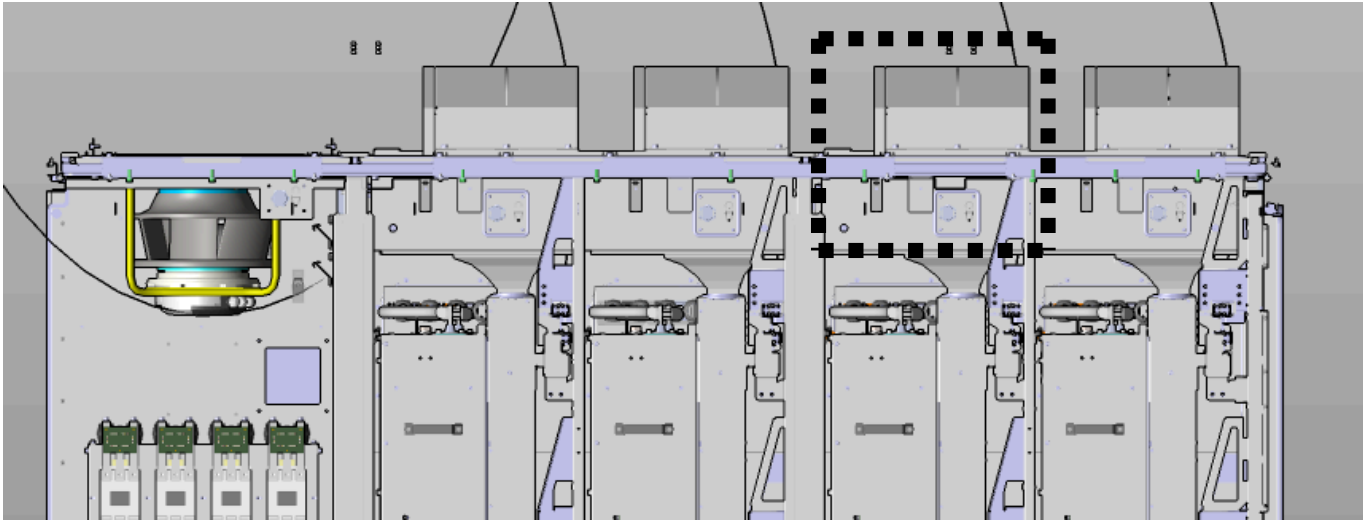
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
PMV4	FAN PRE-ASSEMBLY 4		
PMV4 FAN PRE-ASSEMBLY 4			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV4



PMV4



-XC204.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 4

-XC203 = T CONNECTOR M12 RS485



DESIGNED I+D 1

REVISED I+D 2

APPROVED I+D 3

DATE

REVISION

1.3.11

Project

ESQ_FS4010M10U000012_000

Name

Designation terminals and connectors

Installation:

HEM INVERTER

HEM

A3

Mounting location: +PMV4	FAN PRE-ASSEMBLY 4	++ZMI
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++ZMI	
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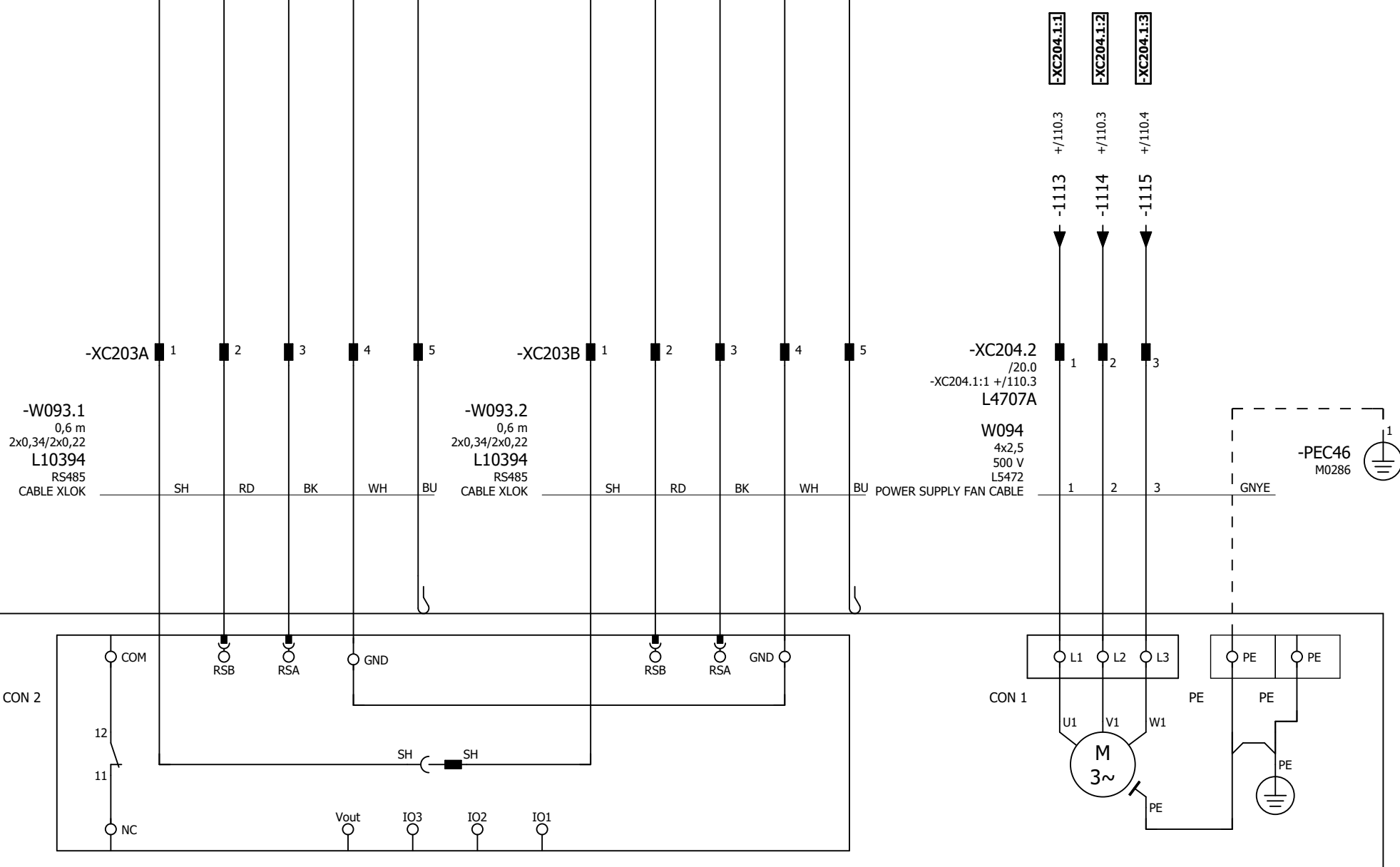
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Project Code:	FS4010M10U10001100410010020101014
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Total number of pages	455
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BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M04	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+PMV4-W093.1	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV4-W093.2	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV4-W094	POWER SUPPLY FAN CABLE	L5472

Plug overview

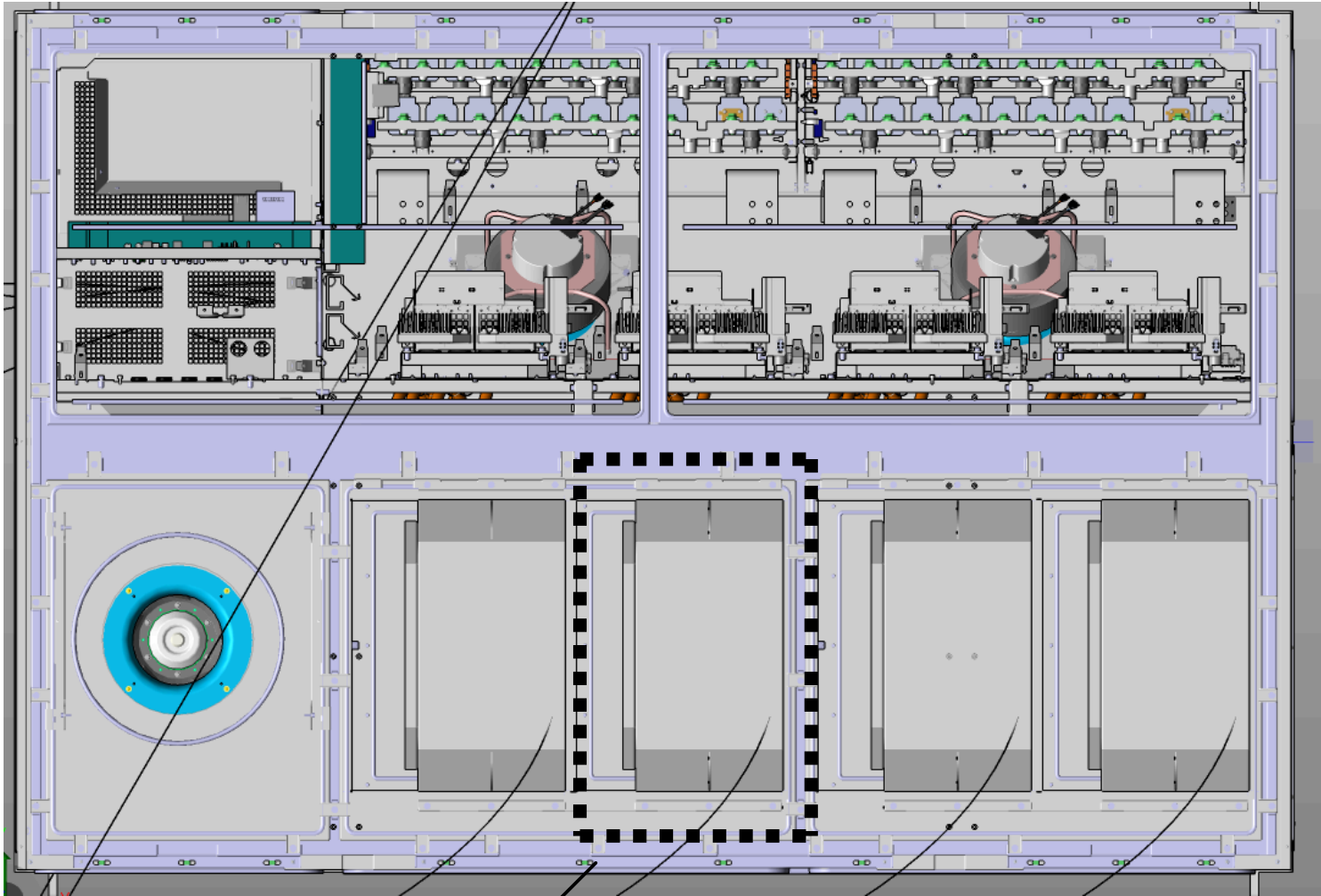
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+PMV4-XC203	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+PMV4-XC203A	SHIELD		
=HEM++ZMI+PMV4-XC203B	SHIELD		
=HEM++ZMI+PMV4-XC204.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 4	L4707A	1
		L5710	3
		L8258	1
		L4897	1



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
PMV5	FAN PRE-ASSEMBLY 5		
PMV5 FAN PRE-ASSEMBLY 5			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV5



PMV5



-XC207.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 5

-XC206 = T CONNECTOR M12 RS485



DESIGNED I+D 1

REVISÉ I+D 2

APPROVED I+D 3

DATE

REVISION

1.3.11

	Project
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ESQ_FS4010M10U000012_000

Name

Designation terminals and connectors

Installation:

HEM INVERTER

HEM

A3

Mounting location: +PMV5	FAN PRE-ASSEMBLY 5	++ZMI
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++ZMI	
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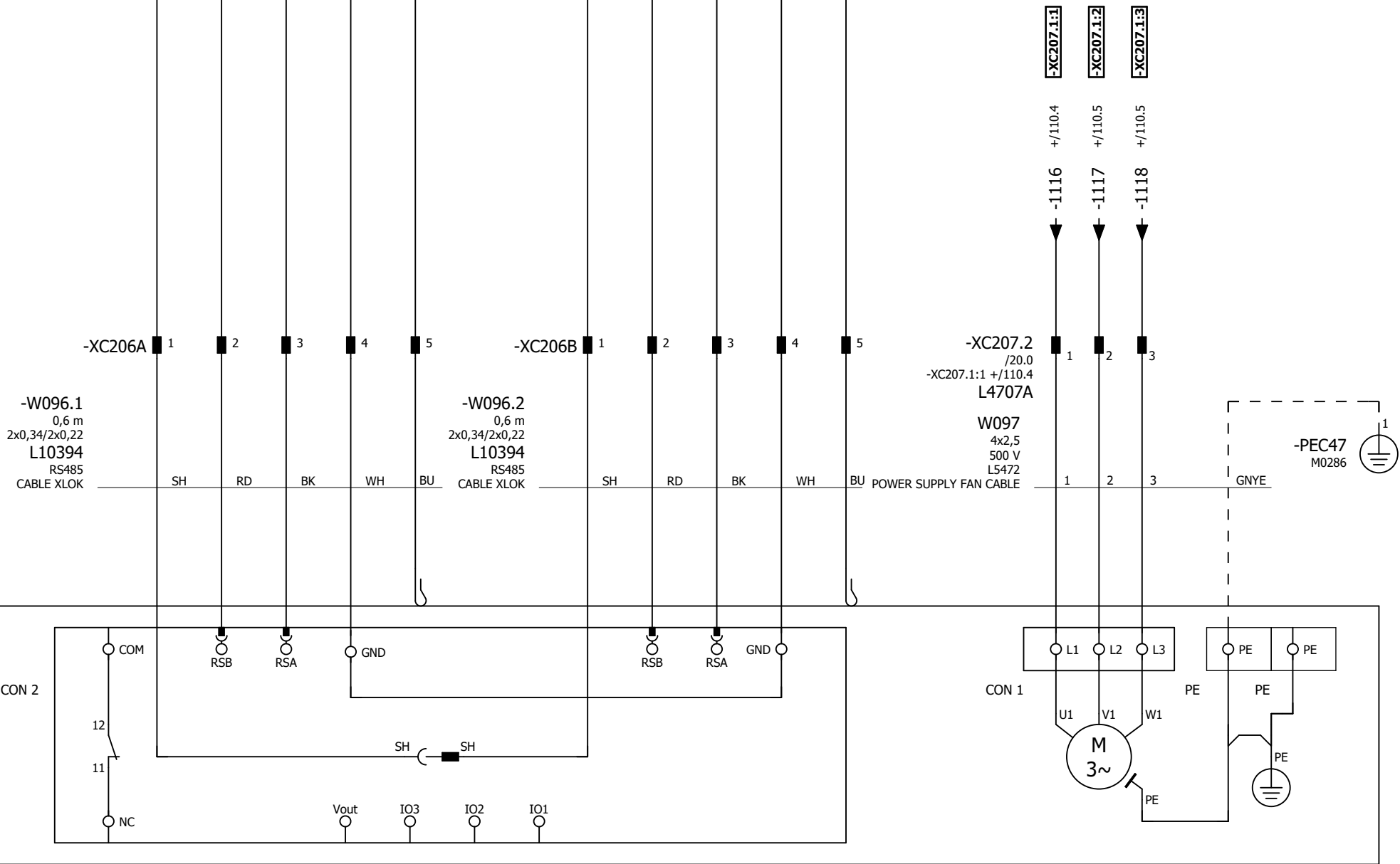
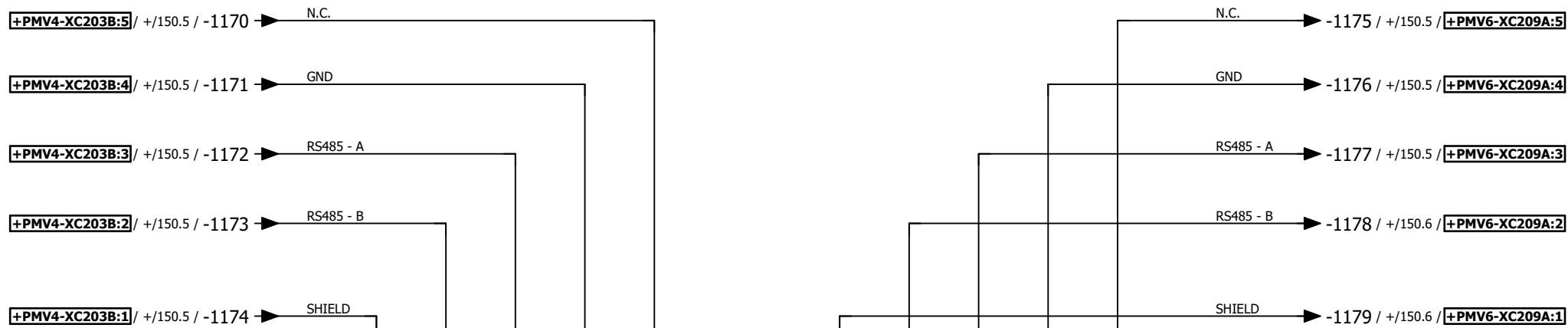
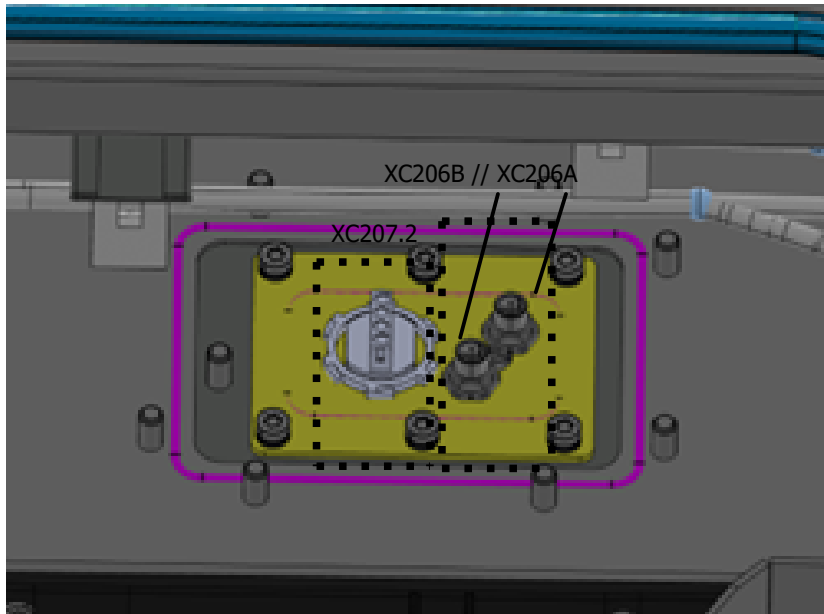
Previous page	10
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Project Code:	FS4010M10U10001100410010020101014
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Total number of pages	455
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BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M05	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+PMV5-W096.1	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV5-W096.2	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV5-W097	POWER SUPPLY FAN CABLE	L5472

Plug overview

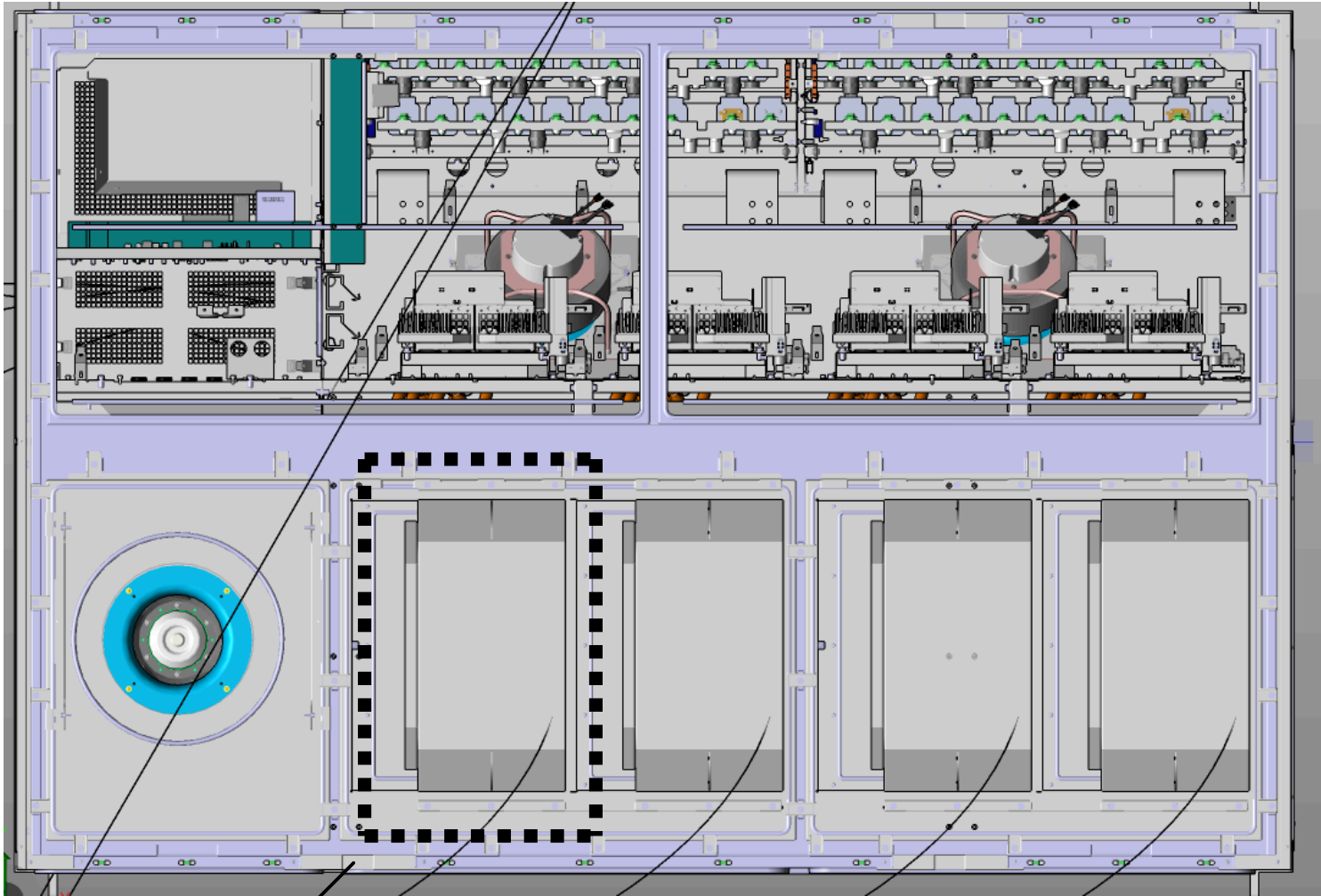
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+PMV5-XC206	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+PMV5-XC206A	SHIELD		
=HEM++ZMI+PMV5-XC206B	SHIELD		
=HEM++ZMI+PMV5-XC207.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 5	L4707A	1
		L5710	3
		L8258	1
		L4897	1



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
PMV6	FAN PRE-ASSEMBLY 6		
PMV6 FAN PRE-ASSEMBLY 6			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV6



PMV6



-XC210.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 6

-XC209 = T CONNECTOR M12 RS485



DESIGNED I+D 1

REVISED I+D 2

APPROVED I+D 3

DATE

REVISION

1.3.11

Project

ESQ_FS4010M10U000012_000

Name

Designation terminals and connectors

Installation:

HEM INVERTER

HEM

A3

Mounting location: +PMV6	FAN PRE-ASSEMBLY 6	++ZMI
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++ZMI	
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Previous page	10
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Next page 200

Project Code:	FS4010M10U10001100410010020101014
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Page 20

Total number of pages
455

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M06	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+PMV6-W099.1	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV6-W099.2	RS485 CABLE XLOK	L10394
=HEM++ZMI+PMV6-W0100	POWER SUPPLY FAN CABLE	L5472

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+PMV6-XC209	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+PMV6-XC209A	SHIELD		
=HEM++ZMI+PMV6-XC209B	SHIELD		
=HEM++ZMI+PMV6-XC210.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 6	L4707A	1
		L5710	3
		L8258	1
		L4897	1



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
AFC	C FILTER CABINET		
AFC C FILTER CABINET			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

-XC215.2 = MATE-N-LOK 12P PANEL CONTACTORS



DESIGNED	I+D 1
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REVISÉ I+D 2

APPROVED I+D 3

DATE _____

REVISION	1.3.11
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Project	
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ESQ_FS4010M10U000012_000

Name	
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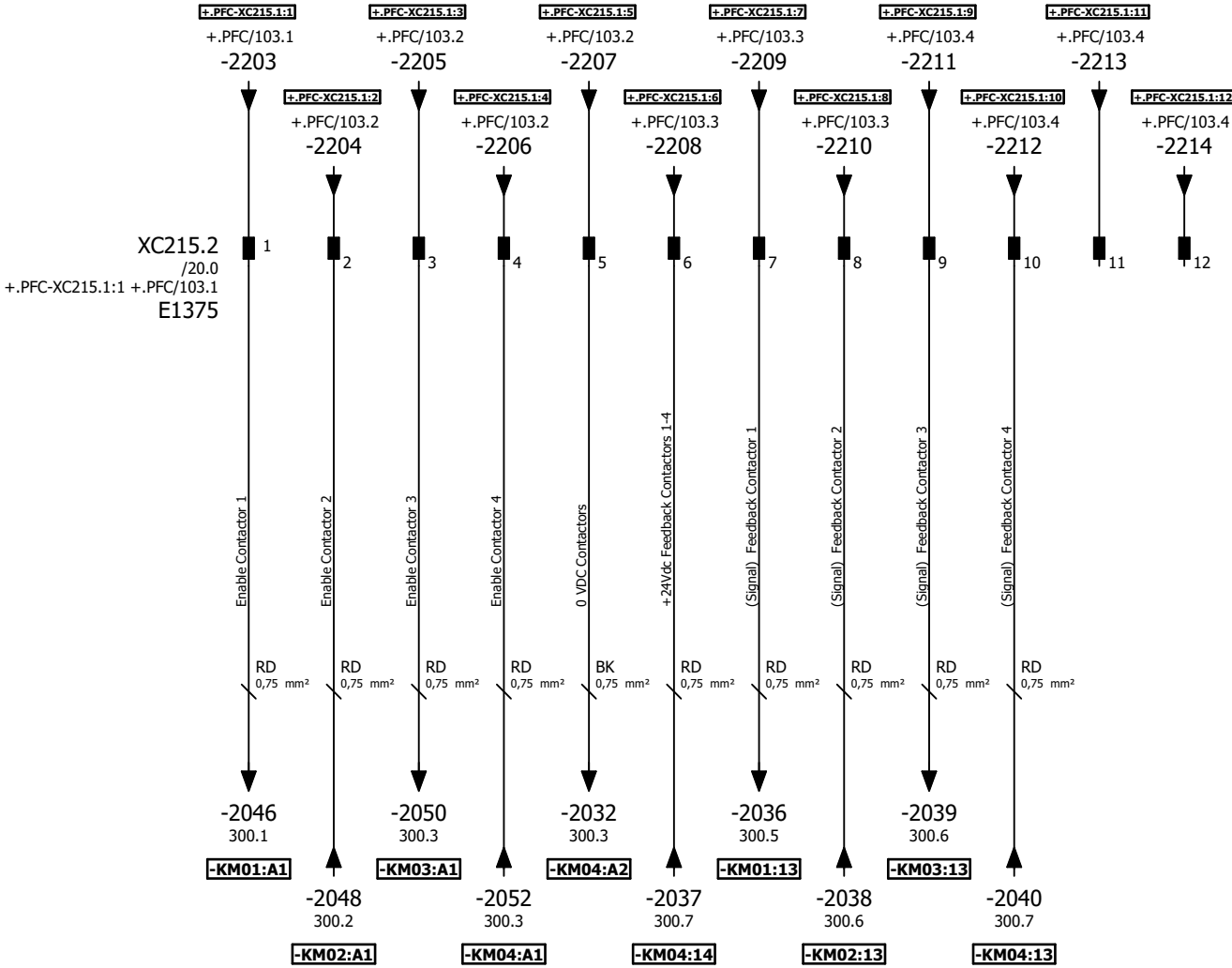
Designation terminals and connectors

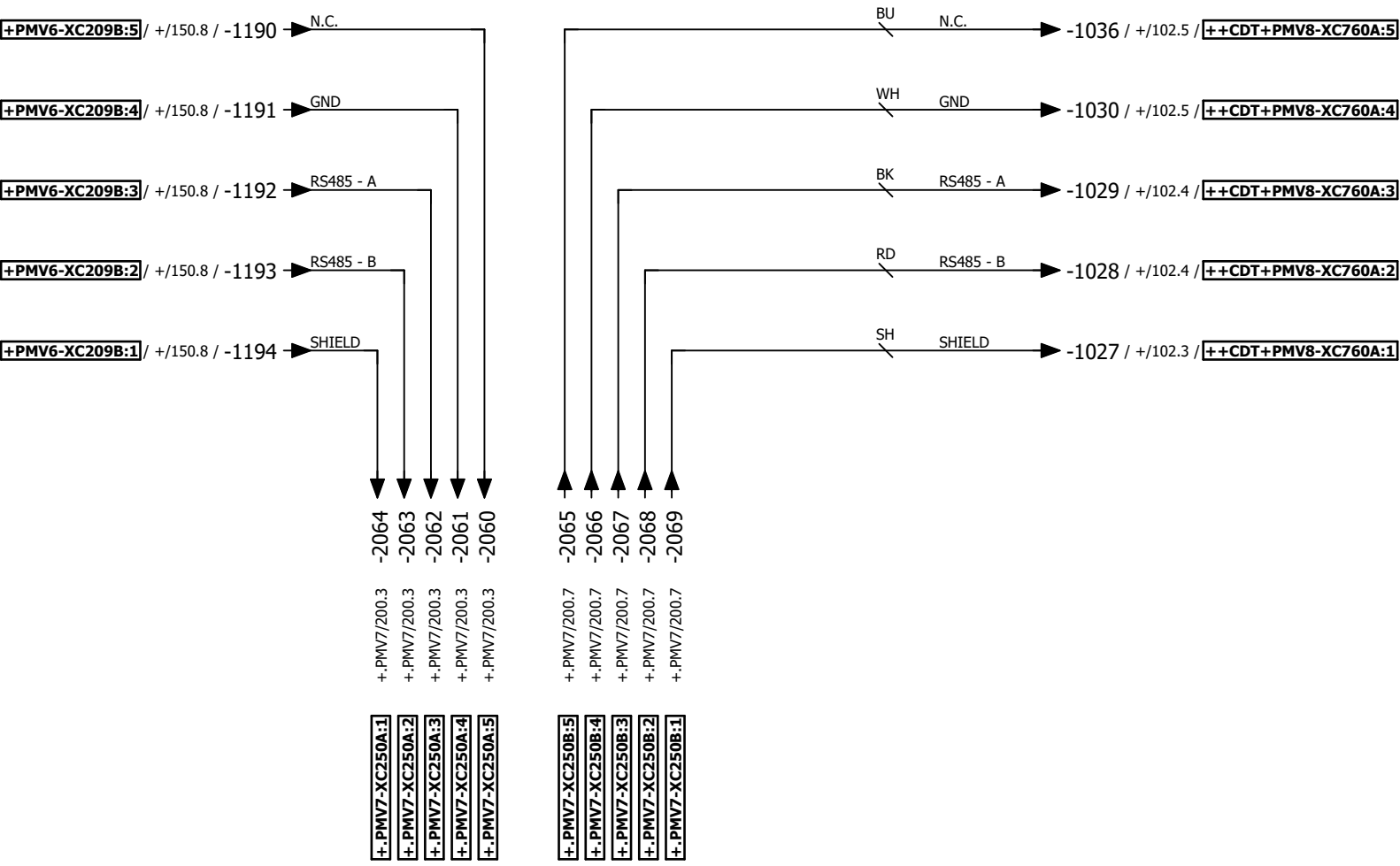
Installation:	HEM INVERTER	HEM
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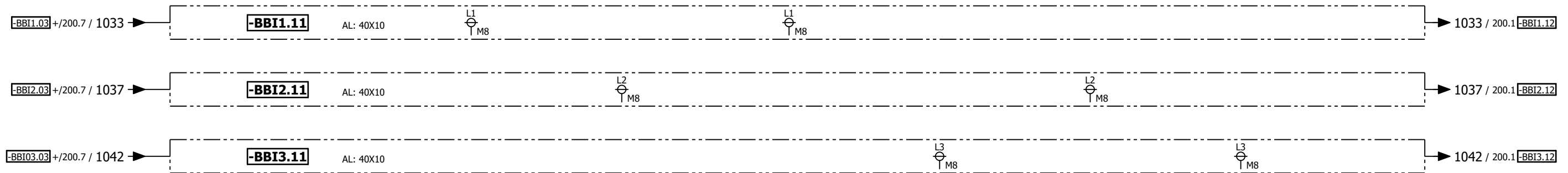
Mounting location: +AFC	C FILTER CABINET	++ZMI	Previous page 1	Next page 100
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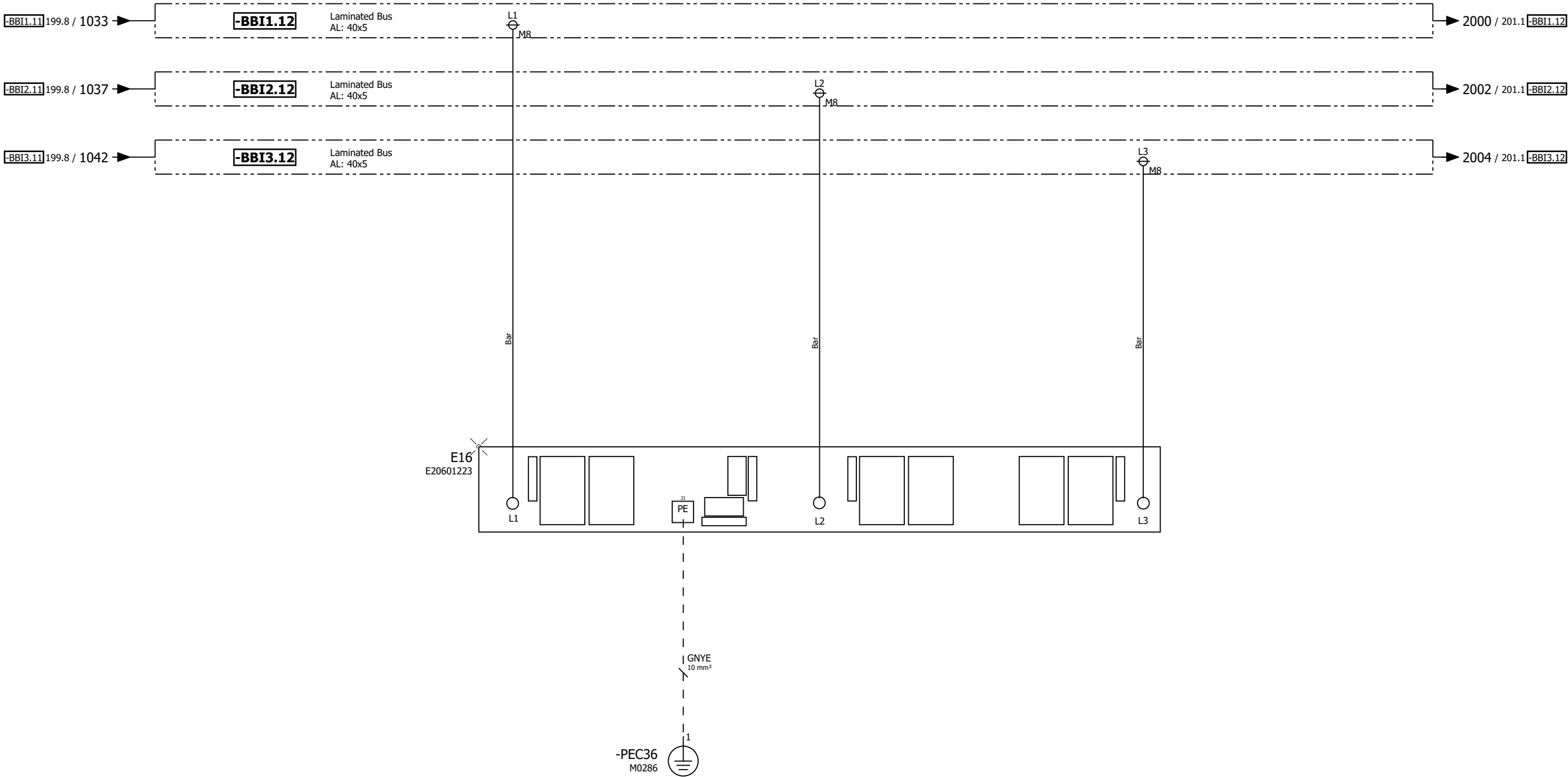
Project Code:	FS4010M10U10001100410010020101014	Page	20	Total number of pages	455
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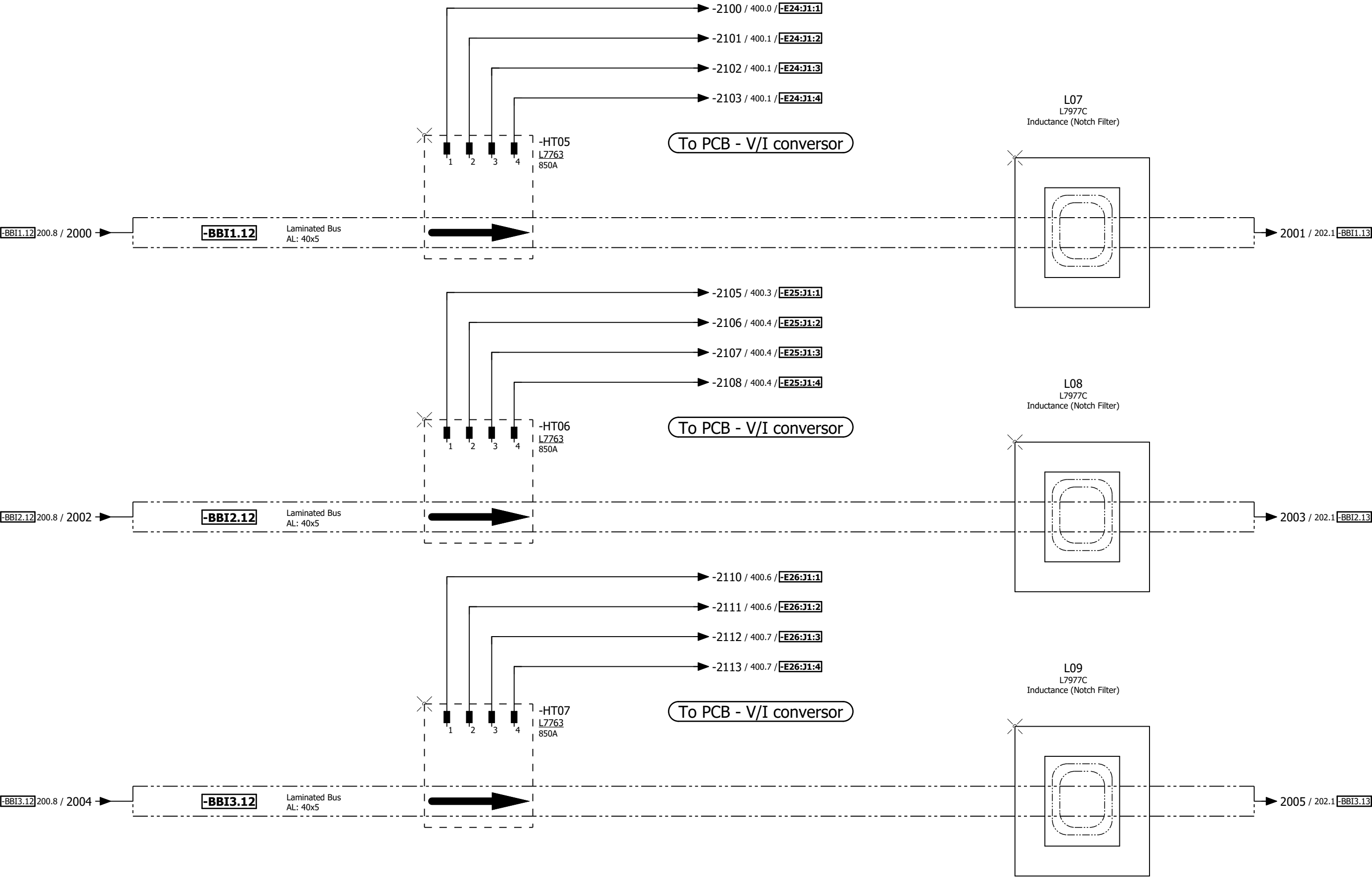
Interconnection AFC - AFC.PFC

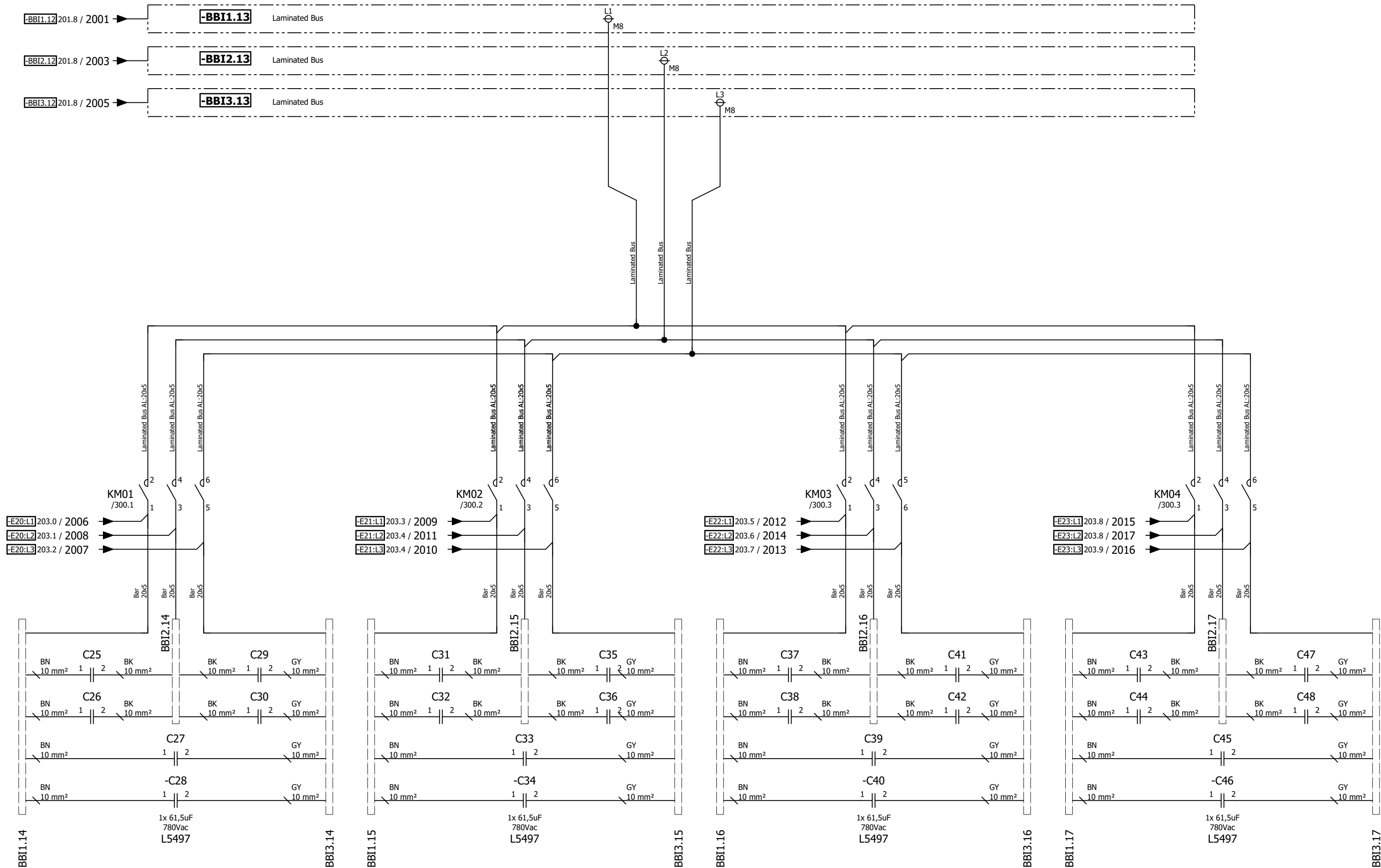




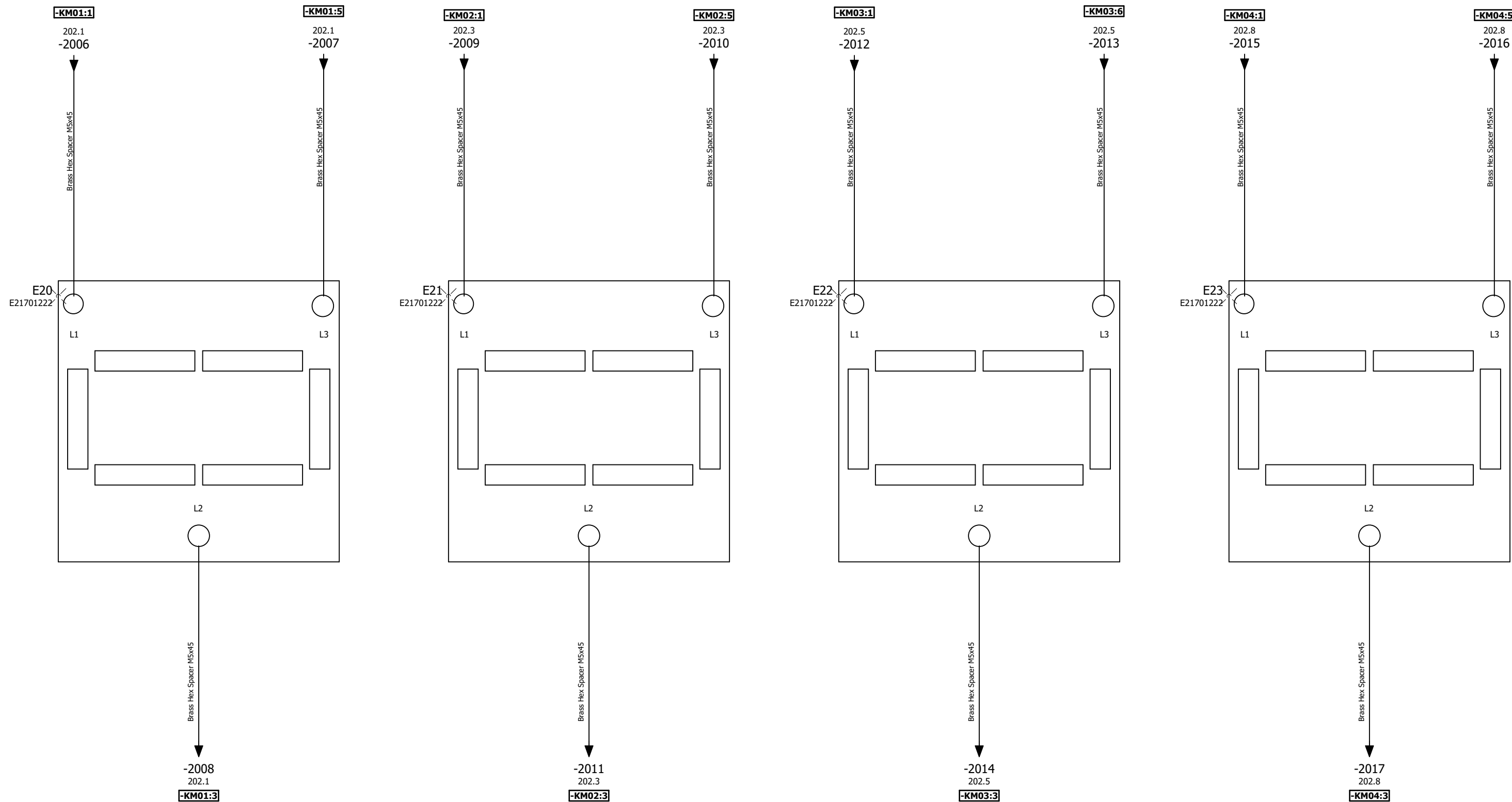


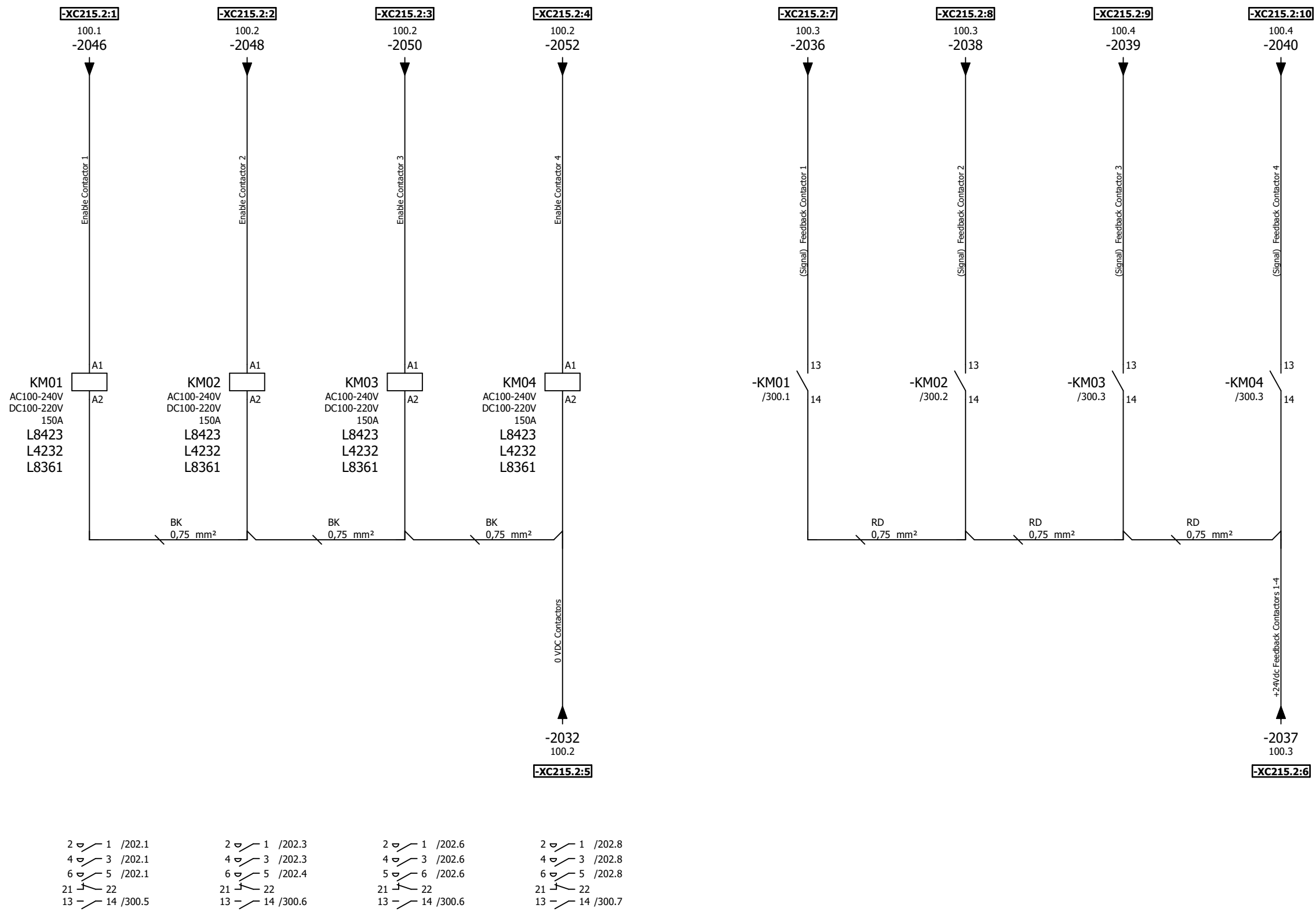


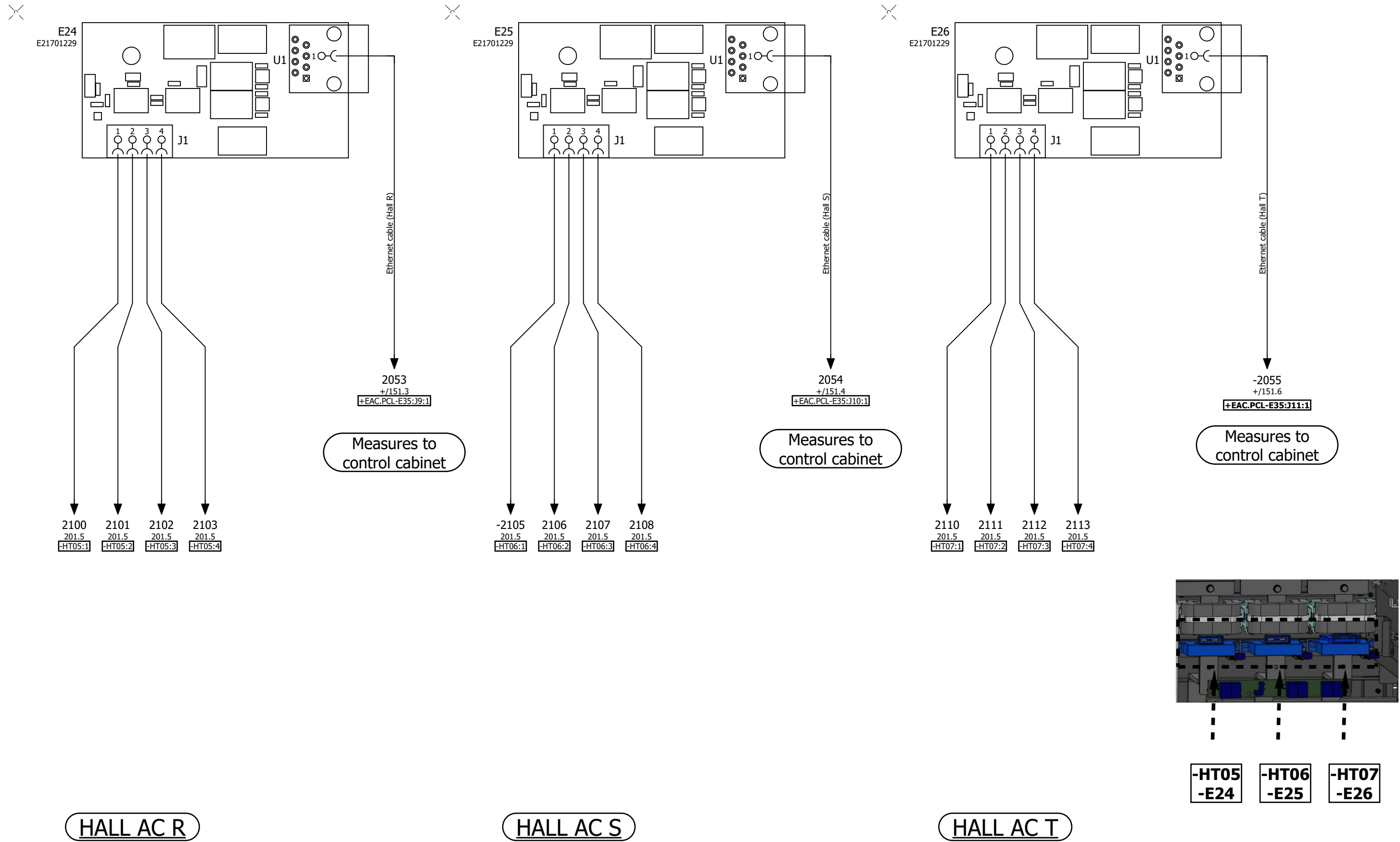
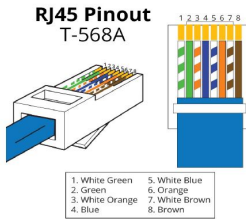




DISCHARGE BOARD RESISTOR LCL







BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C25	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C26	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C27	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C28	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C29	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C30	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C31	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C32	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C33	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C34	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C35	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C36	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C37	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C38	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C39	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C40	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C41	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C42	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C43	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C44	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C45	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C46	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C47	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-C48	L5497	L5497	CAPACITOR 1x 61.5uF 780Vac	1
-E16	PCB E20601223	PCB E20601223	RFI AC HEM GEN3 FILTER PCB BOARD	1
-E20	PCB E21701222	PCB E21701222	PCB DISCHARGE LCL HEM GENIII	1
-E21	PCB E21701222	PCB E21701222	PCB DISCHARGE LCL HEM GENIII	1
-E22	PCB E21701222	PCB E21701222	PCB DISCHARGE LCL HEM GENIII	1
-E23	PCB E21701222	PCB E21701222	PCB DISCHARGE LCL HEM GENIII	1
-E24	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E25	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-E26	PCB E21701229	PCB E21701229	BOARD ACONDICIO. HALL DELTA SIGMA HEM GEN3	1
-HT05	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT06	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-HT07	L7763	HAX 850-S	TRANSDUCER HAX 850-S	1
-KM01	L8423	1364031600	CONTACTOR MC-150A 3P AC100-240V DC100-220V 1a1b	1
-KM01	L4232	83631613009	CONTACTOR ACCESORY "CAPACITOR UNIT"	1
-KM01	L8361	83611634005	ACTUATOR CONTACTOR SUP VARISTOR + RC 100-125VDC	1

-KM02	L8423	1364031600	CONTACTOR MC-150A 3P AC100-240V DC100-220V 1a1b	1
-KM02	L4232	83631613009	CONTACTOR ACCESORY "CAPACITOR UNIT"	1
-KM02	L8361	83611634005	ACTUATOR CONTACTOR SUP VARISTOR + RC 100-125VDC	1
-KM03	L8423	1364031600	CONTACTOR MC-150A 3P AC100-240V DC100-220V 1a1b	1
-KM03	L4232	83631613009	CONTACTOR ACCESORY "CAPACITOR UNIT"	1
-KM03	L8361	83611634005	ACTUATOR CONTACTOR SUP VARISTOR + RC 100-125VDC	1
-KM04	L8423	1364031600	CONTACTOR MC-150A 3P AC100-240V DC100-220V 1a1b	1
-KM04	L4232	83631613009	CONTACTOR ACCESORY "CAPACITOR UNIT"	1
-KM04	L8361	83611634005	ACTUATOR CONTACTOR SUP VARISTOR + RC 100-125VDC	1
-L07	L7977C	IM6987	INDUCTANCIA MONOFÁSICA (Filtro Notch)	1
-L08	L7977C	IM6987	INDUCTANCIA MONOFÁSICA (Filtro Notch)	1
-L09	L7977C	IM6987	INDUCTANCIA MONOFÁSICA (Filtro Notch)	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+AFC-XC215.2	MATE-N-LOK 12P PANEL CONTACTORS	E1375	1
		E1204	10

 POWER ELECTRONICS®	DESIGNED I+D 1	DATE	Project ESQ_FS4010M10U000012_000	Name Plug overview : =HEM++ZMI+AFC-XC215.2 - =HEM++ZMI+AFC-XC215.2	Installation: HEM INVERTER			A3
	REVISED I+D 2	REVISION 1.3.11			Mounting location: +AFC C FILTER CABINET ++ZMI		Previous page + .PFC&/1	Next page + .PFC&/1
	APPROVED I+D 3				Project Code: FS4010M10U10001100410010020101014		Page 3000	Total number of pages 455



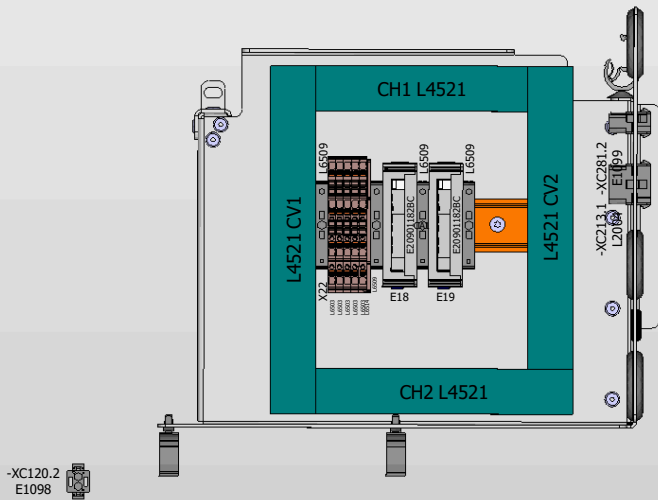
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
AFC.PFC	PANEL FILTER C		
PFC PANEL FILTER C			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

LAYOUT PANEL

Enclosure legend

IME	ERP	Description
CH1;CH2;CV1;CV2	L4521	CABLE DUCT UNEX 42X30 EN U23X
CA1	L1061	DIN RAIL 35x7,5mm
E18;E19	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD



-XC281.2 = MATE-N-LOK 2P PANEL POWER SUPPLY AFC

-X22 = 120Vdc Distribution

-XC213.1 = MATE-N-LOK 4P PANEL TH IND-CROWBAR

-XC215.1 = MATE-N-LOK 12P AERIAL CONTACTORS

-XC120.2 = MATE-N-LOK 2P SPD



DESIGNED I+D 1

REVISÉ I+D 2

APPROVED I+D 3

DATE

REVISION

1.3.11

Project

ESQ_FS4010M10U000012_000

Name

Designation terminals and connectors

Installation:

HEM INVERTER

HEM

A3

Mounting location: +AFC.PFC C FILTER CABINET ++ZMI

Project Code:	FS4010M10U10001100410010020101014
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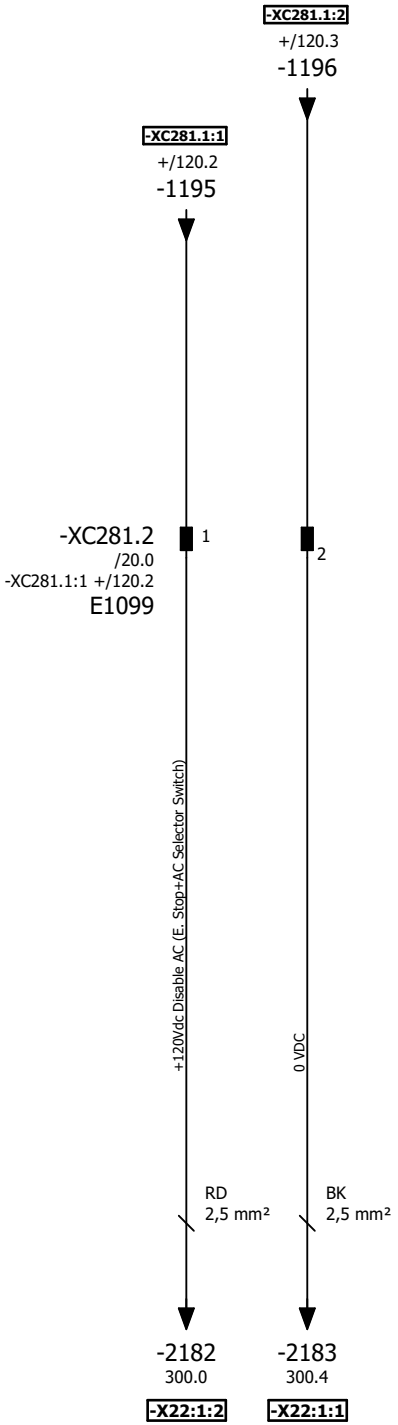
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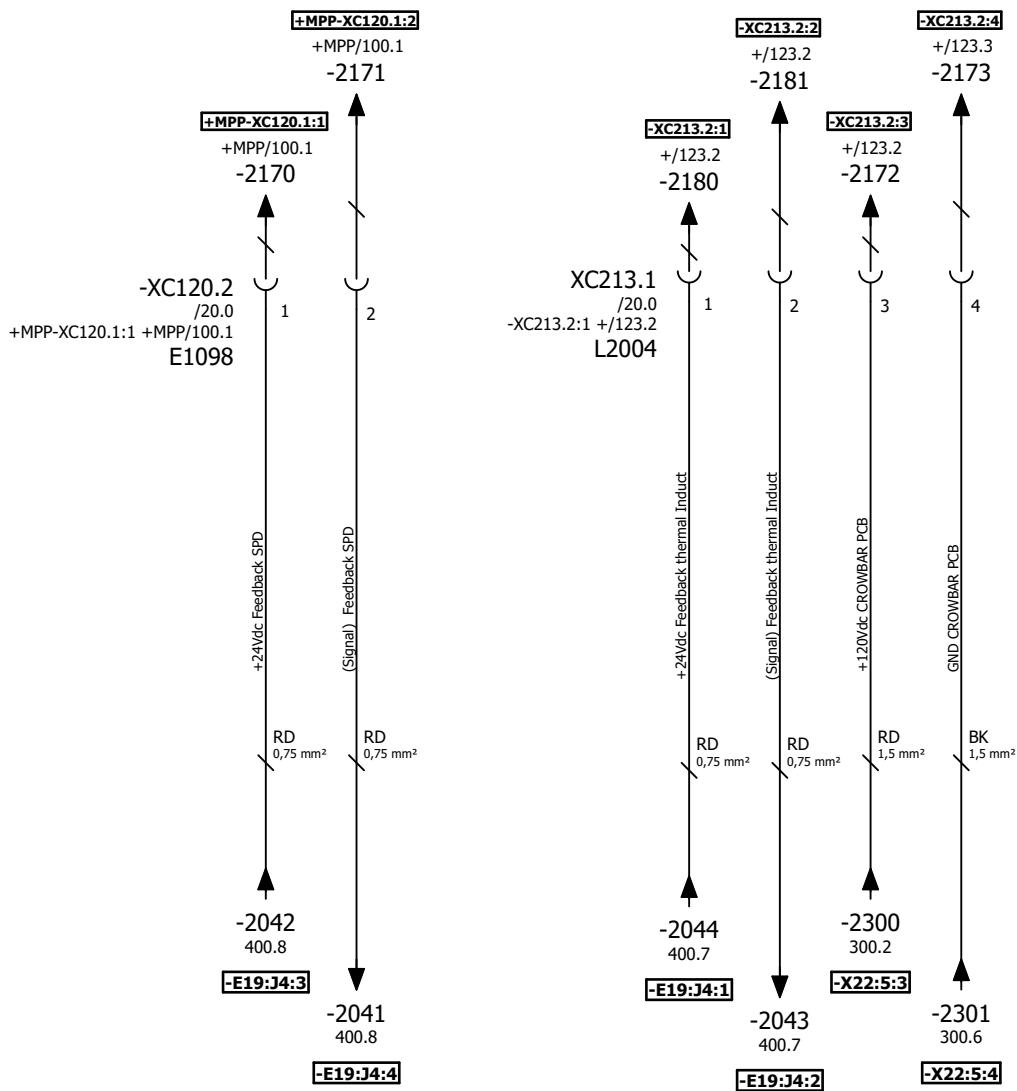
Next page 100

Total number of pages
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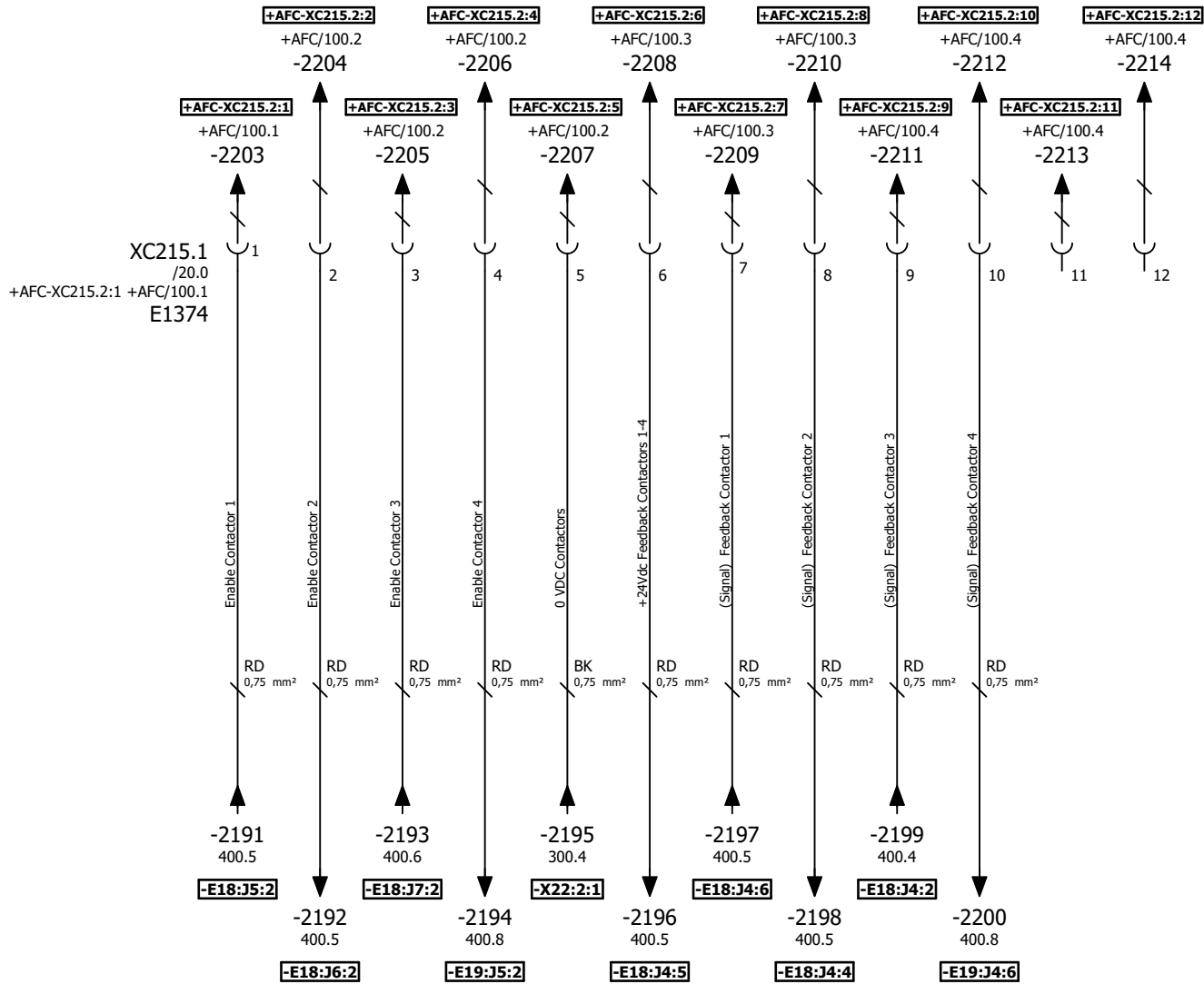
Interconnections - PSA//AFC



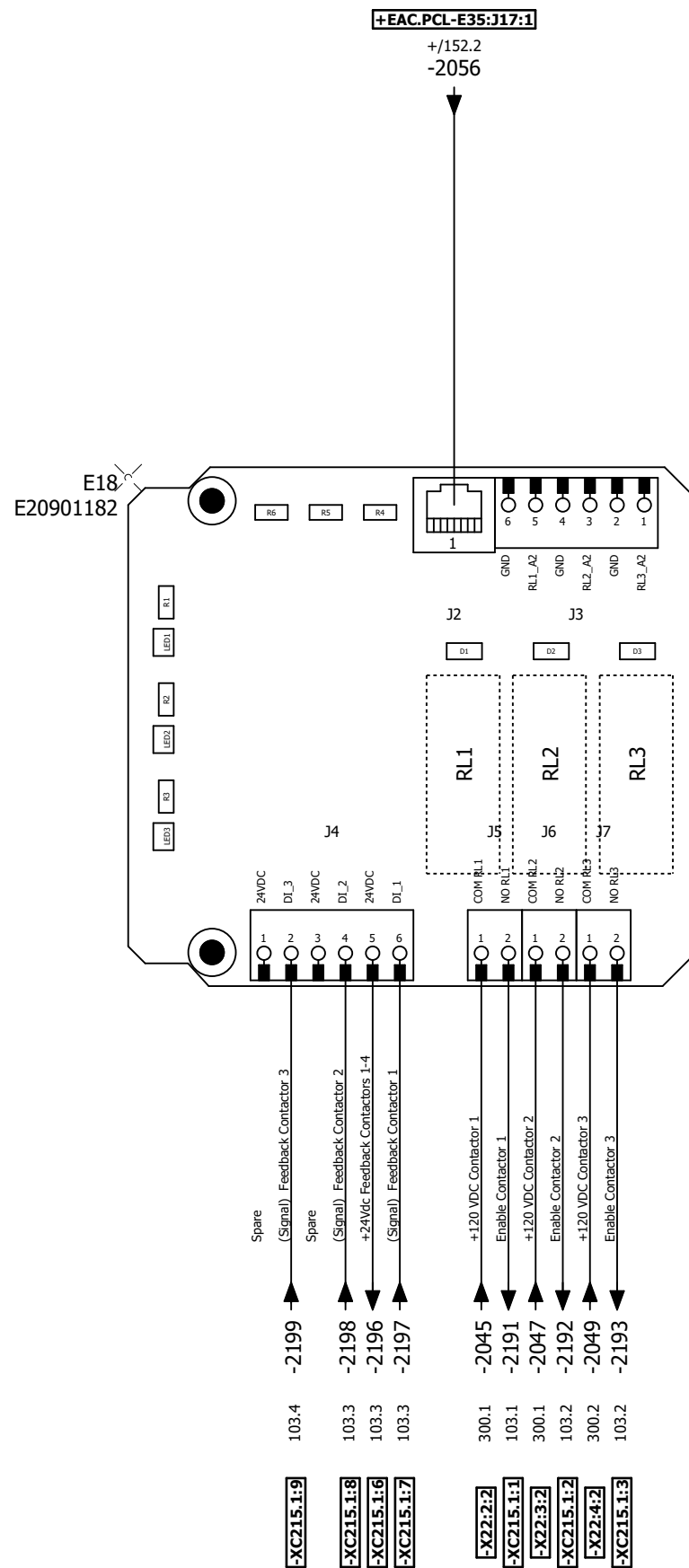
Interconnection SPD/Induct - C Filter



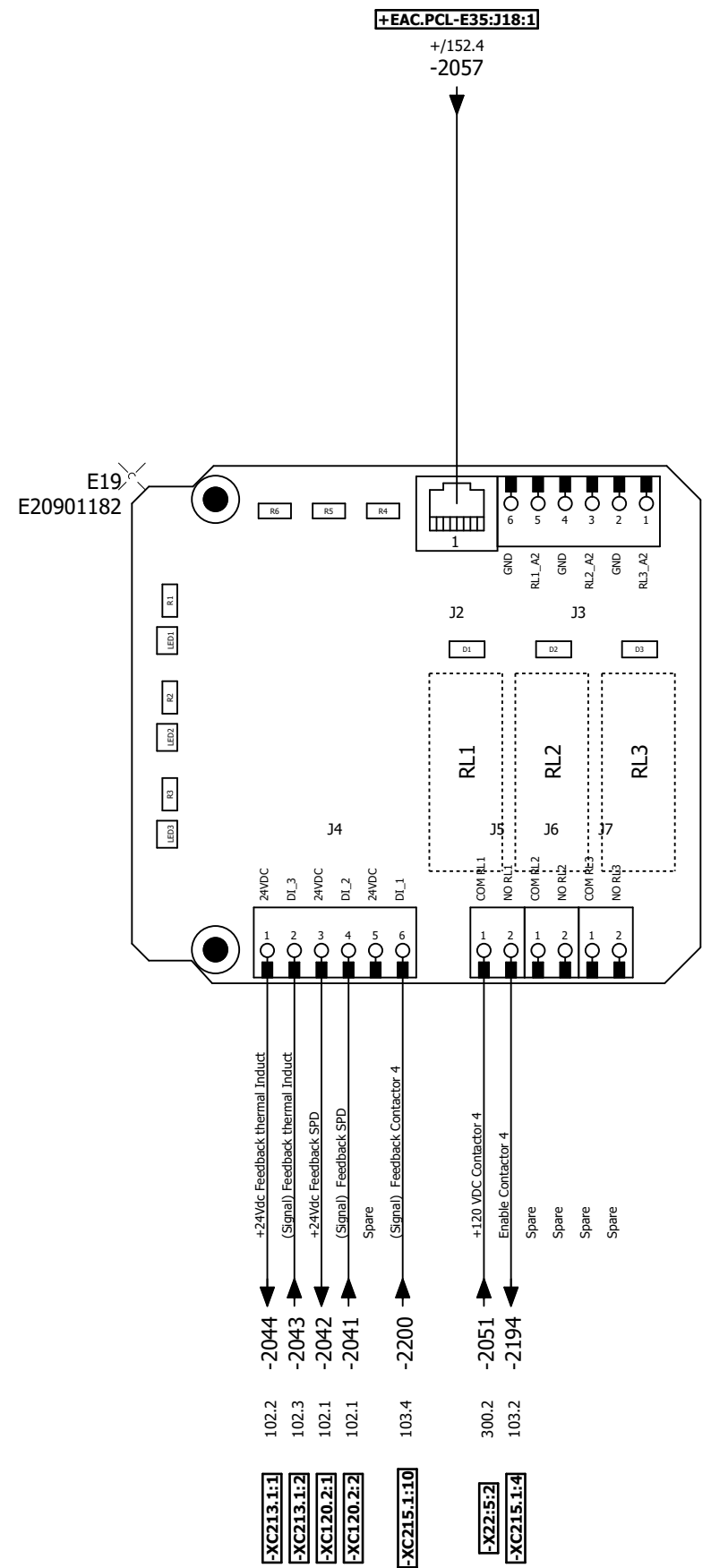
Interconnection AFC - AFC.PFC



PCB - I/O C Filter



PCB - I/O C Filter



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E18	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E18	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E19	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E19	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1

Plug overview

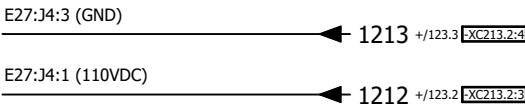
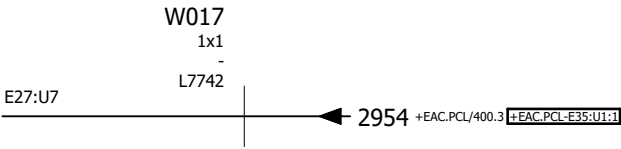
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+AFC.PFC-XC120.2	MATE-N-LOK 2P SPD	E1098	1
		E1203	2
=HEM++ZMI+AFC.PFC-XC213.1	MATE-N-LOK 4P PANEL TH IND-CROWBAR	L2004	1
		E1203	2
=HEM++ZMI+AFC.PFC-XC215.1	MATE-N-LOK 12P AERIAL CONTACTORS	E1374	1
		E1203	10
=HEM++ZMI+AFC.PFC-XC281.2	MATE-N-LOK 2P PANEL POWER SUPPLY AFC	E1099	1
		L5710	2



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
AFC.PCW	CROWBAR ASSEMBLY		
PCW CROWBAR ASSEMBLY			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

The power supply and optical fiber interconnection to the crowbar PCB remain standard in the inverter, so that the equipment is ready to implement the protection in case it is later required in the field.



Cable overview

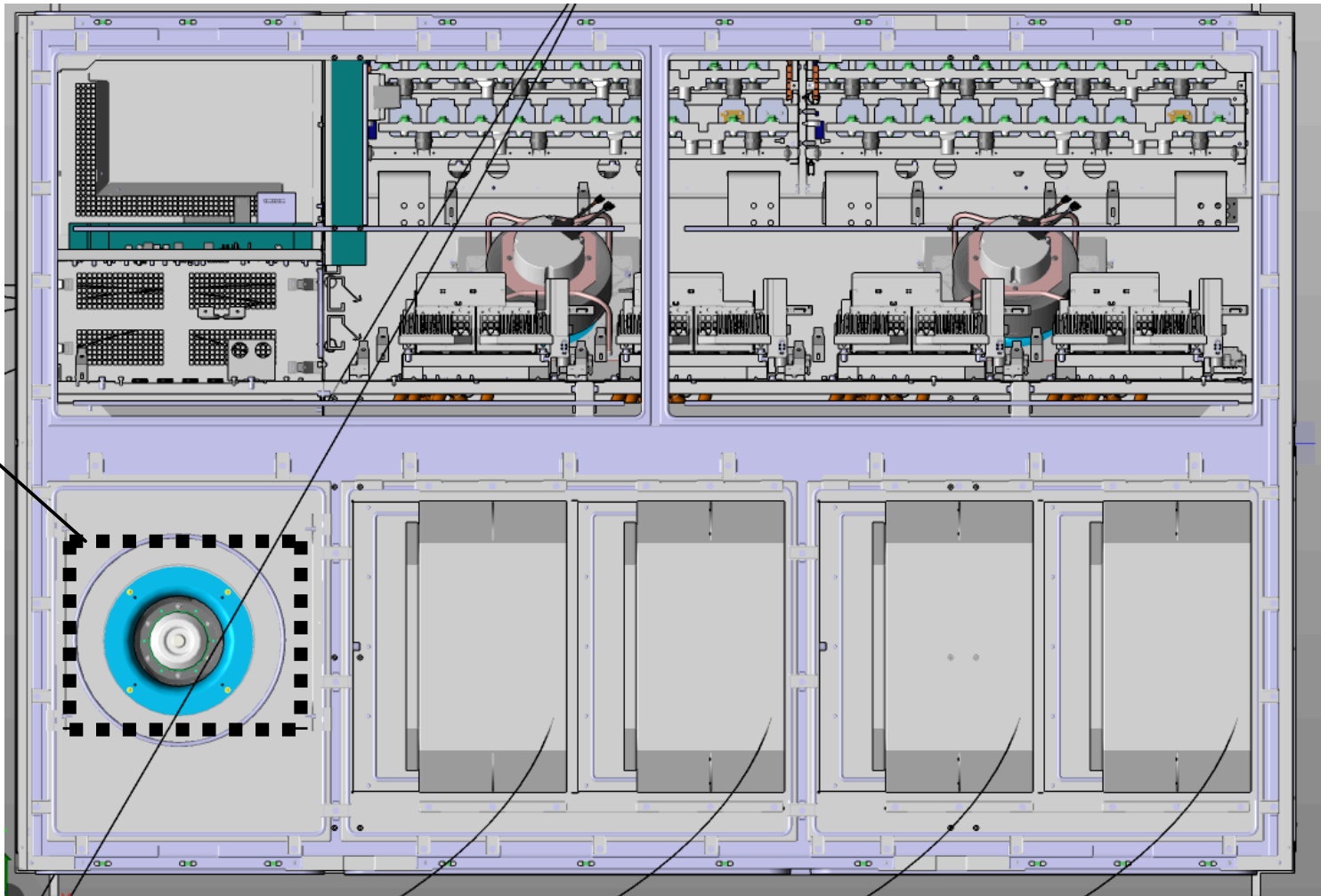
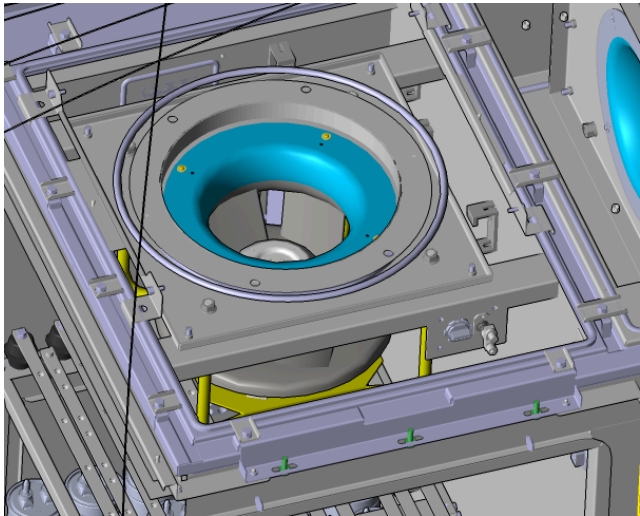
Cable designation	Function text	ERP
=HEM++ZMI+AFC.PCW-W017	crowbar hose	L7742



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
AFC.PMV7	FAN PRE-ASSEMBLY 7		
PMV7 FAN PRE-ASSEMBLY 7			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV7



PMV7

-XC251.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 7

-XC250 = T CONNECTOR M12 RS485



DESIGNED I+D 1

REVISÉ I+D 2

APPROVED I+D 3

DATE _____

REVISION

1.3.11

Project

ESQ_FS4010M10U000012_000

Name

Designation terminals and connectors

Installation:

HEM INVERTER

HEM

A3	
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Mounting location: +AFC.PMV7 C FILTER CABINET ++ZMI

++ZMI

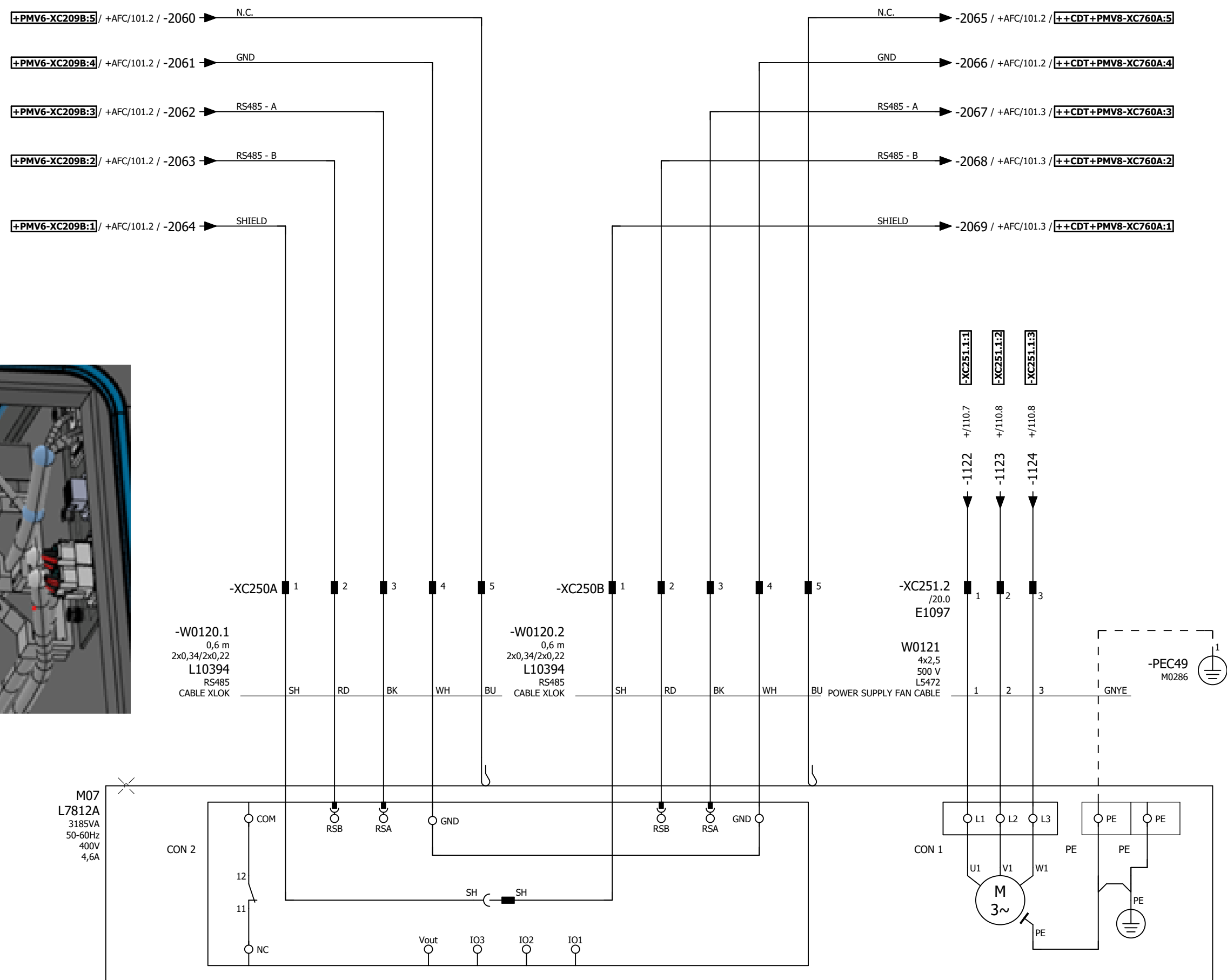
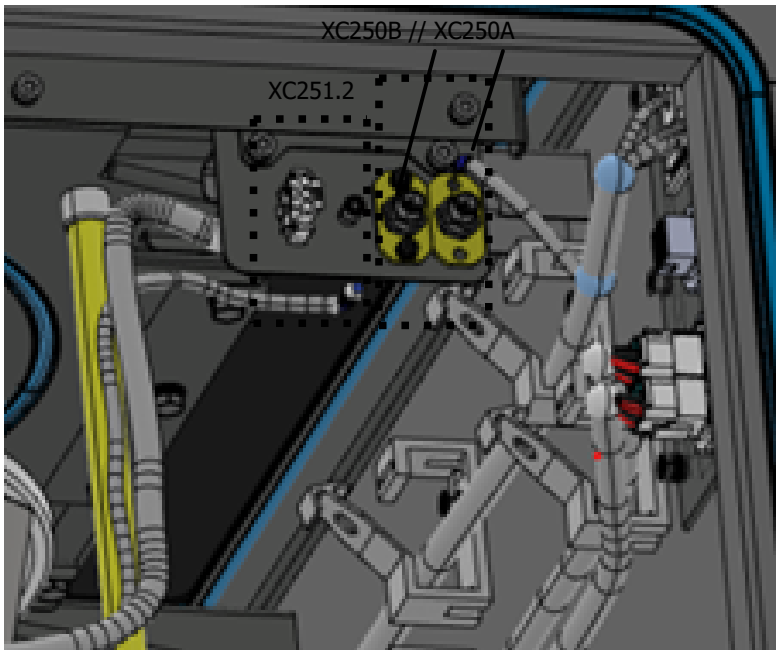
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Next page	200
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Project Code:	FS4010M10U10001100410010020101014
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Page	20
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Total number of pages	455
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BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M07	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+AFC.PMV7-W0120.1	RS485 CABLE XLOK	L10394
=HEM++ZMI+AFC.PMV7-W0120.2	RS485 CABLE XLOK	L10394
=HEM++ZMI+AFC.PMV7-W0121	POWER SUPPLY FAN CABLE	L5472

Plug overview

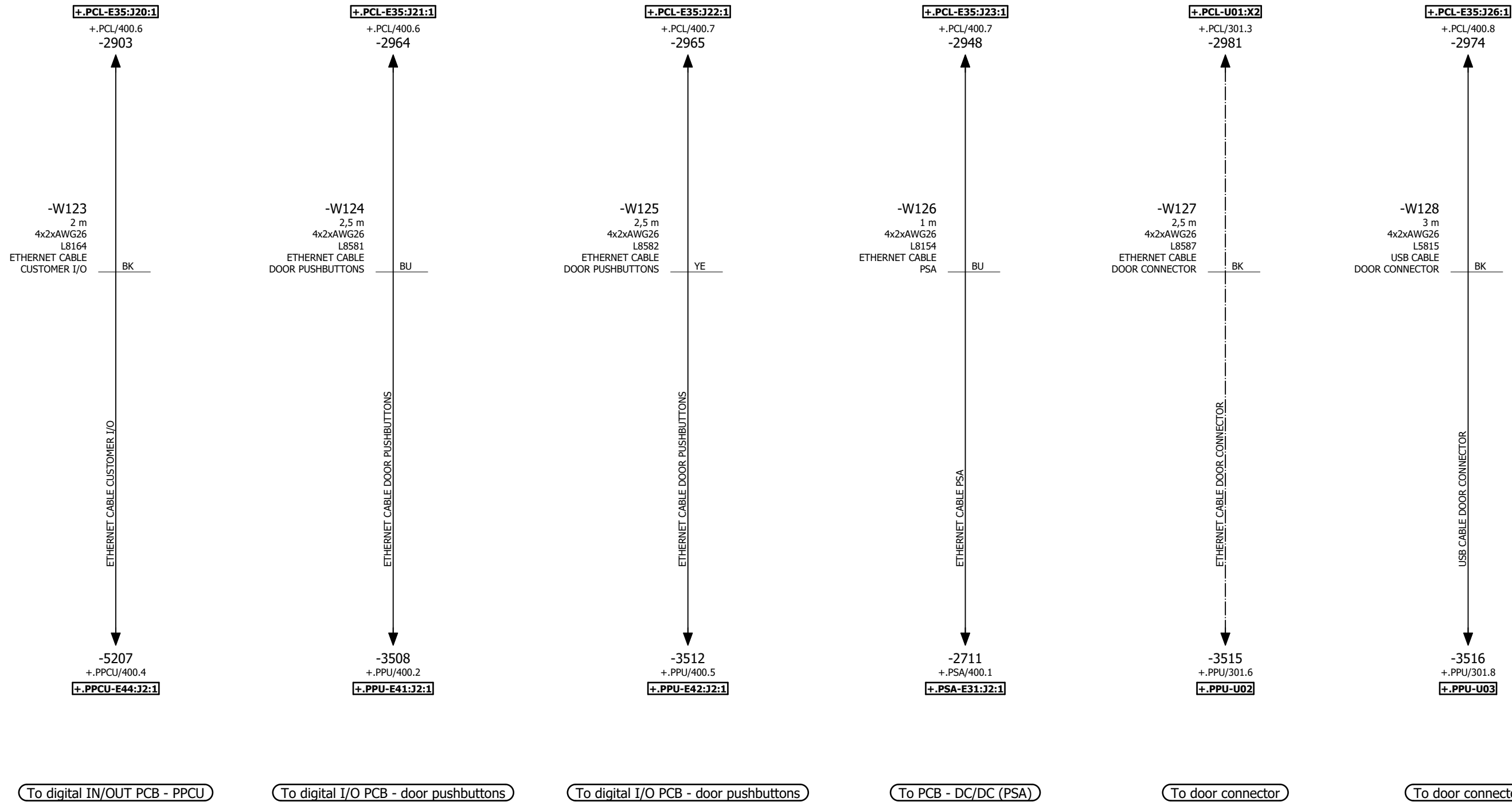
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		ERP	CANTIDAD
=HEM++ZMI+AFC.PMV7-XC250	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+AFC.PMV7-XC250A	SHIELD		
=HEM++ZMI+AFC.PMV7-XC250B	SHIELD		
=HEM++ZMI+AFC.PMV7-XC251.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 7	E1097	1
		L5710	3

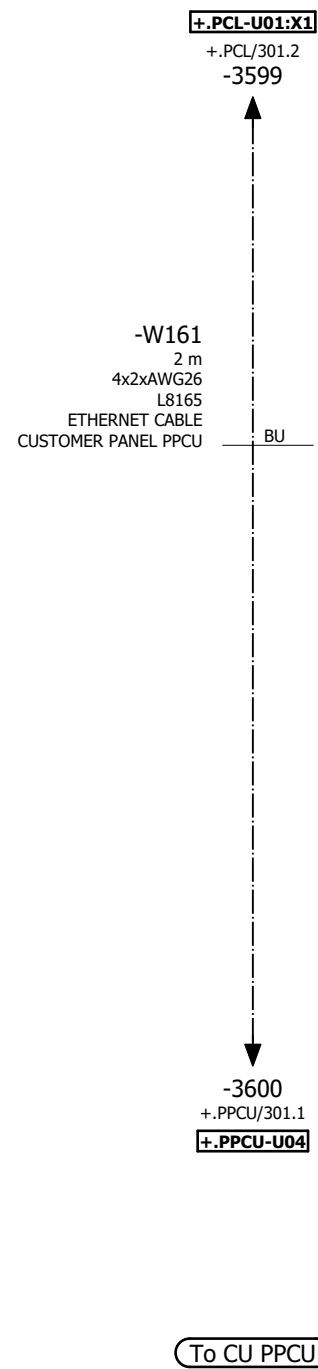


POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

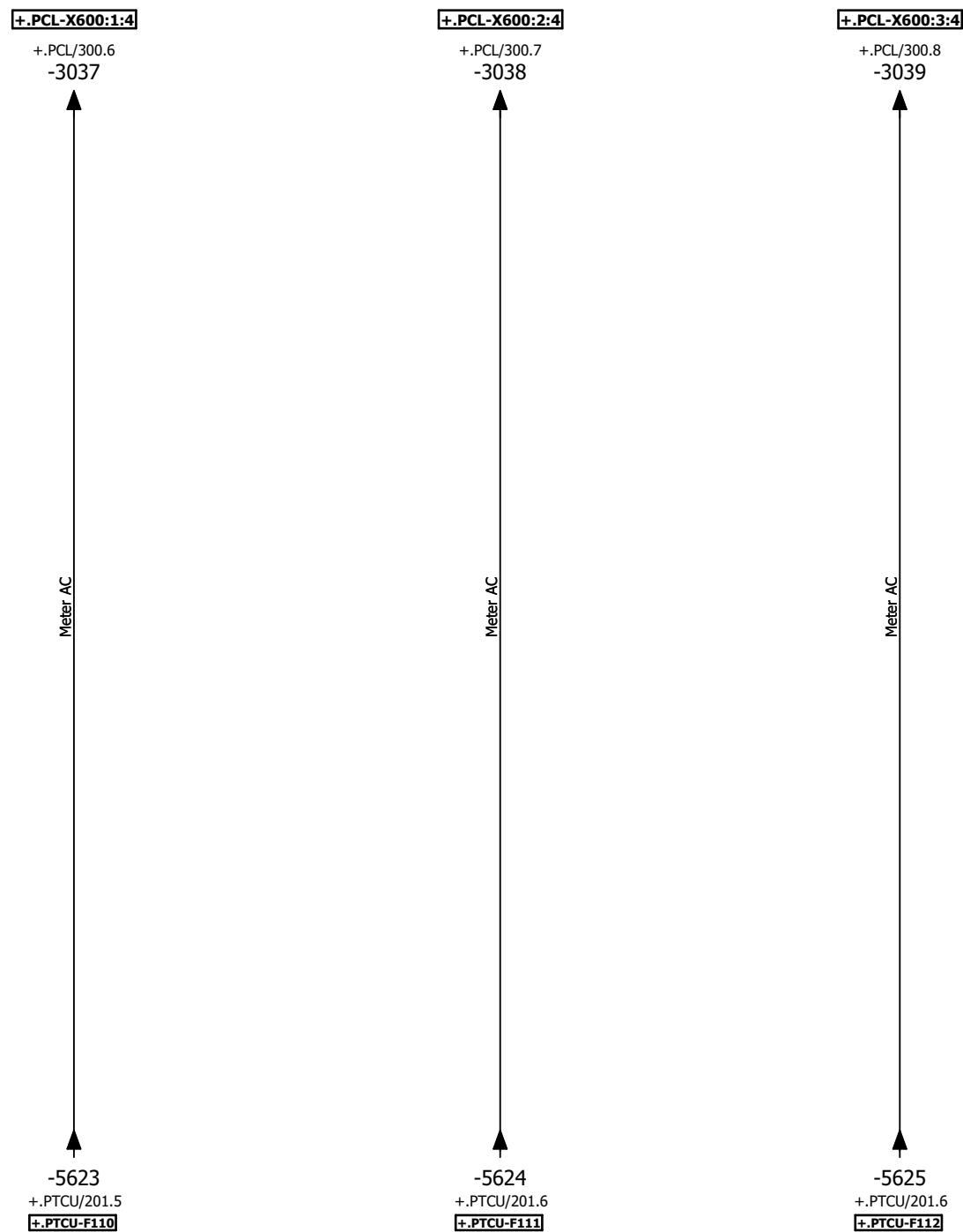
HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
EAC	ENCLOSURE CONTROL CABINET		
EAC ENCLOSURE CONTROL CABINET			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

To central control PCB

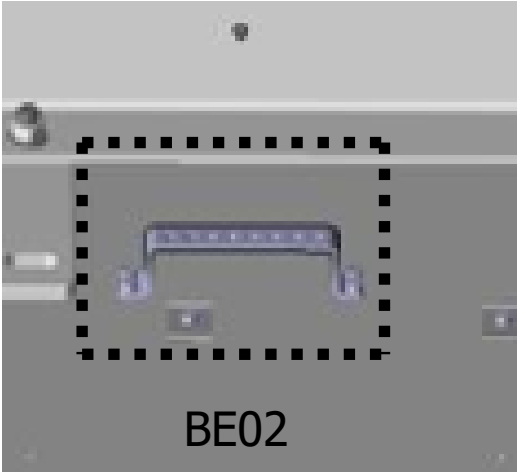




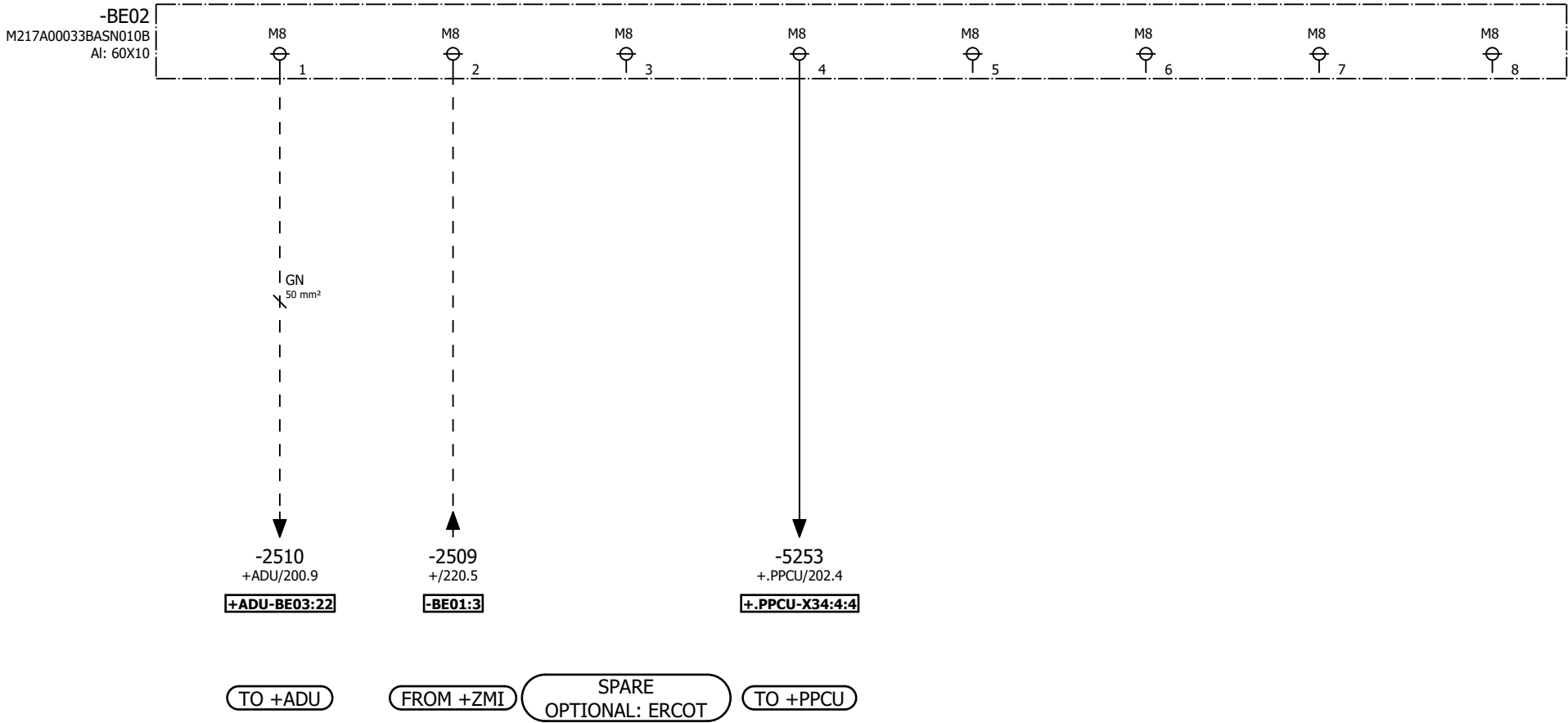
Measure AC

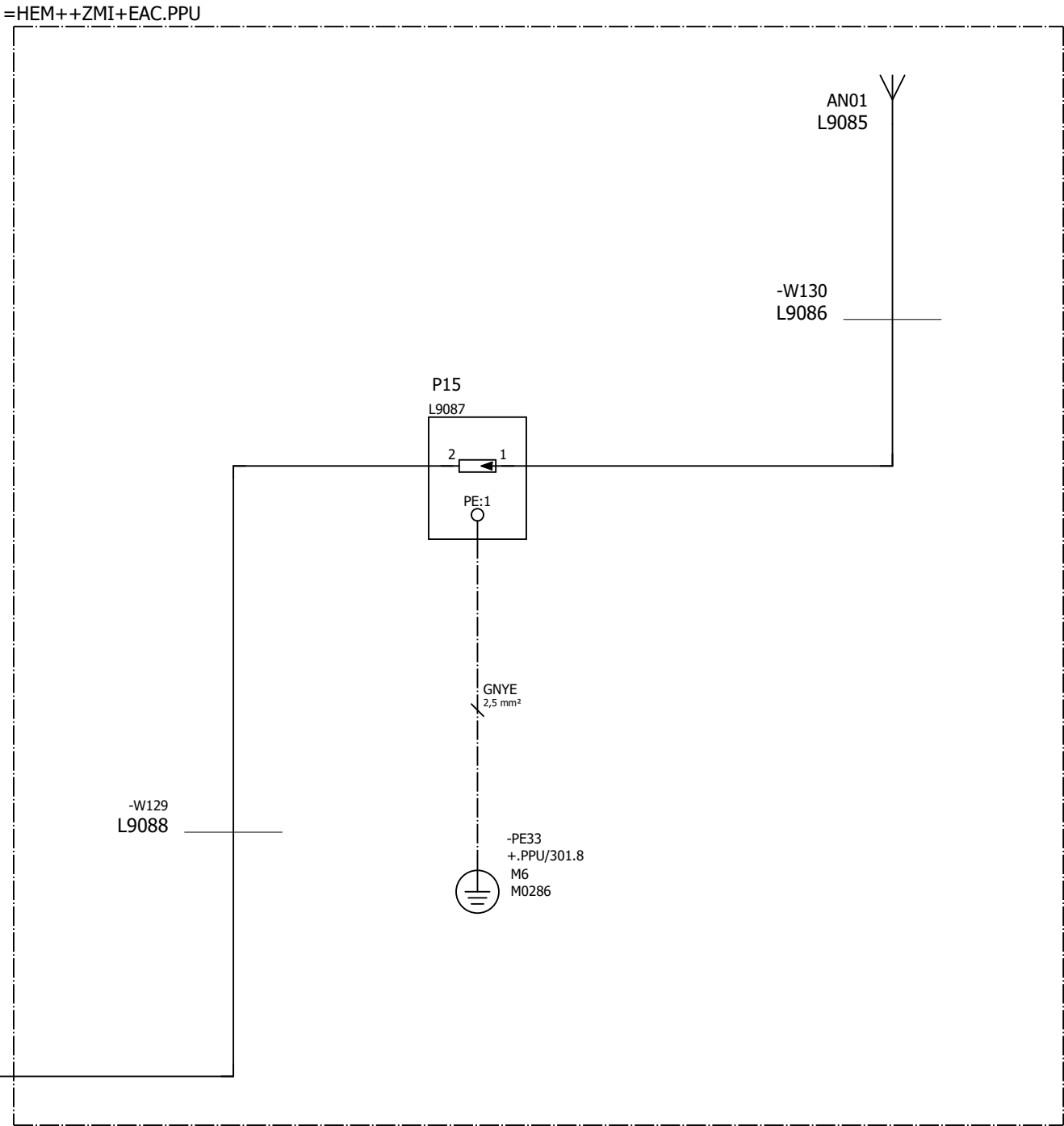
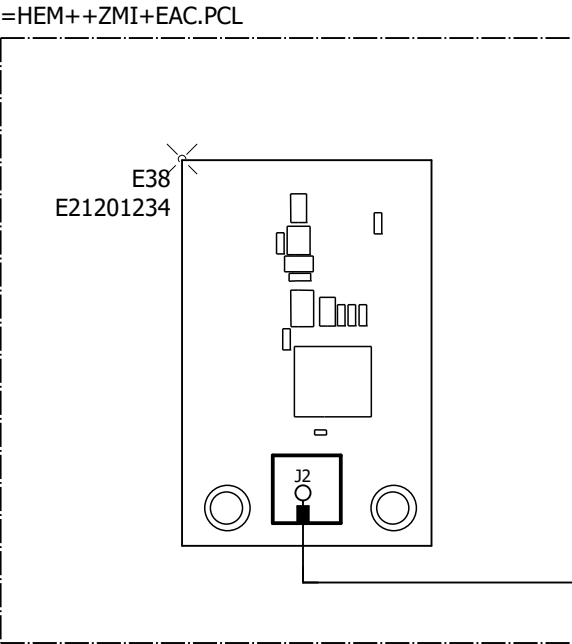
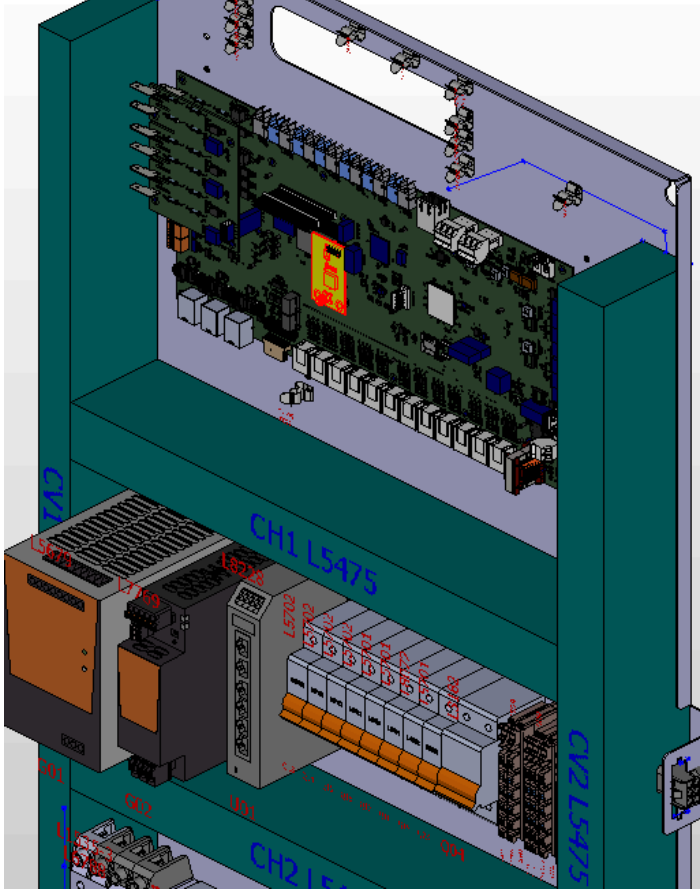


Earth connection



-BE02





BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BE02	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1

Cable overview

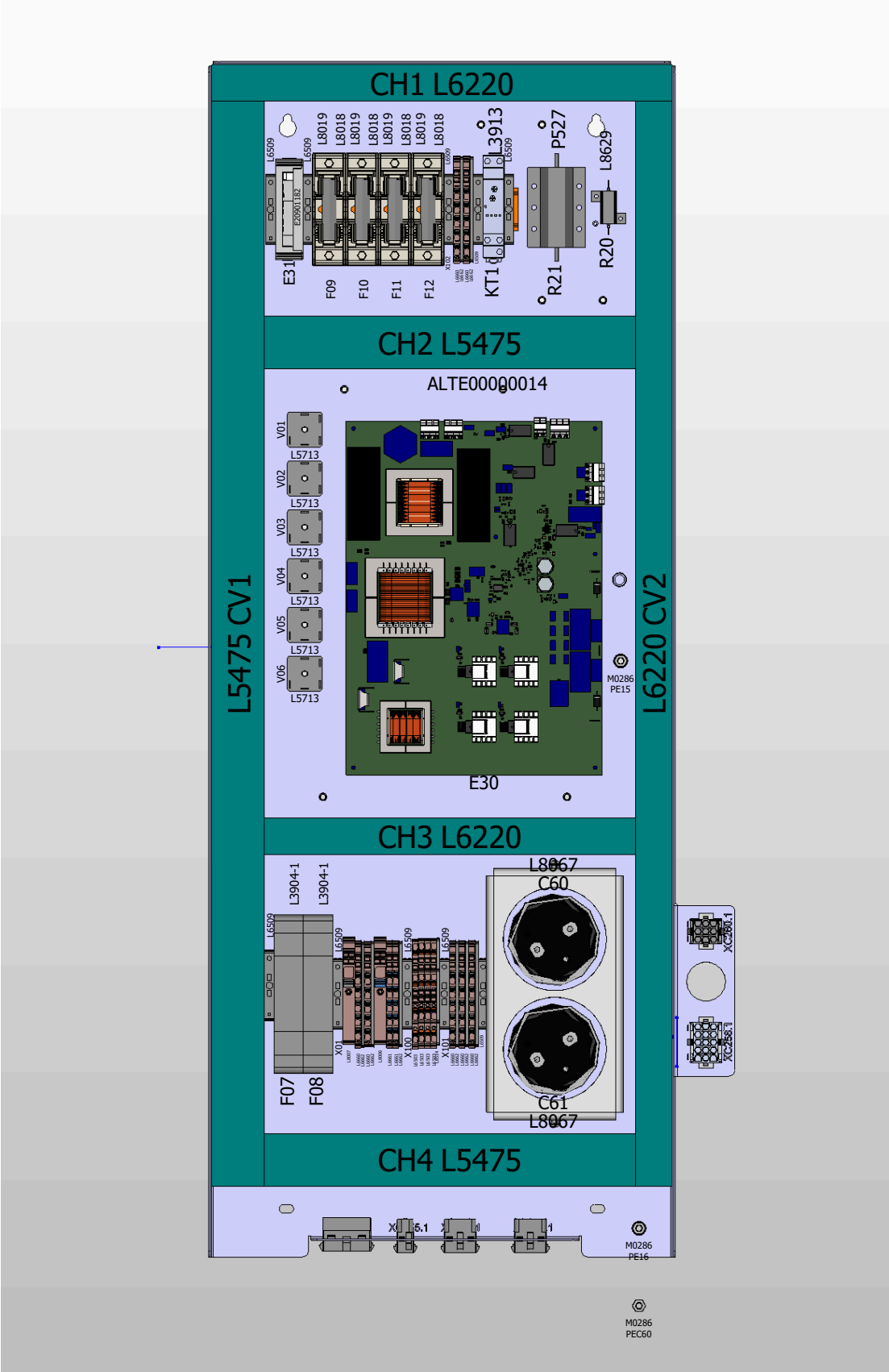
Cable designation	Function text	ERP
=HEM++ZMI+EAC-W123	ETHERNET CABLE CUSTOMER I/O	L8164
=HEM++ZMI+EAC-W124	ETHERNET CABLE DOOR PUSHBUTTONS	L8581
=HEM++ZMI+EAC-W125	ETHERNET CABLE DOOR PUSHBUTTONS	L8582
=HEM++ZMI+EAC-W126	ETHERNET CABLE PSA	L8154
=HEM++ZMI+EAC-W127	ETHERNET CABLE DOOR CONNECTOR	L8587
=HEM++ZMI+EAC-W128	USB CABLE DOOR CONNECTOR	L5815
=HEM++ZMI+EAC-W161	ETHERNET CABLE CUSTOMER PANEL PPCU	L8165



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
EAC.PSA	POWER SUPPLY PANEL		
PSA POWER SUPPLY PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

LAYOUT PANEL

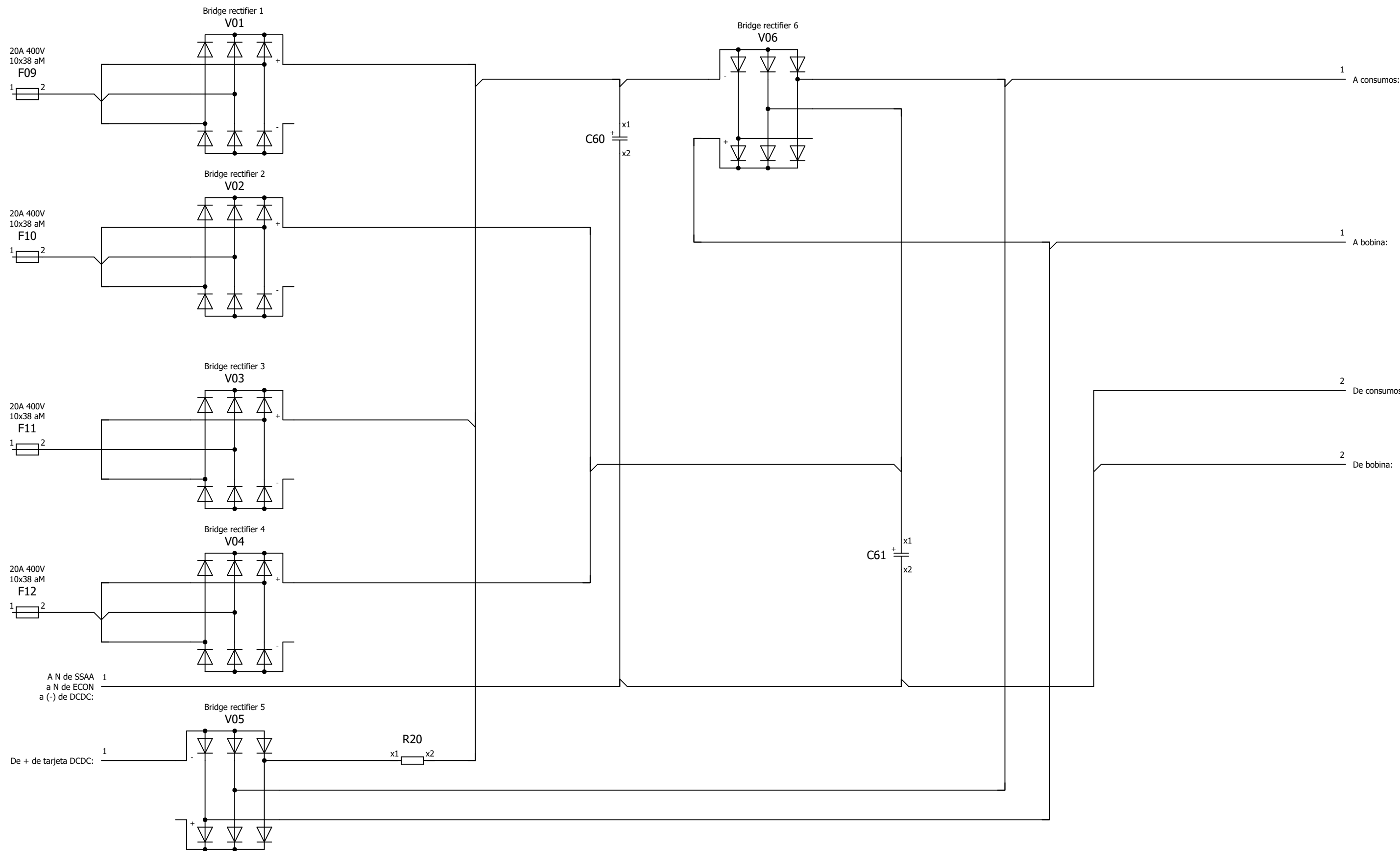


Enclosure legend

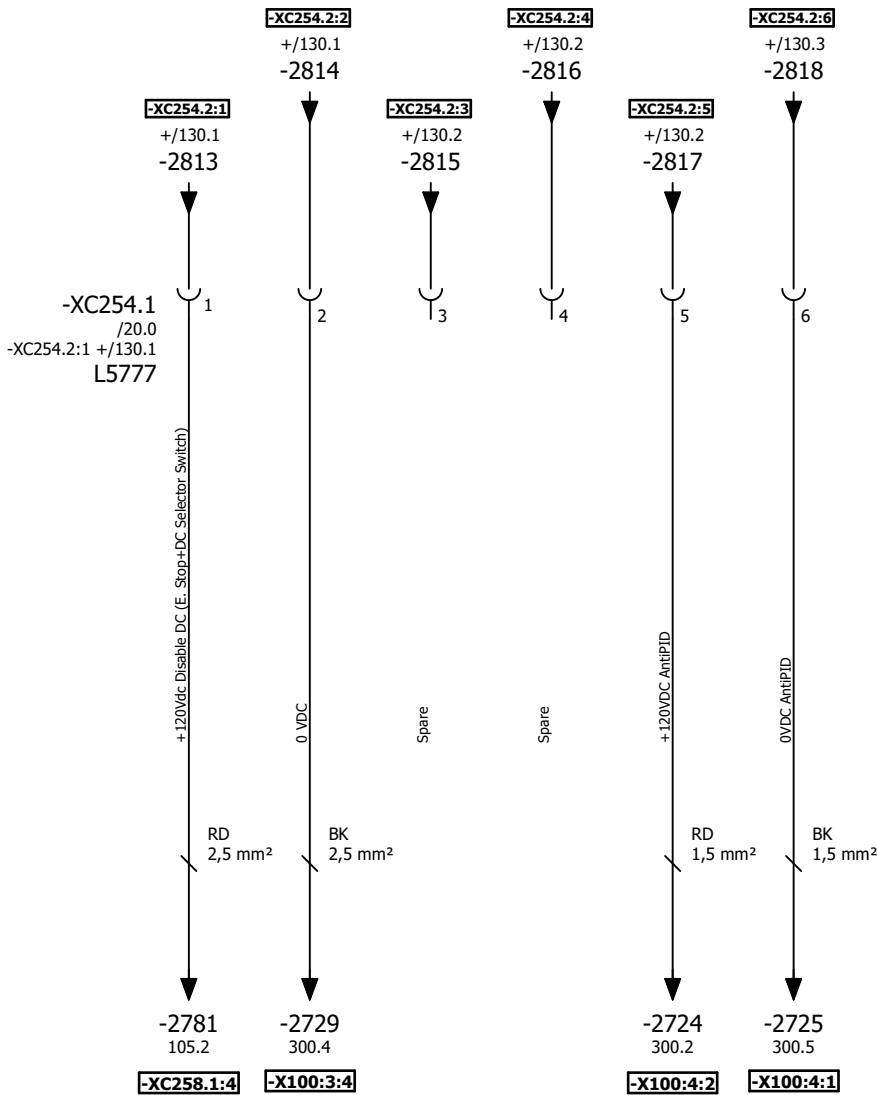
IME	ERP	Description
CH1;CH3;CV2	L6220	UNEX CHANNEL 80X30 IN U23X (60°)
CA1;CA2	L1061	DIN RAIL 35x7,5mm
E31	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
F09-F12	L8019	BASE HOLDER OPENED BAC 10X38
F09-F12	L8018	FUSE 20A 400VAC aM 10X38 120KA
CH2;CH4;CV1	L5475	CABLE DUCT UNEX 80X43 EN U23X GR (60°)
C60;C61	L8067	CAPACITOR 22.000uF/200VDC
V01-V06	L5713	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A
F07;F08	L3904-1	FUSE HOLDER 32A 1500V 1P PML 10x85 DF
R20	L8629	RESISTOR HS-25 1K J M303
F07;F08	L4635	FUSE 4A 1500VDC 10KA 10x85 UL DF
XC282.1	L5778	CONNECTOR PLUG HOUSING 6P MATE-N-LOK
R21	P527	RESISTOR 330R 100W 5%
KT1	L3913	RELAY, WITH PICK-UP DELAY 16A 24-230VAC/VDC 1CO
E30	PCB ALTE00000014	PCB LLC 1600V A 10A 110V

0	1	2	3	4	5	6	7	8	9
-XC253.1 = MATE-N-LOK 12P PANEL 120VDC POWER SUPPLY & LEDS -XC254.1 = MATE-N-LOK 6P PANEL DISABLE DC -XC255.1 = MATE-N-LOK 2P PANEL FEEDBACK THERMAL TRANF. -XC256.1 = MATE-N-LOK 4P PANEL DISABLE AC -XC257.1 = MATE-N-LOK 12P PANEL -XC258.1 = MATE-N-LOK 15P PANEL 120VDC DISTRIBUTION -XC260.1 = MATE-N-LOK 9P PANEL -XC282.1 = AERIAL CONNECTOR - 6P INTERNAL FAULT POWER SUPPLY RELAY -XC362.1 = AERIAL CONNECTOR - 12P External Emergency -XC380.2 = MATE-N-LOK AERIAL CONNECTOR 4P -XC414.1 = FEMALE CONNECTOR TRIPLE-LOCK 120VDC PSA -X01 = Input SSAA/ECON -X100 = 120 Vdc distribution 1 -X101 = 120 Vdc distribution 2 -X102 = Distribution after rectifiers									

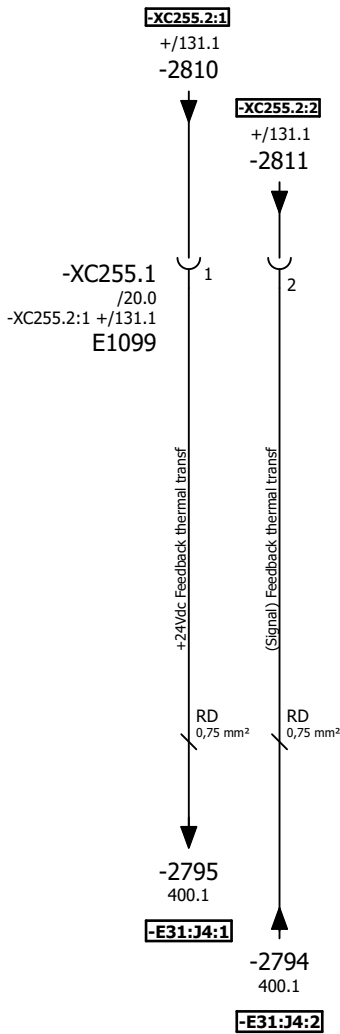
Single Line Diagram



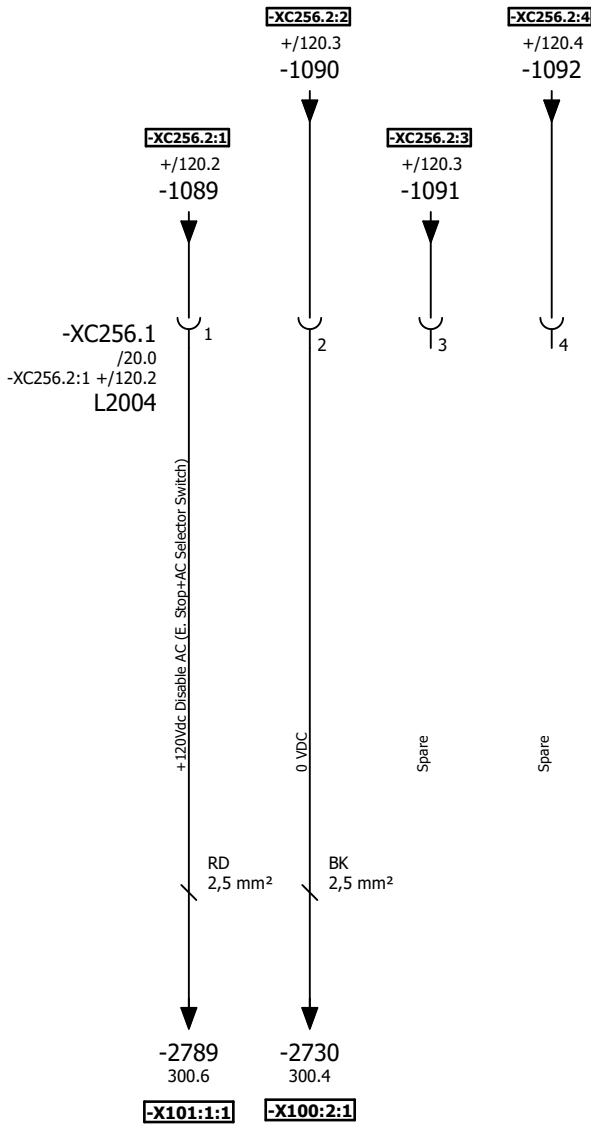
Interconnections - PSA//ADU



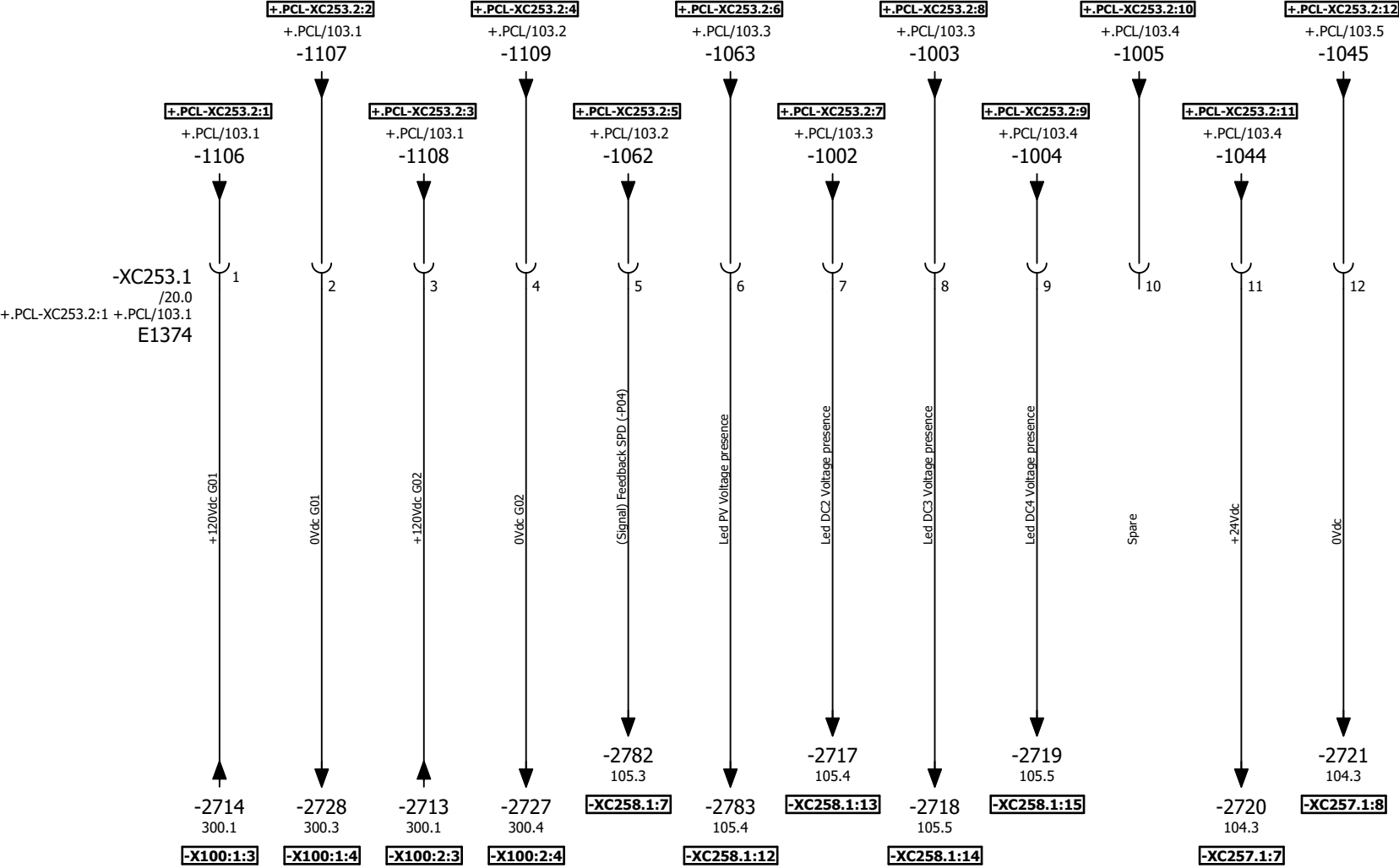
Interconnection SSAA - PSA



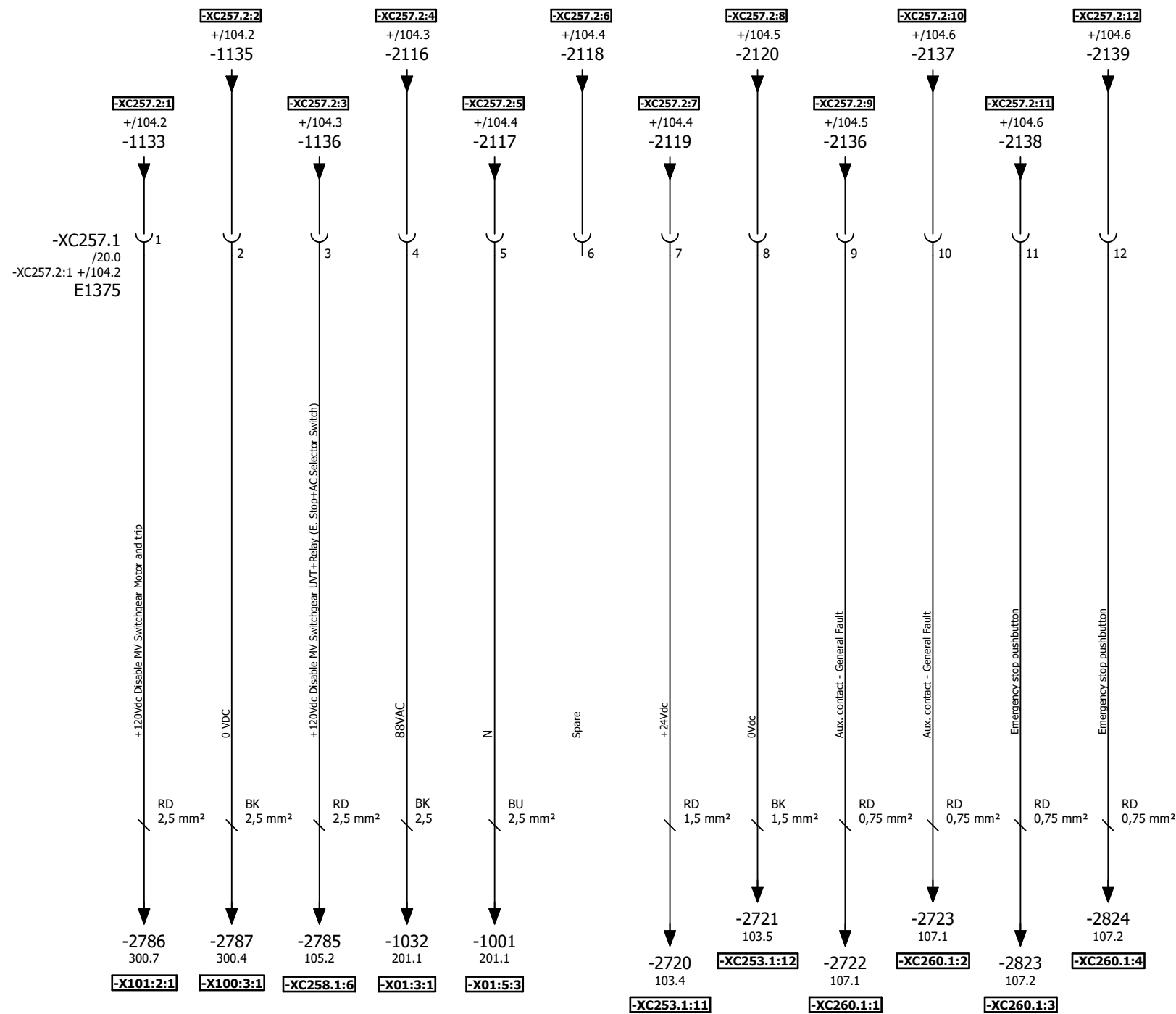
Interconnections - PSA//AFC



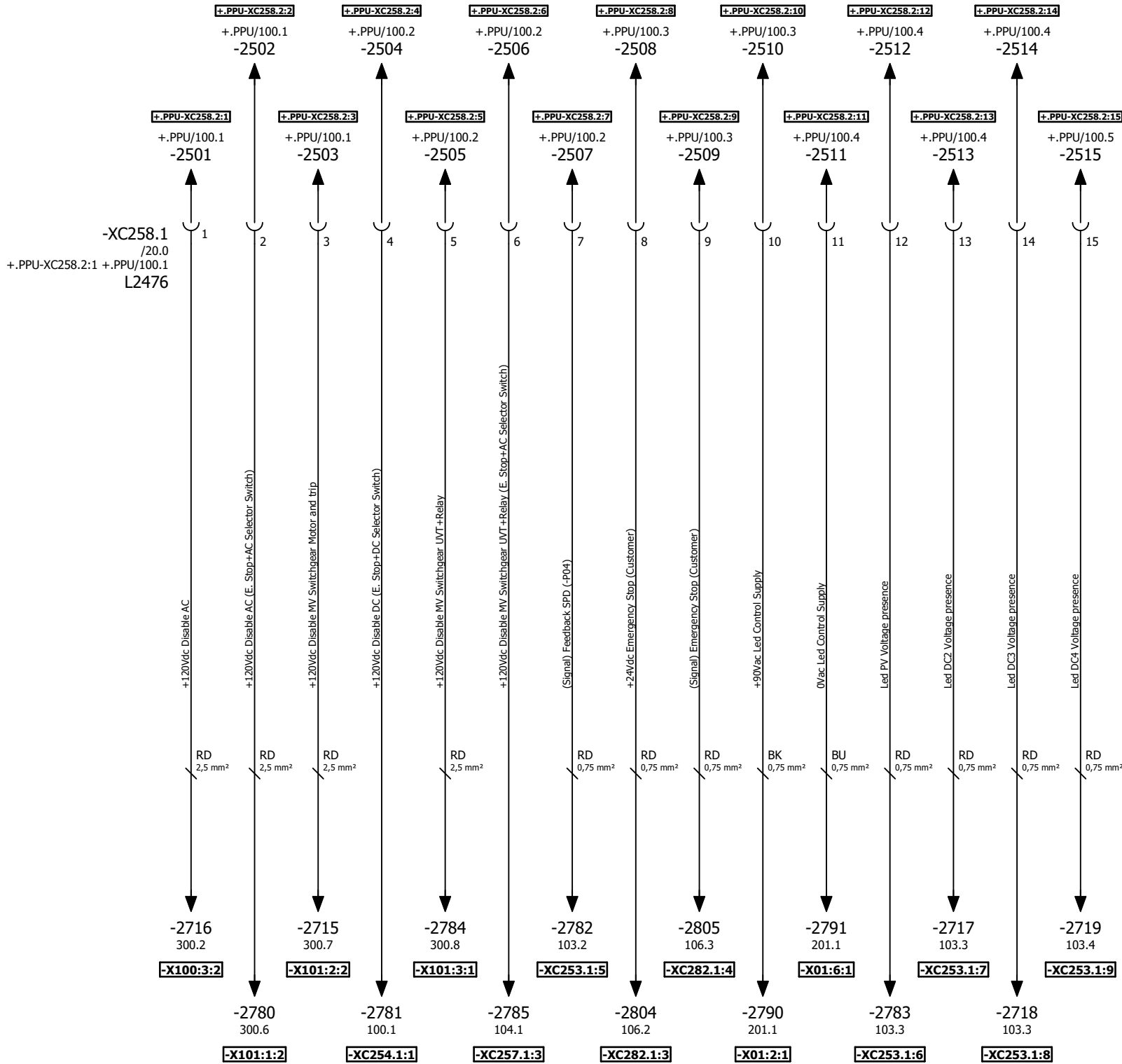
Interconnections PSA//PCL



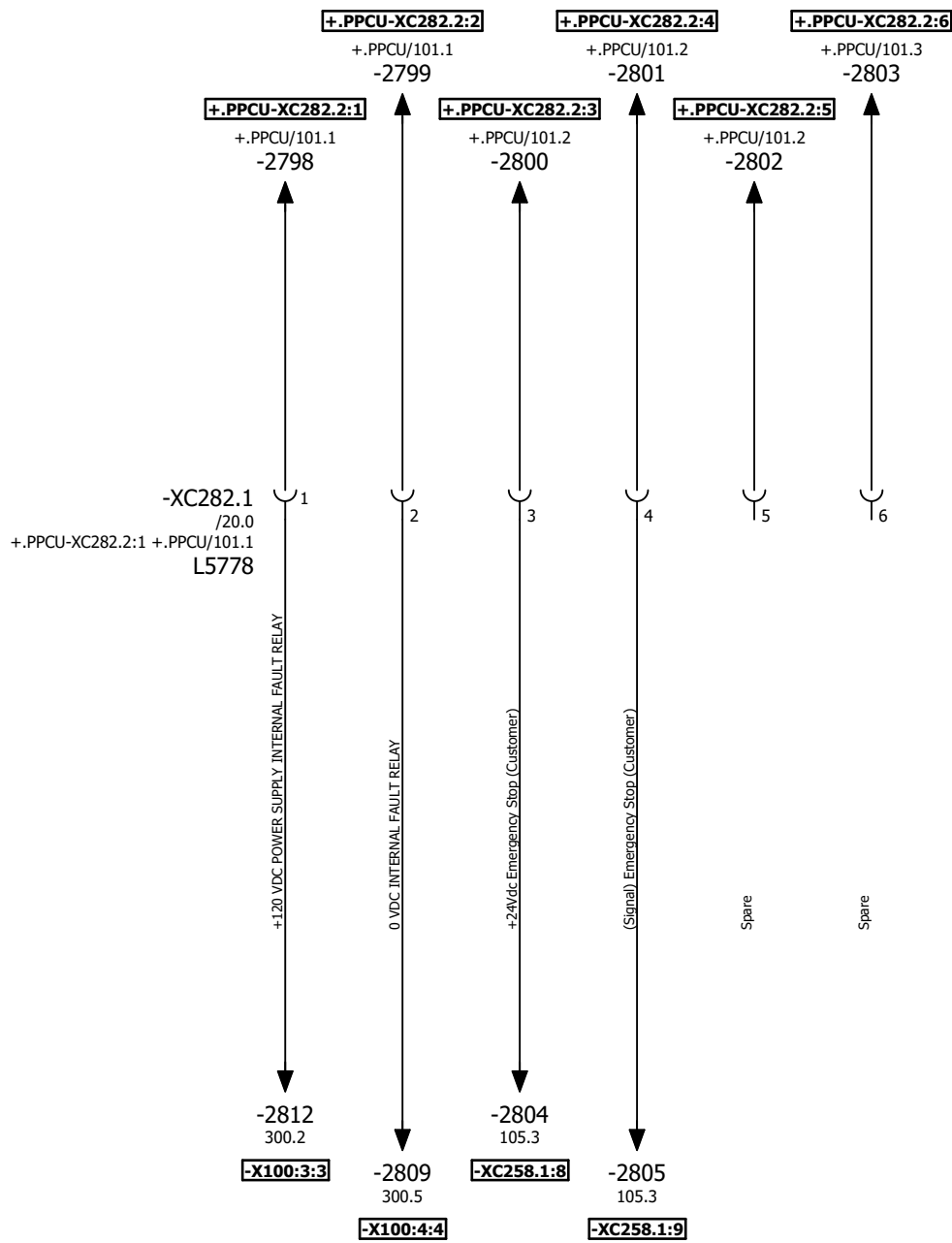
Interconnections PSA//CDT



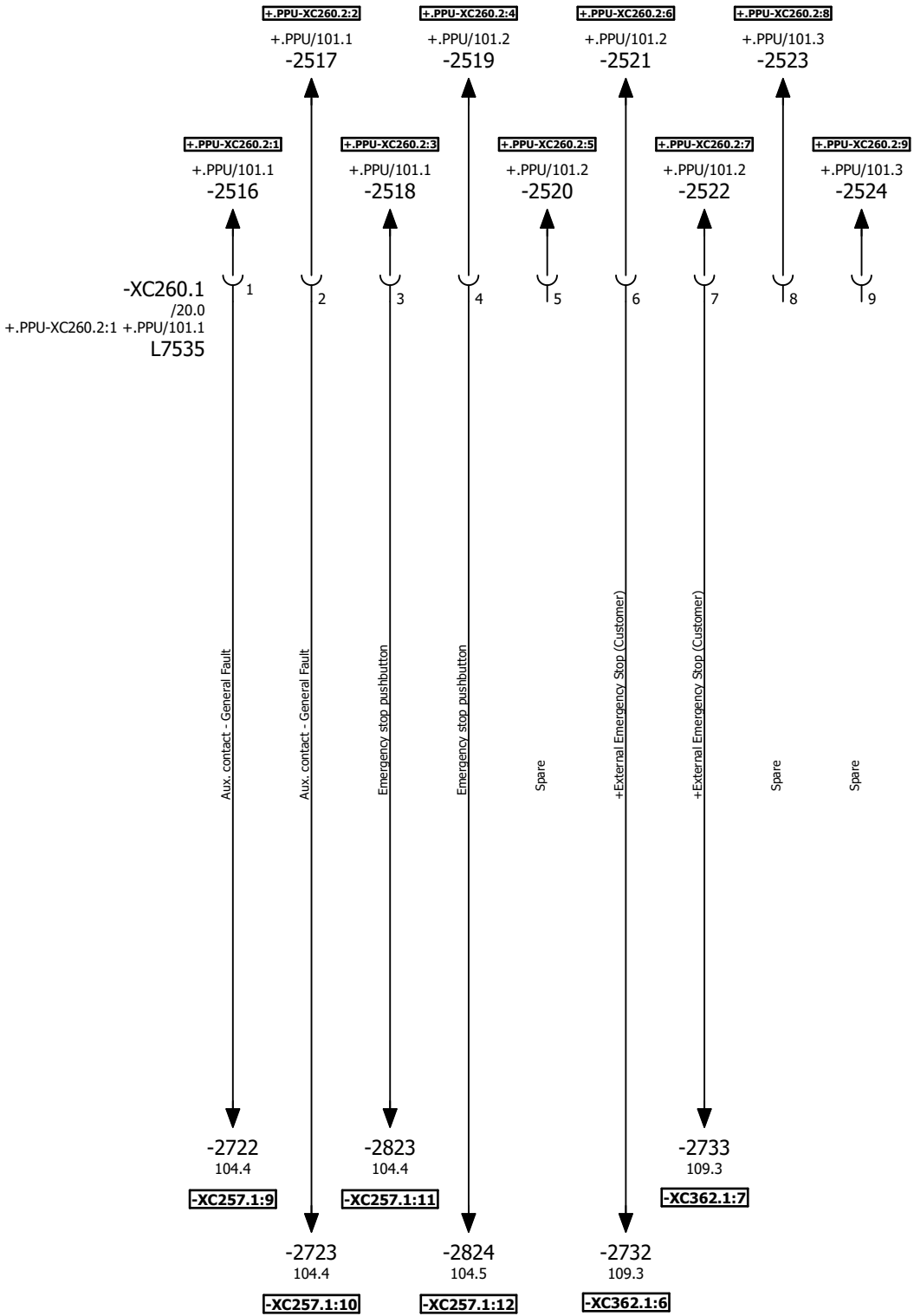
Interconnections - PSA//Door



Interconnections PSA//PPCU

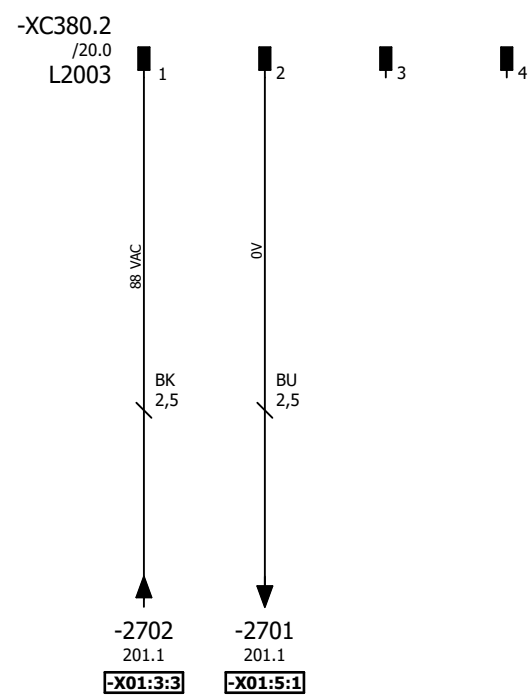


Interconnections - PSA//Door



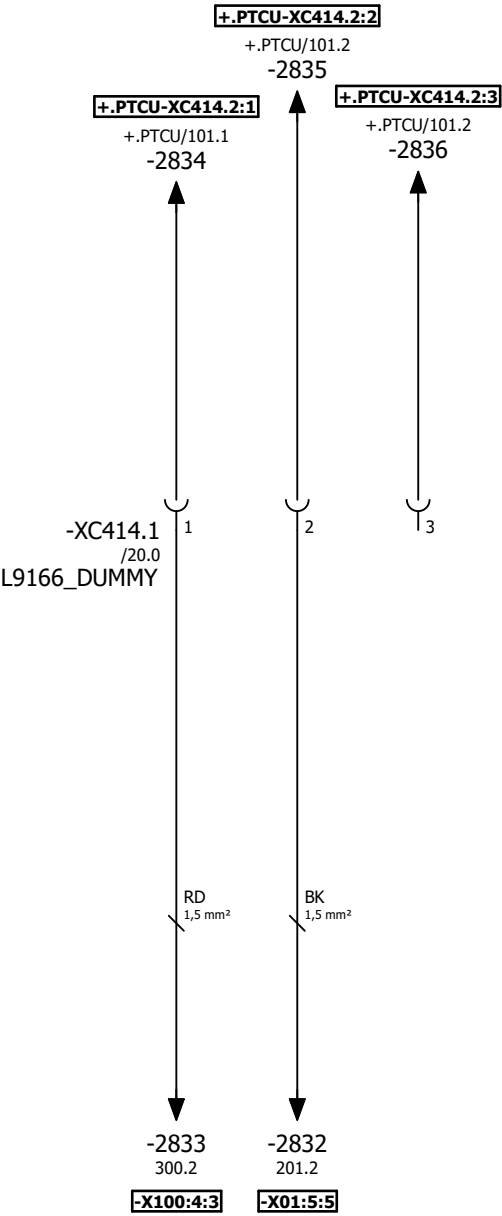
INTERCONNECTIONS PSA/ POCU

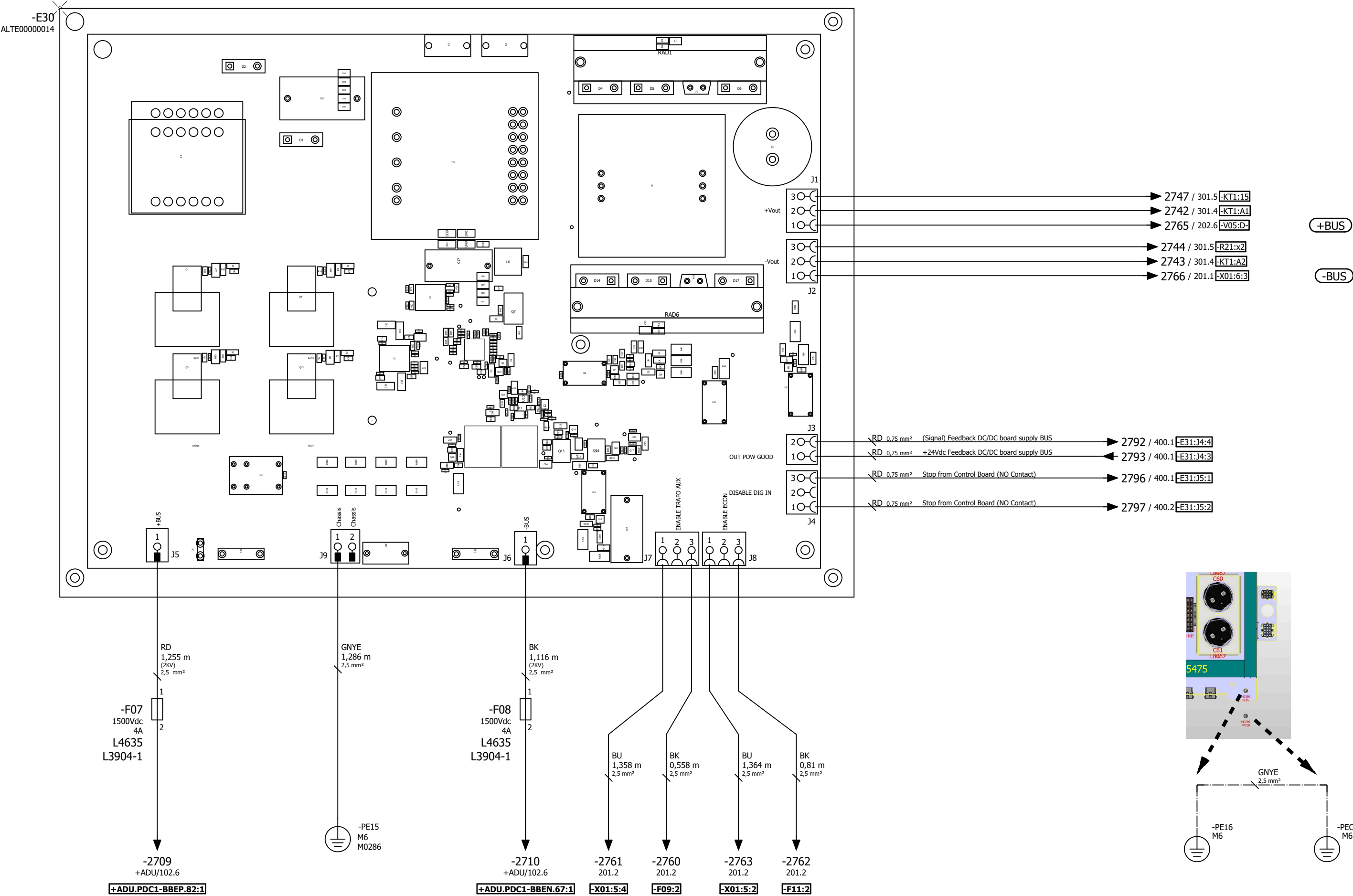
TO POCU

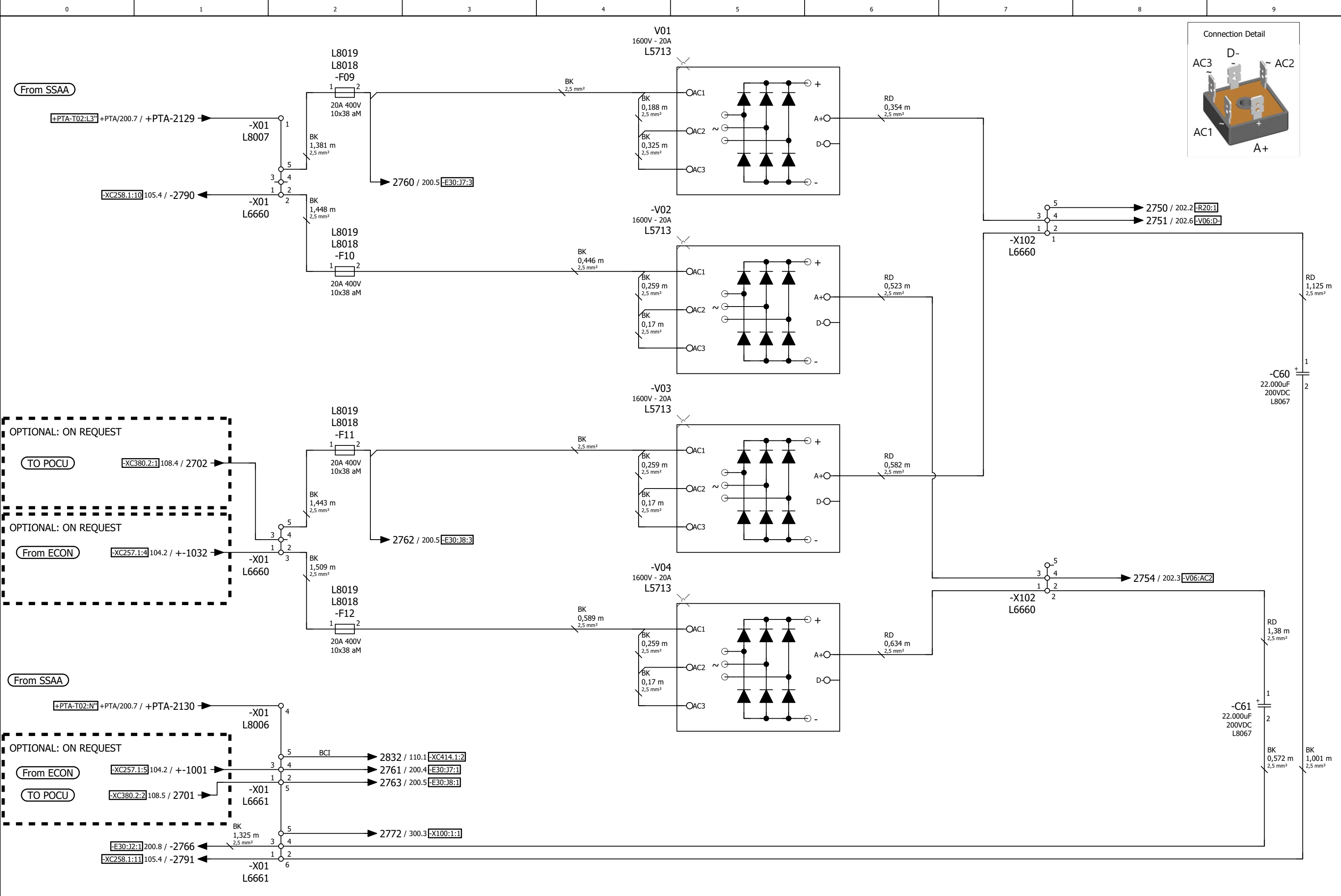


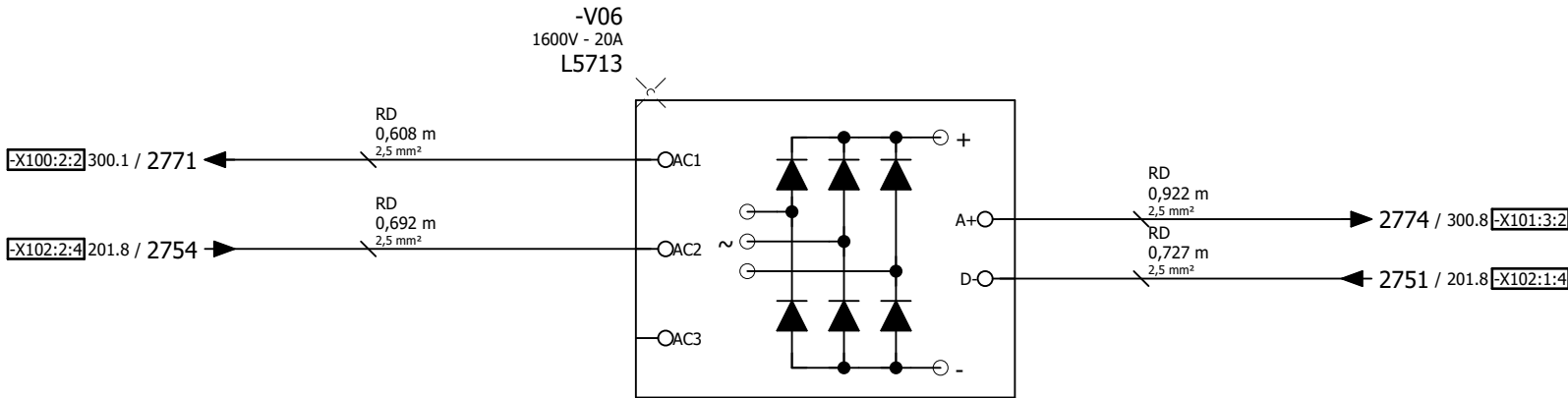
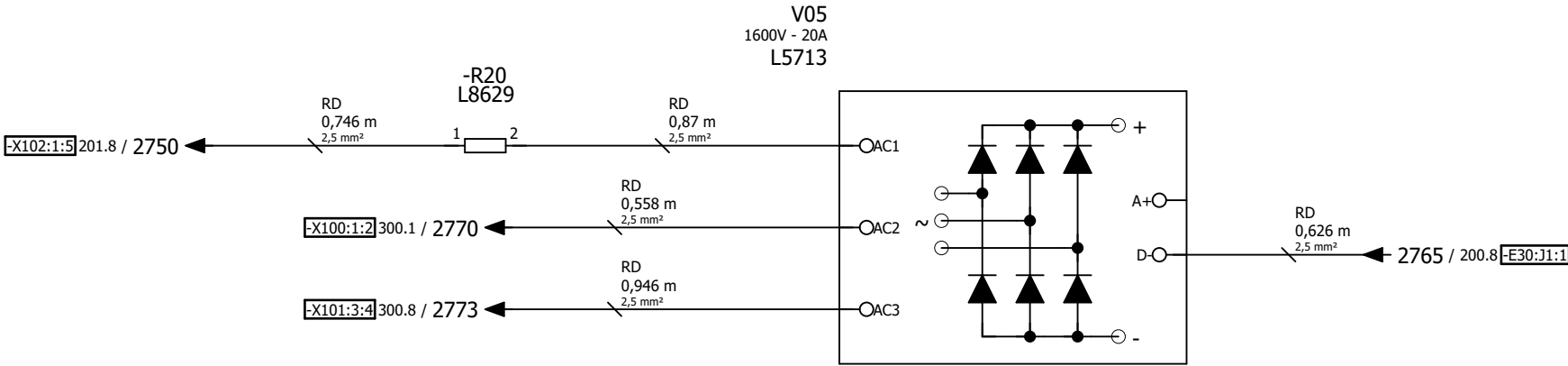
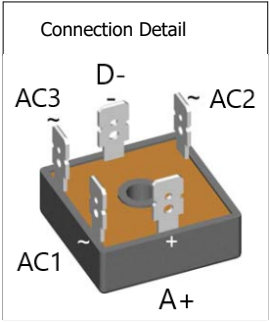
TO PSA

Interconnections PSA//PTCU

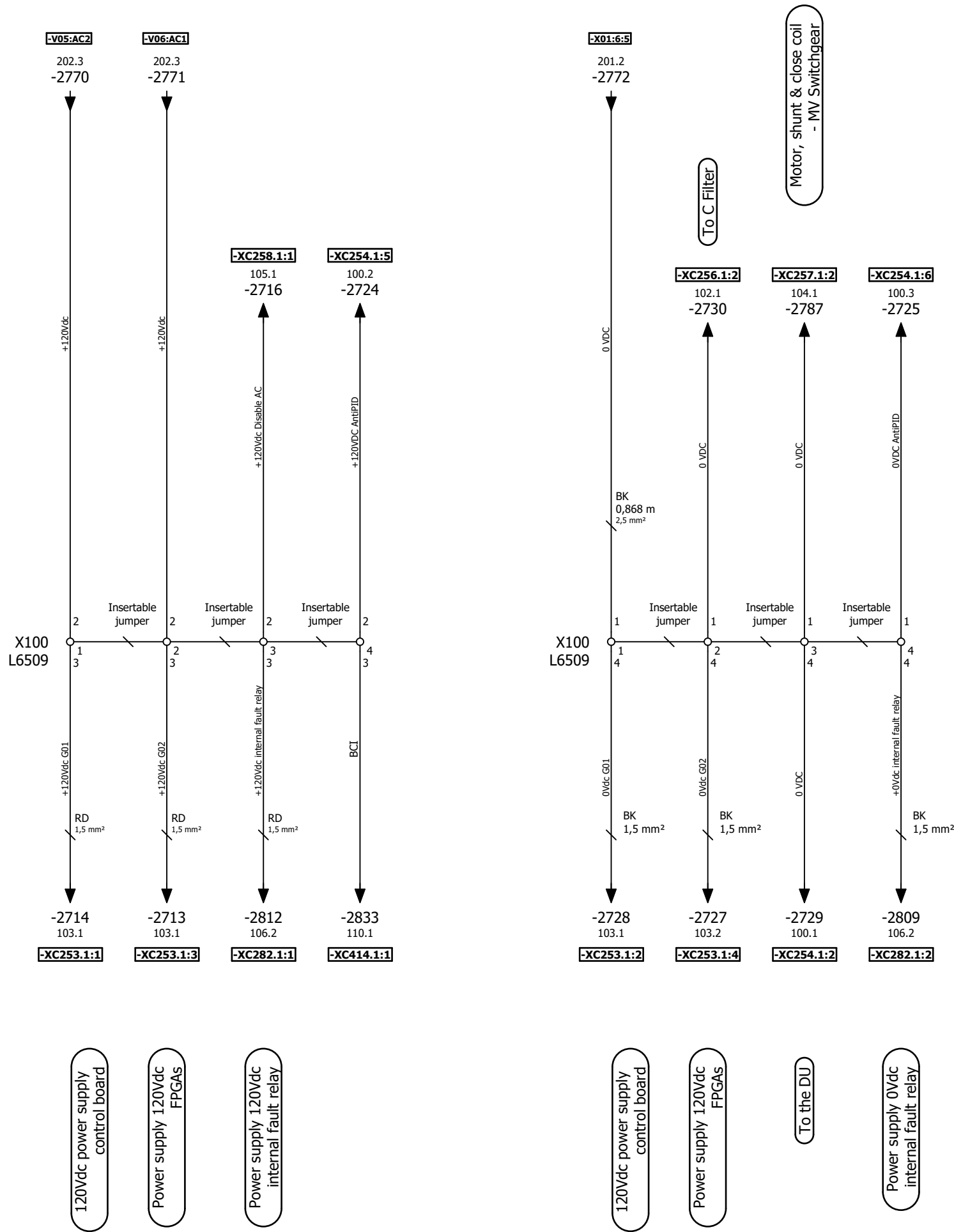




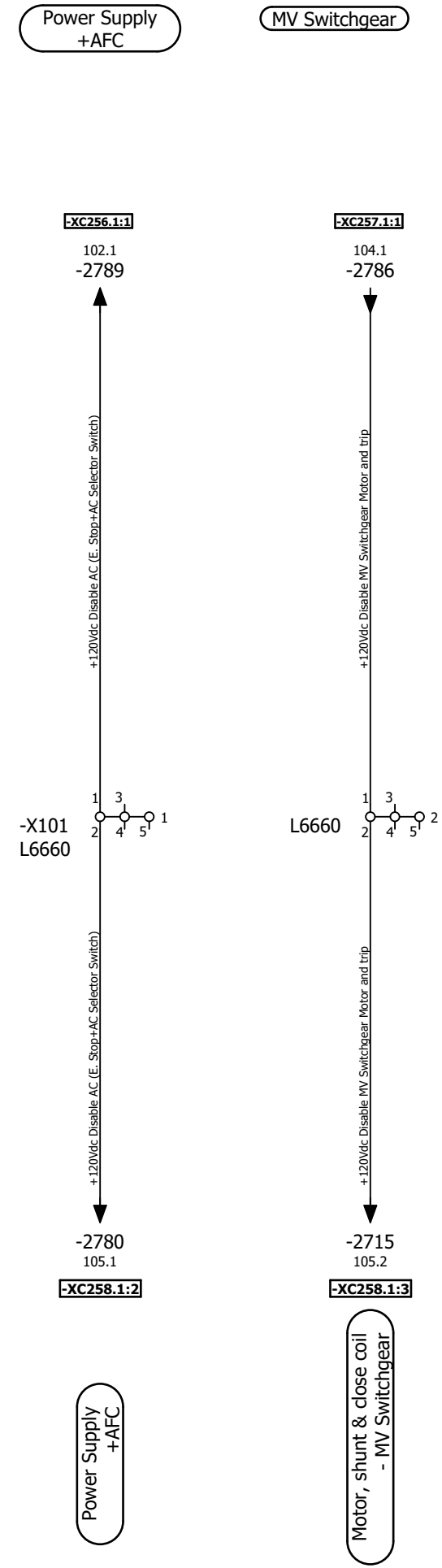




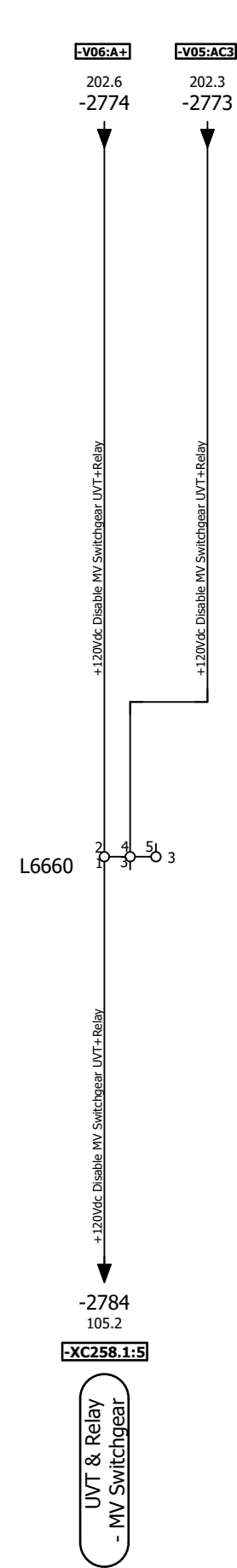
120Vdc Main Bus Distribution

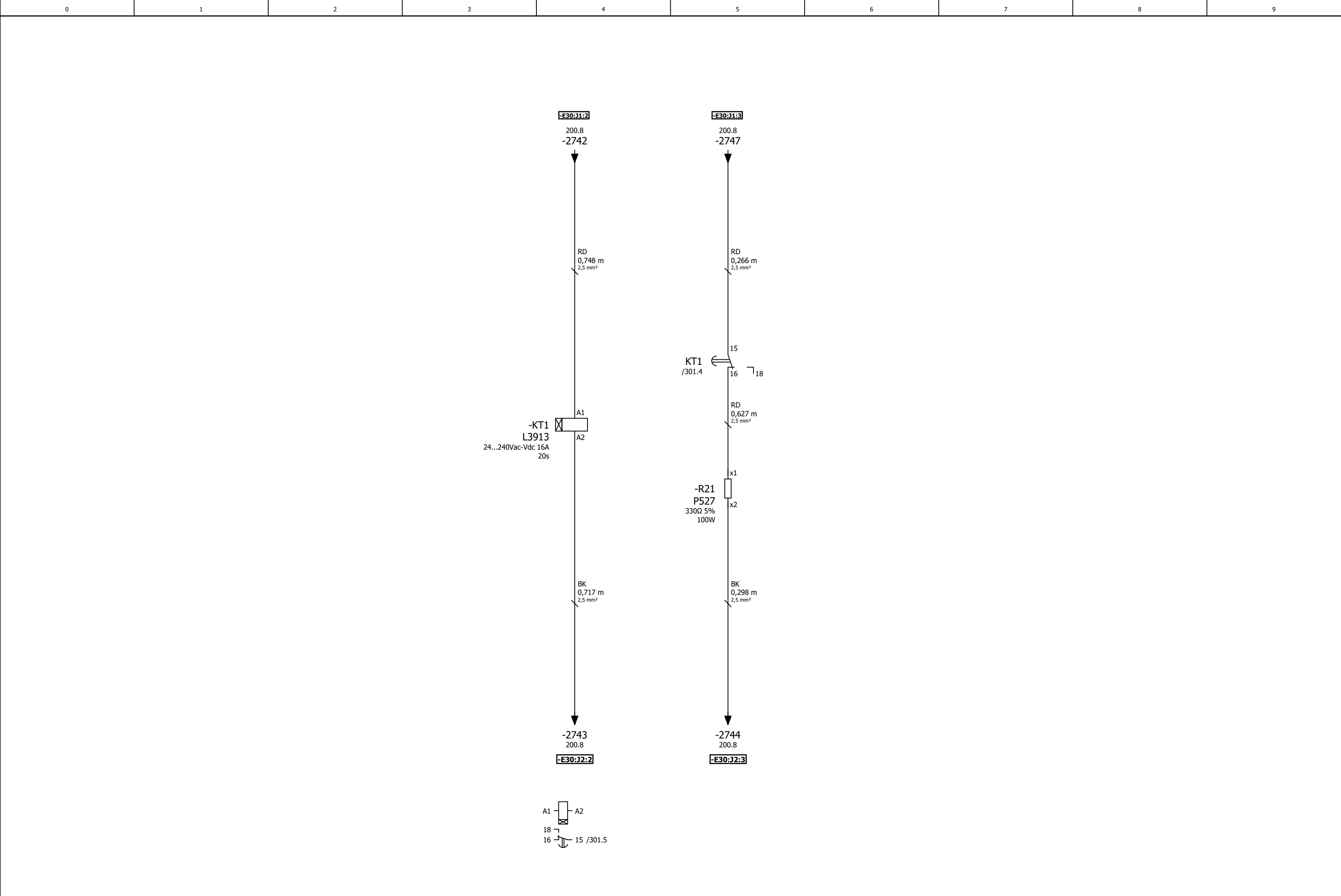


120Vdc Motor/Shunt/Close coil

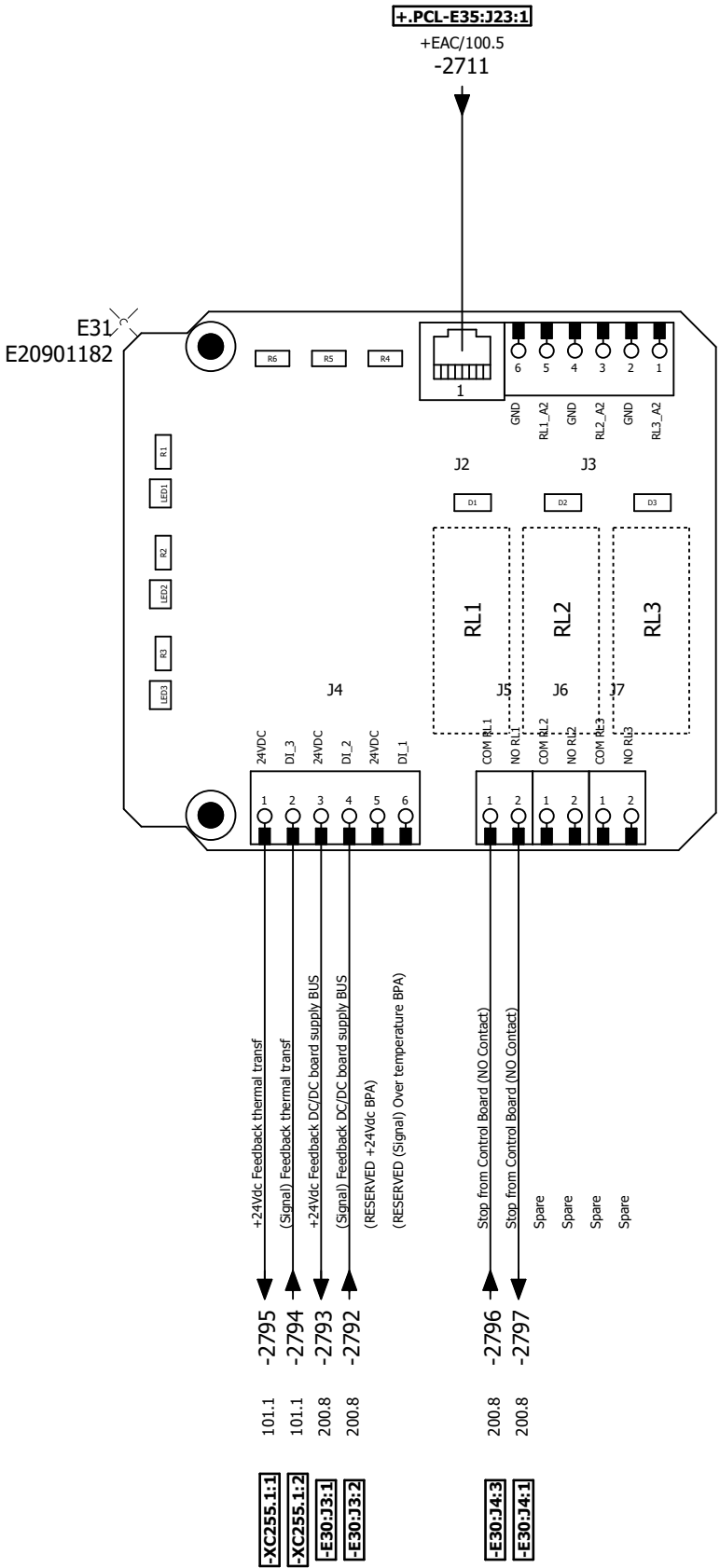


120Vdc UVT MV Switchgear





PCB - DC/DC (PSA)



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-C60	L8067	ALS70A1005NP	CAPACITOR 22.000uF/200VDC	1
-C61	L8067	ALS70A1005NP	CAPACITOR 22.000uF/200VDC	1
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E30	PCB ALTE00000014	PCB ALTE000000014	PCB LLC 1600V A 10A 110V	1
-E31	E20901182AA	E20901182AA	DIGITAL INPUTS/OUTPUTS BOARD	1
-E31	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E31	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-F07	L4635	492205	FUSE 4A 1500VDC 10KA 10x85 UL DF	1
-F07	L3904-1	485701	FUSE HOLDER 32A 1500V 1P PML 10x85 DF	1
-F07	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-F08	L4635	492205	FUSE 4A 1500VDC 10KA 10x85 UL DF	1
-F08	L3904-1	485701	FUSE HOLDER 32A 1500V 1P PML 10x85 DF	1
-F09	L8018	440020	FUSE 20A 400VAC aM 10X38 120KA	1
-F09	L8019	451250	BASE HOLDER OPENED BAC 10X38	1
-F09	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-F10	L8018	440020	FUSE 20A 400VAC aM 10X38 120KA	1
-F10	L8019	451250	BASE HOLDER OPENED BAC 10X38	1
-F11	L8018	440020	FUSE 20A 400VAC aM 10X38 120KA	1
-F11	L8019	451250	BASE HOLDER OPENED BAC 10X38	1
-F12	L8018	440020	FUSE 20A 400VAC aM 10X38 120KA	1
-F12	L8019	451250	BASE HOLDER OPENED BAC 10X38	1
-KT1	L3913	80.11.0.240.0000	RELAY, WITH PICK-UP DELAY 16A 24-230VAC/VDC 1CO	1
-KT1	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-R20	L4264	HS100 10K J	RESISTOR 100W 10K	1
-R20	L8629	HS-25 1K J M303	RESISTOR HS-25 1K J M303	1
-R21	P527	HS100 330R J	RESISTOR 330R 100W 5%	1
-V01	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	1
-V02	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	1
-V03	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	1
-V04	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	1
-V05	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	1
-V06	L5713	VS-36MT-160	RECTIFIER BRIDGE 3pH PUENTE 1600V 35A	1

Plug overview

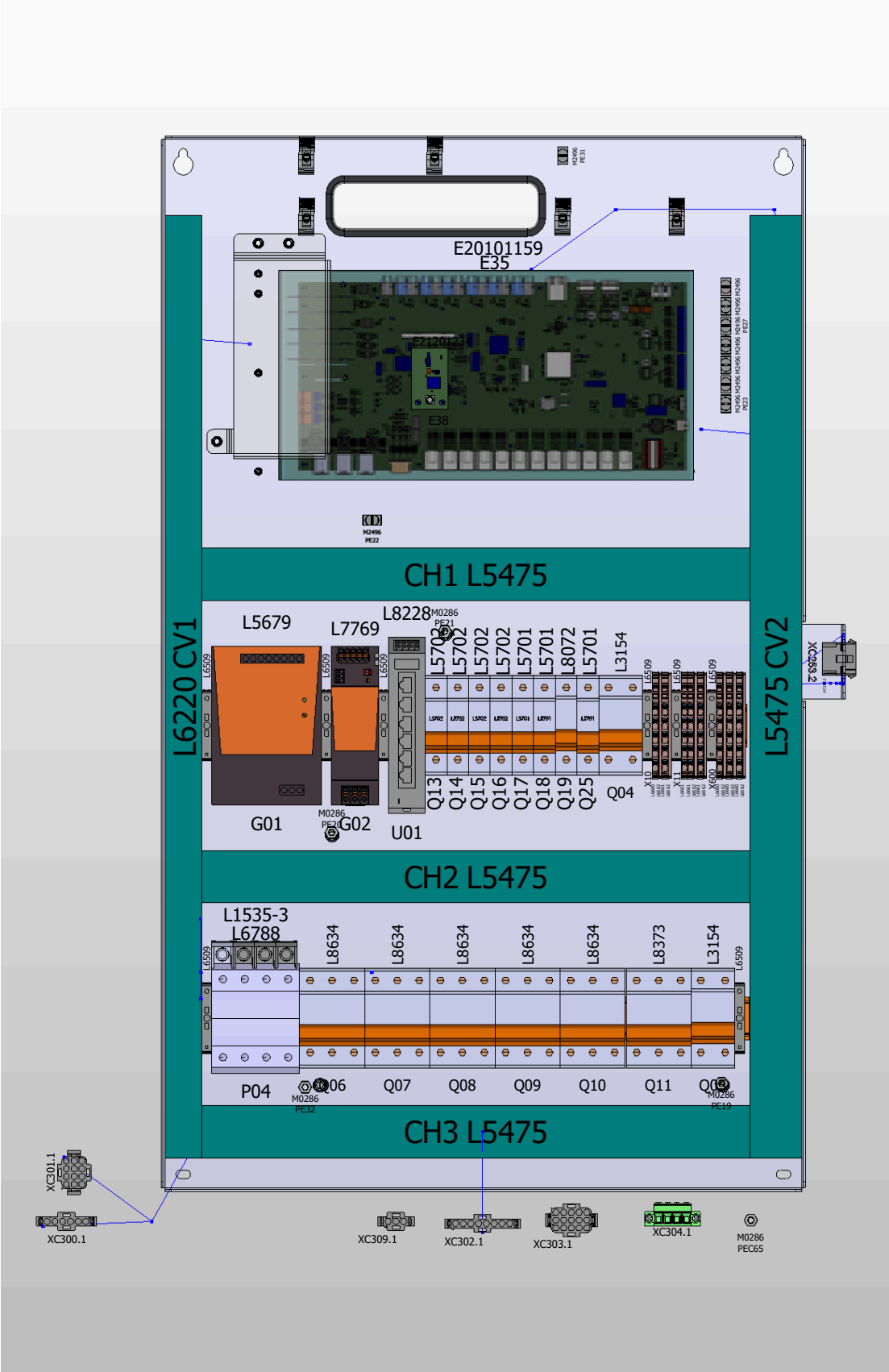
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+EAC.PSA-XC253.1	MATE-N-LOK 12P PANEL 120VDC POWER SUPPLY & LEDS	E1374	1
		E1203	9
=HEM++ZMI+EAC.PSA-XC254.1	MATE-N-LOK 6P PANEL DISABLE DC	L5777	1
		E1203	2
=HEM++ZMI+EAC.PSA-XC255.1	MATE-N-LOK 2P PANEL FEEDBACK THERMAL TRANF.	E1099	1
		E1203	2
=HEM++ZMI+EAC.PSA-XC256.1	MATE-N-LOK 4P PANEL DISABLE AC	L2004	1
		L5709	2
=HEM++ZMI+EAC.PSA-XC257.1	MATE-N-LOK 12P PANEL	E1375	1
		L5709	5
		E1203	4
=HEM++ZMI+EAC.PSA-XC258.1	MATE-N-LOK 15P PANEL 120VDC DISTRIBUTION	L2476	1
		E1203	6
		L5709	6
=HEM++ZMI+EAC.PSA-XC260.1	MATE-N-LOK 9P PANEL	L7535	1
		E1203	2
=HEM++ZMI+EAC.PSA-XC282.1	AERIAL CONNECTOR - 6P INTERNAL FAULT POWER SUPPLY RELAY	L5778	1
		E1203	4
=HEM++ZMI+EAC.PSA-XC362.1	AERIAL CONNECTOR - 12P External Emergency	L9134_DUMMY	1
		L9145	6
=HEM++ZMI+EAC.PSA-XC380.2	MATE-N-LOK AERIAL CONNECTOR 4P	L2003	1
		E1204	4
=HEM++ZMI+EAC.PSA-XC414.1	FEMALE CONNECTOR TRIPLE-LOCK 120VDC PSA	L9166_DUMMY	1
		L9145	2



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
EAC.PCL	CONTROL PANEL		
PCL CONTROL PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

LAYOUT PANEL



Enclosure legend

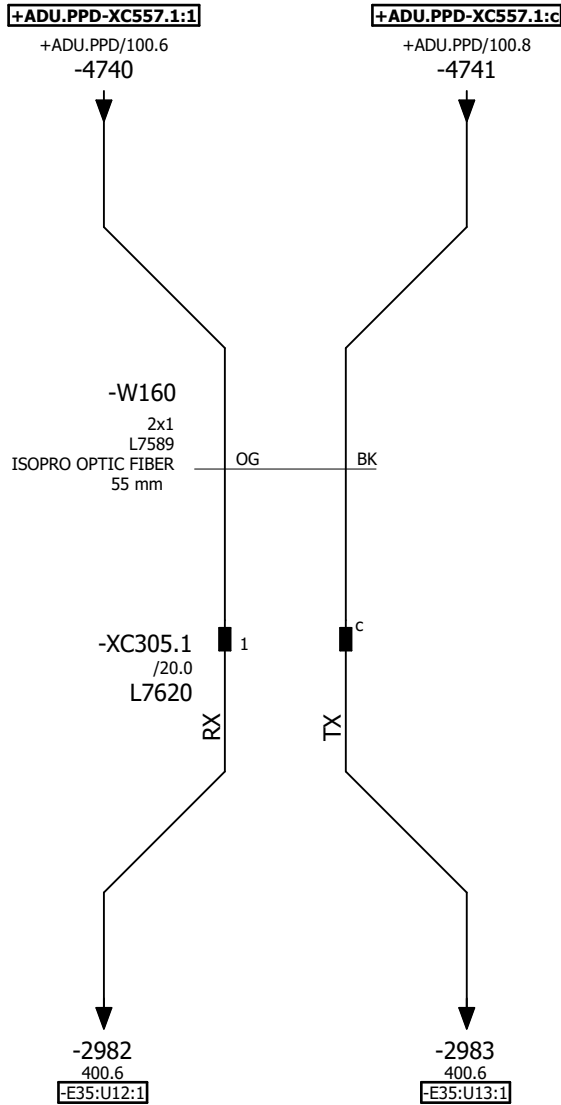
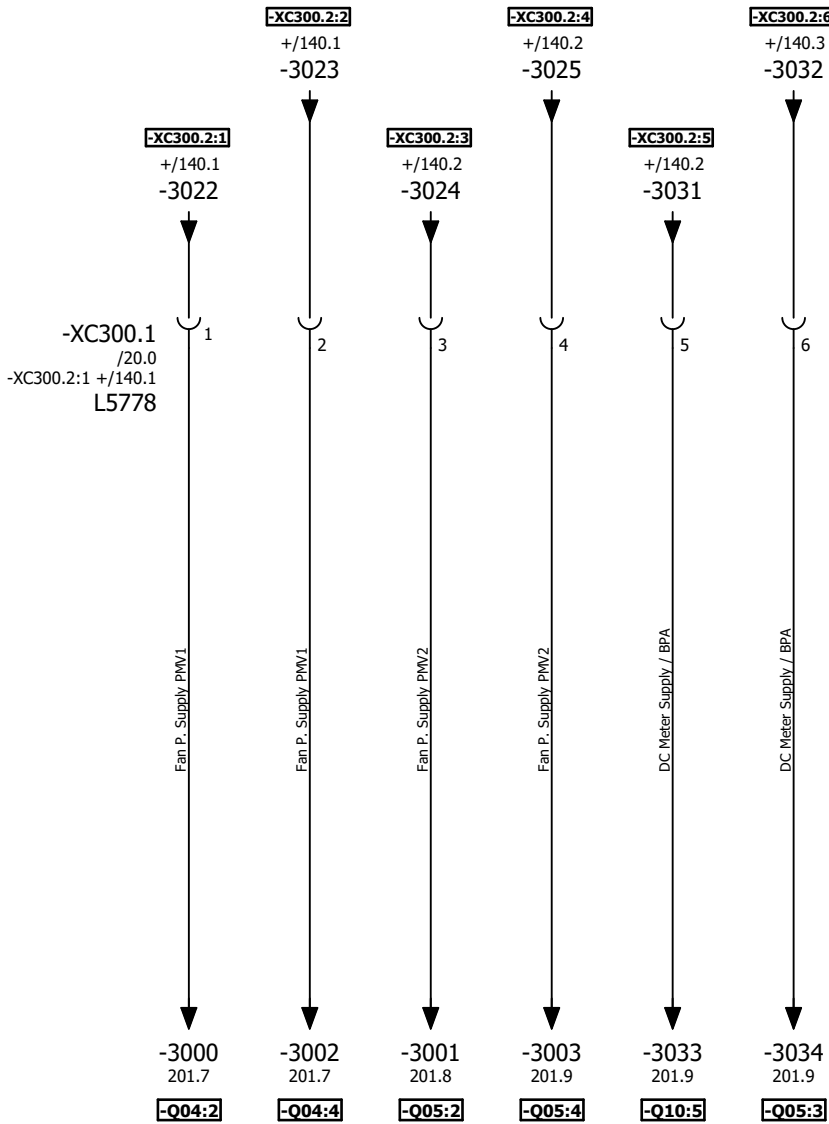
IME	ERP	Description
CV1	L6220	UNEX CHANNEL 80X30 IN U23X (60°)
CH1-CH3;CV2	L5475	CABLE DUCT UNEX 80X43 EN U23X GR (60°)
CA1;CA2	L1061	DIN RAIL 35x7,5mm
G02	L7769	Power supply
G01	L5679	POWER SUPPLY PRO MAX 480W 24V 20A
U01	L8228	SWITCH ETHERNET 10/1000 6 P 70°C
PE22-PE29;PE31	M2496	SCREWED CLAMP MSKL 3-12
Q11	L8373	BK63H 3P C32A 6kA 480Y/277VAC UL1077
Q13-Q16	L5702	CIRCUIT BRAKER 1Ph 4A CURVE "C"
Q17;Q18;Q25	L5701	BKN-b 1P C1A 10KA 240/415VAC
Q19	L8072	BKN-B 1P, B2A, 10KA
P04	L1535-3	Surge arrester DGMU3PY 480 4WGR DEH
P04	L6788	ACCESORY TERMINAL SURGE ARRESTER DEHN
Q06-Q10	L8634	BK63H 3P C10A 6kA 480Y/277VAC UL1077
E35	PCB E20101159	HEM-PLUS CENTRAL CONTROL BOARD
Q04;Q05	L3154	MCB BKN-B 2P C6A (10KA) 415VAC
E38	PCB E21201234	TARJETA CAISO

0	1	2	3	4	5	6	7	8	9
-XC253.2 = MATE-N-LOK 12P PANEL 120VDC POWER SUPPLY 6 LEDS -XC300.1 = MATE-N-LOK AERIAL CONNECTOR FAN SUPPLY -XC301.1 = MATE-N-LOK 12P AERIAL CONNECTOR POWER SUPPLY PPD -XC302.1 = MATE-N-LOK 8P AERIAL CONNECTOR POWER SUPPLY MODULES -XC303.1 = MATE-N-LOK 15P AERIAL CONNECTOR FAN MODULES -XC304.1 = MATE-N-LOK PANEL 6mm2 POWER SUPPLY FANS SKID -XC305.1 = CONNECTOR OPTICAL FIBER DUPLEX ISOPRO -XC307.1 = NTC CONNECTOR - FAN ADMISSION TEMP. -XC308.1 = NTC CONNECTOR - C FILTER TEMP. -XC309.1 = CONNECTOR MATE-N-LOK 3P AC MEASURE -XC401.2 = CONNECTOR BASE MATE-N-LOK PCL SUPPLY METER -X10 = Terminal distribution 24Vdc -X11 = Terminal distribution 24Vdc -X600 = AC Voltage measure Terminal-strip									
									

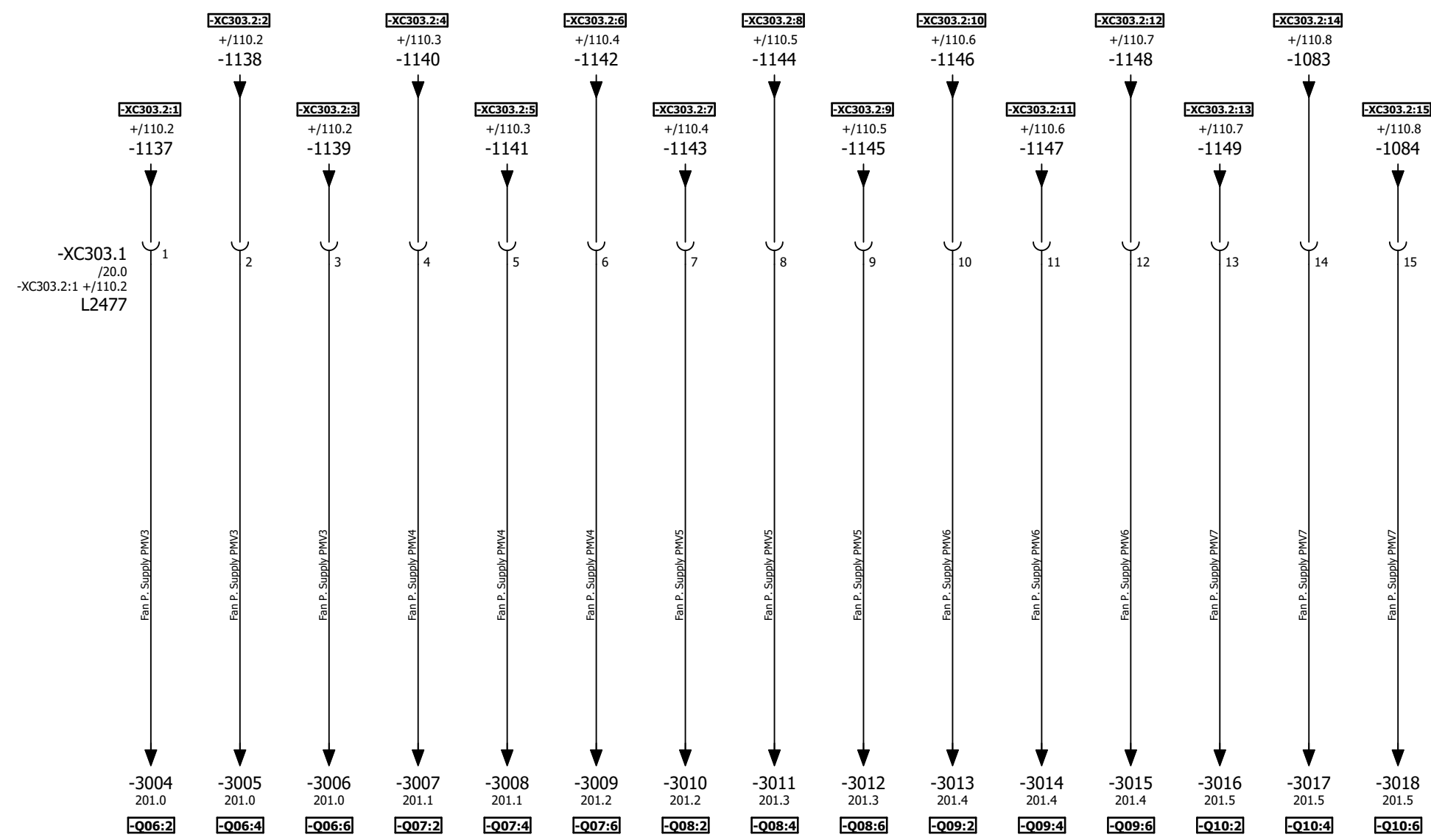
Interconnections - PCL towards ADU

Fan

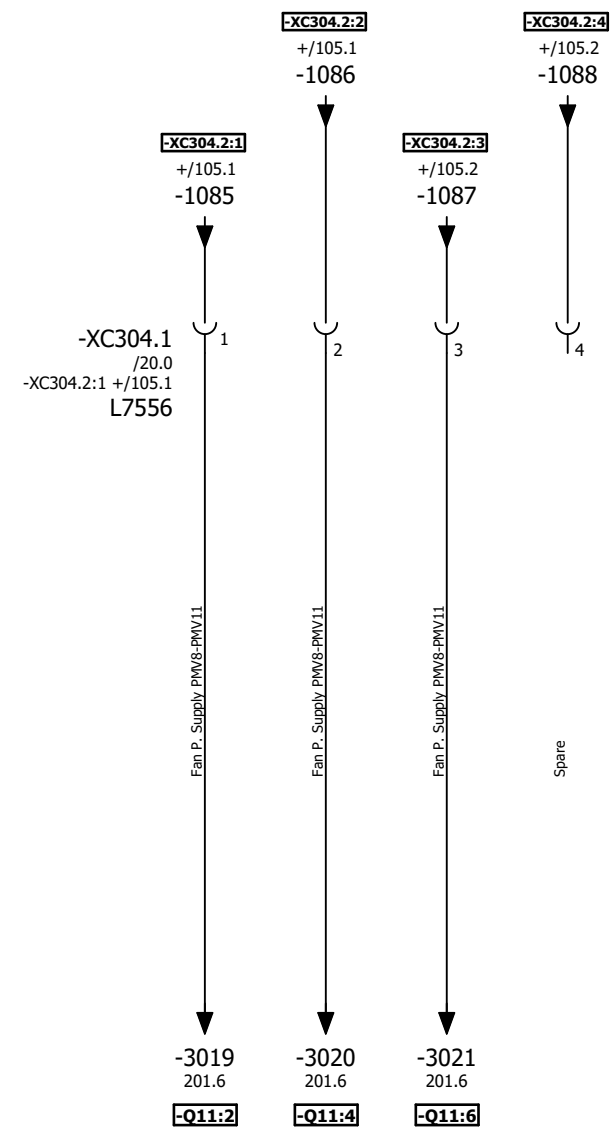
Optical fiber



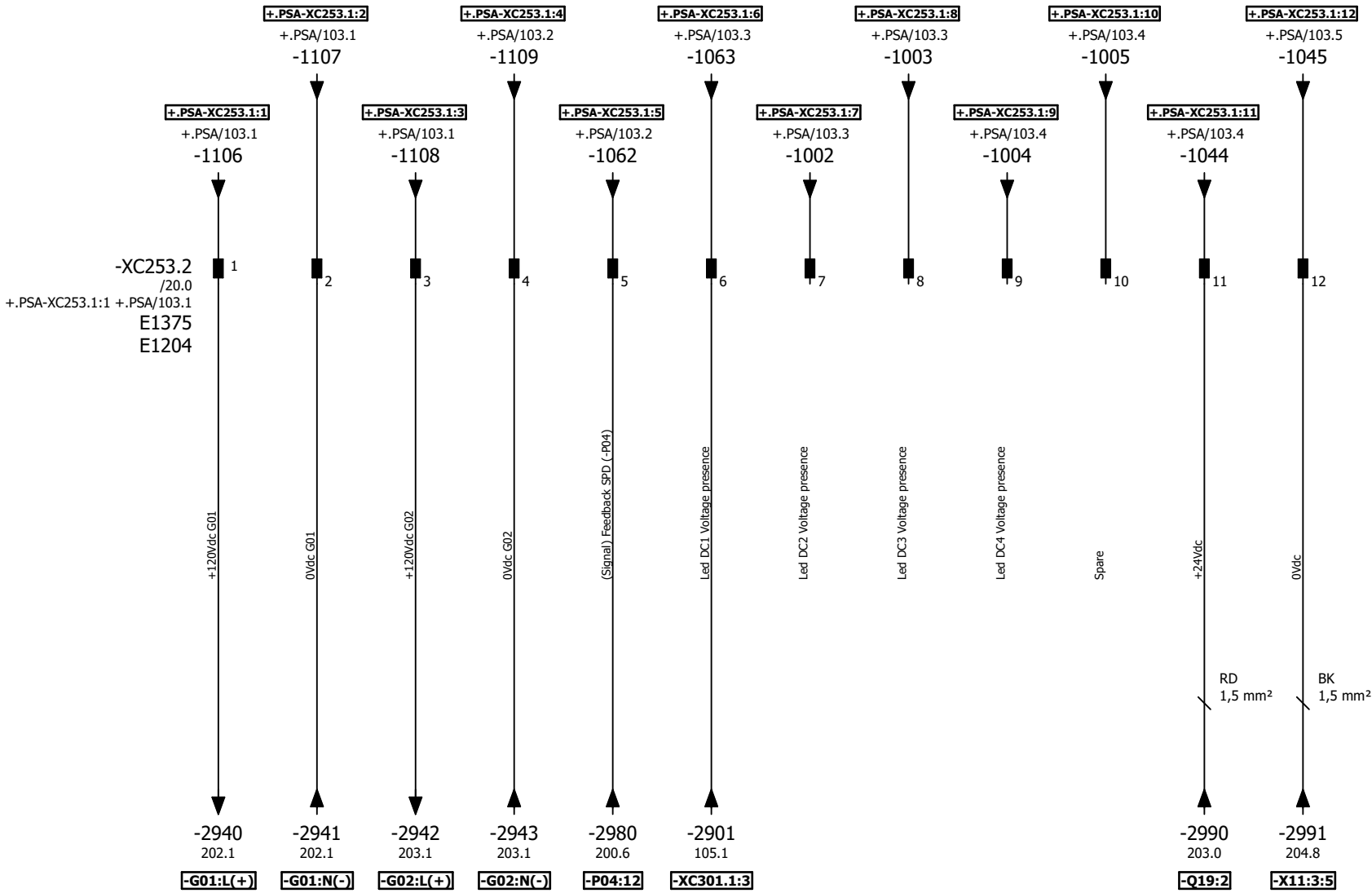
Interconnections - Fan PCL//Modules-AFC



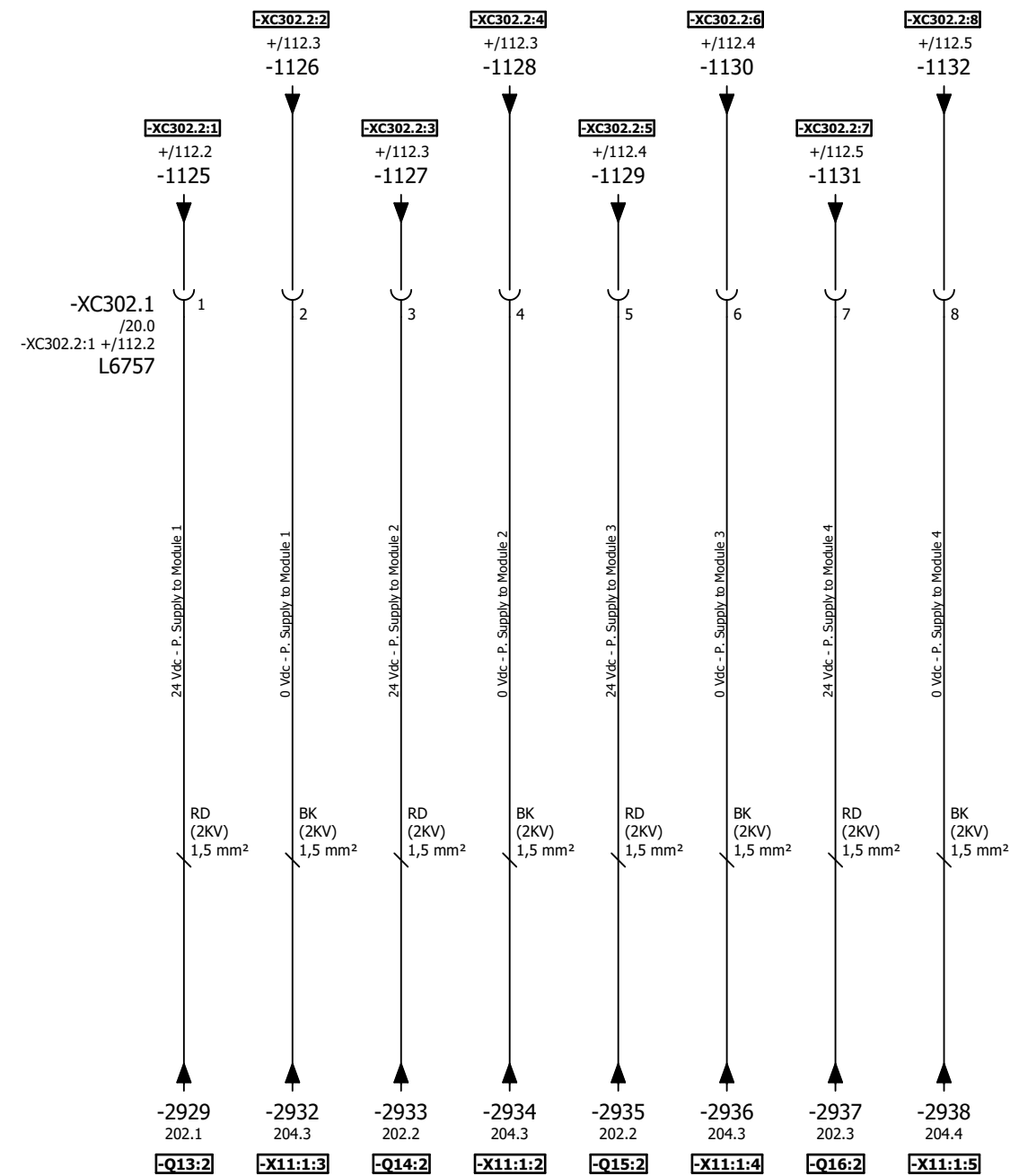
Interconnections - Fan PCL//CDT



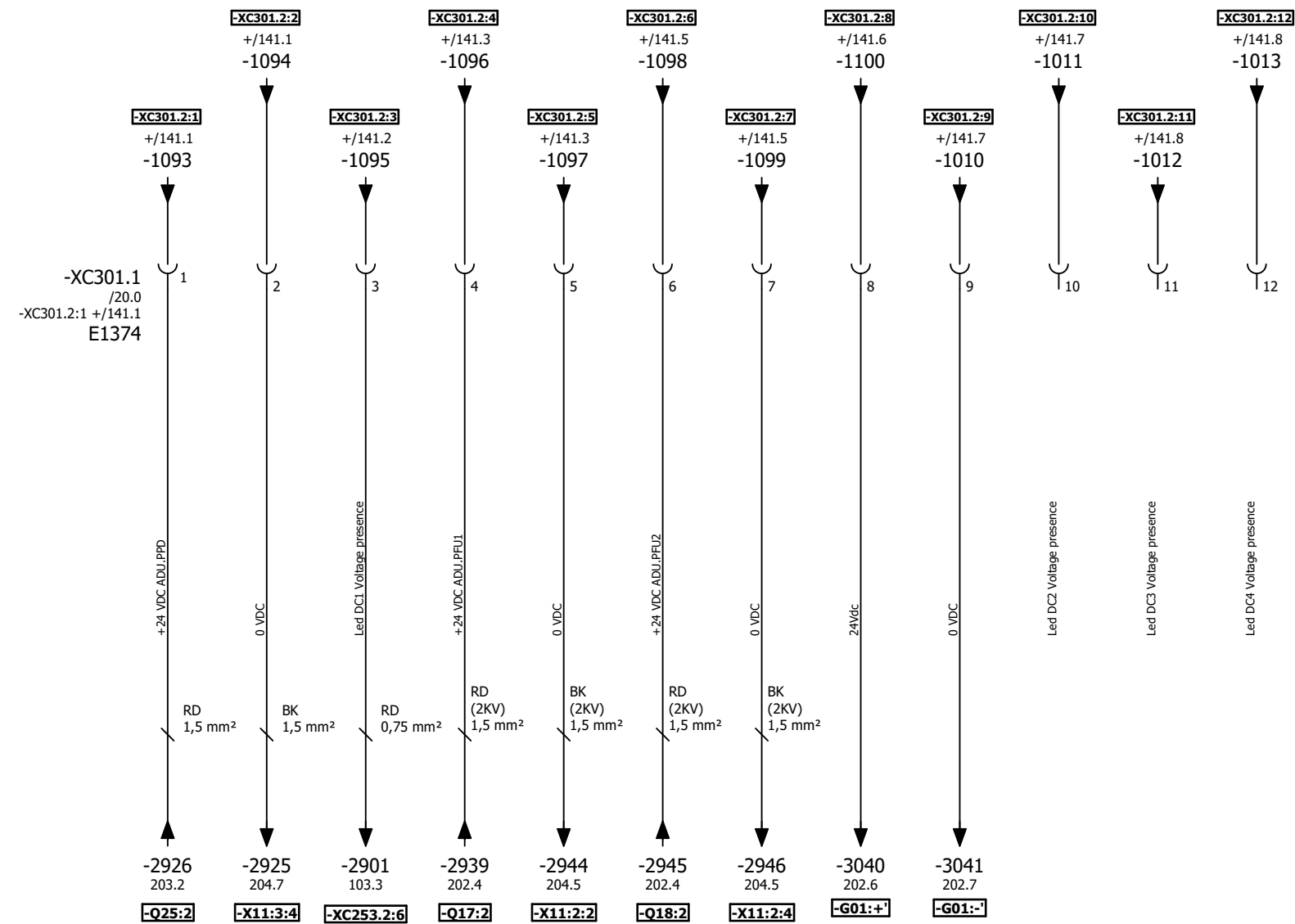
Interconnections PSA//PCL

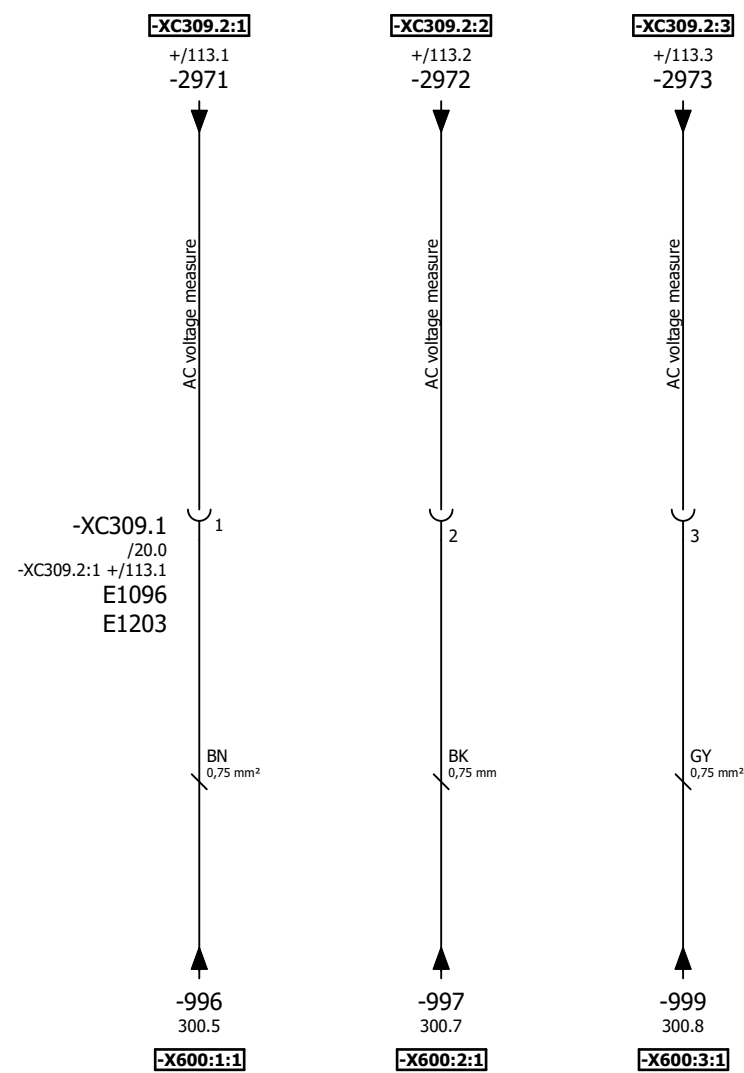


Interconnections PCL/Modules

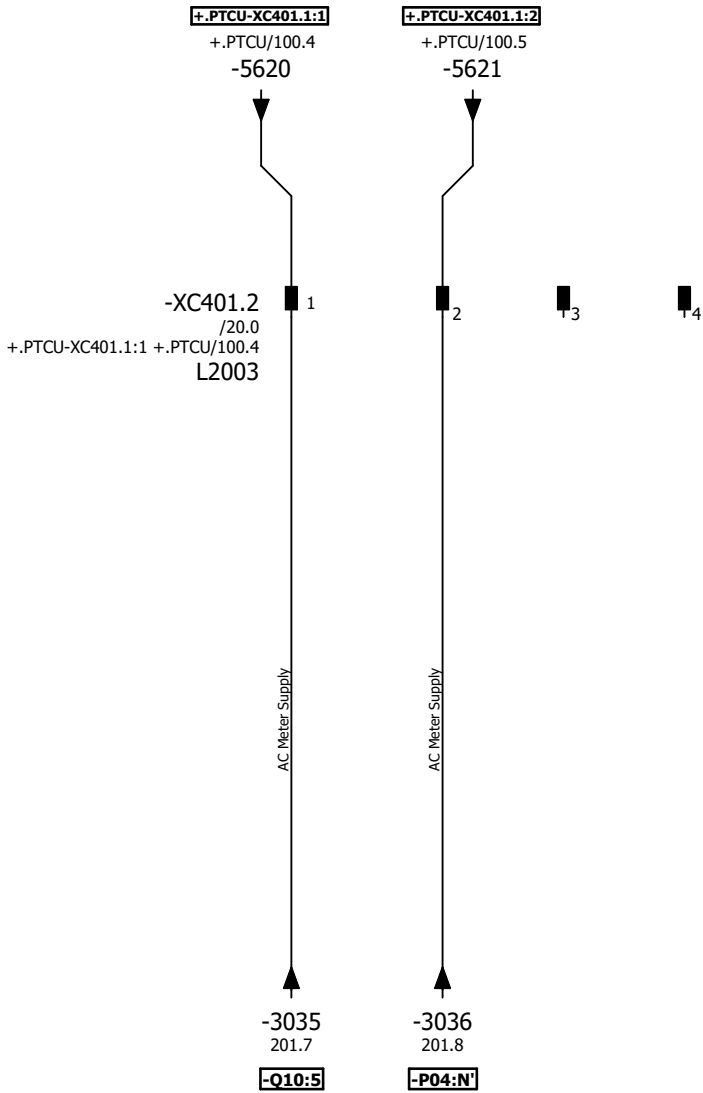


Interconnections PCL//ADU.PPD/PFU1/2

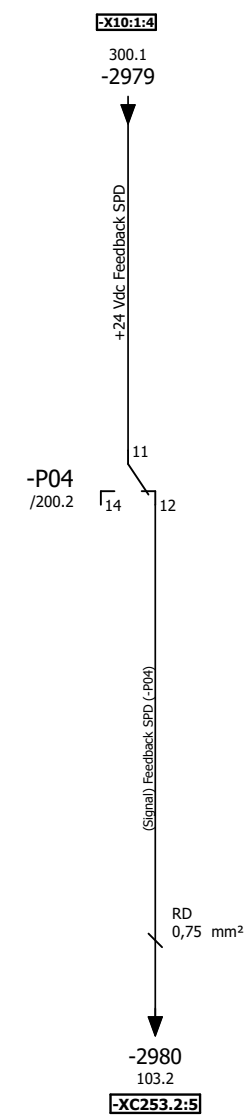
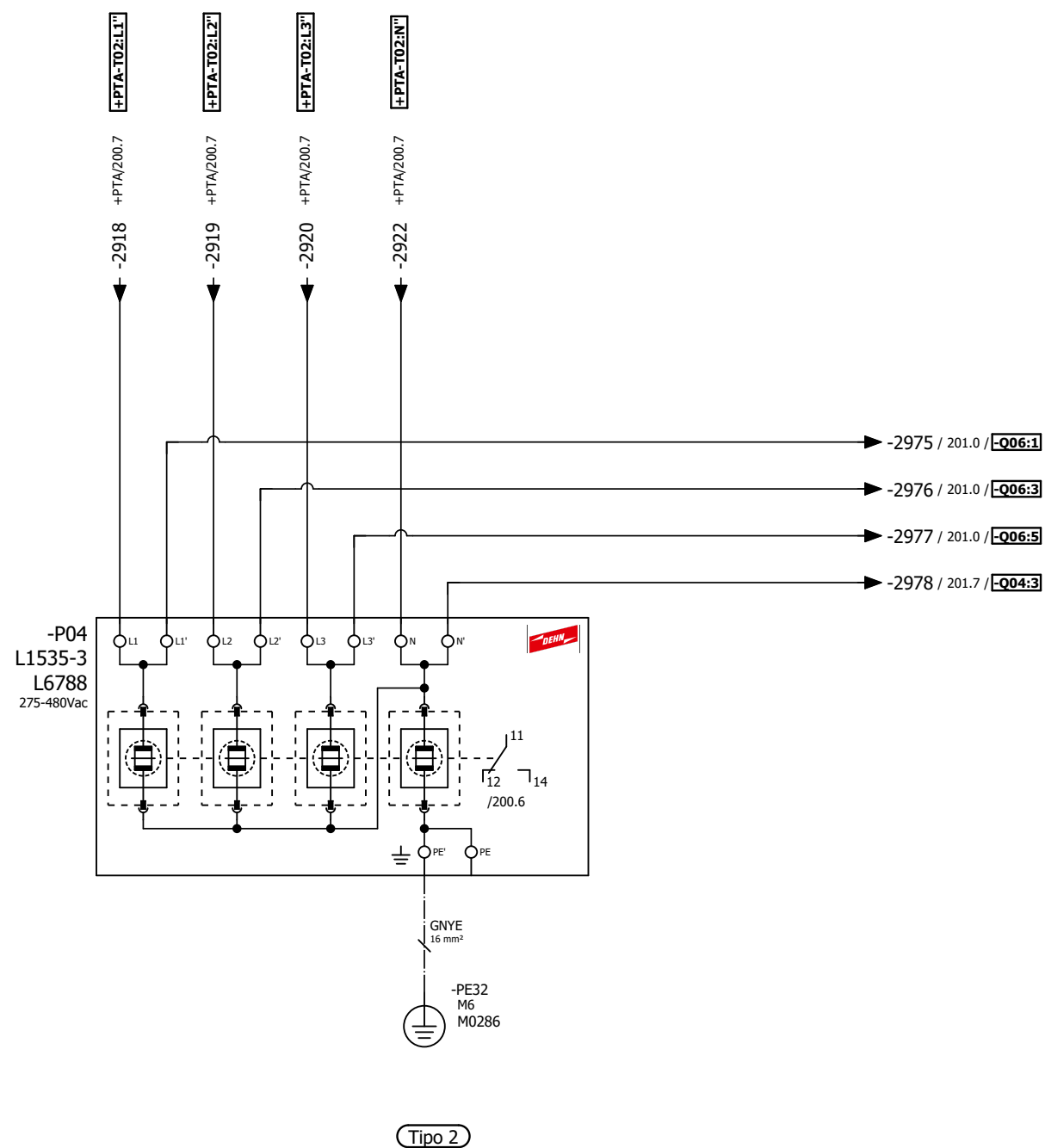


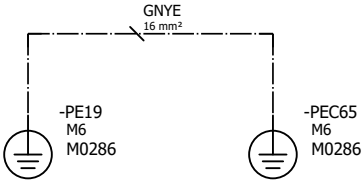
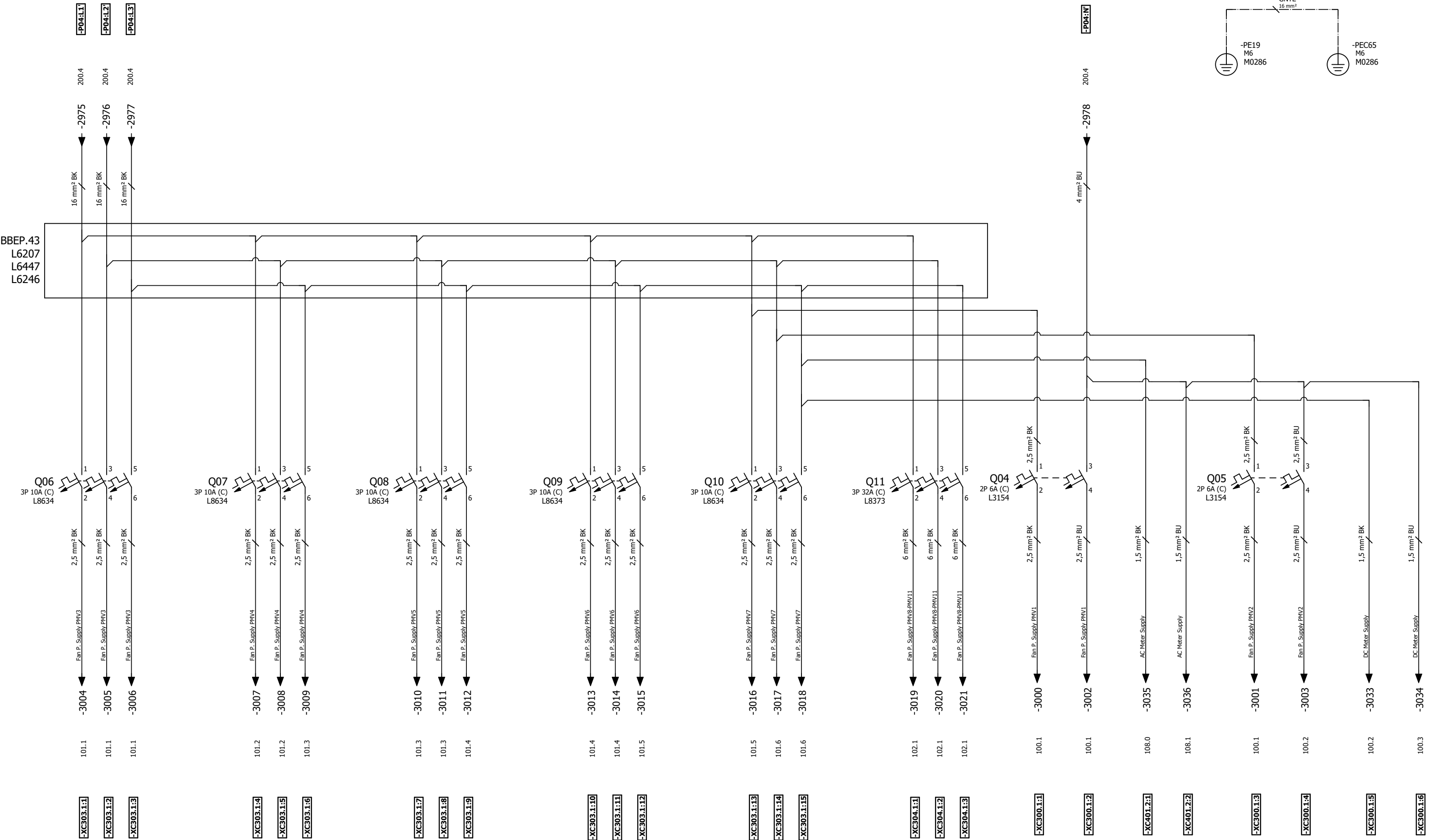


Interconnections PCL//PTCU



From auxiliar transformer





To fan 3
module 4

To fan 4
module 3

To fan 5
module 2

To fan 6
module 1

To fan 7
C filter

To fans
MV skid

To DU fan 1

To AC Meter
Panel PTCU

To DU fan 2

To DC Meter
Panel (-PAD1)
// BPA



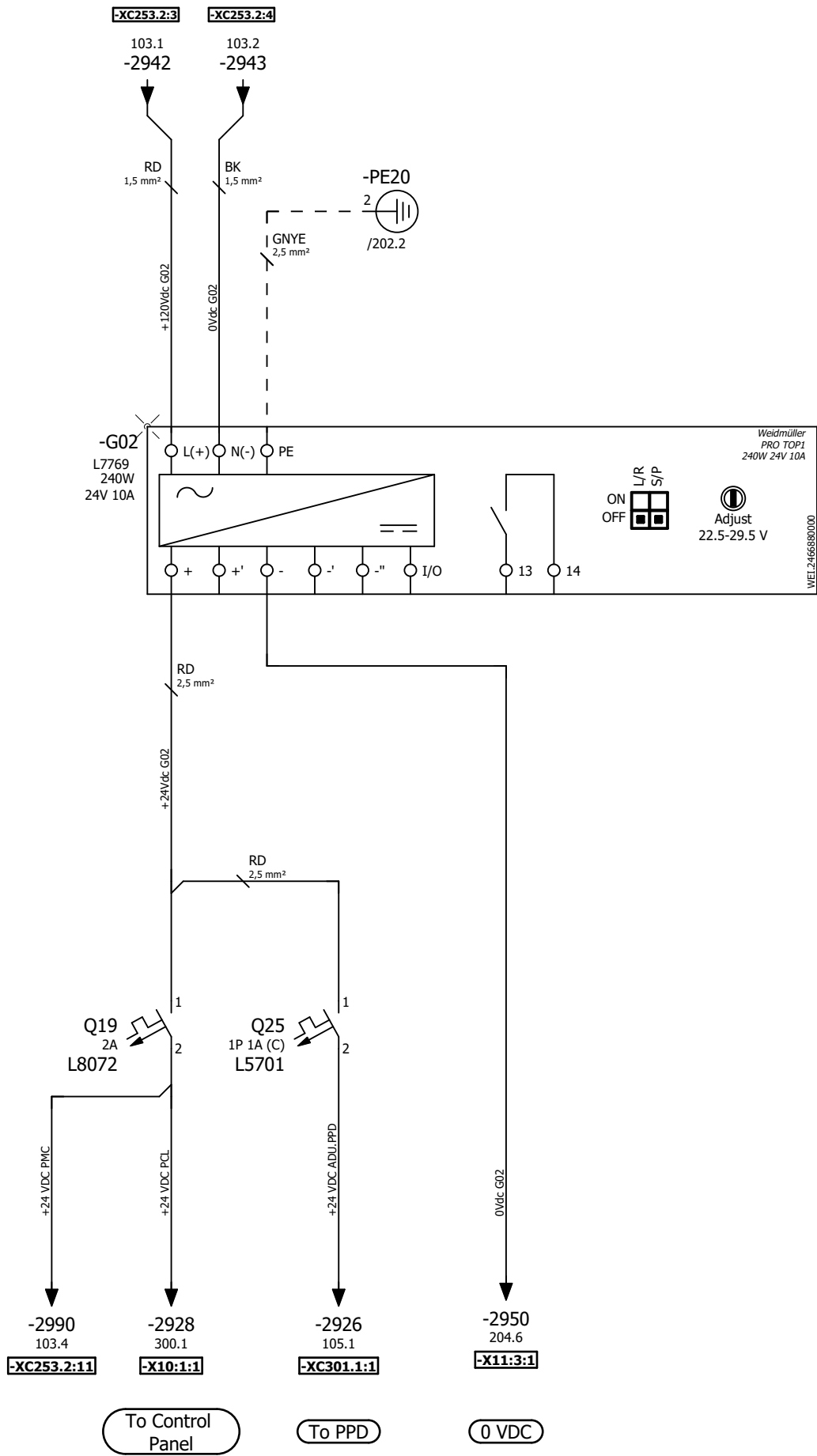
DESIGNED	I+D 1
REVISED	I+D 2
APPROVED	I+D 3

DATE	
REVISION	1.3.11

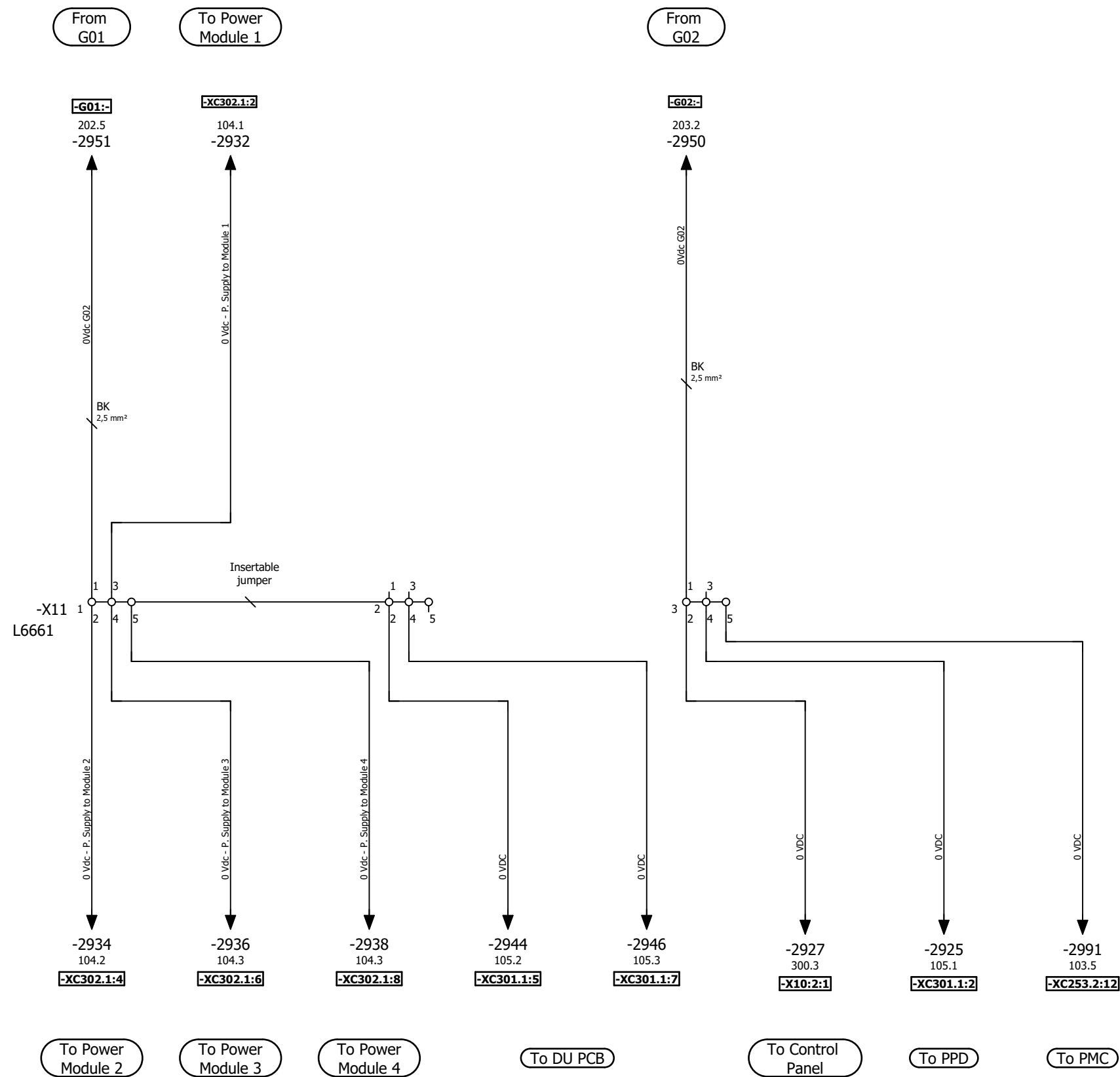
Project	ESQ_FS4010M10U000012_000
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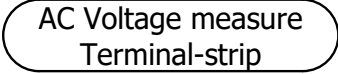
Name	Fan distribution
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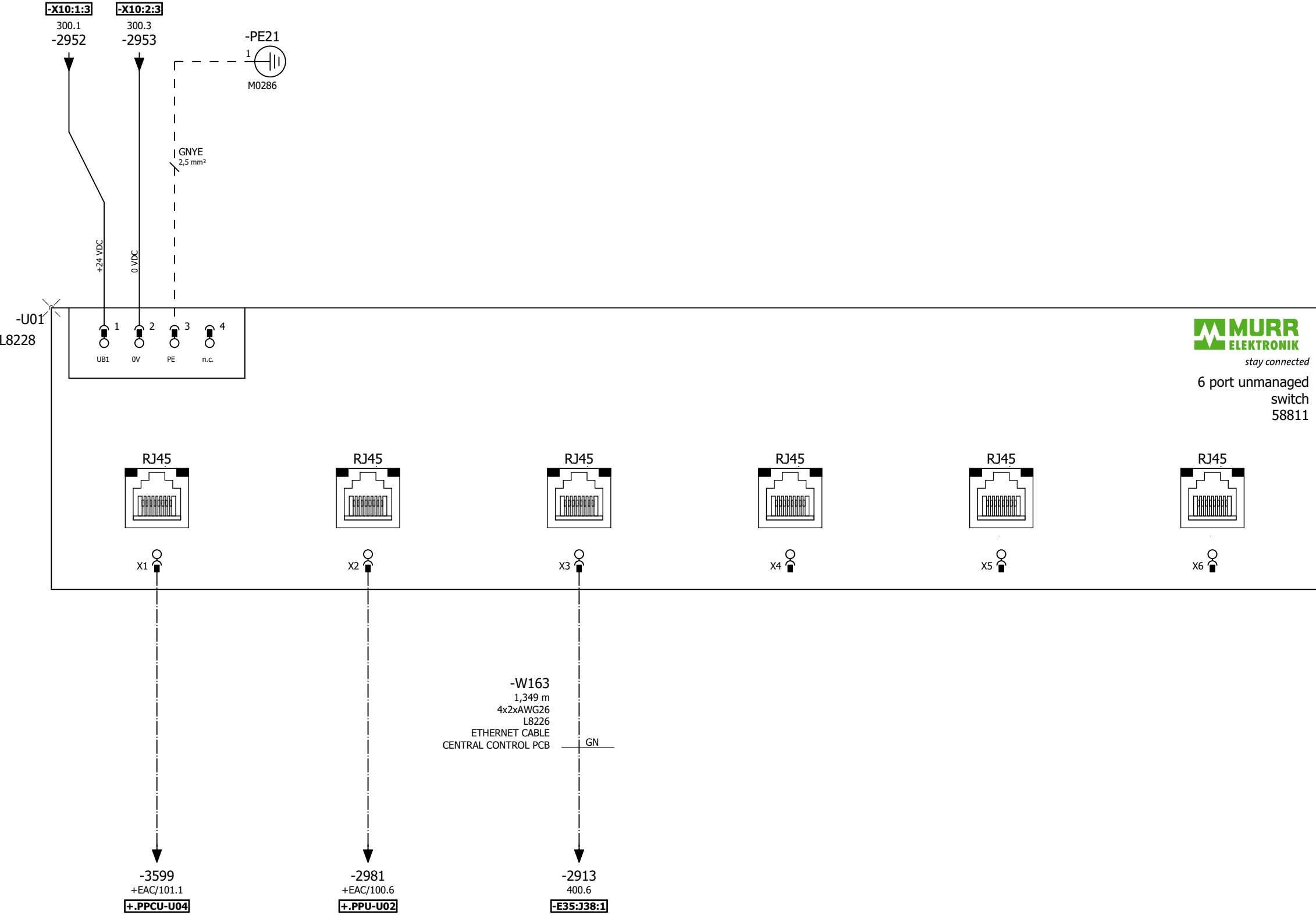
Installation:	HEM INVERTER	HEM	A3
Mounting location: +EAC.PCL	ENCLOSURE CONTROL CABINET	++ZMI	Previous page 200
Project Code:	FS4010M10U10001100410010020101014	Page 201	Next page 202
			Total number of pages 455



24VDC
power
supply







User connector
(PPCU cabinet or PTCU)

Door connector

PCB MAIN CONTROL

Spare --
DC Meter

Spare --
AC Meter

Spare --
BCI

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BBEP.41	L6102	70211413001	CROSS CONECTION MCB BKN 3P BB101	1
-BBEP.43	L6246	220110	220110 PUENTE MAGNETICOS PIA 3X19	1
-BBEP.43	L6447	220122	TRIPOLAR END COVER PIA BRIDGE	2
-BBEP.43	L6207	220131	PIA CABLE CONNECTION TIP 25MM	3
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E35	PCB E20101159	PCB E20101159	HEM-PLUS CENTRAL CONTROL BOARD	1
-E38	PCB E21201234	PCB E21201234	TARJETA CAISO	1
-G01	L5679	1478140000	POWER SUPPLY PRO MAX 480W 24V 20A	1
-G01	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	2
-G02	L7769	2466880000	Power supply	1
-G02	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-P04	L1535-3	908346	Surge arrester DGMU3PY 480 4WGR DEH	1
-P04	L6788	952589	ACCESORY TERMINAL SURGE ARRESTER DEHN	4
-P04	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-PE22	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	2
-PE23	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE24	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE25	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE26	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE27	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE28	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE29	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-PE31	M2496	MSKL 3-12	SCREWED CLAMP MSKL 3-12	1
-Q04	L3154	06110203R0	MCB BKN-B 2P C6A (10KA) 415VAC	1
-Q05	L3154	06110203R0	MCB BKN-B 2P C6A (10KA) 415VAC	1
-Q05	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-Q06	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q07	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q08	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q09	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q10	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q11	L8373	0628032863	BK63H 3P C32A 6kA 480Y/277VAC UL1077	1
-Q13	L5702	06110176R0	CIRCUIT BRAKER 1Ph 4A CURVE "C"	1
-Q14	L5702	06110176R0	CIRCUIT BRAKER 1Ph 4A CURVE "C"	1
-Q15	L5702	06110176R0	CIRCUIT BRAKER 1Ph 4A CURVE "C"	1
-Q16	L5702	06110176R0	CIRCUIT BRAKER 1Ph 4A CURVE "C"	1
-Q17	L5701	06110173R0	BKN-b 1P C1A 10KA 240/415VAC	1

-Q18	L5701	06110173R0	BKN-b 1P C1A 10KA 240/415VAC	1
-Q19	L8072	06110161R0	BKN-B 1P, B2A, 10KA	1
-Q25	L5701	06110173R0	BKN-b 1P C1A 10KA 240/415VAC	1
-U01	L8228	58811	SWITCH ETHERNET 10/1000 6 P 70°C	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+EAC.PCL-W160	ISOPRO OPTIC FIBER	L7589
=HEM++ZMI+EAC.PCL-W163	ETHERNET CABLE CENTRAL CONTROL PCB	L8226

Plug overview

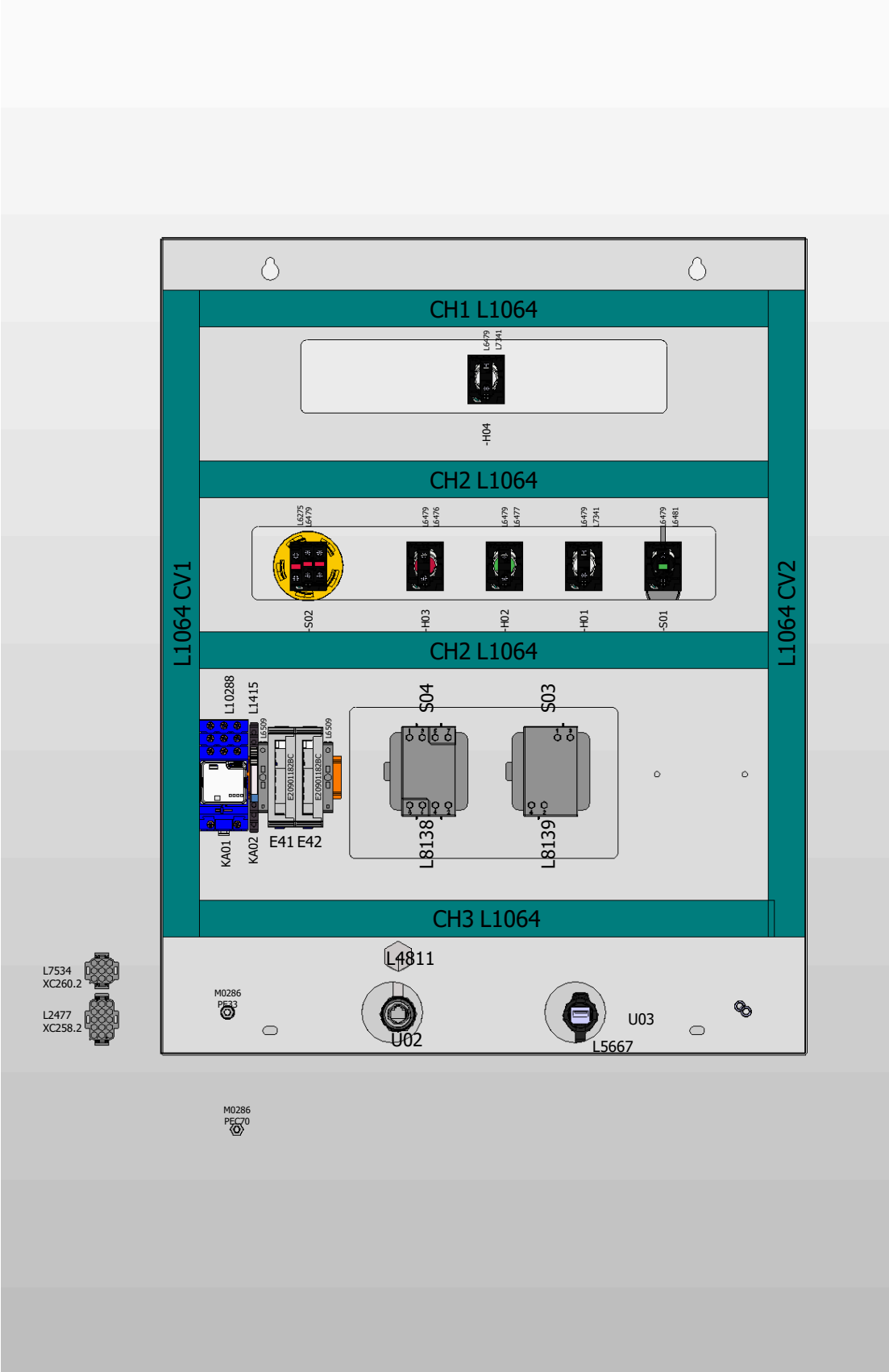
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+EAC.PCL-XC253.2	MATE-N-LOK 12P PANEL 120VDC POWER SUPPLY 6 LEDS	E1375	1
		E1204	9
=HEM++ZMI+EAC.PCL-XC300.1	MATE-N-LOK AERIAL CONNECTOR FAN SUPPLY	L5778	1
		L5709	4
		E1203	2
=HEM++ZMI+EAC.PCL-XC301.1	MATE-N-LOK 12P AERIAL CONNECTOR POWER SUPPLY PPD	E1374	1
		E1203	7
=HEM++ZMI+EAC.PCL-XC302.1	MATE-N-LOK 8P AERIAL CONNECTOR POWER SUPPLY MODULES	L6757	1
		E1203	8
=HEM++ZMI+EAC.PCL-XC303.1	MATE-N-LOK 15P AERIAL CONNECTOR FAN MODULES	L2477	1
		L5709	15
=HEM++ZMI+EAC.PCL-XC304.1	MATE-N-LOK PANEL 6mm2 POWER SUPPLY FANS SKID	L7556	1
		L0040	3
=HEM++ZMI+EAC.PCL-XC305.1	CONNECTOR OPTICAL FIBER DUPLEX ISOPRO	L7620	1
=HEM++ZMI+EAC.PCL-XC307.1	NTC CONNECTOR - FAN ADMISSION TEMP.	L1047	1
		L1051	2
=HEM++ZMI+EAC.PCL-XC308.1	NTC CONNECTOR - C FILTER TEMP.	L1047	1
		L1051	2
=HEM++ZMI+EAC.PCL-XC309.1	CONNECTOR MATE-N-LOK 3P AC MEASURE	E1096	1
		E1203	3
=HEM++ZMI+EAC.PCL-XC401.2	CONNECTOR BASE MATE-N-LOK PCL SUPPLY METER	L2003	1
		E1204	4



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HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
EAC.PPU	PUSHBUTTONS PANEL		
PPU PUSHBUTTONS PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

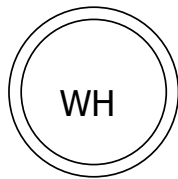
LAYOUT PANEL



Enclosure legend

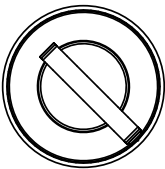
IME	ERP	Description
H01;H04	L7341	LED MODULE, WHITE
CH1-CH3;CV1;CV2	L1064	CABLE DUCT UNEX 42X30 EN U23X
H01-H04;S01;S02	L6479	HOLDER
CA1	L1061	DIN RAIL 35x7,5mm
E41;E42	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
U02	L4811	INSTALLATION SOCKET RJ45 CAT5E IP65
S01	L6480	KEY-OPERATED SWITCH RONIS, O-I
H01;H04	L6475	INDICATOR LIGHT, WHITE
H02	L6474	INDICATOR LIGHT, GREEN
H03	L6473	INDICATOR LIGHT, RED
S02	L6483	EM. STOP MUSHROOM PUSHBUTTON, 40MM, RED
S02	L6276	PROTECTIVE COLLAR FOR PADLOCKS
S02	L6275	CONTACT MODULE 1NC
H03	L6476	LED MODULE, RED
H02	L6477	LED MODULE, GREEN
S01	L6481	CONTACT MODULE 1NO
	L6482	DUST COVER
KA02	L1415	RELAY 24VDC + SLIM BASE
KA01	L10288	RELÉ 24Vdc 3PDT 16A SERIE 62.33 + BASE 92.03
S04	L8138	LOTO AC DESCRIPCION
S03	L8139	LOTO DC DESCRIPCION

H04



DC VOLTAGE
PRESENCE

S01



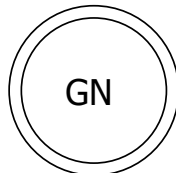
OFF/ON

H01



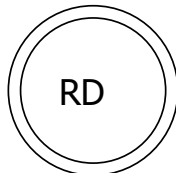
AC VOLTAGE
PRESENCE

H02



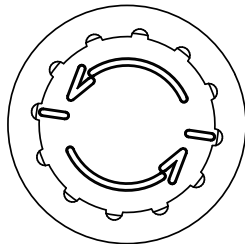
RUN

H03



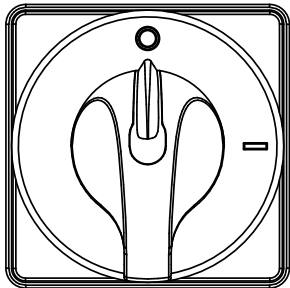
FAULT / WARNING

S02



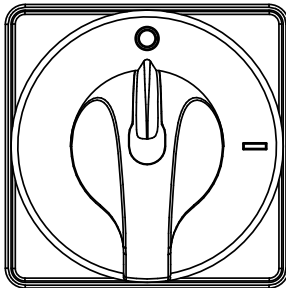
EMERGENCY
STOP

S03



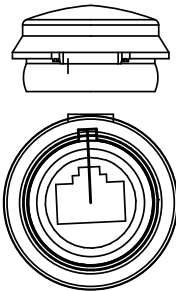
DC DISCONNECT

S04



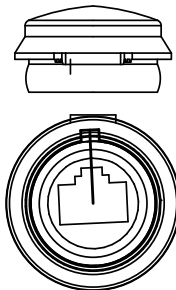
AC DISCONNECT

U03



DISPLAY

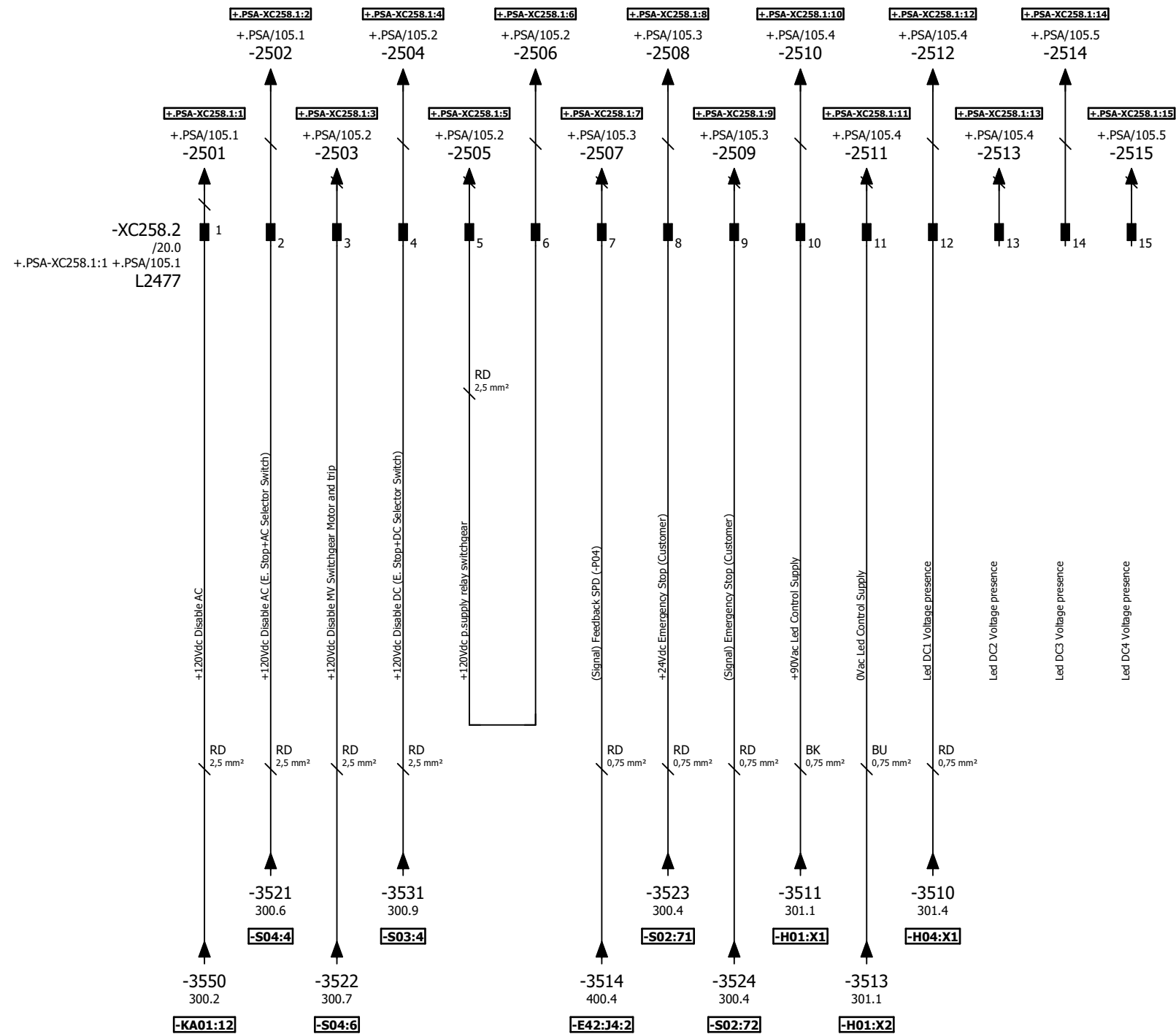
U02



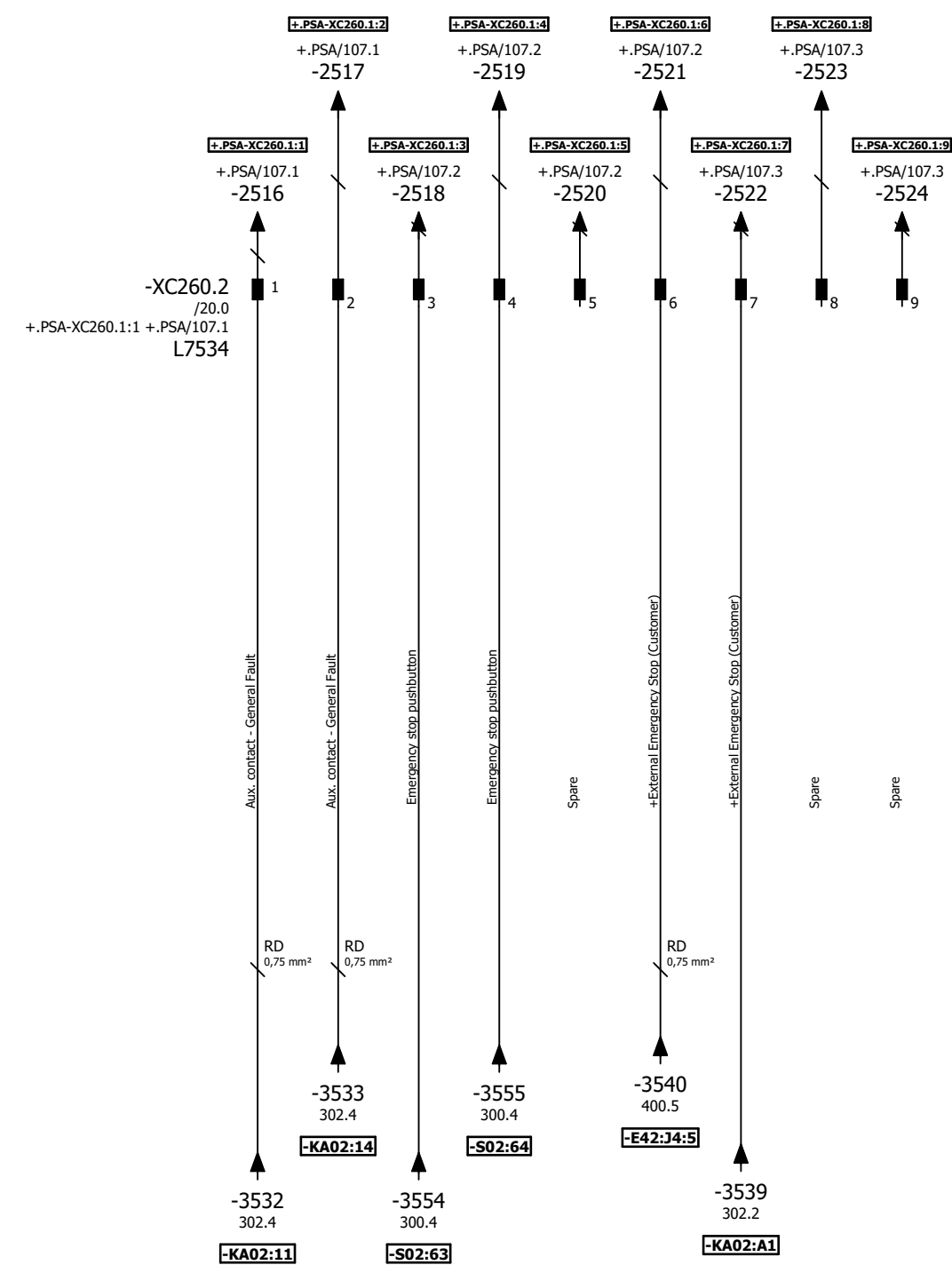
ETHERNET

0	1	2	3	4	5	6	7	8	9
-XC258.2 = MATE-N-LOK 15P AERIAL 120VDC DISTRIBUTION -XC260.2 = MATE-N-LOK 9P AERIAL									

Interconnections - Control Cabinet//Door



Interconnections - Control Cabinet//Door

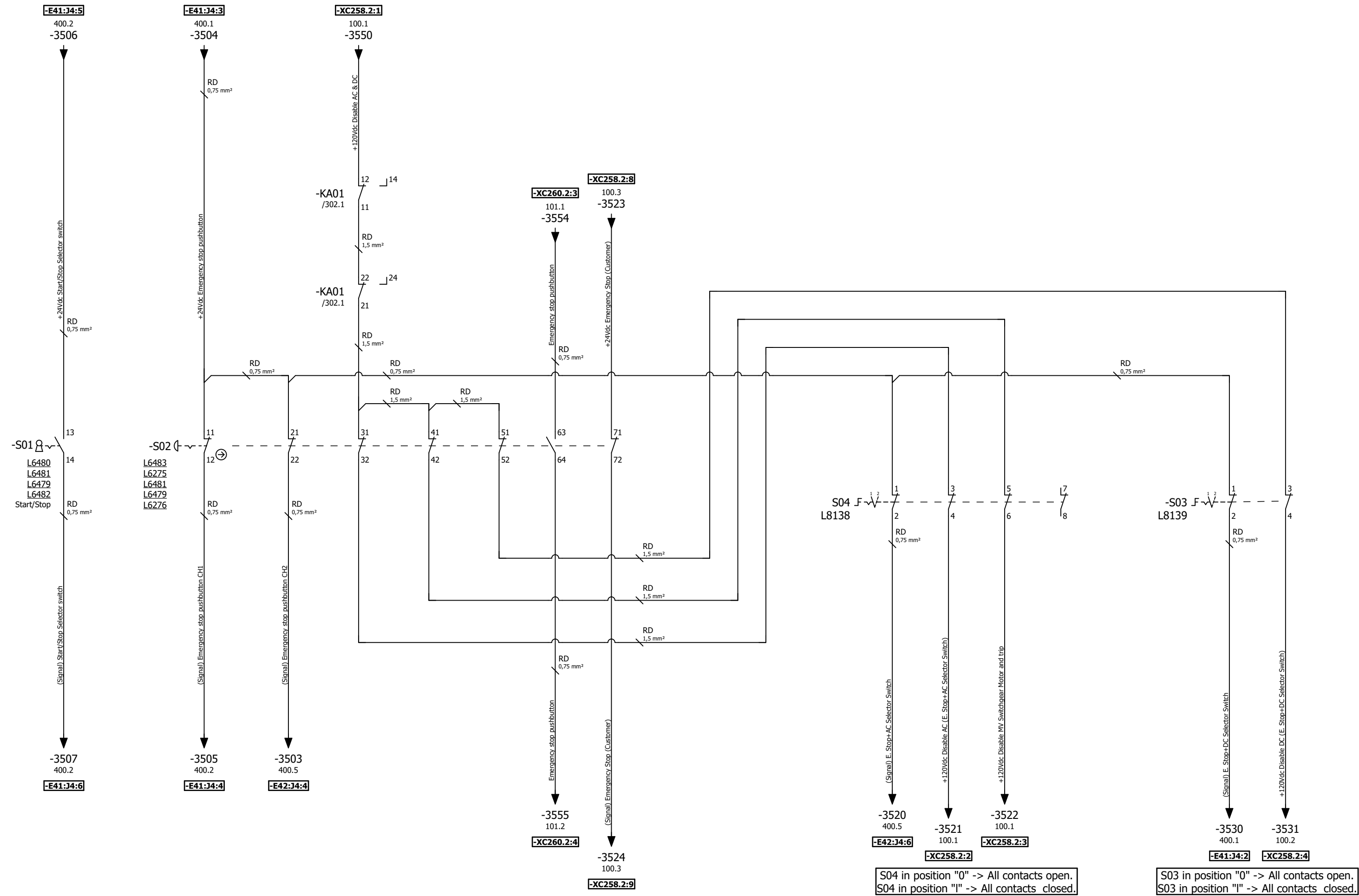


Start/Stop
Selector switch

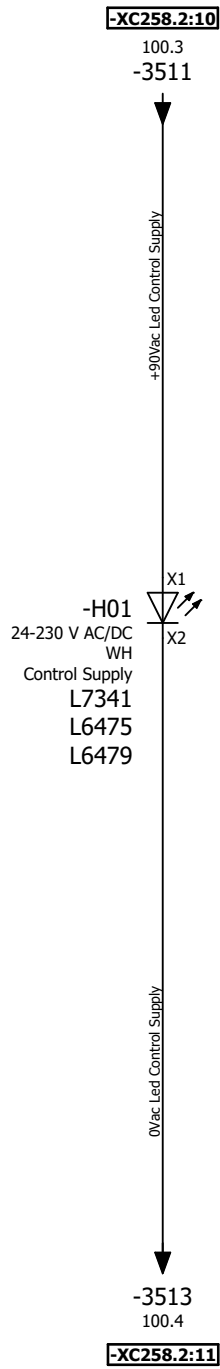
Emergency stop pushbutton

AC Selector Switch

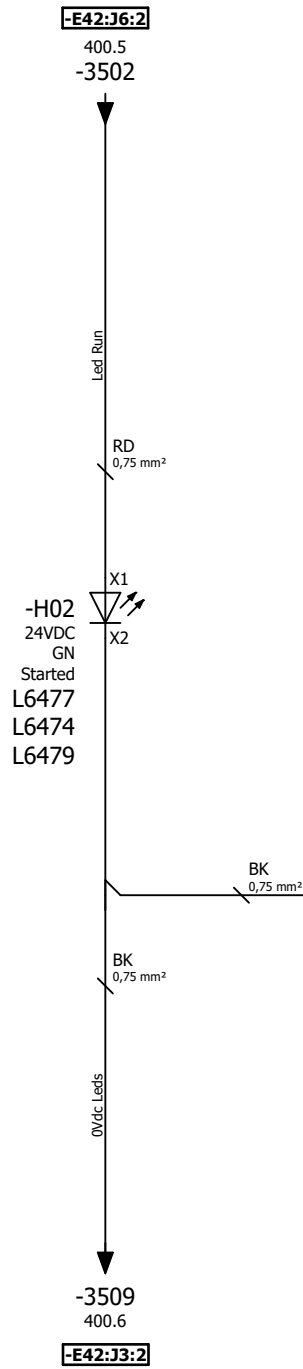
DC Selector Switch



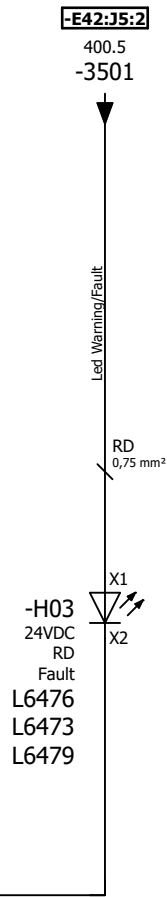
Ctrl Supply Led



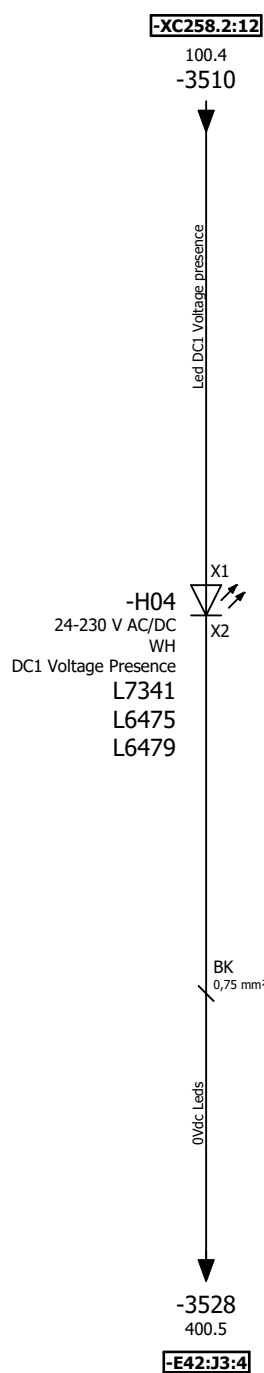
Run Led



Fault led



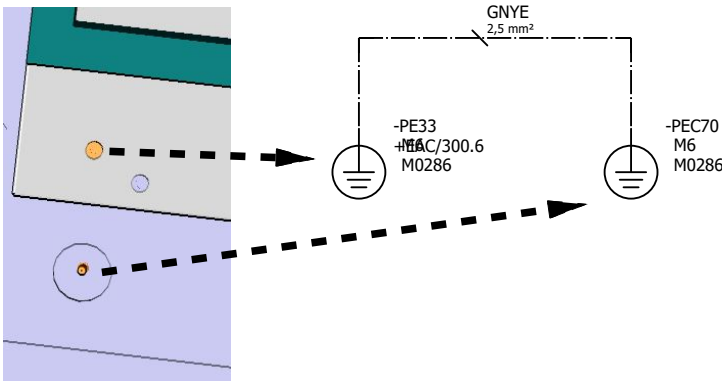
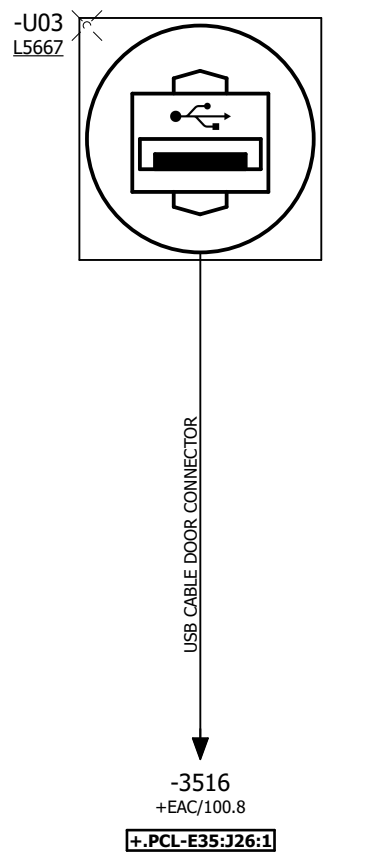
DC1 voltage presence



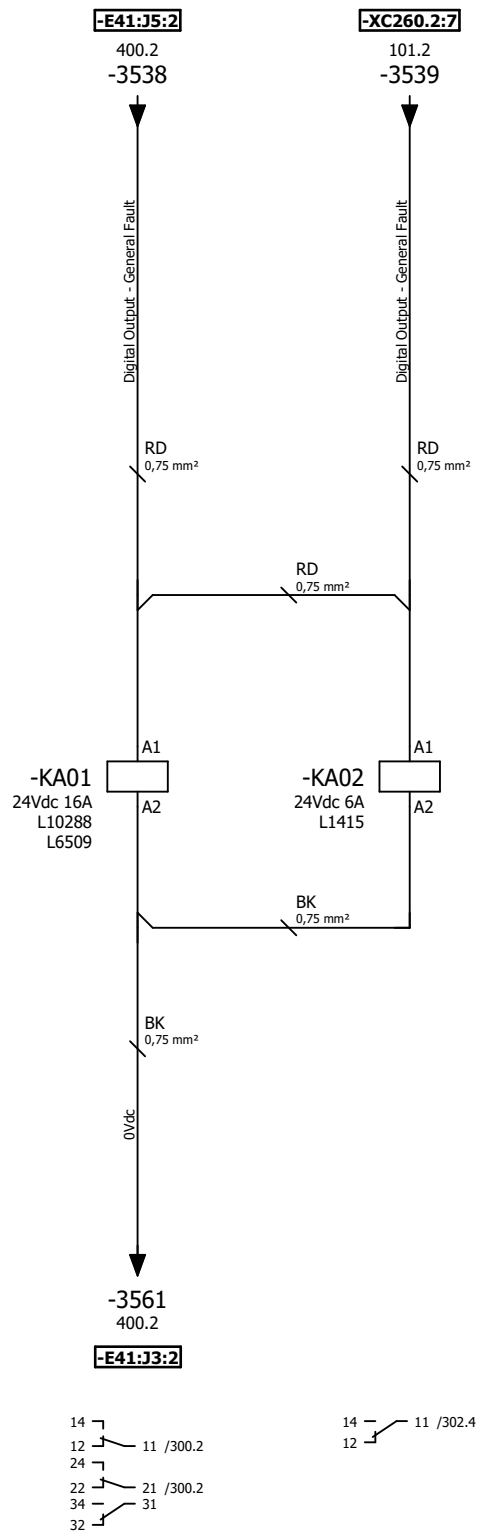
Ethernet Connection



Display Connection



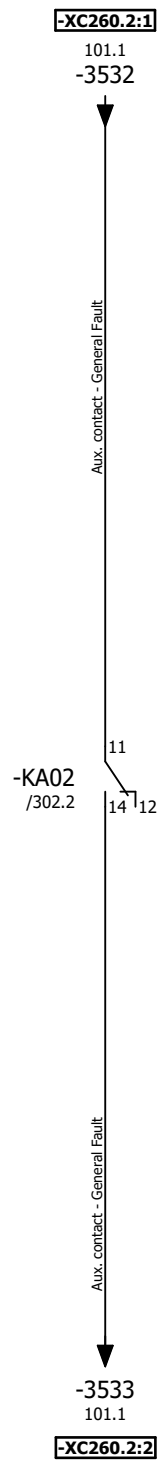
Digital Output
General Fault



AUX. RELAY
GENERAL FAULT

AUX. RELAY
GENERAL FAULT

Aux. Contact
General Fault



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-AN01	L9085	000118009358	MTX-Antenna Screwed ESFERA IP67 GPS SMA F	1
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E41	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E41	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E42	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-H01	L7341	3SU1401-1BH60-1AA0	LED MODULE, WHITE	1
-H01	L6475	3SU1001-6AA60-0AA0	INDICATOR LIGHT, WHITE	1
-H01	L6479	3SU1500-0AA10-0AA0	HOLDER	1
-H02	L6477	3SU1401-1BB40-1AA0	LED MODULE, GREEN	1
-H02	L6474	3SU1001-6AA40-0AA0	INDICATOR LIGHT, GREEN	1
-H02	L6479	3SU1500-0AA10-0AA0	HOLDER	1
-H03	L6476	3SU1401-1BB20-1AA0	LED MODULE, RED	1
-H03	L6473	3SU1001-6AA20-0AA0	INDICATOR LIGHT, RED	1
-H03	L6479	3SU1500-0AA10-0AA0	HOLDER	1
-H04	L7341	3SU1401-1BH60-1AA0	LED MODULE, WHITE	1
-H04	L6475	3SU1001-6AA60-0AA0	INDICATOR LIGHT, WHITE	1
-H04	L6479	3SU1500-0AA10-0AA0	HOLDER	1
-KA01	L10288	38010282	RELÉ 24Vdc 3PDT 16A SERIE 62.33 + BASE 92.03	1
-KA01	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-KA02	L1415	1122770000	RELAY 24VDC + SLIM BASE	1
-P15	L9087	000118015086	MTX-RF Lightning Arrestor N F - N M	1
-PMHEMGEN340	L3714A	1724151687	SELF-ADHESIVE LABEL LM MT300 17x9 GE	17
-PMHEMGEN340	V1290	V1290	STICKER EARTH 15x15	1
-S01	L6480	3SU1000-4CF11-0AA0	KEY-OPERATED SWITCH RONIS, O-I	1
-S01	L6481	3SU1400-1AA10-1BA0	CONTACT MODULE 1NO	1
-S01	L6479	3SU1500-0AA10-0AA0	HOLDER	1
-S01	L6482	3SU1900-0EB10-0AA0	DUST COVER	1
-S02	L6483	3SU1000-1HB20-0AA0	EM. STOP MUSHROOM PUSHBUTTON, 40MM, RED	1
-S02	L6275	3SU1400-1AA10-1CA0		19
-S02	L6481	3SU1400-1AA10-1BA0	CONTACT MODULE 1NO	1
-S02	L6479	3SU1500-0AA10-0AA0	HOLDER	1
-S02	L6276	3SU1950-0DX30-0AA0	PROTECTIVE COLLAR FOR PADLOCKS	1
-S03	L8139	ZFC16E2H1A2	LOTO DC DESCRIPCION	1
-S04	L8138	ZFC16E2H1A4	LOTO AC DESCRIPCION	1
-U02	L4811	4000-73000-0010000	INSTALLATION SOCKET RJ45 CAT5E IP65	1
-U03	L5667	1018840000	CABLE GLAND USB A - USB A IP65	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+EAC.PPU-W129		L9088
=HEM++ZMI+EAC.PPU-W130		L9086

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+EAC.PPU-XC258.2	MATE-N-LOK 15P AERIAL 120VDC DISTRIBUTION	L2477	1
		E1204	6
		L5710	6
=HEM++ZMI+EAC.PPU-XC260.2	MATE-N-LOK 9P AERIAL	L7534	1
		E1204	2

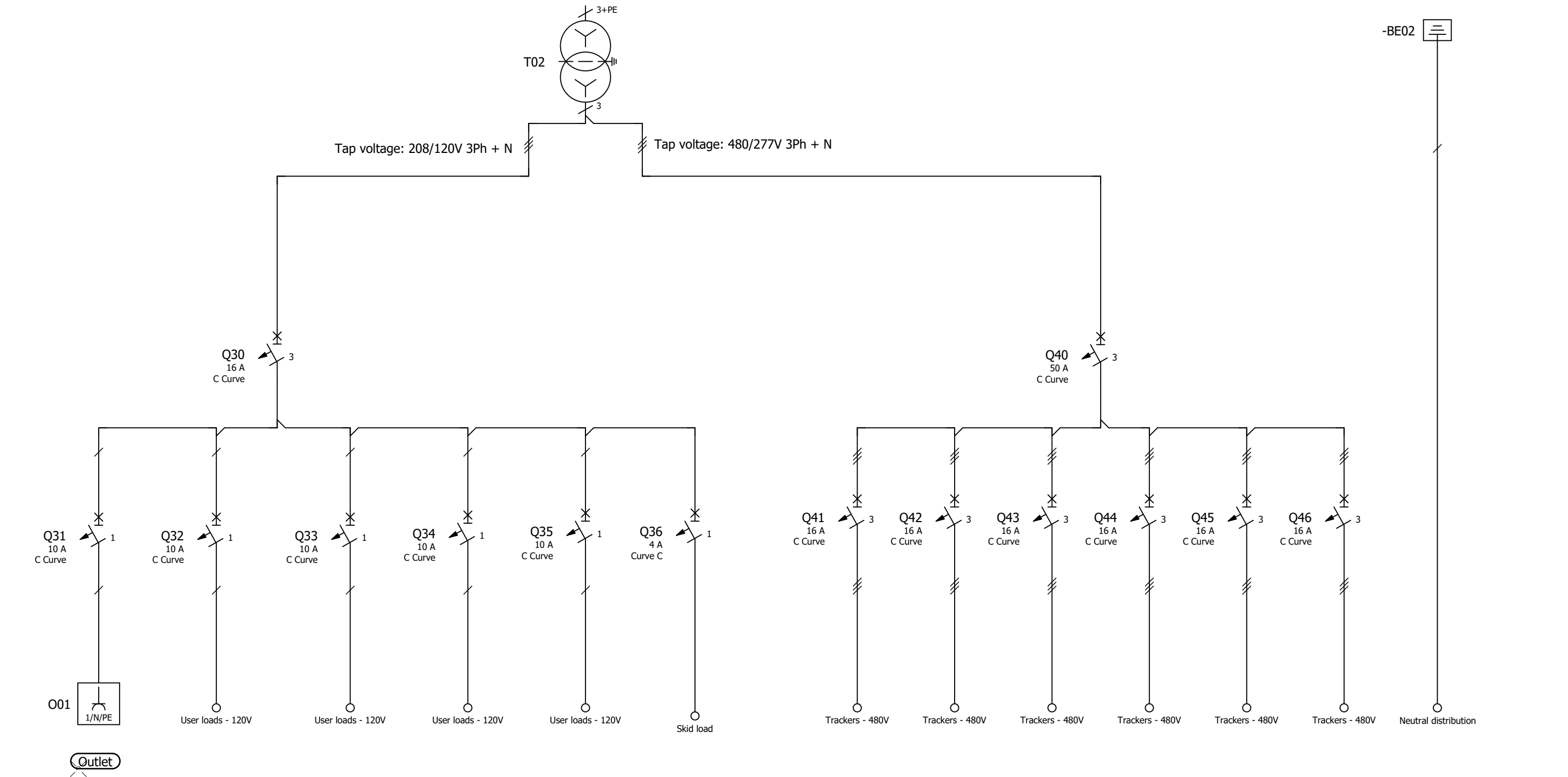


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CV35 SALIDA 30

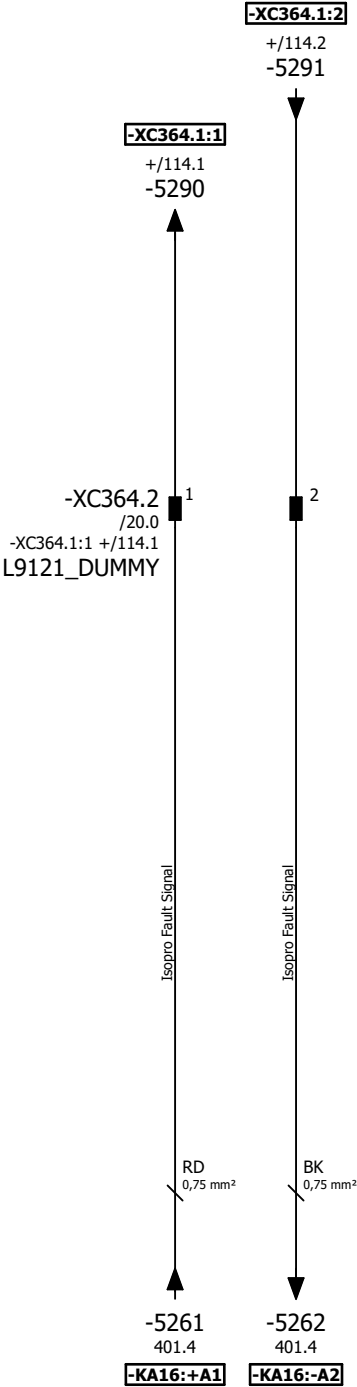
HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
EAC.PPCU	MAIN CUSTOMER PANEL		
PPCU MAIN CUSTOMER PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

0	1	2	3	4	5	6	7	8	9
-XC282.2 = PANEL CONNECTOR - 6P INTERNAL FAULT POWER SUPPLY RELAY -XC362.2 = PANEL CONNECTOR - 12P SPARE -XC364.2 = TRIPLE-LOK 2P ISOPRO FAULT SIGNAL -X34 = DSITRIBUTION TERMINAL -X35 = CUSTOMER TERMINALS (SIGNAL) -X36 = CUSTOMER TERMINALS (LOADS) -X37 = CUSTOMER TERMINALS (TRACKERS) -X39 = TERMINALS FOR ERCOT/JAPAN									
									

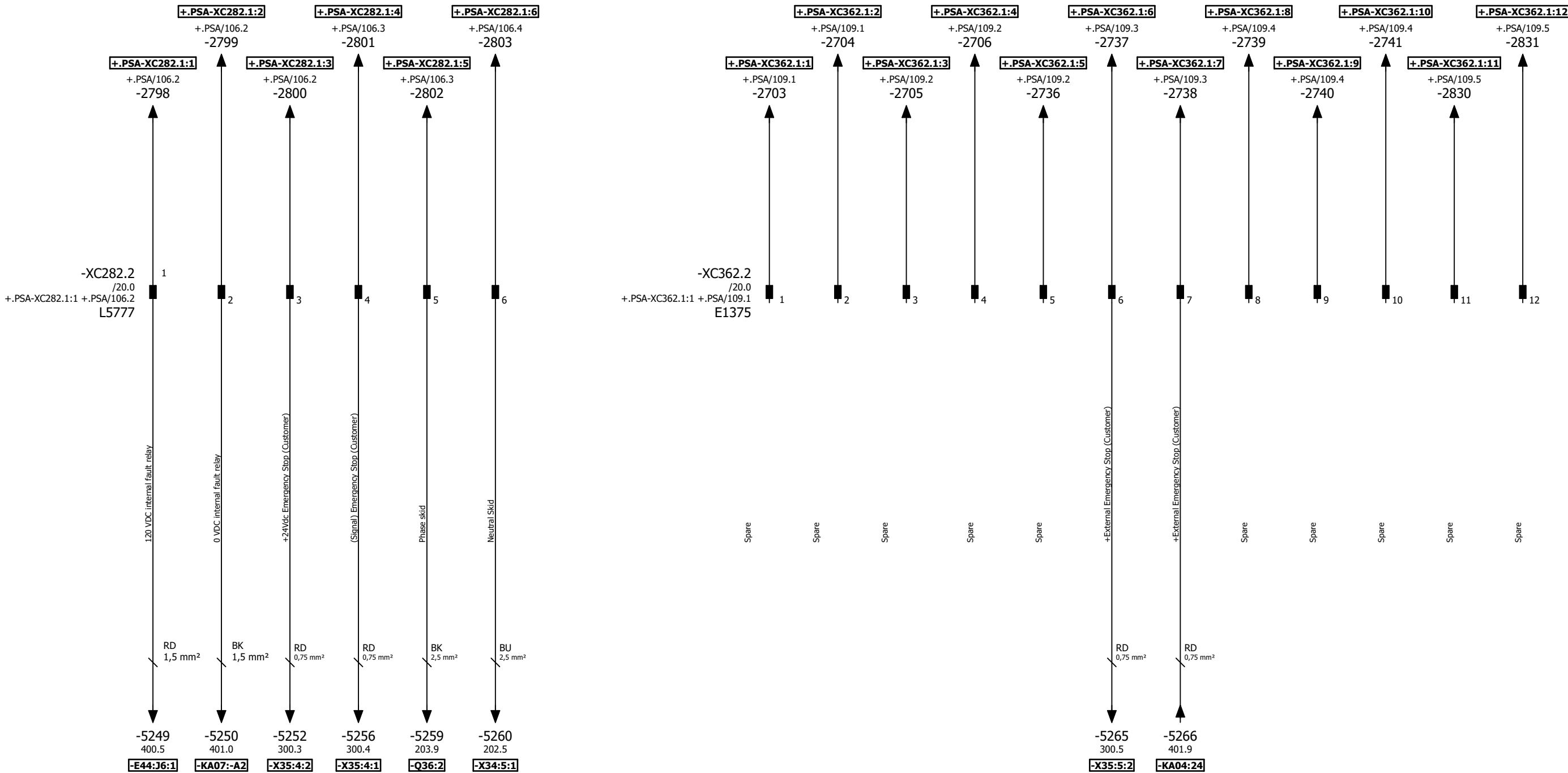
Advanced UL



Interconnections PPCU//PPD

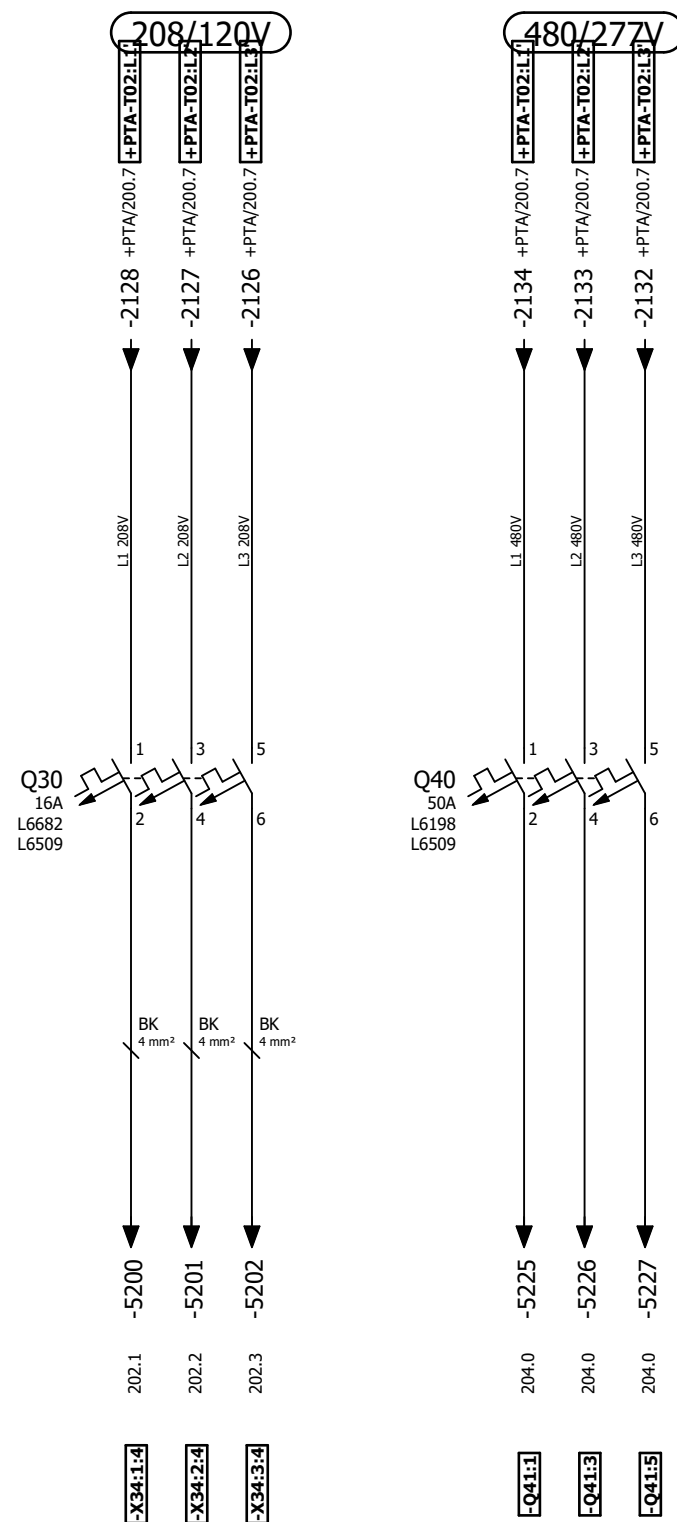


Interconnections PSA//PPCU



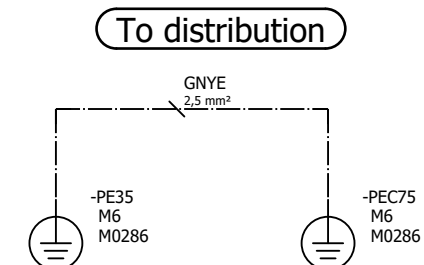
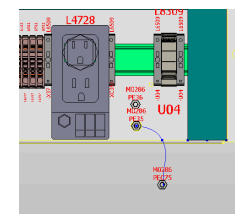
Power supply advanced UL

From secondary SSAA

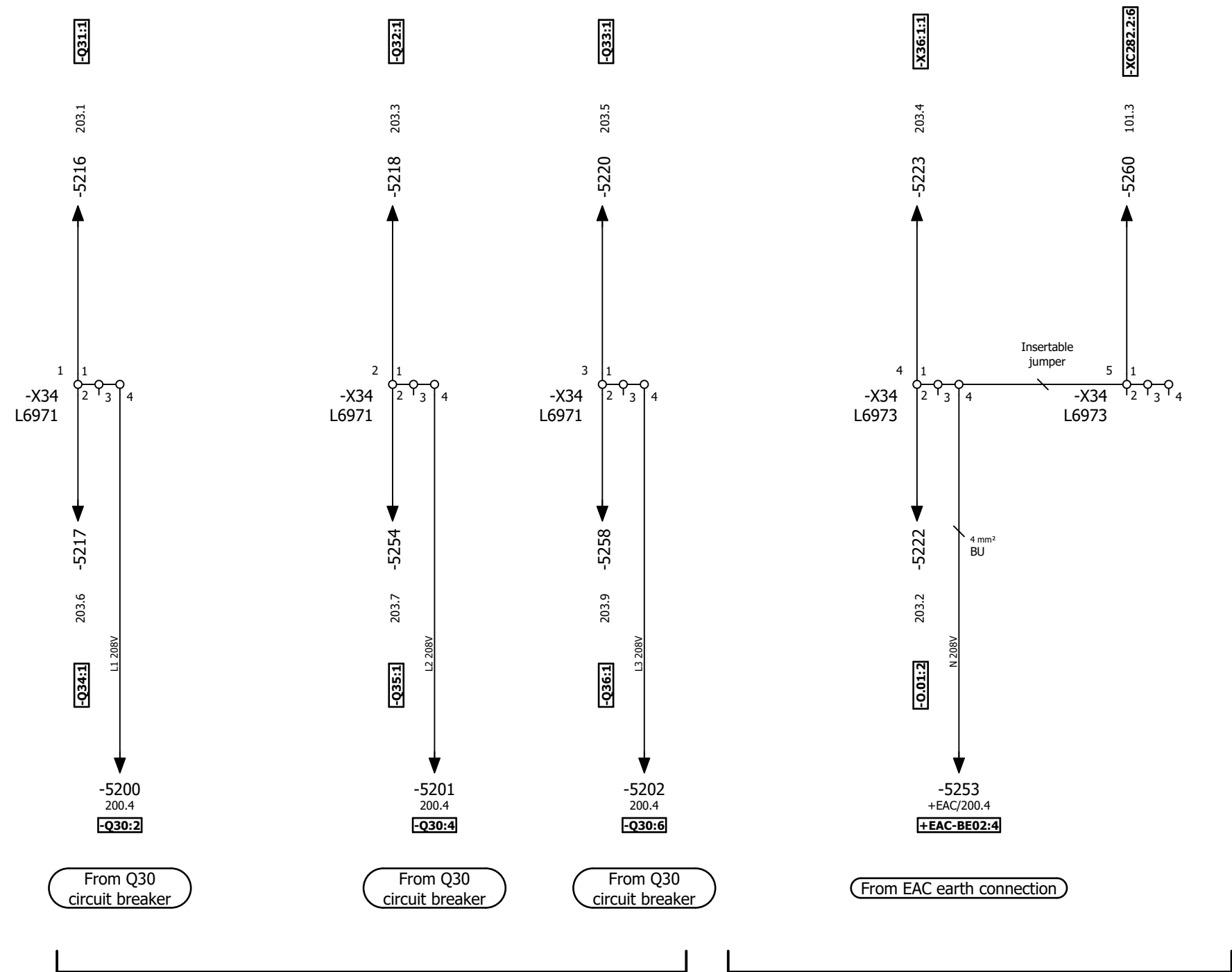


To distribution
400/230V

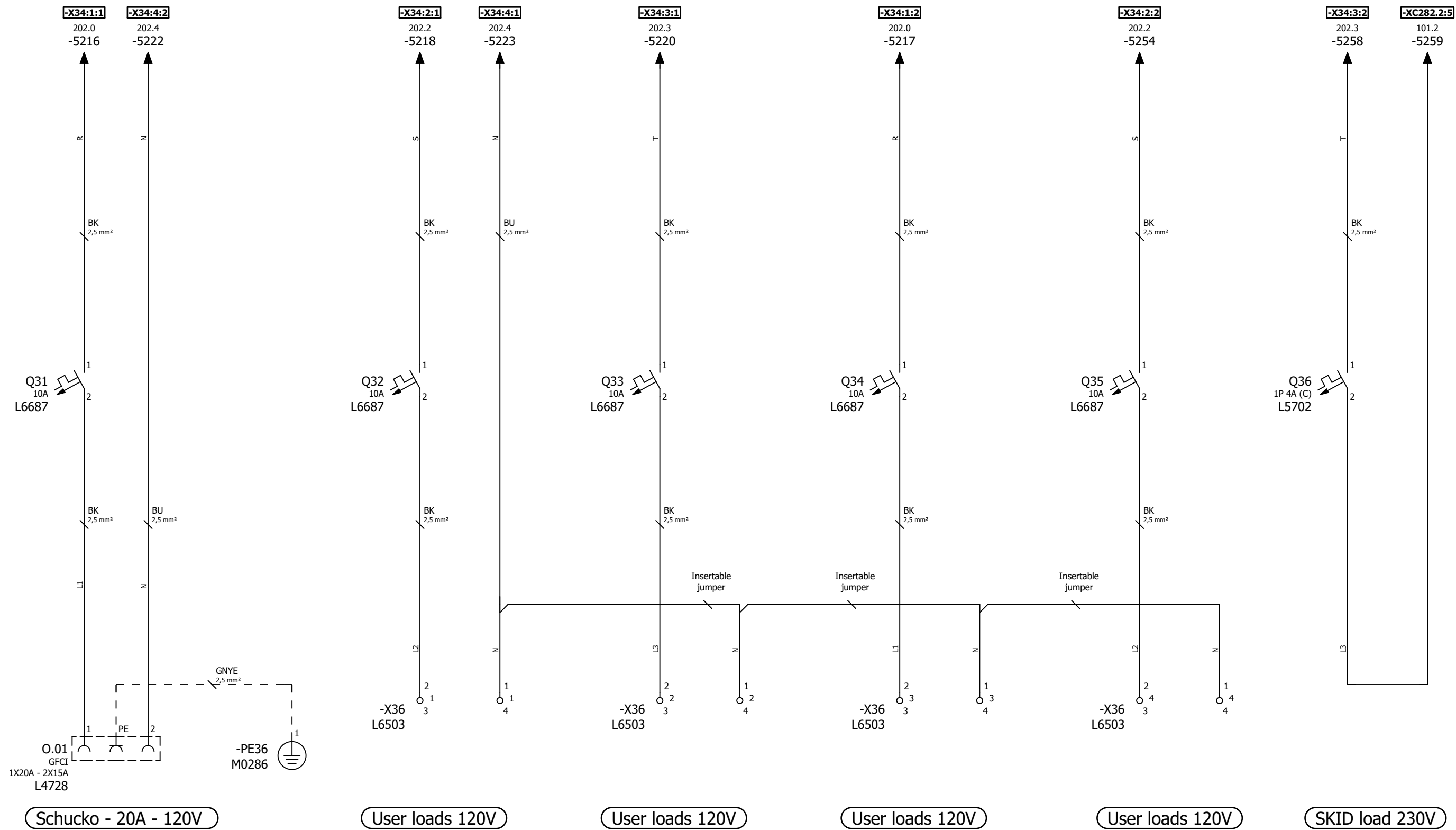
To Q41-Q46

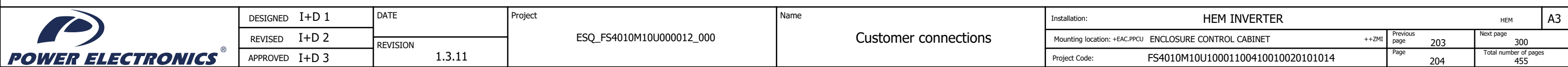


Distribution 208/120V



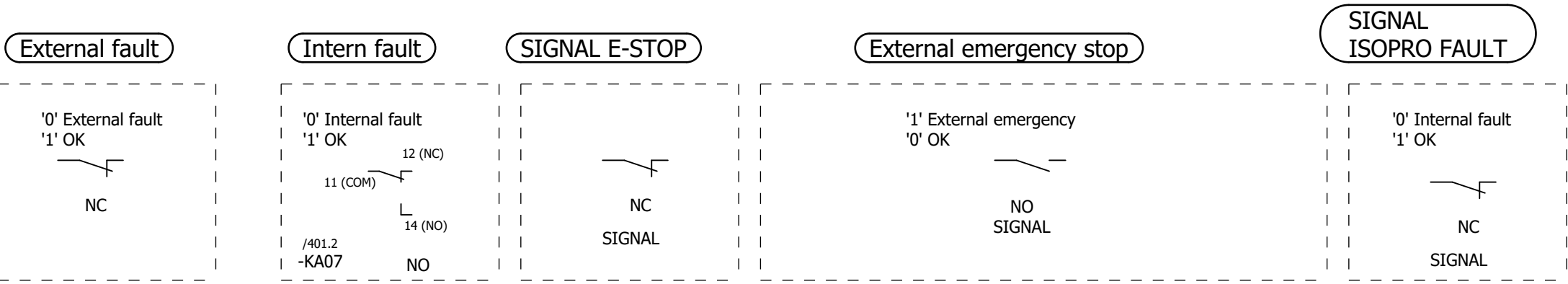
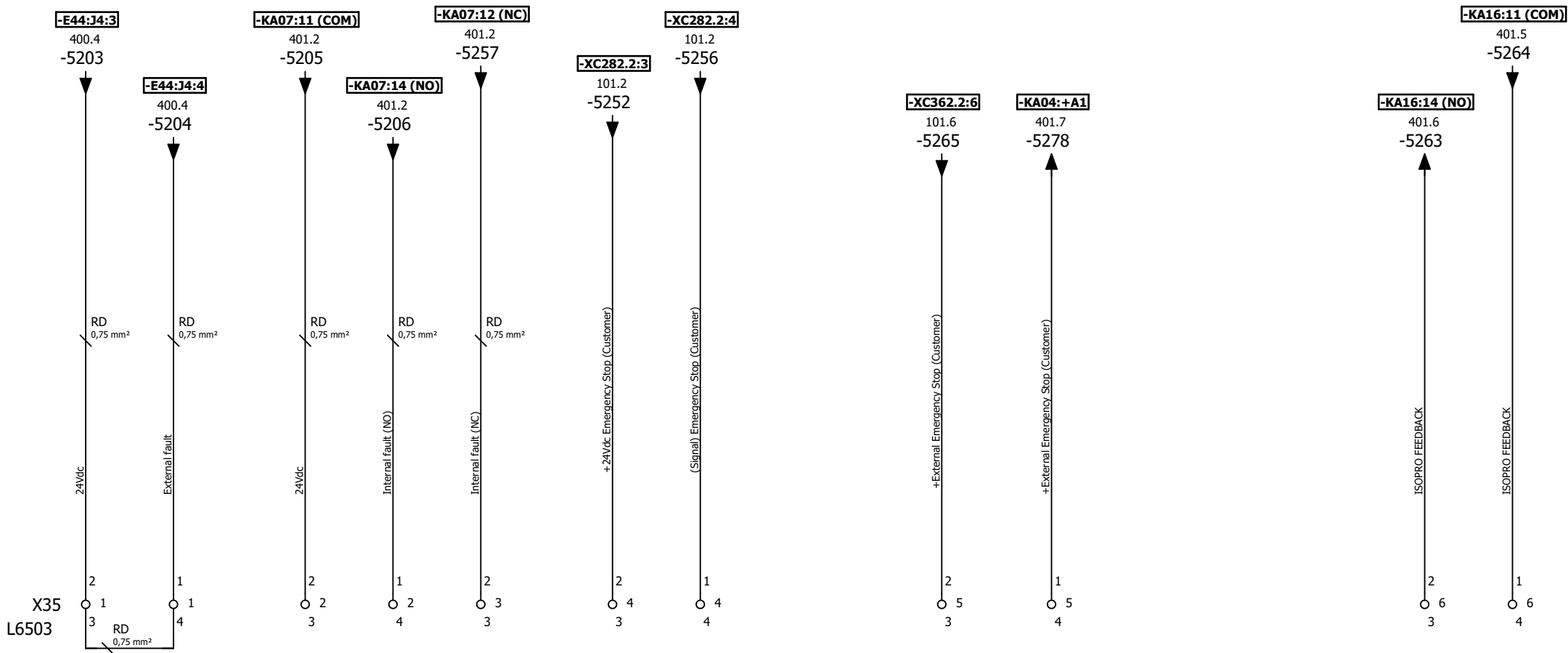
Customer connections



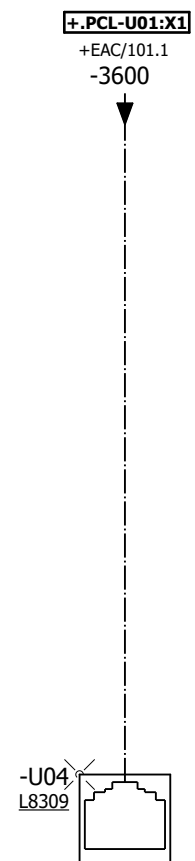


Customer signals

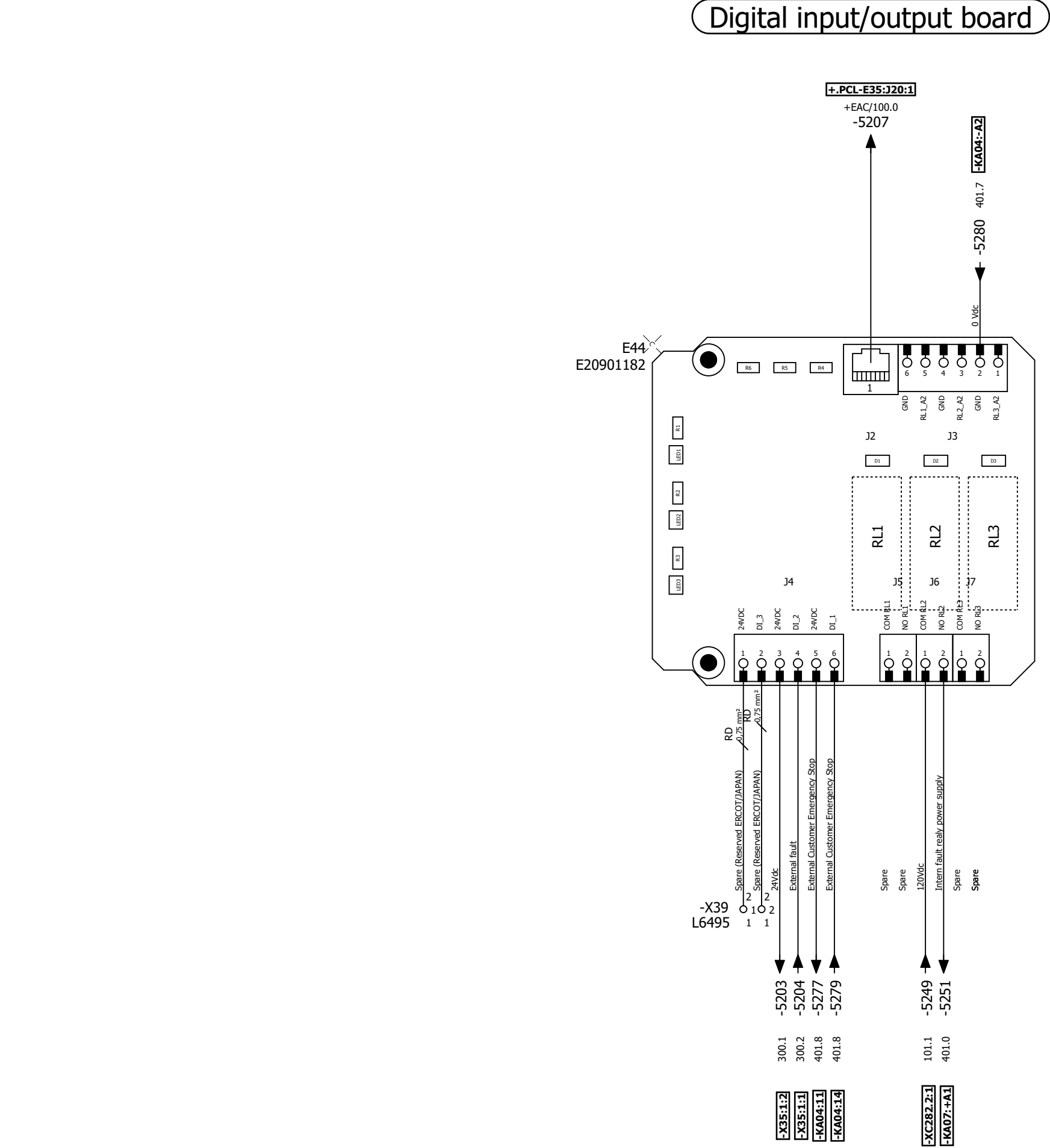
Digital input/output board



Customer signals



Borna RJ45



A3

Internal fault relay

Isopro fault relay

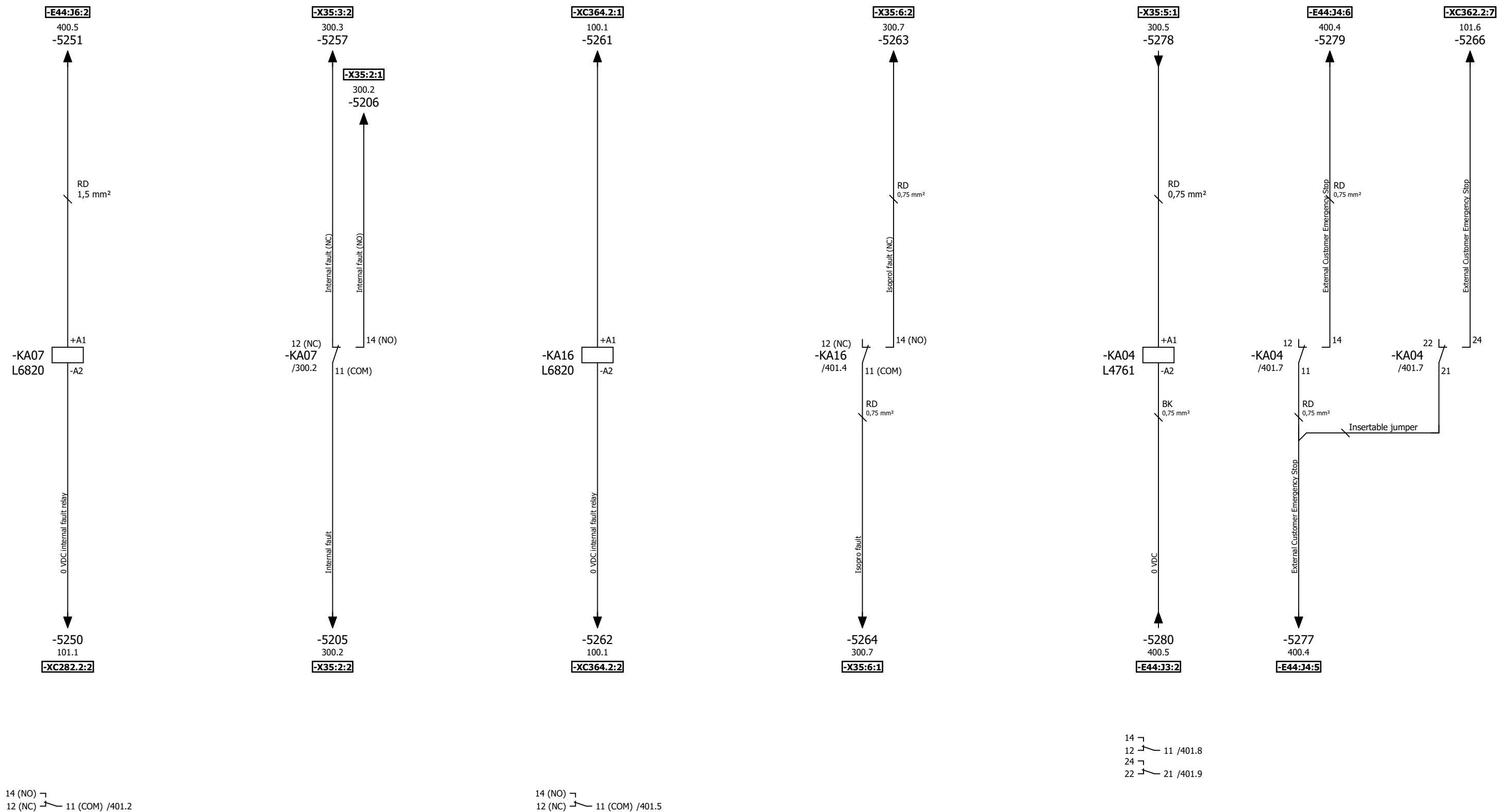
External
Customer Emergency

To digital output PCB

To customer connections

To PPD Isopro

To customer connections



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA3	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E44	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E44	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	2
-KA04	L4761	1123490000	RELAY 24VDC + BASE 2NO+2NC	1
-KA07	L6820	1122850000	RELAY, 24...230 V UC; 6 A;	1
-KA16	L6820	1122850000	RELAY, 24...230 V UC; 6 A;	1
-O.01	L4728	67981	SOCKET DOUBLE GFCI DIN RAIL	1
-O.01	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-Q30	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-Q30	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	2
-Q31	L6687	0626043563	CIRCUIT BREAKER,1P,10A, 277VAC,10kA	1
-Q32	L6687	0626043563	CIRCUIT BREAKER,1P,10A, 277VAC,10kA	1
-Q33	L6687	0626043563	CIRCUIT BREAKER,1P,10A, 277VAC,10kA	1
-Q34	L6687	0626043563	CIRCUIT BREAKER,1P,10A, 277VAC,10kA	1
-Q35	L6687	0626043563	CIRCUIT BREAKER,1P,10A, 277VAC,10kA	1
-Q36	L5702	06110176R0	CIRCUIT BRAKER 1Ph 4A CURVE "C"	1
-Q40	L6198	0628036963	CIRCUIT BREAKER, 3P, 50A, 480Y/277VAC ,10kA	1
-Q40	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-Q41	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-Q41	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-Q42	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-Q43	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-Q44	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-Q45	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-Q46	L6682	L6682	CIRCUIT BREAKER, 3P, 16A, 480Y/277VAC ,10kA	1
-U04	L8309	8946920000	Feed-through plug-in connector RJ45	1
-U04	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	2

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+EAC.PPCU-XC282.2	PANEL CONNECTOR - 6P INTERNAL FAULT POWER SUPPLY RELAY	L5777	1
		E1204	4
=HEM++ZMI+EAC.PPCU-XC362.2	PANEL CONNECTOR - 12P SPARE	E1375	1
=HEM++ZMI+EAC.PPCU-XC364.2	TRIPLE-LOK 2P ISOPRO FAULT SIGNAL	L9121_DUMMY	1
		L9146	1



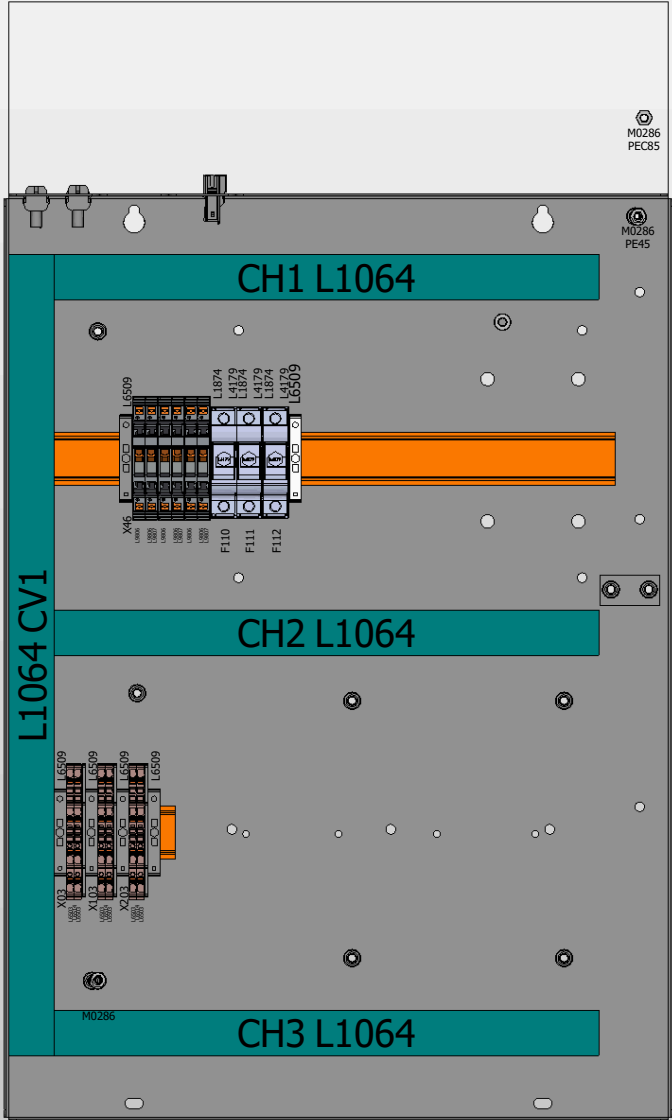
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
EAC.PTCU	CONTROL USER PANEL		
PTCU CONTROL USER PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

LAYOUT PANEL

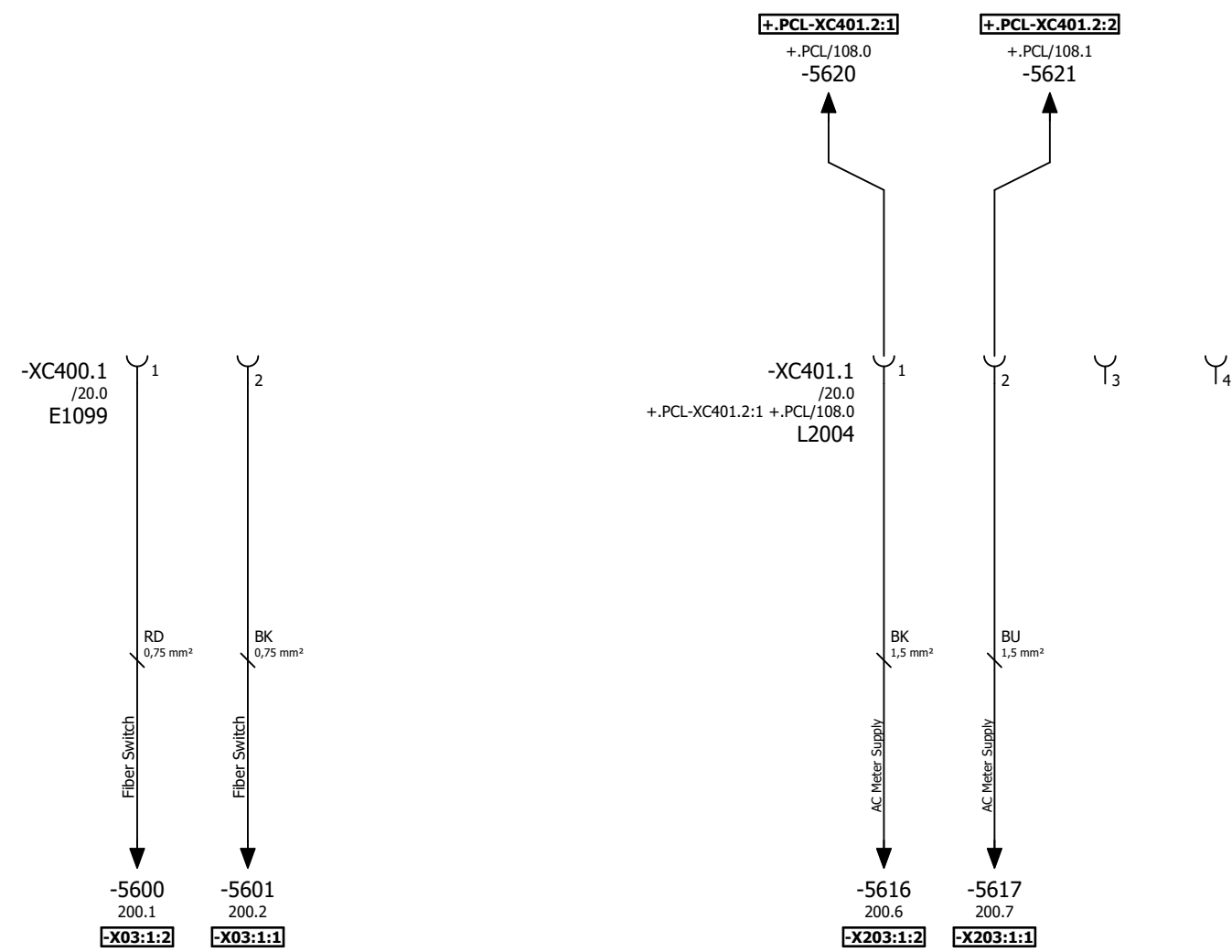
Enclosure legend

IME	ERP	Description
CA1;CA2	L1061	DIN RAIL 35x7,5mm
CH1-CH3;CV1	L1064	CABLE DUCT UNEX 42X30 EN U23X
F110-F112	L1874	FUSE BASE
F110-F112	L4179	FUSE 2A 690VAC gR 10X38 120KA



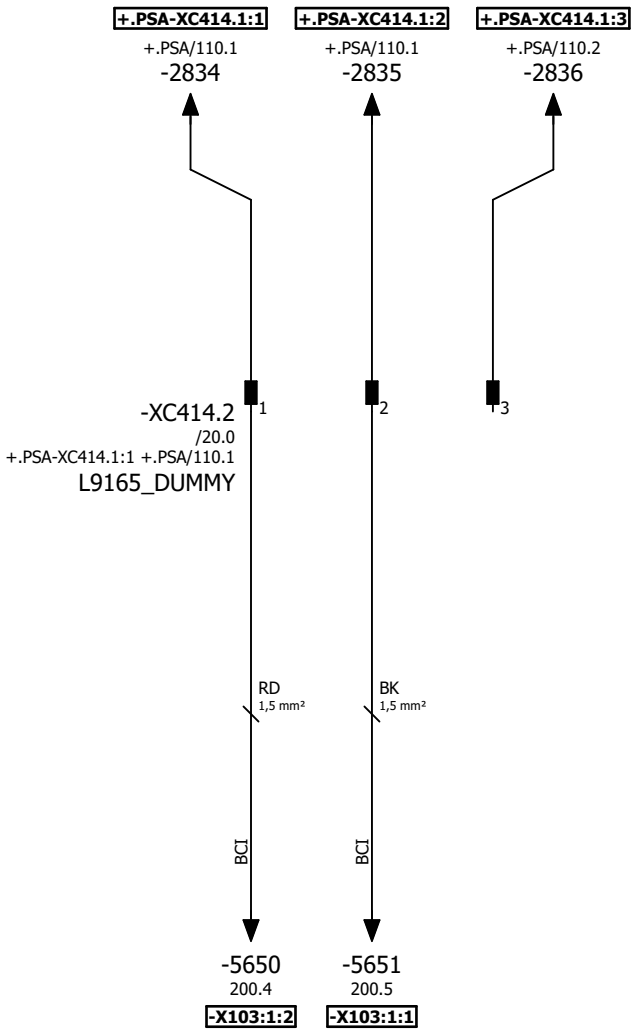
0	1	2	3	4	5	6	7	8	9
-XC400.1 = CONNECTOR AERIAL MATE-N-LOK PCL SUPPLY -XC401.1 = CONNECTOR AERIAL MATE-N-LOK PCL SUPPLY -XC414.2 = MALE CONNECTOR TRIPLE-LOCK 120VDC PSA -X03 = Distribution 24Vdc PTCU -X103 = Distribution 120Vdc PTCU -X203 = Distribution 230Vac PTCU -X46 = Terminal block for current transformers									
</									

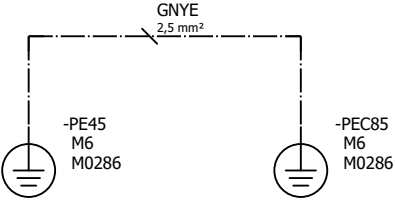
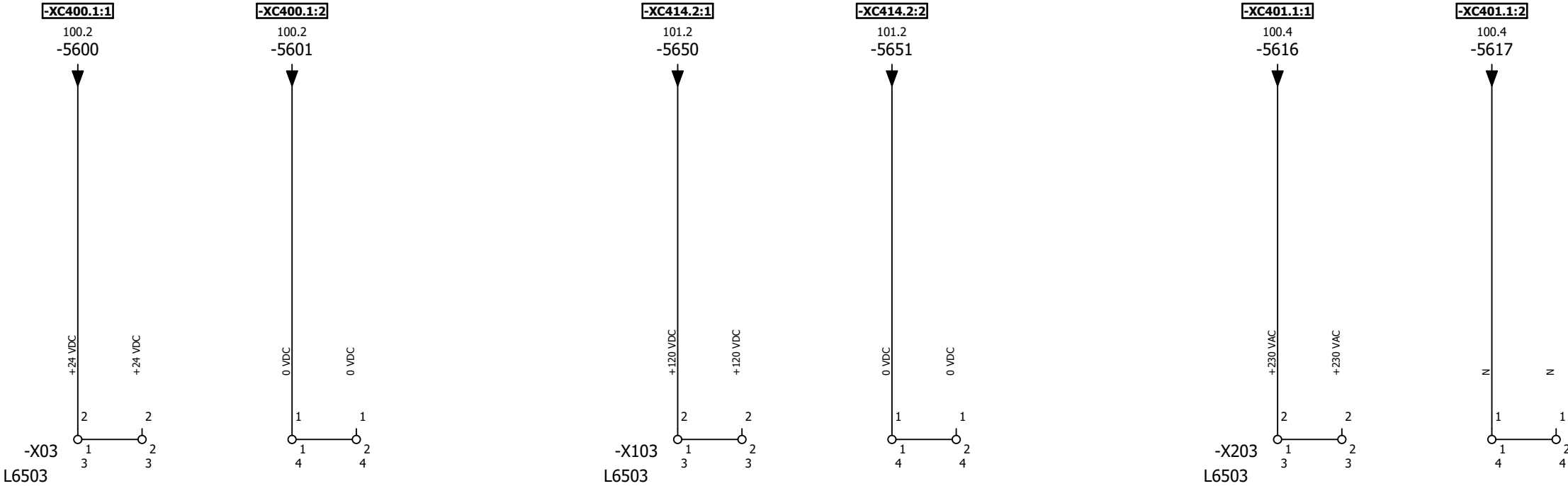
Interconnections PTCU//PCL

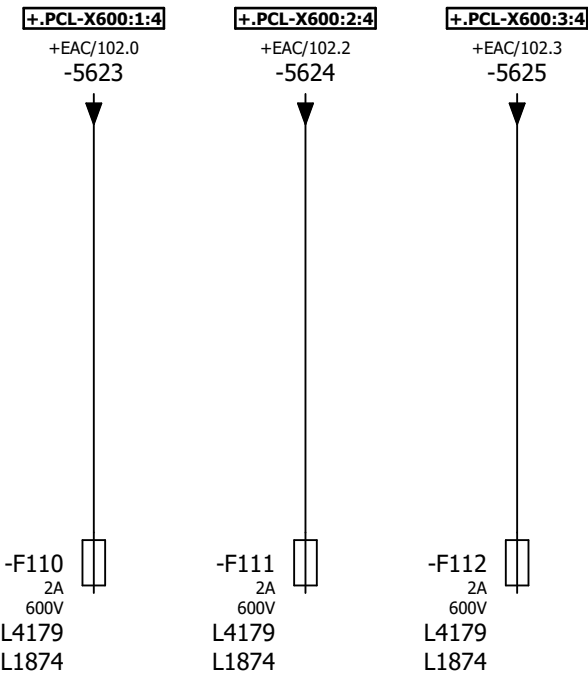
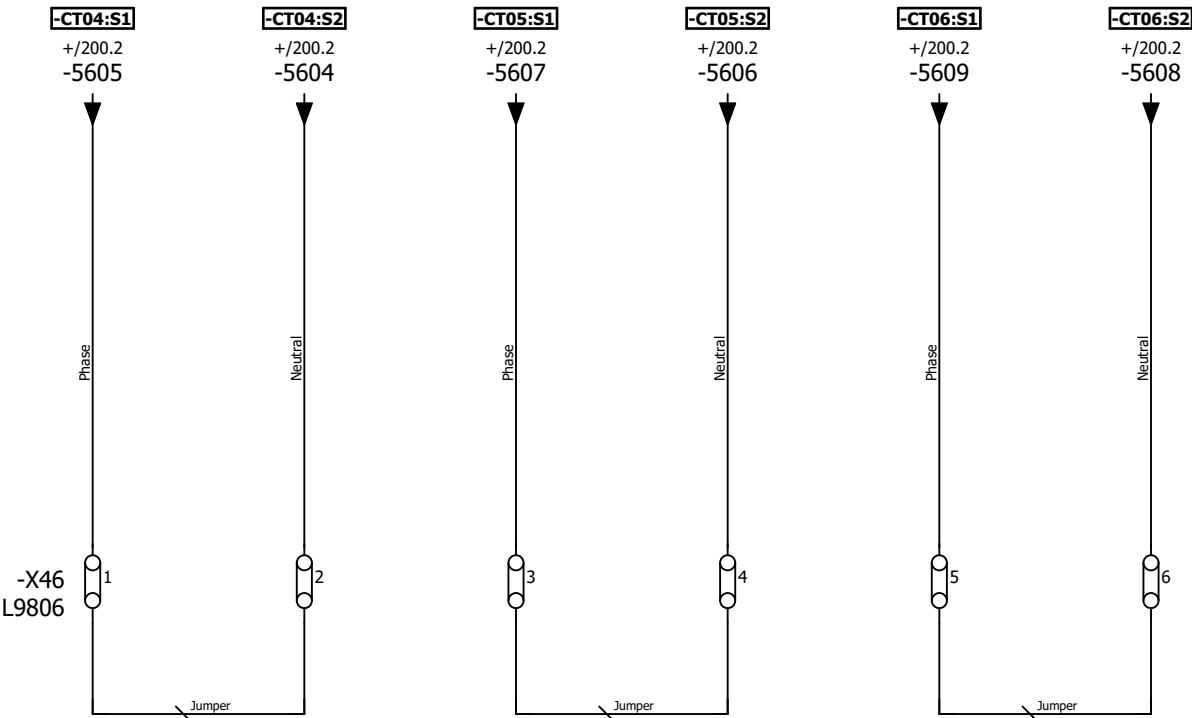


Interconnections optional

BCI







TERMINAL BLOCK FOR CURRENT TRANSFORMERS

Note: If the current transformers are not connected to the meter insert jumpers between the following terminal blocks X46:1-2, 3-4 y 5-6.

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA1	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA2	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-F110	L4179	FR10GR69V2	FUSE 2A 690VAC gR 10X38 120KA	1
-F110	L1874	CHPV1U	FUSE BASE	1
-F111	L4179	FR10GR69V2	FUSE 2A 690VAC gR 10X38 120KA	1
-F111	L1874	CHPV1U	FUSE BASE	1
-F112	L4179	FR10GR69V2	FUSE 2A 690VAC gR 10X38 120KA	1
-F112	L1874	CHPV1U	FUSE BASE	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+EAC.PTCU-XC400.1	CONNECTOR AERIAL MATE-N-LOK PCL SUPPLY	E1099	1
		E1203	2
=HEM++ZMI+EAC.PTCU-XC401.1	CONNECTOR AERIAL MATE-N-LOK PCL SUPPLY	L2004	1
		E1203	4
=HEM++ZMI+EAC.PTCU-XC414.2	MALE CONNECTOR TRIPLE-LOCK 120VDC PSA	L9165_DUMMY	1
		L9146	2

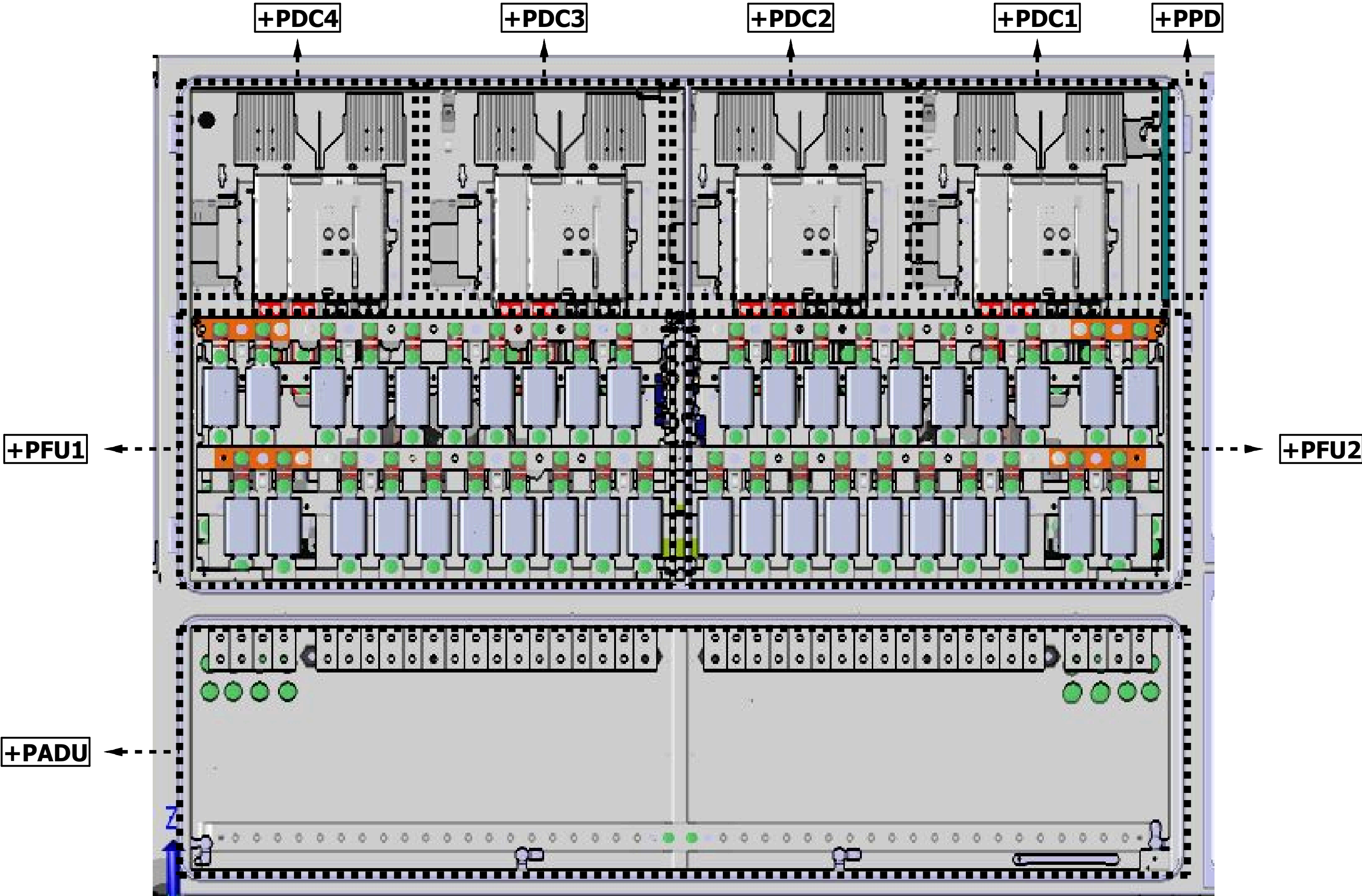


POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

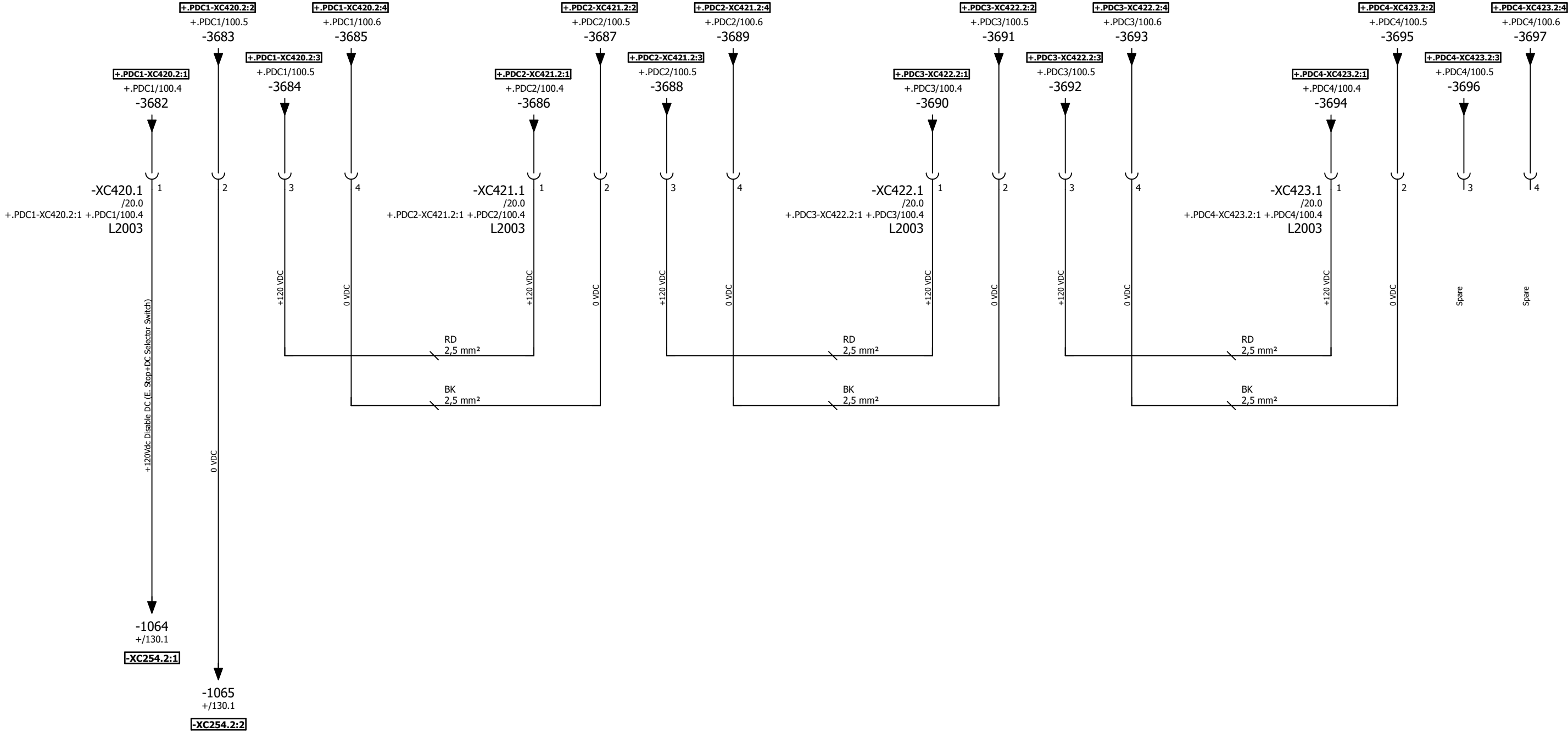
HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU	DU CABINET		
ADU DU CABINET			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

General layout

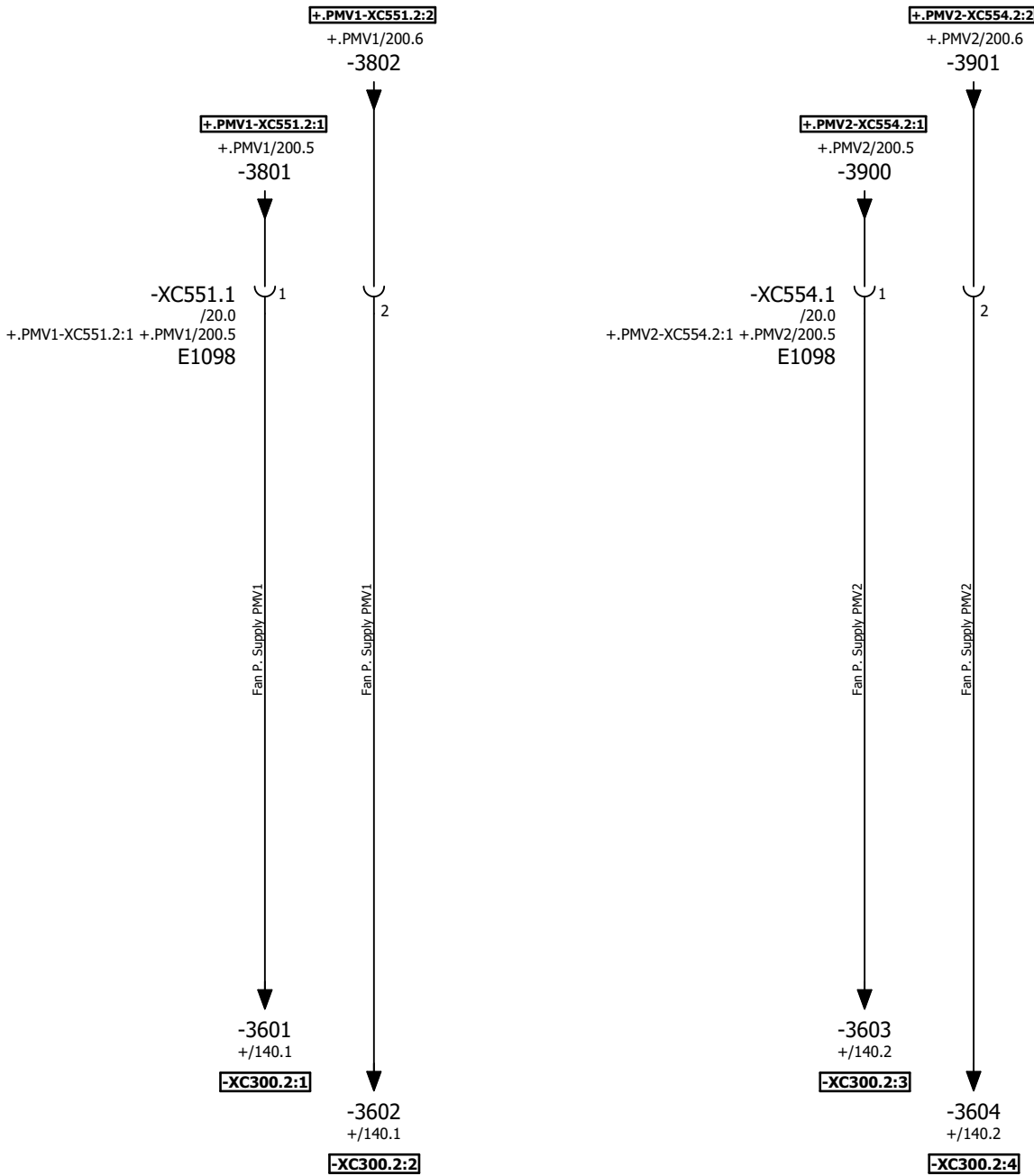
IMAGE AS A GUIDANCE MODE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

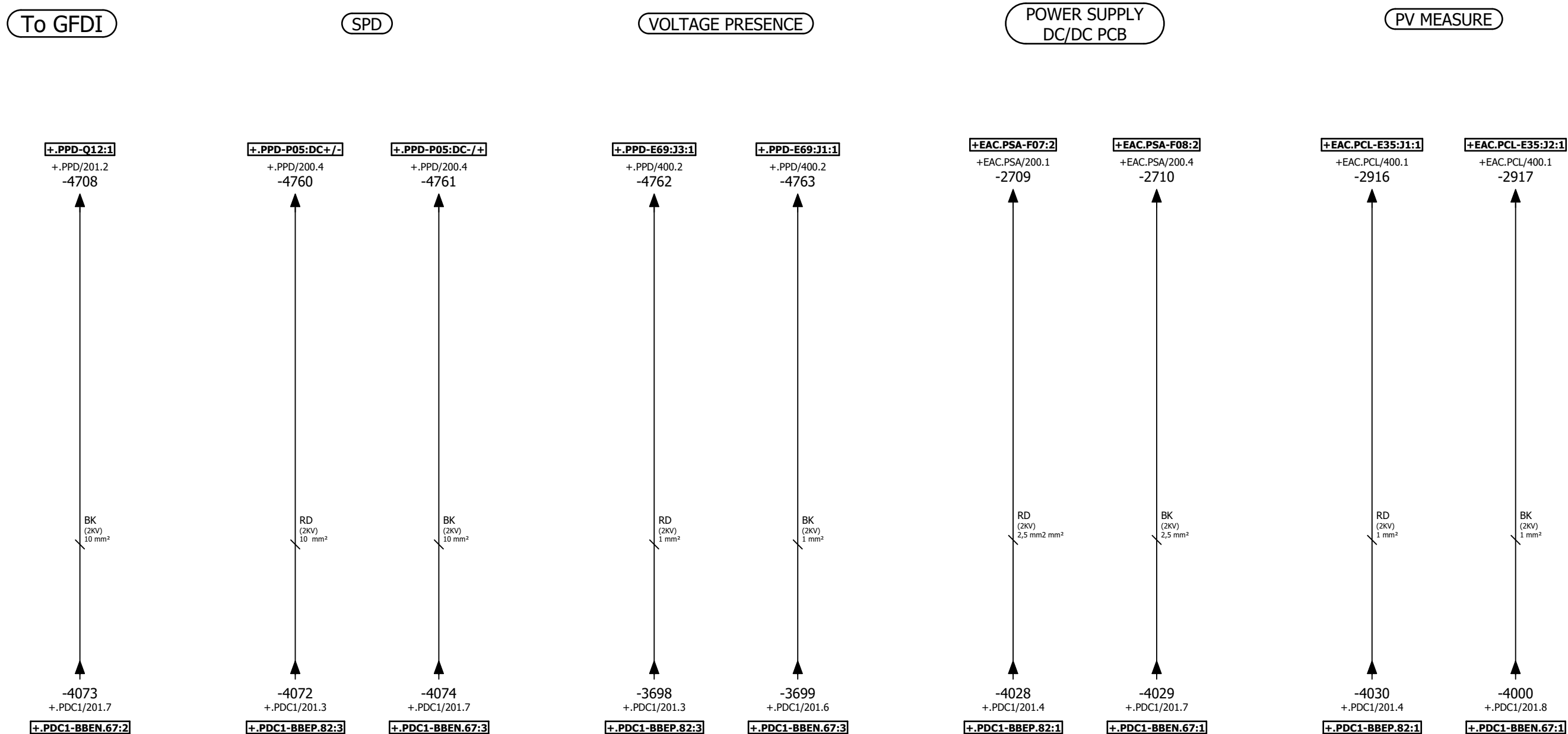


0	1	2	3	4	5	6	7	8	9
-XC420.1 = MATE-N-LOK AERIAL 4P DC MODULE 1 DISTRIBUTION -XC421.1 = MATE-N-LOK AERIAL 4P DC MODULE 2 DISTRIBUTION -XC422.1 = MATE-N-LOK AERIAL 4P DC MODULE 3 DISTRIBUTION -XC423.1 = MATE-N-LOK AERIAL 4P DC MODULE 4 DISTRIBUTION -XC551.1 = MATE-N-LOK AERIAL 2P DU FANS -XC554.1 = MATE-N-LOK AERIAL 2P DU FANS -XC424.1 = CONNECTOR OPTICAL FIBER DUPLEX MEASURES DU -XC416.1 = TRIPLE-LOK 2P ISOPRO FAULT SIGNAL									



Interconnections - Fan-DC Meter PCL//ADU

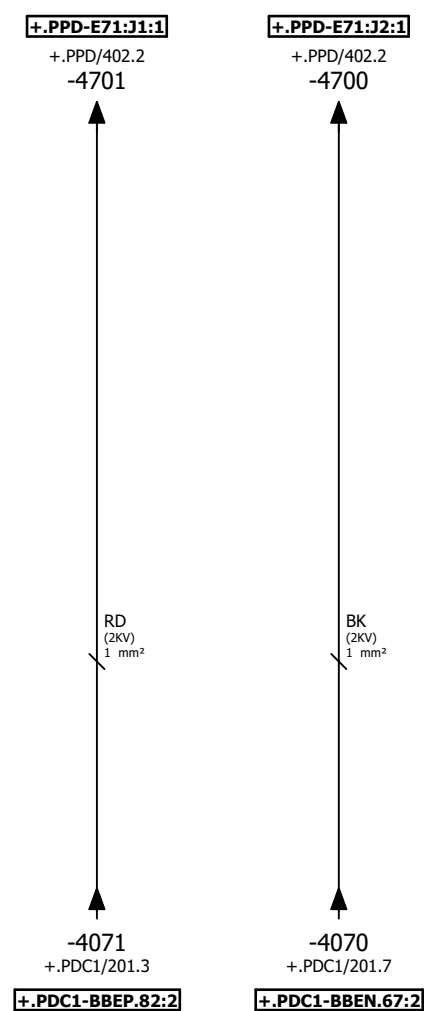




FROM DC DISCONNECTOR PRE-ASSEMBLY 1 (+PDC1)

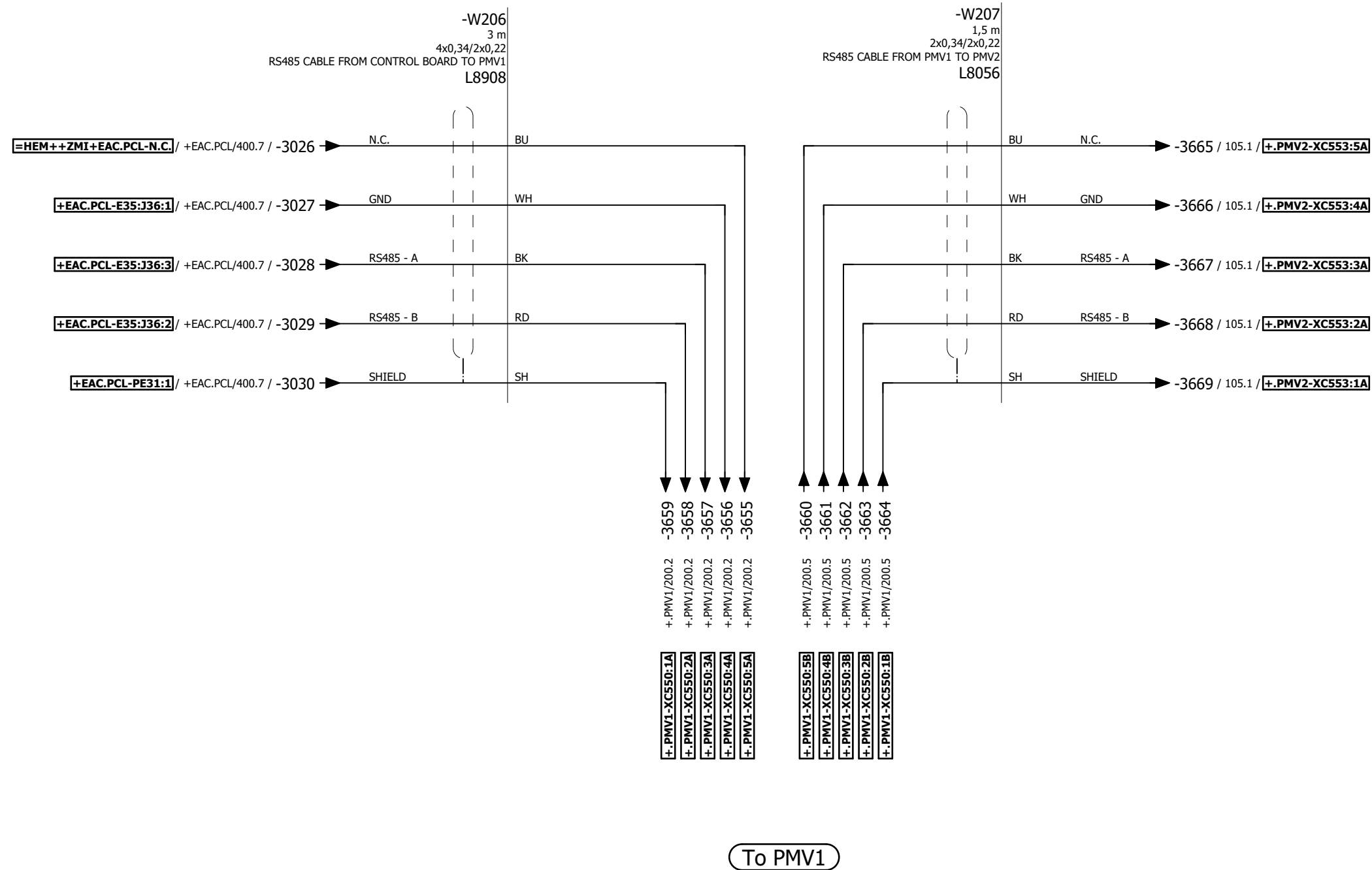
PV STRINGS

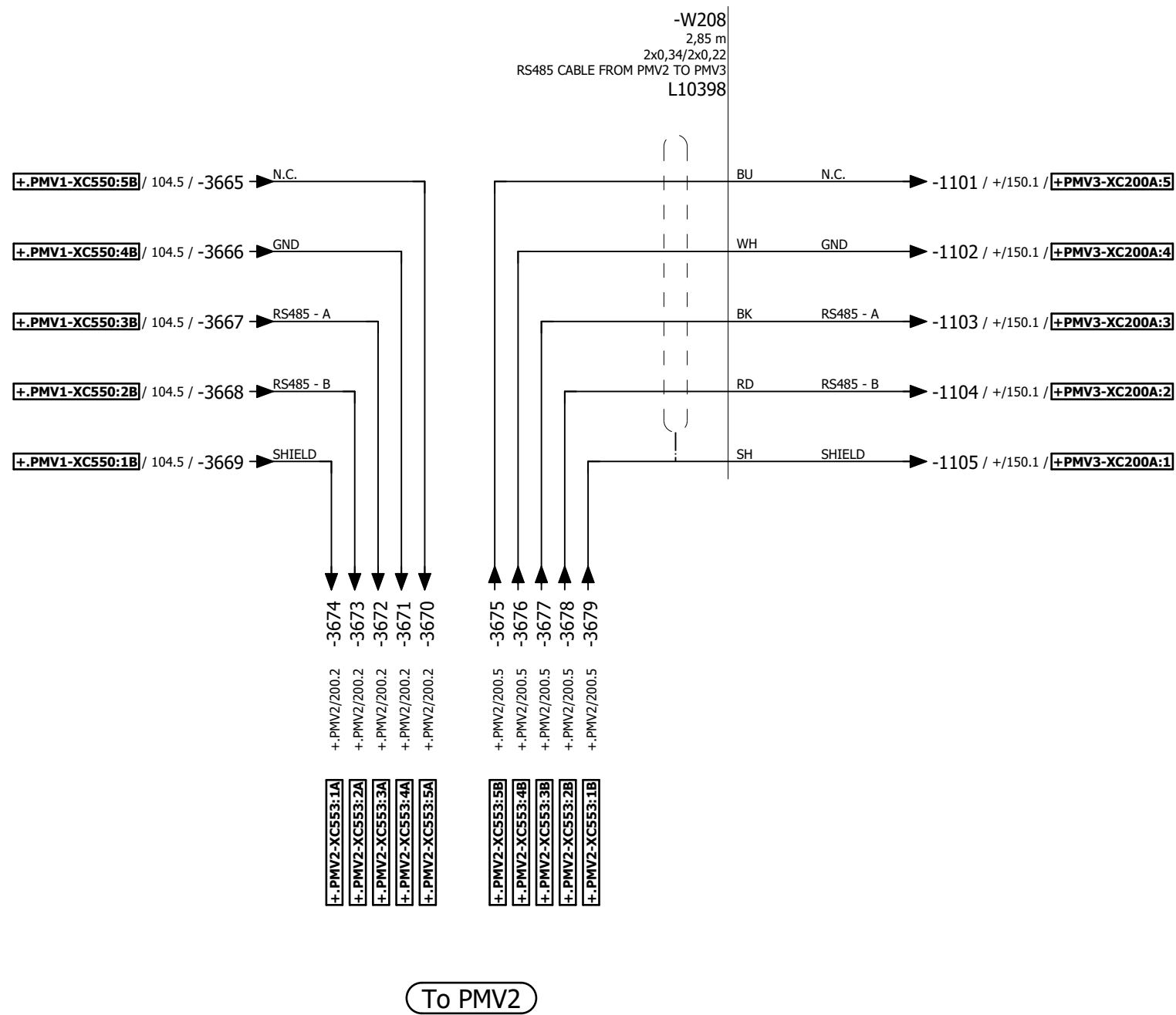
TO DC PROTECTION PANEL (+PPD)



FROM DC DISCONNECTOR
PRE-ASSEMBLY 1 (+PDC1)

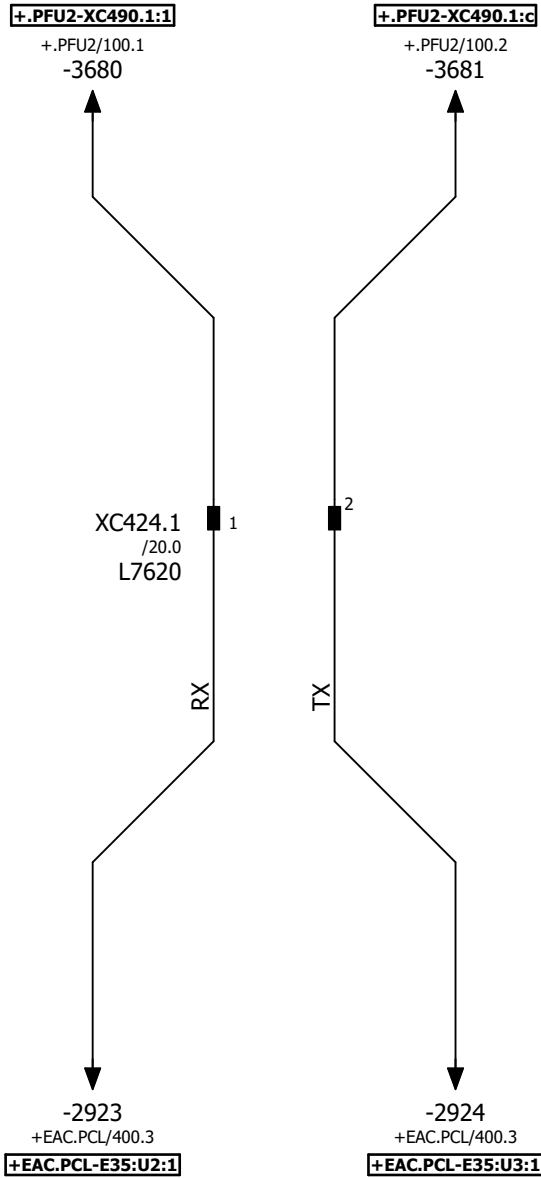
FROM DC/AC MODULES SIDE





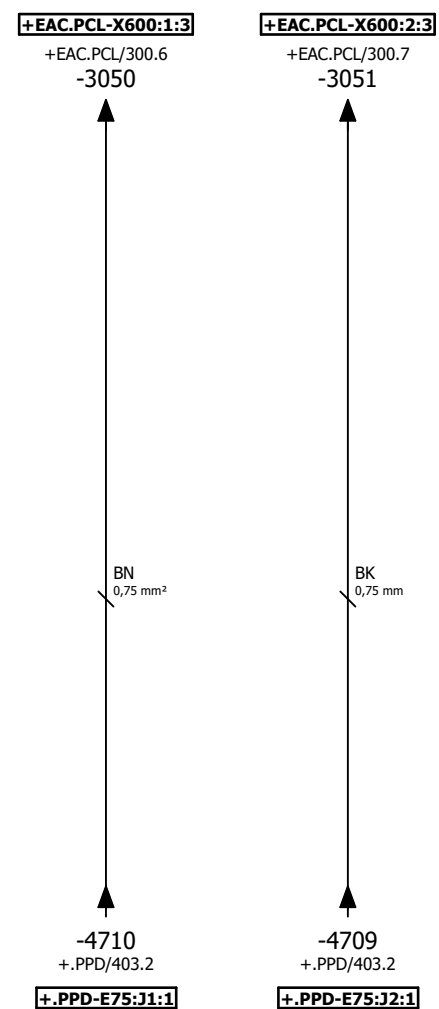
Interconnections F.O.
PCL//ADU

TO PCB DU CURRENT MEASURES



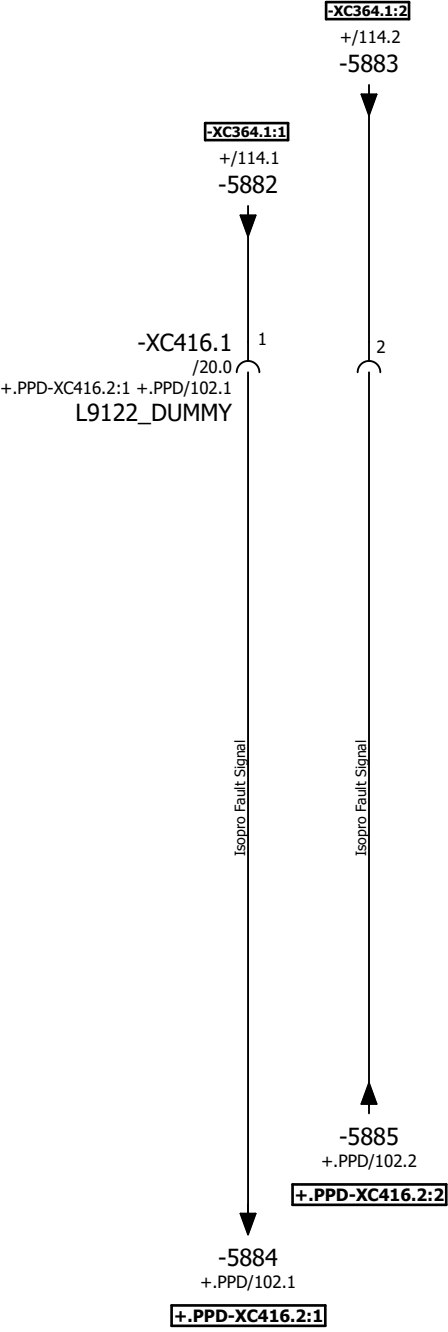
TO PCB E35 (U2,U3)

AC MEASURE FROM CONTROL PANEL (+PCL)

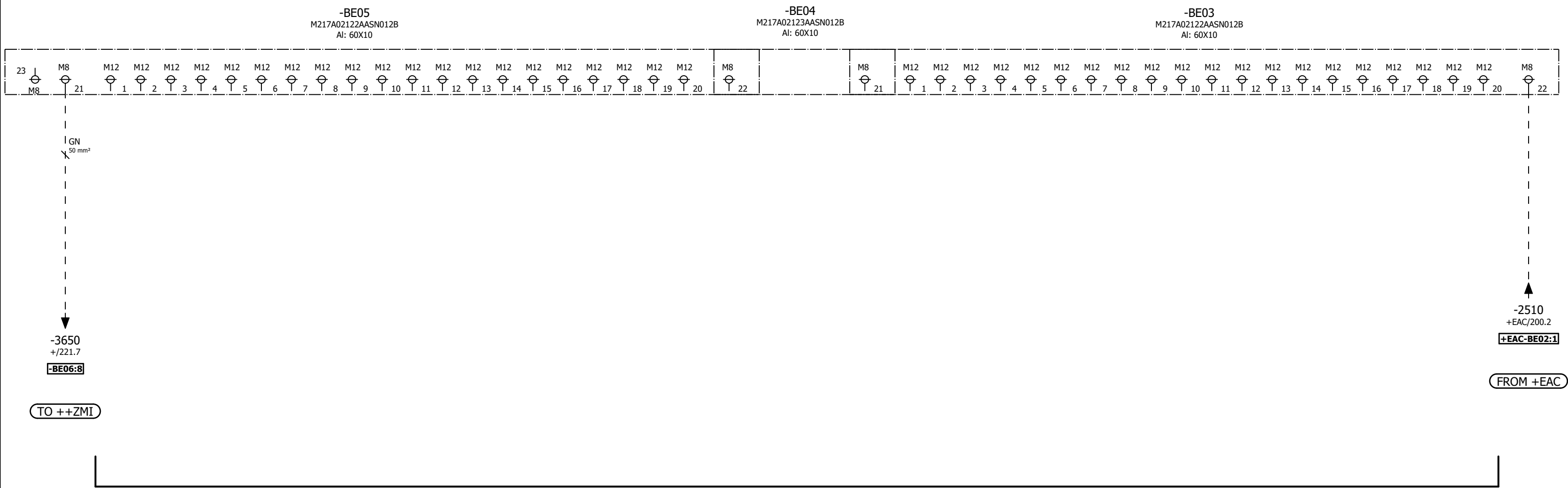


TO DC PROTECTION PANEL (+PPD)

Interconnections PPD//PCDU



Earth connection



EARTH CONNECTION FROM PV SYSTEM AND BESS SYSTEM

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BE03	M217A02122AASN012B	M217A02122AASN012B	EARTH BUSBAR 20 HOLES M12	1
-BE04	M217A02123AASN012B	M217A02123AASN012B	EARTH BUSBAR JOINT	1
-BE05	M217A02122AASN012B	M217A02122AASN012B	EARTH BUSBAR 20 HOLES M12	1

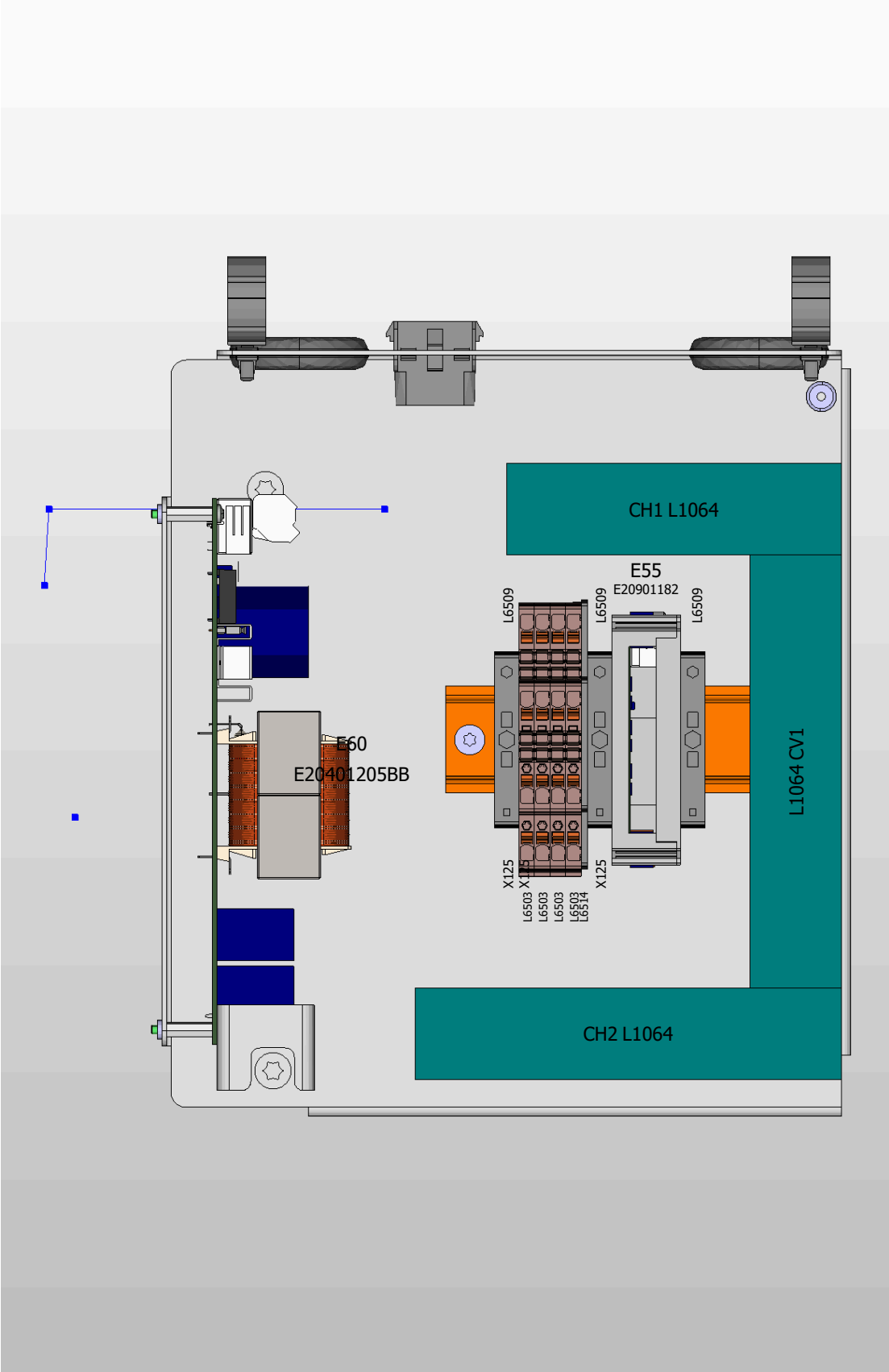
Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+ADU-W206	RS485 CABLE FROM CONTROL BOARD TO PMV1	L8908
=HEM++ZMI+ADU-W207	RS485 CABLE FROM PMV1 TO PMV2	L8056
=HEM++ZMI+ADU-W208	RS485 CABLE FROM PMV2 TO PMV3	L10398

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU-XC416.1	TRIPLE-LOK 2P ISOPRO FAULT SIGNAL	L9122_DUMMY	1
		L9145	2
=HEM++ZMI+ADU-XC420.1	MATE-N-LOK AERIAL 4P DC MODULE 1 DISTRIBUTION	L2003	1
		L5709	4
=HEM++ZMI+ADU-XC421.1	MATE-N-LOK AERIAL 4P DC MODULE 2 DISTRIBUTION	L2003	1
		L5709	4
=HEM++ZMI+ADU-XC422.1	MATE-N-LOK AERIAL 4P DC MODULE 3 DISTRIBUTION	L2003	1
		L5709	4
=HEM++ZMI+ADU-XC423.1	MATE-N-LOK AERIAL 4P DC MODULE 4 DISTRIBUTION	L2003	1
		L5709	4
=HEM++ZMI+ADU-XC424.1	CONNECTOR OPTICAL FIBER DUPLEX MEASURES DU	L7620	1
=HEM++ZMI+ADU-XC551.1	MATE-N-LOK AERIAL 2P DU FANS	E1098	1
		L5709	2
=HEM++ZMI+ADU-XC554.1	MATE-N-LOK AERIAL 2P DU FANS	E1098	1
		L5709	2

LAYOUT PANEL



Enclosure legend

IME	ERP	Description
E60	E20401205BB	SOFT LOAD BOARD
CA1	L1061	DIN RAIL 35x7,5mm
E55	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
CH1;CH2;CV1	L1064	CABLE DUCT UNEX 42X30 EN U23X

0	1	2	3	4	5	6	7	8	9
-XC470.1 = CONNECTOR DISCONNECTOR SWITCH 1 - OPERATING -XC471.1 = CONNECTOR DISCONNECTOR SWITCH 1 - AUX. -XC420.2 = MATE-N-LOK PANEL 4P DC MODULE 1 DISTRIBUTION -X125 = Distribution 120 Vdc									



TO MODULE 1

++MOI+ETR-E02:J13:1

++MOI+/102.0
-038



-4021
400.1

-E55:J2:1

++MOI+ETR-E02:J11:1

++MOI+/102.1
-039



-4022
400.4

-E60:J2:1

+ADU-XC420.1:2

+ADU/100.1
-3683

+ADU-XC420.1:4

+ADU/100.2
-3685

+ADU-XC420.1:1

+ADU/100.1
-3682

+ADU-XC420.1:3

+ADU/100.1
-3684

1

2

3

4

-XC420.2
/20.0
+ADU-XC420.1:1 +ADU/100.1
L2004

+120 VDC

0 VDC

+120 VDC

0 VDC

-4060
301.1

-4061
301.3

-4062
301.1

-4063
301.3

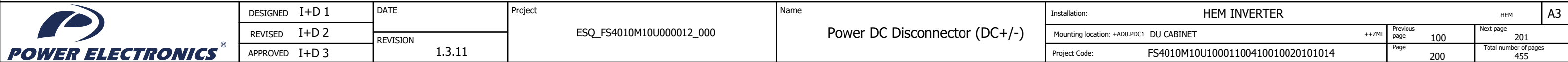
-X125:1:2

-X125:1:1

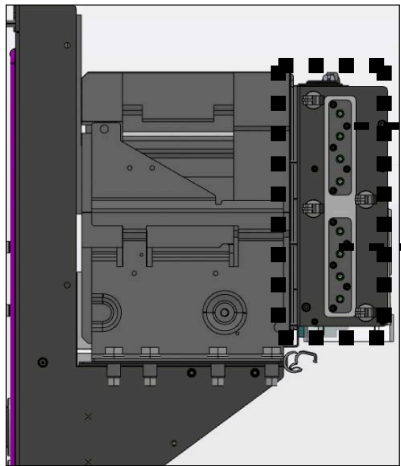
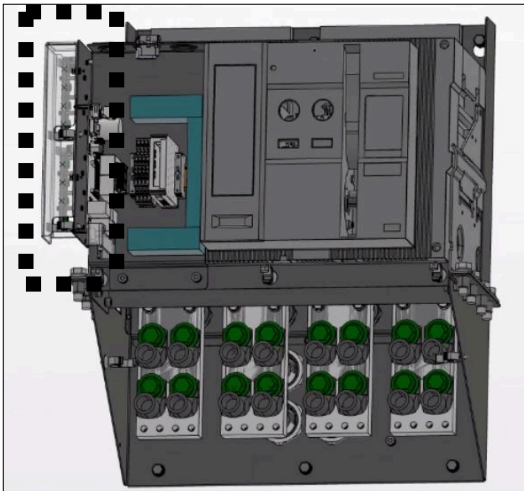
-X125:2:2

-X125:2:1

FROM PCB'S
DISCONNECTOR



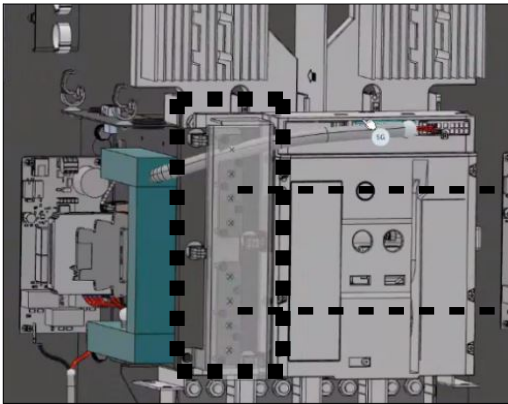
SWITCH DISCONNECTOR 2500A



BBEN.67

BBEP.82

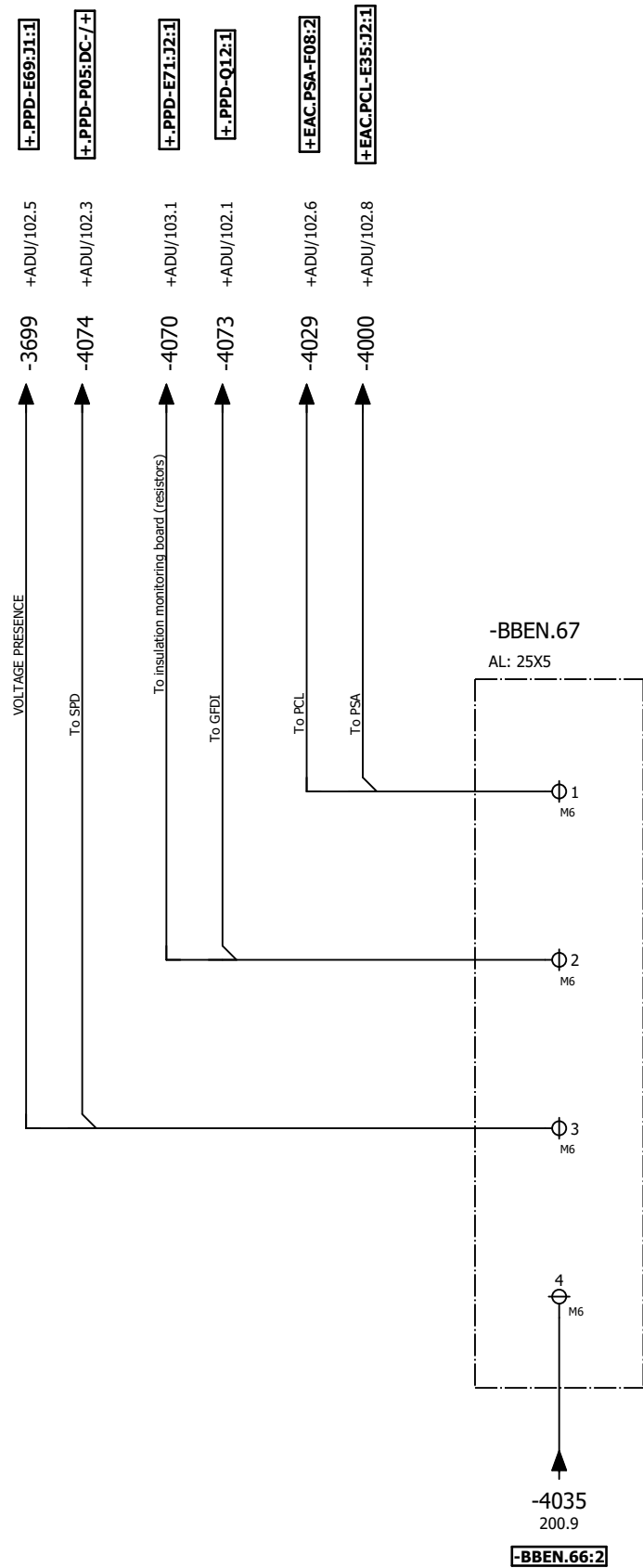
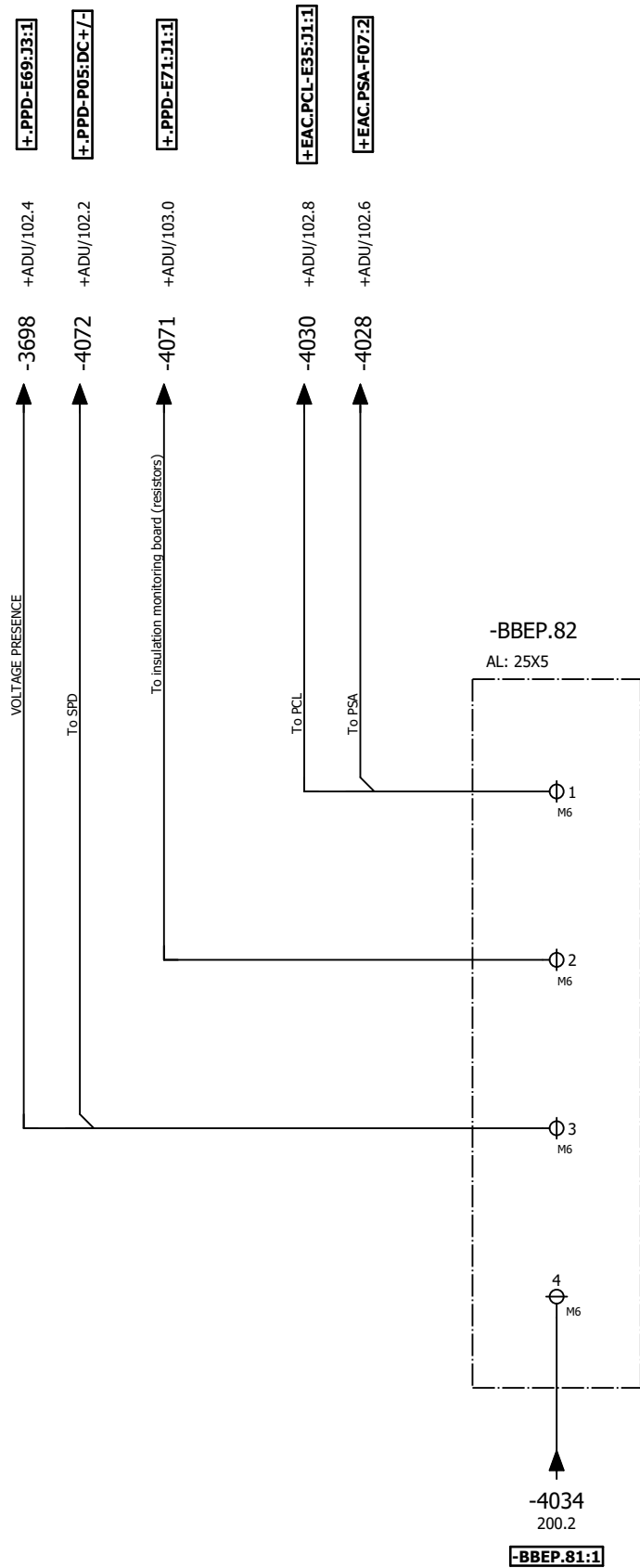
SWITCH DISCONNECTOR 1600A

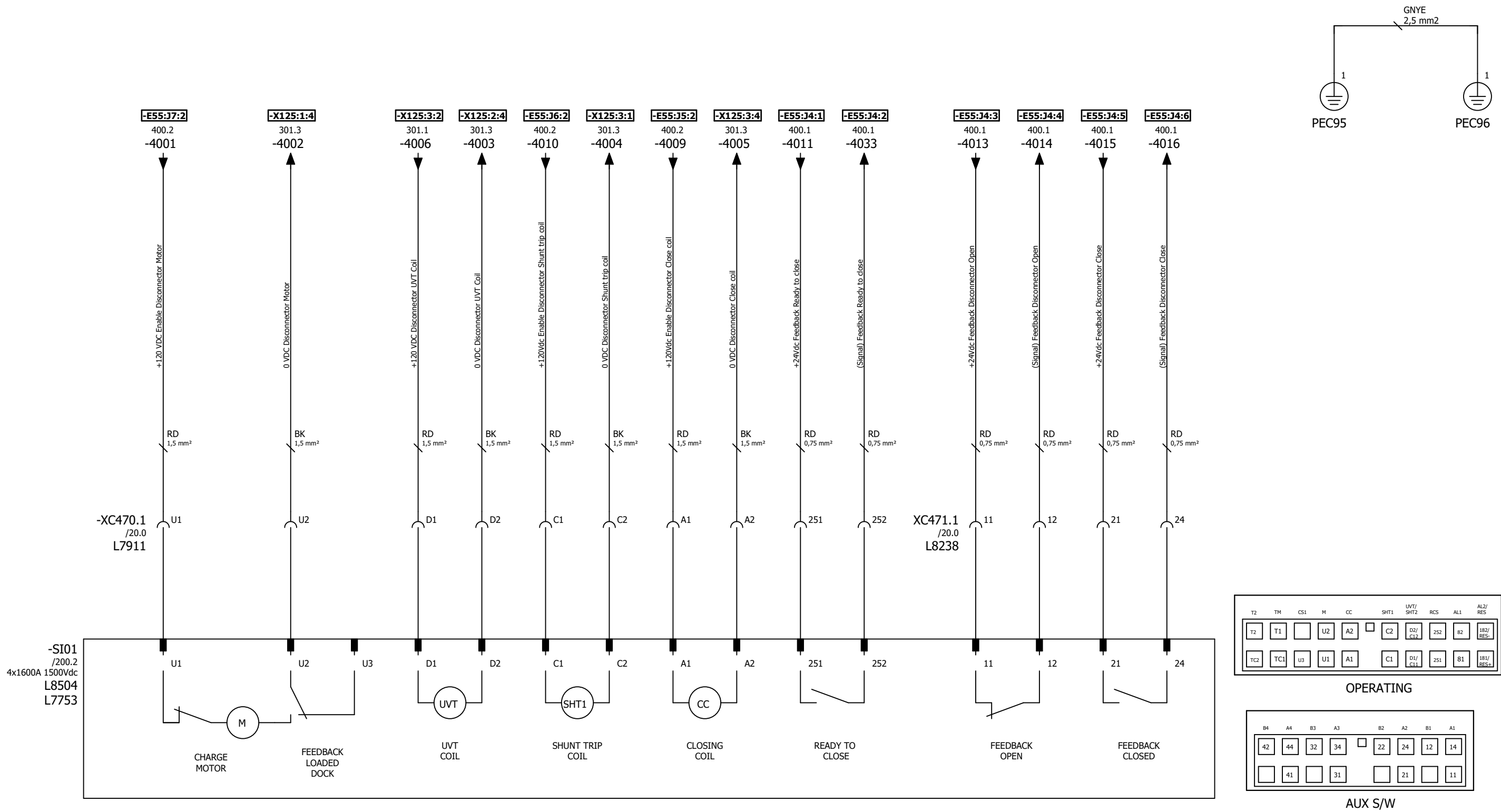


BBEN.67

BBEP.82

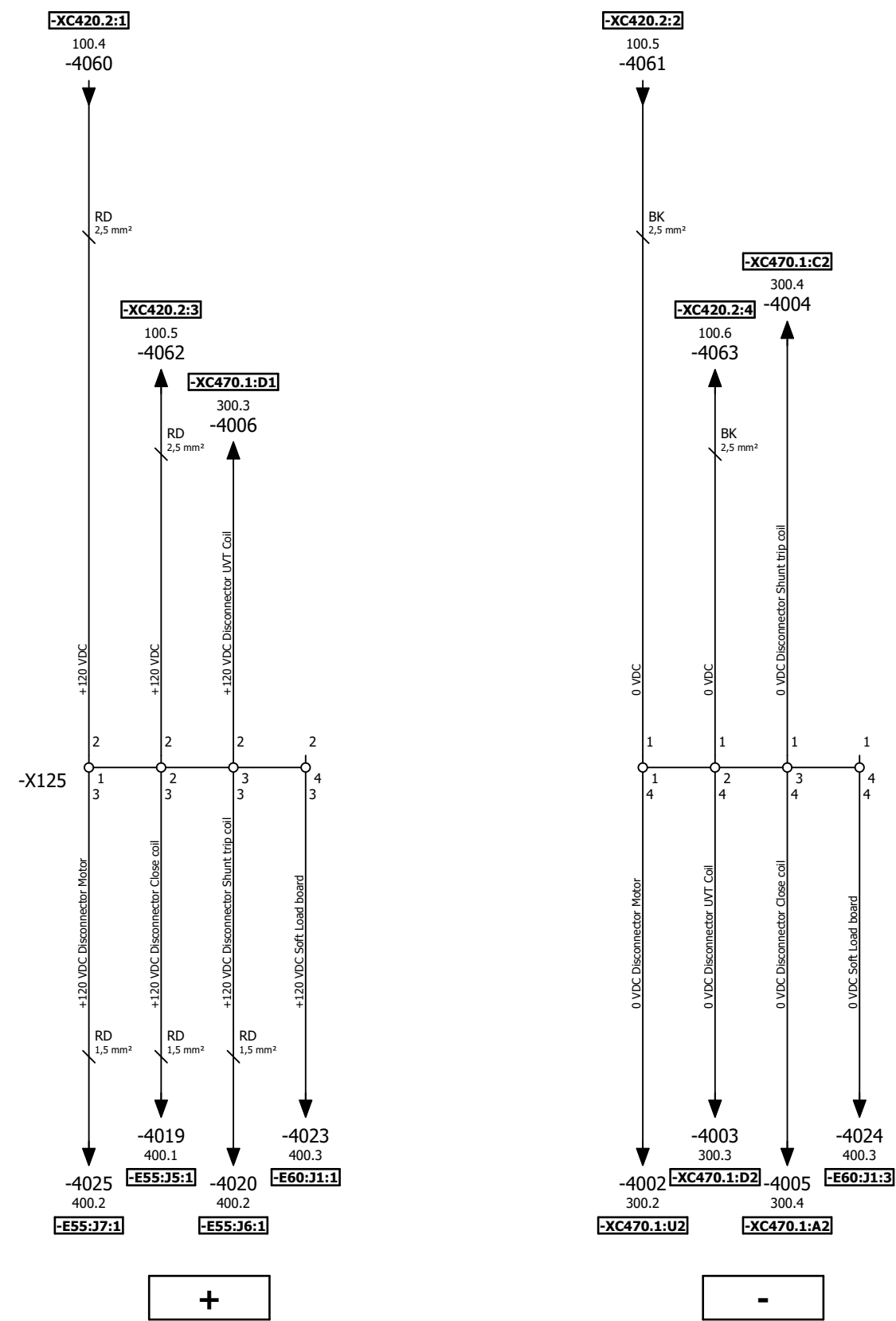
THE IMAGE MAY VARY DEPENDING ON THE DISCONNECTOR SWITCH MODEL.
THE CONNECTIONS AND IME OF THE PLATES ARE IDENTICAL.



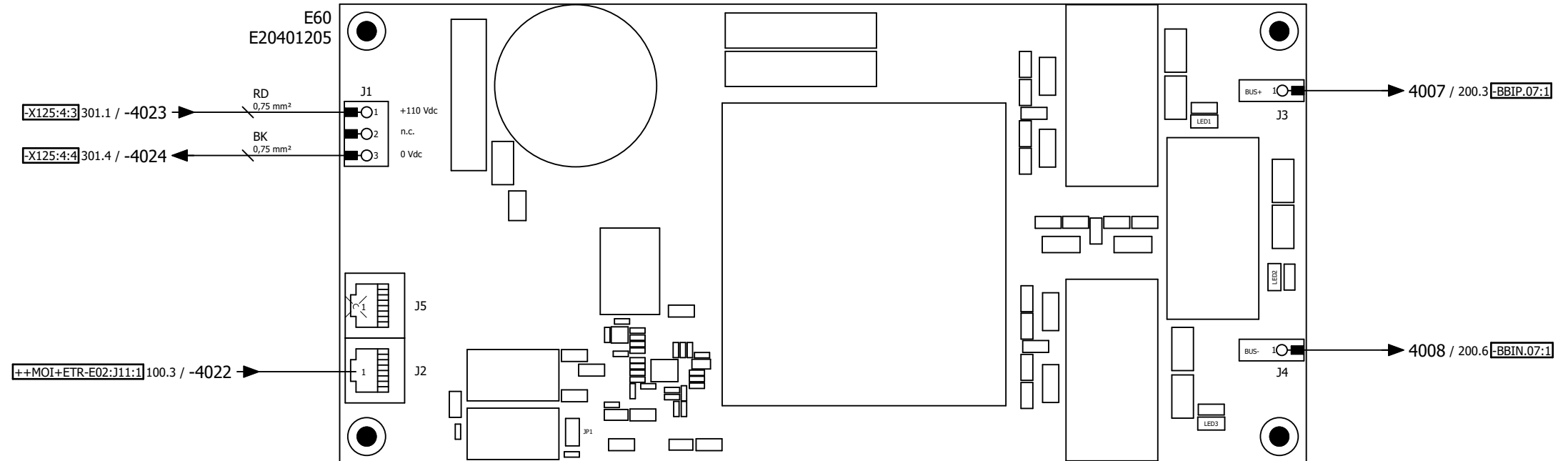
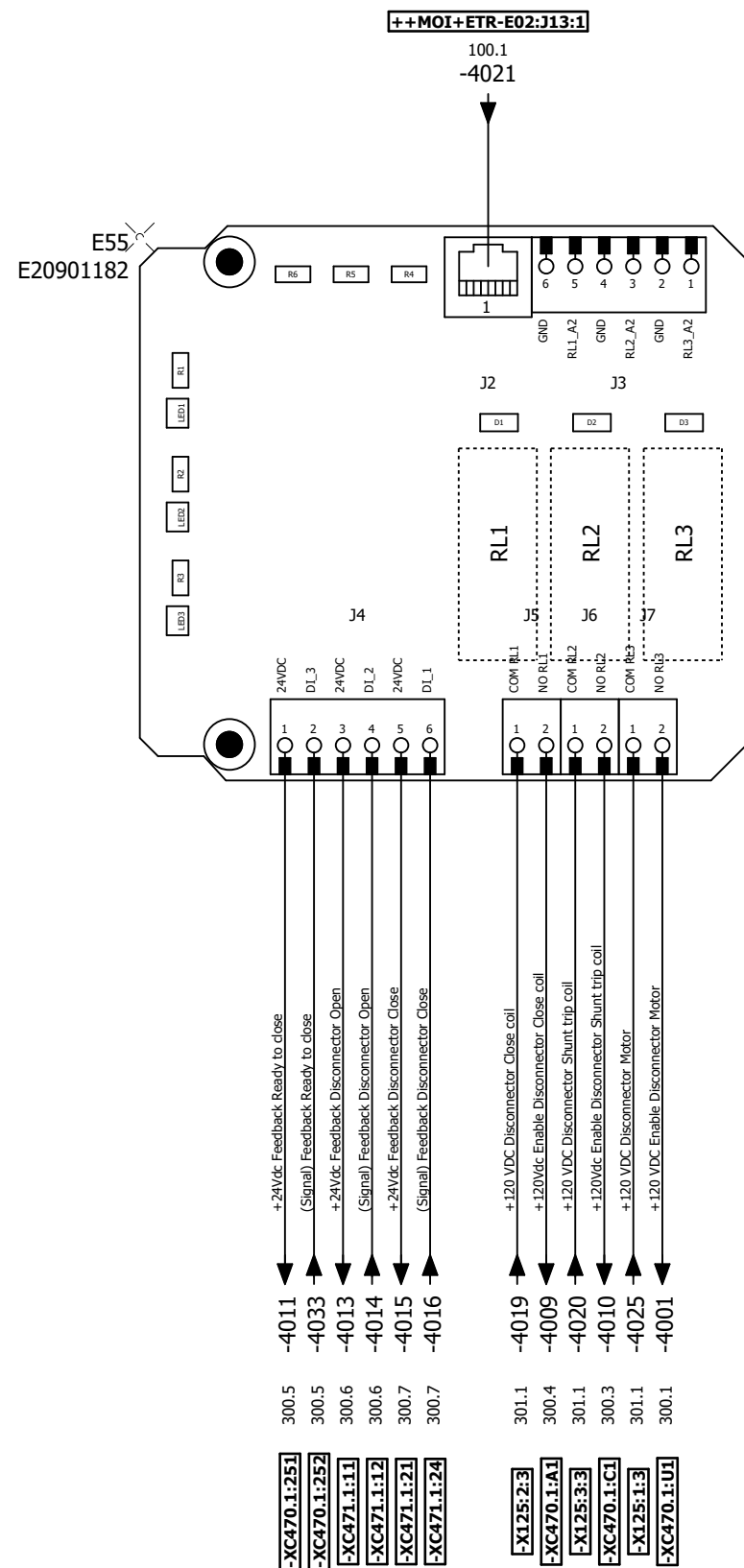


This diagram is based on "CONNECTED" position of a circuit breaker and Opening, Motor charging, Releasing of locking plate should be normal condition.

Distribution 120Vdc



PCB - I/O DC DISCONNECTOR



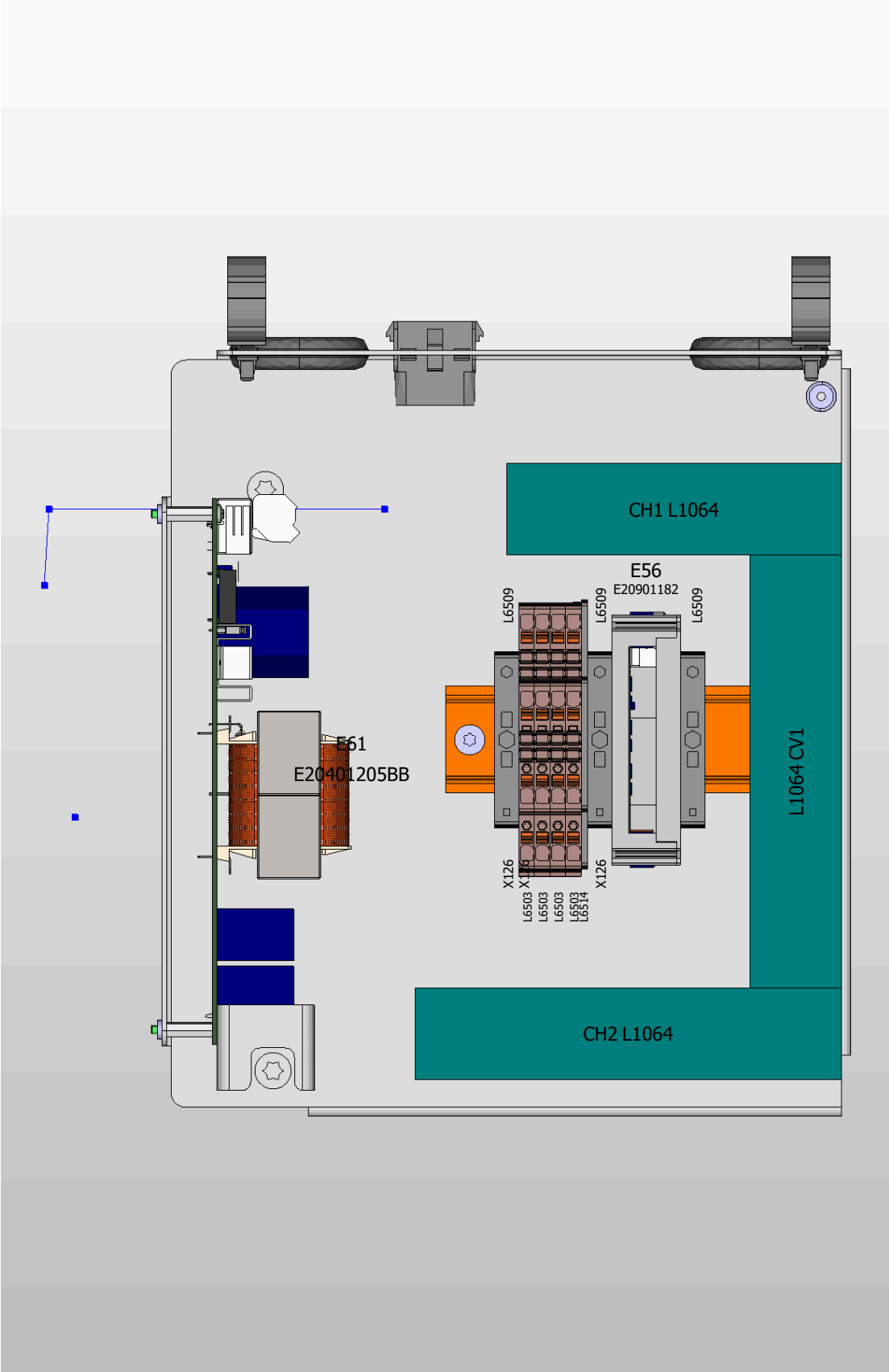
BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E55	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E55	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E60	E20401205BB	E20401205BB	SOFT LOAD BOARD	1
-SI01	L8504	2950009300	4x1600A-C-P IEC DISCONNECTOR SWITCH	1
-SI01	L7753	70223472602	Short bars + heatsink for DDV-16C4-00P	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PDC1-XC420.2	MATE-N-LOK PANEL 4P DC MODULE 1 DISTRIBUTION	L2004	1
		L5710	4
=HEM++ZMI+ADU.PDC1-XC470.1	CONNECTOR DISCONNECTOR SWITCH 1 - OPERATING	L7911	1
		L7912	10
=HEM++ZMI+ADU.PDC1-XC471.1	CONNECTOR DISCONNECTOR SWITCH 1 - AUX.	L8238	1
		L7912	4

LAYOUT PANEL



Enclosure legend

IME	ERP	Description
E61	E20401205BB	SOFT LOAD BOARD
CA1	L1061	DIN RAIL 35x7,5mm
E56	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
CH1;CH2;CV1	L1064	CABLE DUCT UNEX 42X30 EN U23X

0	1	2	3	4	5	6	7	8	9
-XC475.1 = CONNECTOR DISCONNECTOR SWITCH 2 - OPERATING -XC476.1 = CONNECTOR DISCONNECTOR SWITCH 2 - AUX. -XC421.2 = MATE-N-LOK PANEL 4P DC MODULE 2 DISTRIBUTION -X126 = Distribution 120 Vdc									



TO MODULE 2

++MOI-MOD2:E02:J13

++MOI+/102.3
-040



I/O PCB - DC DISCONNECTOR



-4121
400.1

-E56:J2:1

++MOI-MOD2:E02:J11

++MOI+/102.4
-041



SOFT LOAD



-4122
400.4

-E61:J2:1

+ADU-XC421.1:1

+ADU/100.3
-3686



1

-XC421.2
/20.0
+ADU-XC421.1:1 +ADU/100.3
L2004

+120 VDC



-4160
301.1

-X126:1:2

+ADU-XC421.1:2

+ADU/100.3
-3687



2

0 VDC



-4162
301.3

-X126:1:1

+ADU-XC421.1:3

+ADU/100.4
-3688



3

+120 VDC



-4161
301.1

-X126:2:2

+ADU-XC421.1:4

+ADU/100.4
-3689



4

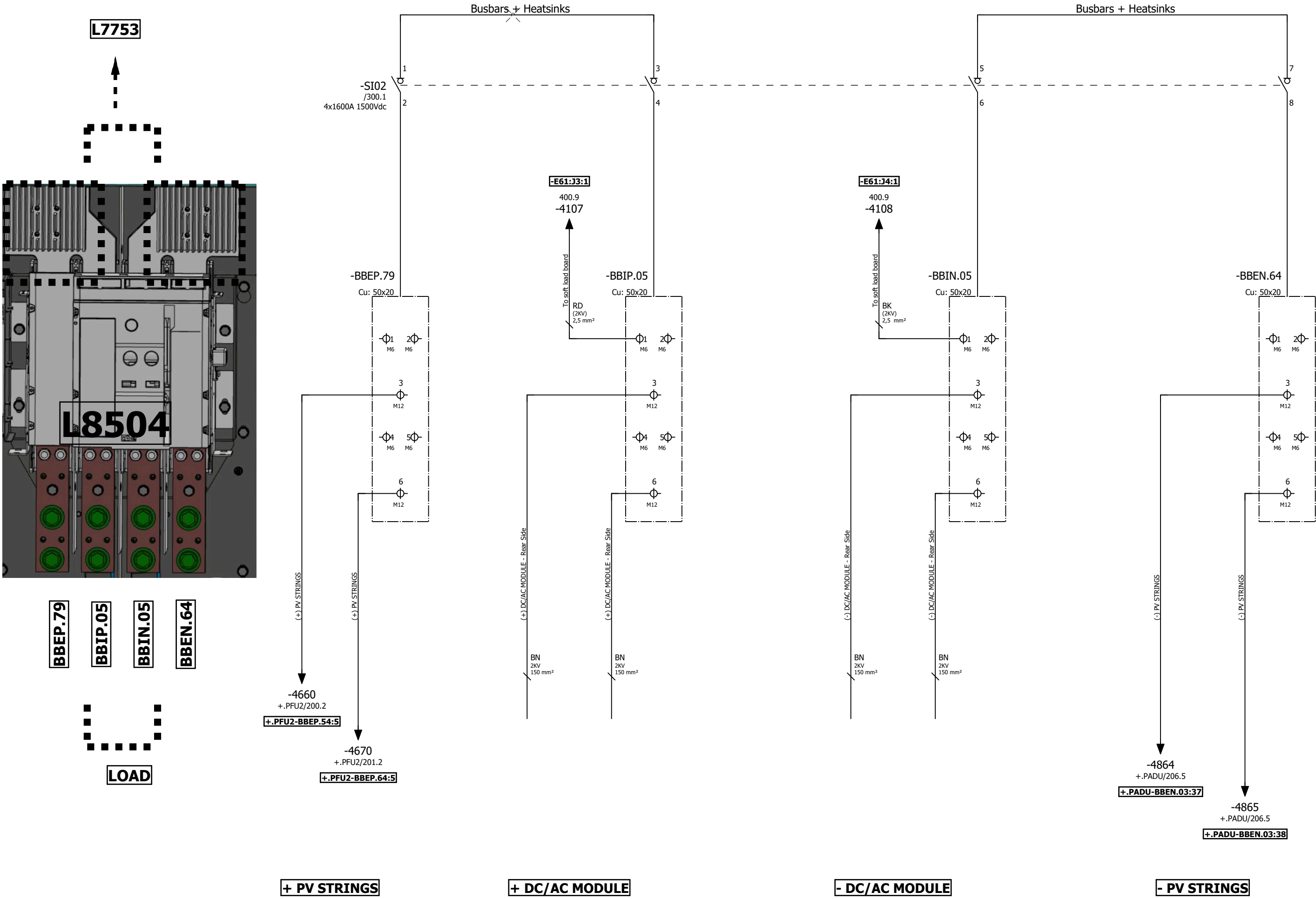
0 VDC

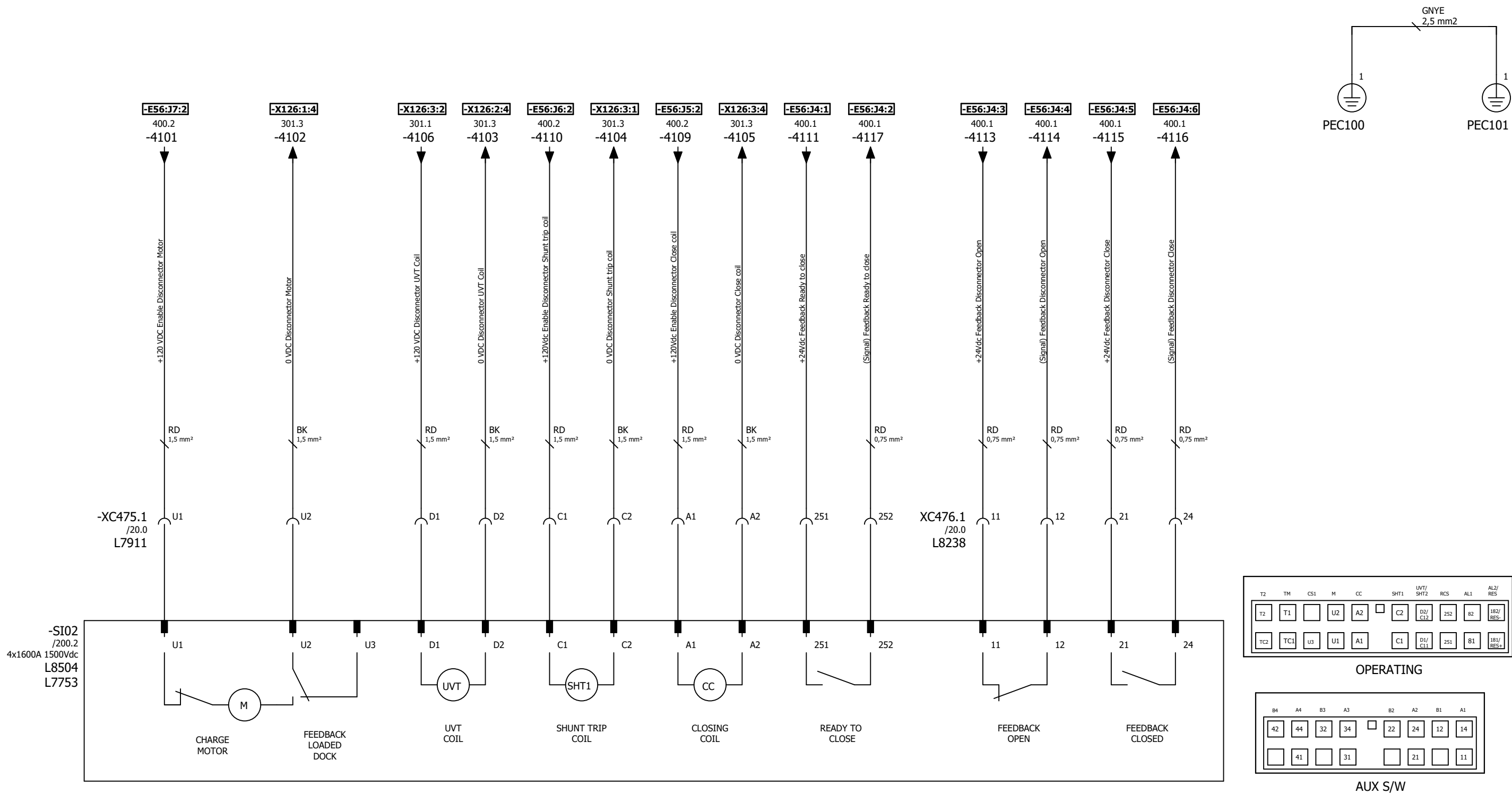


-4163
301.3

-X126:2:1

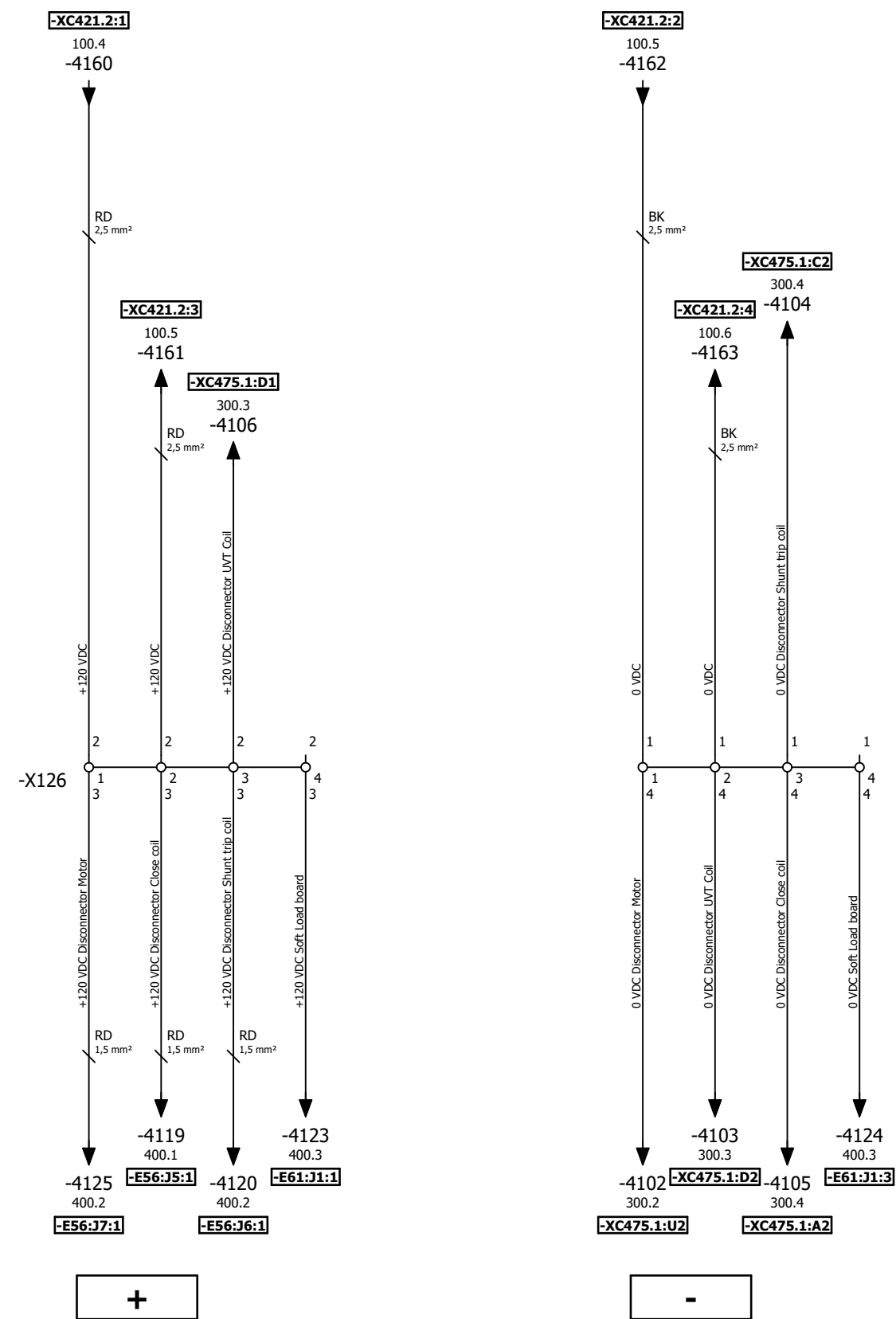
FROM PCB'S
DISCONNECTOR S002



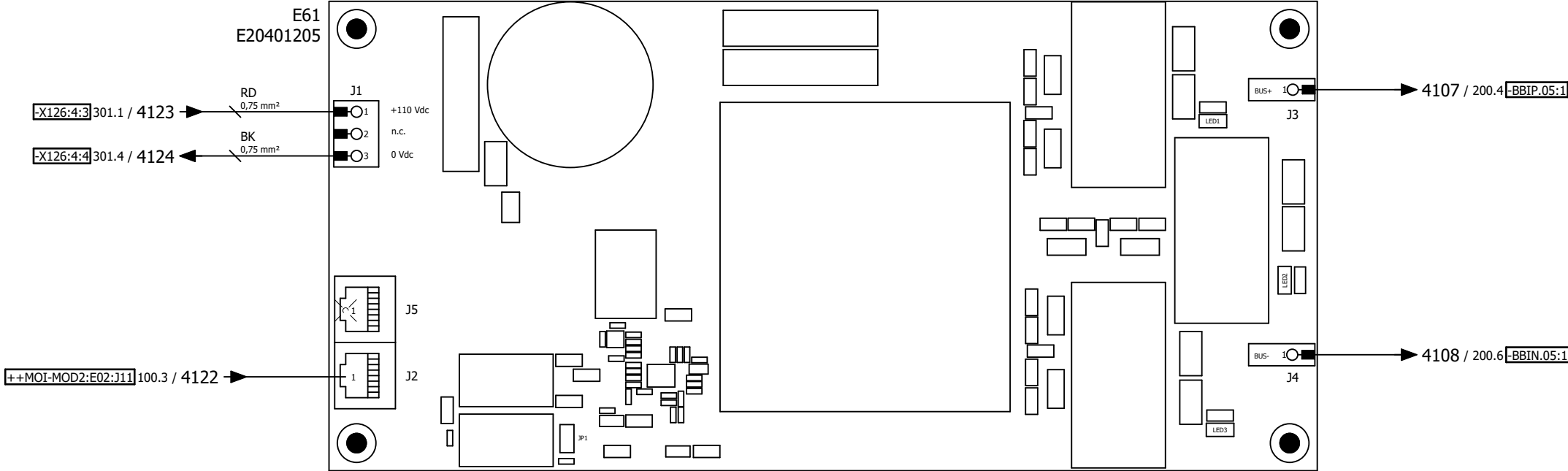
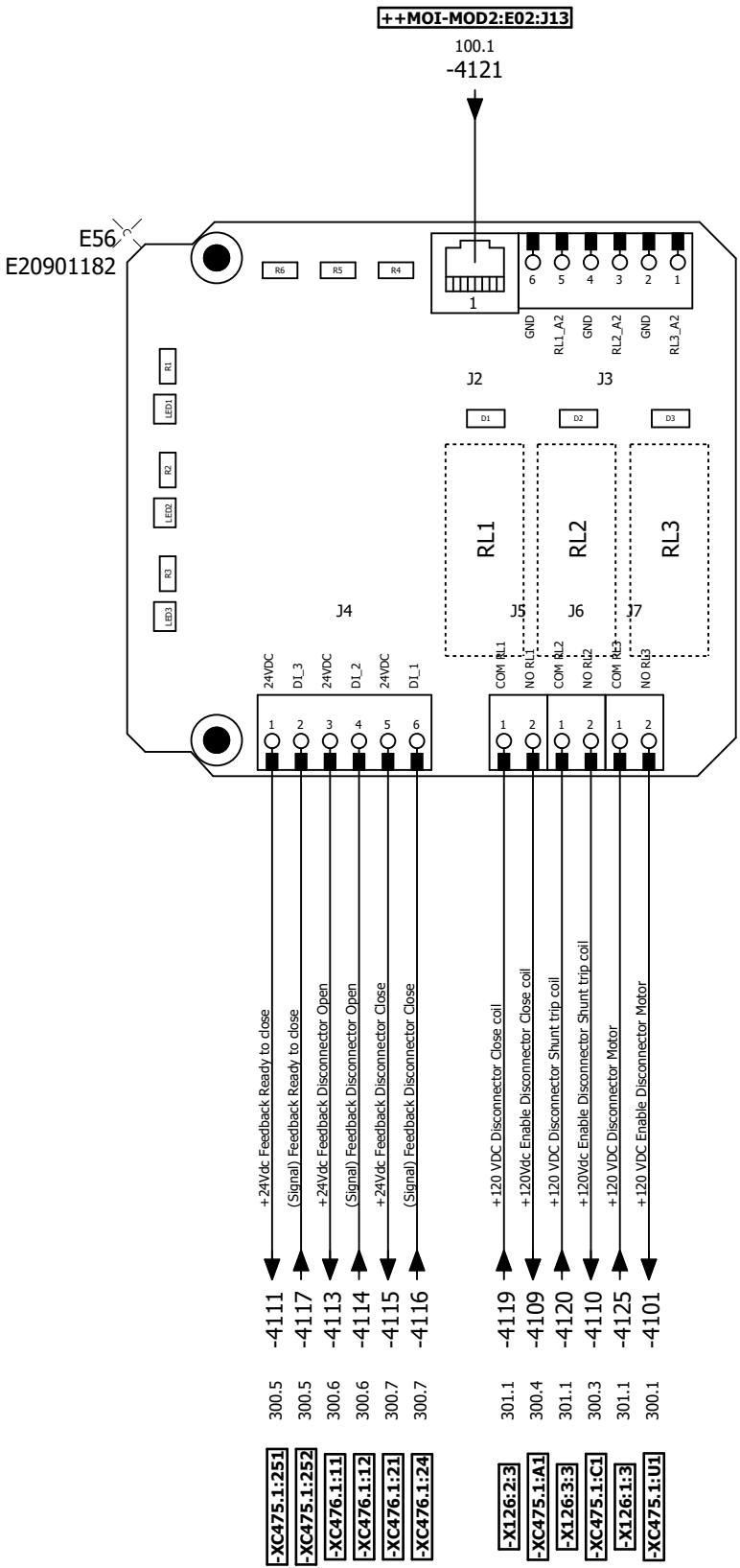


This diagram is based on "CONNECTED" position of a circuit breaker and Opening, Motor charging, Releasing of locking plate should be normal condition.

Distribution 120Vdc



PCB - I/O DC DISCONNECTOR



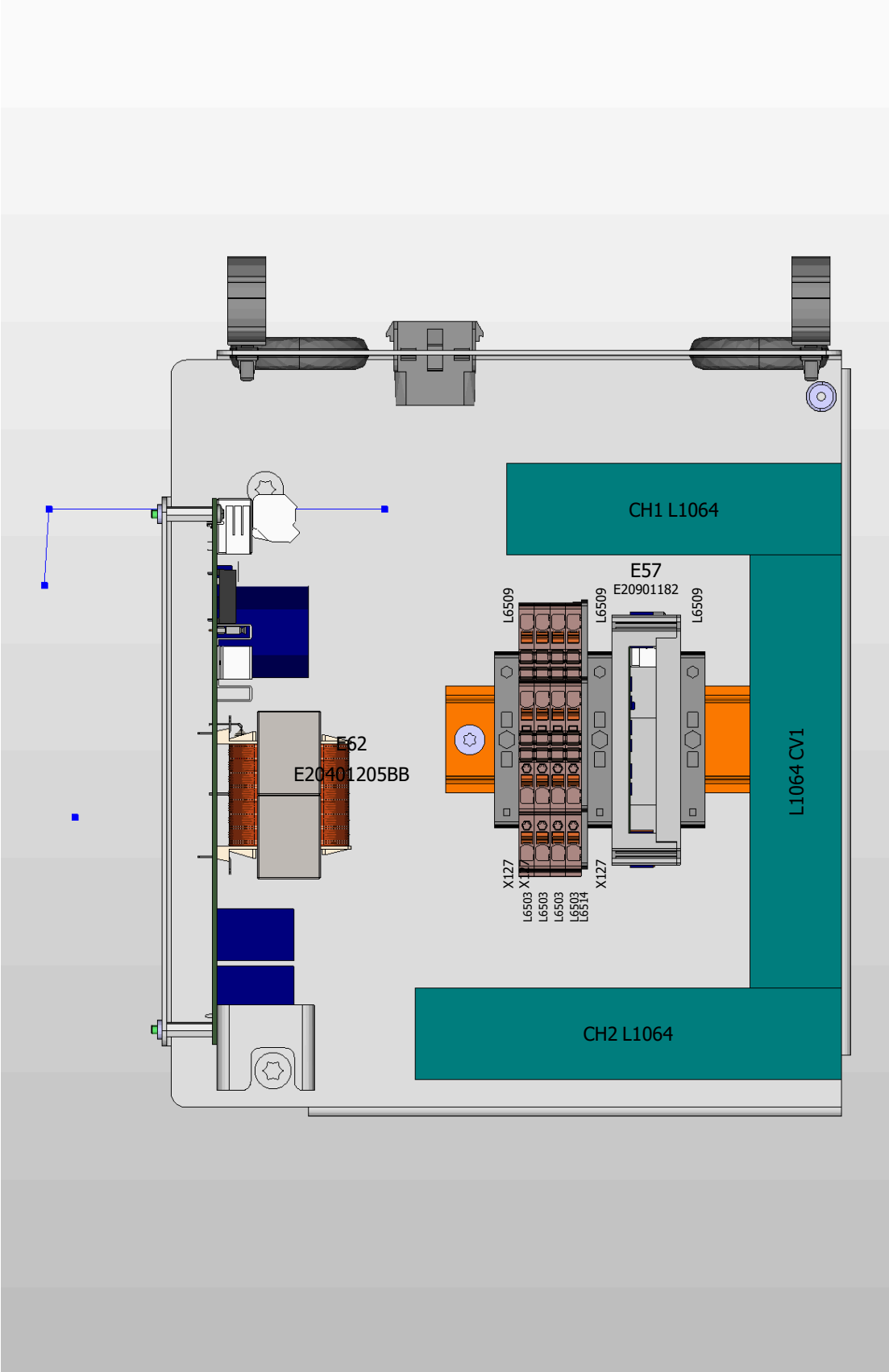
BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E56	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E56	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E61	E20401205BB	E20401205BB	SOFT LOAD BOARD	1
-SI02	L8504	2950009300	4x1600A-C-P IEC DISCONNECTOR SWITCH	1
-SI02	L7753	70223472602	Short bars + heatsink for DDV-16C4-00P	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PDC2-XC421.2	MATE-N-LOK PANEL 4P DC MODULE 2 DISTRIBUTION	L2004	1
		L5710	4
=HEM++ZMI+ADU.PDC2-XC475.1	CONNECTOR DISCONNECTOR SWITCH 2 - OPERATING	L7911	1
		L7912	10
=HEM++ZMI+ADU.PDC2-XC476.1	CONNECTOR DISCONNECTOR SWITCH 2 - AUX.	L8238	1
		L7912	4

LAYOUT PANEL



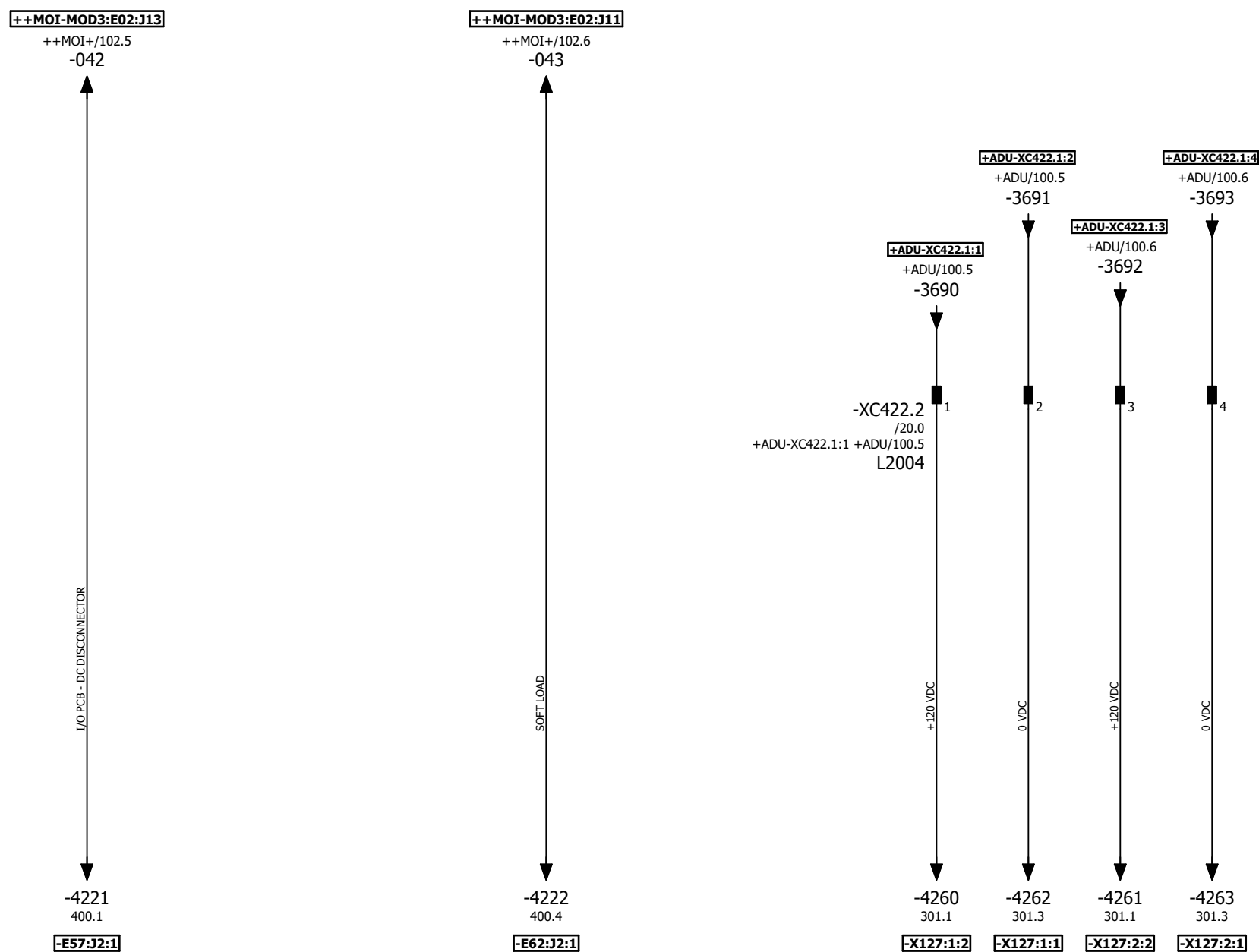
Enclosure legend

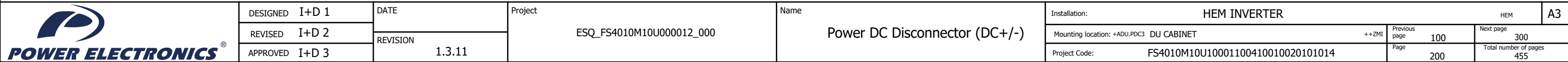
IME	ERP	Description
E62	E20401205BB	SOFT LOAD BOARD
CA1	L1061	DIN RAIL 35x7,5mm
E57	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
CH1;CH2;CV1	L1064	CABLE DUCT UNEX 42X30 EN U23X

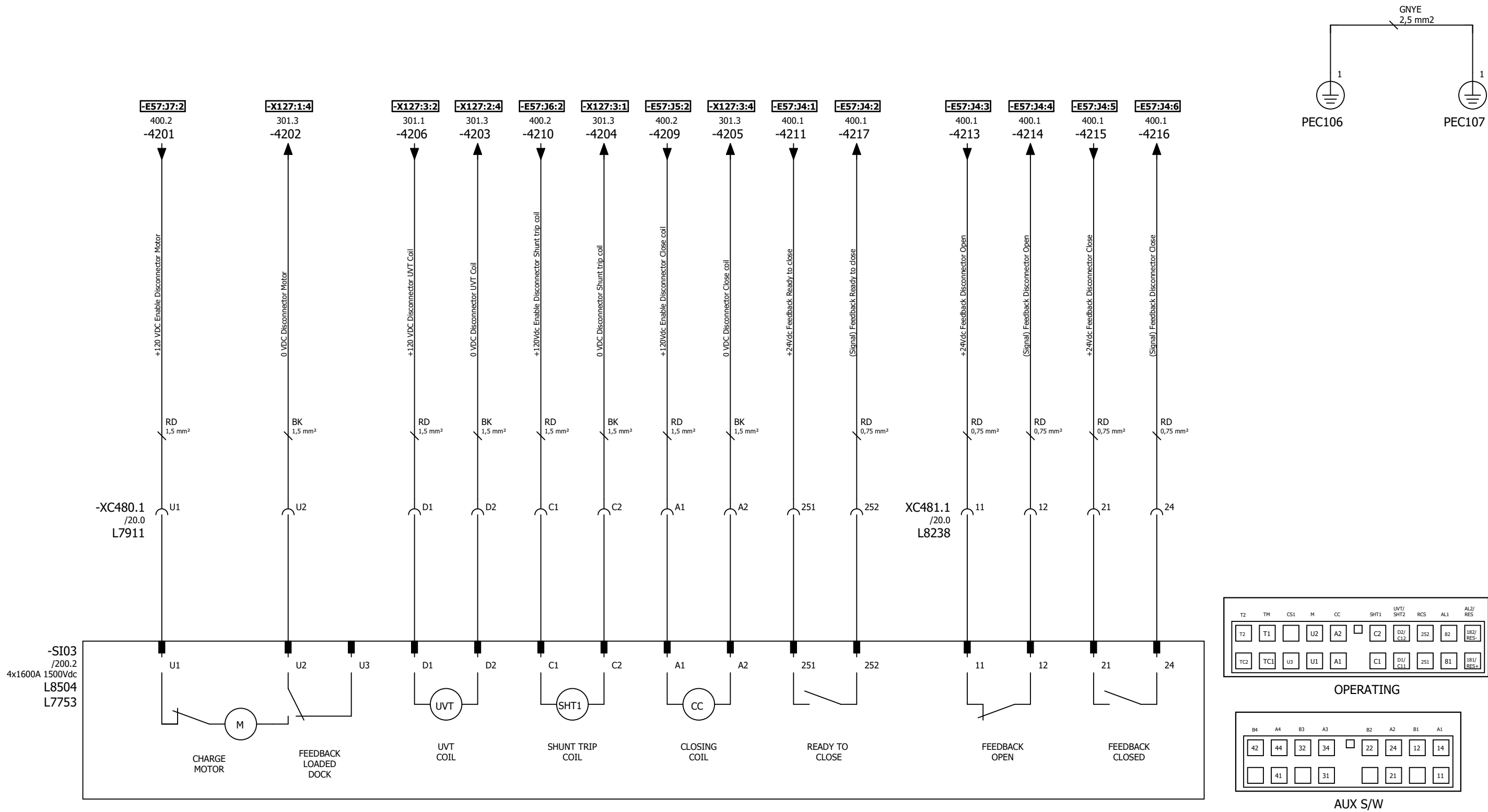
0	1	2	3	4	5	6	7	8	9
-XC480.1 = CONNECTOR DISCONNECTOR SWITCH 3 - OPERATING -XC481.1 = CONNECTOR DISCONNECTOR SWITCH 3 - AUX. -XC422.2 = MATE-N-LOK PANEL 4P DC MODULE 3 DISTRIBUTION -X127 = Distribution 120 Vdc									



TO MODULE 3

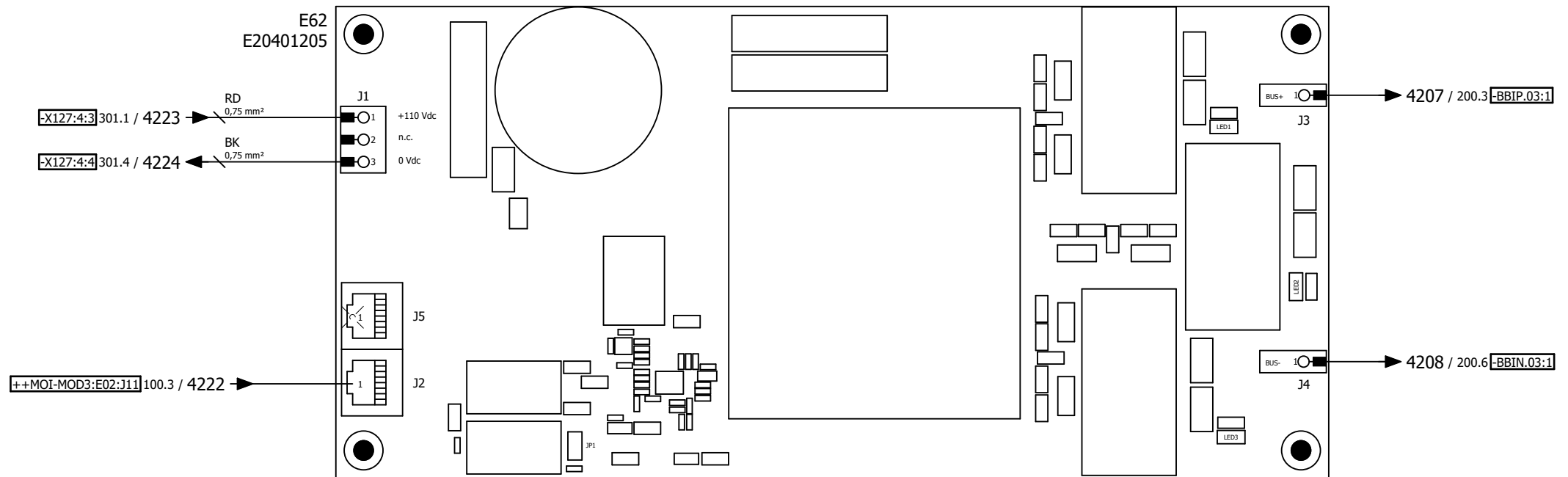
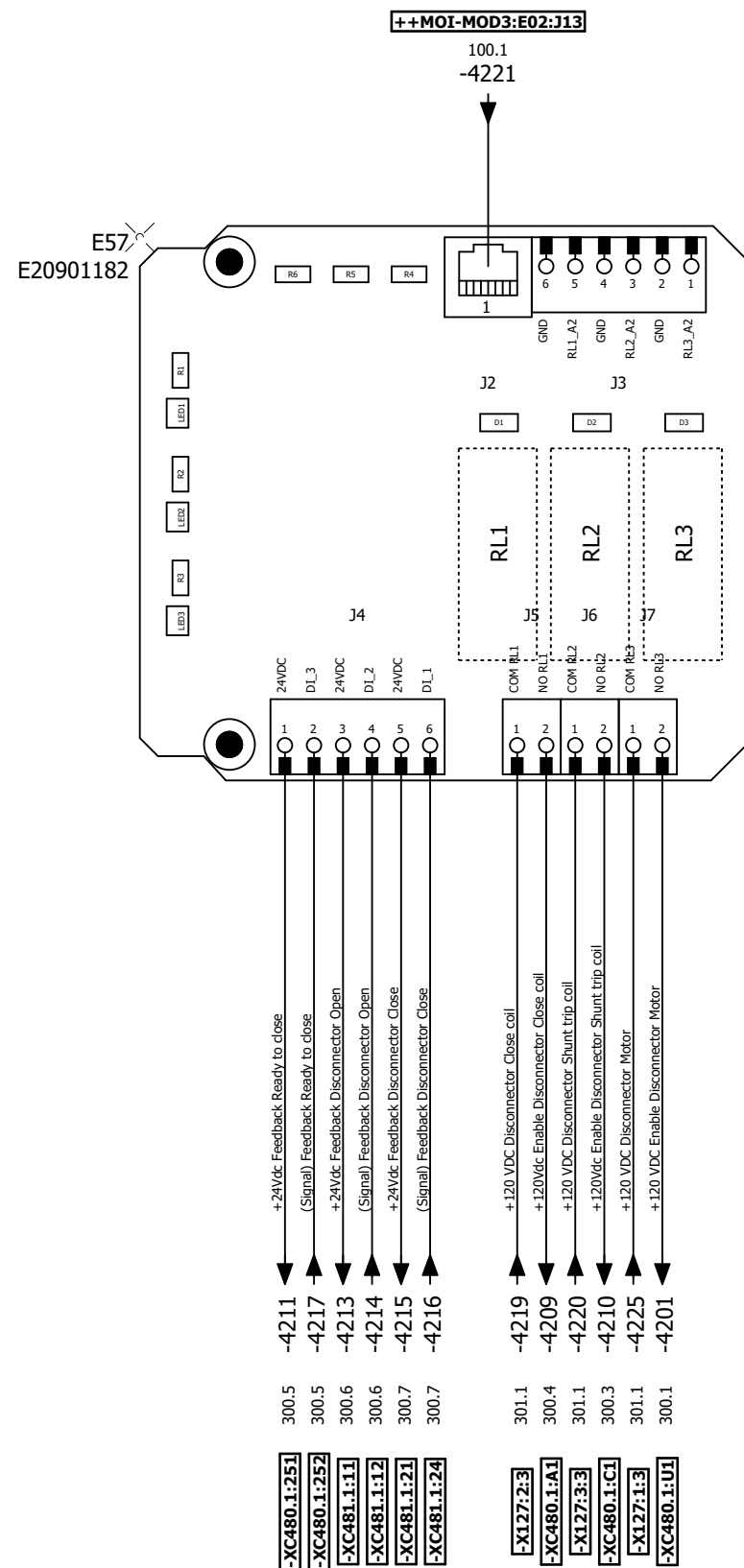






This diagram is based on "CONNECTED" position of a circuit breaker and Opening, Motor charging, Releasing of locking plate should be normal condition.

PCB - I/O DC DISCONNECTOR



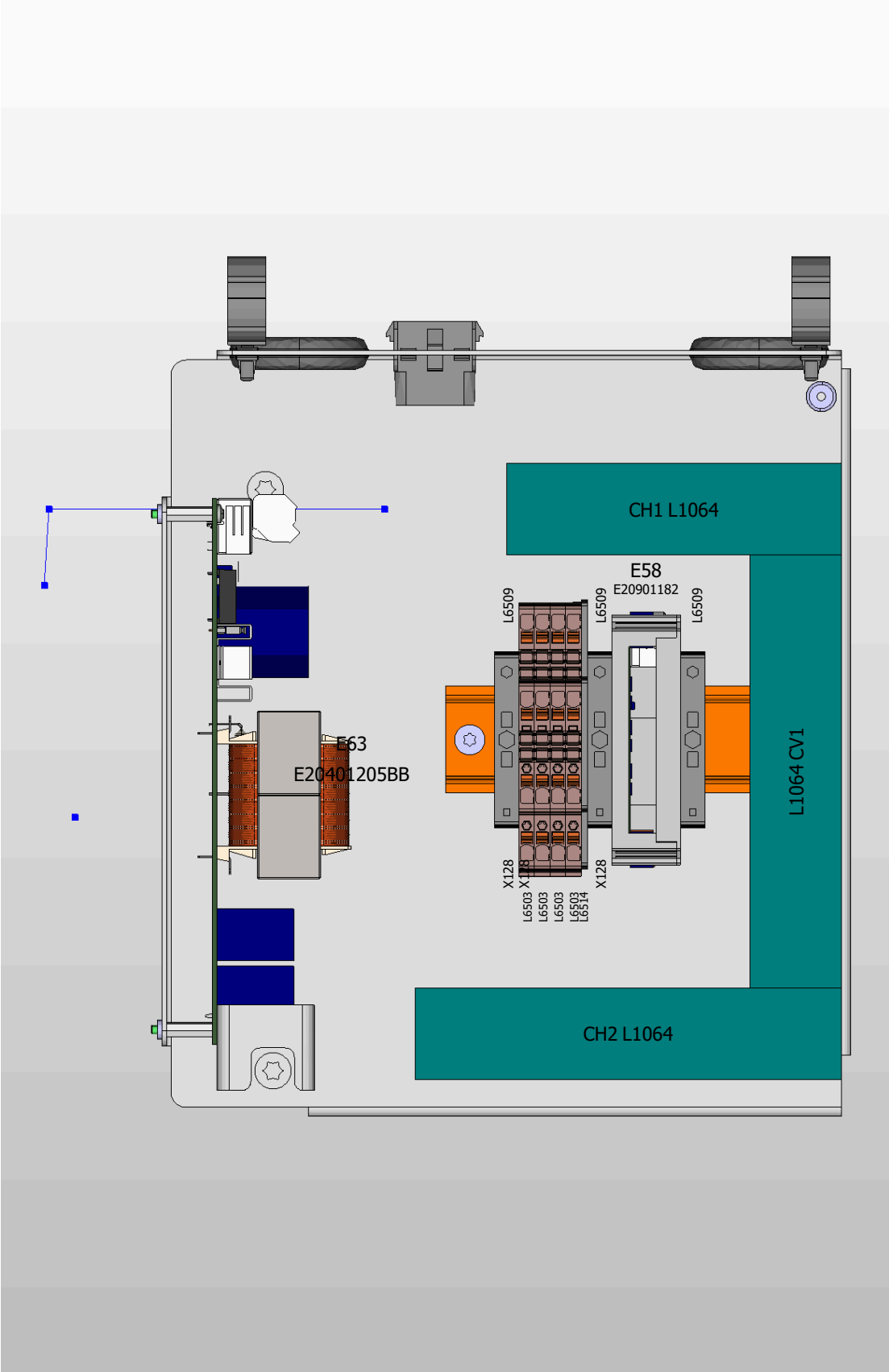
BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E57	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E57	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E62	E20401205BB	E20401205BB	SOFT LOAD BOARD	1
-SI03	L8504	2950009300	4x1600A-C-P IEC DISCONNECTOR SWITCH	1
-SI03	L7753	70223472602	Short bars + heatsink for DDV-16C4-00P	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PDC3-XC422.2	MATE-N-LOK PANEL 4P DC MODULE 3 DISTRIBUTION	L2004	1
		L5710	4
=HEM++ZMI+ADU.PDC3-XC480.1	CONNECTOR DISCONNECTOR SWITCH 3 - OPERATING	L7911	1
		L7912	10
=HEM++ZMI+ADU.PDC3-XC481.1	CONNECTOR DISCONNECTOR SWITCH 3 - AUX.	L8238	1
		L7912	4

LAYOUT PANEL



Enclosure legend

IME	ERP	Description
E63	E20401205BB	SOFT LOAD BOARD
CA1	L1061	DIN RAIL 35x7,5mm
E58	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
CH1;CH2;CV1	L1064	CABLE DUCT UNEX 42X30 EN U23X

-XC485.1 = CONNECTOR DISCONNECTOR SWITCH 4 - OPERATING

-X128 = Distribution 120 Vdc

-XC486.1 = CONNECTOR DISCONNECTOR SWITCH 4 - AUX.

-XC423.2 = MATE-N-LOK PANEL 4P DC MODULE 4 DISTRIBUTION



TO MODULE 4

++MOI-MOD4:E02:J13

++MOI+/102.8
-044



I/O PCB - DC DISCONNECTOR



-4321
400.1

-E58:J2:1

++MOI-MOD4:E02:J11

++MOI+/102.9
-045



SOFT LOAD



-4322
400.4

-E63:J2:1

+ADU-XC423.1:1

+ADU/100.7
-3694



-XC423.2
/20.0
+ADU-XC423.1:1 +ADU/100.7
L2004



1

+120 VDC

-4360
301.1

-X128:1:2

+ADU-XC423.1:2

+ADU/100.8
-3695



2

0 VDC

-4362
301.3

-X128:1:1

+ADU-XC423.1:3

+ADU/100.8
-3696



3

+120 VDC

-4361
301.1

-X128:2:2

+ADU-XC423.1:4

+ADU/100.8
-3697



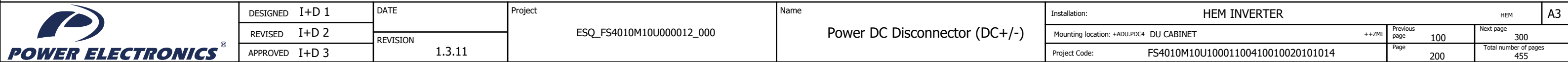
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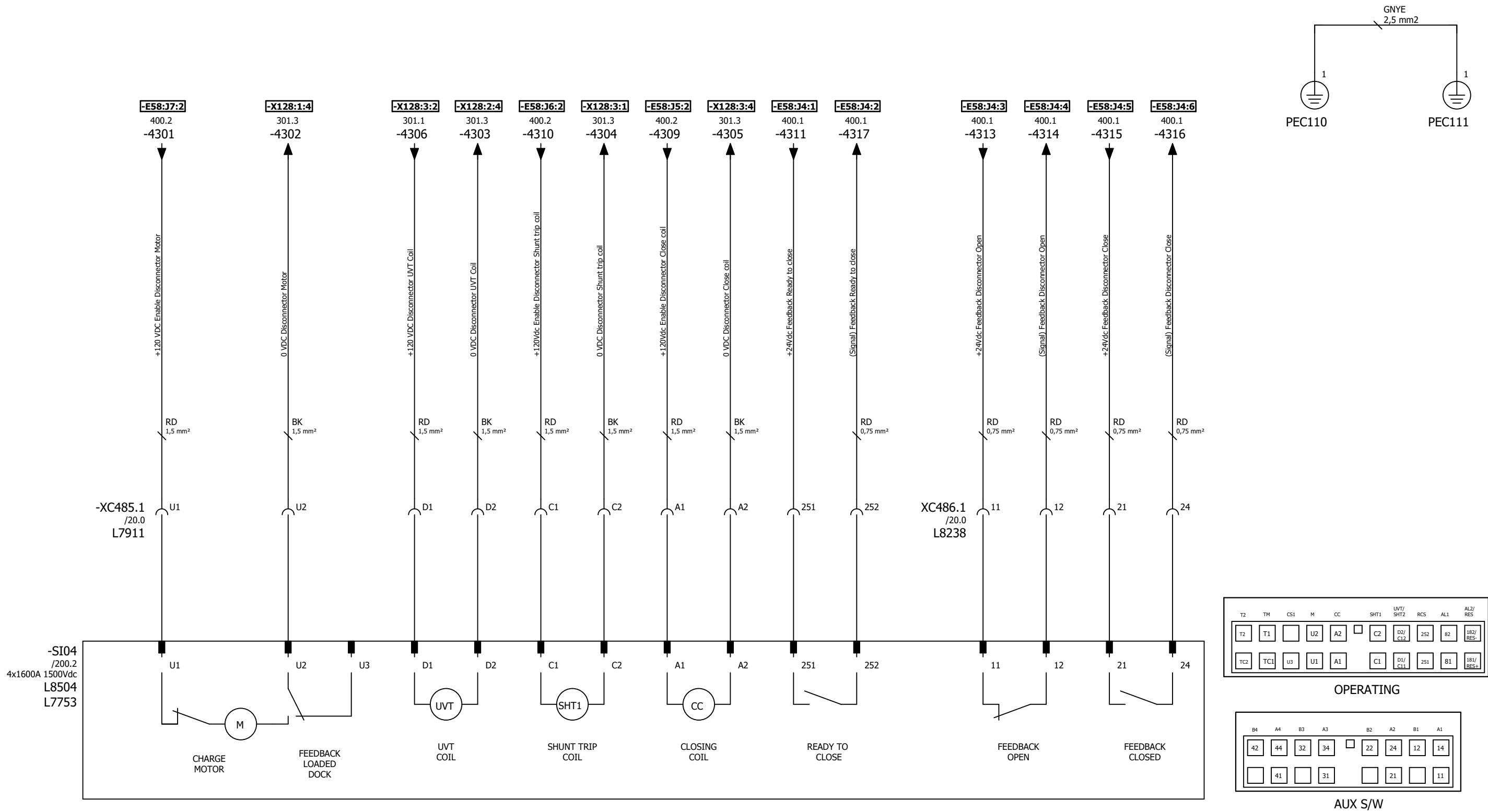
0 VDC

-4363
301.3

-X128:2:1

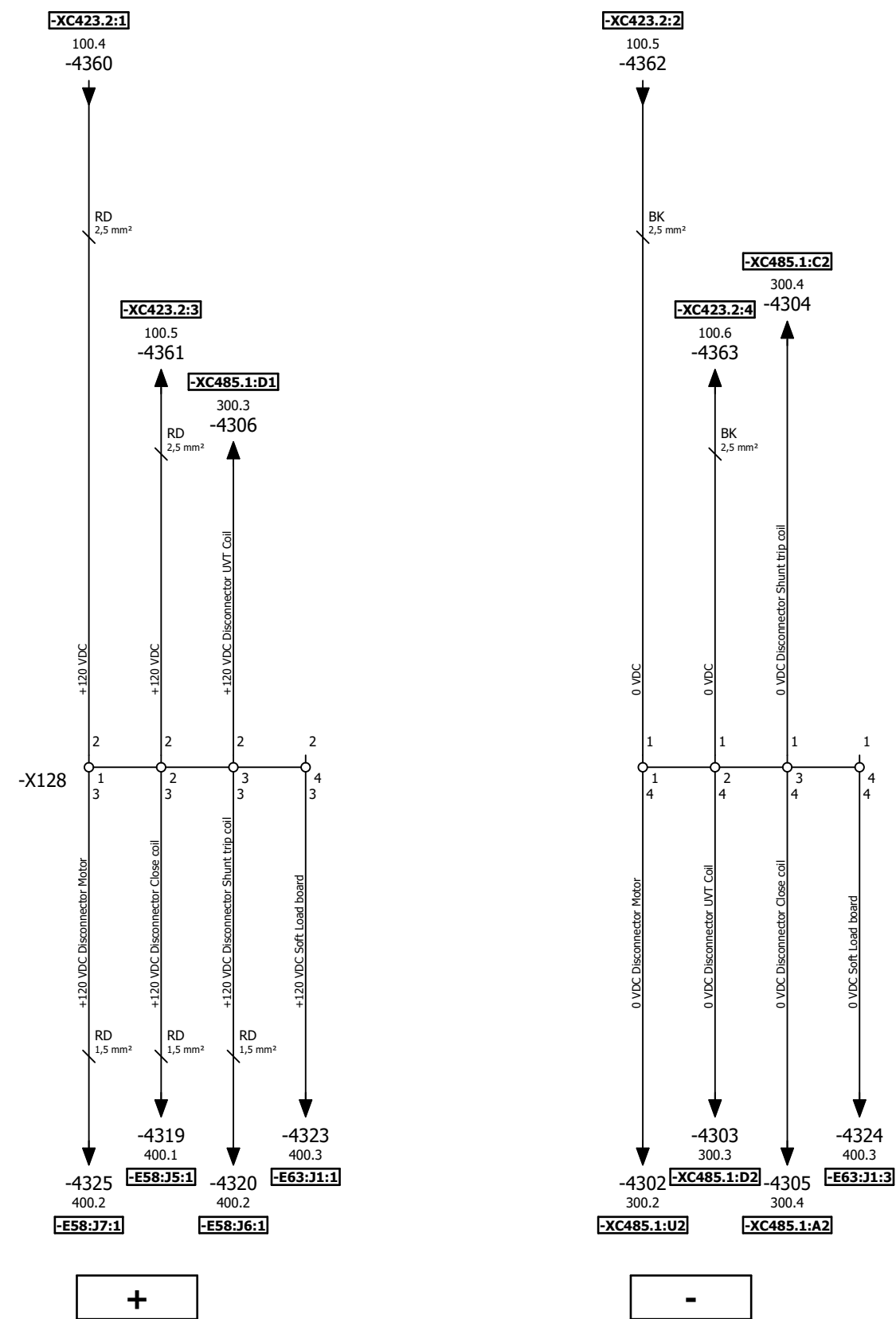
FROM PCB'S
DISCONNECTOR S004



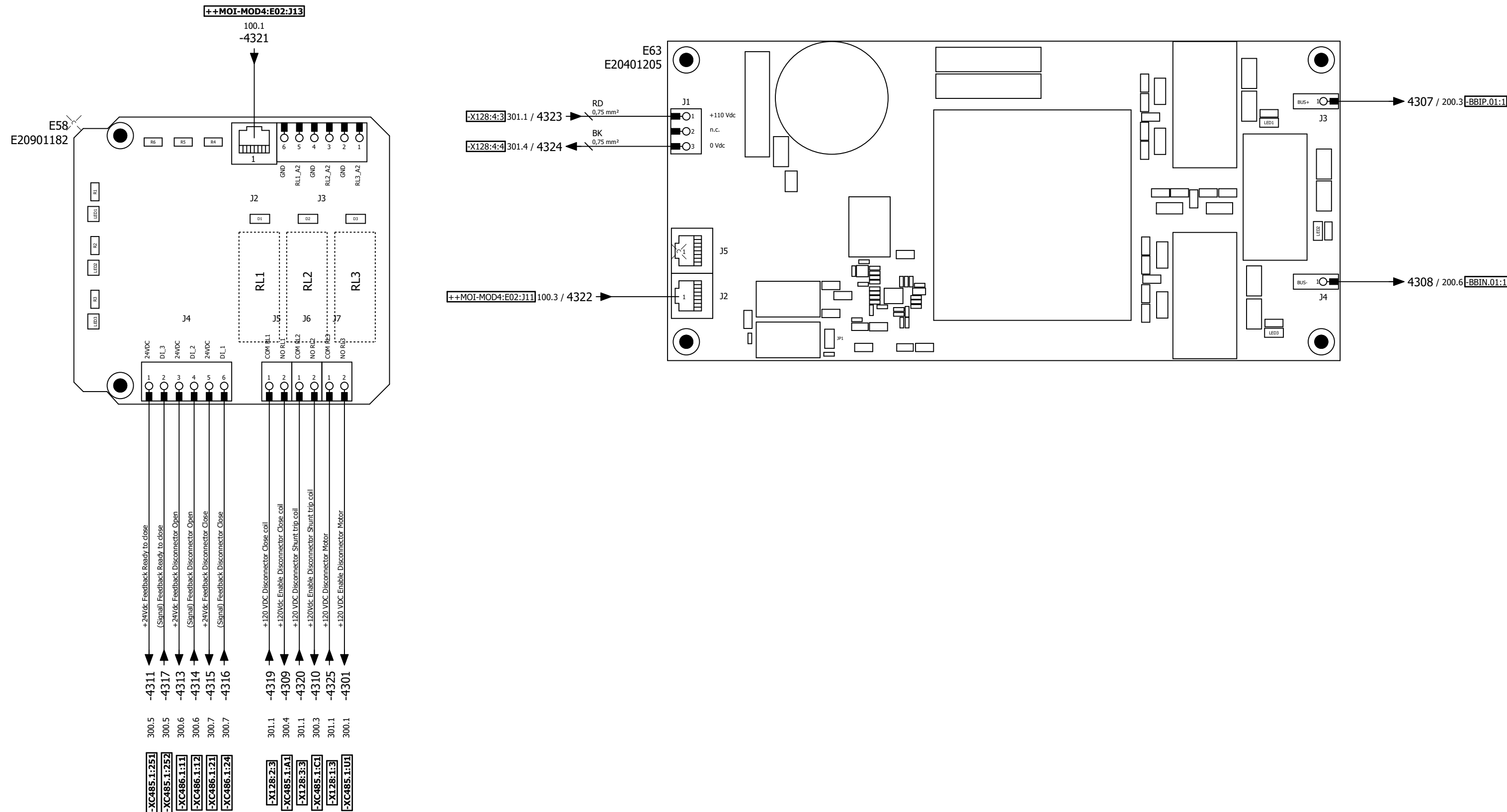


This diagram is based on "CONNECTED" position of a circuit breaker and Opening, Motor charging, Releasing of locking plate should be normal condition.

Distribution 120Vdc



PCB - I/O DC DISCONNECTOR



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E58	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E58	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E63	E20401205BB	E20401205BB	SOFT LOAD BOARD	1
-SI04	L8504	2950009300	4x1600A-C-P IEC DISCONNECTOR SWITCH	1
-SI04	L7753	70223472602	Short bars + heatsink for DDV-16C4-00P	1

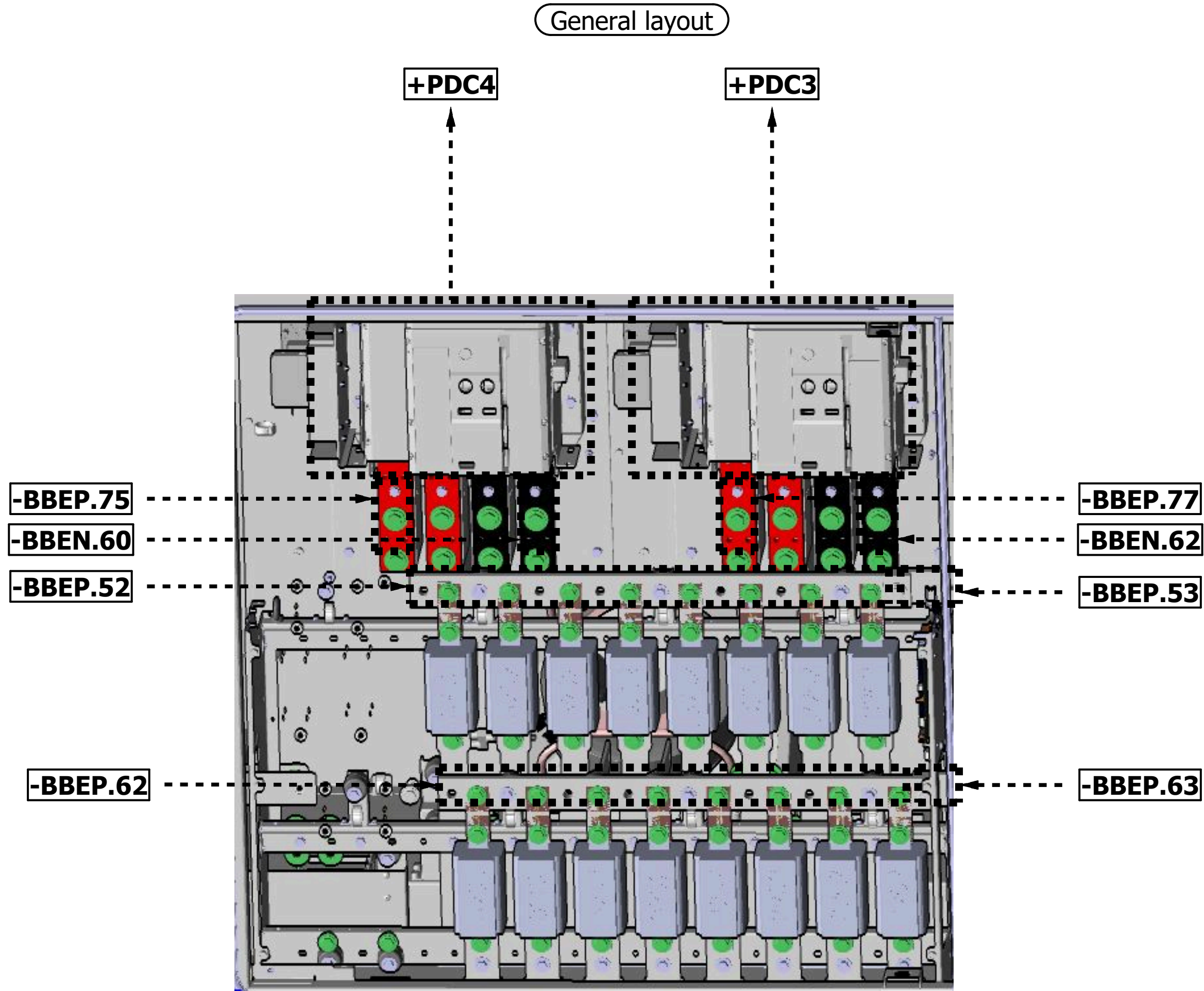
Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PDC4-XC423.2	MATE-N-LOK PANEL 4P DC MODULE 4 DISTRIBUTION	L2004	1
		L5710	4
=HEM++ZMI+ADU.PDC4-XC485.1	CONNECTOR DISCONNECTOR SWITCH 4 - OPERATING	L7911	1
		L7912	10
=HEM++ZMI+ADU.PDC4-XC486.1	CONNECTOR DISCONNECTOR SWITCH 4 - AUX.	L8238	1
		L7912	4

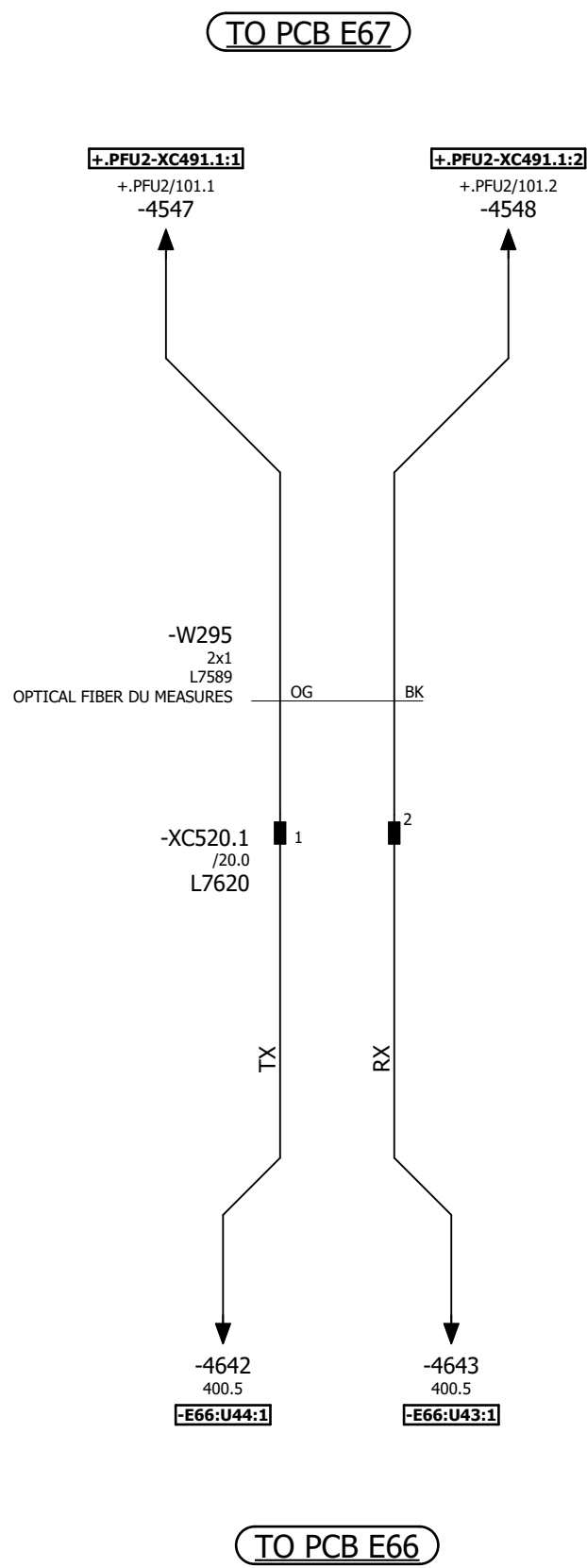


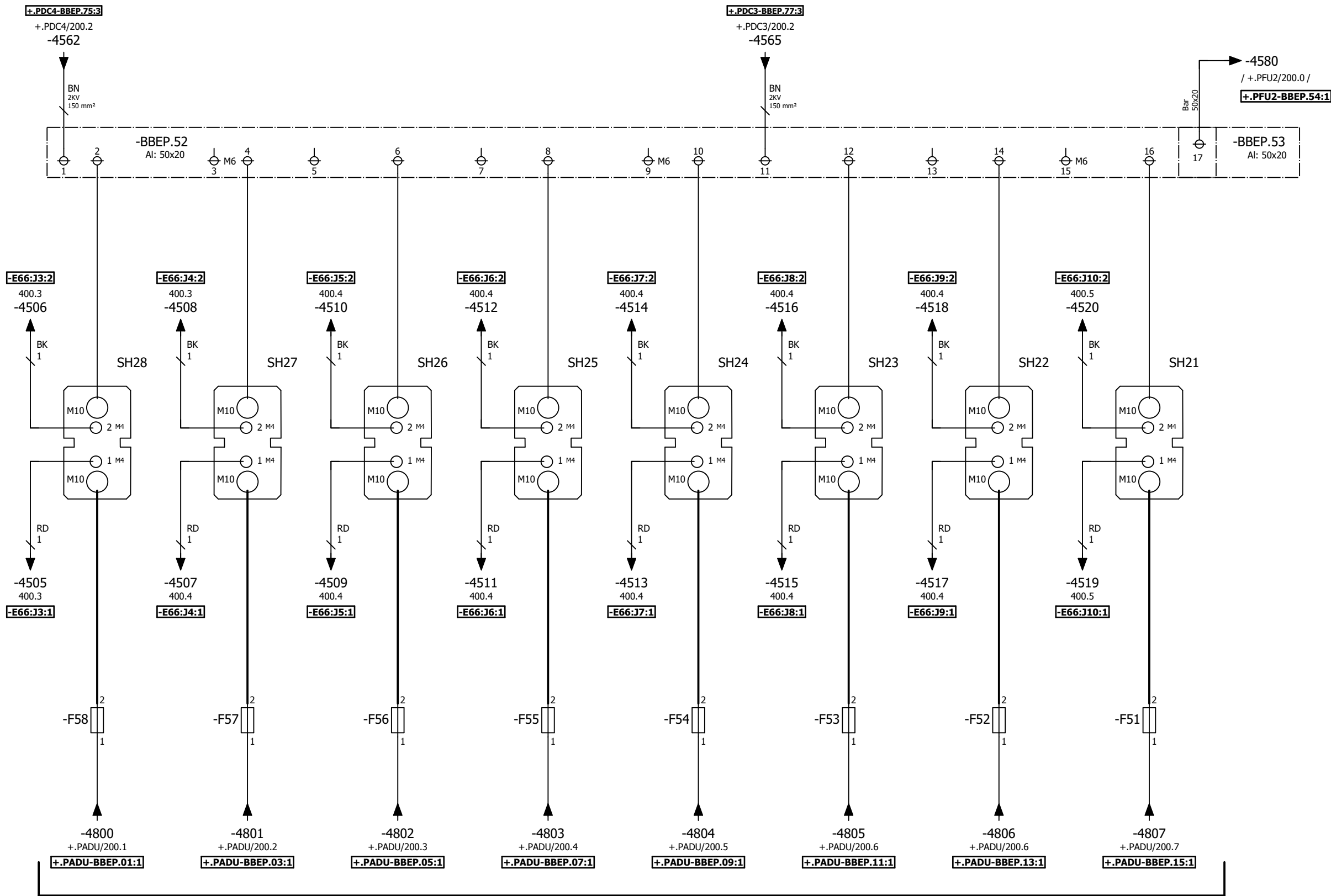
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

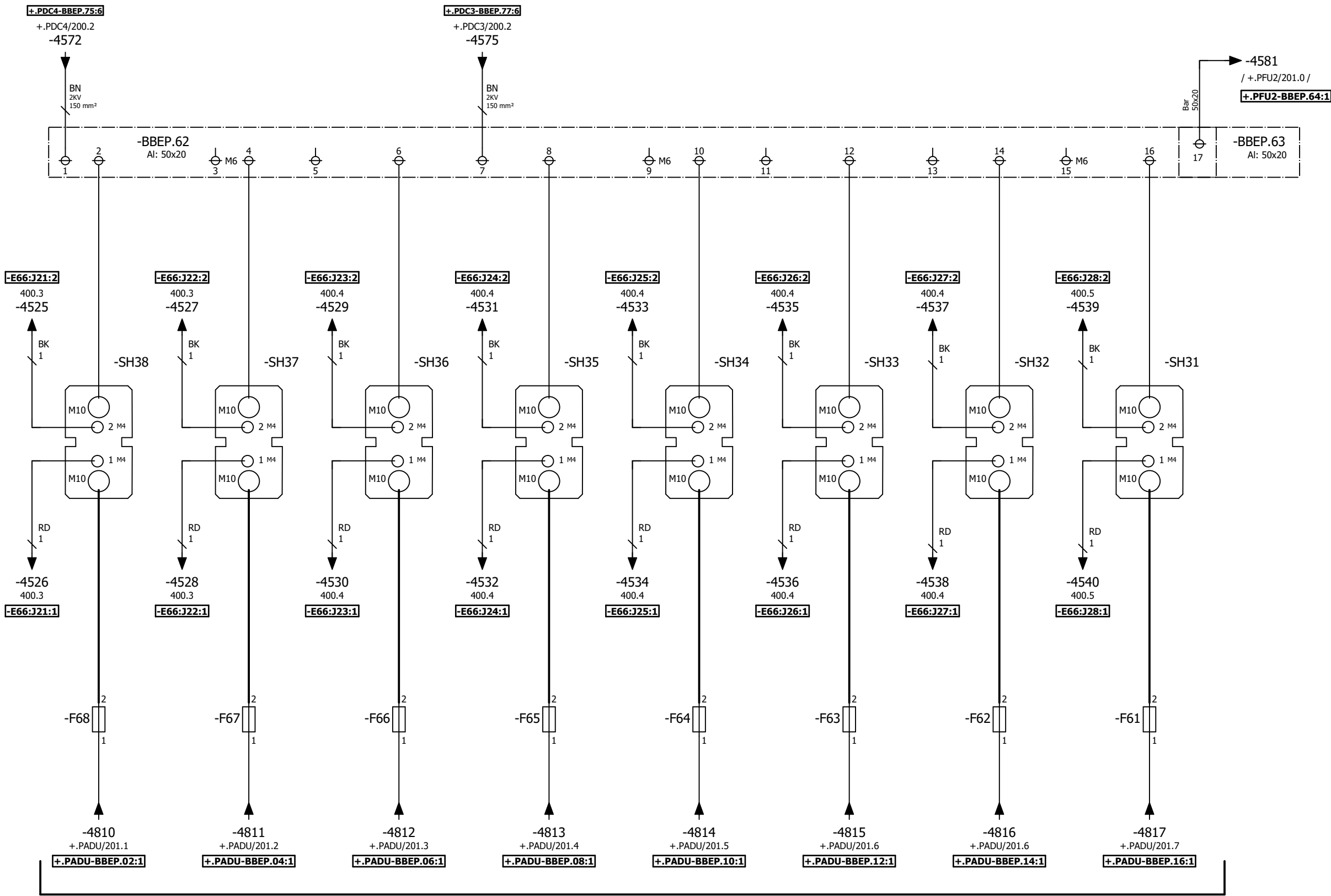
HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU.PFU1	FUSES PRE-ASSEMBLY 1		
PFU1 FUSES PRE-ASSEMBLY 1			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		



-XC520.1 = CONNECTOR OPTICAL FIBER DUPLEX DU MEASURES



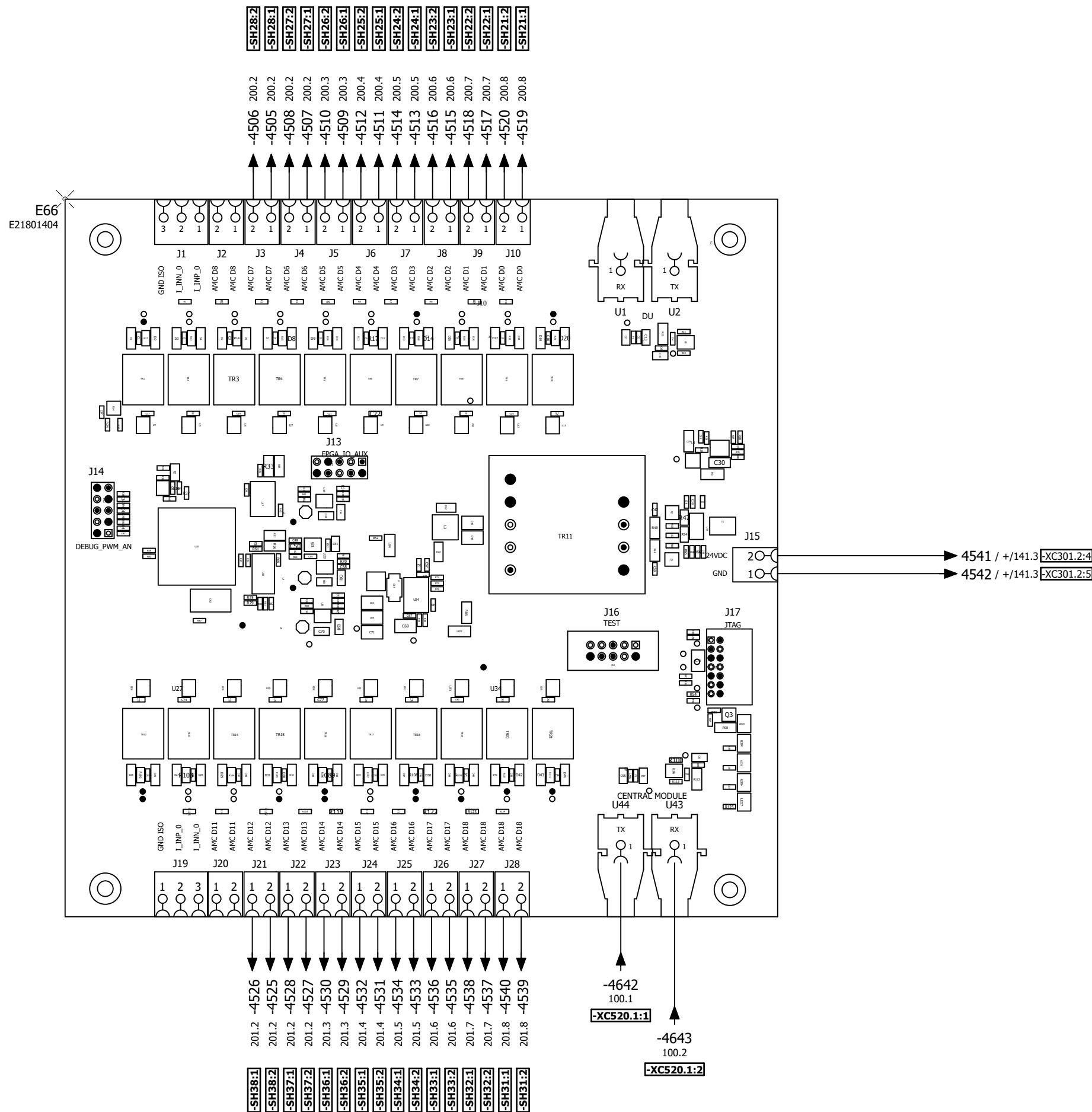




FROM PV SYSTEM

FROM -PADU

PCB DU CURRENT MEASURES



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-E66	PCB E21801204	PCB E21801204	DU MEASURES PCB	1
-SH21		8535	SHUNT RESISTANCE 36W	1
-SH22		8535	SHUNT RESISTANCE 36W	1
-SH23		8535	SHUNT RESISTANCE 36W	1
-SH24		8535	SHUNT RESISTANCE 36W	1
-SH25		8535	SHUNT RESISTANCE 36W	1
-SH26		8535	SHUNT RESISTANCE 36W	1
-SH27		8535	SHUNT RESISTANCE 36W	1
-SH28		8535	SHUNT RESISTANCE 36W	1
-SH31		8535	SHUNT RESISTANCE 36W	1
-SH32		8535	SHUNT RESISTANCE 36W	1
-SH33		8535	SHUNT RESISTANCE 36W	1
-SH34		8535	SHUNT RESISTANCE 36W	1
-SH35		8535	SHUNT RESISTANCE 36W	1
-SH36		8535	SHUNT RESISTANCE 36W	1
-SH37		8535	SHUNT RESISTANCE 36W	1
-SH38		8535	SHUNT RESISTANCE 36W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+ADU.PFU1-W295	OPTICAL FIBER DU MEASURES	L7589

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PFU1-XC520.1	CONNECTOR OPTICAL FIBER DUPLEX DU MEASURES	L7620	1

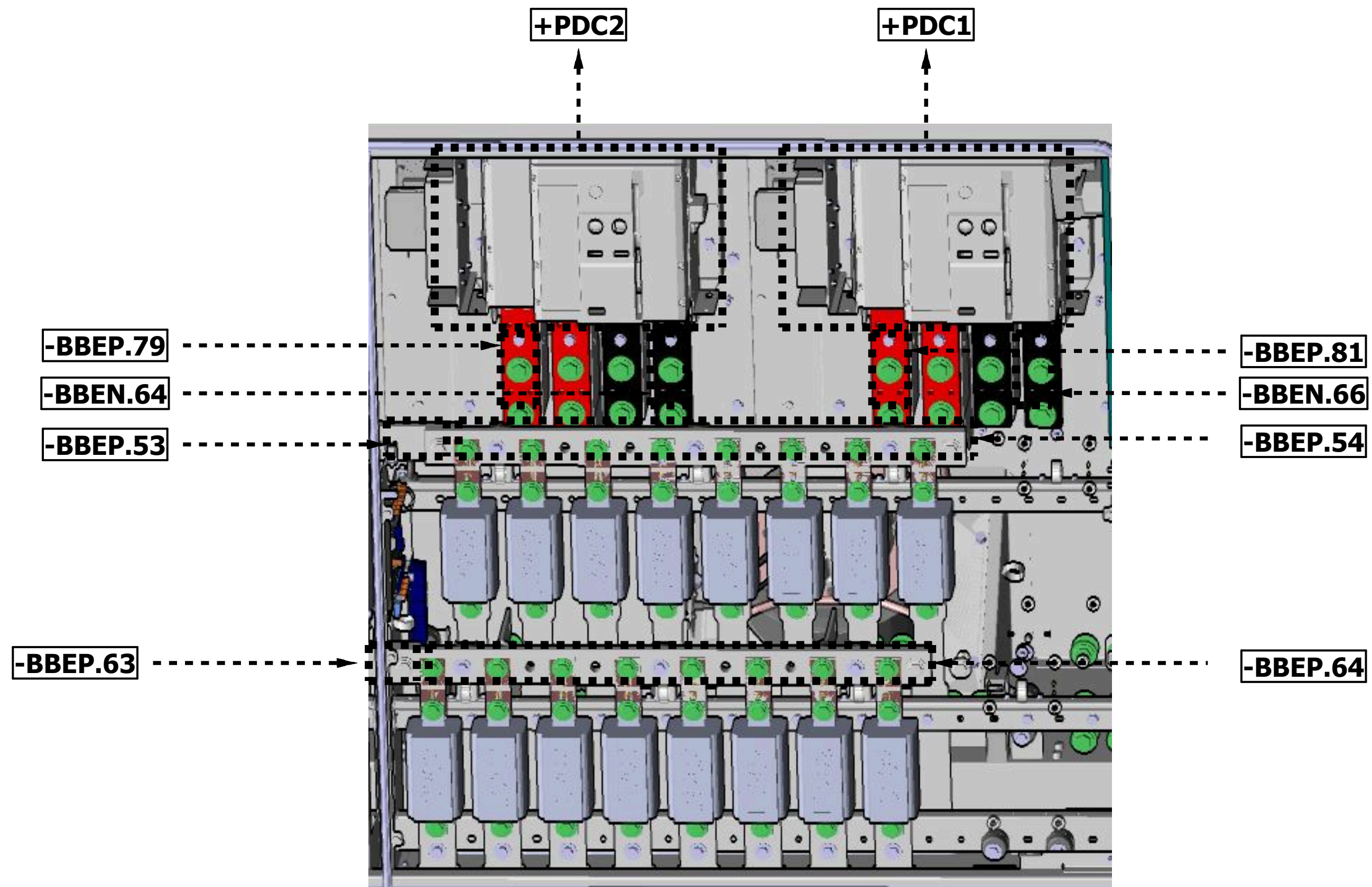
 POWER ELECTRONICS®	DESIGNED I+D 1	DATE	Project ESQ_FS4010M10U000012_000	Name Plug overview : =HEM++ZMI+ADU.PFU1-XC520.1 - =HEM++ZMI+ADU.PFU1-XC520.1	Installation: HEM INVERTER				HEM	A3
	REVISED I+D 2	REVISION 1.3.11			Mounting location: +ADU.PFU1 DU CABINET		++ZMI	Previous page +&CAB/2000	Next page +&PFU2&/1	
	APPROVED I+D 3				Project Code: FS4010M10U10001100410010020101014		Page 3000	Total number of pages 455		



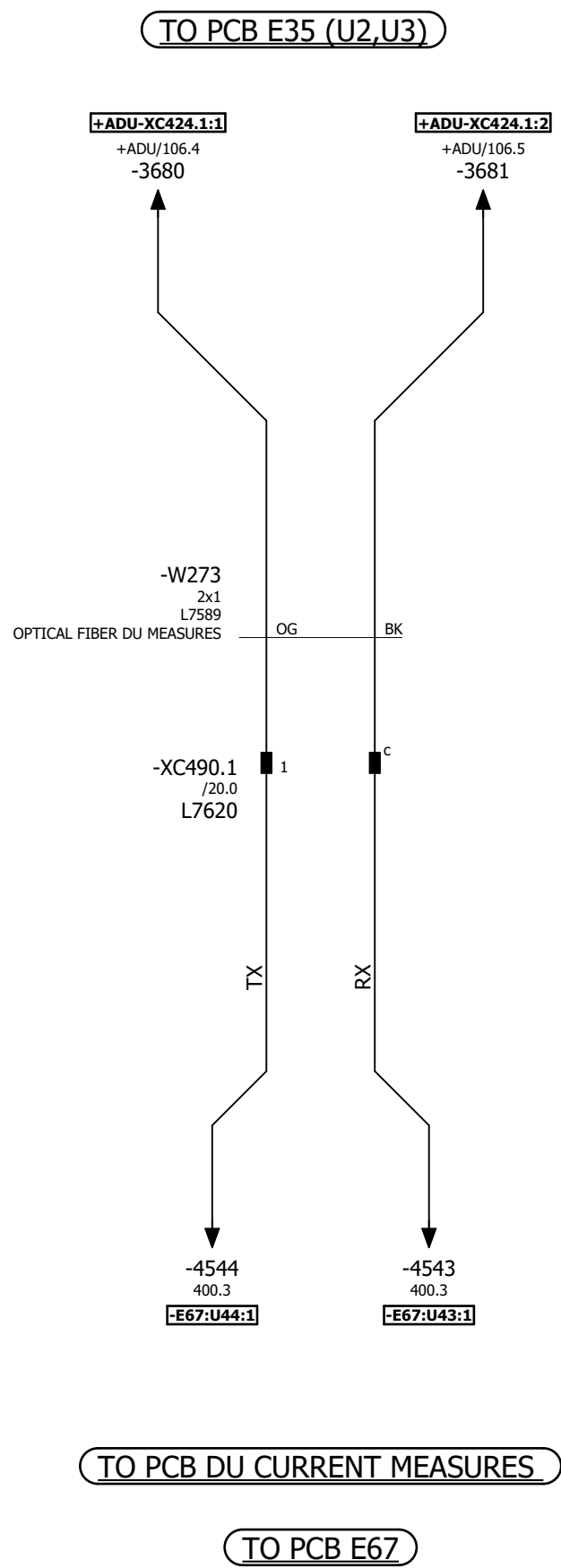
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

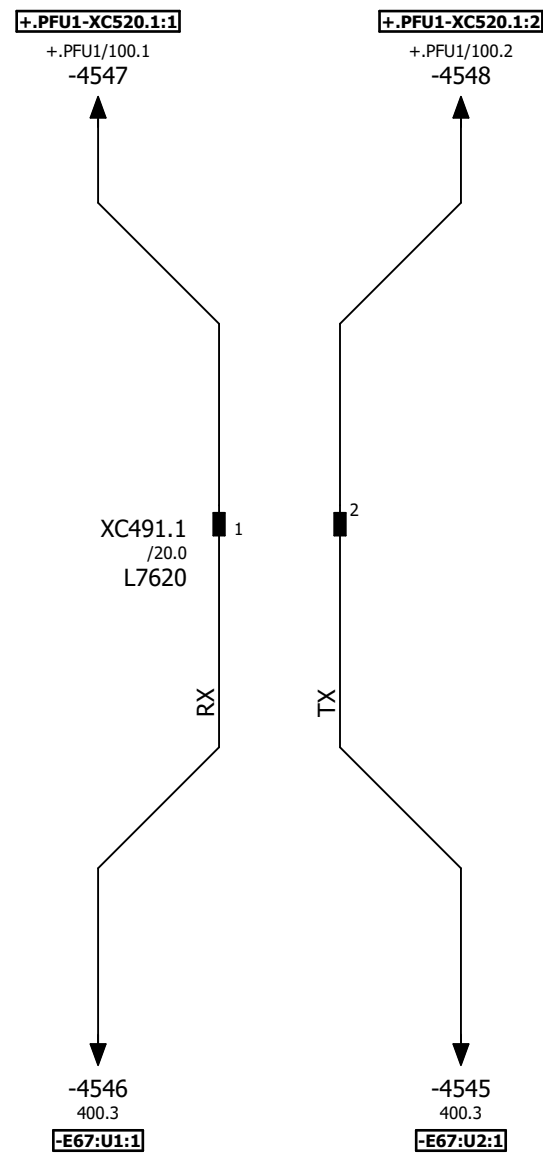
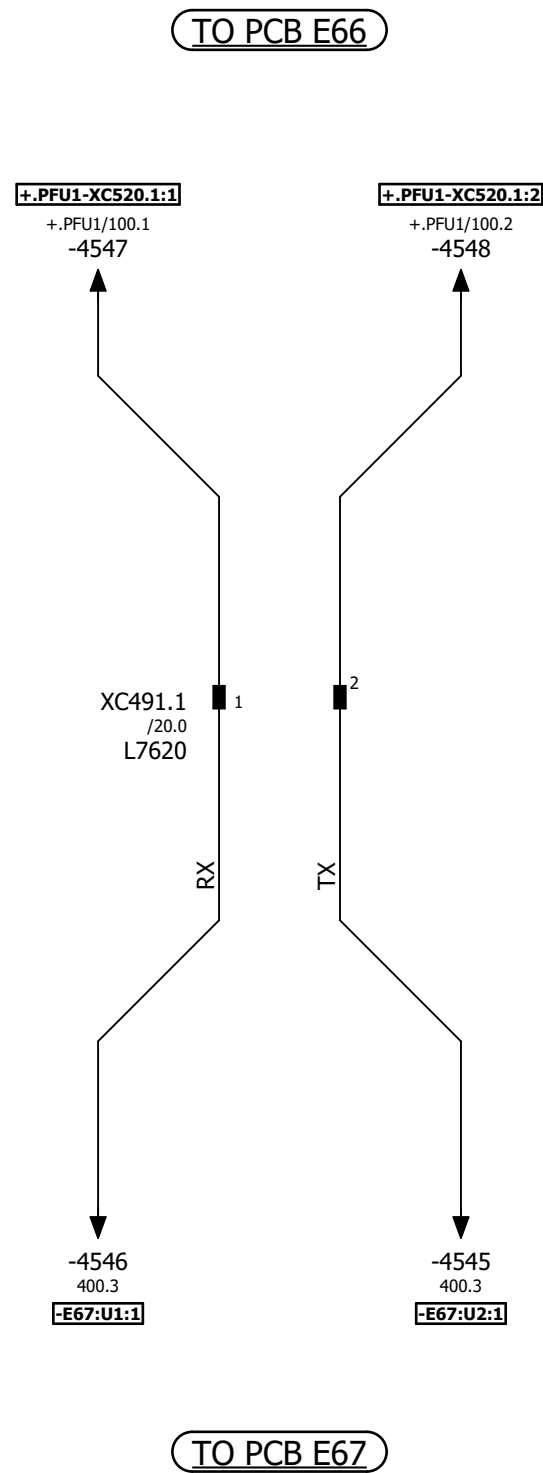
HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU.PFU2	FUSES PRE-ASSEMBLY 2		
PFU2 FUSES PRE-ASSEMBLY 2			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

General layout



0	1	2	3	4	5	6	7	8	9
-XC490.1 = CONNECTOR OPTICAL FIBER DUPLEX DU MEASURES -XC491.1 = CONNECTOR OPTICAL FIBER DUPLEX DU MEASURES									





DESIGNED I+D 1	DATE
REVISED I+D 2	REVISION
APPROVED I+D 3	1.3.11

REVISED	I+D 2
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APPROVED I+D 3

DATE	
REVISION	1.3.11

REVISION	1.3.11
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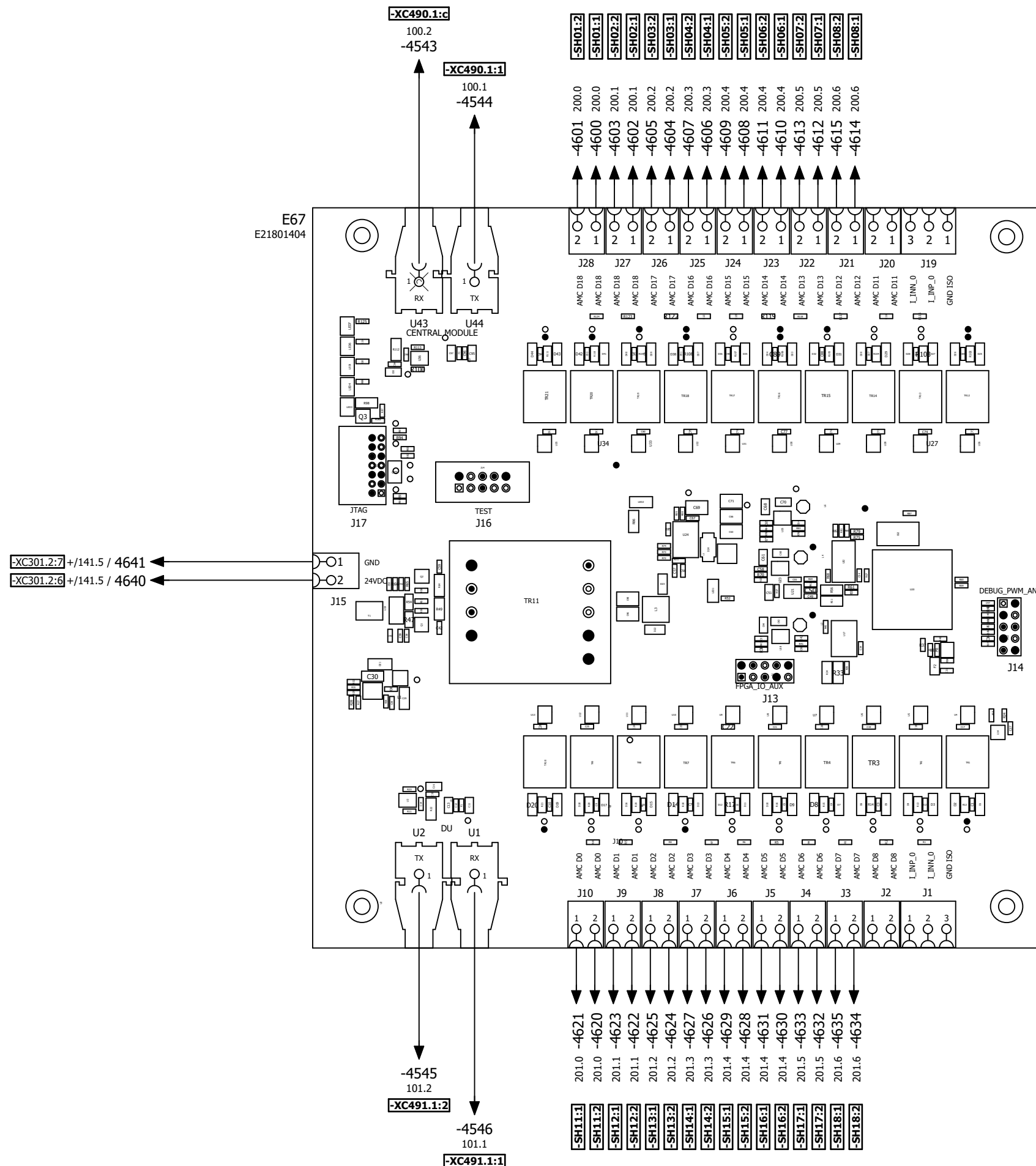
Project	ESQ_FS4010M10U000012_000
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Name	Interconnections F.O. PFU1//PFU2
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Installation:		HEM INVERTER		HEM	A3
Mounting location:		DU CABINET	ADU.PFU2	Previous page 100	Next page 200
Project Code:		FS4010M10U10001100410010020101014		Page 101	Total number of pages 455

Mounting location:	DU CABINET	ADU.PFU2	Previous page 100	Next page 200
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Project Code:	FS4010M10U10001100410010020101014	Page 101	Total number of pages 455
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Filter RFI DC

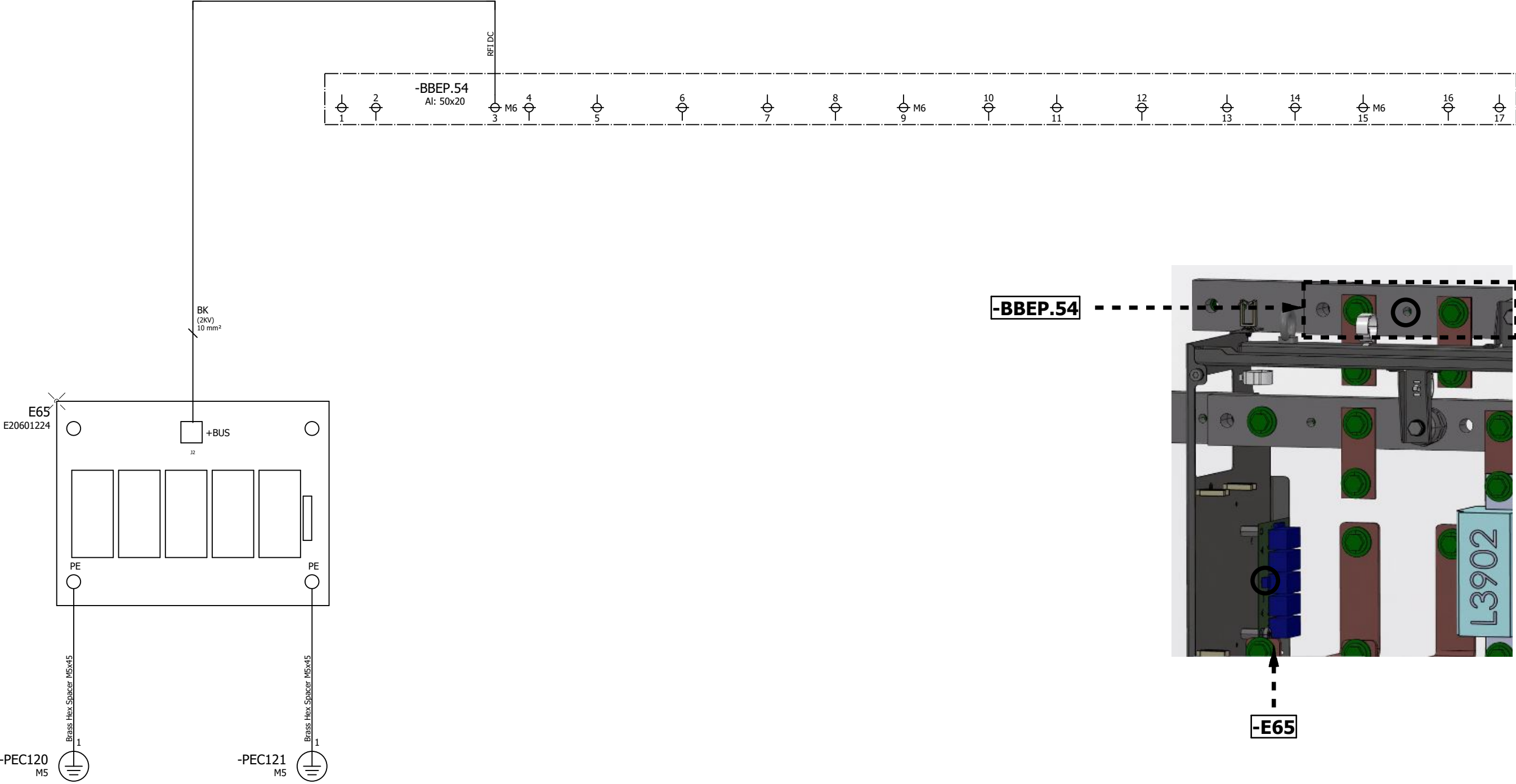


IMAGE AS A GUIDANCE INDICATING CONNECTION POINTS
(MAY VARY DEPENDING ON EACH CUSTOMER'S DU CONFIGURATION).

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-E65	E20601224AA	E20601224AA	RFI DC HEM GENIII PCB BOARD	1
-E67	PCB E21801204	PCB E21801204	DU MEASURES PCB	1
-SH01		8535	SHUNT RESISTANCE 36W	1
-SH02		8535	SHUNT RESISTANCE 36W	1
-SH03		8535	SHUNT RESISTANCE 36W	1
-SH04		8535	SHUNT RESISTANCE 36W	1
-SH05		8535	SHUNT RESISTANCE 36W	1
-SH06		8535	SHUNT RESISTANCE 36W	1
-SH07		8535	SHUNT RESISTANCE 36W	1
-SH08		8535	SHUNT RESISTANCE 36W	1
-SH11		8535	SHUNT RESISTANCE 36W	1
-SH12		8535	SHUNT RESISTANCE 36W	1
-SH13		8535	SHUNT RESISTANCE 36W	1
-SH14		8535	SHUNT RESISTANCE 36W	1
-SH15		8535	SHUNT RESISTANCE 36W	1
-SH16		8535	SHUNT RESISTANCE 36W	1
-SH17		8535	SHUNT RESISTANCE 36W	1
-SH18		8535	SHUNT RESISTANCE 36W	1

Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+ADU.PFU2-W273	OPTICAL FIBER DU MEASURES	L7589

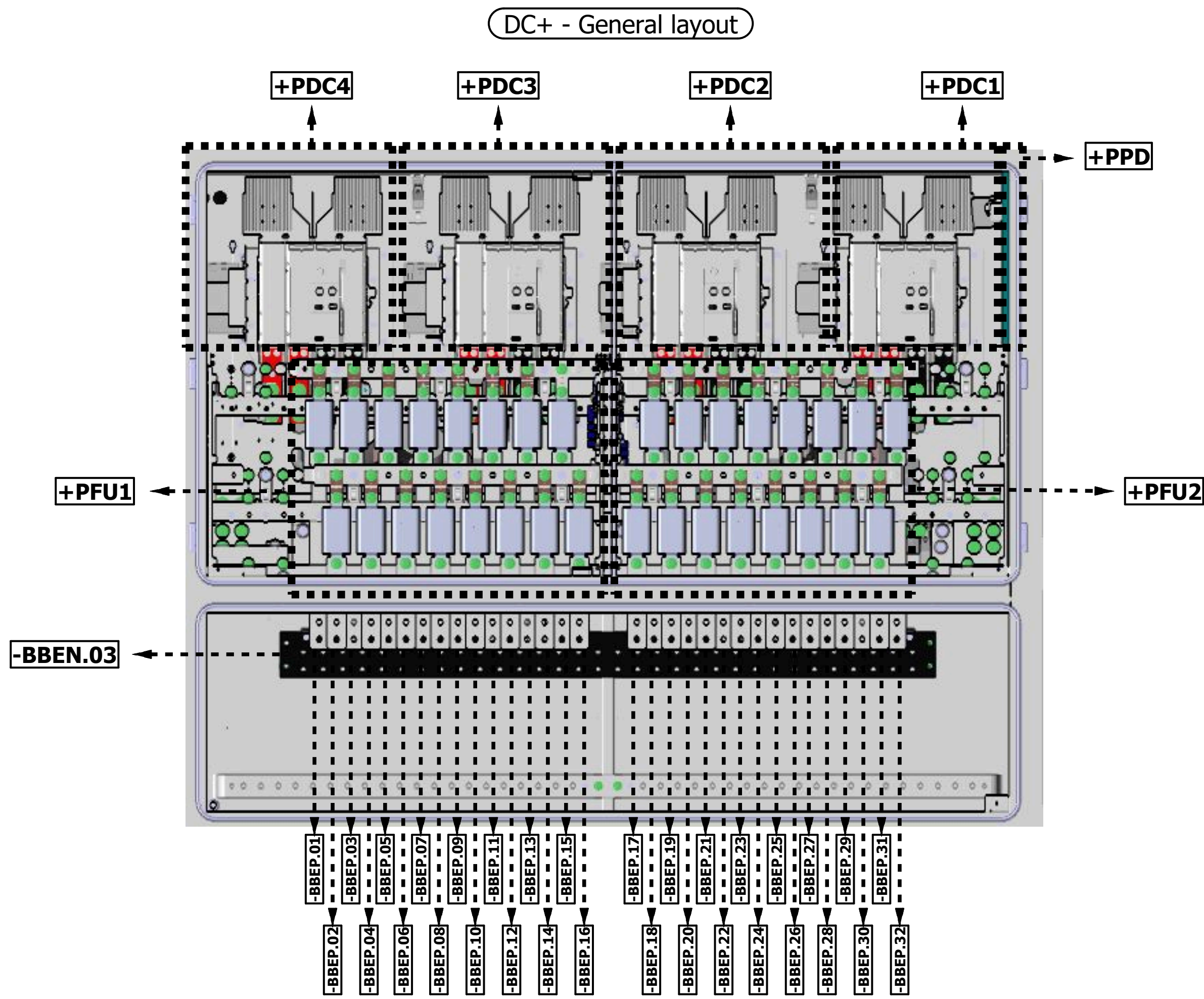
Plug overview

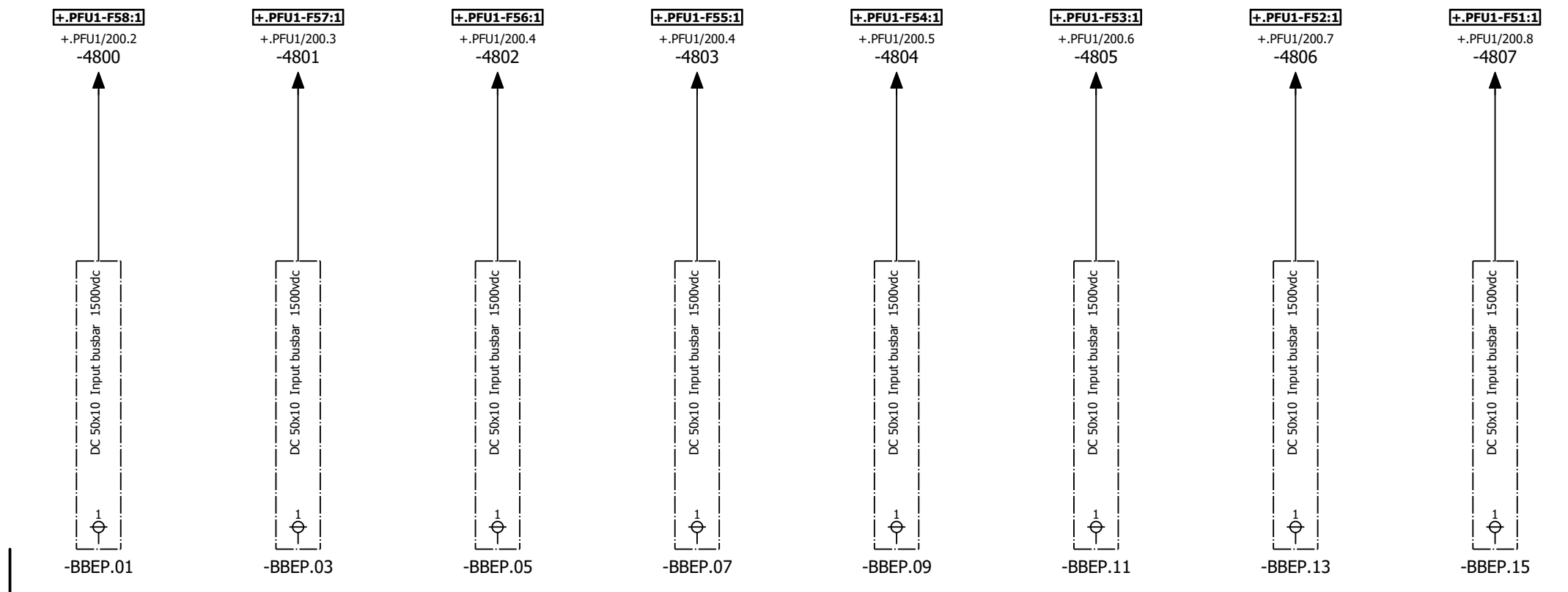
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PFU2-XC490.1	CONNECTOR OPTICAL FIBER DUPLEX DU MEASURES	L7620	1
=HEM++ZMI+ADU.PFU2-XC491.1	CONNECTOR OPTICAL FIBER DUPLEX DU MEASURES	L7620	1



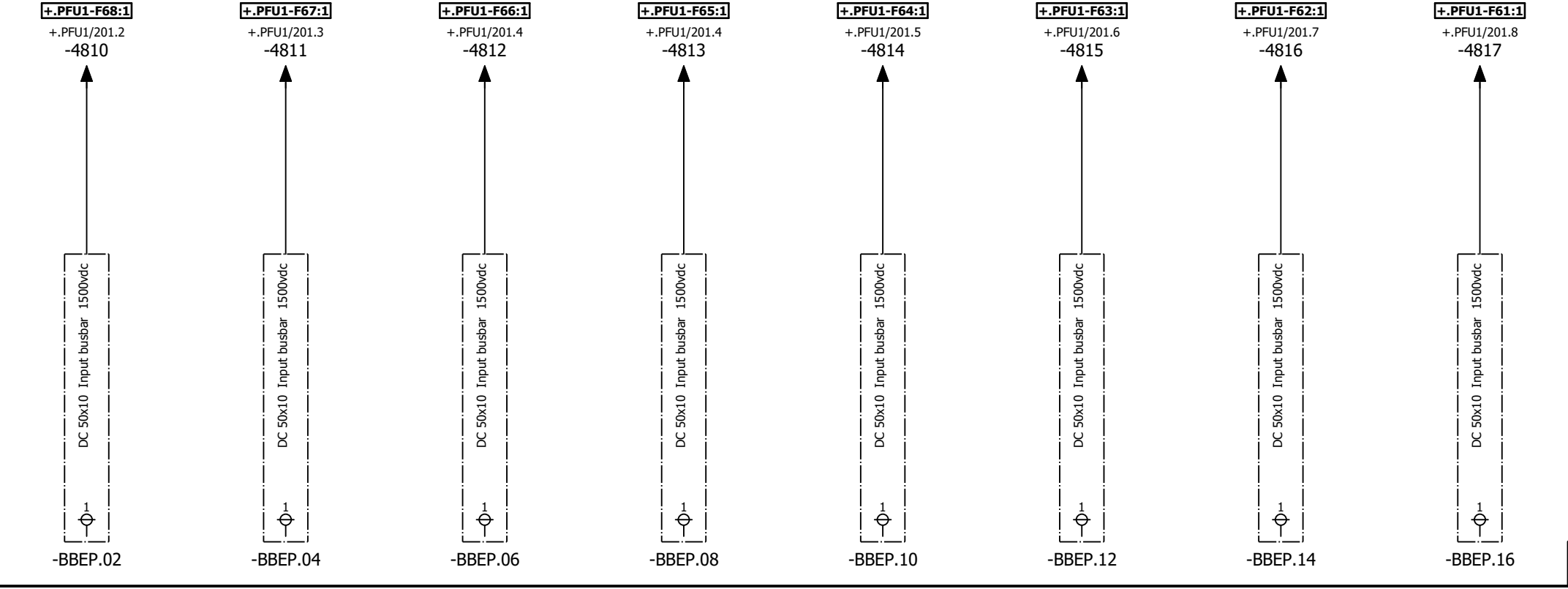
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU.PADU	CUSTOMER PV INPUT		
PADU CUSTOMER PV INPUT			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

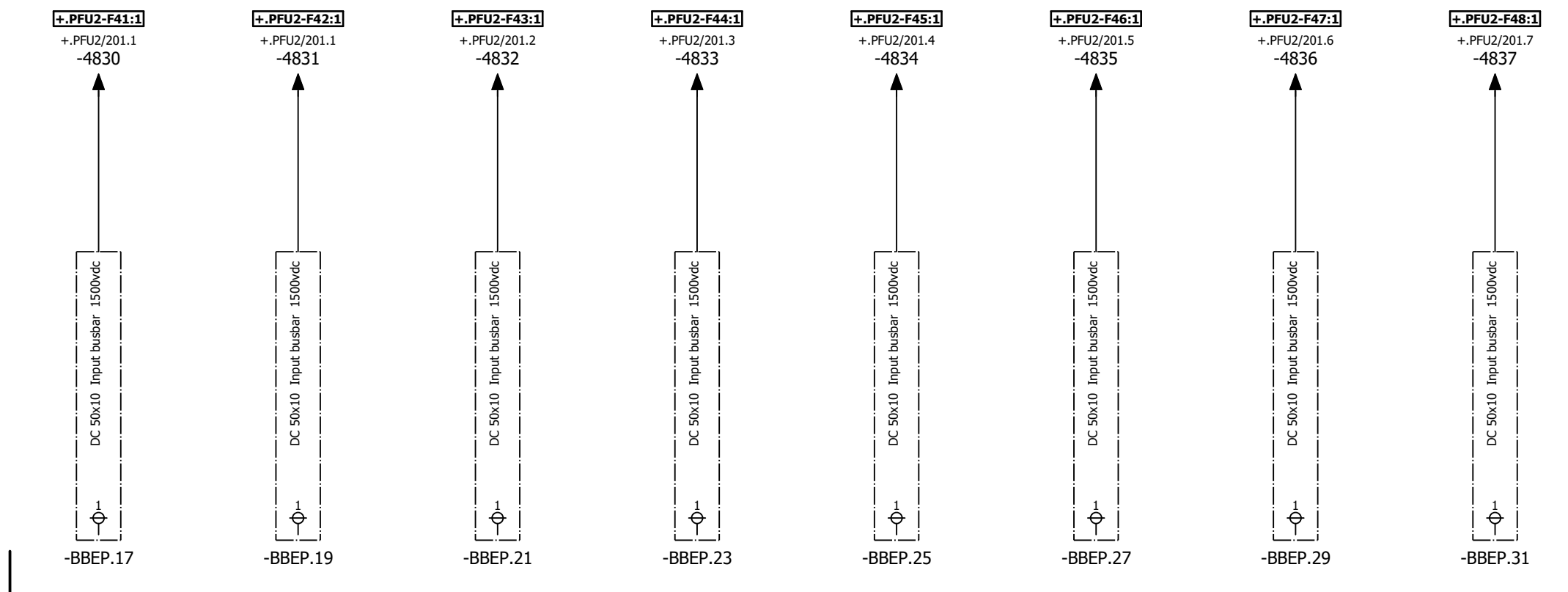




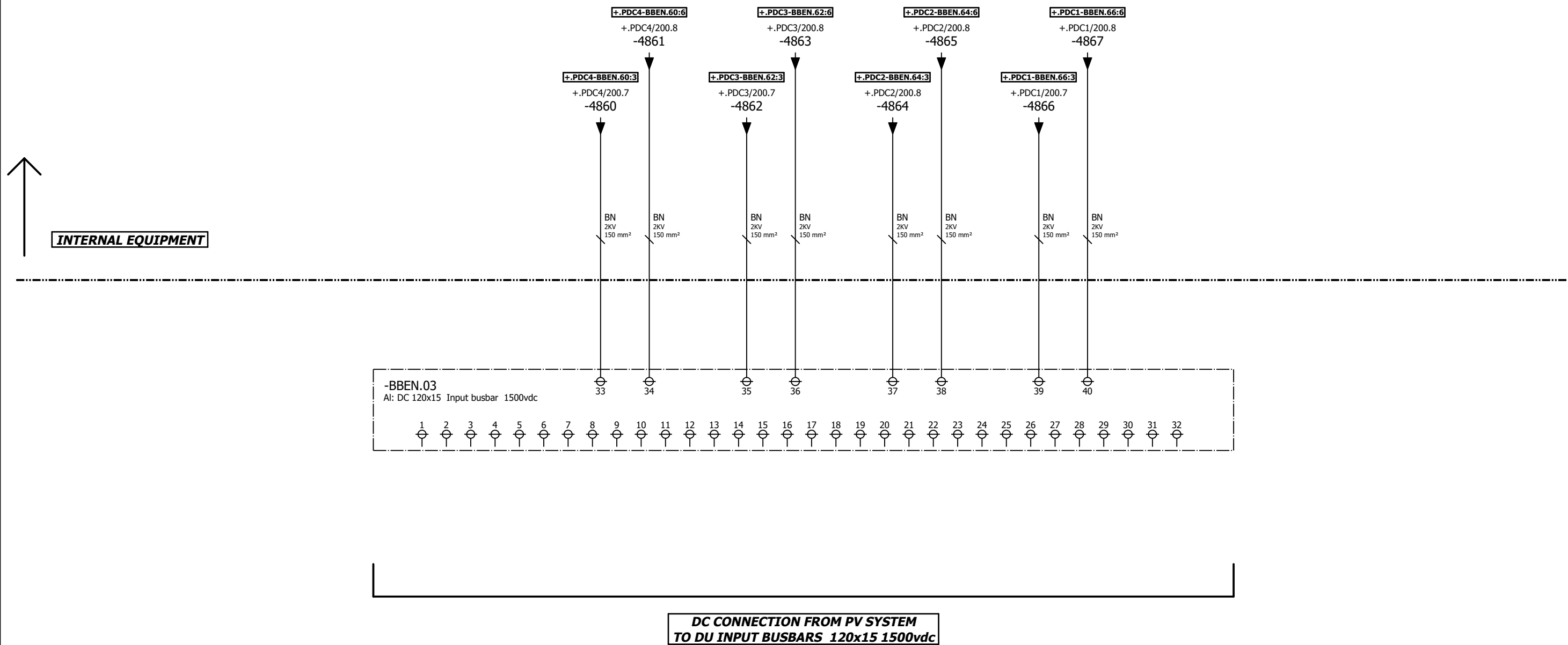
**DC CONNECTION FROM PV SYSTEM
TO DU INPUT BUSBARS 50x10 1500vdc**



DC CONNECTION FROM PV SYSTEM
TO DU INPUT BUSBARS 50x10 1500vdc



**DC CONNECTION FROM PV SYSTEM
TO DU INPUT BUSBARS 50x10 1500vdc**

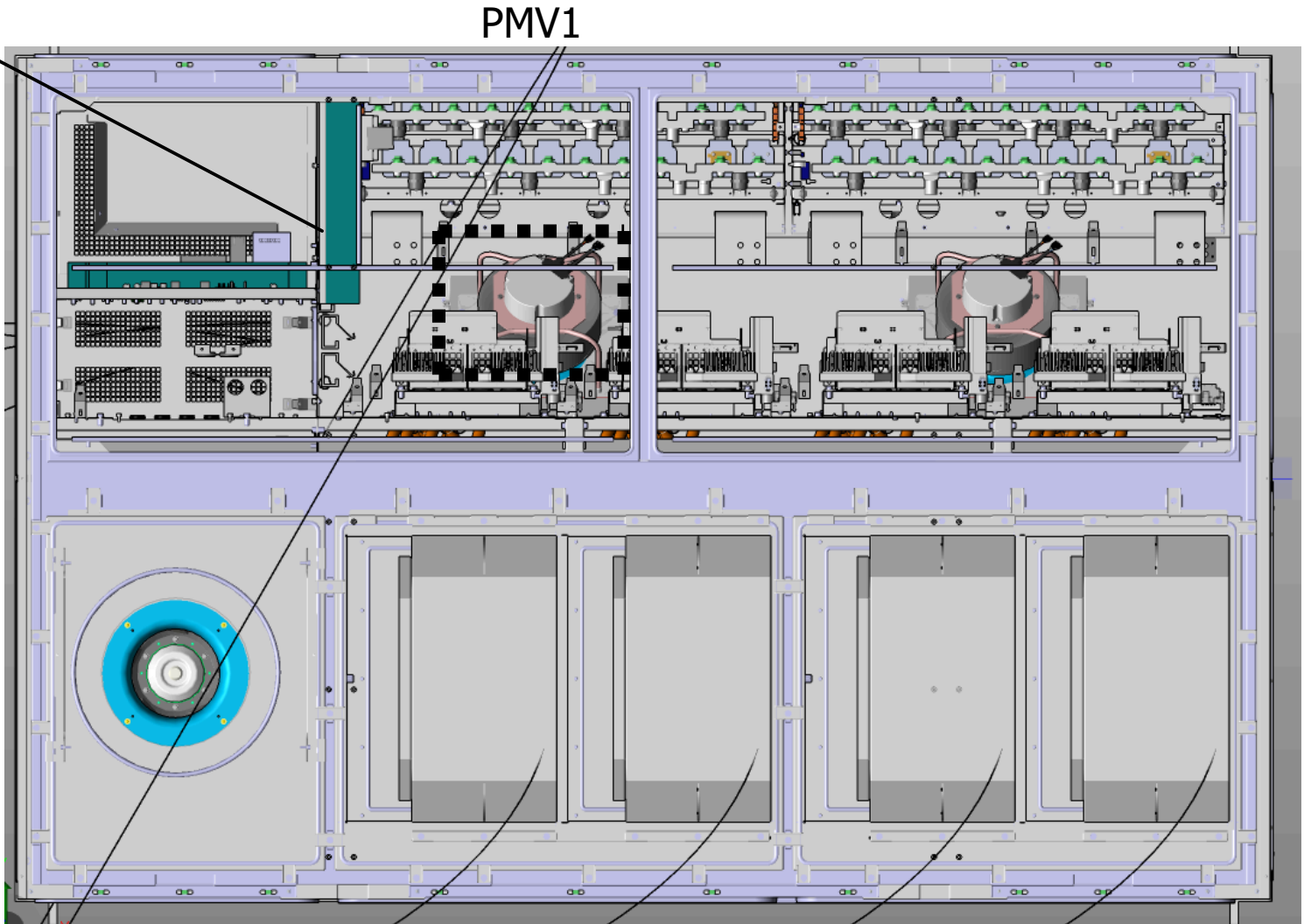
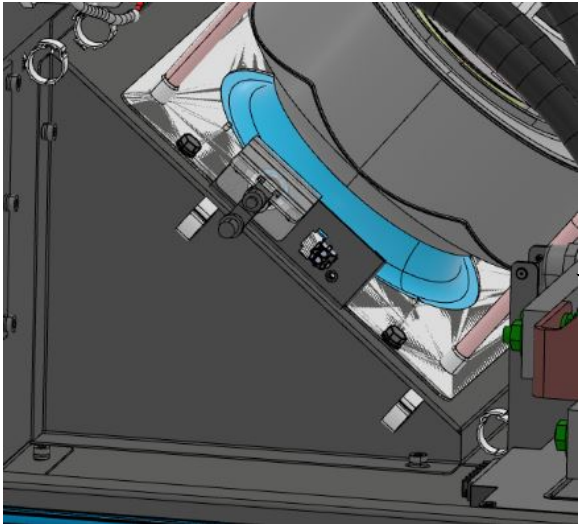




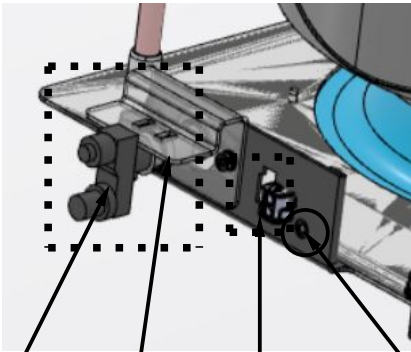
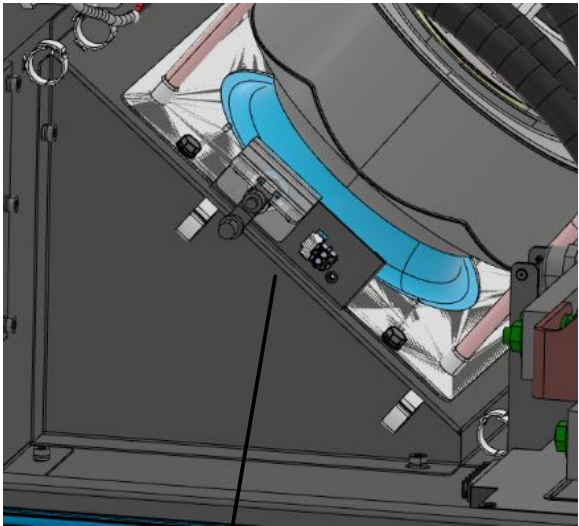
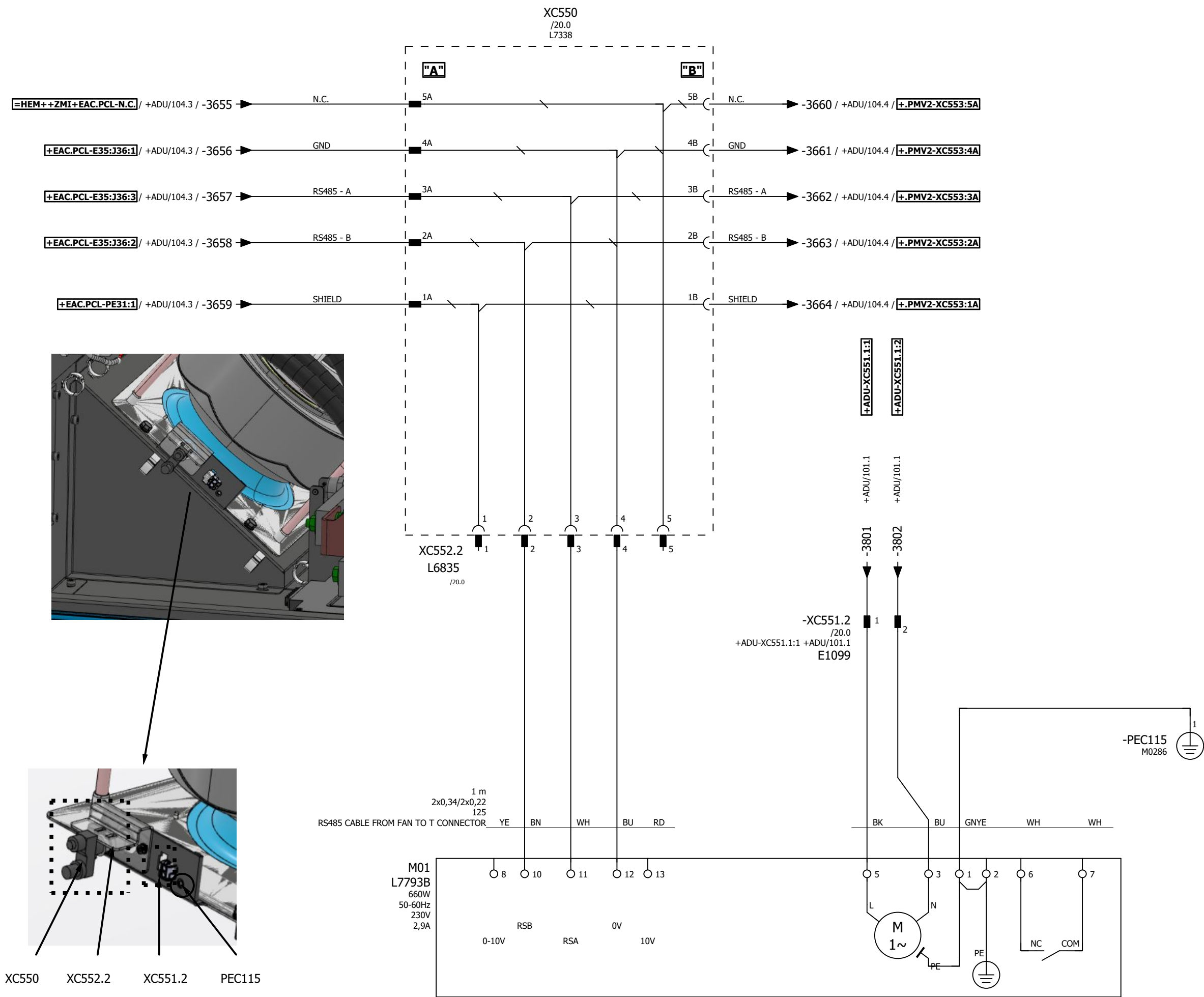
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU.PMV1	FAN PRE-ASSEMBLY 1		
PMV1 FAN PRE-ASSEMBLY 1			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV1



0	1	2	3	4	5	6	7	8	9
-XC550 = T CONNECTOR M12 RS485 -XC551.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 1 -XC552.2 = MALE CONNECTOR M12 - TEMPORARY (PREASSEMBLY FAN CABLE)									



XC550 XC552.2 XC551.2 PEC115

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M01	L7793B	FALTA CONFIRMAR CODIGO CON EBM	VENTILADOR CENT. 2100M3/H 230VAC 660W	1

Plug overview

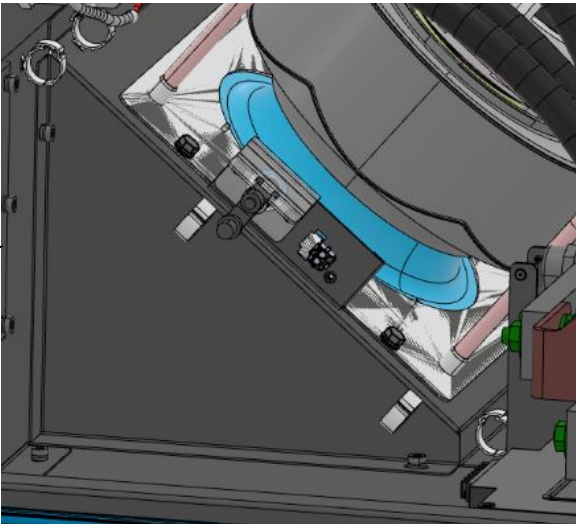
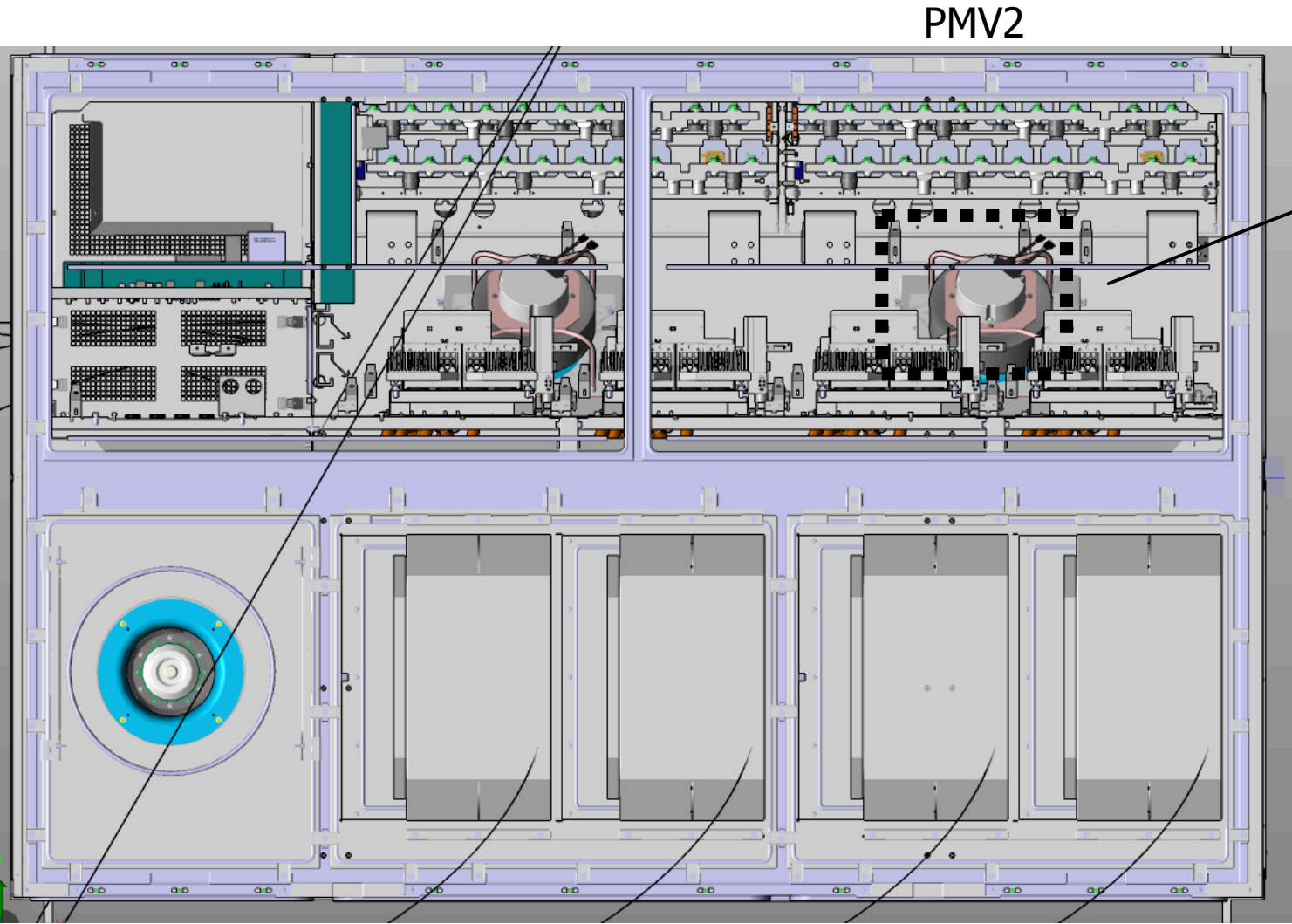
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PMV1-XC550	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+ADU.PMV1-XC551.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 1	E1099	1
		L5710	2
=HEM++ZMI+ADU.PMV1-XC552.2	MALE CONNECTOR M12 - TEMPORARY (PREASSEMBLY FAN CABLE)	L6835	1



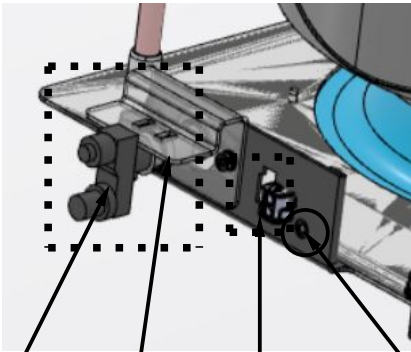
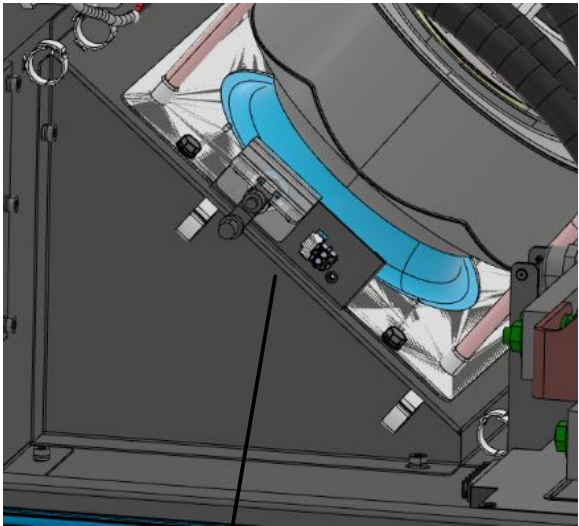
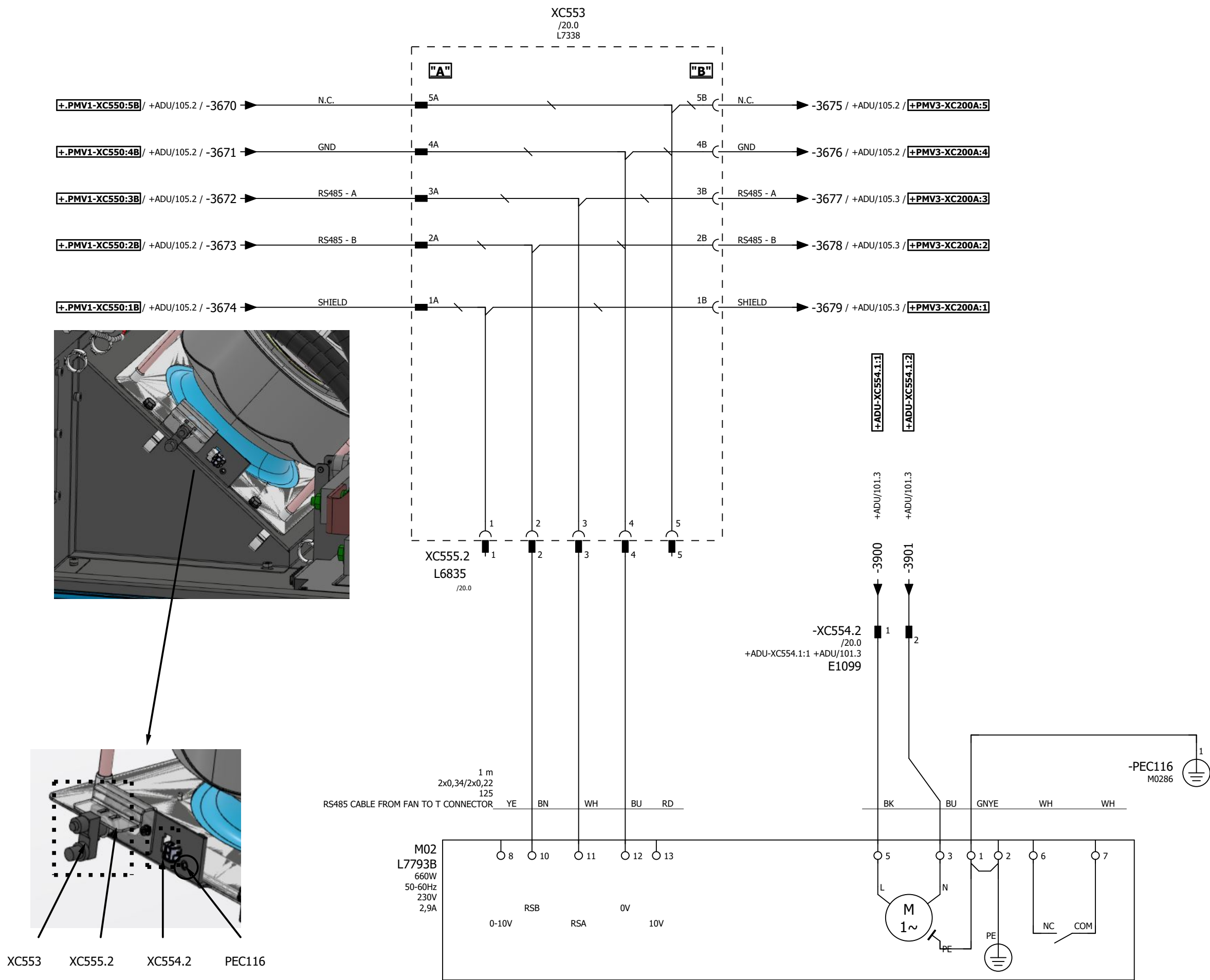
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU.PMV2	FAN PRE-ASSEMBLY 2		
PMV2 FAN PRE-ASSEMBLY 2			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV2



0	1	2	3	4	5	6	7	8	9
-XC553 = T CONNECTOR M12 RS485 -XC554.2 = MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 2 -XC555.2 = MALE CONNECTOR M12 - TEMPORARY (PREASSEMBLY FAN CABLE)									



XC553 XC555.2 XC554.2 PEC116

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M02	L7793B	FALTA CONFIRMAR CODIGO CON EBM	VENTILADOR CENT. 2100M3/H 230VAC 660W	1

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PMV2-XC553	T CONNECTOR M12 RS485	L7338	1
=HEM++ZMI+ADU.PMV2-XC554.2	MATE-N-LOCK BASE POWER SUPPLY FAN ENGINE 2	E1099	1
		L5710	2
=HEM++ZMI+ADU.PMV2-XC555.2	MALE CONNECTOR M12 - TEMPORARY (PREASSEMBLY FAN CABLE)	L6835	1



DESIGNED	I+D 1
REVISED	I+D 2
APPROVED	I+D 3

DATE
REVISION
1.3.11

Project
ESQ_FS4010M10U000012_000

Name
Plug overview : =HEM++ZMI+ADU.PMV2-XC553 - =HEM++ZMI+ADU.PMV2-XC555.2

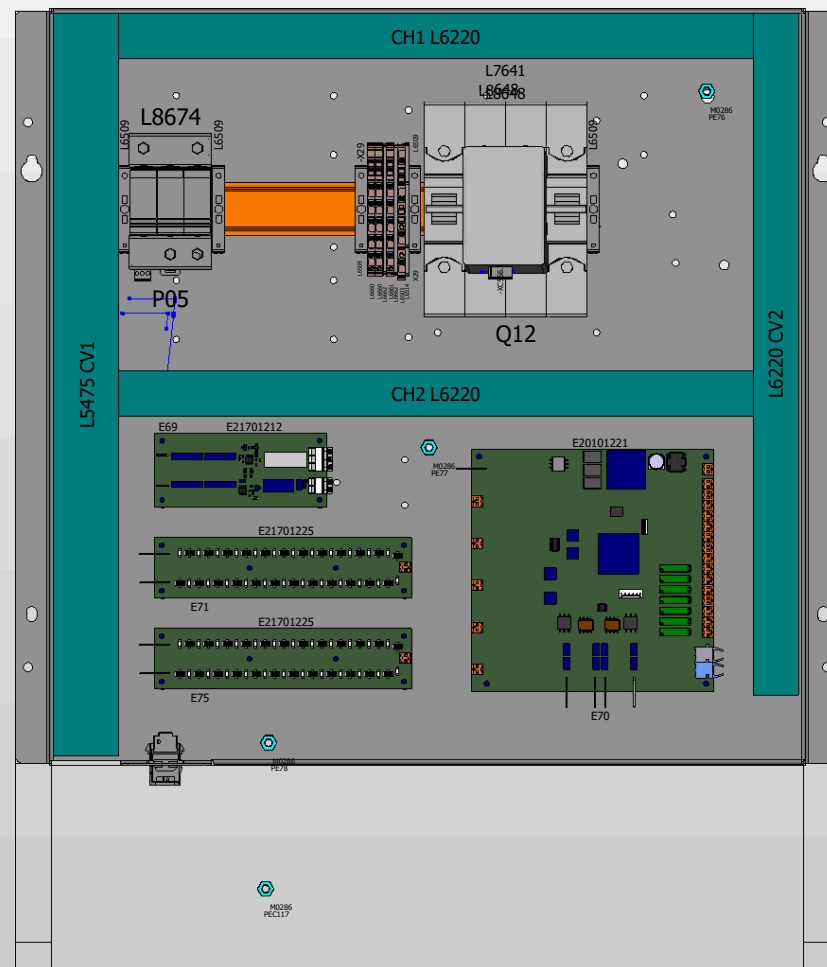
Installation:	HEM INVERTER	HEM	A3
Mounting location: +ADU.PMV2 DU CABINET	++ZMI	Previous page &MAT/1000	Next page +.PPD&/1
Project Code: FS4010M10U10001100410010020101014	Page 3000	Total number of pages 455	



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
ZMI	INVERTER ASSEMBLY AREA		
ADU.PPD	DC PROTECTIONS PANEL		
PPD DC PROTECTIONS PANEL			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

LAYOUT PANEL



Enclosure legend

IME	ERP	Description
Q12	L7641	High performance circuit breaker für GFDI-Anwendungen tested acc. to UL489B
CV1	L5475	CABLE DUCT UNEX 80X43 EN U23X GR (60°)
CH1;CH2;CV2	L6220	UNEX CHANNEL 80X30 IN U23X (60°)
CA1	L1061	DIN RAIL 35x7,5mm
E69	PCB E21701212	BUS SIGNAL BOARD
Q12	L8648	Remote Switching Unit for S800
E71;E75	PCB E21701225	INSULATION MONITORING BOARD (RESISTOR)
P05	L8674	DS50PVS-1500/51 SURGE PROTECTOR DEV.
E70	PCB E20101221	INSULATION MONITORING BOARD 2 CH

-XC047.2 = MATE-N-LOK 9P PANEL CONNECTOR PCL-PPD

-X29 = Distribution 24Vdc

-XC557.1 = CONNECTOR OPTICAL FIBER DUPLEX ISOPRO

-XC416.2 = TRIPLE-LOK 2P ISOPRO FAULT SIGNAL

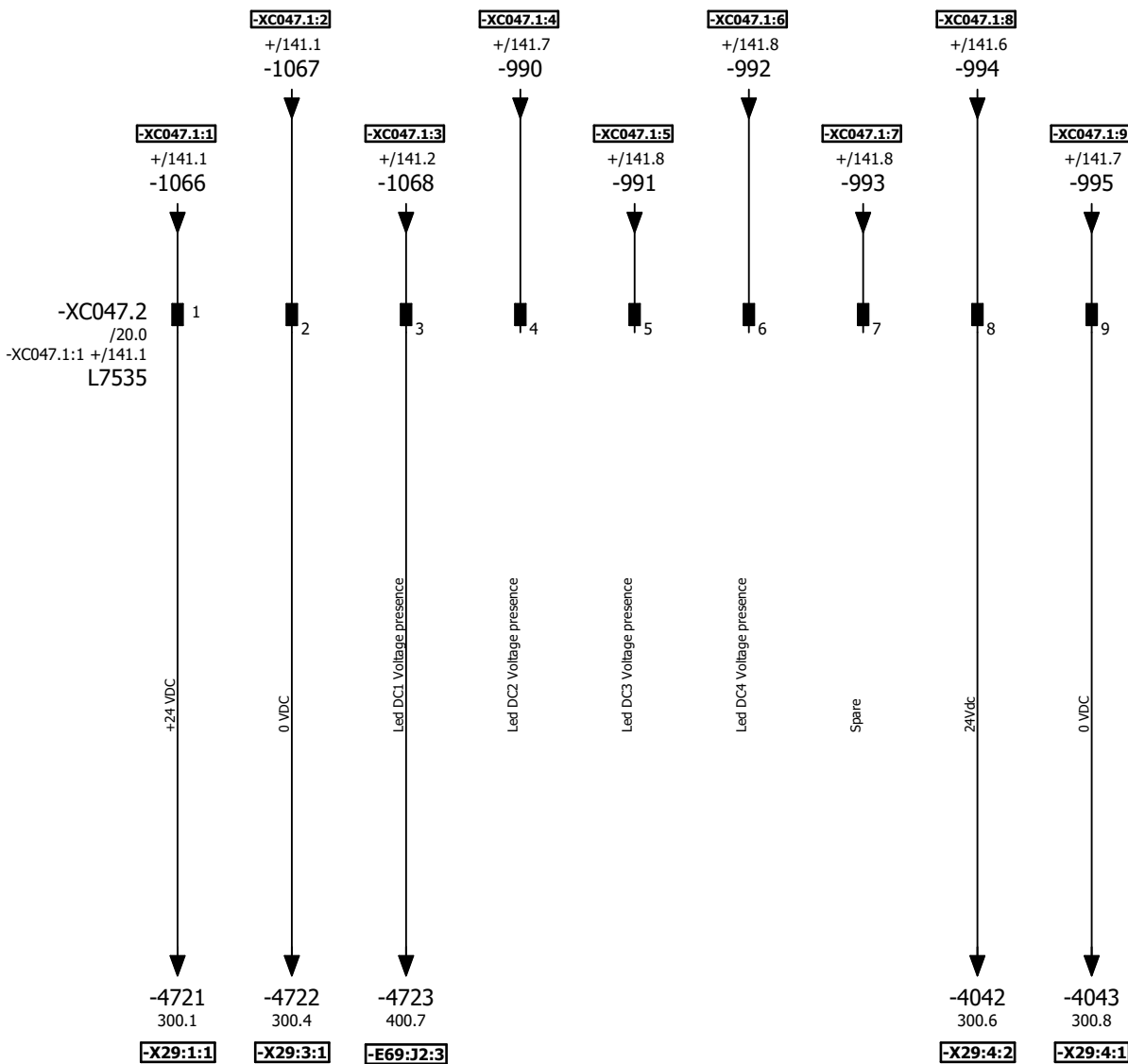


Optional Connectors

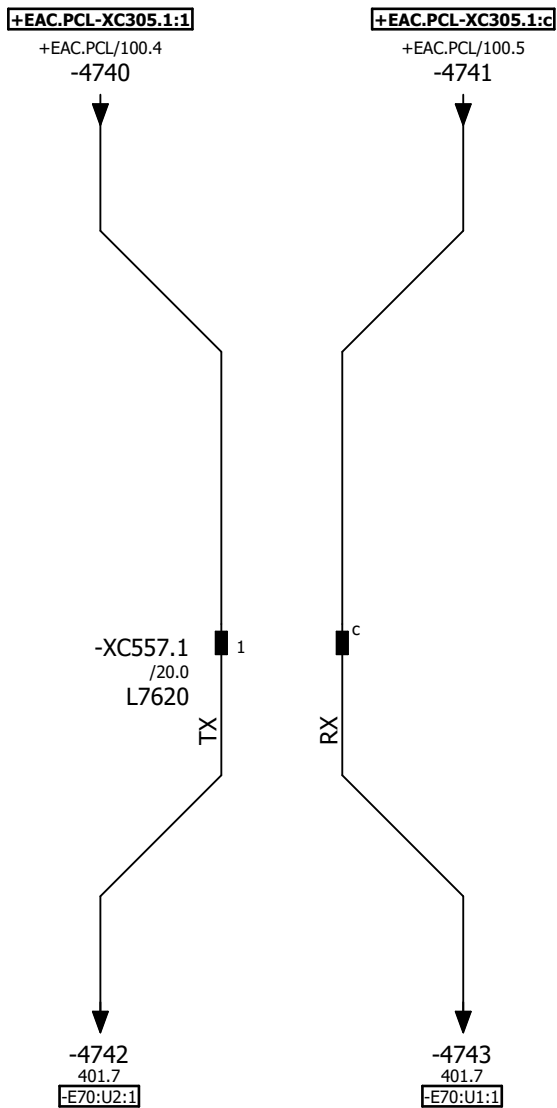
-XC556.1 = GFDI CONNECTOR

Interconnections - EAC.PCL towards ADU.PPD

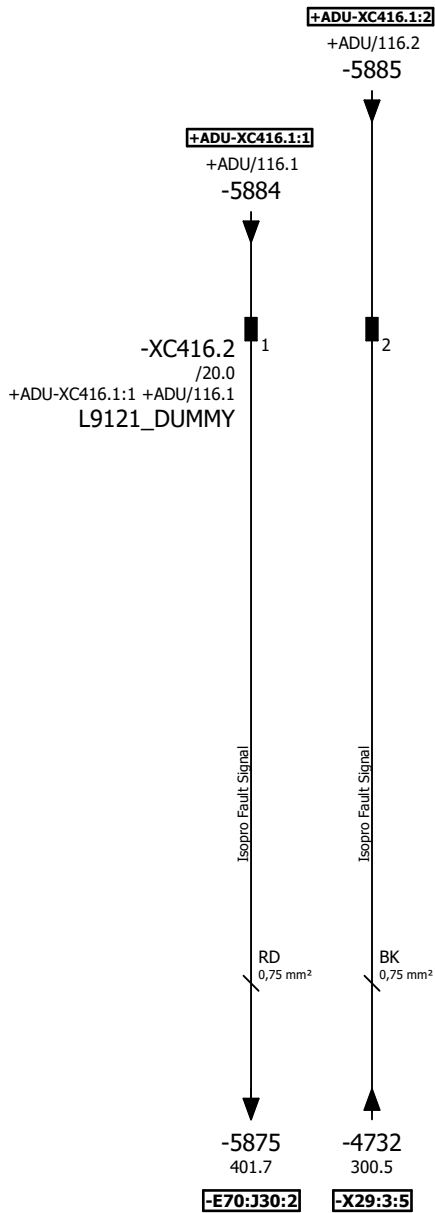
Power supply 24Vdc



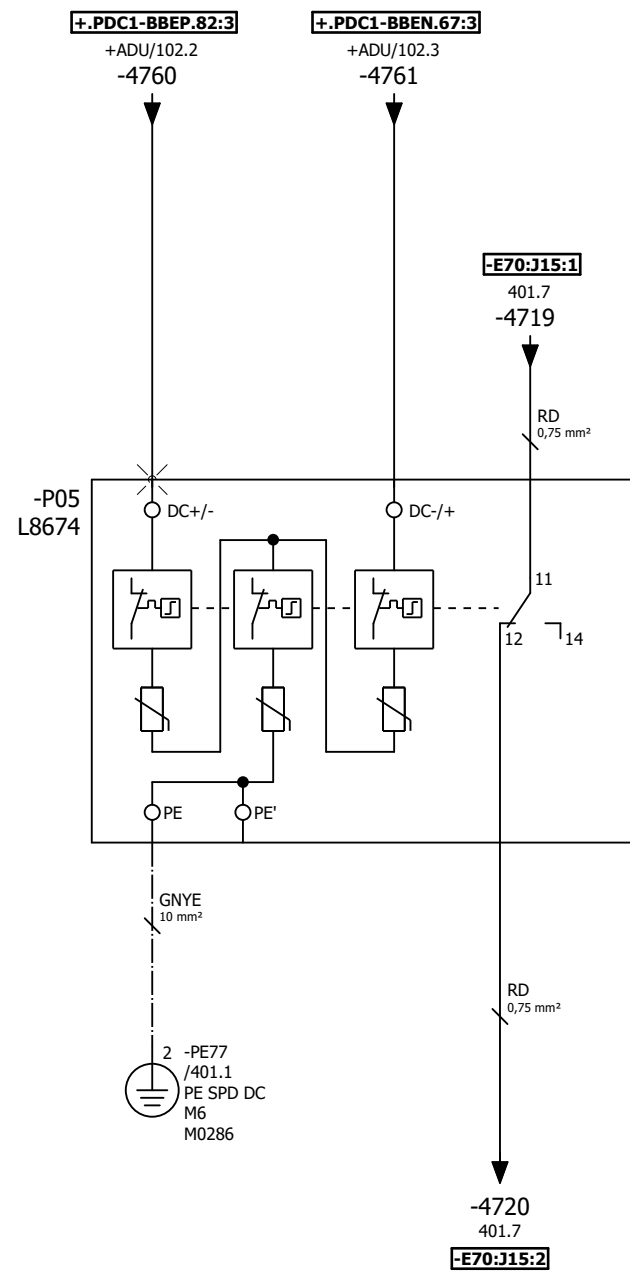
Optical fiber



Interconnections PPD//PCDU



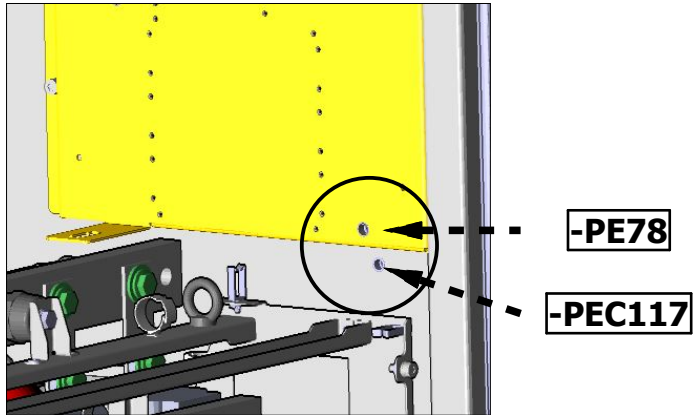
FROM PV-



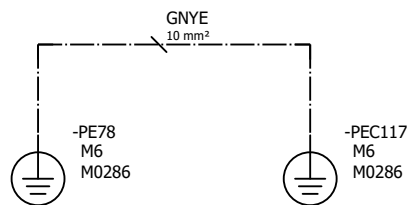
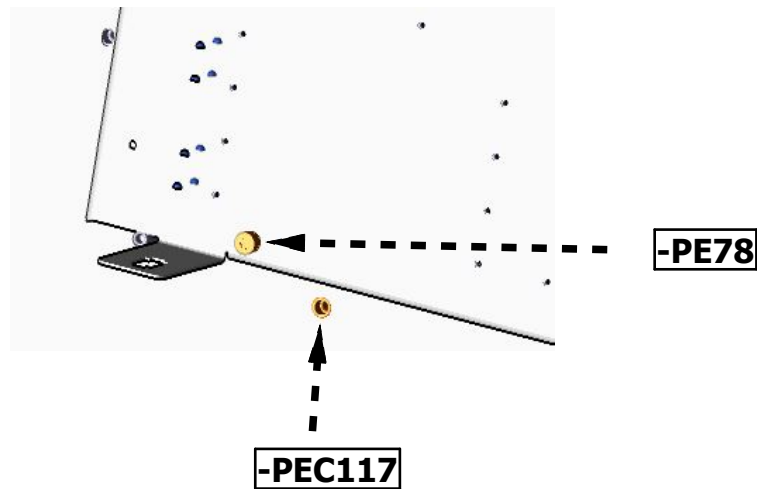
SPD - Type II

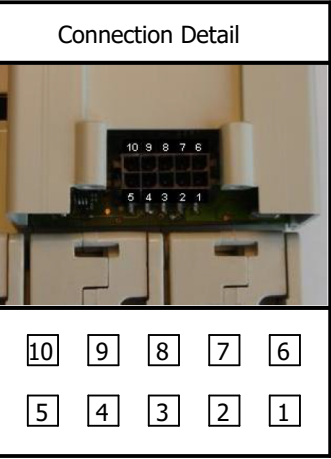
THE IMAGE MAY VARY DEPENDING ON
THE DISCONNECTOR SWITCH MODEL.
THE CONNECTIONS AND IME OF THE PLATES ARE IDENTICAL.

SWITCH DISCONNECTOR 2500A - ADVANCED PANEL

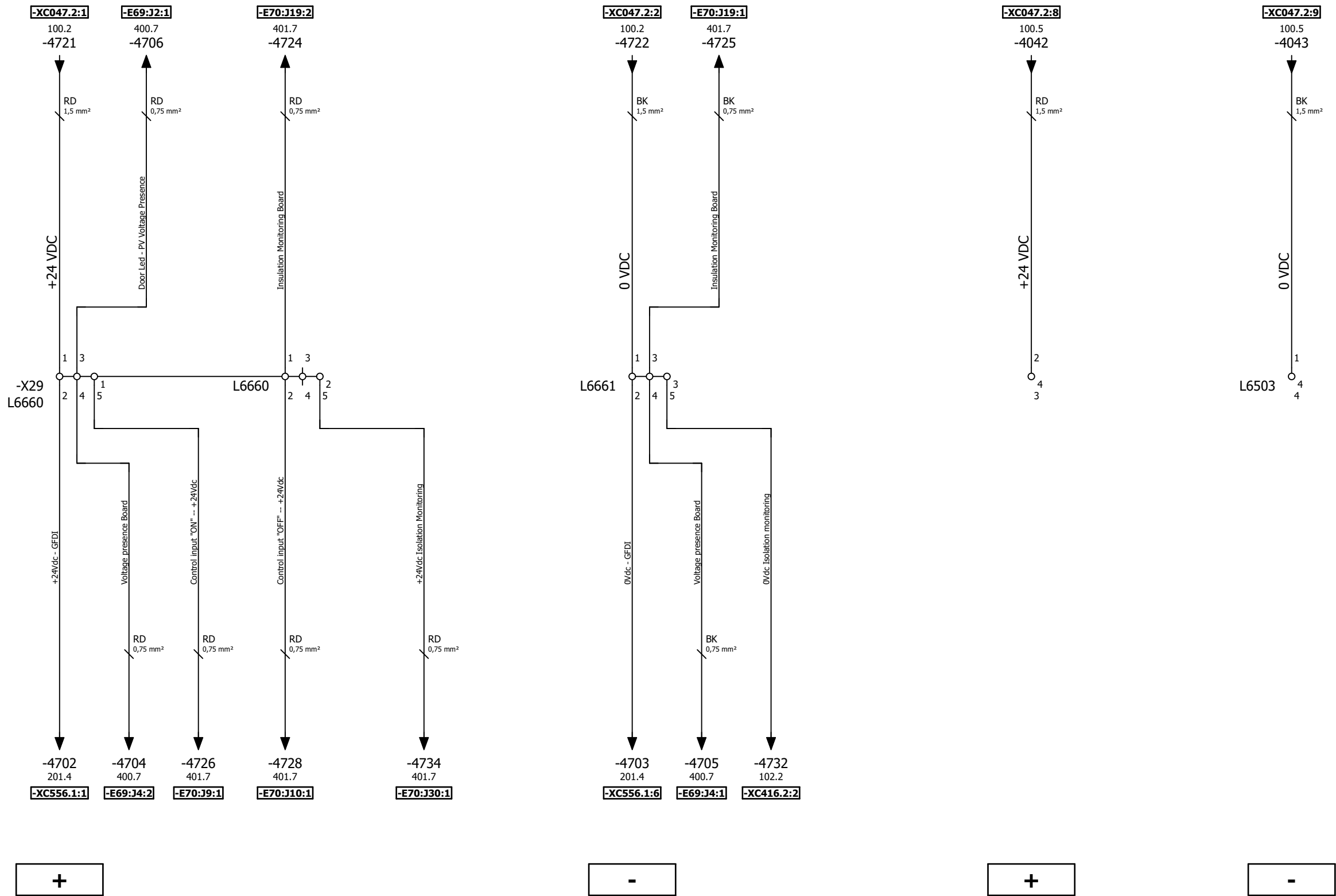


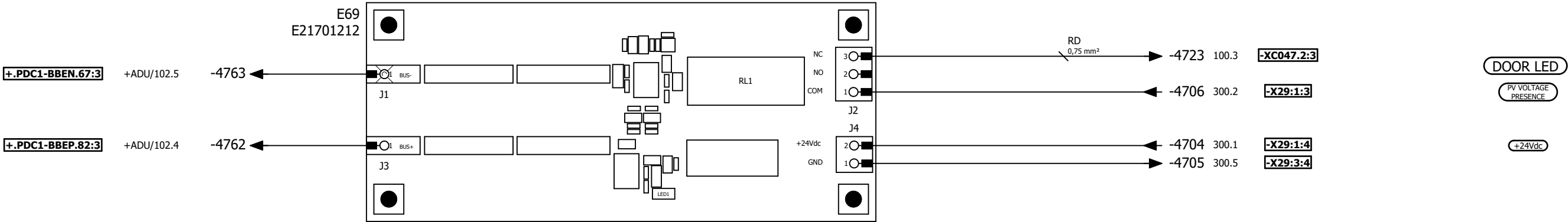
SWITCH DISCONNECTOR 1600A -- BASIC PANEL





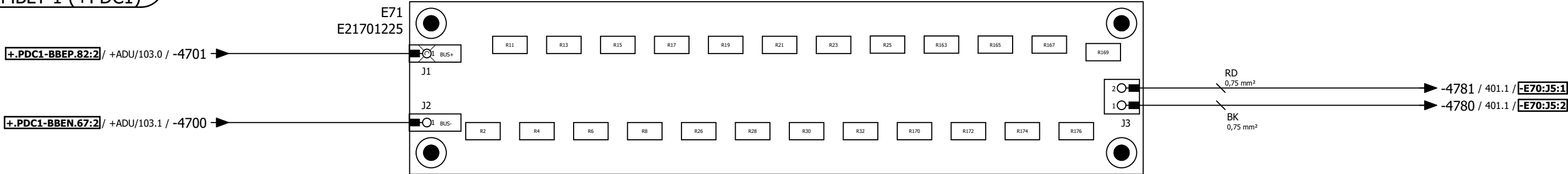
DISTRIBUTION 24VDC



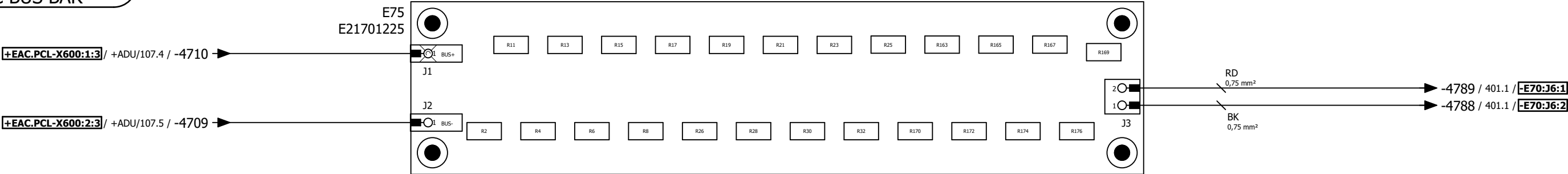




FROM DC DISCONNECTOR
PRE-ASSEMBLY 1 (+PDC1)



AC MEASURE FROM
AC BUS BAR



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E69	PCB E21701212	PCB E21701212	BUS SIGNAL BOARD	1
-E70	PCB E20101221	PCB E20101221	INSULATION MONITORING BOARD 2 CH	1
-E71	PCB E21701225	PCB E21701225	INSULATION MONITORING BOARD (RESISTOR)	1
-E75	PCB E21701225	PCB E21701225	INSULATION MONITORING BOARD (RESISTOR)	1
-P05	L8674	480571	DS50PVS-1500/51 SURGE PROTECTOR DEV.	1
-P05	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	2
-Q12	L7641	2CCF019592R0001	High performance circuit breaker für GFDI-Anwendungen tested acc. to UL489B	1
-Q12	L8648	2CCS800900R0511	Remote Switching Unit for S800	1
-Q12	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1

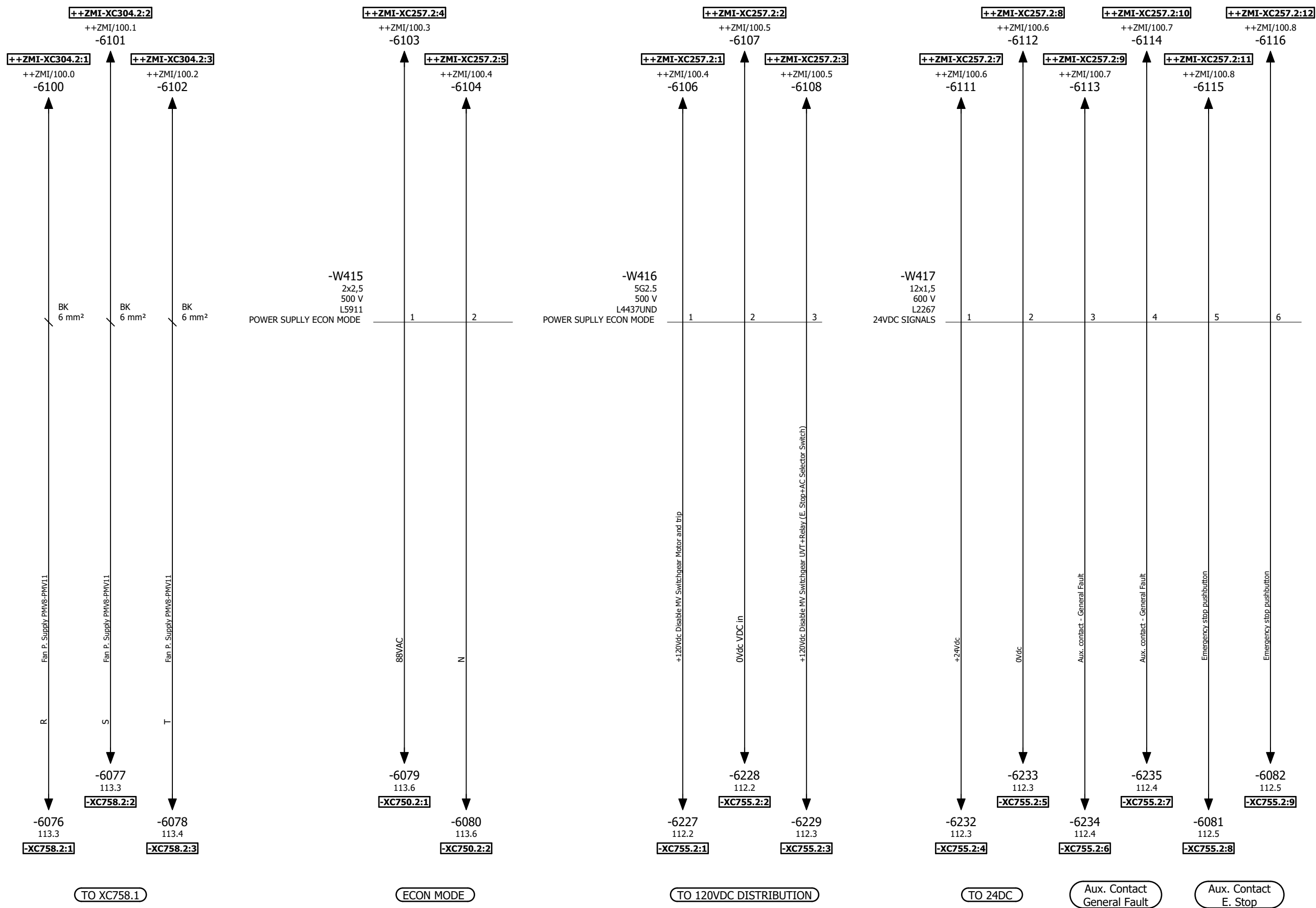
Cable overview

Cable designation	Function text	ERP
=HEM++ZMI+ADU.PPD-W315	+24Vdc - GFDI	L8135

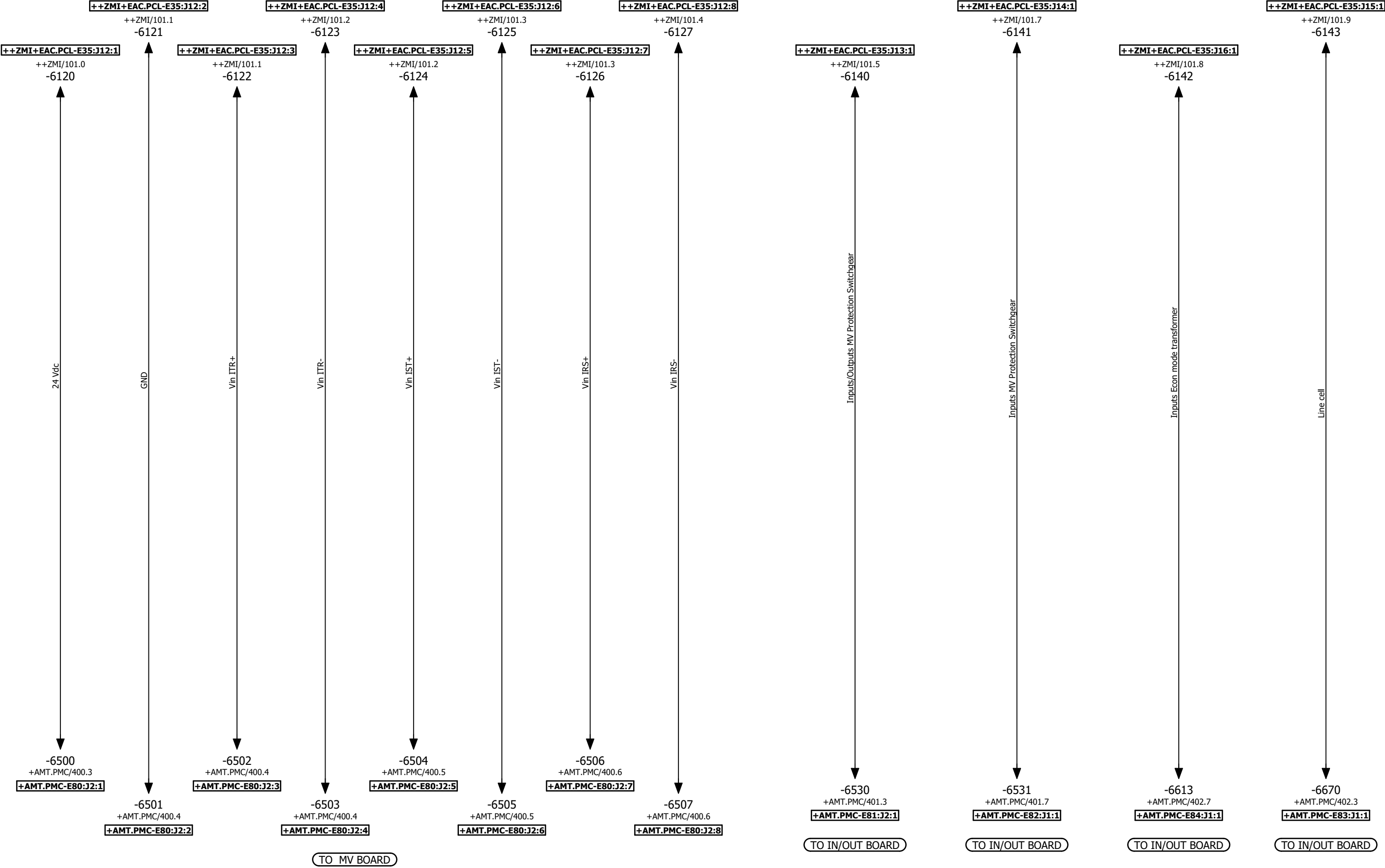
Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++ZMI+ADU.PPD-XC047.2	MATE-N-LOK 9P PANEL CONNECTOR PCL-PPD	L7535	1
		E1204	3
=HEM++ZMI+ADU.PPD-XC416.2	TRIPLE-LOK 2P ISOPRO FAULT SIGNAL	L9121_DUMMY	1
		L9144	1
=HEM++ZMI+ADU.PPD-XC556.1	GFDI CONNECTOR	L8133	1
		L8132	8
=HEM++ZMI+ADU.PPD-XC557.1	CONNECTOR OPTICAL FIBER DUPLEX ISOPRO	L7620	1

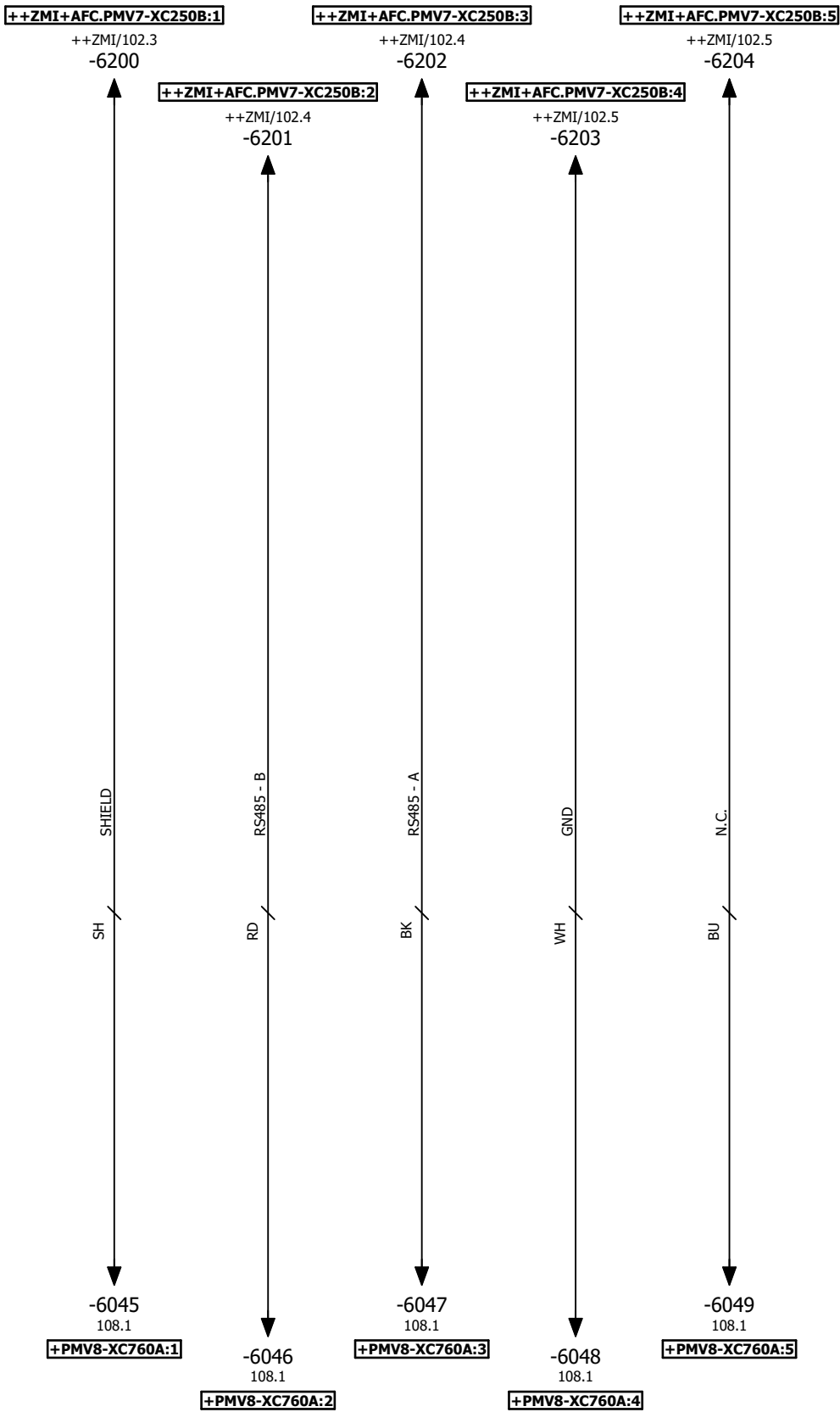
Interconnections ZMI/CDT - P. Supply



Interconnections ZMI/CDT - Comms/Measures

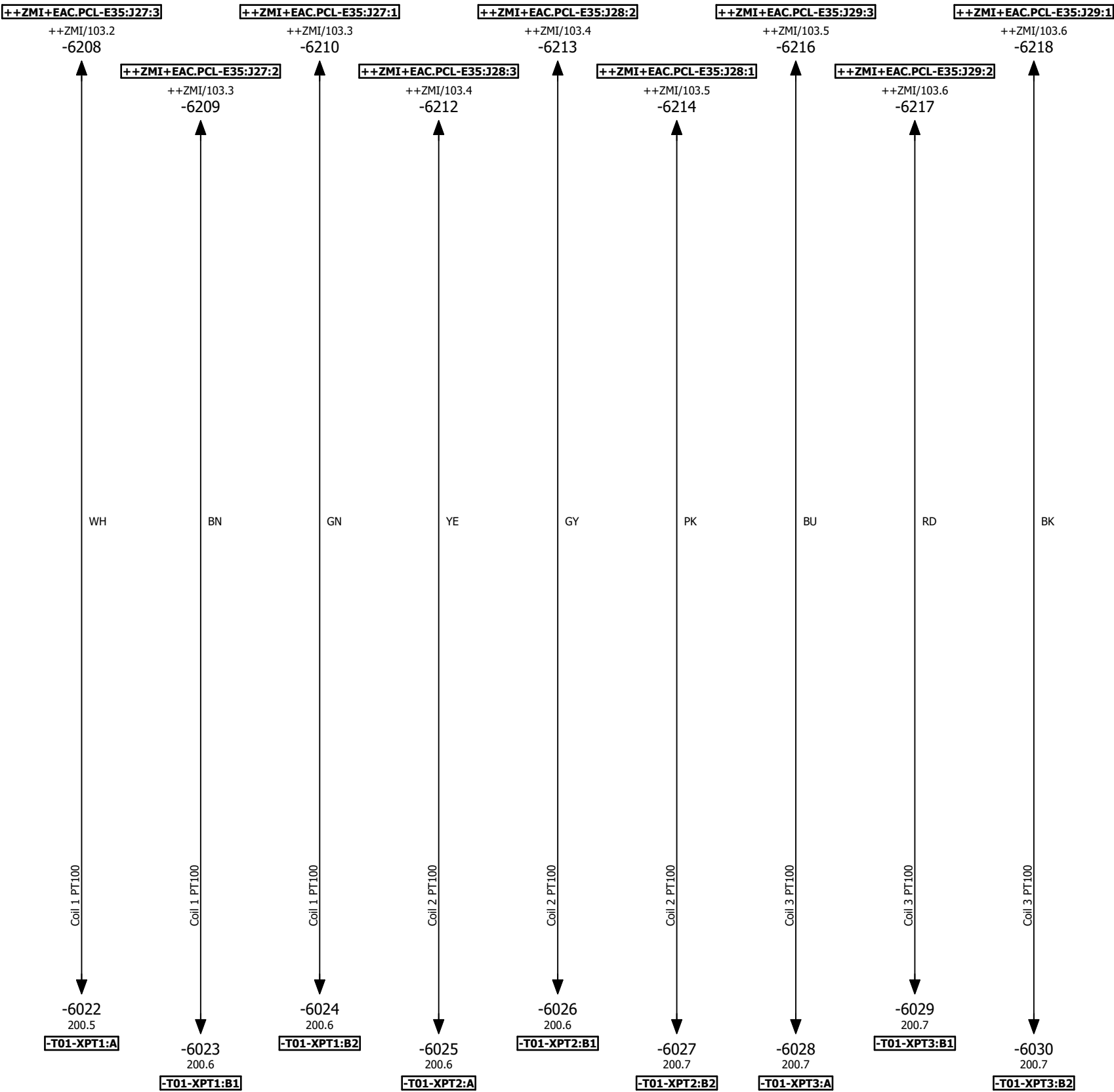


Interconnections ZMI/CDT - Comms/Measures

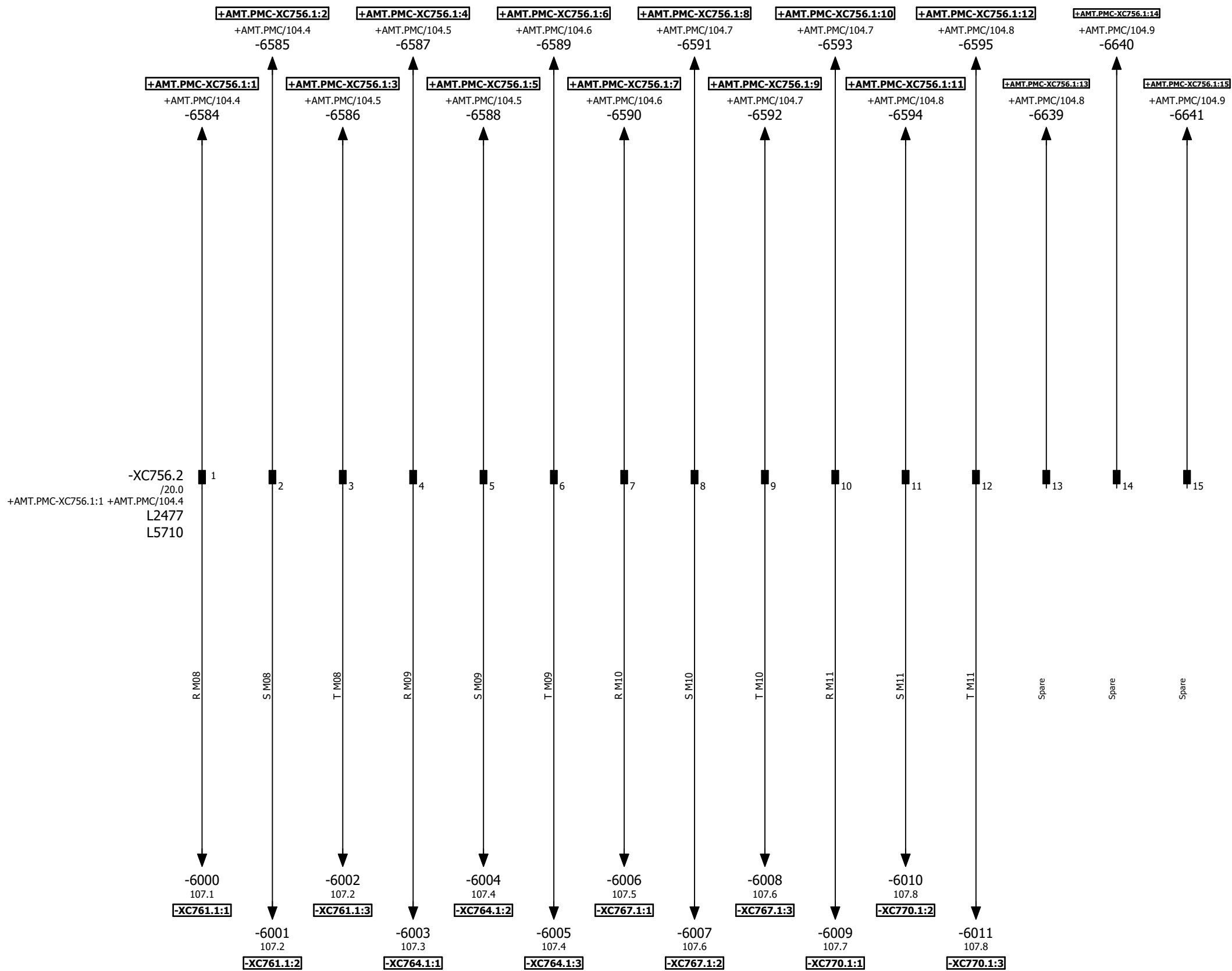


TO RS485 CONNECTION

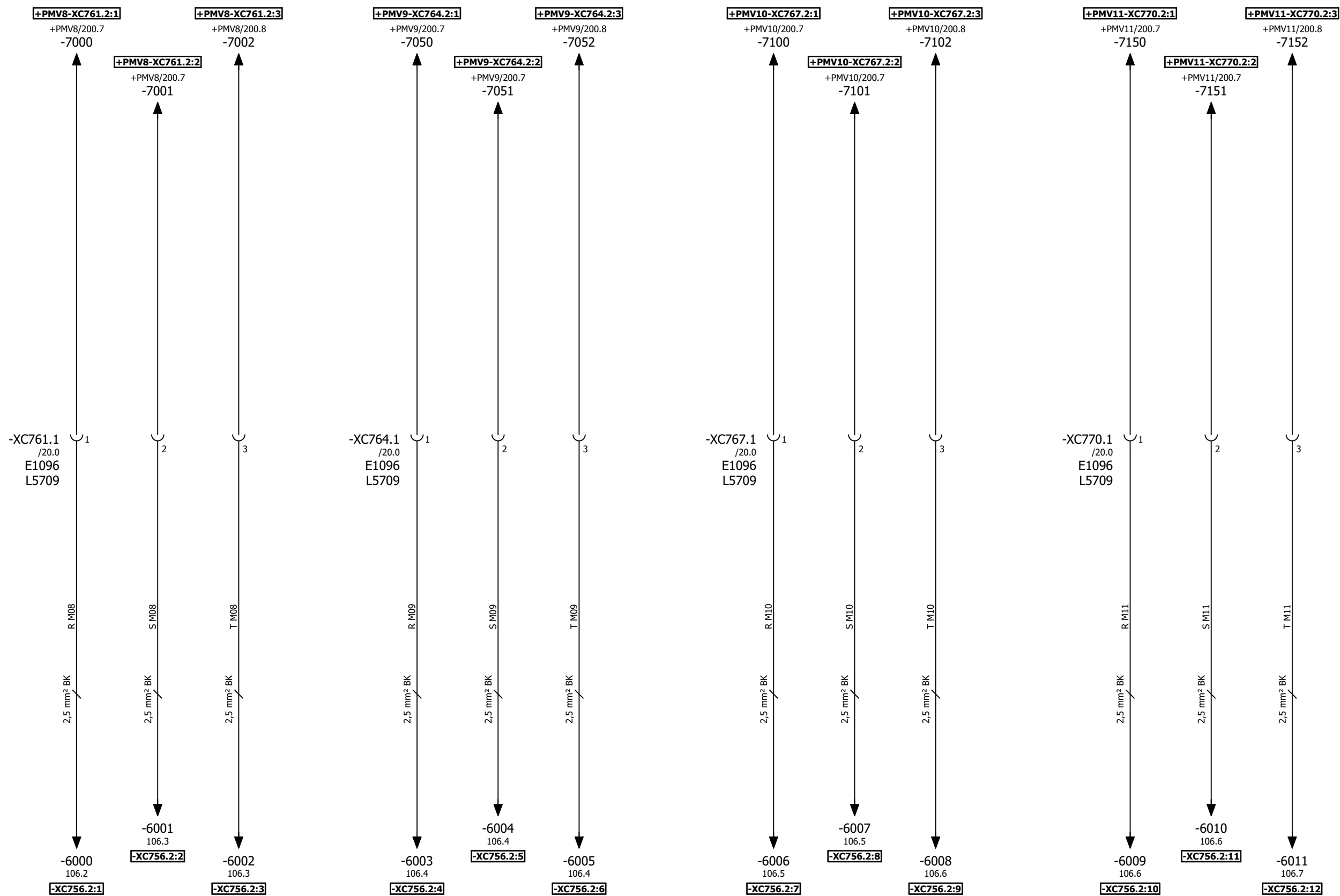
Interconnections ZMI/CDT - Comms/Measures



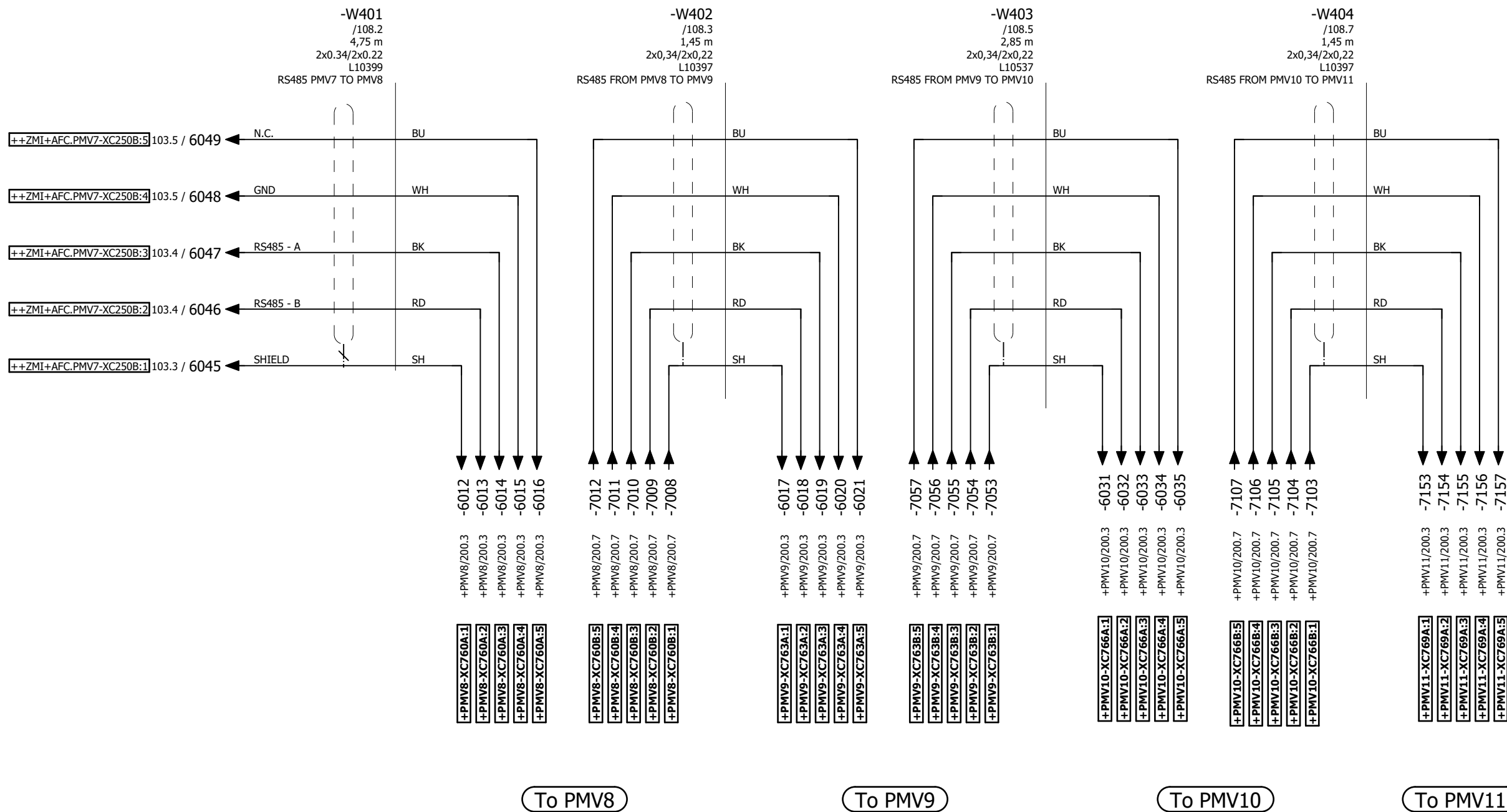
Interconnections - Power supply Fans



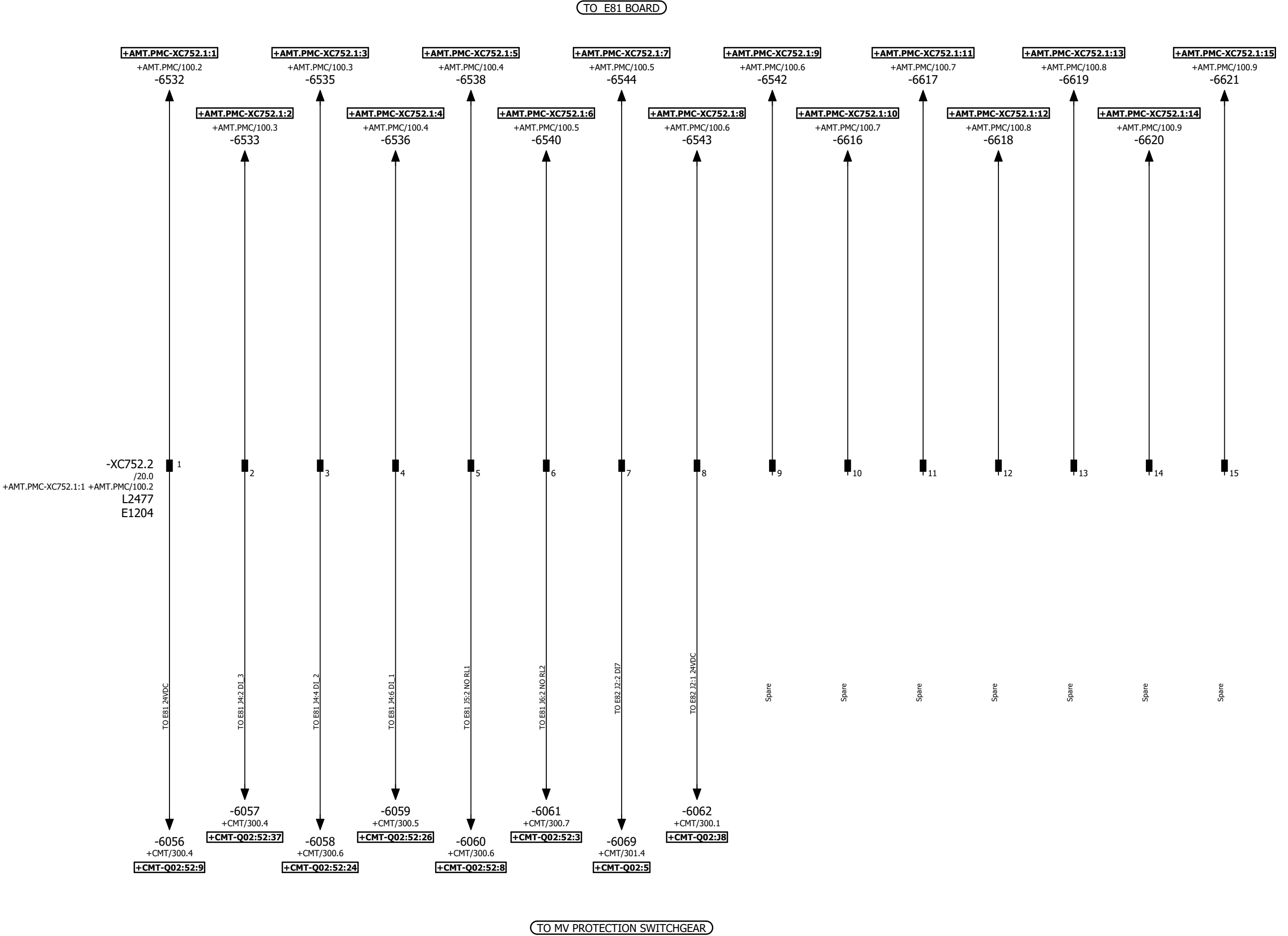
Interconnections - Power supply Fans



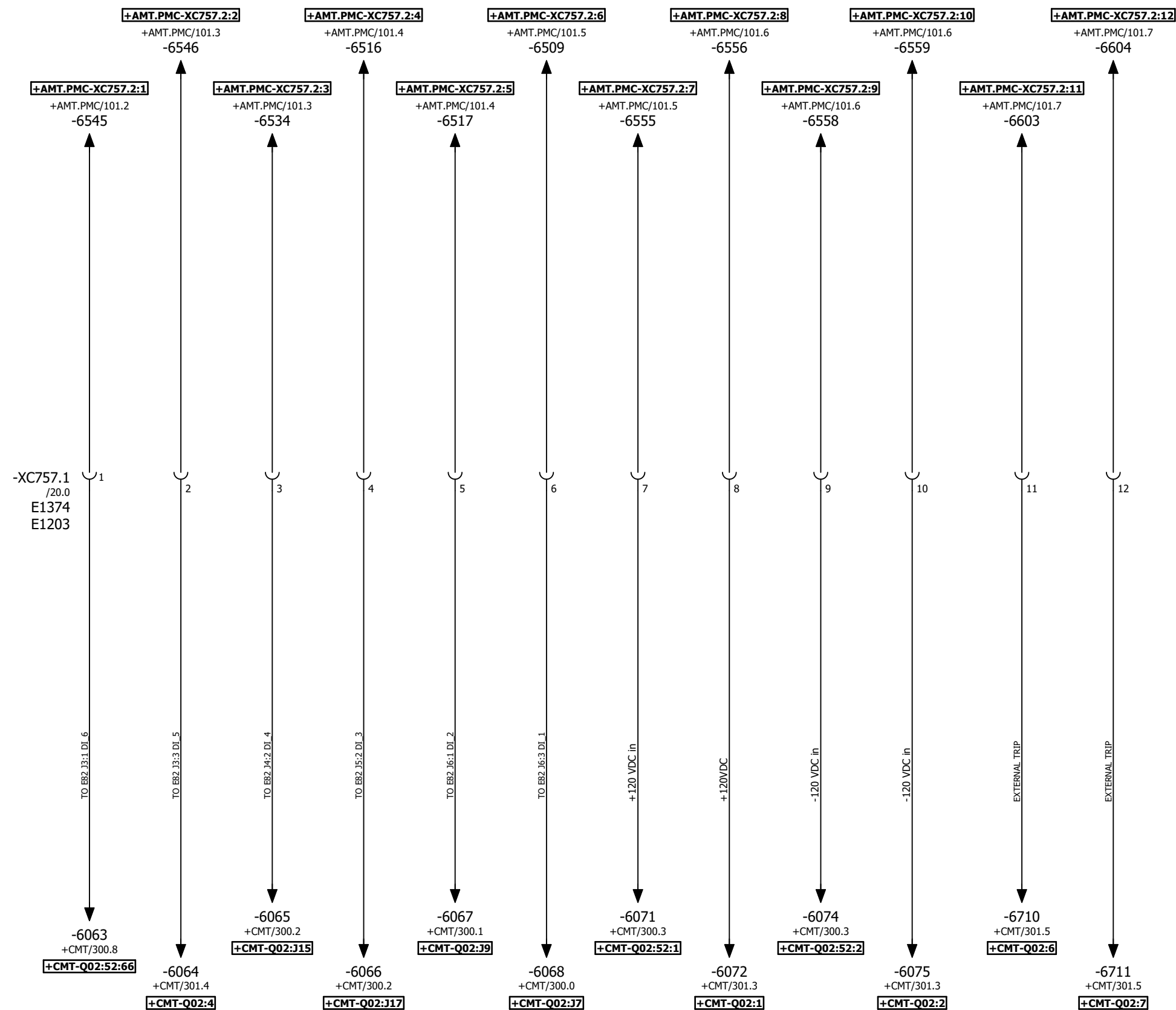
Interconnections communication fans



Interconnections - I/O board

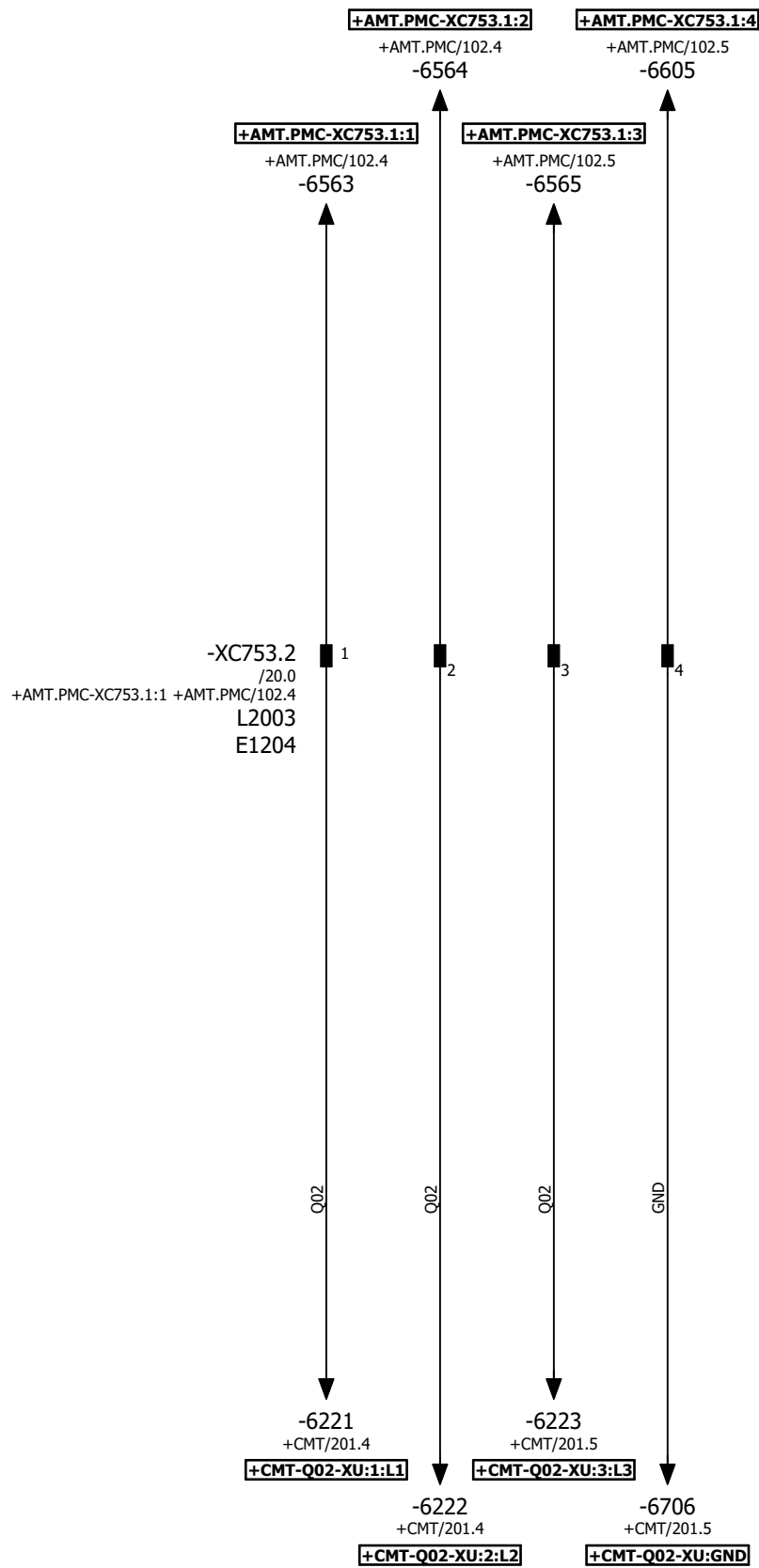


INTERCONNECTIONS PROTECTION SWITCHGEAR

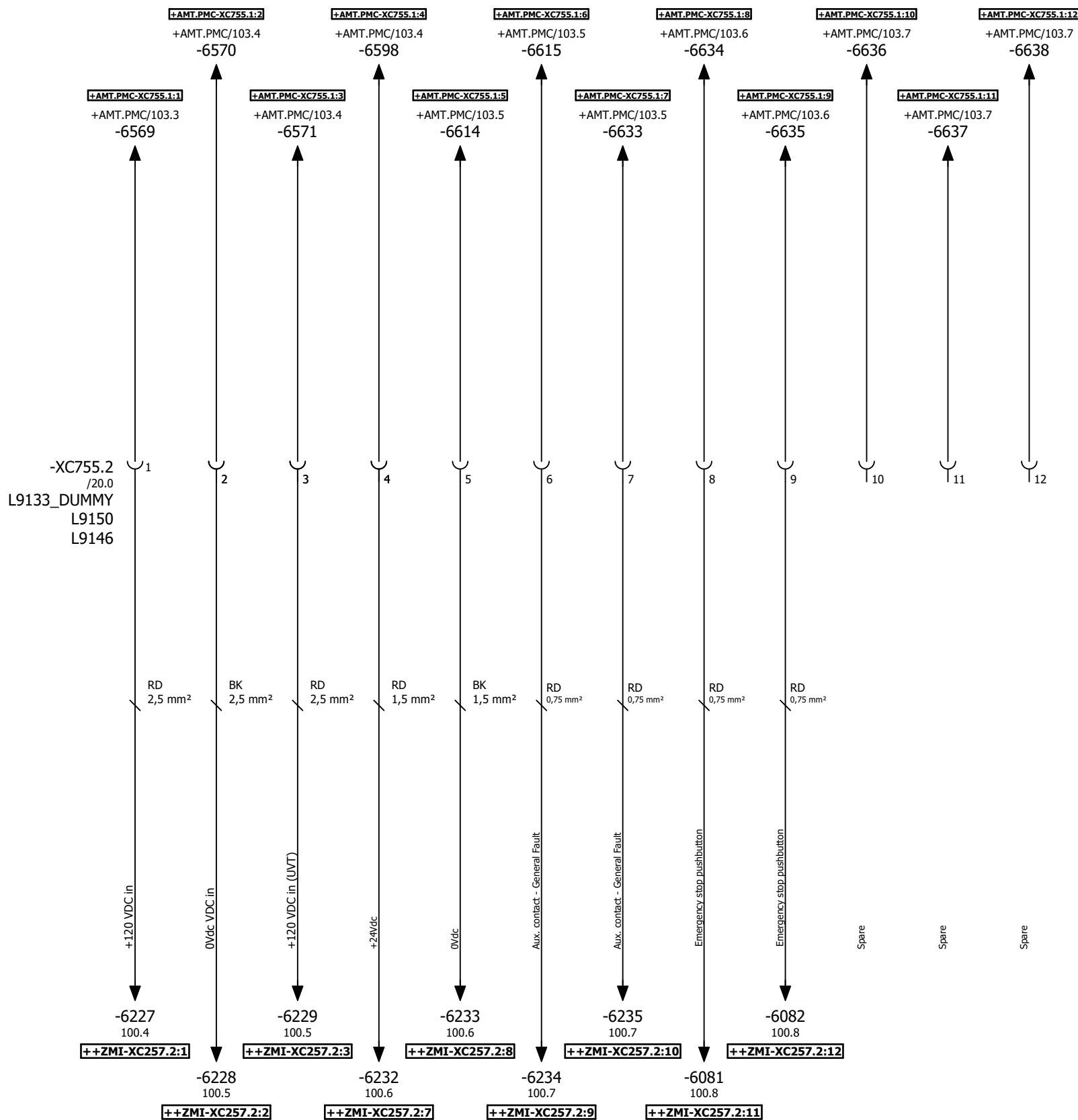


TO MV PROTECTION SWITCHGEAR

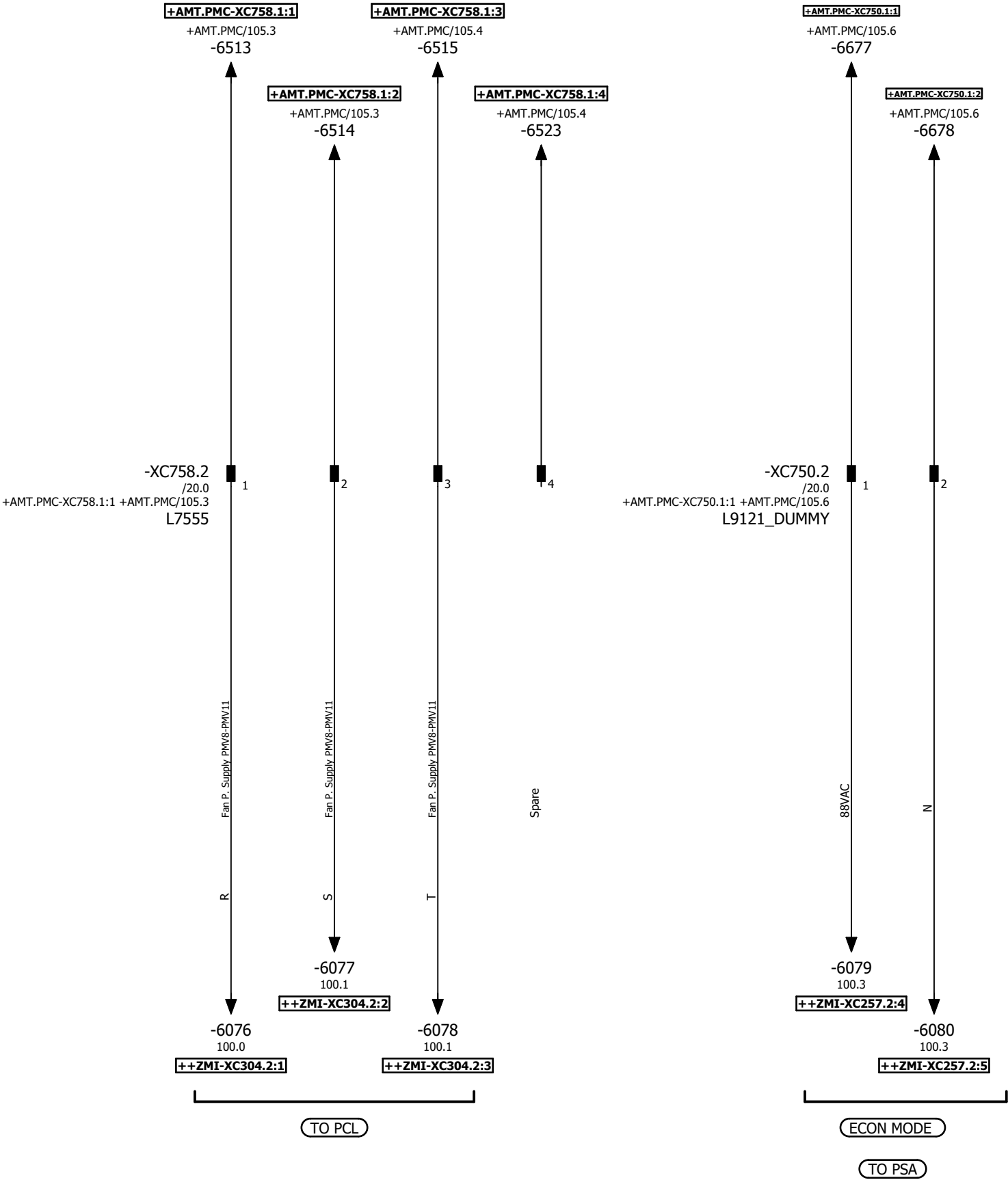
INTERCONNECTIONS AC MEASUREMENT

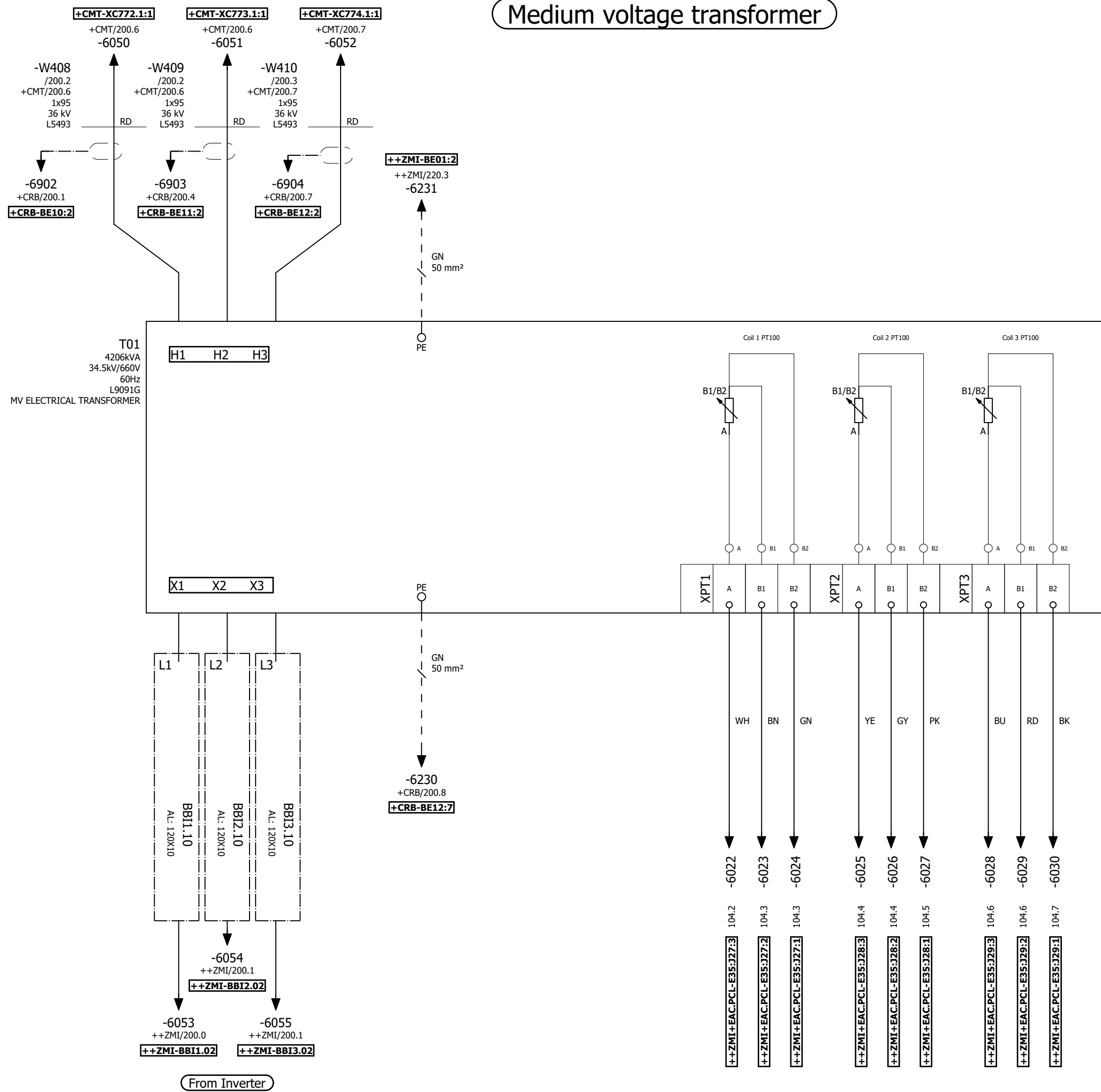


INTERCONNECTIONS 120 VDC



Interconnections - 430VAC power supply + Econ mode





BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-T01	L9091G	L9091G	TRAFO HEM PLUS 4200KVA 34,5KVAC 2000M ±10%	1

Cable overview

Cable designation	Function text	ERP
=HEM++CDT+-W401	RS485 PMV7 TO PMV8	L10399
=HEM++CDT+-W402	RS485 FROM PMV8 TO PMV9	L10397
=HEM++CDT+-W403	RS485 FROM PMV9 TO PMV10	L10537
=HEM++CDT+-W404	RS485 FROM PMV10 TO PMV11	L10397
=HEM++CDT+-W408	U TRAFO	L5493
=HEM++CDT+-W409	V TRAFO	L5493
=HEM++CDT+-W410	W TRAFO	L5493
=HEM++CDT+-W415	POWER SUPLLY ECON MODE	L5911
=HEM++CDT+-W416	POWER SUPLLY ECON MODE	L4437UND
=HEM++CDT+-W417	24VDC SIGNALS	L2267

Plug overview

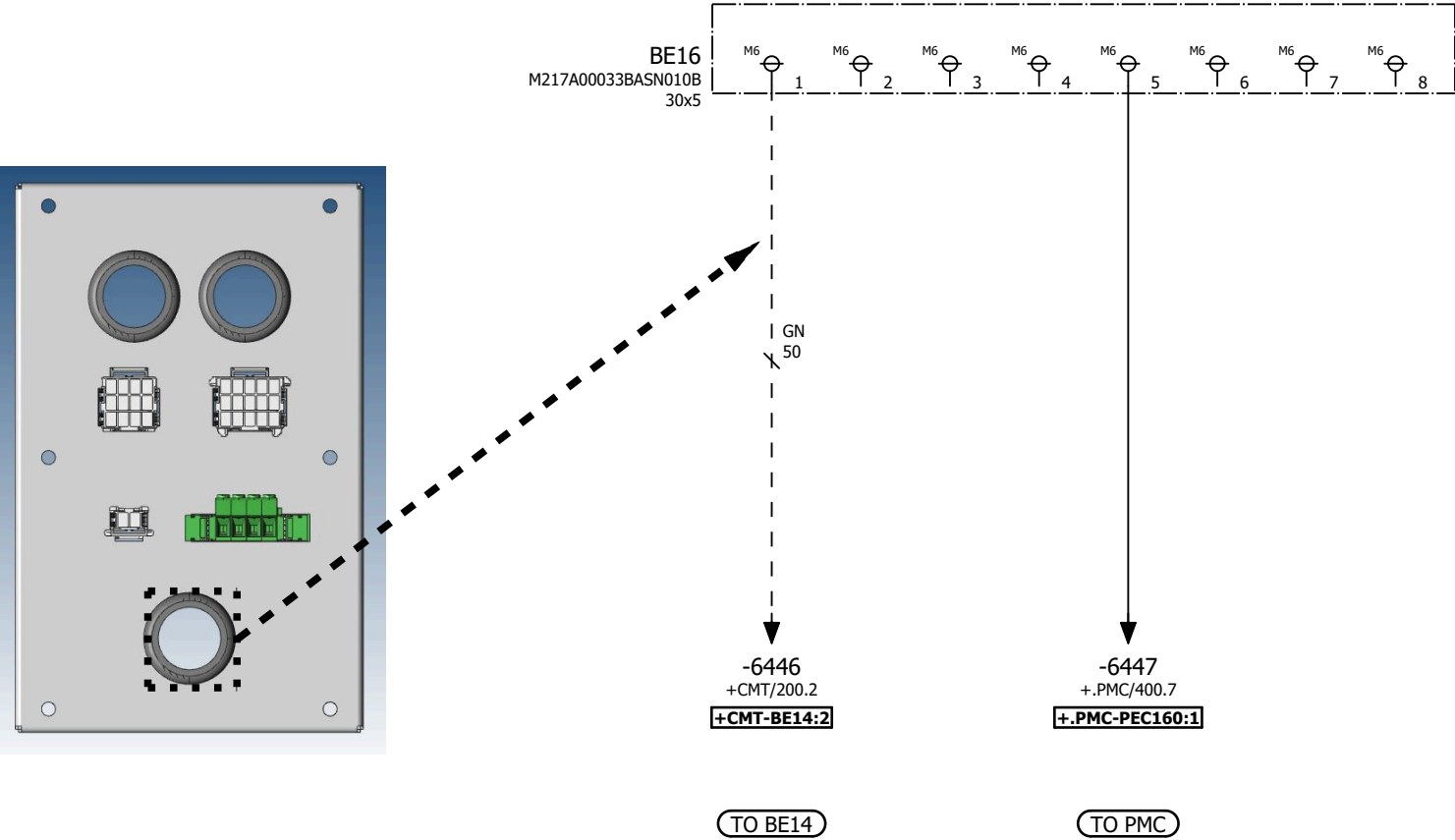
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+-XC750.2	TRIPLE-LOK CONNECTOR AERIAL 2P ECON MODE	L9121_DUMMY	1
		L9150	2
=HEM++CDT+-XC752.2	MATE-N-LOK AERIAL CONNECTOR 15P DIG IN/OUT	L2477	1
		E1204	9
=HEM++CDT+-XC753.2	MATE-N-LOK AERIAL CONNECTOR 4P MV BOARD	L2003	1
		E1204	4
=HEM++CDT+-XC755.2	MATE-N-LOK AERIAL CONNECTOR 12P (120VDC)	L9133_DUMMY	1
		L9150	5
		L9146	6
=HEM++CDT+-XC756.2	MATE-N-LOK MATRIX AERIAL FANS 15P	L2477	1
		L5710	12
=HEM++CDT+-XC757.1	MATE-N-LOK AERIAL CONNECTOR 12P	E1374	1
		E1203	10
		L5709	2
=HEM++CDT+-XC758.2	HARTING CONNECTOR 6P FANS POWER SUPPLY + ECON MODE	L7555	1
		L8022	3
=HEM++CDT+-XC761.1	MATE-N LOK BASE FEMALE CONNECTOR FAN 8	E1096	1
		L5709	3
=HEM++CDT+-XC764.1	MATE-N LOK BASE FEMALE CONNECTOR FAN 9	E1096	1
		L5709	3
=HEM++CDT+-XC767.1	MATE-N LOK BASE FEMALE CONNECTOR FAN 10	E1096	1
		L5709	3
=HEM++CDT+-XC770.1	MATE-N LOK BASE FEMALE CONNECTOR FAN 11	E1096	1
		L5709	3



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
CDT	MV TRANSFORMER CUBICLE		
AMT	MEDIUM VOLTAGE ENCLOSURE		
AMT MEDIUM VOLTAGE ENCLOSURE			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PE connections



BILL OF MATERIALS

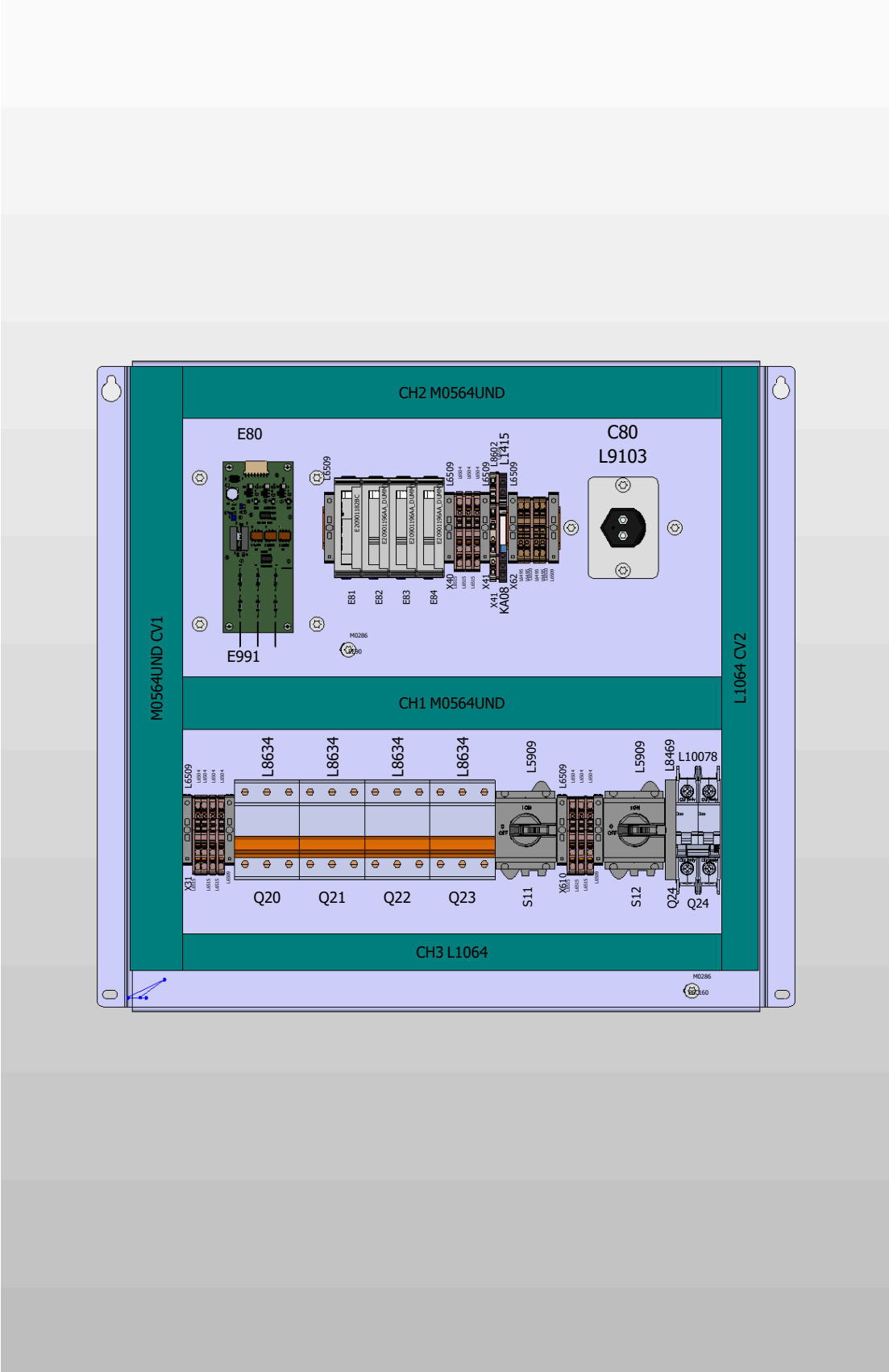
Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BE16	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
CDT	MV TRANSFORMER CUBICLE		
AMT.PMC	PANEL SWITCHGEAR MANEUVER		
<u>PMC</u> <u>PANEL SWITCHGEAR MANEUVER</u>			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

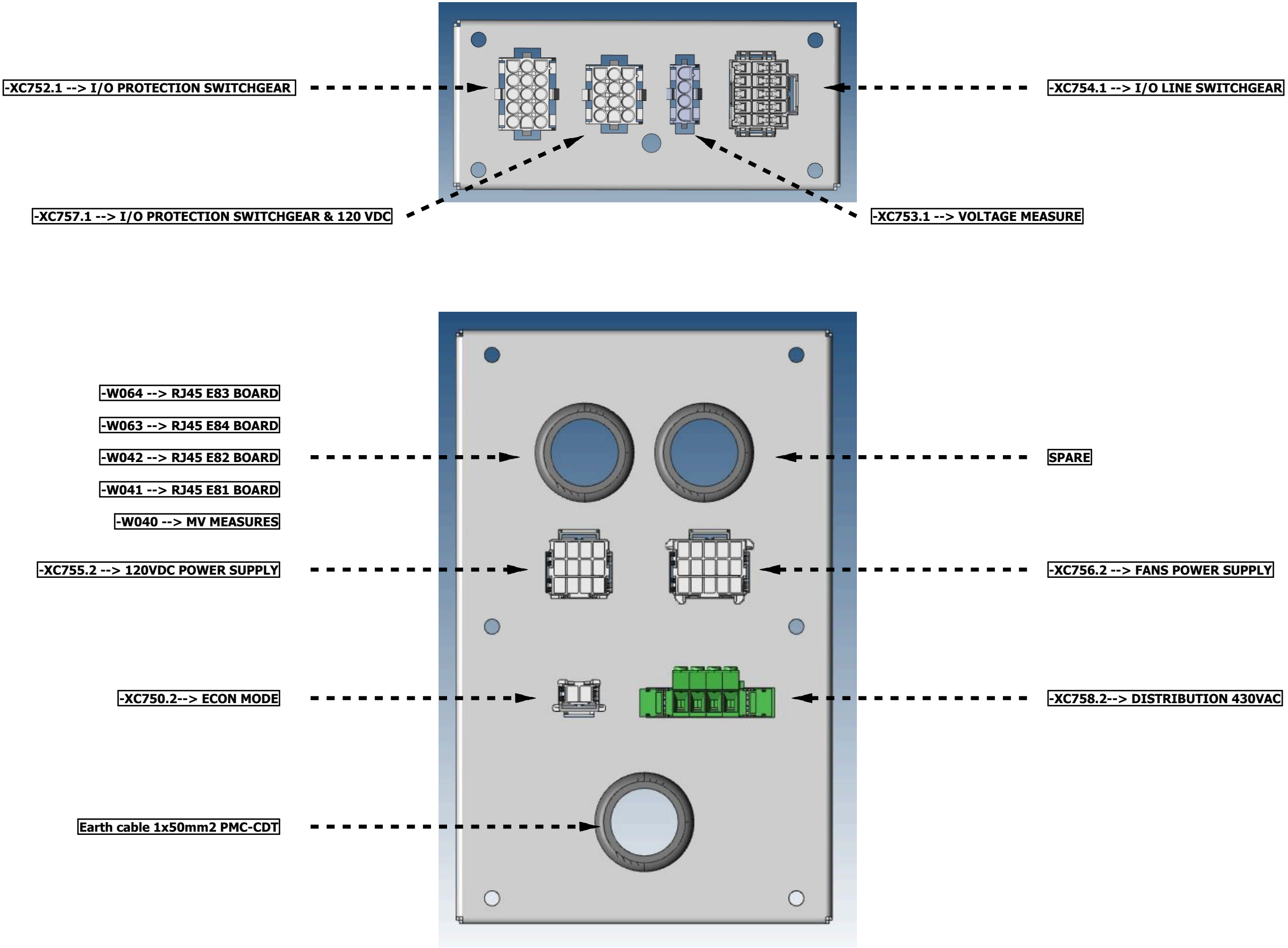
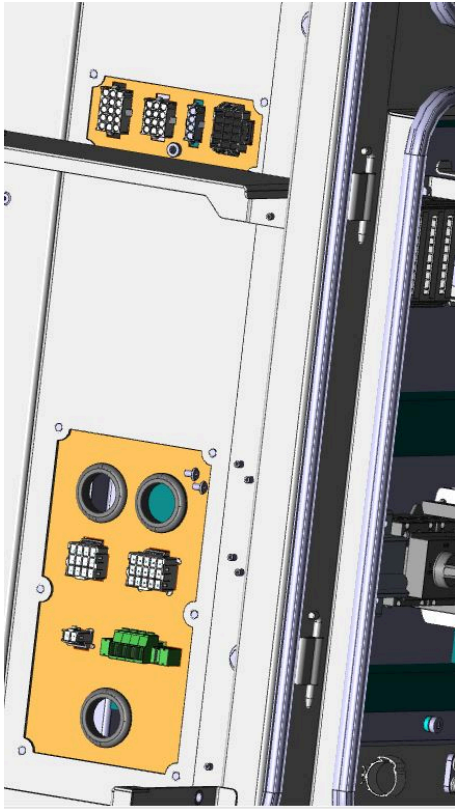
LAYOUT PANEL



Enclosure legend

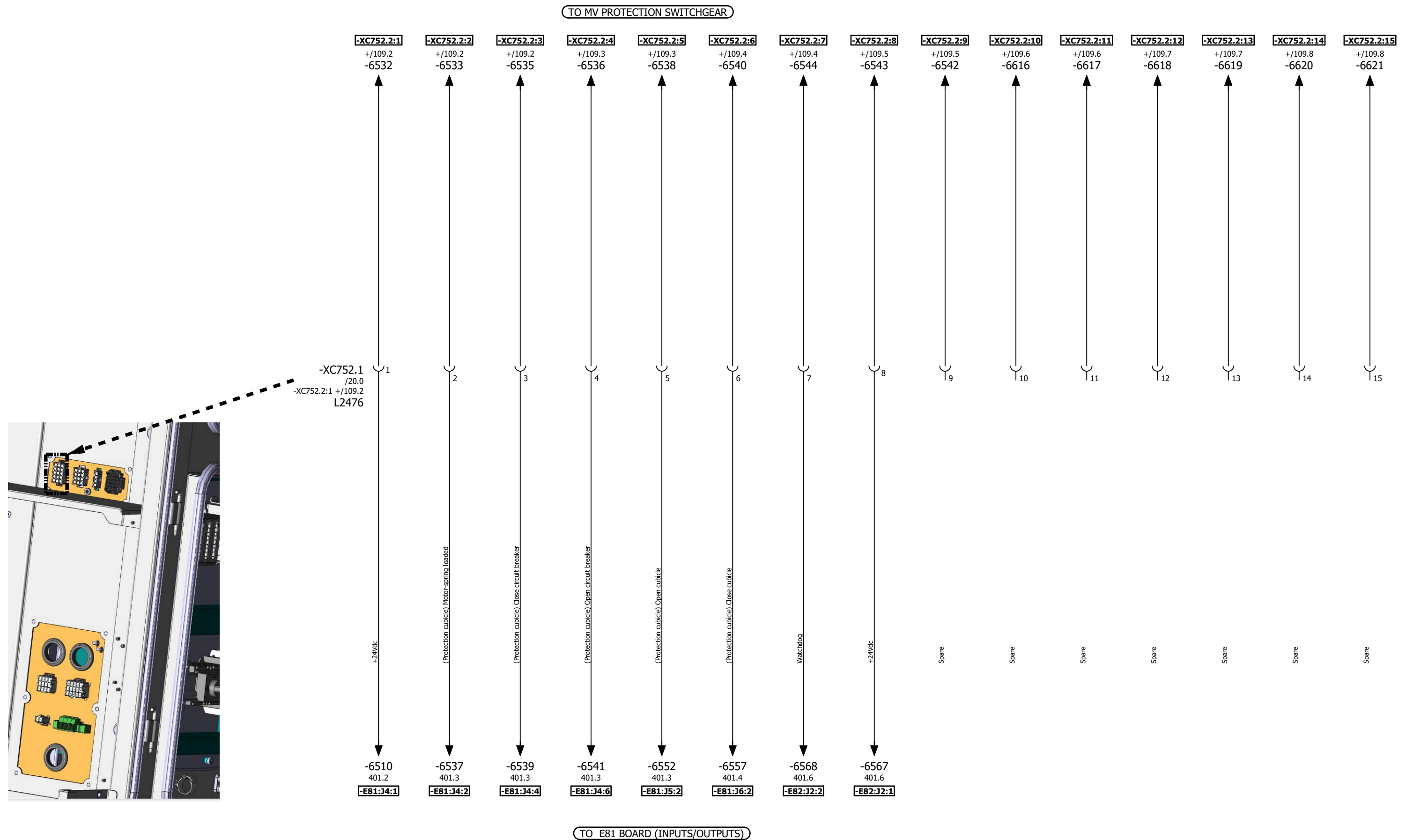
IME	ERP	Description
CH3;CV2	L1064	CABLE DUCT UNEX 42X30 EN U23X
CA1;CA2	L1061	DIN RAIL 35x7,5mm
E80	PCB E991	MEASUREMENT MV PCB PHASES TO 4-20MA HEM
E81	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD
E82-E84	PCB E20901196	DIGITAL INPUTS BOARD
CH1;CH2;CV1	M0564UND	CABLE DUCT UNEX 42X30 EN U23X
S11;S12	L5909	SWITCH DISCONNECTOR 3P 20A 690VAC
Q24	L8469	CONT. AUX. AX-HU CIRCUIT BREAKER BK63HU
Q20-Q23	L8634	BK63H 3P C10A 6ka 480Y/277VAC UL1077
Q24	L10078	BK63HU 2P D16A 240/277/480Y, 10KA LS
KA08	L1415	RELAY 24VDC + SLIM BASE
C80	L9103	CAPACITOR 2200uF 100VDC

Interconnection PMC and CDT

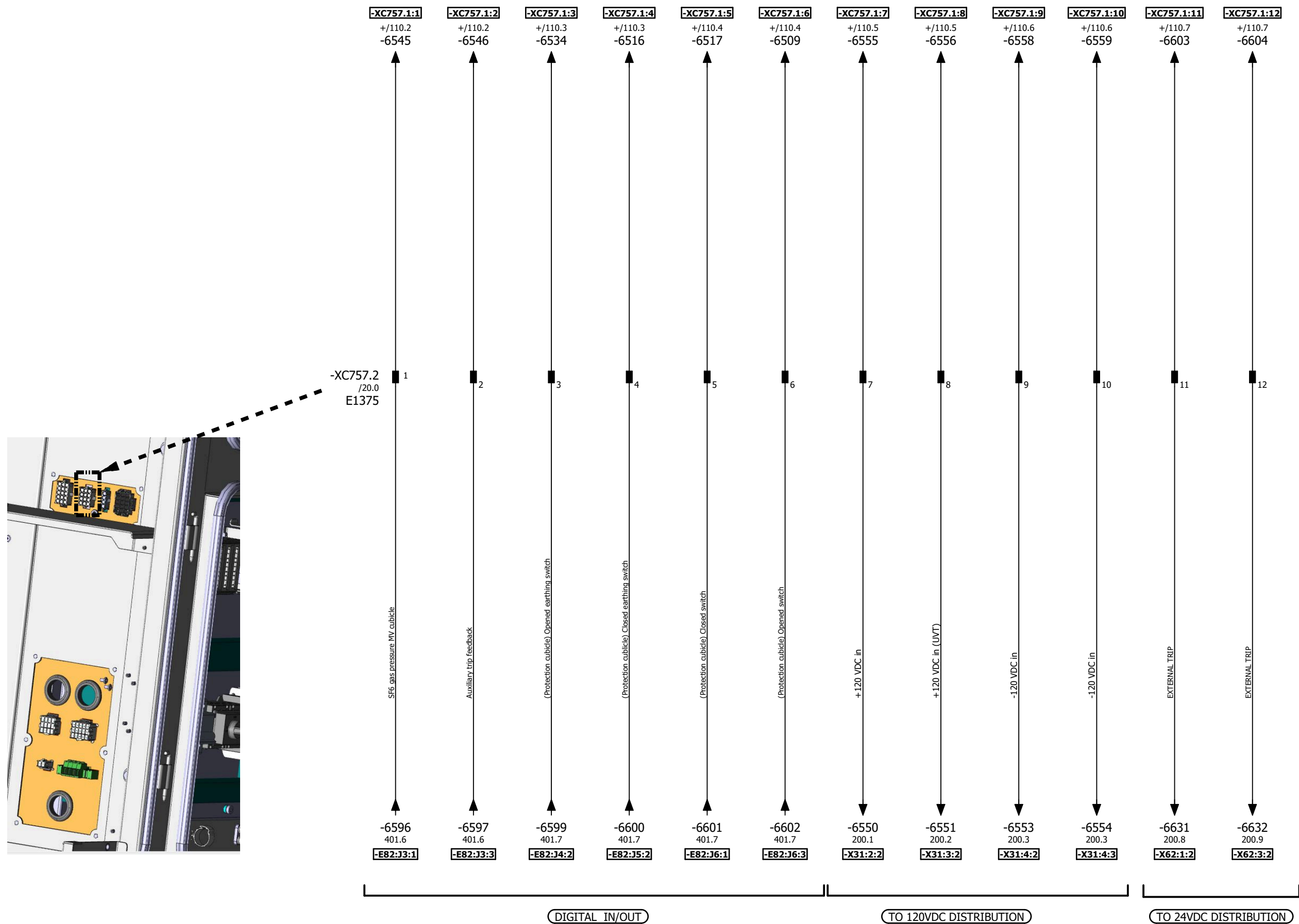


0	1	2	3	4	5	6	7	8	9
-XC752.1 = MATE-N-LOK CONNECTOR 15P DIG IN/OUT PCB -XC753.1 = MATE-N-LOK CONNECTOR BASE 4P MV BOARD -XC750.1 = TRIPLE-LOK CONNECTOR AERIAL 2P ECON MODE -XC754.1 = TRIPLE-LOK CONNECTOR BASE 15P LINE CELL 1 & 2 -XC755.1 = TRIPLE-LOK CONNECTOR AERIAL 12P 120VDC -XC756.1 = TRIPLE-LOK AERIAL FANS 15P -XC757.2 = MATE-N-LOK CONNECTOR 12P -XC758.1 = AERIAL CONNECTOR 4P FANS POWER SUPPLY 6MM2 -XC792.2 = MATE-N-LOK AERIAL OUPUT POWER SUPPLY ECON MODE -X31 = DISTRIBUCIÓN 120VDC -X40 = 24VDC DISTRIBUTION -X41 = TERMINAL BLOCK DIODE 24VDC -X62 = Terminal 24Vdc ext trip switchgear MT -X610 = Terminal LOTO voltage measure									

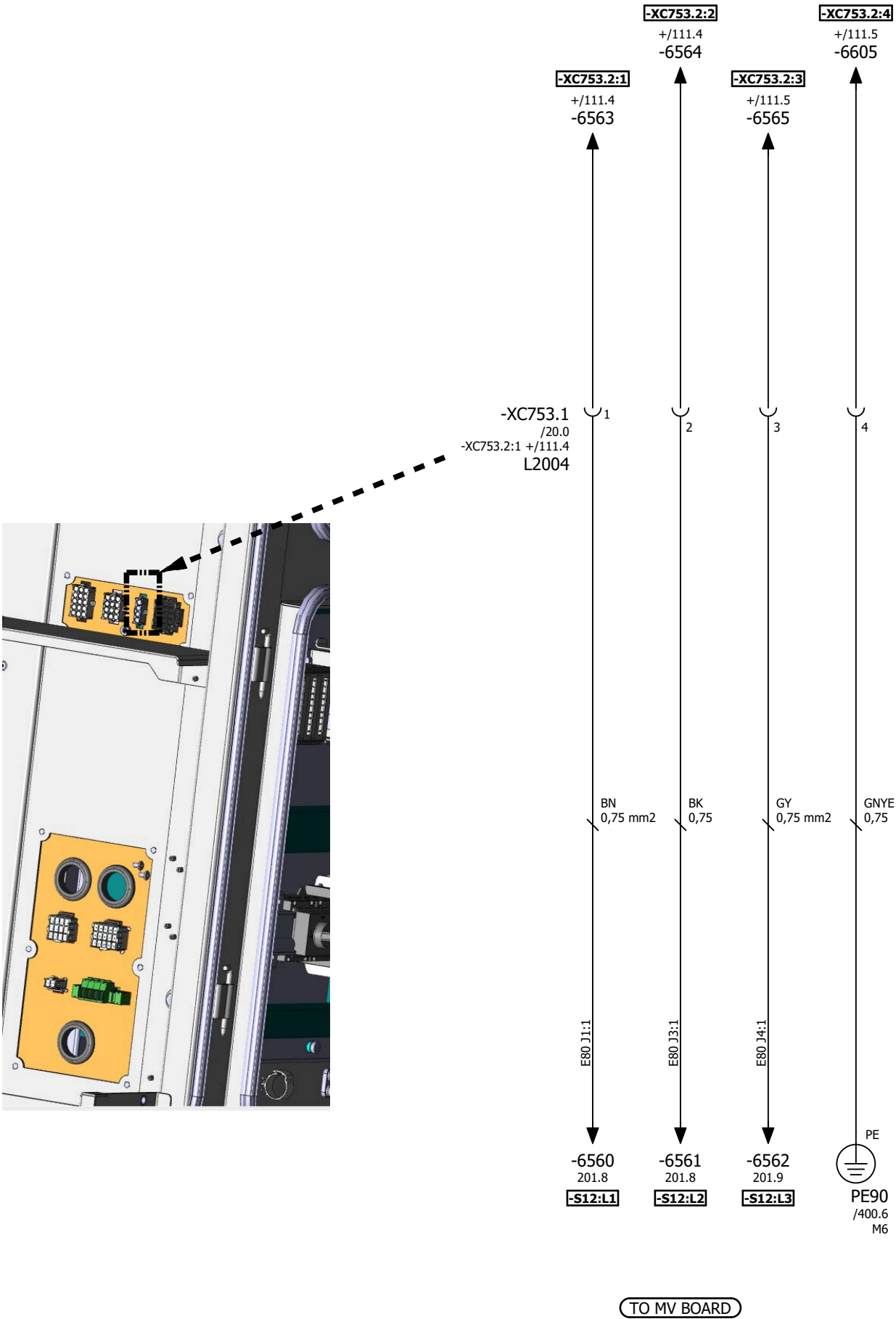
INTERCONNECTIONS PROTECTION SWITCHGEAR



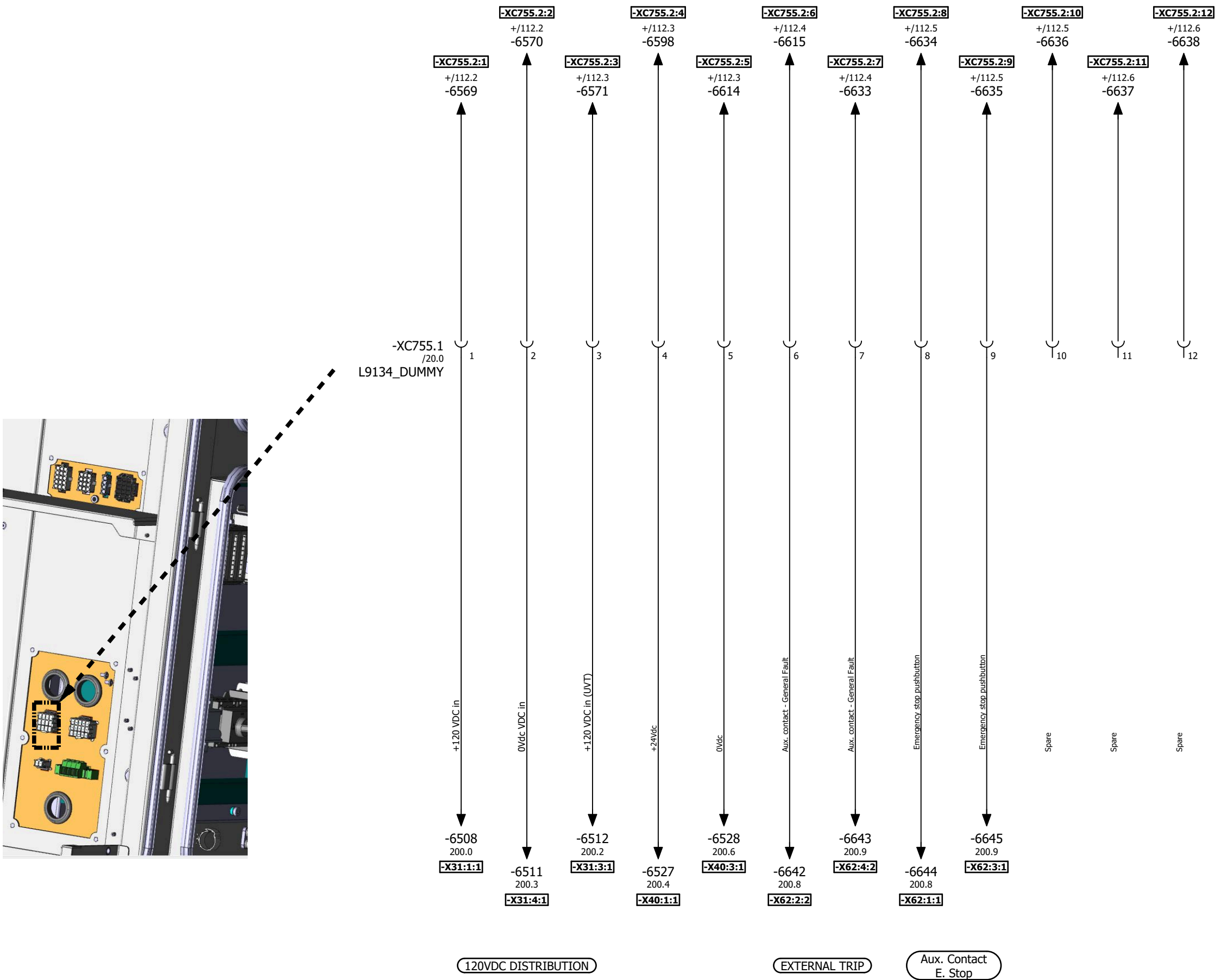
INTERCONNECTIONS PROTECTION SWITCHGEAR



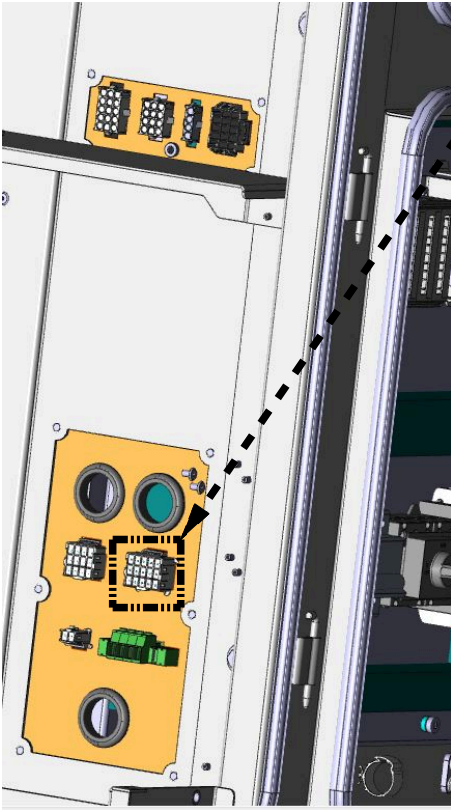
INTERCONNECTIONS AC MEASUREMENT



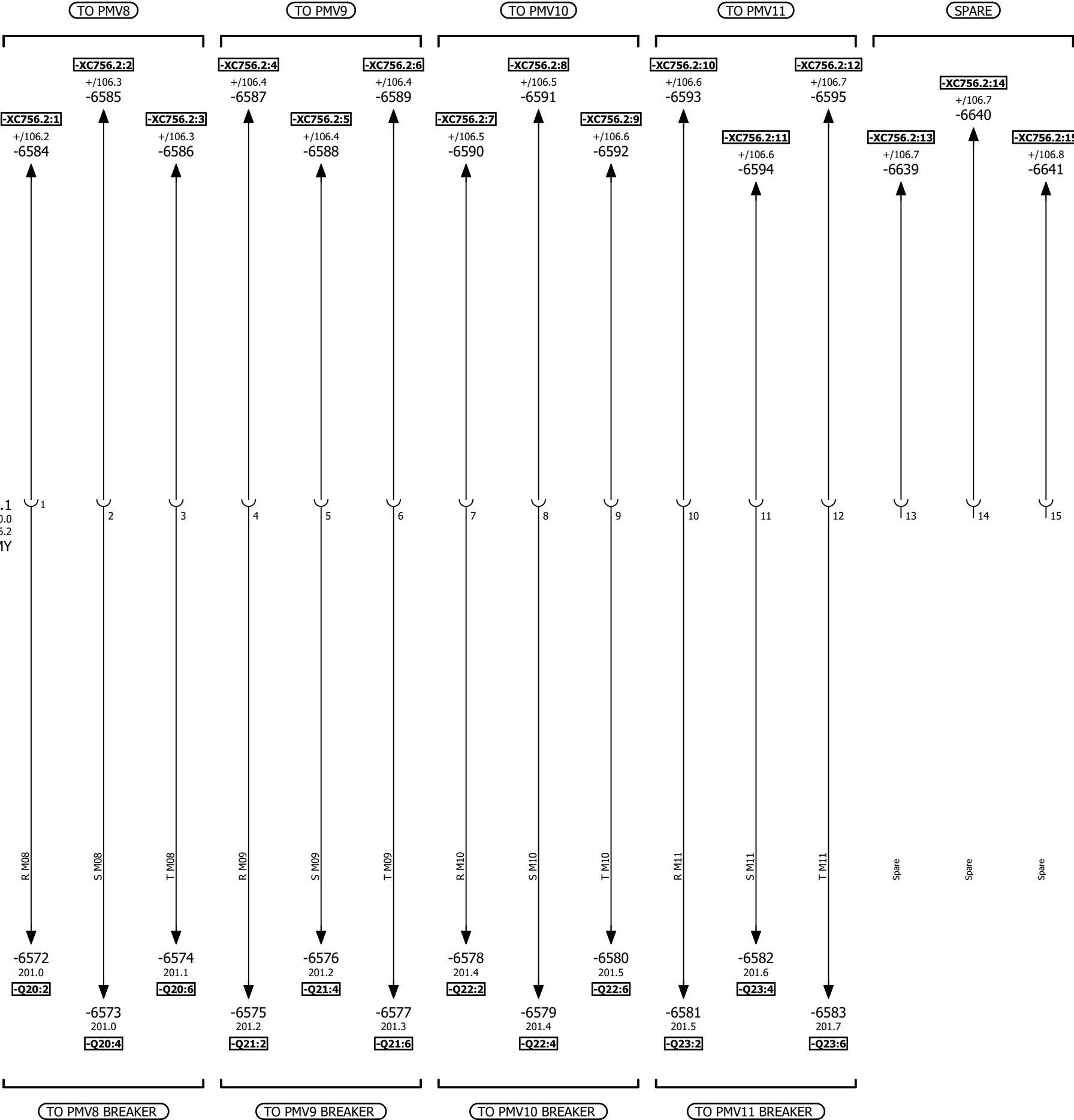
INTERCONNECTIONS 120VDC + ECON MODE



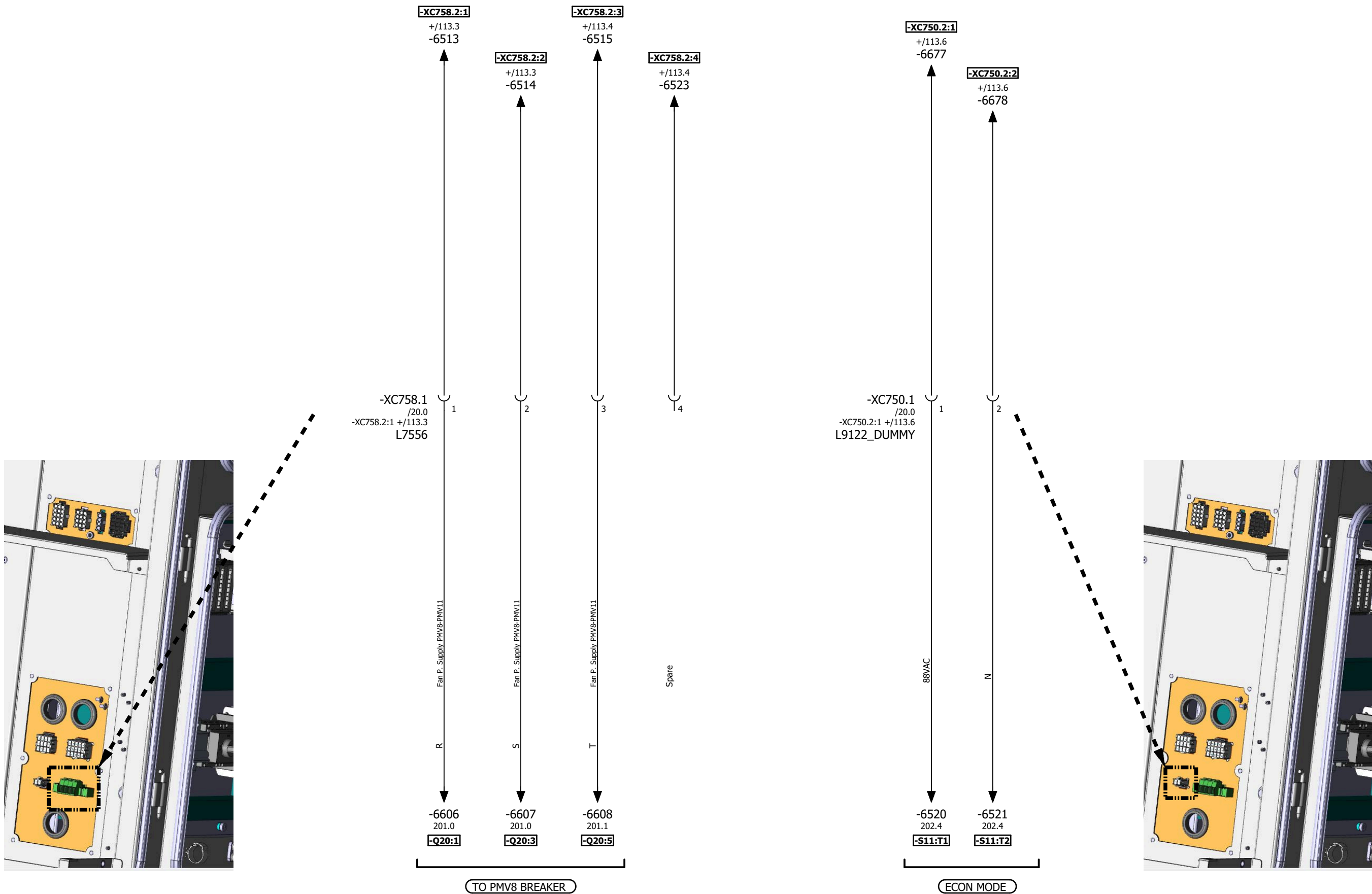
Interconnections - Pwer supply Fans



-XC756.1
/20.0
-XC756.2:1 +/-106.2
L9136_DUMMY

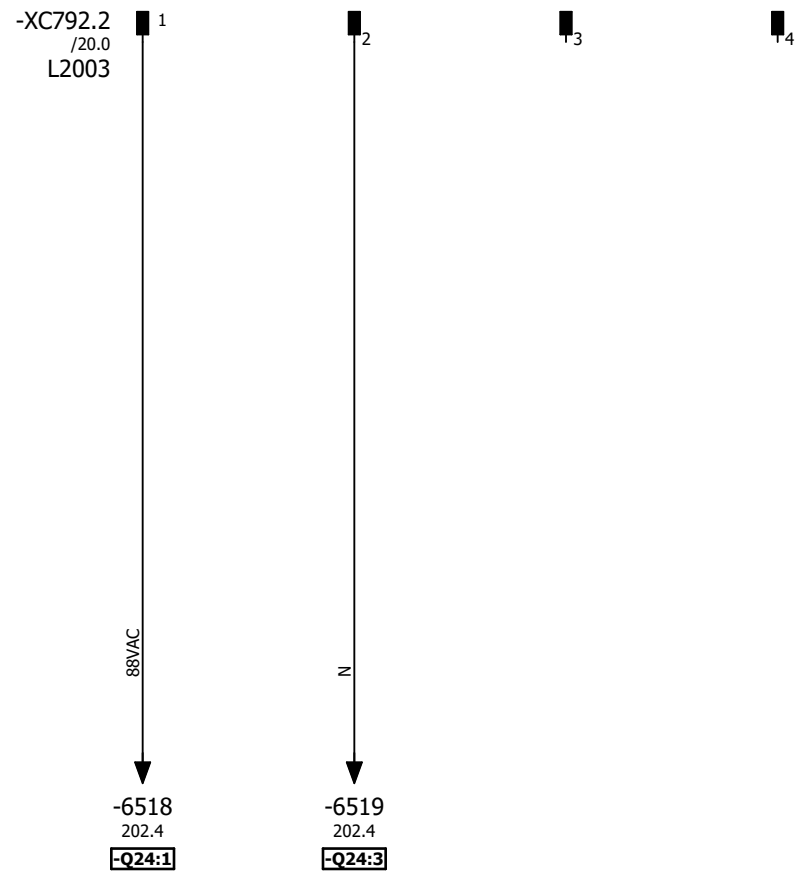
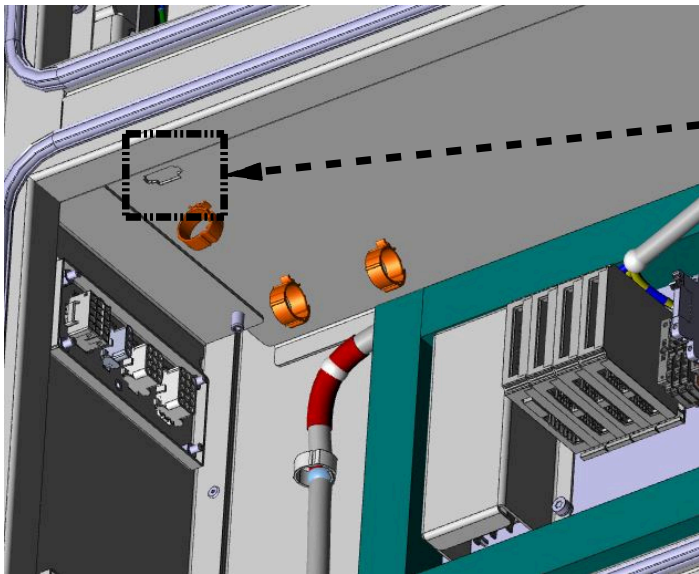


Interconnections - 430VAC power supply + Econ mode



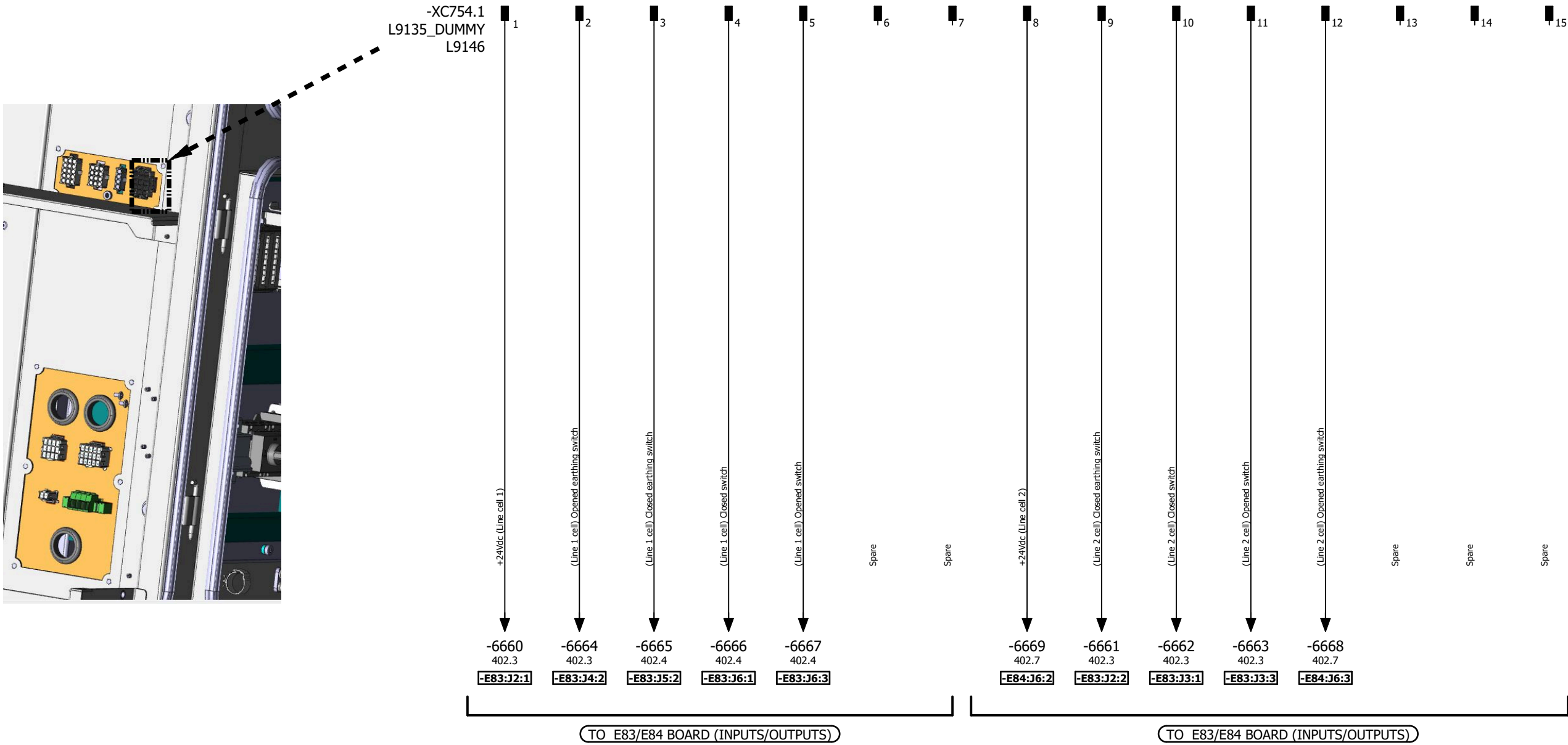
Interconnections CEM/PMC

TO ECON MODE SECONDARY

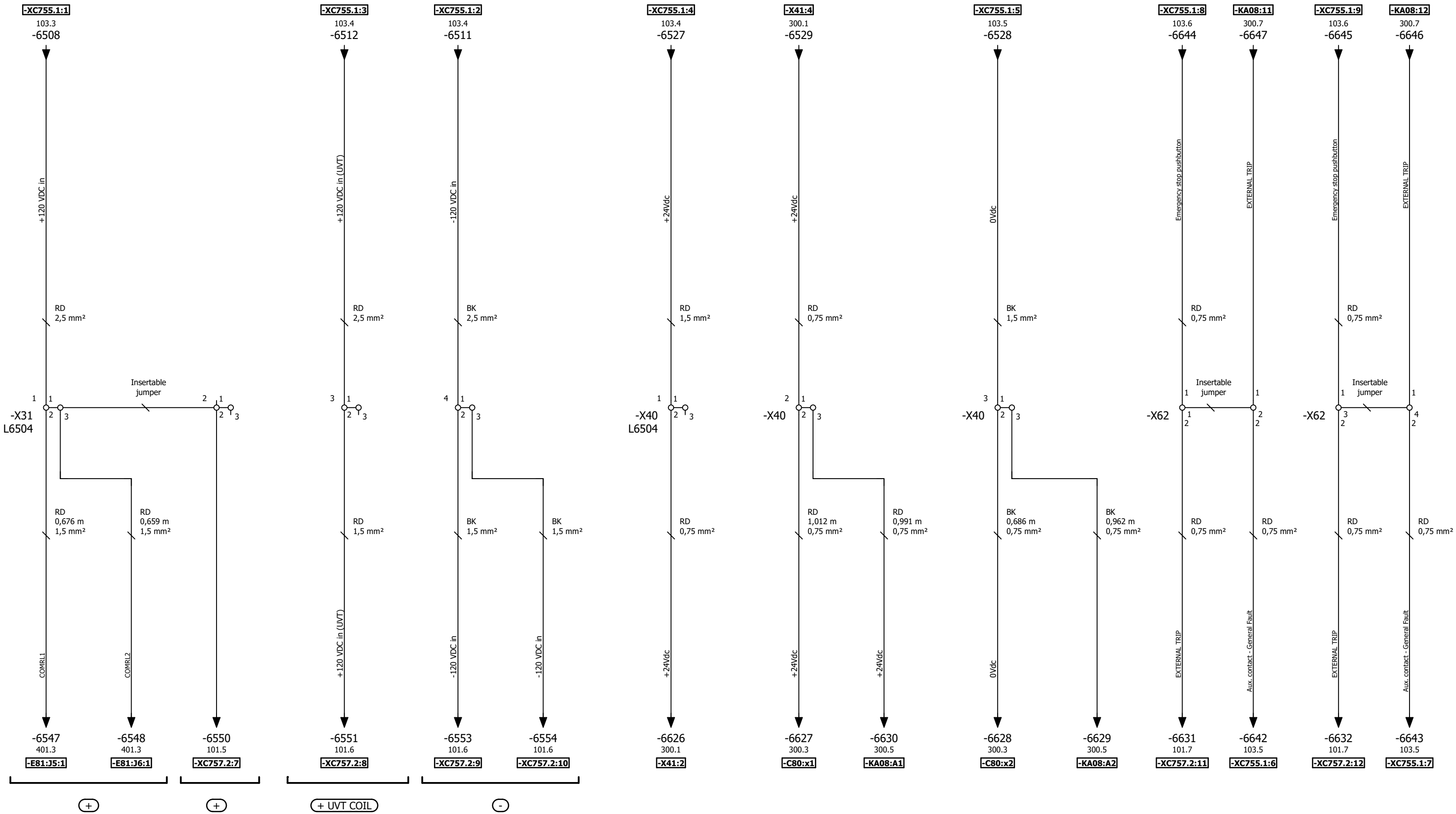


TO PMC

Interconnections - Spare Line cell (VLL)



Distribution 120VDC/24VDC

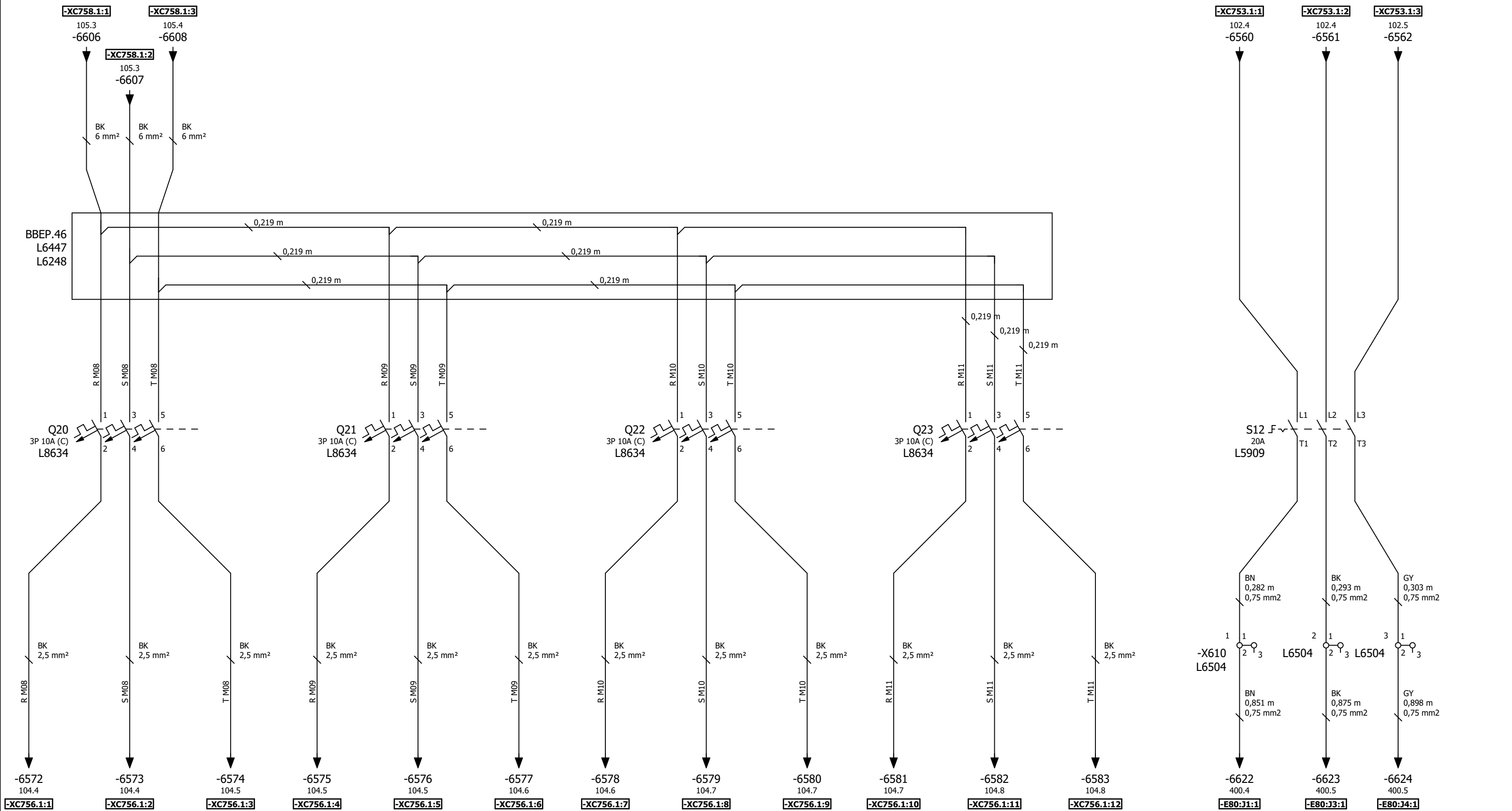


Distribution 120Vdc

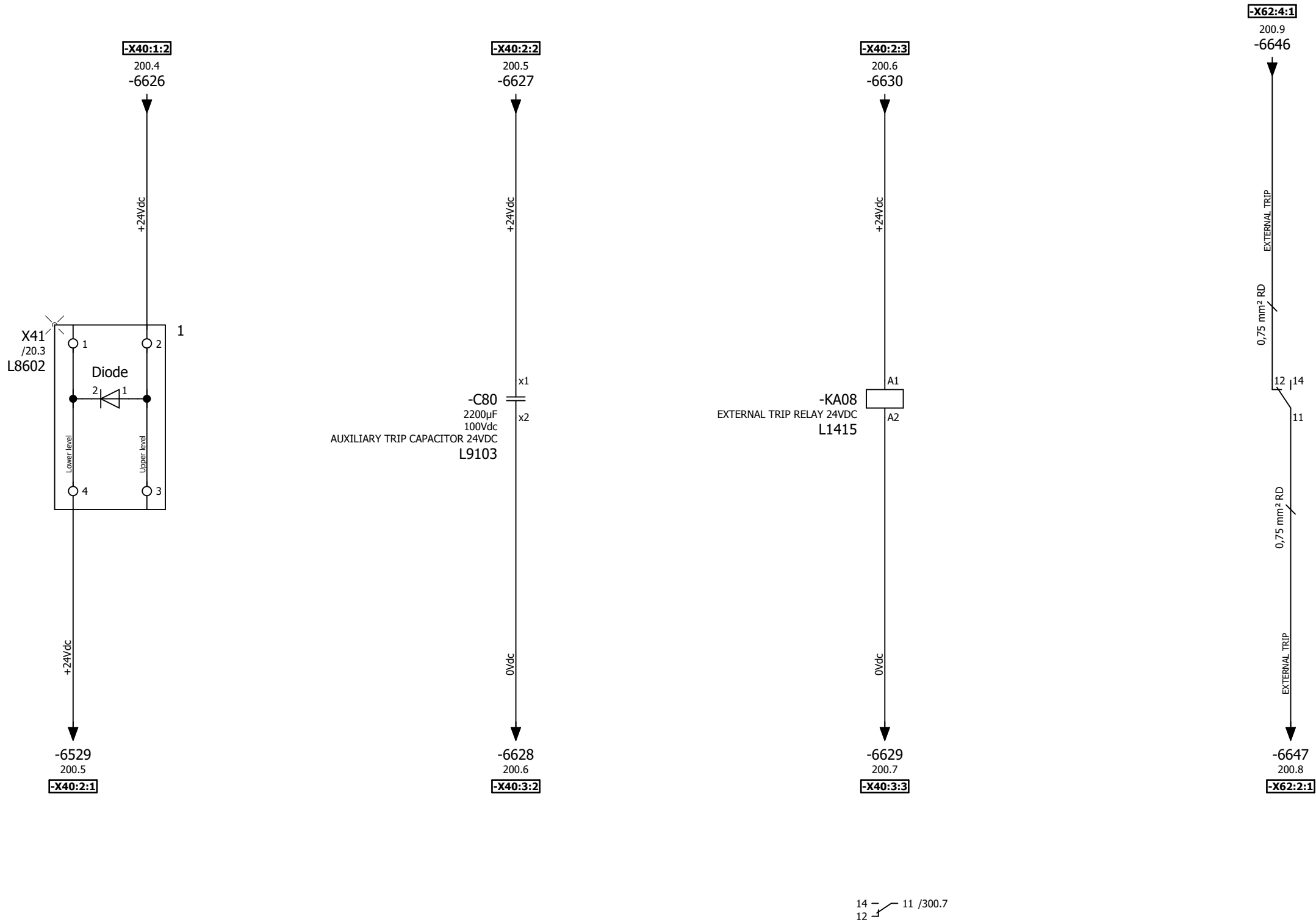
Distribution 24Vdc

EXTERNAL TRIP

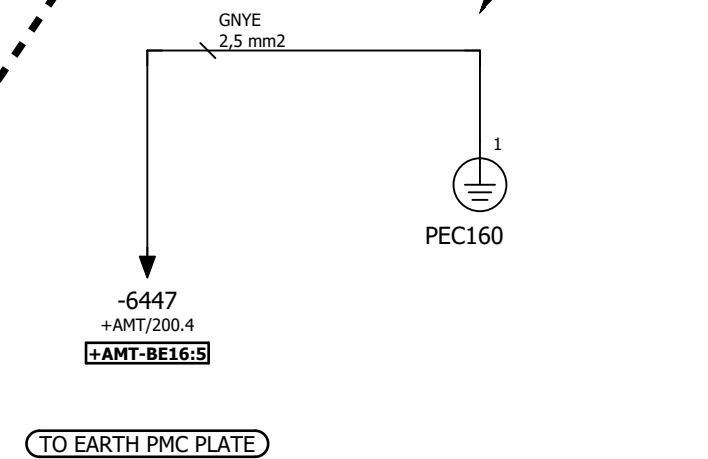
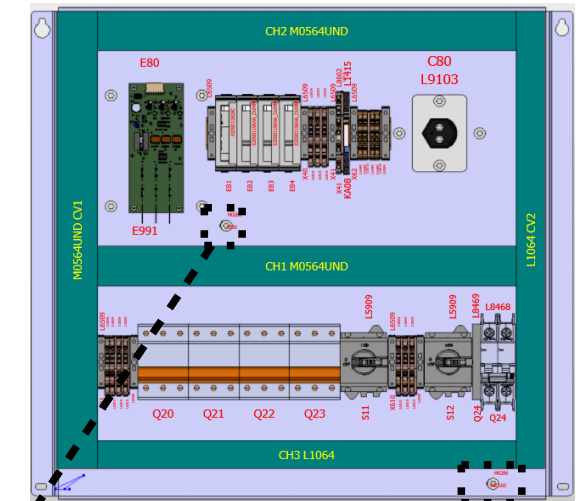
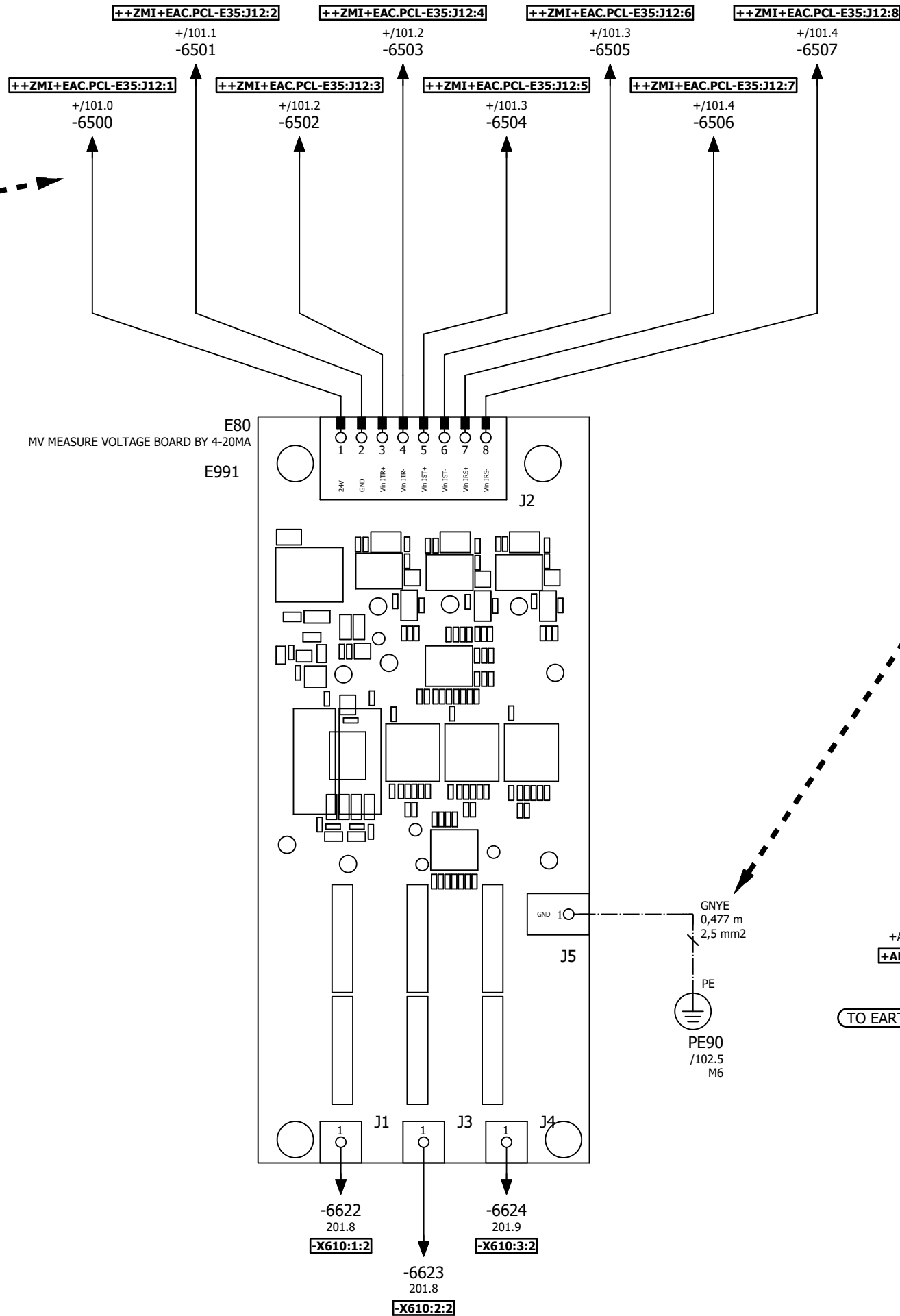
General protections



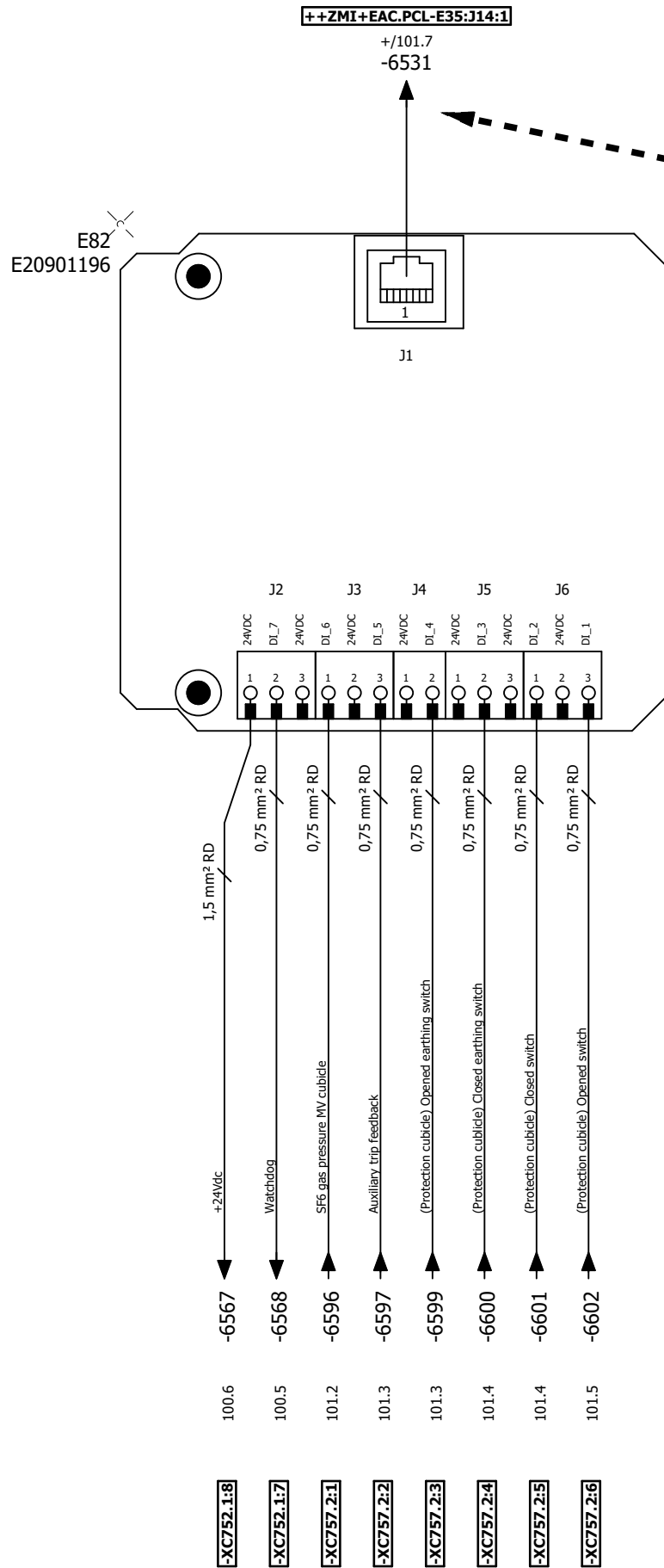
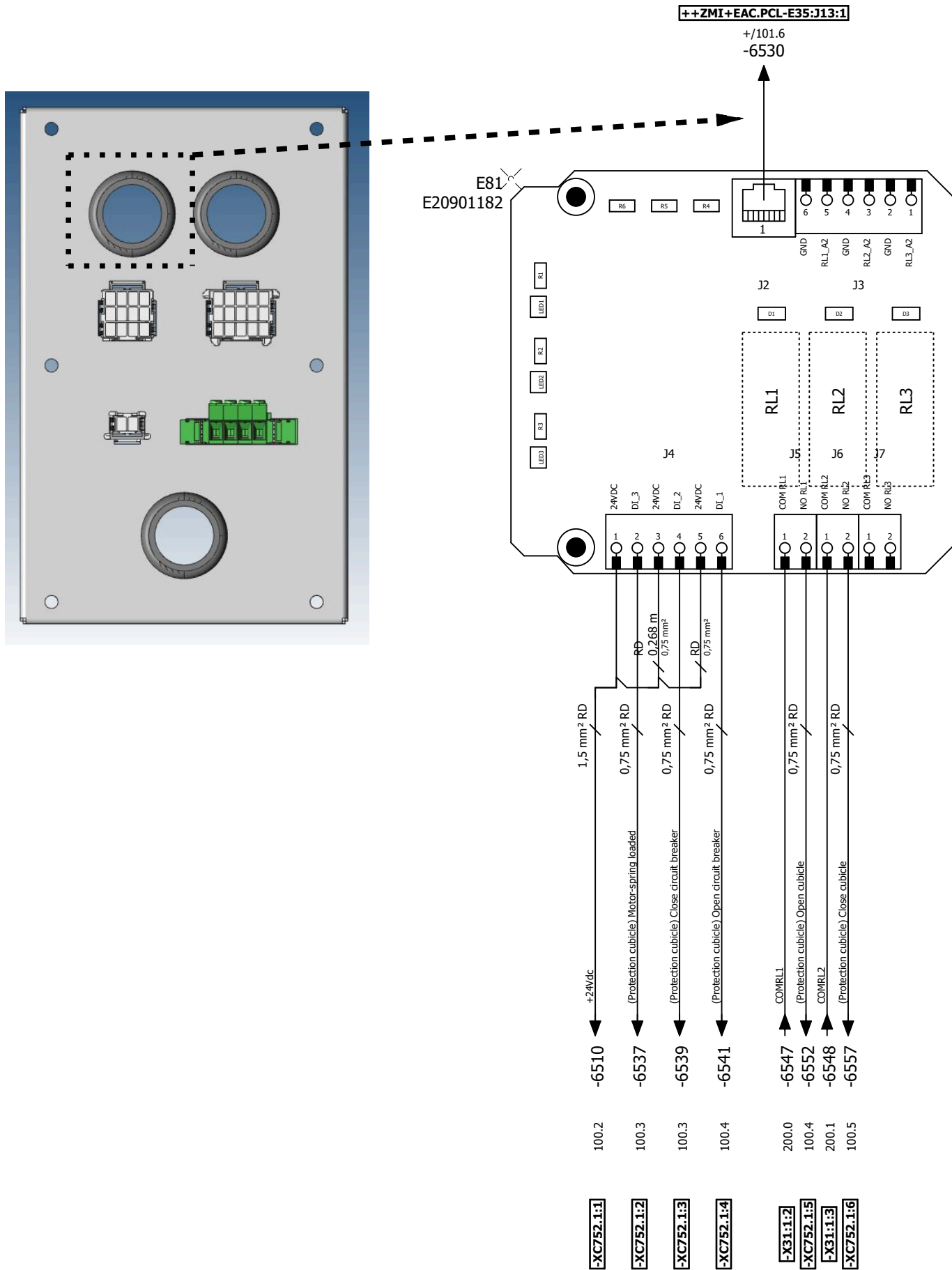
Auxiliary trip 24Vdc MV switchgear



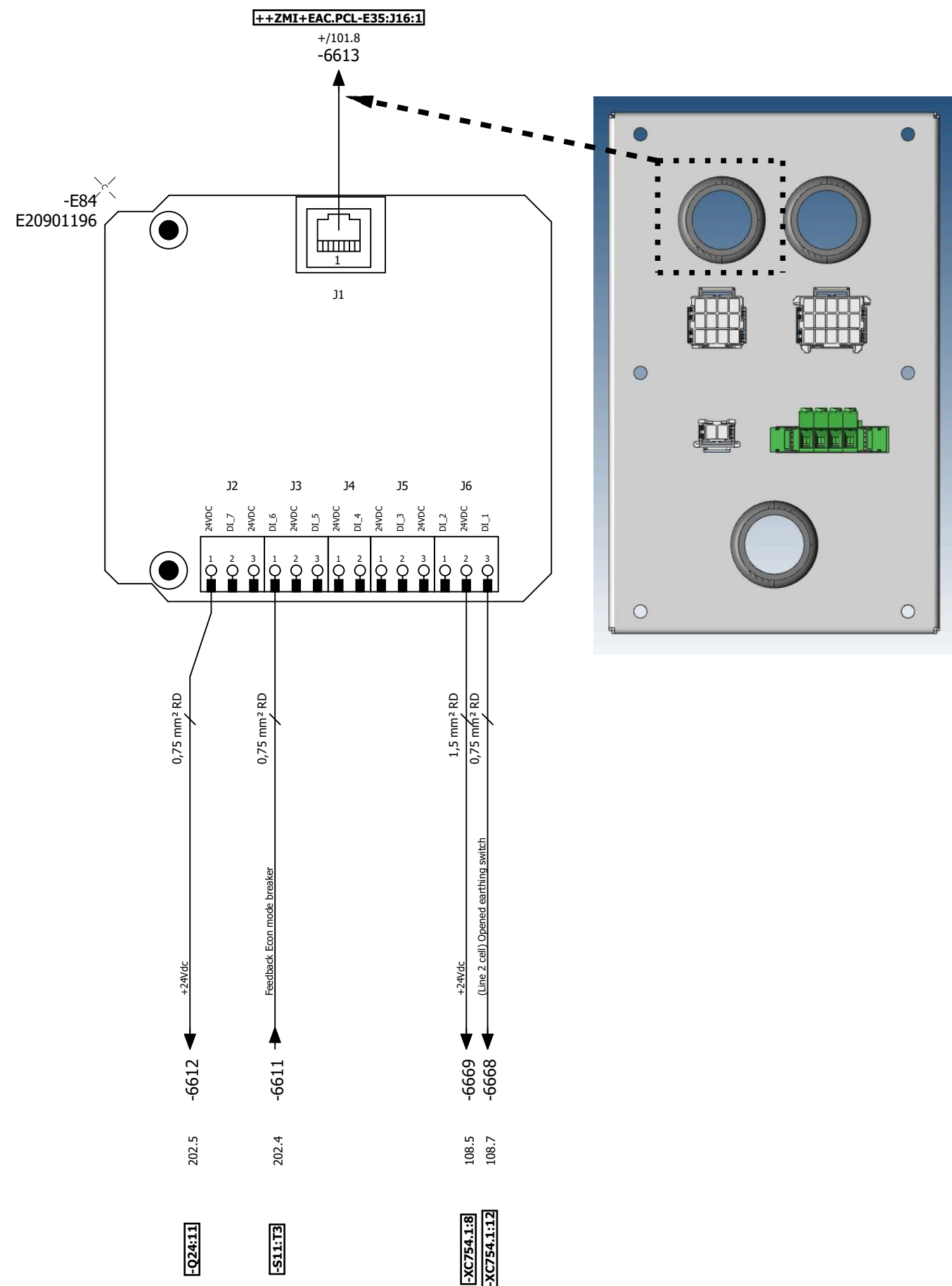
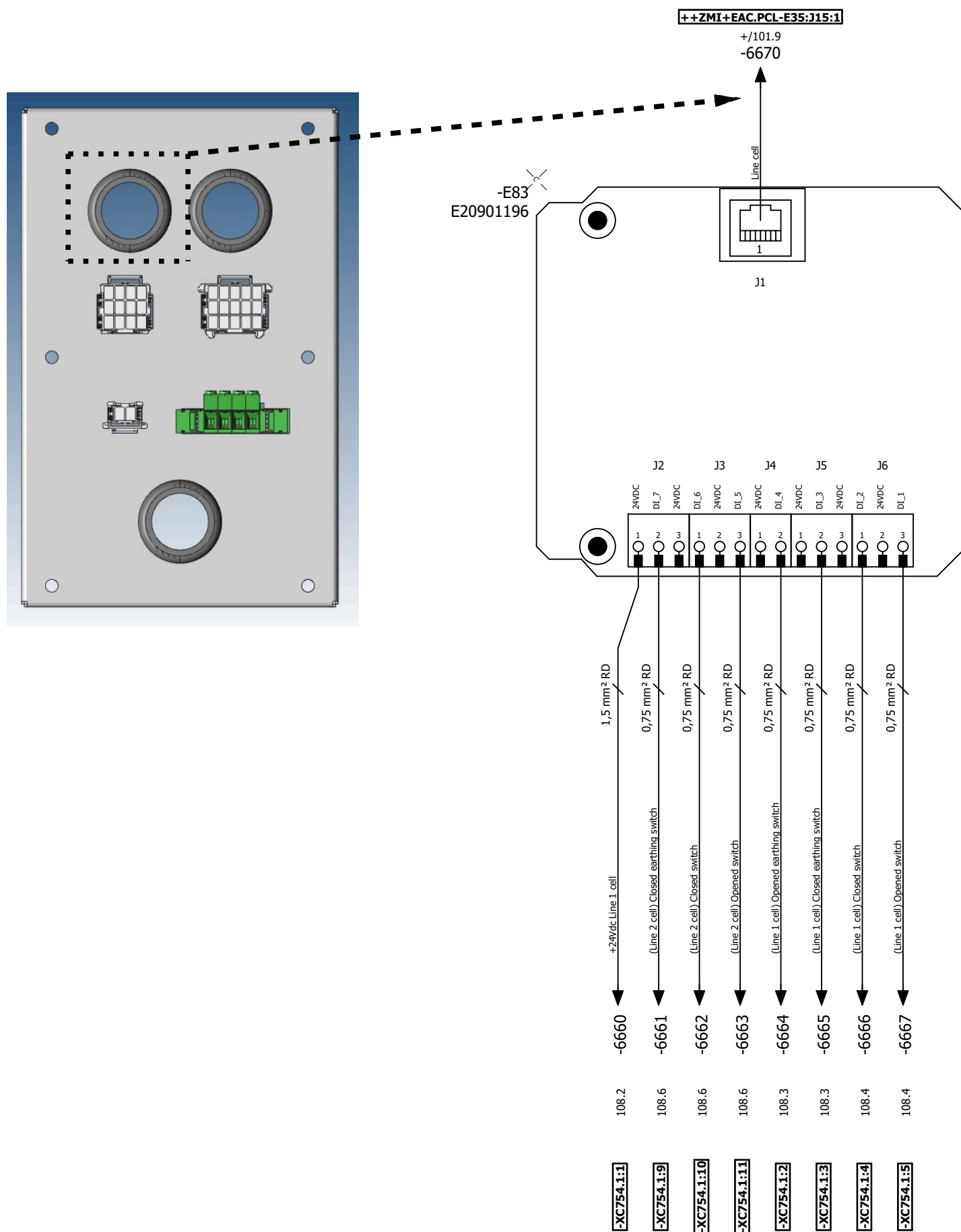
MV measure PCB



Inputs/outputs distributor boards



Inputs/outputs distributor board (IV)



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BBEP.46	L6248	220115	JUMPER PIA 3X19 FORK 63A 10mm2 690VAC	1
-BBEP.46	L6447	220122	TRIPOLAR END COVER PIA BRIDGE	2
-C80	L9103	K01100222M0M0E060	CAPACITOR 2200uF 100VDC	1
-CA1	L1061	0514500000	DIN RAIL 35x7,5mm	1
-CA2	L1061	0514500000	DIN RAIL 35x7,5mm	1
-E80	PCB E991	PCB E991	MEASUREMENT MV PCB PHASES TO 4-20MA HEM	1
-E81	PCB E20901182	PCB E20901182	DIGITAL INPUTS/OUTPUTS BOARD	1
-E81	L6509	1991920000	TERMINAL ACCESSORY PUSH IN FINAL FIXATION	1
-E82	PCB E20901196	PCB E20901196	DIGITAL INPUTS BOARD	1
-E83	PCB E20901196	PCB E20901196	DIGITAL INPUTS BOARD	1
-E84	PCB E20901196	PCB E20901196	DIGITAL INPUTS BOARD	1
-KA08	L1415	1122770000	RELAY 24VDC + SLIM BASE	1
-PMHEMGEN315B	L6517	1527730000	Cross-connector (terminal)	1
-PMHEMGEN315B	L1691	1806120000	TERMINAL MARKER	5
-PMHEMGEN315B	L1748	1640740000	LABEL N° TERMINAL WS 8/5 PLUS MC NEUTRAL	31
-PMHEMGEN315B	L3715A	1607720000	LABEL ESO 7 A4 WS	5
-PMHEMGEN315B	L3714A	1724151687	SELF-ADHESIVE LABEL LM MT300 17x9 GE	14
-PMHEMGEN315B	L6248	220115	JUMPER PIA 3X19 FORK 63A 10mm2 690VAC	1
-PMHEMGEN315B	L6447	220122	TRIPOLAR END COVER PIA BRIDGE	2
-PMHEMGEN315B	V1290	V1290	STICKER EARTH 15x15	2
-Q20	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q21	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q22	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q23	L8634	0628032463	BK63H 3P C10A 6kA 480Y/277VAC UL1077	1
-Q24	L10078	0627047663	BK63HU 2P D16A 240/277/480Y, 10KA LS	1
-Q24	L8469	0641005363	CONT. AUX. AX-HU CIRCUIT BREAKER BK63HU	1
-S11	L5909	ZBK20SMAHN1A3	SWITCH DISCONNECTOR 3P 20A 690VAC	1
-S12	L5909	ZBK20SMAHN1A3	SWITCH DISCONNECTOR 3P 20A 690VAC	1

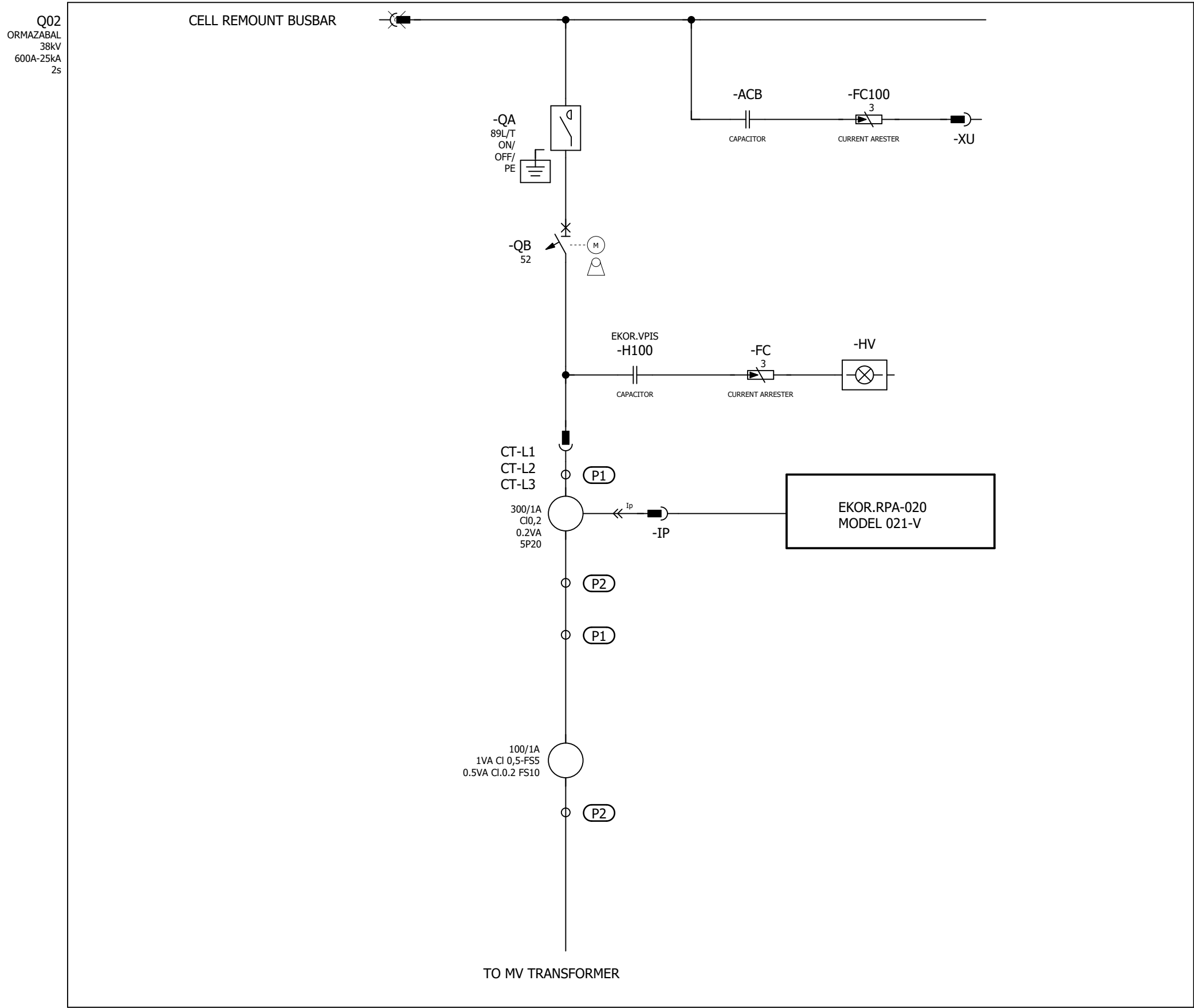
Plug overview

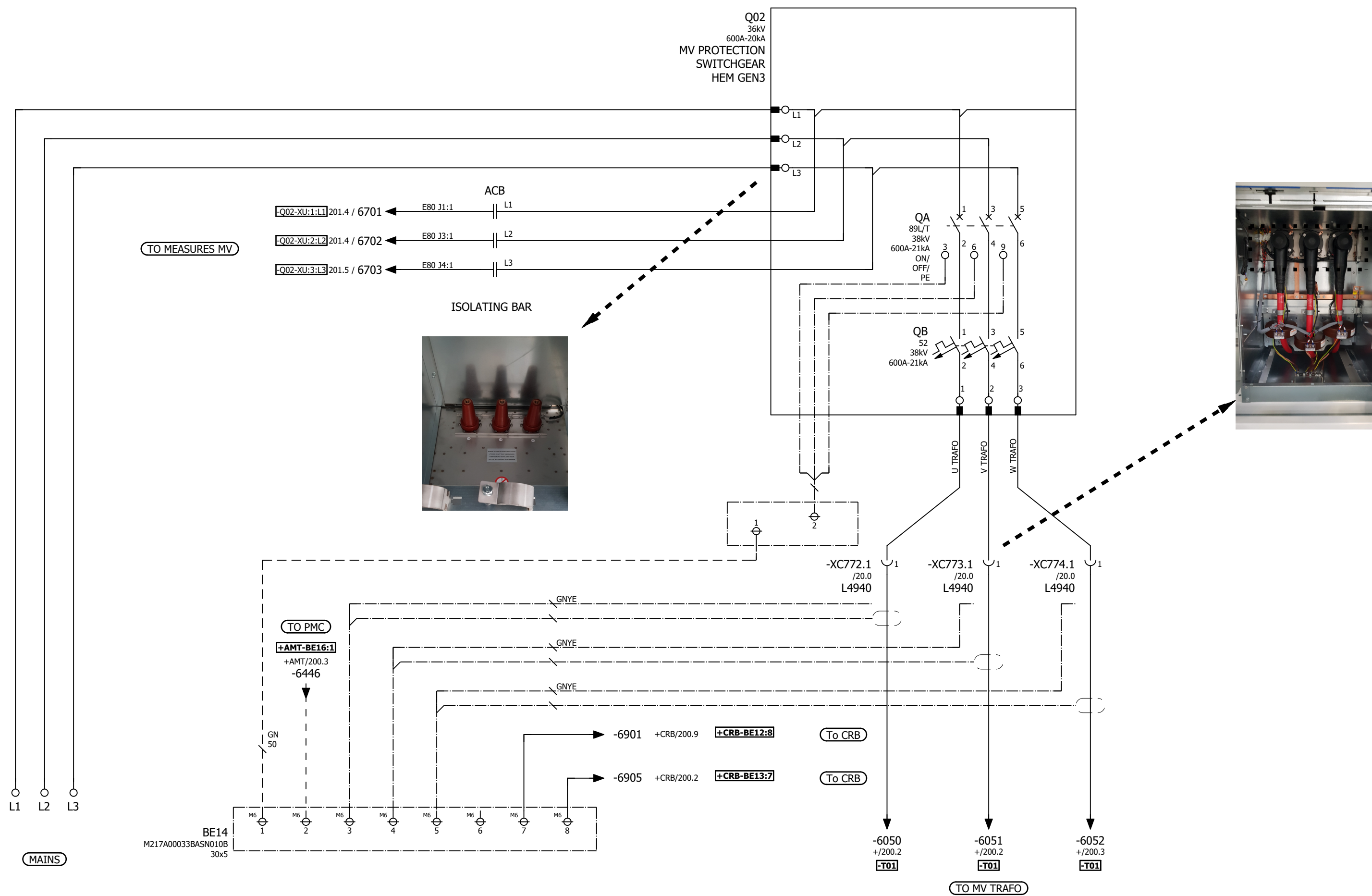
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+AMT.PMC-XC750.1	TRIPLE-LOK CONNECTOR AERIAL 2P ECON MODE	L9122_DUMMY	1
		L9149	2
=HEM++CDT+AMT.PMC-XC752.1	MATE-N-LOK CONNECTOR 15P DIG IN/OUT PCB	L2476	1
		E1203	9
=HEM++CDT+AMT.PMC-XC753.1	MATE-N-LOK CONNECTOR BASE 4P MV BOARD	L2004	1
		E1203	4
=HEM++CDT+AMT.PMC-XC754.1	TRIPLE-LOK CONNECTOR BASE 15P LINE CELL 1 & 2	L9135_DUMMY	1
		L9146	10
=HEM++CDT+AMT.PMC-XC755.1	TRIPLE-LOK CONNECTOR AERIAL 12P 120VDC	L9134_DUMMY	1
		L9149	3
		L9145	6
=HEM++CDT+AMT.PMC-XC756.1	TRIPLE-LOK AERIAL FANS 15P	L9136_DUMMY	1
		L9149	12
=HEM++CDT+AMT.PMC-XC757.2	MATE-N-LOK CONNECTOR 12P	E1375	1
		E1204	12
=HEM++CDT+AMT.PMC-XC758.1	AERIAL CONNECTOR 4P FANS POWER SUPPLY 6MM2	L7556	1
		L0040	3
=HEM++CDT+AMT.PMC-XC792.2	MATE-N-LOK AERIAL OUPUT POWER SUPPLY ECON MODE	L2003	1
		L5710	2

0	1	2	3	4	5	6	7	8	9

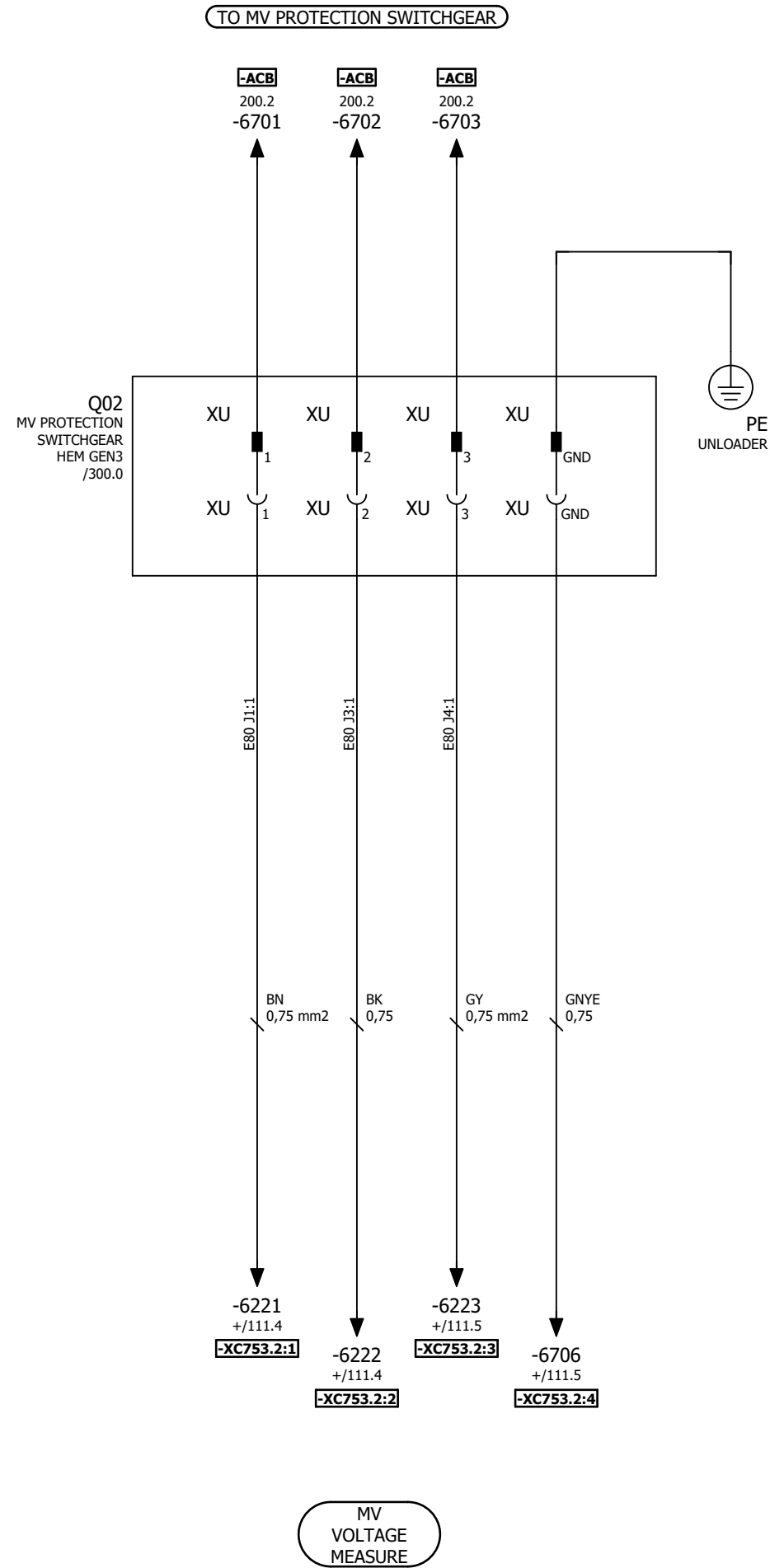
MV Single line diagram

VACUUM CIRCUIT BREAKER (V)

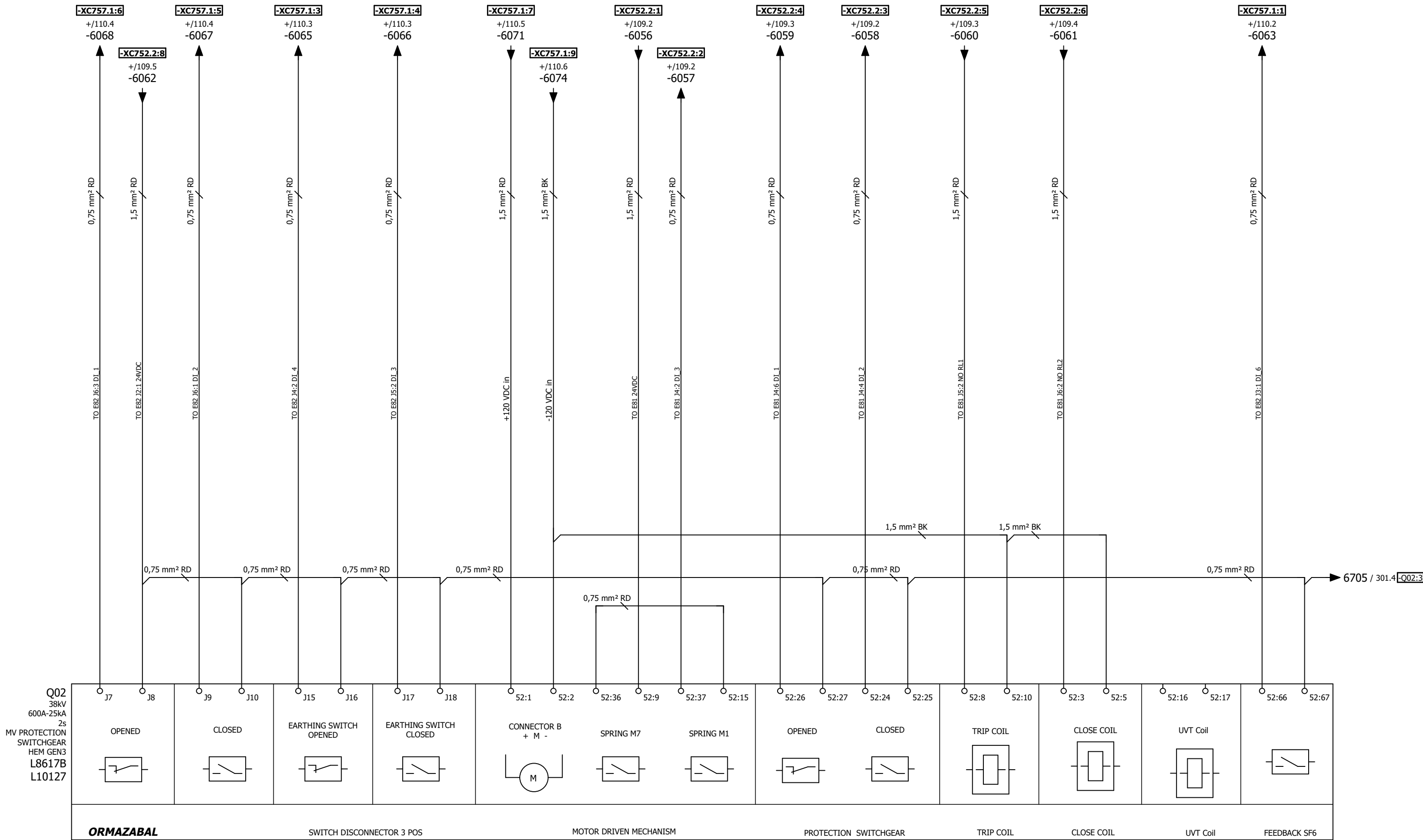




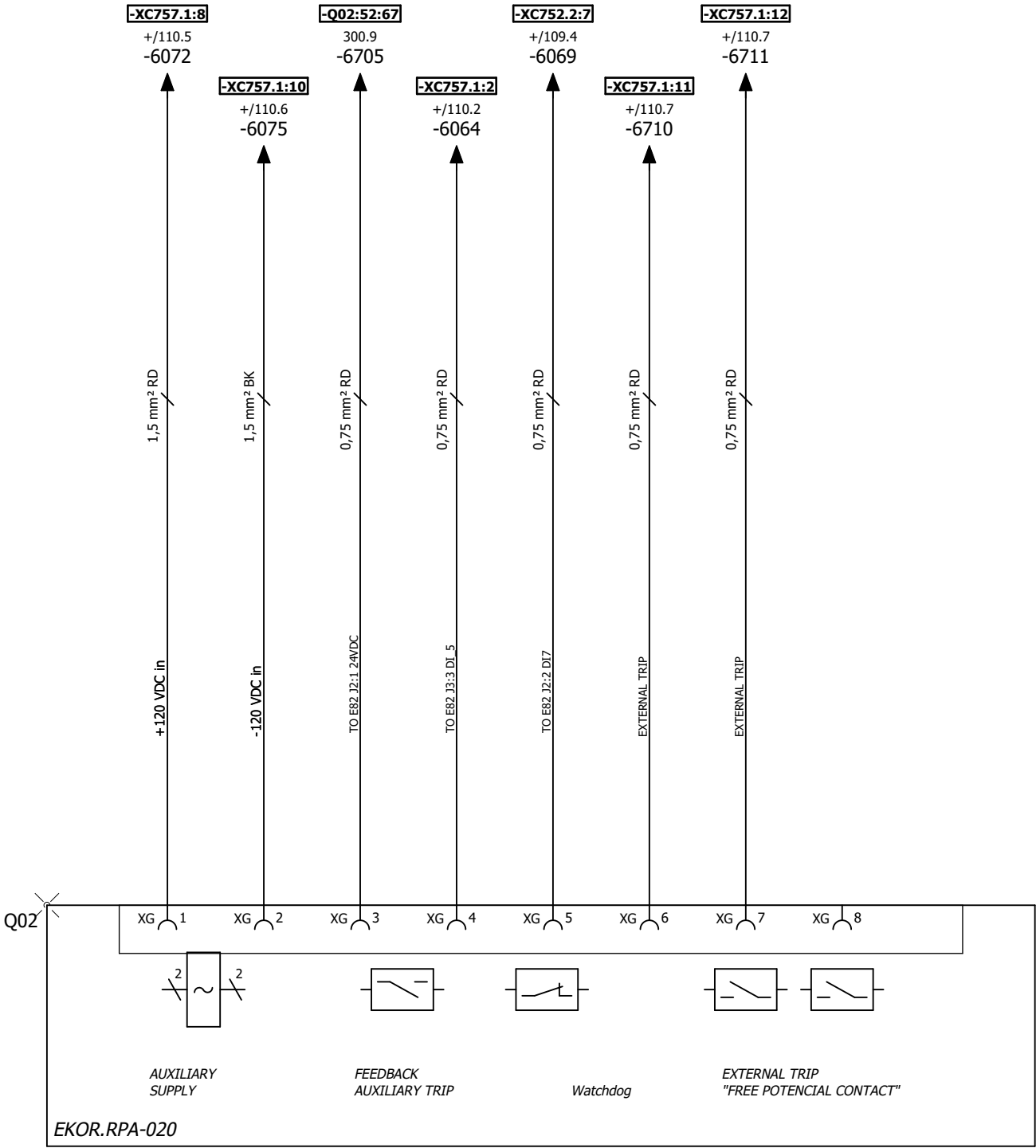
MV protection switchgear



MV protection switchgear



MV protection switchgear



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BE14	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1
-Q02	L8617B	L8617B	SWITCHGEAR HEM GEN3 V 38KV 600A 25KA/2S IAC	1
-Q02	L10127	C40R	4 NUMBER PADLOCK NUMERICAL WITHOUT KEY	1

Plug overview

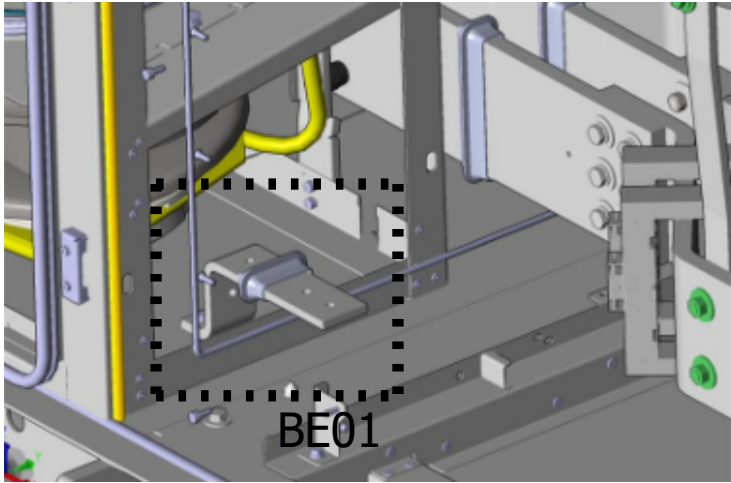
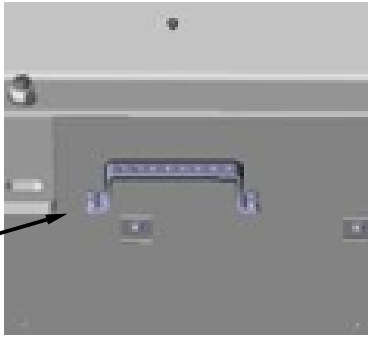
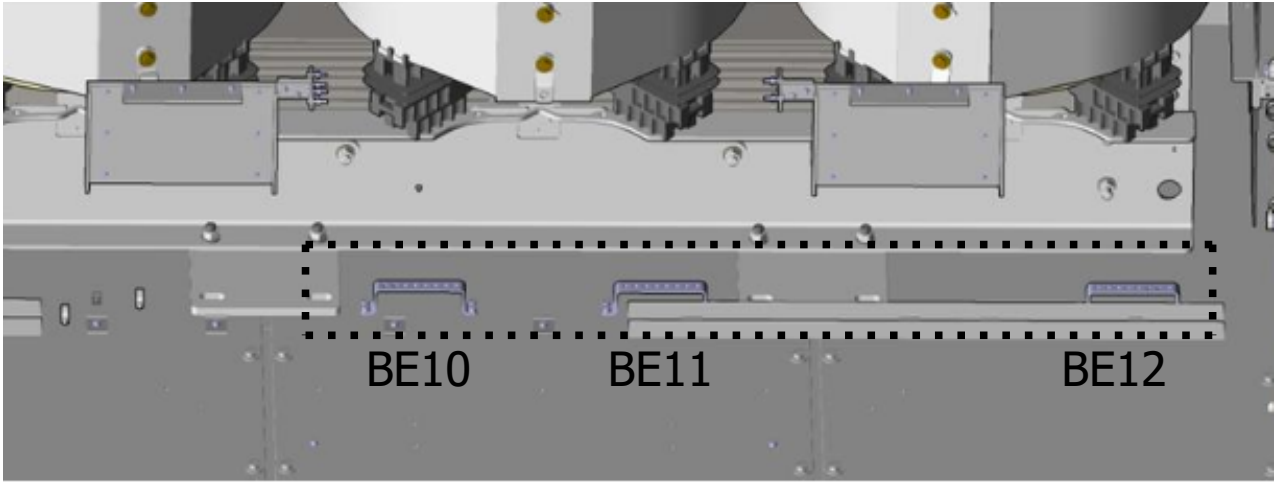
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+CMT-XC772.1	T-SHAPED SEPARABLE CONNECTOR 36KV 630A 95-150mm²	L4940	1
=HEM++CDT+CMT-XC773.1	T-SHAPED SEPARABLE CONNECTOR 36KV 630A 95-150mm²	L4940	1
=HEM++CDT+CMT-XC774.1	T-SHAPED SEPARABLE CONNECTOR 36KV 630A 95-150mm²	L4940	1
=HEM++CDT+CMT-XU	CONNECTOR MEASURE MT VOLTAGE WITHIN ORM CELL		



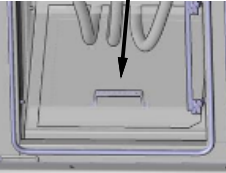
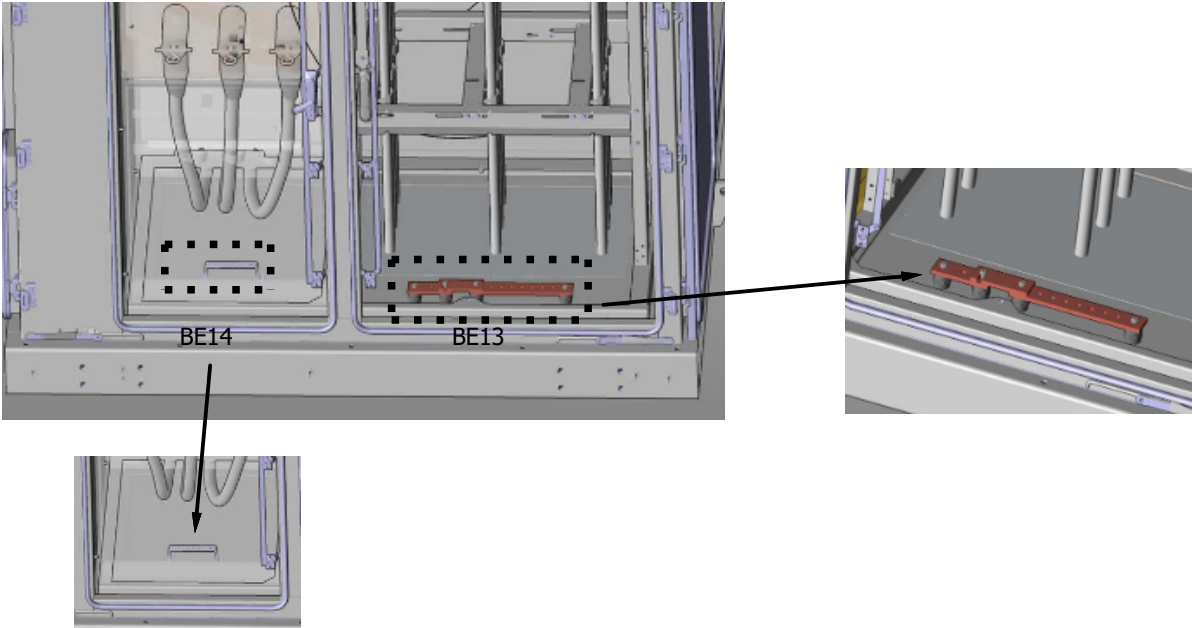
POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
CDT	MV TRANSFORMER CUBICLE		
CRB	CELL REMOUNT BUSBAR		
CRB CELL REMOUNT BUSBAR			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

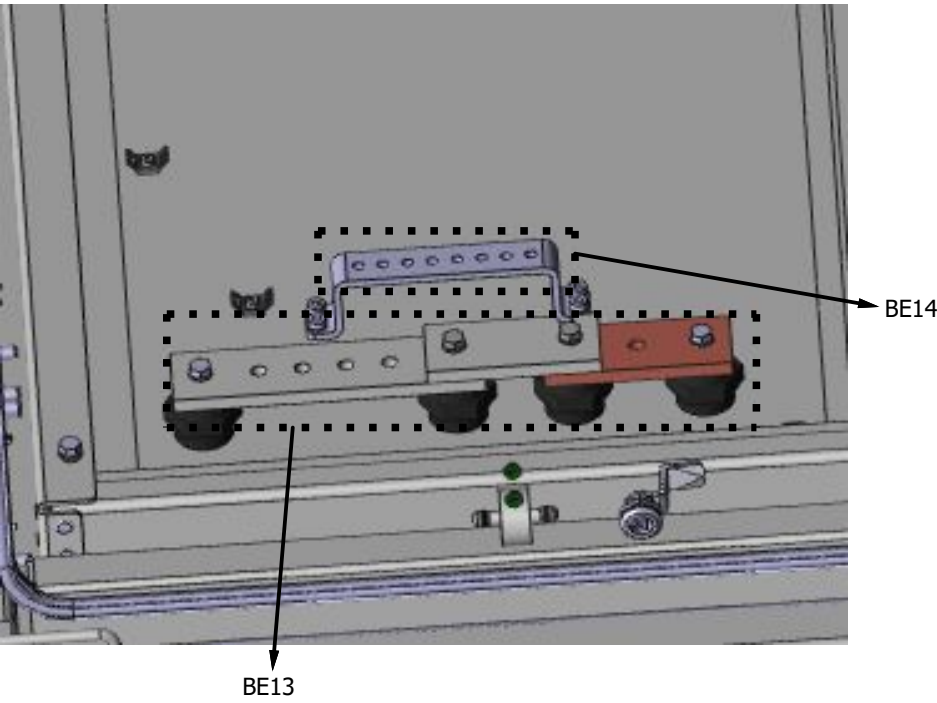
EARTH PLATES



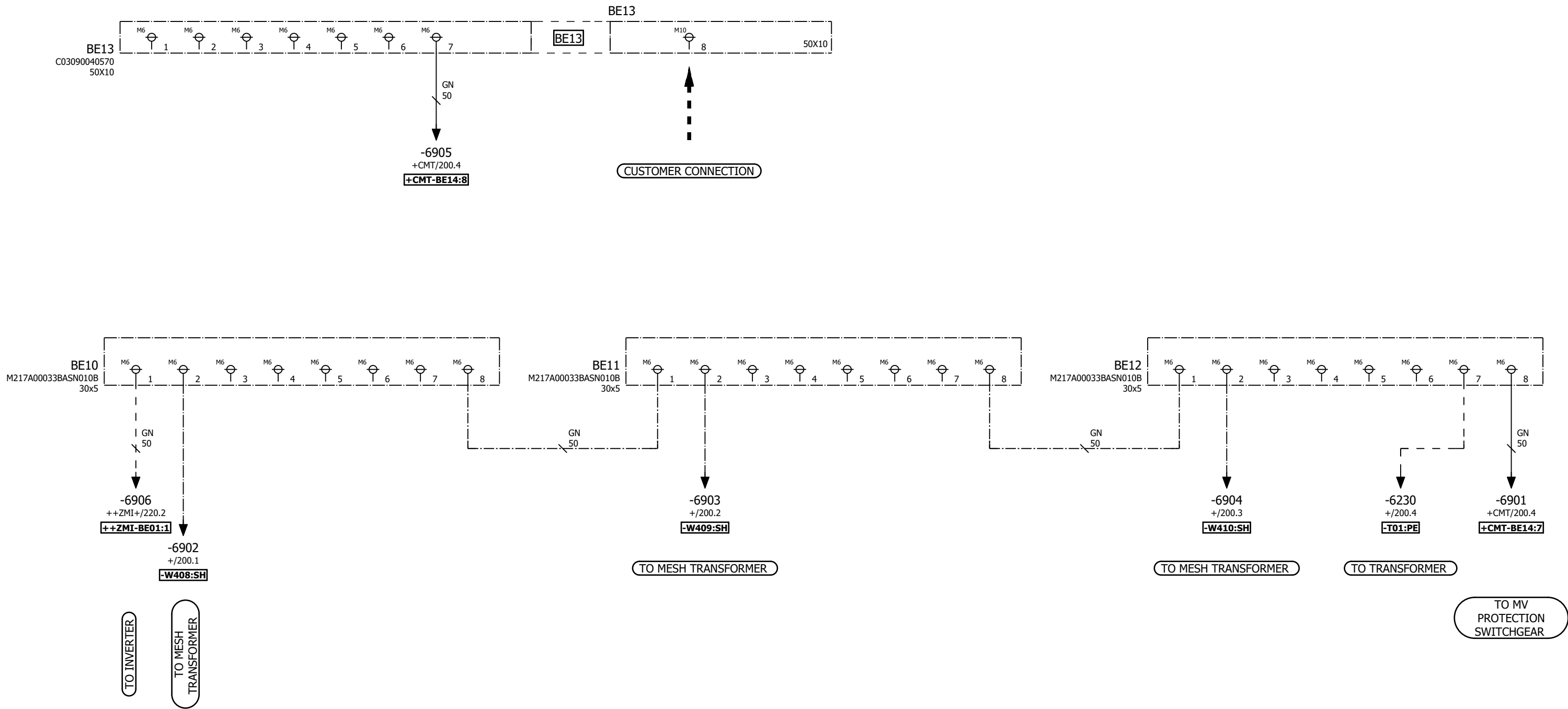
UL OPTION



IEC OPTION



PE connections



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-BE10	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1
-BE11	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1
-BE12	M217A00033BAS N010B	M217A00033BASN010B	EARTH BUSBAR 8 HOLES M6	1
-BE13	C03090040570	C03090040570	SECTIONABLE EARTH BUSBAR 8 LEADS M10 DRILL	1



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
CDT	MV TRANSFORMER CUBICLE		
PMV8	FAN PRE-ASSEMBLY 8		
PMV8 FAN PRE-ASSEMBLY 8			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV8

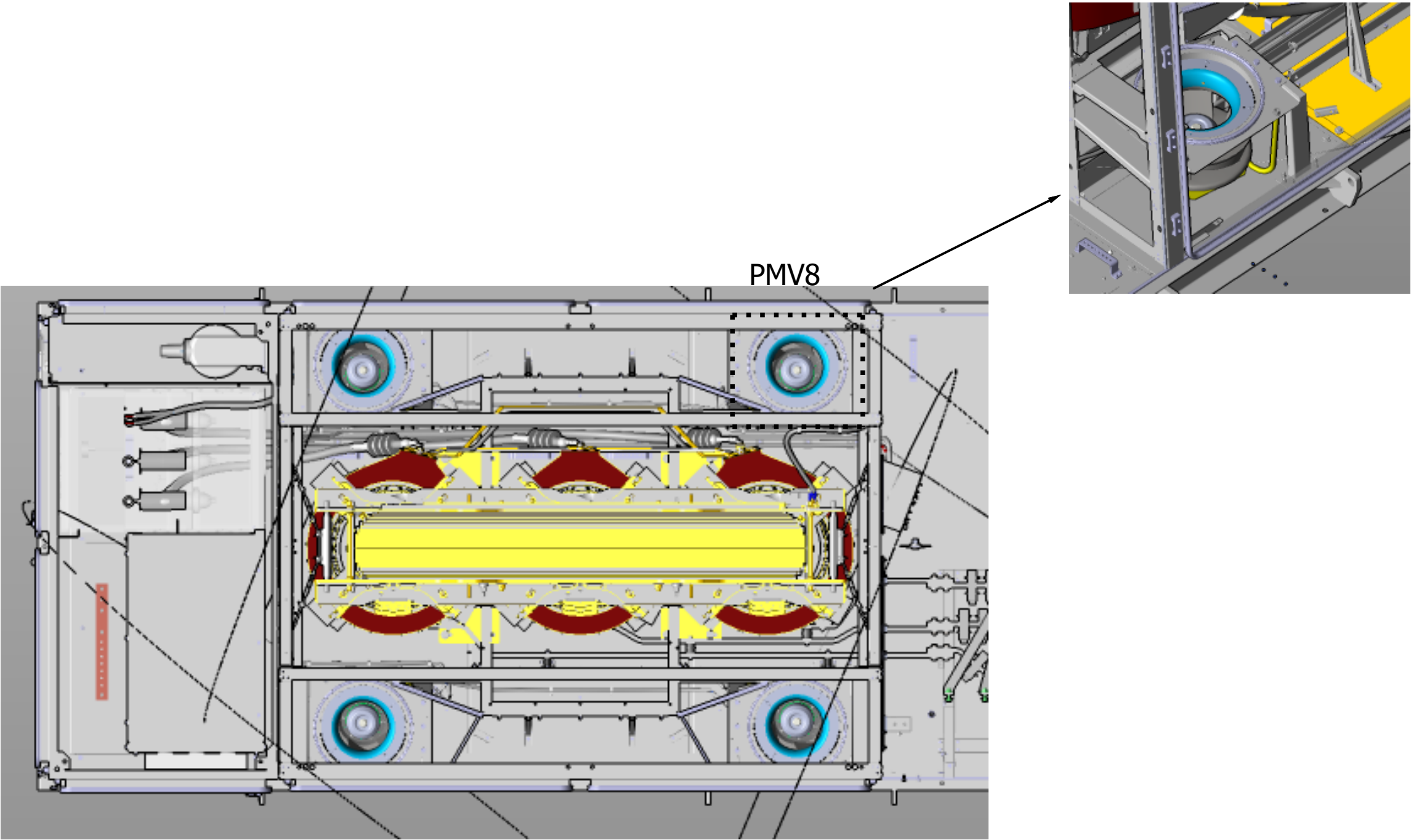


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

0	1	2	3	4	5	6	7	8	9

BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M08	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++CDT+PMV8-W0440.1	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV8-W0440.2	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV8-W0441	POWER SUPPLY FAN CABLE	L5472

Plug overview

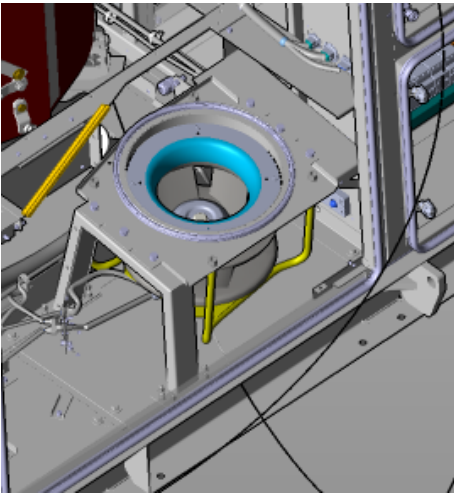
Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+PMV8-XC760	T CONNECTOR M12 RS485	L7338	1
=HEM++CDT+PMV8-XC760A	SHIELD		
=HEM++CDT+PMV8-XC760B	SHIELD		
=HEM++CDT+PMV8-XC761.2	MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 8	E1097	1
		L5710	3



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
CDT	MV TRANSFORMER CUBICLE		
PMV9	FAN PRE-ASSEMBLY 9		
PMV9 FAN PRE-ASSEMBLY 9			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV9



PMV9

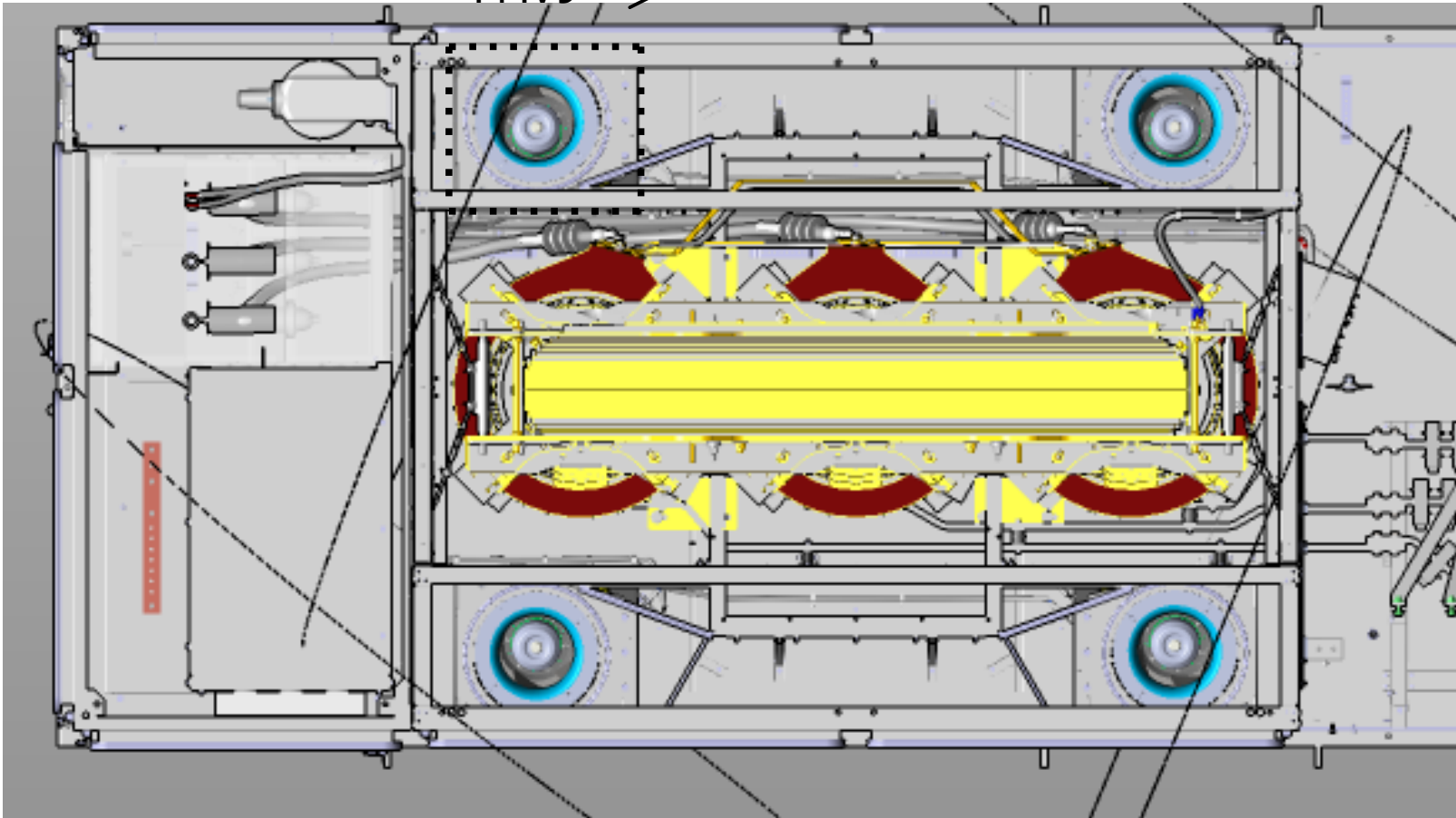
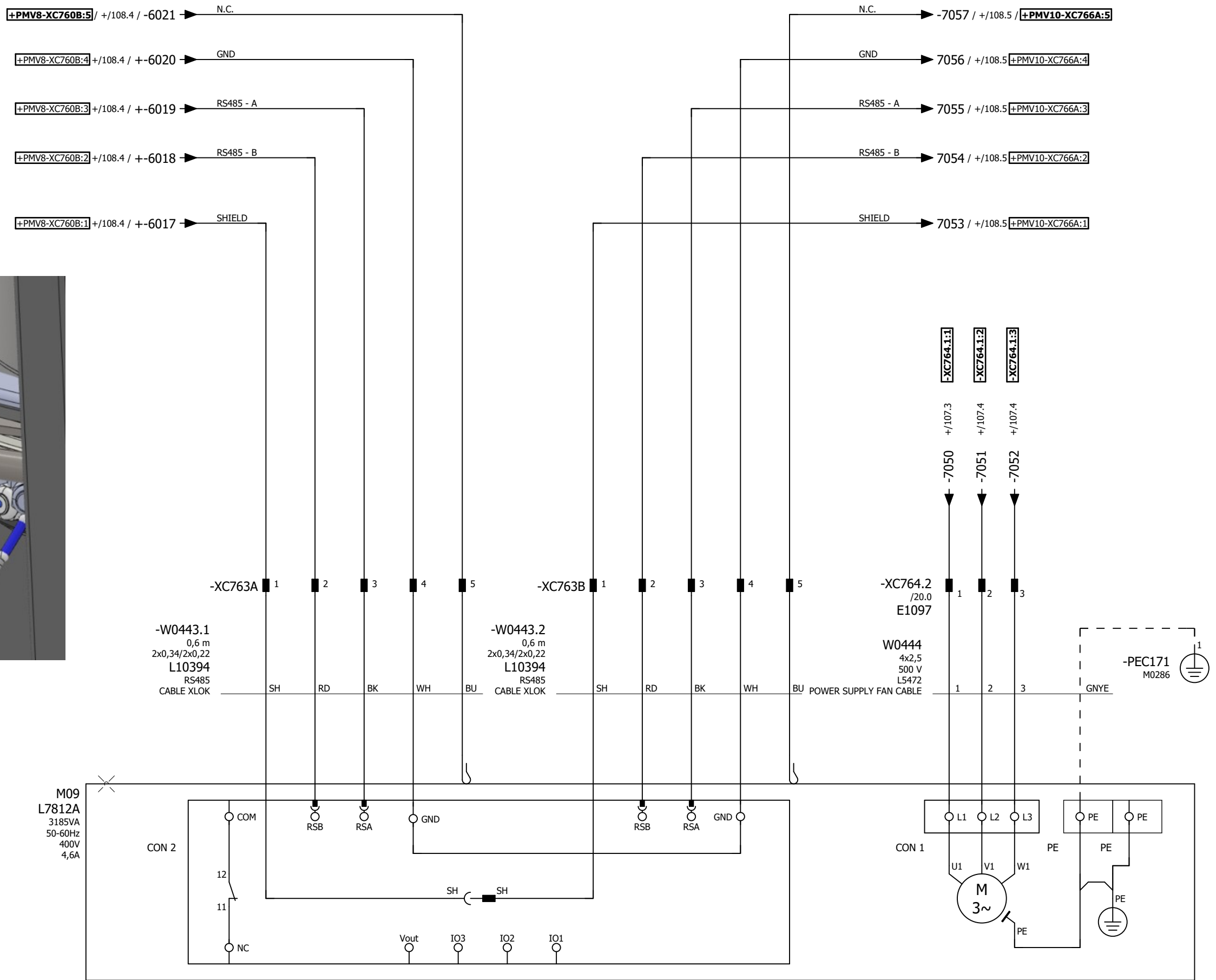
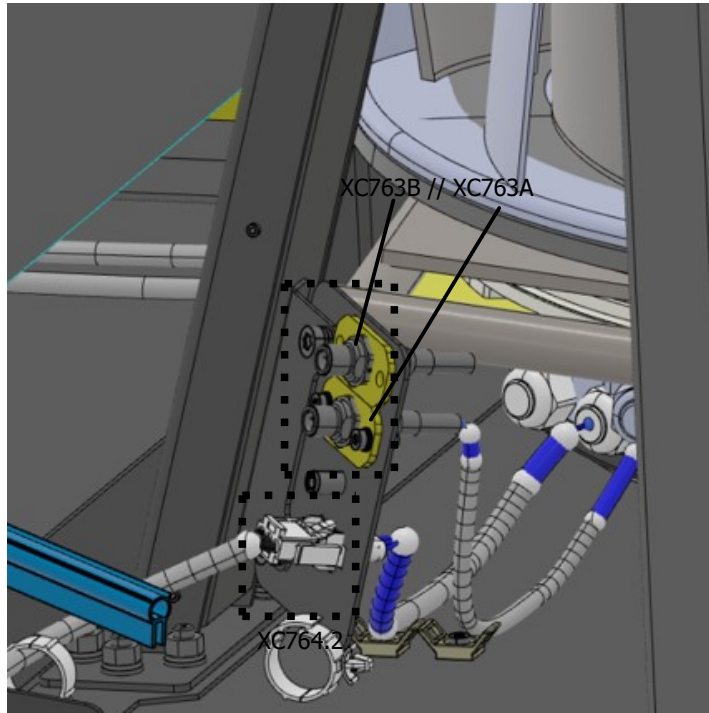


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

0	1	2	3	4	5	6	7	8	9
-XC764.2 = MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 9 -XC763 = T CONNECTOR M12 RS485									



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M09	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++CDT+PMV9-W0443.1	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV9-W0443.2	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV9-W0444	POWER SUPPLY FAN CABLE	L5472

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+PMV9-XC763	T CONNECTOR M12 RS485	L7338	1
=HEM++CDT+PMV9-XC763A	SHIELD		
=HEM++CDT+PMV9-XC763B	SHIELD		
=HEM++CDT+PMV9-XC764.2	MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 9	E1097	1
		L5710	3



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM	HEM INVERTER		
CDT	MV TRANSFORMER CUBICLE		
PMV10	FAN PRE-ASSEMBLY 10		
PMV10 FAN PRE-ASSEMBLY 10			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV10

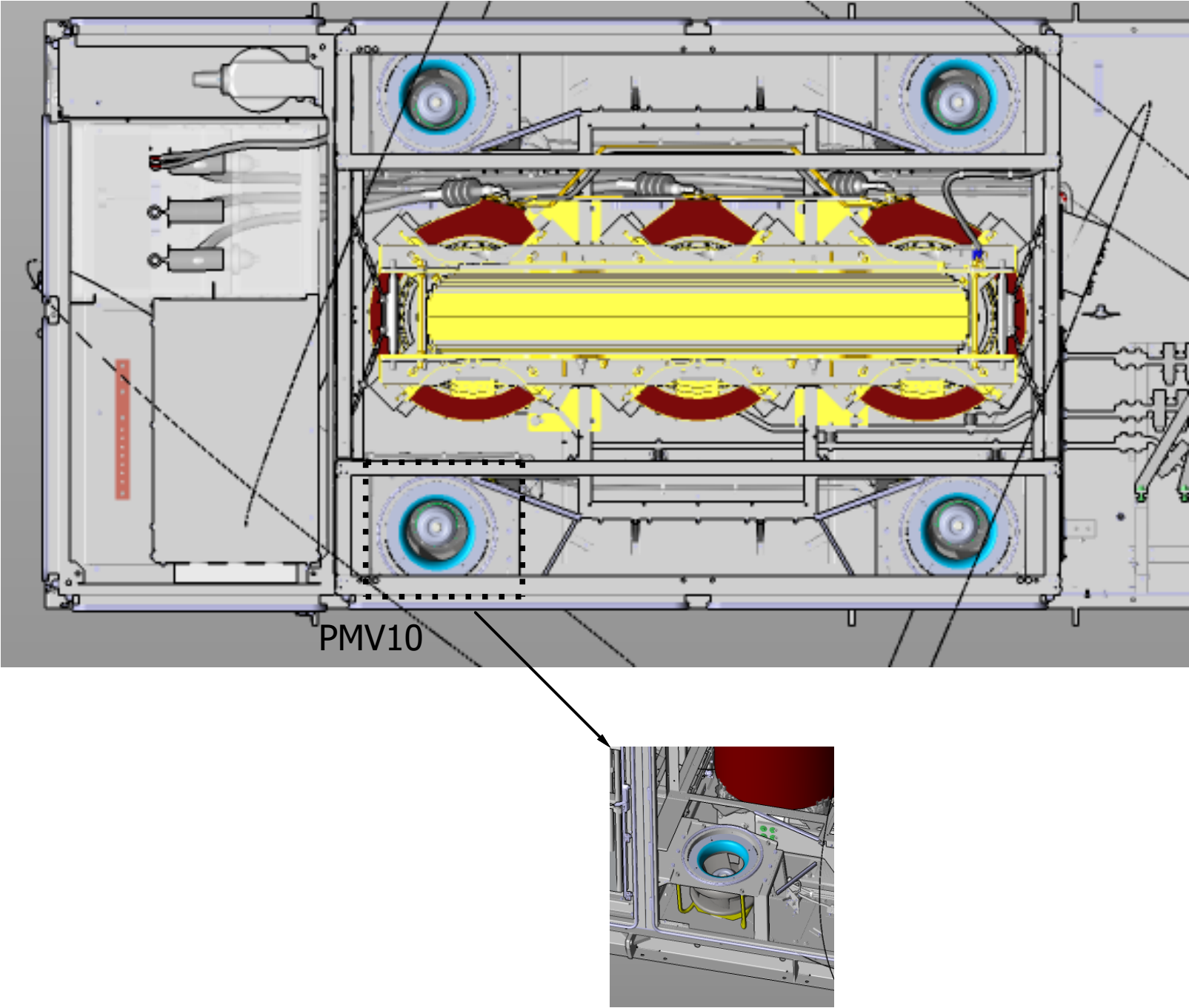
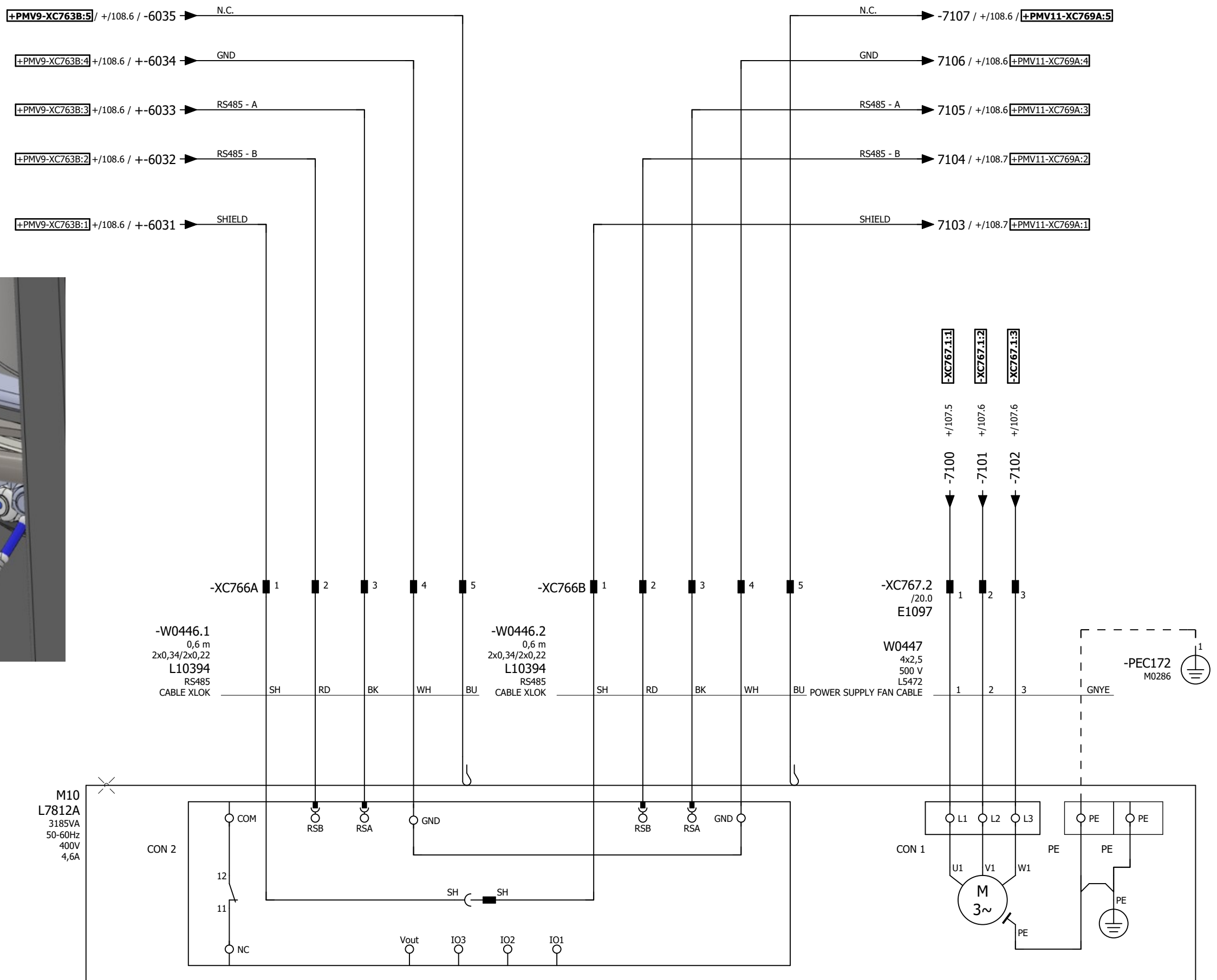
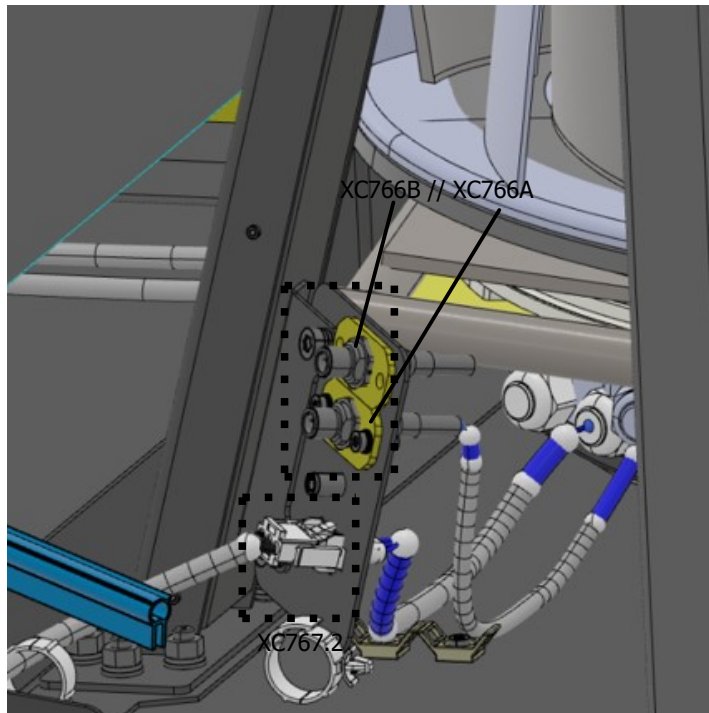


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

0	1	2	3	4	5	6	7	8	9
-XC767.2 = MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 10 -XC766 = T CONNECTOR M12 RS485									



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M10	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++CDT+PMV10-W0446.1	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV10-W0446.2	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV10-W0447	POWER SUPPLY FAN CABLE	L5472

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+PMV10-XC766	T CONNECTOR M12 RS485	L7338	1
=HEM++CDT+PMV10-XC766A	SHIELD		
=HEM++CDT+PMV10-XC766B	SHIELD		
=HEM++CDT+PMV10-XC767.2	MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 10	E1097	1
		L5710	3



POWER ELECTRONICS
POLÍGONO PLA DE CARRASES B.
CV35 SALIDA 30

HEM		HEM INVERTER	
CDT		MV TRANSFORMER CUBICLE	
PMV11		FAN PRE-ASSEMBLY 11	
PMV11 FAN PRE-ASSEMBLY 11			
Created on	06/02/2024	Creator	I+D 1
Modification date	12/09/2024		

PRE-ASSEMBLY LAYOUT PMV11

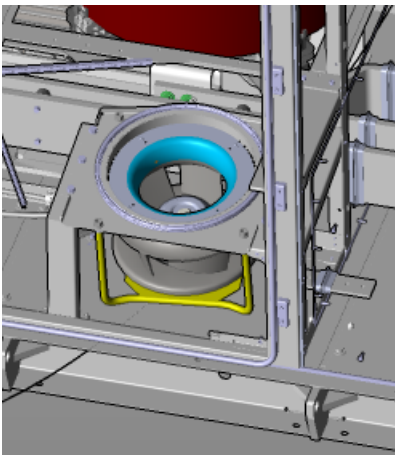
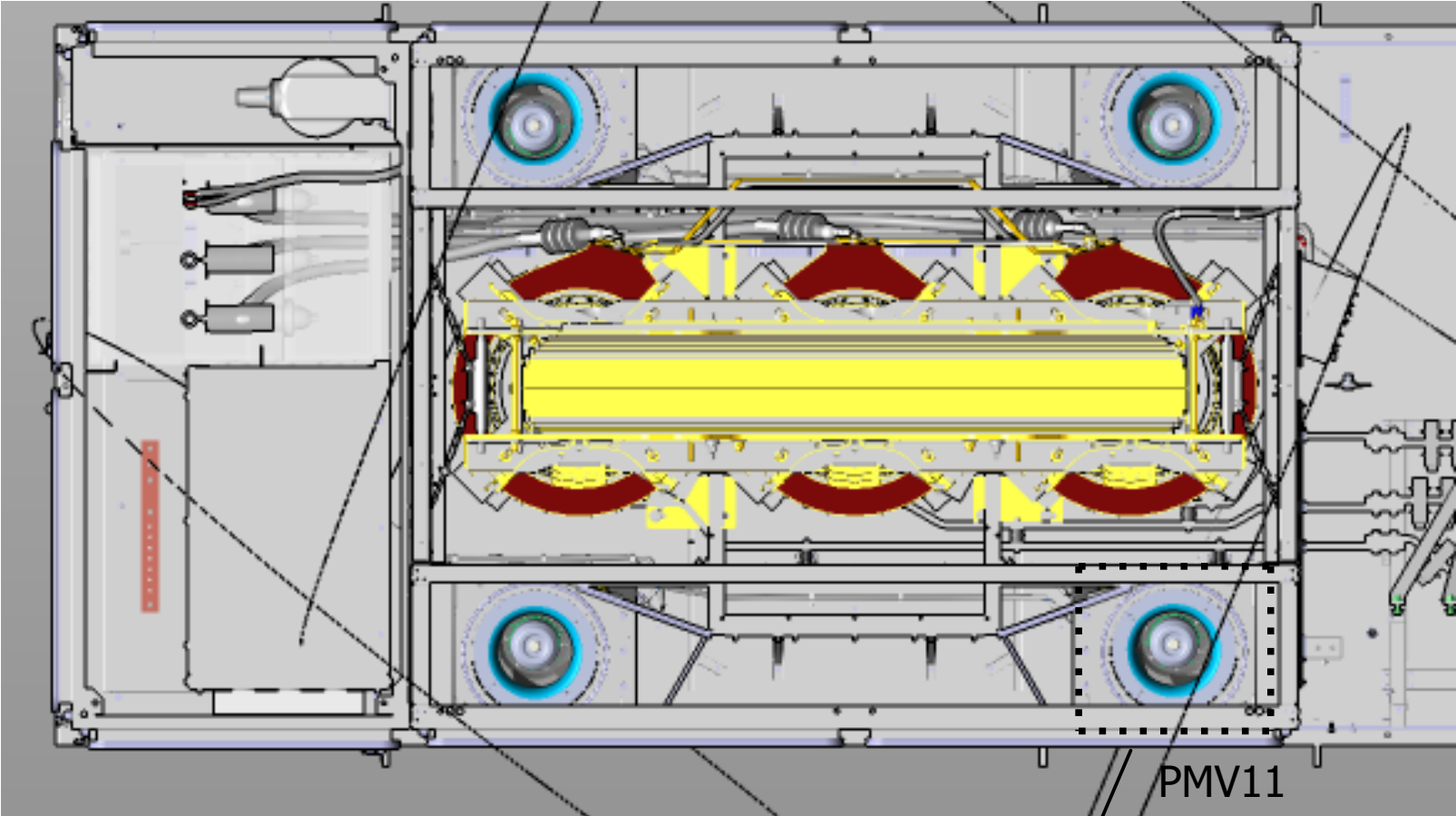


IMAGE AS A GUIDANCE INDICATING EXISTING MOUNTING SITES
(MAY VARY ACCORDING TO THE CONFIGURATION OF EACH CUSTOMER).

0	1	2	3	4	5	6	7	8	9
-XC770.2 = MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 11 -XC771.1 = END LINE RESISTANCE RS485 -XC769 = T CONNECTOR M12 RS485									



BILL OF MATERIALS

Identifier (DT)	ERP number	Manufacturer's reference	Description	Number of units meters / units
-M11	L7812A	A CONFIRMAR CON EBM	FAN CENT. 4455M3/H 400VAC 2950W	1

Cable overview

Cable designation	Function text	ERP
=HEM++CDT+PMV11-W0449.1	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV11-W0449.2	RS485 CABLE XLOK	L10394
=HEM++CDT+PMV11-W0450	POWER SUPPLY FAN CABLE	L5472

Plug overview

Plug designation	Function text	Plug	
		ERP	CANTIDAD
=HEM++CDT+PMV11-XC769	T CONNECTOR M12 RS485	L7338	1
=HEM++CDT+PMV11-XC769A	SHIELD		
=HEM++CDT+PMV11-XC769B	SHIELD		
=HEM++CDT+PMV11-XC770.2	MATE-N-LOK BASE POWER SUPPLY FAN ENGINE 11	E1097	1
		L5710	3
=HEM++CDT+PMV11-XC771.1	END LINE RESISTANCE RS485	L10400	1